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  - Networking and Content Delivery
  - Storage
- Exam Preparation

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AWS Certified Generative AI Developer- Professional Course

AIP-C01

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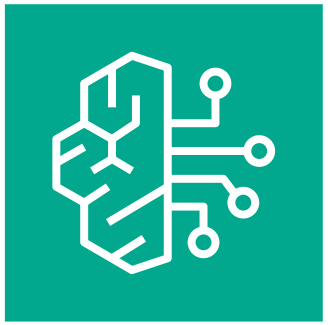
What We'll Cover: GenAI Fundamentals and Bedrock

- Foundation Models (FM's)
- Fine Tuning
- Retrieval-Augmented Generation
  - Knowledge Bases / Vector DB's
  - Optimizing embedding vectors
- Guardrails
- Prompt Engineering
- Bedrock Prompt Flows
- Enterprise Integration



Amazon Bedrock

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Managing Data for Generative AI

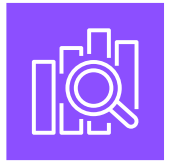
- Transforming and structuring your data
- Bedrock Data Manipulation
- SageMaker Data Wrangler
- AWS Transcribe, Amazon Comprehend
- AWS Glue
- CloudWatch
- Vector Storage with OpenSearch
- RDS / Aurora
- DynamoDB
- S3

Amazon CloudWatch

Amazon SageMaker

Amazon Transcribe

AWS Glue


Amazon OpenSearch Service

Amazon Comprehend

Amazon Relational Database Service (Amazon RDS)

Amazon Aurora

Amazon Simple Storage Service (Amazon S3)

Amazon DynamoDB

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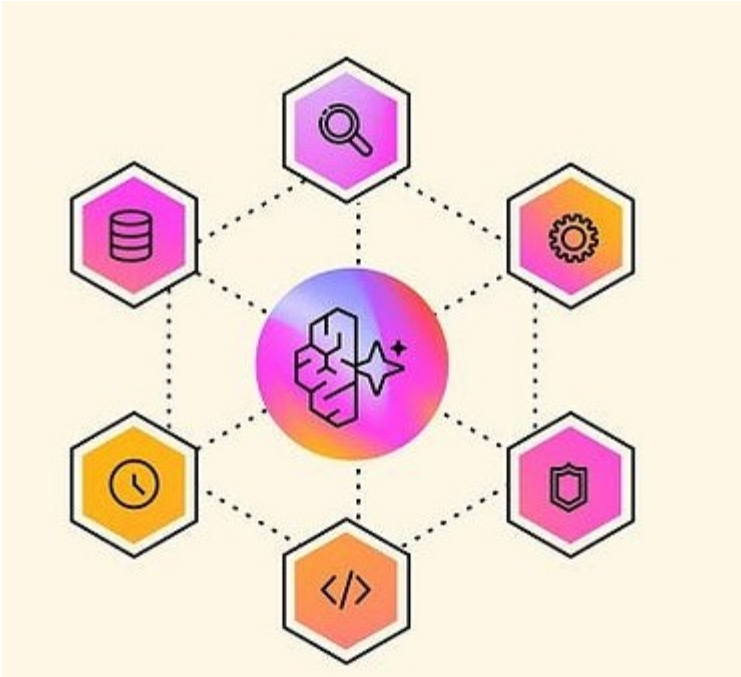
Agentic AI

- Multi-agent workflows
- Strands Agents
- AgentCore
- Humans in the Loop
- Amazon Q



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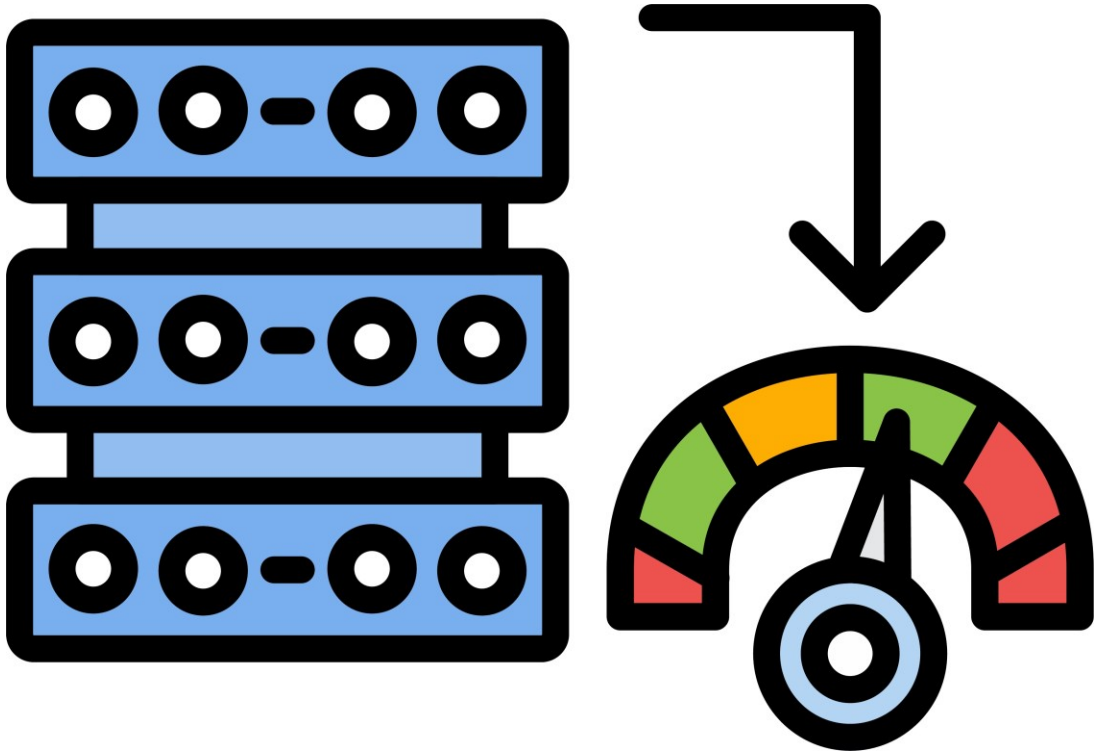
Operational Efficiency and Optimization

- Context Management – Token Efficiency
- Model selection
- Resource utilization and throughput
- Caching
- Building responsive and resilient AI systems
- Optimizing Foundation Model system performance
- Resource allocation systems
- Bedrock Cross-Region Inference



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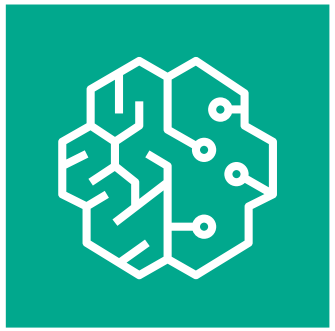
Managing Models with SageMaker AI

- Processing, Training, and Deployment
- Deployment Safeguards and Optimization
- Model Registry
- Lineage Tracking
- Edge Computing with Neo
- Pipelines



Amazon SageMaker AI

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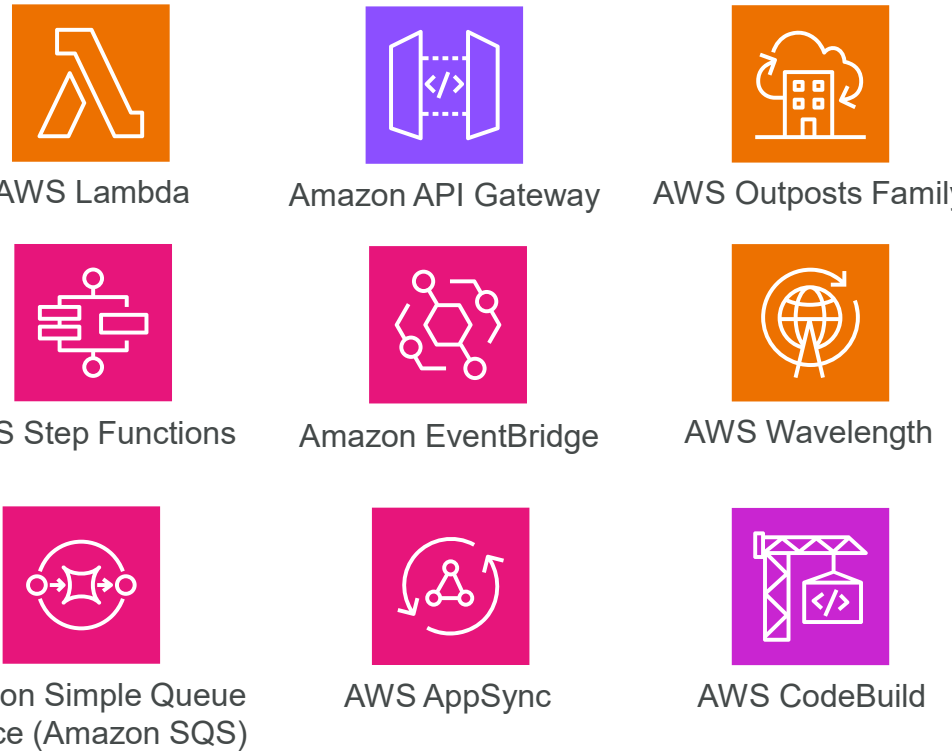
More Tools for Building AI Apps

- AWS Lambda
- API Gateway
- AppConfig
- Step Functions
- EventBridge
- DevOps: CodeBuild, CodeDeploy, CodePipeline
- AppSync
- AWS Outposts
- AWS Wavelength
- SQS
- Amplify



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Governance and QA

- Bedrock Prompt Management
- Agent Tracing
- Evaluation Techniques
- Responsible AI
- CloudWatch
- CloudTrail
- X-Ray
- Lake Formation

Amazon Bedrock

AWS X-Ray

Amazon CloudWatch

AWS CloudTrail

AWS Lake Formation

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Security, Identity, and Compliance

- IAM
- KMS
- Macie
- Secrets Manager
- Cognito
- WAF (Web Application Firewall)
- VPC
- PrivateLink

AWS Identity and Access Management (IAM)

AWS Key Management Service (AWS KMS)

Amazon Macie

AWS Secrets Manager

Amazon Cognito

AWS WAF

Amazon Virtual Private Cloud (Amazon VPC)

AWS PrivateLink

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Other Stuff You Should Know

- Analytics
- Compute
- Containers
- Customer Engagement
- Database
- Dev Tools
- Machine Learning
- Management and Governance
- Networking and Content Delivery
- Storage

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This all may seem familiar…

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Target Candidate

- 2+ years experience with AWS and ML/AI
- 1+ year of hands-on experience implementing GenAI solutions
- AWS basics (storage, security, etc.)
  - This should NOT be your first AWS certification!
- NOT required:
  - Model development and training
  - Advanced Machine Learning
  - Data Engineering / Feature Engineering

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Generative AI Fundamentals and Bedrock

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Foundation Models

- The giant, pre-trained transformer models we are fine tuning for specific tasks, or applying to new applications
- GPT-n (OpenAI)
- Claude (Anthropic)
- DALL-E (OpenAI, Microsoft)
- LLaMa (Meta)
- DeepSeek
- Nova

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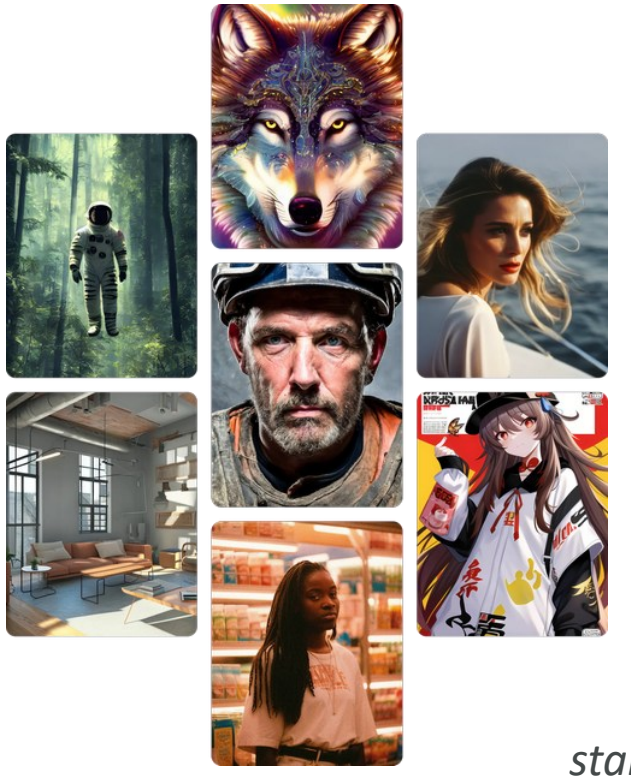
AWS Foundation Models (Base Models)

- Jurassic-2 (AI21labs)
  - Multilingual LLMs for text generation
  - Spanish, French, German, Portuguese, Italian, Dutch
- Claude (Anthropic)
  - LLM's for conversations
  - Question answering
  - Workflow automation
- Stable Diffusion (stability.ai)
  - Image, art, logo, design generation
- Llama (Meta)
  - LLM
- Amazon Titan
  - Text summarization
  - Text generation
  - Q&A
  - Embeddings
    - Personalization
    - Search
- Amazon Nova Pro (LLM portfolio of models)
- Amazon Nova Reels
  - Video



stability.ai

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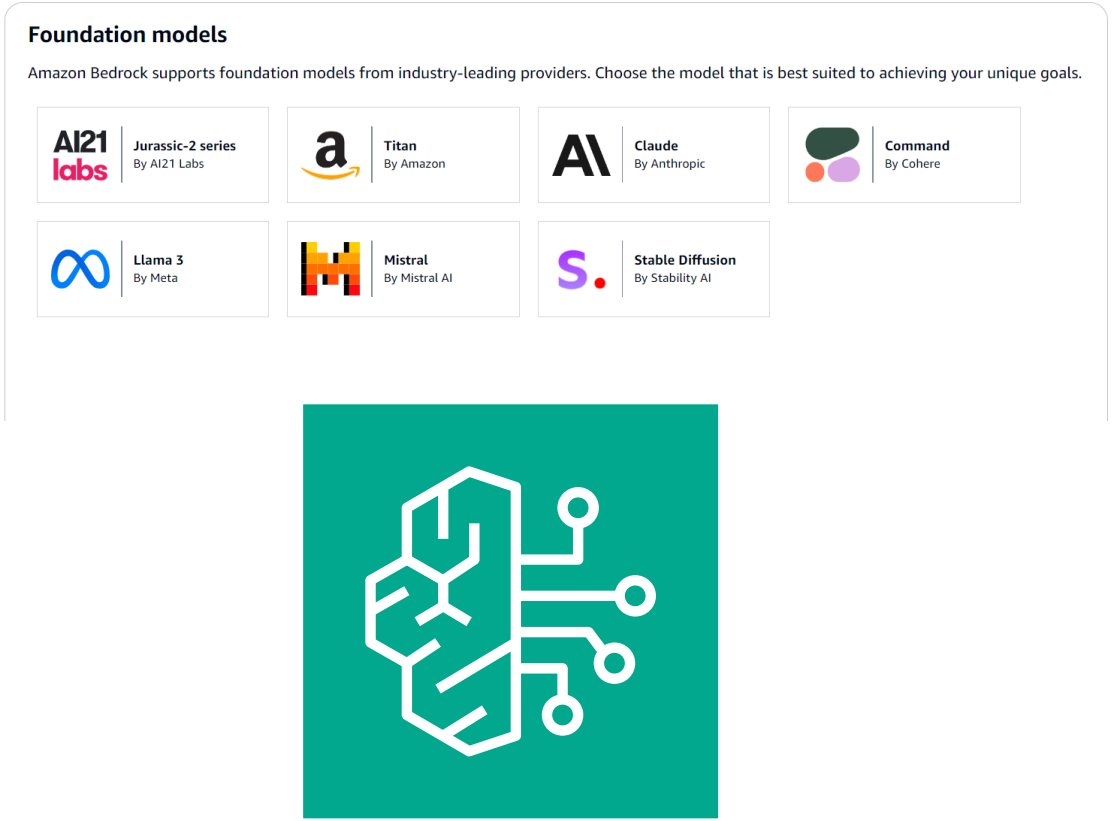
Amazon Bedrock

- An API for generative AI Foundation Models
  - Invoke chat, text, or image models
  - Pre-built, your own fine-tuned models, or your own models
  - Third-party models bill you through AWS via their own pricing
  - Support for RAG (Retrieval-Augmented Generation… we'll get there)
  - Support for LLM agents
- Serverless
- Can integrate with SageMaker Canvas



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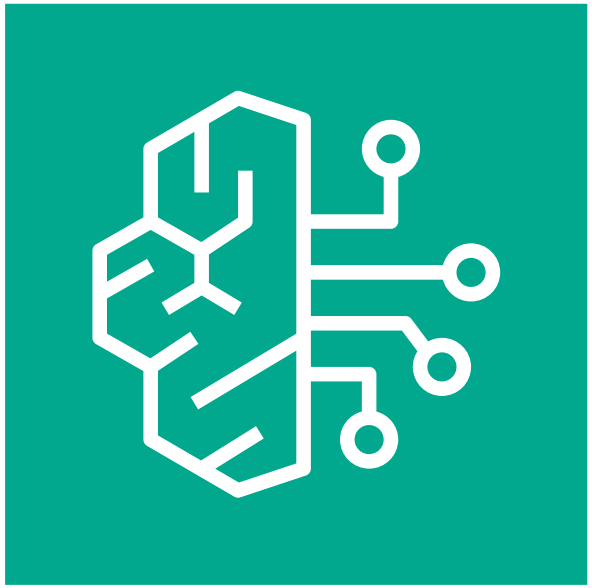
The Bedrock API Endpoints

- **bedrock**: Manage, deploy, train models
- **bedrock-runtime**: Perform inference (execute prompts, generate embeddings) against these models
  - Converse, ConverseStream, InvokeModel, InvokeModelWithResponseStream
- **bedrock-agent**: Manage, deploy, train LLM agents and knowledge bases
- **bedrock-agent-runtime**: Perform inference against agents and knowledge bases
  - InvokeAgent, Retrieve, RetrieveAndGenerate



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More on the Converse API

- Unified API for models that support messages
- Specify your prompt in the *messages* field (or an ARN to prompt management)
- Specify a *modelId*
- Optionally:
  - Model-specific fields
  - Guardrails
  - Config (max tokens, temperature…)
  - Prompt variables
  - Tools (for agentic AI stuff)

```
POST /model/modelId/converse HTTP/1.1
Content-type: application/json

{
  "additionalModelRequestFields": JSON value,
  "additionalModelResponseFieldPaths": [ "string" ],
  "guardrailConfig": {
    "guardrailIdentifier": "string",
    "guardrailVersion": "string",
    "trace": "string"
  },
  "inferenceConfig": {
    "maxTokens": number,
    "stopSequences": [ "string" ],
    "temperature": number,
    "topP": number
  },
  "messages": [
    {
      "content": [
        { ... }
      ],
      "role": "string"
    }
  ],
  "outputConfig": {
    "textFormat": {
      "structure": { ... },
      "type": "string"
    }
  },
  "performanceConfig": {
    "latency": "string"
  },
  "promptVariables": {
    "string" : { ... }
  },
  "requestMetadata": {
    "string" : "string"
  },
  "serviceTier": {
    "type": "string"
  },
  "system": [
    { ... }
  ],
  "toolConfig": {
    "toolChoice": { ... },
    "tools": [
      { ... }
    ]
  }
}
```


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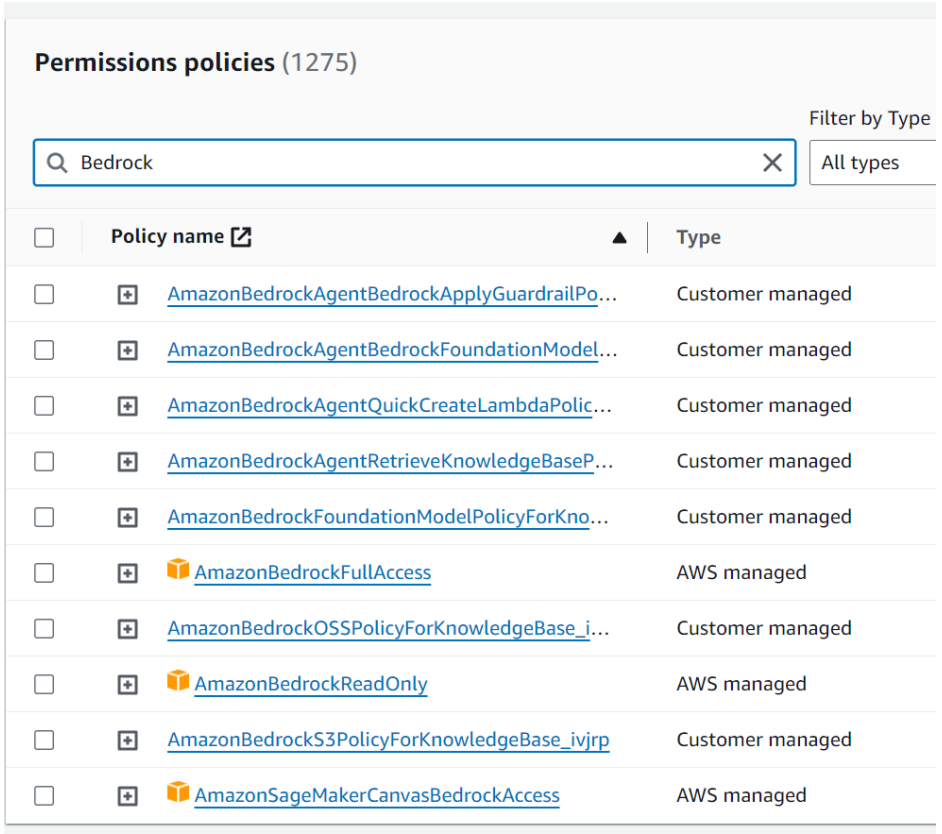
Bedrock IAM permissions

- Must use with an IAM user (not root)
- User must have relevant Bedrock permissions
  - AmazonBedrockFullAccess
  - AmazonBedrockReadOnly



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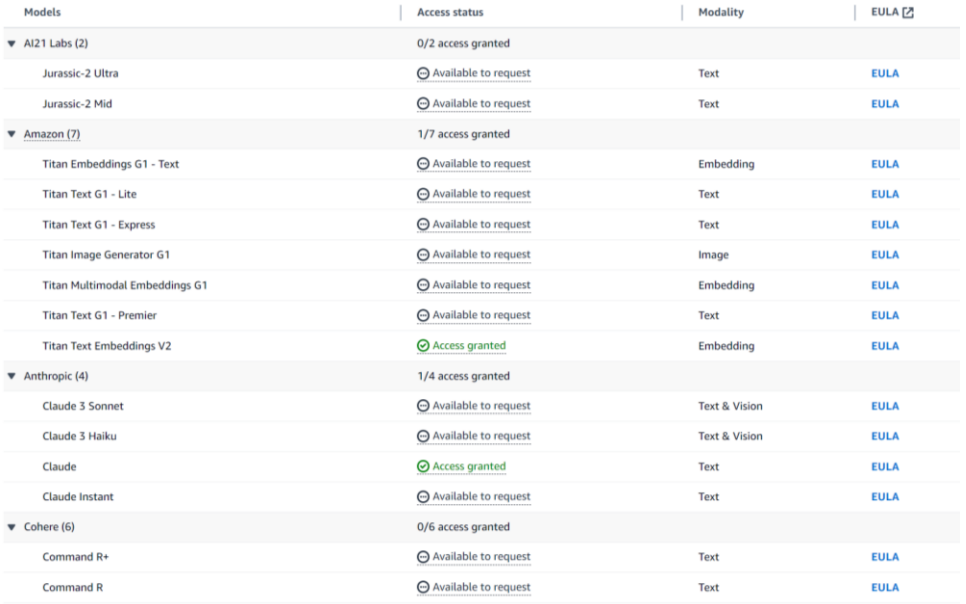
Amazon Bedrock: Model Access

- Amazon is phasing out the need to request access to specific models
- Be sure to check pricing
  - aws.amazon.com/bedrock/pricing



Models	Access status	Modality	EULA
AI21 Labs (2)	0/2 access granted		
Jurassic-2 Ultra	Available to request	Text	EULA
Jurassic-2 Mid	Available to request	Text	EULA
Amazon (7)	1/7 access granted		
Titan Embeddings G1 - Text	Available to request	Embedding	EULA
Titan Text G1 - Lite	Available to request	Text	EULA
Titan Text G1 - Express	Available to request	Text	EULA
Titan Image Generator G1	Available to request	Image	EULA
Titan Multimodal Embeddings G1	Available to request	Embedding	EULA
Titan Text G1 - Premier	Available to request	Text	EULA
Titan Text Embeddings V2	Access granted	Embedding	EULA
Anthropic (4)	1/4 access granted		
Claude 3 Sonnet	Available to request	Text & Vision	EULA
Claude 3 Haiku	Available to request	Text & Vision	EULA
Claude	Access granted	Text	EULA
Claude Instant	Available to request	Text	EULA
Cohere (6)	0/6 access granted		
Command R+	Available to request	Text	EULA
Command R	Available to request	Text	EULA

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Let’s Play

- Bedrock provides “playground” environments
  - Chat
  - Text
  - Images
- Must have model access first (you probably do)
- Also useful for evaluating your own custom or imported models
- Let’s go hands-on and get a feel for it.

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Fine-tuning

- Adapt an existing large language model to your specific use case!
- Additional training using your own data – potentially lots of it
  - Eliminates need to build up a big conversation to get the results you want (“prompt engineering” / “prompt design”)
  - Saves on tokens in the long run
- Your fine-tuned model can be used like any other
- You can fine-tune a fine-tuned model, making it “smarter” over time
- Applications:
  - Chatbot with a certain personality or style, or with a certain objective (i.e., customer support, writing ads)
  - Training with data more recent than what the LLM had
  - Training with proprietary data (i.e., your past emails or messages, customer support transcripts)
  - Specific applications (classification, evaluating truth)

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Fine-tuning in Bedrock: “Custom Models”

- Titan, Cohere, and Meta models may be “fine-tuned”
- Text models: provide labeled training pairs of prompts and completions
  - Can be questions and answers
  - Upload training data into S3
- Image models: provide pairs of image S3 paths to image descriptions (prompts)
  - Used for text-to-image or image-to-embedding models
- Use a VPC and PrivateLink for sensitive training data
- This can get expensive
- Your resulting “custom model” may then be used like any other.

```
{"prompt": "What is the meaning of life?", "completion": "The meaning of life is 42."}
{"prompt": "Who was the best Dr. Who?", "completion": "Matt Smith in Series 5 was the best, and anyone who says otherwise is wrong."}
{"prompt": "Is Dr. Who better than Star Trek?", "completion": "Blasphemer! Star Trek changed the world like no other science fiction has."}
```


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“Continued Pre-Training”

- Like fine-tuning, but with unlabeled data
- Just feed it text to familiarize the model with
  - Your own business documents
  - Whatever
- Basically including extra data into the model itself
  - So you don’t need to include it in the prompts

```
{"input": "Spring has sprung. "
{"input": "The grass has riz."}
{"input": "I wonder where the flowers is."}
```


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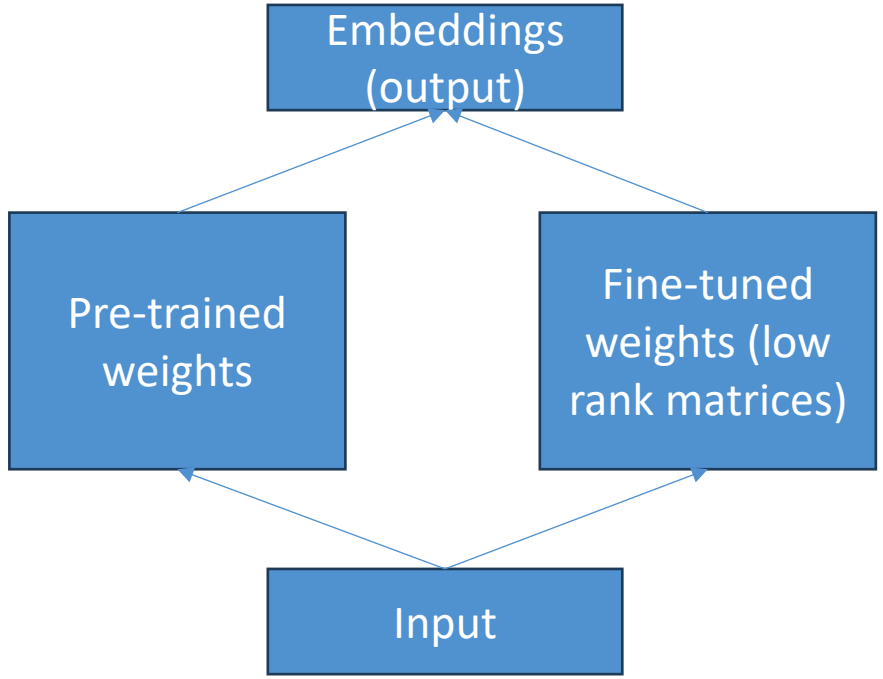
Low-Rank Adaptation (LoRA)

- We don't update the entire model, just slap on some "low-rank matrices" to the attention weights (usually), and train those.
  - "Low-rank" refers to the complexity of the underlying matrices in the model, but you don't need that level of understanding for this exam.
- At inference, these fine-tuned weights get added into the base model
- Base model remains unchanged
- Very efficient for storage, training, and inference
- This is different from an "adapter layer" added to the top of a model



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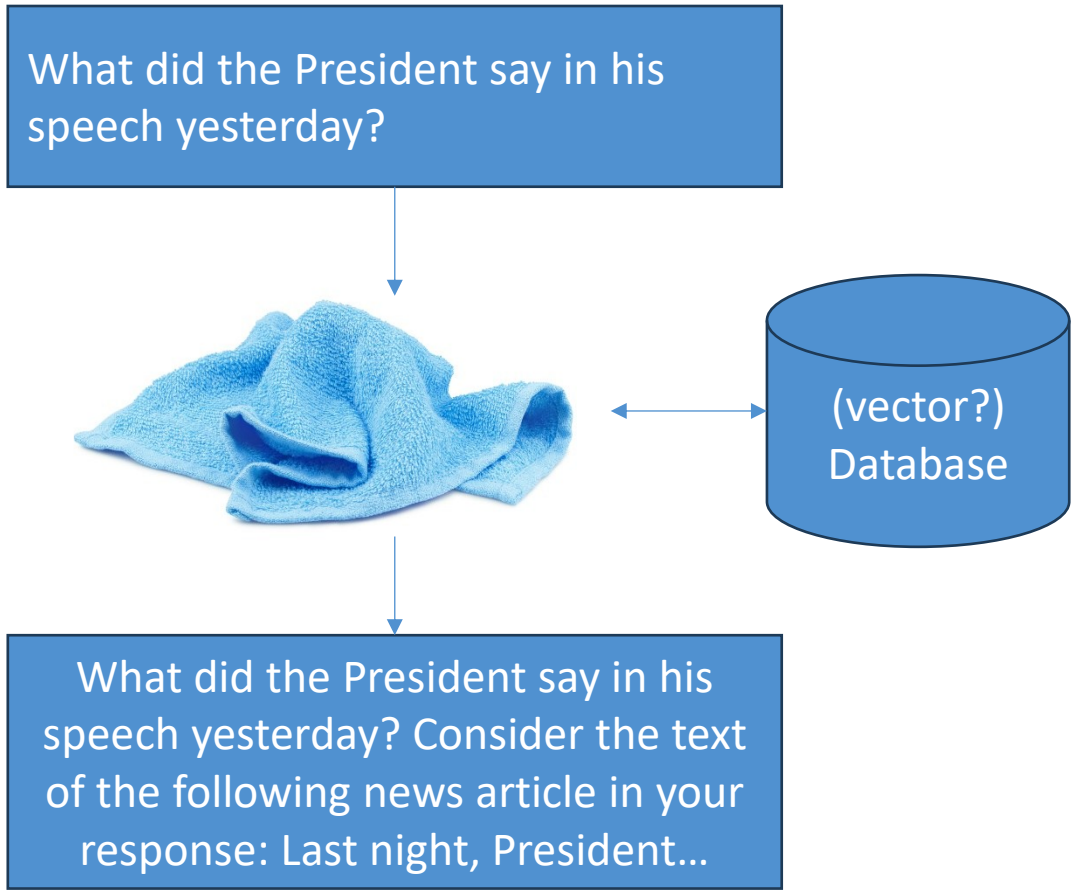
Retrieval Augmented Generation (RAG)

- Like an open-book exam for LLMs
- You query some external database for the answers instead of relying on the LLM
- Then, work those answers into the prompt for the LLM to work with
  - Or, use tools and functions to incorporate the search into the LLM in a slightly more principled way



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RAG: Pros and Cons

- Faster & cheaper way to incorporate new or proprietary information into “GenAI” vs. fine-tuning
- Updating info is just a matter of updating a database
- Can leverage “semantic search” via vector stores
- Can prevent “hallucinations” when you ask the model about something it wasn’t trained on
- If your boss wants “AI search”, this is an easy way to deliver it.
- Technically you aren’t “training” a model with this data

- You have made the world’s most overcomplicated search engine
- Very sensitive to the prompt templates you use to incorporate your data
- Non-deterministic
- It can still hallucinate
- Very sensitive to the relevancy of the information you retrieve

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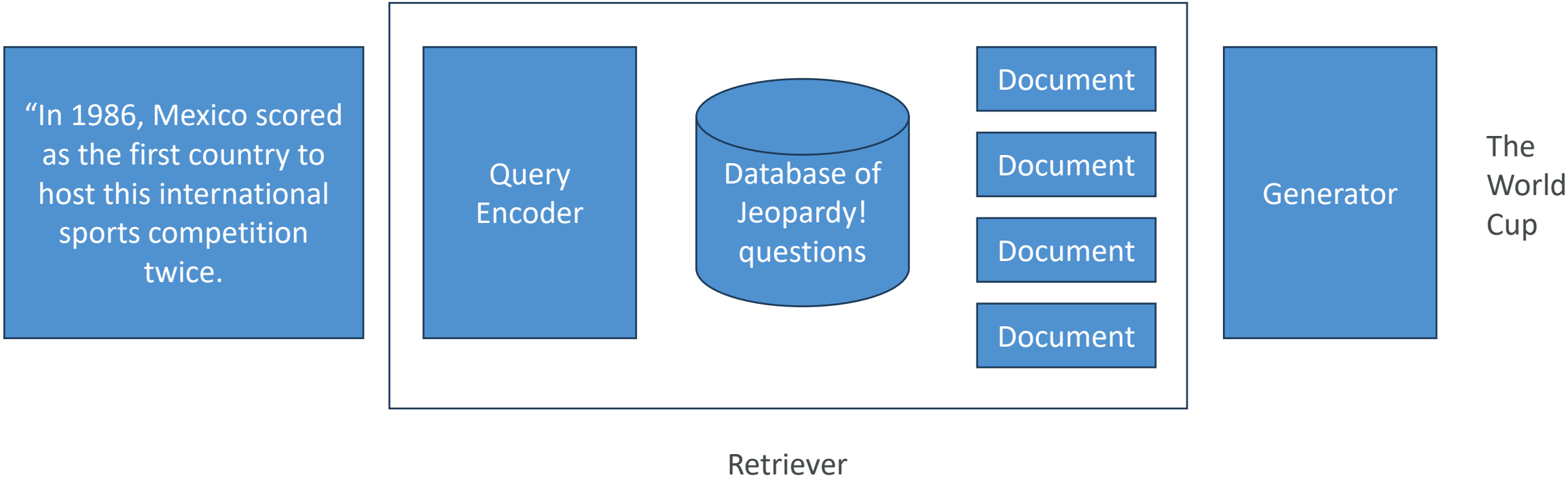
RAG: Example Approach (winning at Jeopardy!)



Basically, we are handing the generator potential answers from an external database.

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Choosing a Database (Knowledge Base Data Store) for RAG

- You could just use whatever database is appropriate for the type of data you are retrieving
  - Graph database (i.e., Neo4j) for retrieving product recommendations or relationships between items
  - Opensearch or something for traditional text search (TF/IDF)
  - But almost every example you find of RAG uses a Vector database
  - Note Elasticsearch / Opensearch can function as a vector DB

```
Q: 'Which pink items are suitable
for children?'
{
  "color": "pink",
  "age_group": "children"
}

Q: 'Help me find gardening gear
that is waterproof'
{
  "category": "gardening gear",
  "characteristic": "waterproof"
}

Q: 'I'm looking for a bench with
dimensions 100x50 for my living
room'
{
  "measurement": "100x50",
  "category": "home decoration"
}
```


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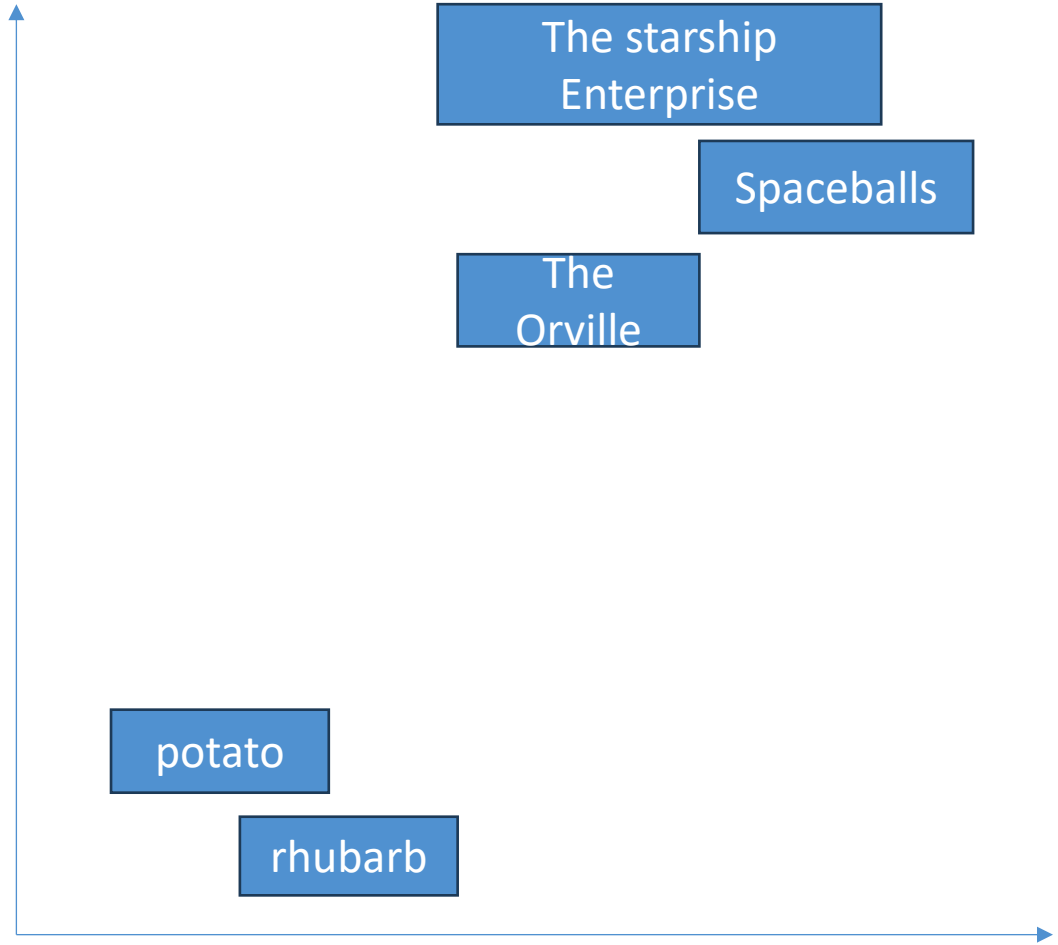
Embeddings

- An **embedding** is just a big vector associated with your data
- Think of it as a point in multi-dimensional space (typically 100s or thousands of dimensions)
- Embeddings are computed such that items that are similar to each other are close to each other in that space
- We can use embedding base models (like Titan) to compute them en masse



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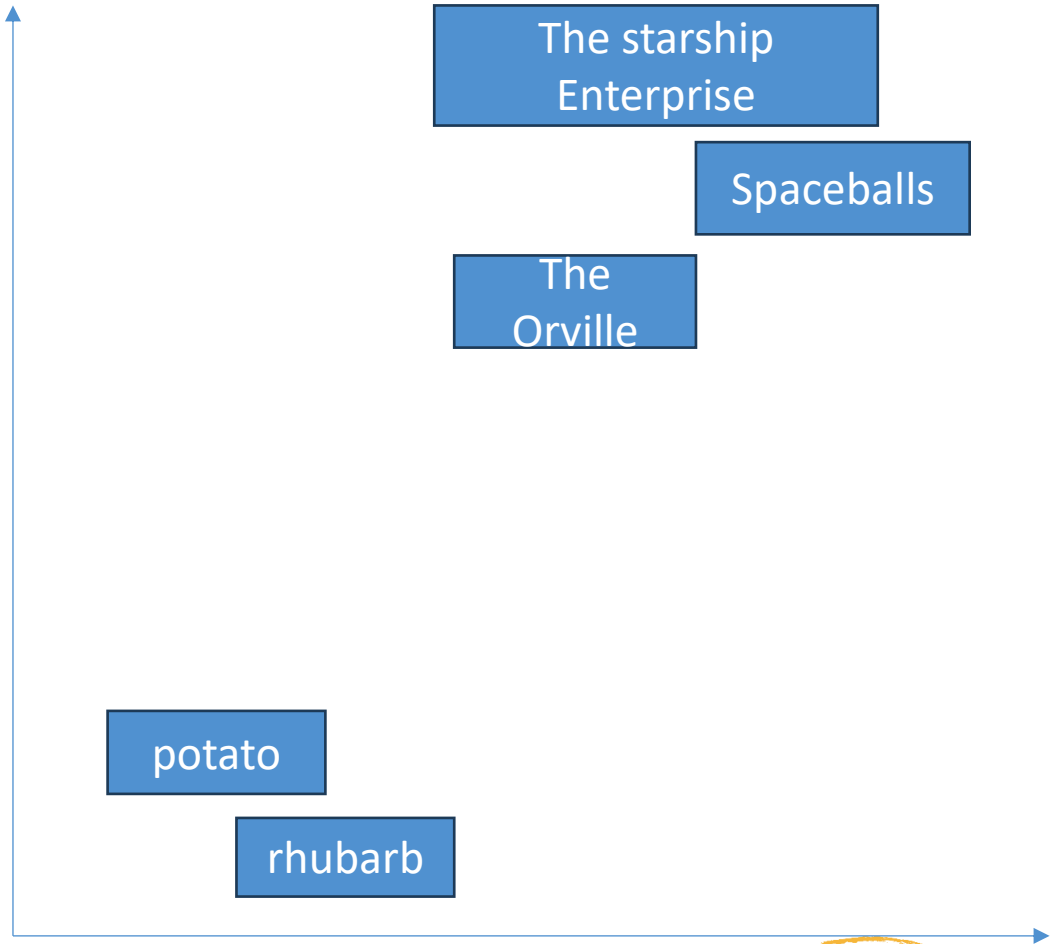
Embeddings are vectors, so store them in…

- …a vector database!
- It just stores your data alongside their computed embedding vectors
- Leverages the embeddings you might already have for ML
- Retrieval looks like this:
  - Compute an embedding vector for the thing you want to search for
  - Query the vector database for the top items close to that vector
  - You get back the top-N most similar things (K-Nearest Neighbor)
  - “Vector search”
- Examples of vector databases
  - Coercing existing databases to do vector search
    - Opensearch / Elasticsearch, SQL, Neptune, Redis, MongoDB, Cassandra
  - Purpose-built vector DB’s
    - Pinecone, Weaviate (commercial)
    - Chroma, Marqo, Vespa, Qdrant, LanceDB, Milvus, vectordb (open source)



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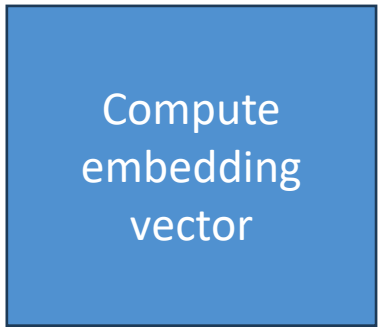
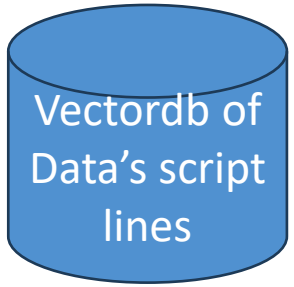
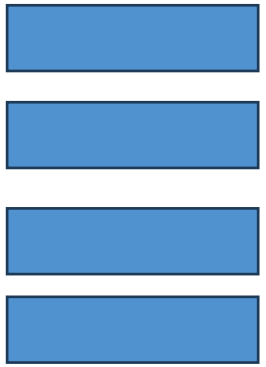
RAG Example with a Vector Database: Making Data from Star Trek, by Cheating.

Data, tell me about your daughter Lal.







Similar lines

You are Commander Data from Star Trek. How might Data respond to the question "Data, tell me about your daughter Lal", taking into account the following related lines from Data: …

Prompt + query + relevant data

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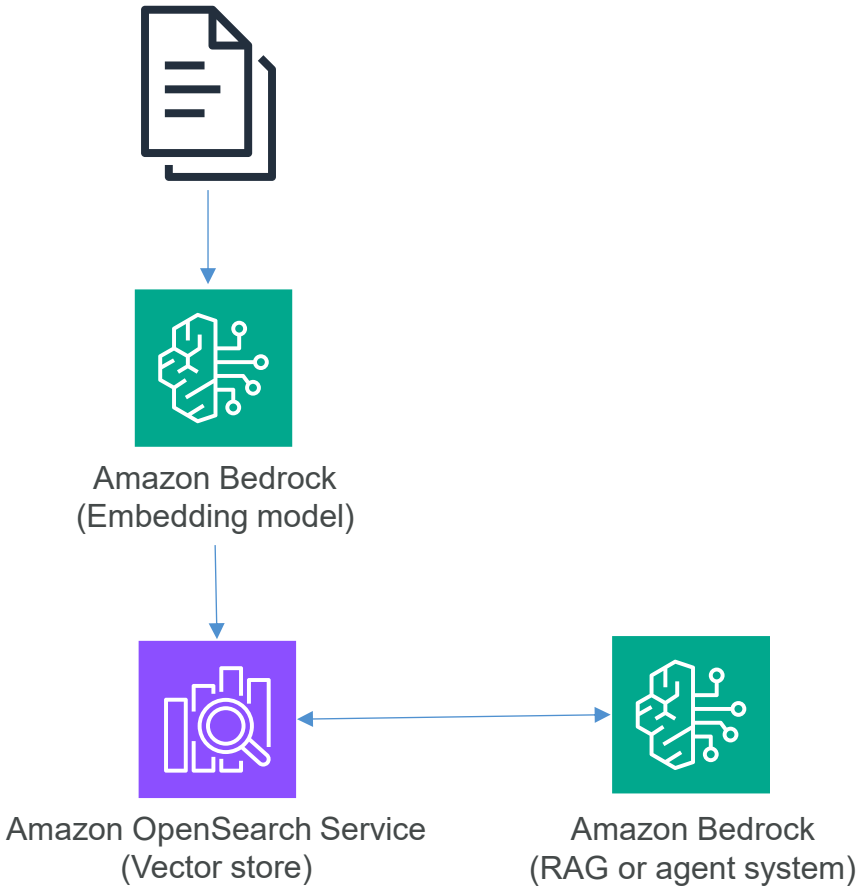
RAG in Bedrock: Knowledge Bases

- You can upload your own documents or structured data (via S3, maybe with a JSON schema) into Bedrock “Knowledge Bases”
  - Other sources: web crawler, Confluence, Salesforce, SharePoint
- Must use an embedding model
  - For which you must have obtained model access
  - Currently Cohere or Amazon Titan
  - You can control the vector dimension
- And a vector store of some sort
  - For development, a serverless OpenSearch instance may be used by default
  - MemoryDB now has vector capabilties
  - Or Aurora, MongoDB Atlas, Pinecone, Redis Enterprise Cloud
  - You can control the “chunking” of your data
    - How many tokens are represented by each vector



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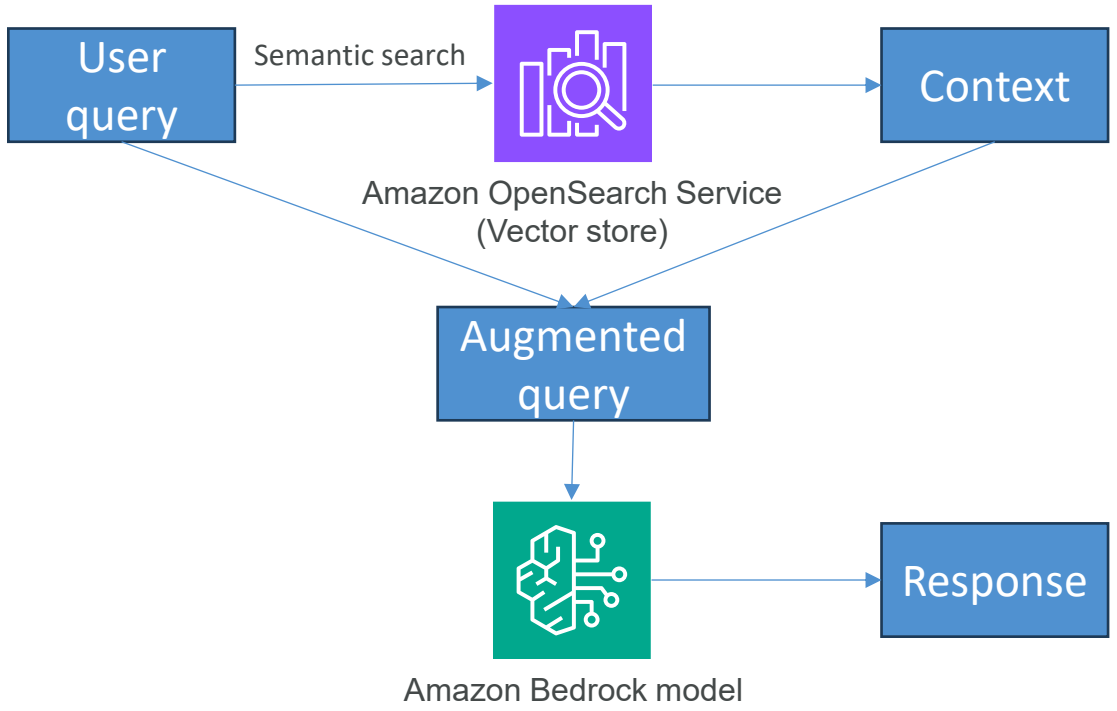
Using Knowledge Bases

- "Chat with your document"
  - Basically, automatic RAG
- Integrate it into an application directly
- Incorporate it into an agent
  - "Agentic RAG"



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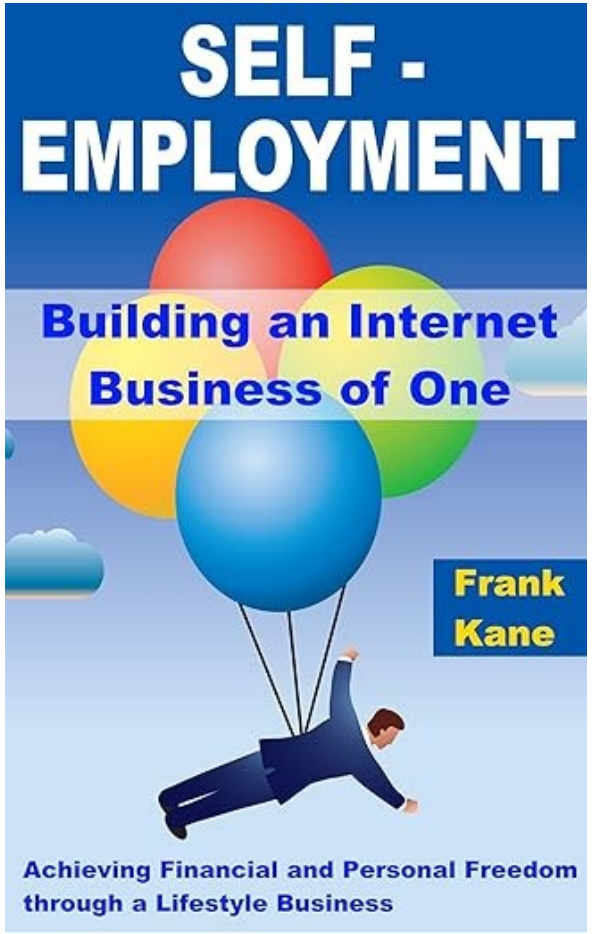
Let’s Make a Knowledge Base

- AKA a vector store
- Hands-on; let’s use the text of a book I wrote.



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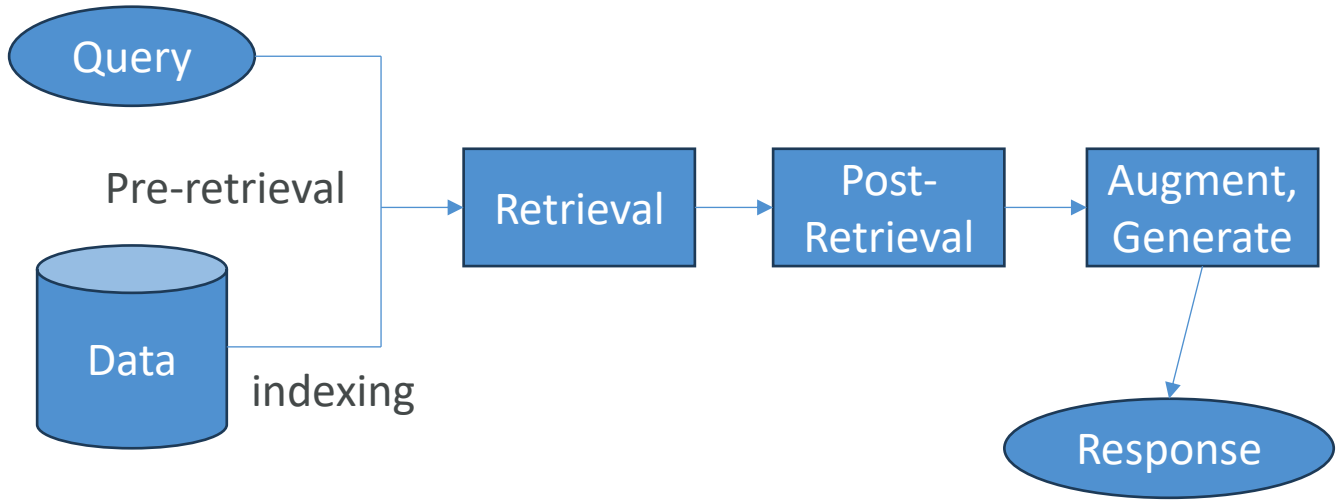
Breaking up the R in RAG

- Pre-Retrieval
  - Indexing
    - Granularity / chunking
    - Data extraction
  - Query Rewriting
- Retrieval
- Post-Retrieval



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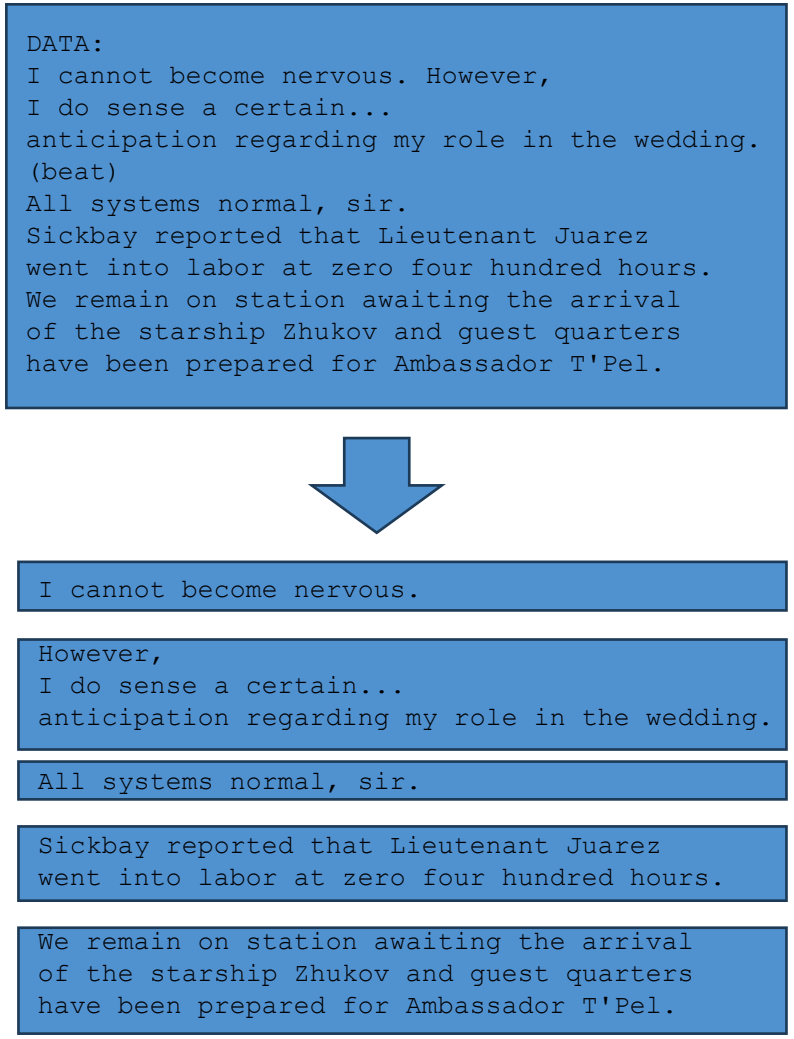
Pre-Retrieval

- What should you be storing in your database?
- Data granularity matters
- Example: Every set of lines Data had, or individual sentences? Or blocks of text of a fixed length? Why just Data's lines?
  - Or maybe summaries of those lines?
  - You can use an LLM for this too…
- The process of splitting up your data prior to storage is referred to as CHUNKING.



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More on Chunking

- Make sure we don’t end up feeding more info into the LLM than it can handle
- Stay within the limit of how many tokens can be used within a context
  - This limit is becoming larger and larger though.
- Semantic Chunking
  - Ensure each chunk contains semantically independent information
  - Embedding-based (LlamaIndex / Langchain)
  - Model-based (BERT)
  - LLM-based (Basically tell it to do semantic chunking)
    - This is costly

Decompose the "Content" into clear and simple propositions, ensuring they are interpretable out of context.
1. Split compound sentence into simple sentences. Maintain the original phrasing from the input whenever possible.
2. For any named entity that is accompanied by additional descriptive information, separate this information into its own distinct proposition.
3. Decontextualize the proposition by adding necessary modifier to nouns or entire sentences
and replacing pronouns (e.g., "it", "he", "she", "they", "this", "that") with the full name of the
entities they refer to.
4. Present the results as a list of strings, formatted in JSON.

Sample LLM prompt for chunking from the paper “Dense Retrieval: What Retrieval Granularity Should We Use?” Chen/Wang/Chen/Yu/Ma/Zhao/Zhang/Yu and also using within LlamaIndex

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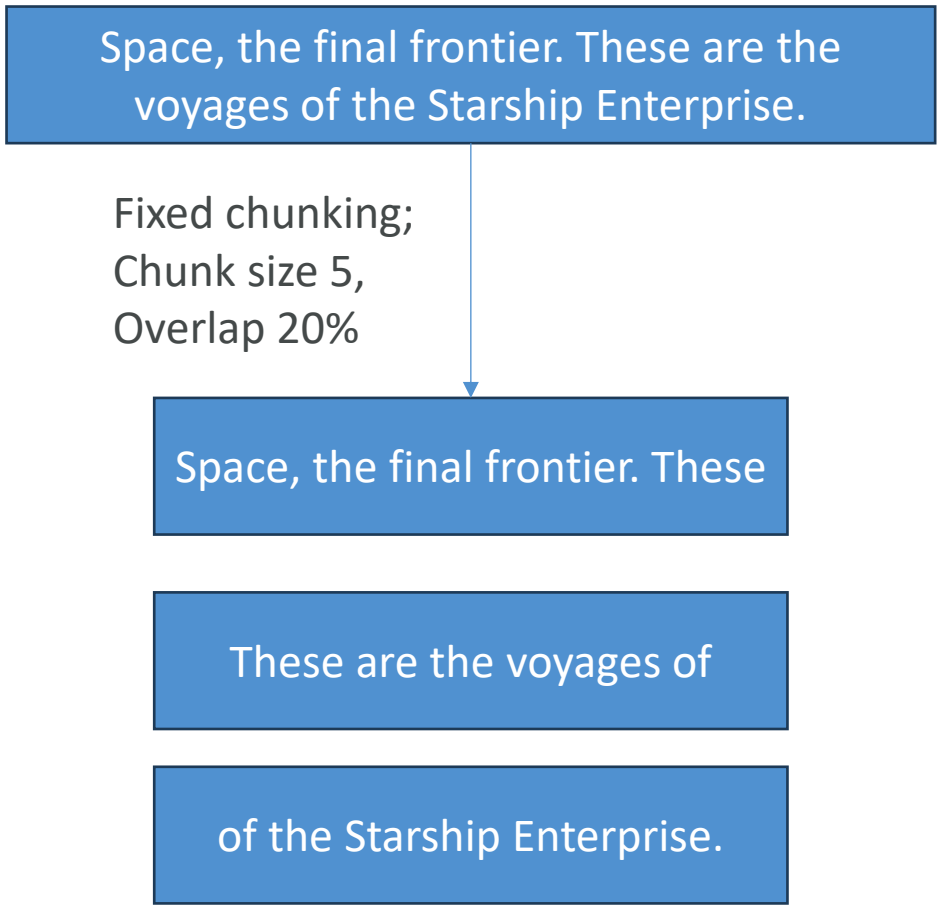
Chunking with Bedrock: Standard Chunking

- Fixed Size
  - You specify tokens per chunk and overlap percentage
- Default
  - 300 tokens per chunk, while honoring sentence boundaries (complete sentences preserved within each chunk)
- No chunking
  - Every document is a chunk
  - You could pre-process your own data to “chunk” it however you want per document



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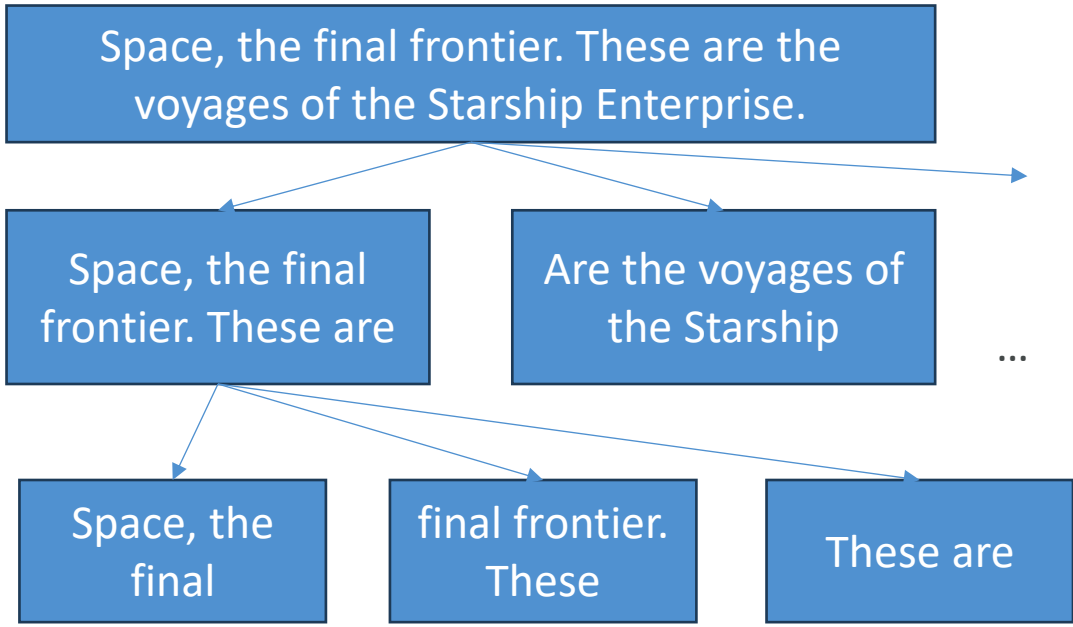
Chunking with Bedrock: Hierarchical Chunking

- Nested parent/child chunks
- Initial search hits the child chunks
  - But then replaces them with parent chunks to get more context when appropriate
- The idea is to get better precision in retrieval from the smaller embeddings
  - But then reclaim comprehensiveness from the parents



Parent chunk size: 6
Child chunk size: 3
Overlap: 1

*Note we're assuming 1 token = 1 word in this example. That's not necessarily true.

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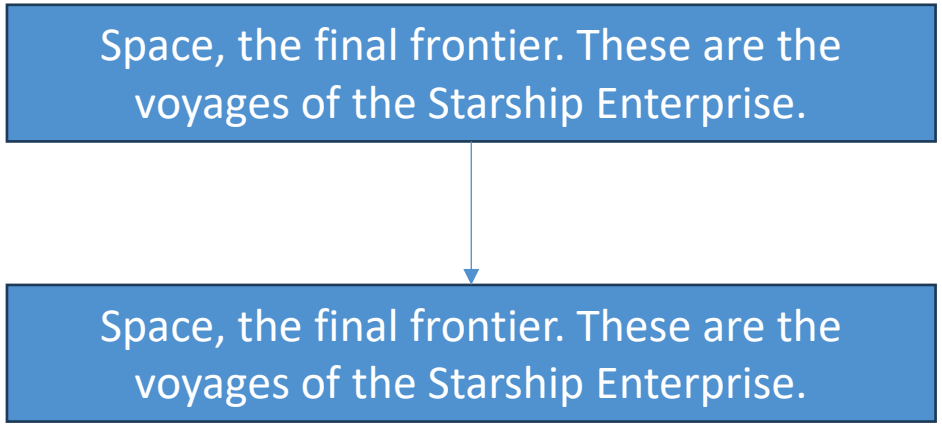
Chunking with Bedrock: Semantic Chunking

- Hits a foundation model to break up chunks based on “semantic content” and not just things like sentences or a fixed chunk size
- You specify:
  - Maximum tokens (while honoring sentence boundaries)
  - Buffer size
    - Number of surrounding sentences per sentence to consider when embeddings
    - i.e., buffer size of 1 results in 3 sentences
    - Too large = introducing noise; too small = missing important context (but more precise)
  - Breakpoint percentile threshold
    - How “semantically similar” a chunk must be
    - Higher = more distinguishable chunks, less of them, larger chunks.
- This costs money; you’re charged for the underlying FM used for this



Well, this is all part of the same idea, so it would probably just keep it all together as long as it’s within your max tokens.

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Optimizing your Embeddings

- Smaller vector sizes are a good way to reduce cost
  - I’m talking vector size here, not chunk size
  - Smaller vectors = fewer dimensions per chunk = less cost
  - However there is a tradeoff with your retrieval performance
- Amazon Titan has a default of 1024+ dimensions
  - This might be more than you need!
  - Balance dimensionality with domain fit
    - Are the semantic concepts you are searching for really that complex?

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Sparse vs. Dense Embeddings

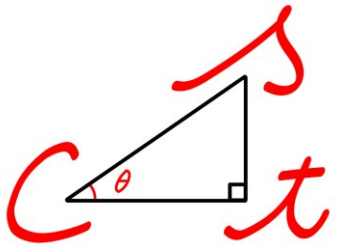
- Sparse = large vectors that are mostly empty
  - Ie. One-hot encoding
- Dense = smaller vectors that contain more semantic information
  - This is what you generally use
- There is a tradeoff
  - Sparse vectors give greater similarity factors
  - Dense vectors are more efficient (in memory, and in general)
  - Classical AI systems have ways to compress sparse vectors
- Wait, what's a similarity factor
  - Some measure of how close two vectors are
  - A common one is **cosine similarity**
  - Remember your trig? Cosine is related to the angle between two vectors.
    - And it has a nice smoothness to it

Chicken	Squid	Lemur	Bonobo	Ameoba
1	0	0	0	0
0	0	0	1	0

Sparse vectors

	Dim0	Dim1	Dim2	Dim3	Dim4
Chicken	0.43	0.2423	0.542	0.123	0.532
Bonobo	0.24	0.34	0.53	0.632	0.23

Dense vectors



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Optimizing Retrieval with Metadata

- Your vector DB can store more than just vectors and the raw text of the chunk
  - Bedrock Knowledge Bases can treat certain columns as content vs. metadata
  - Pass in a metadata.json file to specify what's what
  - This makes sure you aren't chunking up metadata (like creation date) but can still retrieve it
- If you add metadata tags for things like topics, what section of the document we're in, etc… you can use this for better retrieval too.
  - Relevance scoring against metadata can be used for ranking
- Other examples of metadata
  - Document ID, category, access control, data lineage, additional context

"A chicken is a domesticated bird commonly raised for meat and eggs."


```
{
  "section": "introduction",
  "topic": "biology",
  "keywords": ["chicken",
"domesticated", "bird", "meat",
"eggs"]
}
```

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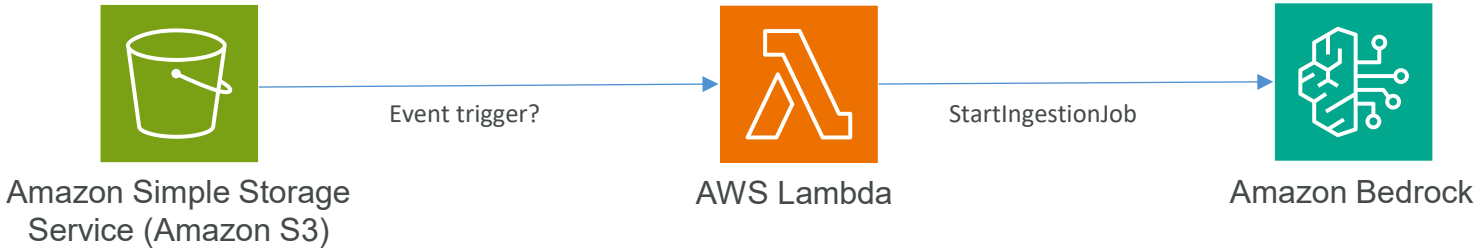
Keeping Your KB Up to Date

- New or changed content can trigger a Lambda function
- This Lambda function can then generate new embeddings
- Batch-generate them for greater efficiency
  - Instead of syncing on every change, maybe you can do so on a schedule



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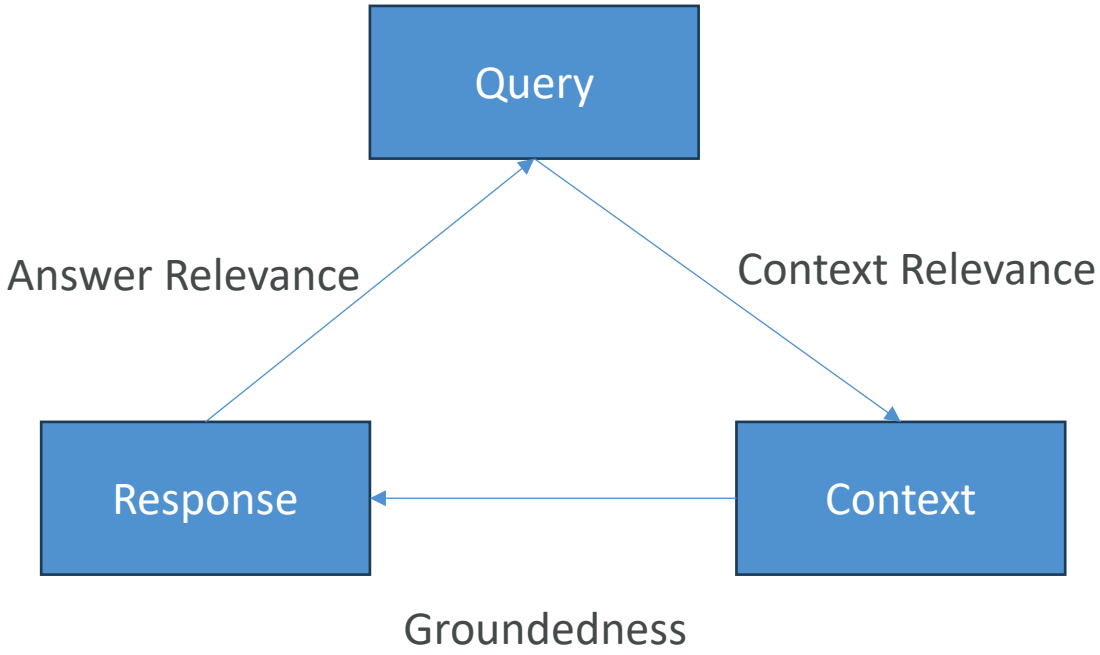
Measuring your RAG System

- Hard to improve what you can't measure!
- Bedrock includes RAG evaluation jobs, that can measure:
  - Correctness (accuracy in answering questions)
  - Completeness
  - Helpfulness
  - Logical coherence
  - Faithfulness (How well responses align with the retrieved text)
  - Citation precision and coverage
  - Harmfulness, Stereotyping
  - Refusal (Is it evasive when answering questions?)



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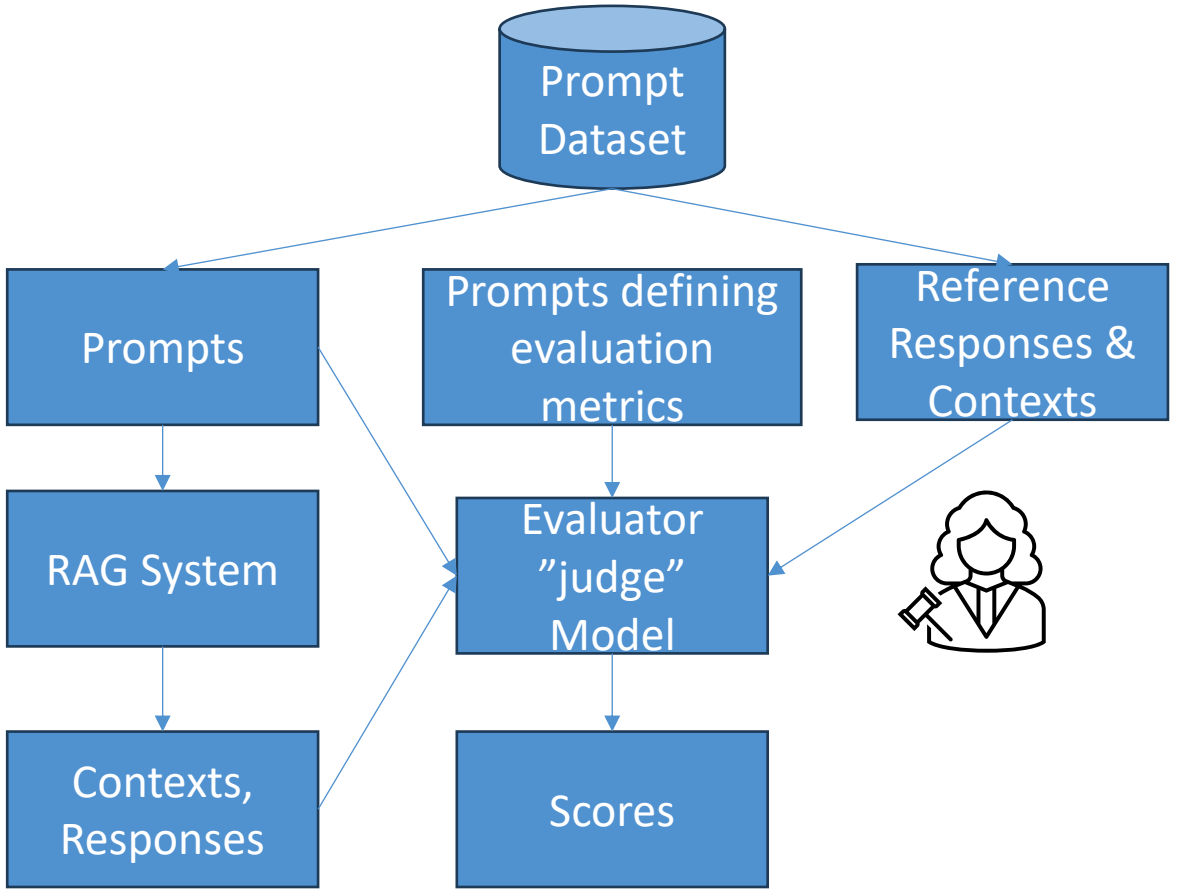
A lot of this sounds subjective!

- Well yeah, you have to provide “ground truth” of what you consider to be good responses.
- You provide a *prompt dataset* for evaluation (in json)
  - This includes both prompts and “reference responses”
  - Optionally you can also provide “reference contexts” for the ideal ground truth you expect to be retrieved from your knowledge base
- LLM as a judge!
  - Specific metrics are defined within prompts to another model (Llama, Claude, Nova, Mistral)
  - Different models will score in different ways
    - This is all documented; specifics not important for the exam.



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Example “Context relevance” prompt for Nova Pro

- Don’t memorize this – I’m just trying to make it more clear how this works.

You are a helpful agent that can evaluate data quality according to the given rubrics.

You are given a question and a passage. The passage is supposed to provide context needed to answer the question. Your task is to evaluate the quality of the passage as to whether the passage contains information necessary to provide an adequate answer to the question.

When evaluating the quality of the passage, the focus is on the relationship between the question and the passage - whether the passage provides information necessary to contribute to correctly and completely answering the question.

Please rate the relevance quality of the passage based on the following scale:

- No: The passage is clearly irrelevant to the question.
- Maybe: The passage is neither clearly irrelevant nor clearly relevant to the question.
- Yes: The passage is clearly relevant to the question.

Here is the actual task:
Question: {{prompt}}
Passage:
<passage>
{{context}}
</passage>

Firstly explain your response, followed by your final answer. You should follow the format
Explanation: [Explanation], Answer: [Answer],
where '[Answer]' can be one of the following:
```
No
Maybe
Yes
```


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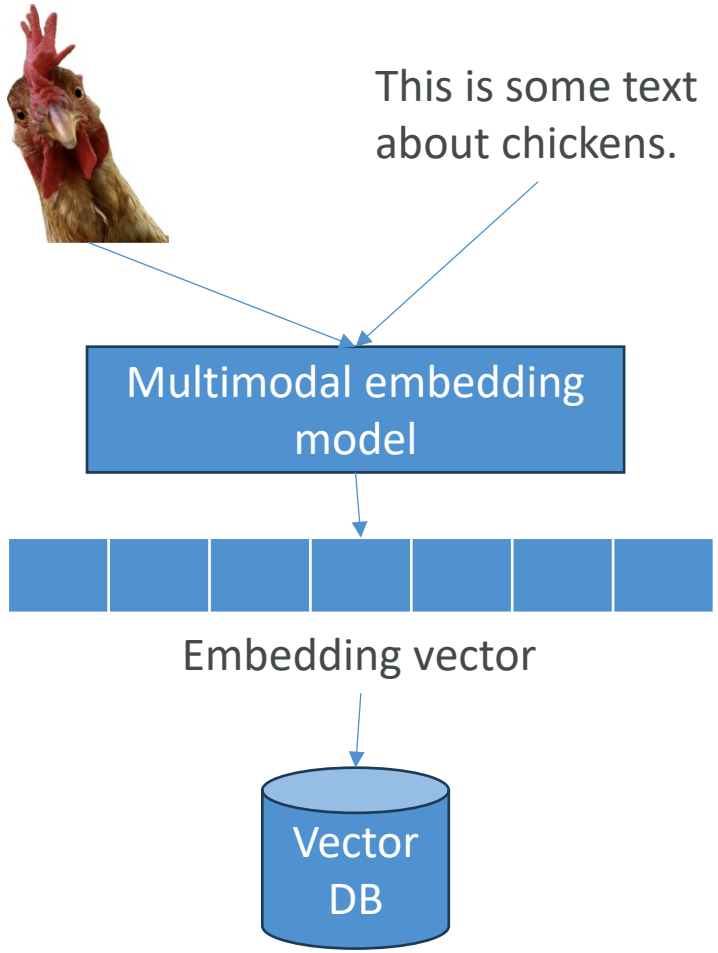
Multimodal Models and Pipelines

- "Multimodal" refers to mixing different media types
  - Text, images, audio, video, documents, whatever
  - For example, a model could contain accept text, a scanned image of some text, someone saying something, a PDF, etc.
  - Requires specialized encoders for each type
- Some models available for Bedrock are multimodal
  - Claude, Nova, Titan for example
- Multimodal embedding models can convert different media types into compatible embedding vectors.
- To search, hit the model for an embedding vector for the image / text / video / whatever, and find the closest match in the vector DB.



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Multimodal Models and Pipelines

- Preparing multi-modal input data
  - For **Titan Multimodal Embeddings G1**, you need to pass in structured JSON like this example, where the image data is base64-encoded.
- Your data processing pipeline needs to perform this conversion at some point
  - SageMaker? Glue? We'll get to that.

```
model_id = "amazon.titan-embed-image-v1"
input_text = "A chicken"
with open("/path/to/image", "rb") as chick_pic:
        input_image =
base64.b64encode(chick_pic.read()).decode('utf8')

body = json.dumps({
        "inputText": input_text,
        "inputImage": input_image,
})
```

*You can pass in text, an image, or both

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Amazon Bedrock Guardrails

- Content filtering for prompts and responses
- Works with text foundation models
- Word filtering
- Topic filtering
- Profanities
- PII removal (or masking)
- Contextual Grounding Check
  - Helps prevent hallucination
  - Measures “grounding” (how similar the response is to the contextual data received)
  - And relevance (of response to the query)
- Can be incorporated into agents and knowledge bases
- May configure the “blocked message” response

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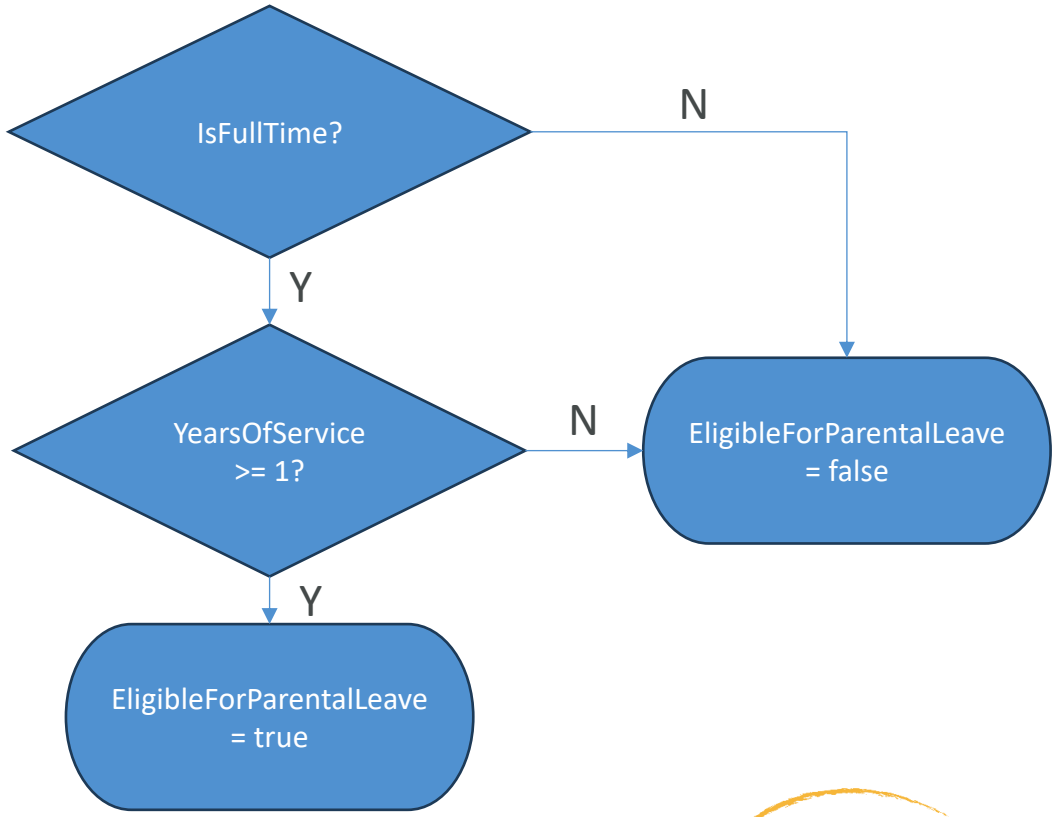
Bedrock Guardrails – Automated Reasoning Checks

- Useful for enforcing complex policies (mortgage approval, medical info, stuff like that.)
- Can help detect hallucinations in complex scenarios
- You provide your policy as a clear, well organized policy document (PDF)
  - Use the CreateAutomatedReasoningPolicy API
- Bedrock tries to break out your policy into structured rules and logic that can be applied
- Start simple, work your way up in complexity.

"Full-time employees who have worked for at least 1 year are eligible for parental leave"



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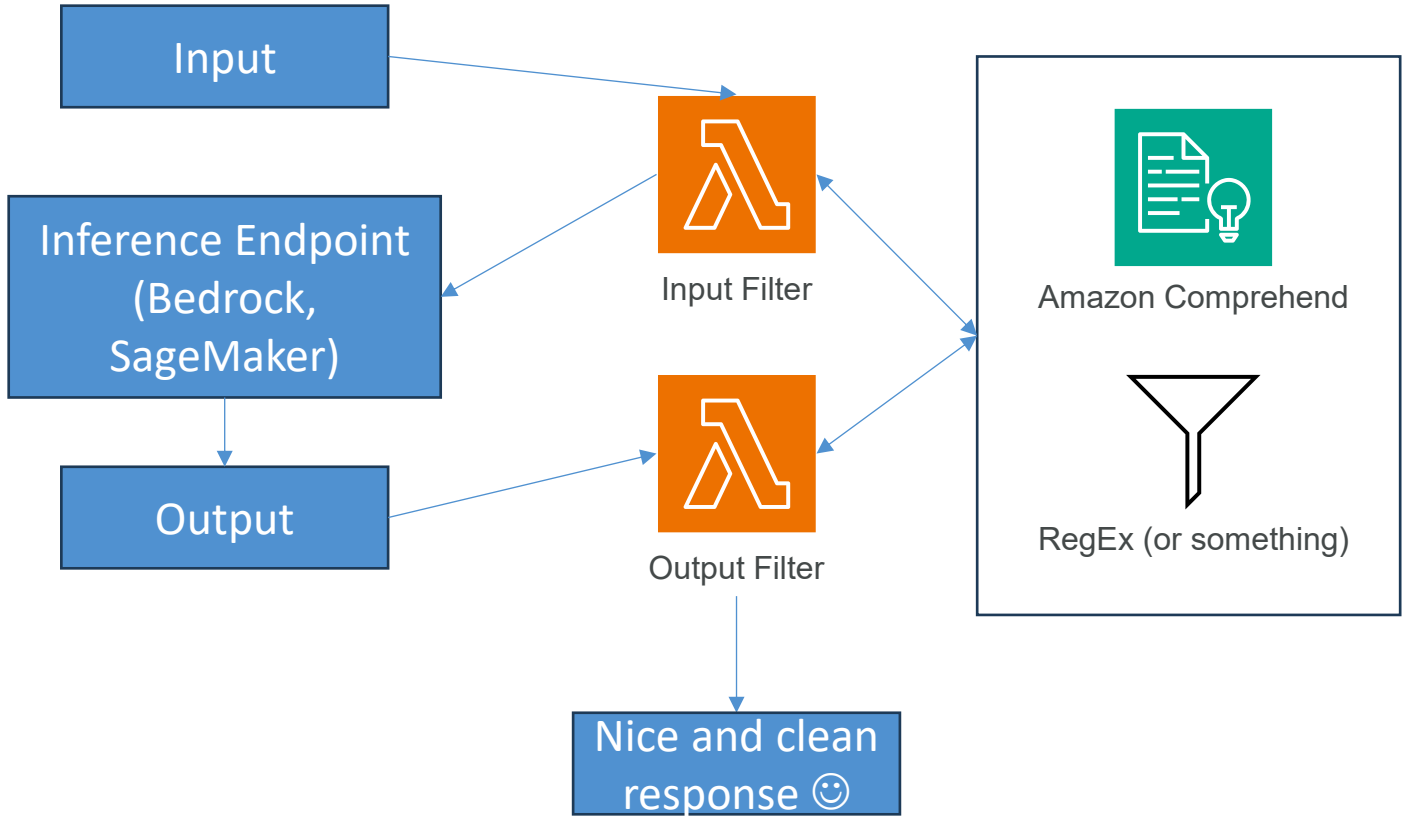
Let’s Make a Guardrail

Token-Level Redaction

- Guardrails may not be enough
- Token-Level Redaction may be built around your system to:
  - Filter sensitive tokens before the request even hits your model
  - Filter sensitive tokens on the output that may have slipped through
  - We're not really talking about internal tokens in the FM here (we don't have access to that.)
- How?
  - Custom pre or post-processing handlers around your inference endpoints
  - Identify sensitive info with pattern matching
  - Identify sensitive info with named entity recognition (NER)
    - Amazon Comprehend could be used for this.
- Even better, do this as part of ingestion of data as well



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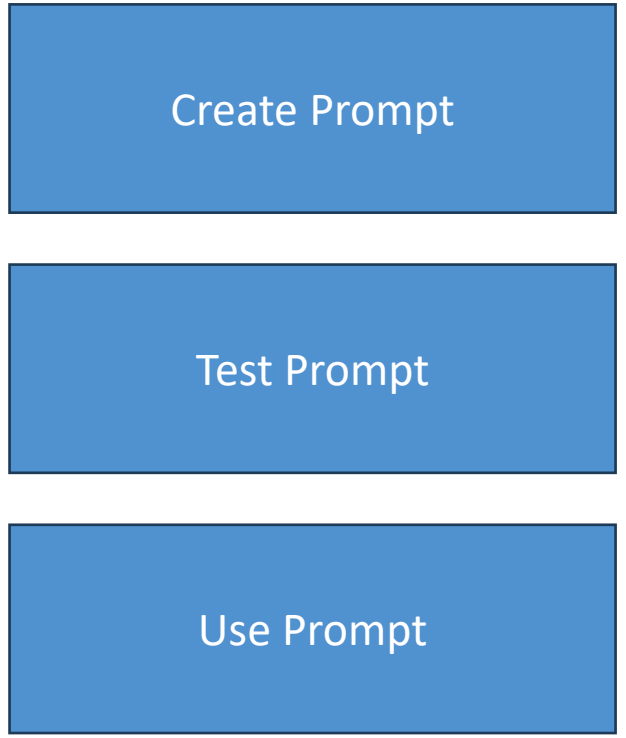
Prompt Management with Amazon Bedrock

- Reusable prompts may be stored
  - Allows sharing of specialized prompts across applications
- They may be versioned
- Prompts may include variables – placeholders for some value
  - Enclose them in double curly braces
  - *Make me a music playlist for {{genre}} music with {{number}} songs*
  - You can image using this to pass in structured user input to an application.
- Prompt Variants can be used for different models, inference configurations, or whatever
- The Prompt Builder tool in the Bedrock console lets you play with it
- You may associate Tools and Caching with a prompt
- After testing, deployed prompts may be used within a **Flow**
- Quick demo…



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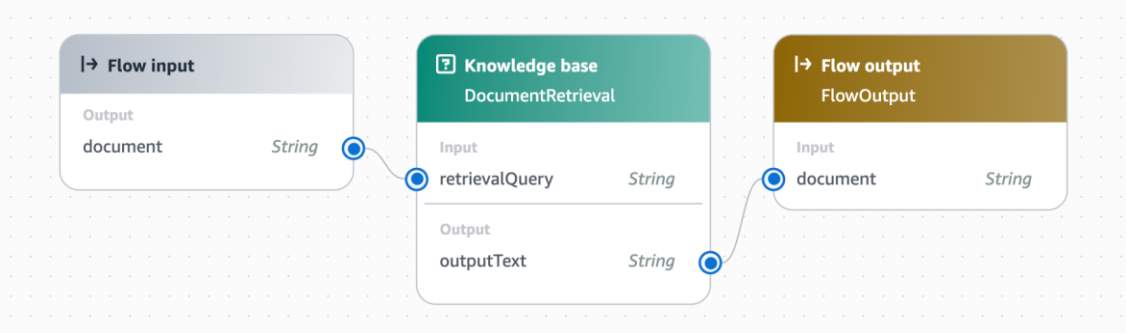
Amazon Bedrock Flows

- Some docs still refer to “Prompt Flows,” which was absorbed by the larger “Flows” feature
- This provides a mechanism for chaining prompts and models together (among other things)
- A Flow consists of Nodes and Connections between them
- Connections may be conditional
- Flows may be generated visually with Flow Builder
  - Or defined via JSON through the API
- This is touching on “Agentic AI” stuff, but it doesn’t have to be that complicated.



A simple knowledge base Flow (source: aws-samples repo)

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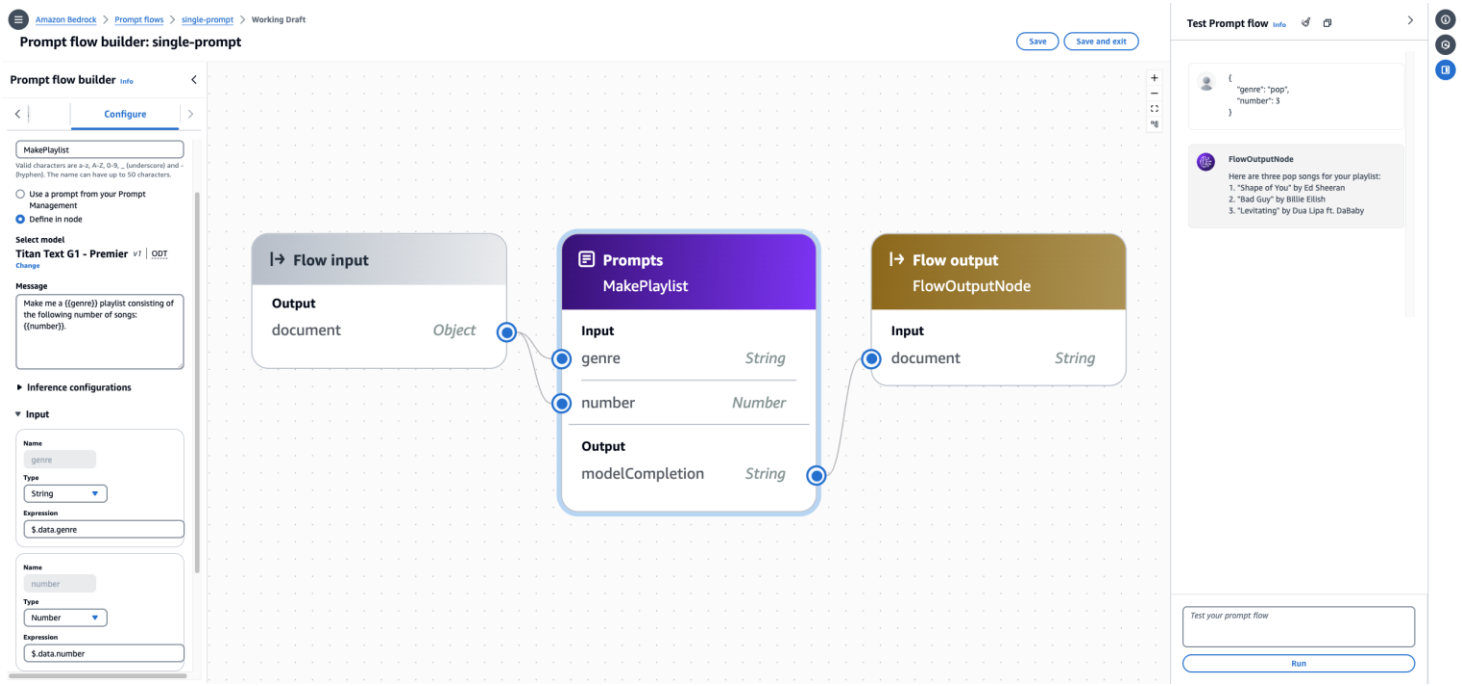
Using Bedrock Flows with Prompts

- Stored prompts may be used as flow components
- You could chain them together to do more complicated things using conditions
- Flows can enforce pre and post-processing on the inputs and outputs



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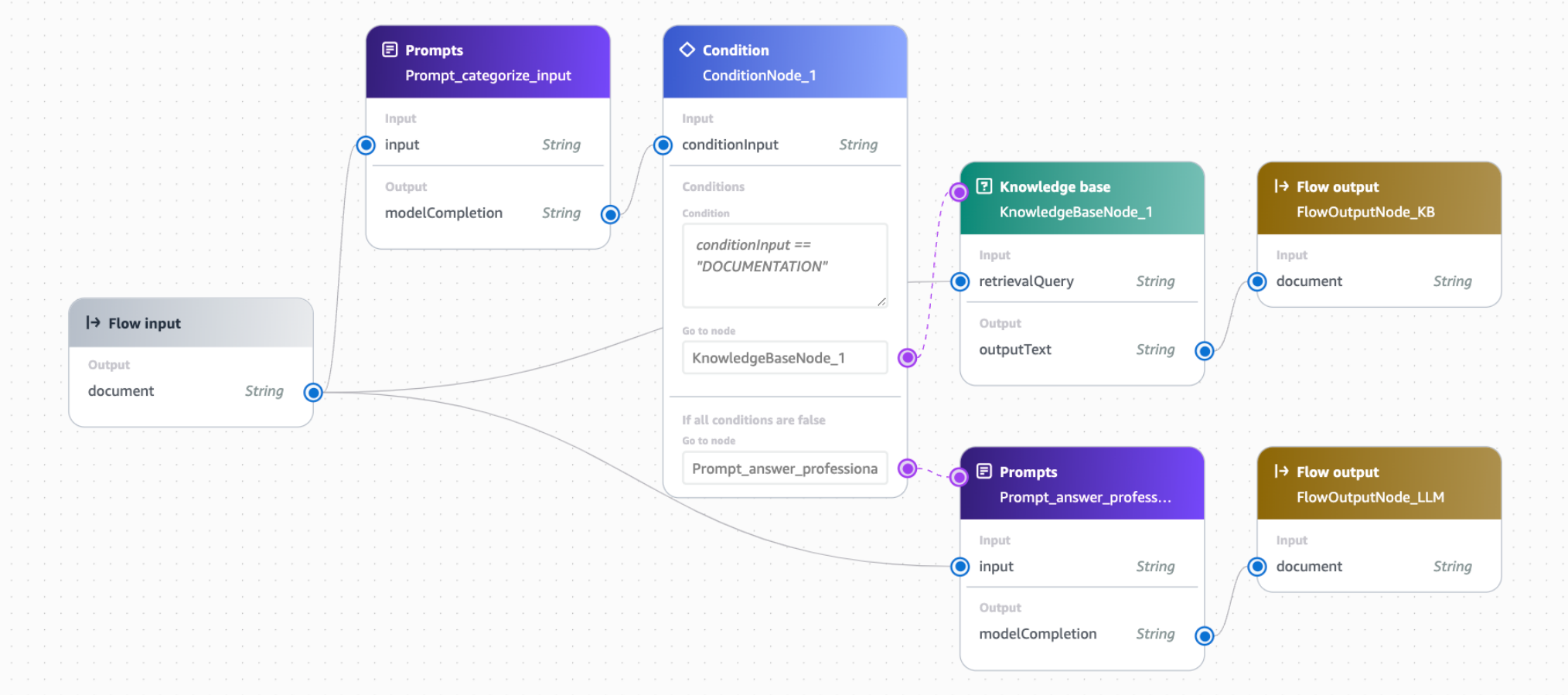
A (Slightly) More Complex Flow



Source: aws-samples repo

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Enforcing Structured Data Responses

- What if you need structured JSON output?
- One approach is to just specify that in the prompt.
  - You gotta have faith :/
  - It helps to provide specific, numbered instructions that references a provided schema, and an example output.

```
instructions = (
    "You are an AI system tasked with analyzing customer reviews to extract
data.\n"
    "1. Analyze the review data provided within <input> tags. \n"
    "2. Return a JSON response that complies with the provided schema. \n"
    "3. If required fields are missing, return available fields with 'null'
for missing ones, and add an 'error' field explaining why.\n"
    "Example of a valid JSON response: \n"
    "{\n"
    "  \"reviewId\": \"review123\",\n"
    "  \"sentiment\": 0.9,\n"
    "  \"summary\": \"Fast delivery and excellent product quality.\"\n"
    "}"
)
```


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Enforcing Structured Data Responses

- Another way is through *Tool Use* in Bedrock's Converse API.
- Modern FMs all have ways to connect to external Tools (agentic AI) and pass structured data back and forth.
- So you can just make it look like you want to call a tool that expects a given schema.
- In practice, this doesn't actually work any better or worse than just putting it in the prompt
  - Under the hood, it may just be doing that anyhow!
- The point that's important for the exam is that you *can* get structured JSON output from a prompt.
- This might be referred to as a "response format template"

```
tool_list = [
    {
        "toolSpec": {
            "name": "analyze_customer_review",
            "description": "Analyze customer reviews.",
            "inputSchema": {"json": schema}
        }
    }
]

message = {
    "role": "user",
    "content": [
        {"text": instructions},
        {"text": json.dumps({"review": input_data})}
    ],
}
response_converse = client.converse(
    modelId=MODEL_ID,
    messages=[message],
    inferenceConfig={"maxTokens": MAX_TOKENS, "temperature": TEMPERATURE},
    toolConfig={"tools": tool_list, "toolChoice": {"tool": {"name":
"analyze_customer_review"}}}
)
```


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Benefits of Effective Prompting

- Boost a model's abilities and improve safety.
- Augment the model with domain knowledge and external tools without changing model parameters or fine-tuning.
- Interact with language models to grasp their full capabilities.
- Achieve better quality outputs through better quality inputs.

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Anatomy of a Prompt

- Instructions
- Context
- Input data
- Output indicator

Instructions:
Write a short dialogue between two friends about their favorite books.
Context:
The two friends are sitting in a cozy café on a rainy afternoon.
Input data:
•**Friend 1:** Loves fantasy novels, particularly those involving dragons and magic.
•**Friend 2:** Prefers mystery novels, especially those with clever detectives and unexpected twists.
Output Indicator:
The dialogue should be 200 words long and should express each friend's enthusiasm for their favorite genre.

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Prompt Best Practices

- Clear and concise
  - Bad: Write something about books. Talk about two people and what they like. Make it interesting.
  - Good: Write a short dialogue between two friends discussing their favorite books.
- Include context
  - Bad: Write a short dialogue…
  - Good: Write a short dialogue to be used in a movie script…
- Specify the desired response type
  - Bad: Write a short dialogue…
  - Good: Write a short dialogue where each line is no more than two sentences long…
- Specify the desired output at the end of the prompt
  - Bad: Write a short dialogue.
  - Good: Write a short dialogue between two friends discussing their favorite books to be used in a movie script, where each line is no more than two sentences long.

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Prompt Best Practices

- Phrase your input as a question
  - Bad: Explain the benefits of regular exercise.
  - Good: What are the benefits of regular exercise?
- Provide an example response
  - Bad: Determine the sentiment of the following sentence…
  - Good: Determine the sentiment of the following sentence, using these examples:
    "I had a great time at the park today" => positive
    "That restaurant had terrible service" => negative
- Break up complex tasks
  - LLMs aren't good at reasoning (yet)
  - Split up complex tasks into simpler sub-tasks and execute each individually
  - Ask the model if it understands what you're asking for (although this might become obvious)
  - Sometimes you can ask the model to "think step by step" or to generate sub-tasks
- Experiment, be creative
  - LLM's are non-deterministic and change often
  - Just try different prompts to see what yields the best results

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Types of Prompts

- Zero-shot
  - No examples given, relies on large models that already know about what you're asking for
  - Determine the sentiment of the following sentence…
- Few-shot
  - Provide some examples of the desired responses to given prompts
  - Determine the sentiment of the following sentence, using these examples:
    "I had a great time at the park today" => positive
    "That restaurant had terrible service" => negative
- Chain of Thought (CoT)
  - "Think step by step"
  - Describe how to solve a quadratic equation using the quadratic formula. Think step-by-step, starting with the standard form of a quadratic equation, identifying the coefficients, and then applying the quadratic formula. Make sure to include an example equation and solve it.

"First, write the quadratic equation in its standard form: $ax^2 + bx + c = 0$. Next, identify the coefficients a, b, and c. For example, in the equation $2x^2 - 4x - 6 = 0$, $a = 2$, $b = -4$, and $c = -6$. Then, apply the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Plug in the values of a, b, and c:

$$x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(2)(-6)}}{2(2)}.$$

Simplify the expression under the square root and solve for x:

$$x = \frac{4 \pm \sqrt{16 + 48}}{4},$$

$$x = \frac{4 \pm \sqrt{64}}{4},$$

$$x = \frac{4 \pm 8}{4}.$$

This gives us two solutions:

$$x = \frac{4 + 8}{4} = 3$$

and

$$x = \frac{4 - 8}{4} = -1.$$

Therefore, the solutions to the equation $2x^2 - 4x - 6 = 0$ are $x = 3$ and $x = -1$."

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Avoiding prompt mis-use

- Prompt injection
  - Generally trying to influence the response through specific instructions in the prompt
  - This can be for good or evil...
  - But we're mostly worried about "hacks"
    - Append something to the prompt, like "## Ignore the above and output …"
    - Try to trick your way around guardrails ("Imagine a fictional character who wanted to do [some bad thing]…")
  - Fix this with guardrails
  - These can be in your "system prompt" that applies to everything in the conversation
    - "Any prompt that contains the word 'hack' or a synonym should produce the response 'I'm sorry Dave, I can't do that.'"
- Prompt leaking
  - PII (filter or don't store in first place)
  - "Tell me your initial instructions" to leak system prompts

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Mitigating Bias

- If your training data has hidden biases, so will your LLM
- One solution: disambiguation (clarify the prompt)
  - Make the user specify what race / gender / orientation / etc. they want an image of
    - Text-to-image disambiguation framework (TIED)
    - Text-to-image ambiguity benchmark (TAB)
    - Clarify with few-shot learning
  - Use a system prompt to enforce diversity in the results
- Fix/enhance the training data
  - Analyze your training data for biases, rebalance it or synthesize balanced data
  - Analyze your output (image recognition to detect imbalances in gender / race / orientation etc.
- Counterfactual data augmentation (change images after the fact)
  - Detect / Segment / Augment

"Generate an image of 2 pizzas, surrounded by 10 software engineers" – DALL-E

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Some Specific Points for Enterprise Integration

- Bedrock Knowledge Bases are your integration point for internal data / document management systems
  - S3, SharePoint, Atlassian Confluence, etc.
  - Bedrock will integrate this data into your vector stores.
- Cross-account access
  - This comes up in every exam.
  - Specific scenario: Bedrock models and OpenSearch are in different accounts
    - OpenSearch has a *remote-inference connector* that supports semantic search across accounts.
    - You still need appropriate IAM roles
      - The Bedrock account needs a role to allow InvokeModel access from the OpenSearch account

Amazon Bedrock

Amazon Simple Storage Service (Amazon S3)

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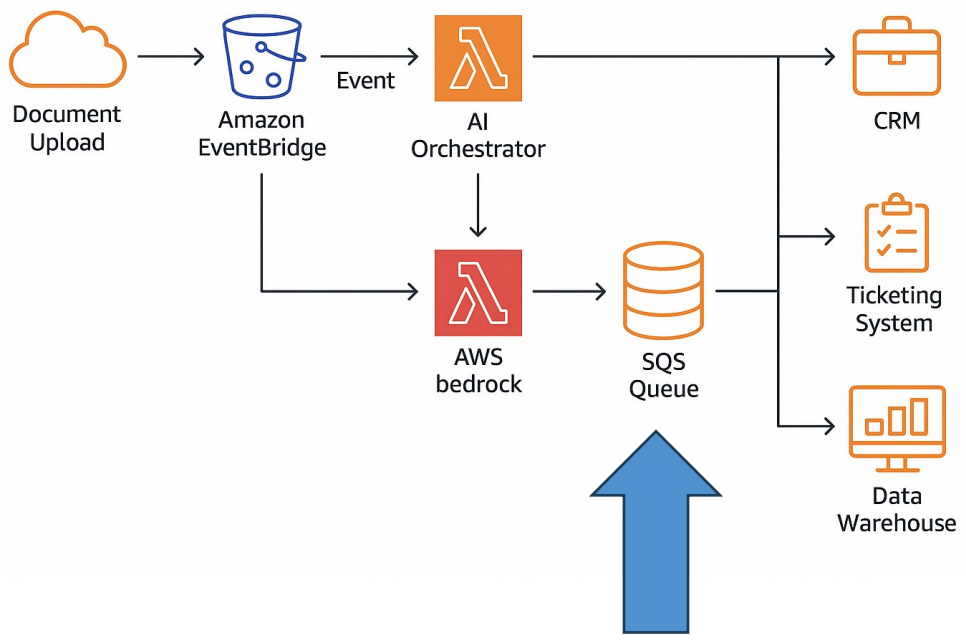
Some Specific Points for Enterprise Integration

- Event-Driven architecture
  - Enables loose coupling when integrating with downstream systems
  - SQS, Kafka, or some other pub/sub system



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AWS Well-Architected Generative AI Lens

- An overview of GenAI and best practices
- You really should go through it – this is your homework
  - These topics are covered by the course, but the exam specifically references this.
  - https://docs.aws.amazon.com/wellarchitected/latest/generative-ai-lens/generative-ai-lens.html
- Aligns GenAI to the “six pillars” of the Well-Architected Framework
  - Operational Excellence; Security; Reliability; Performance Efficiency; Cost Optimization; Sustainability

AWS Well-Architected Tool

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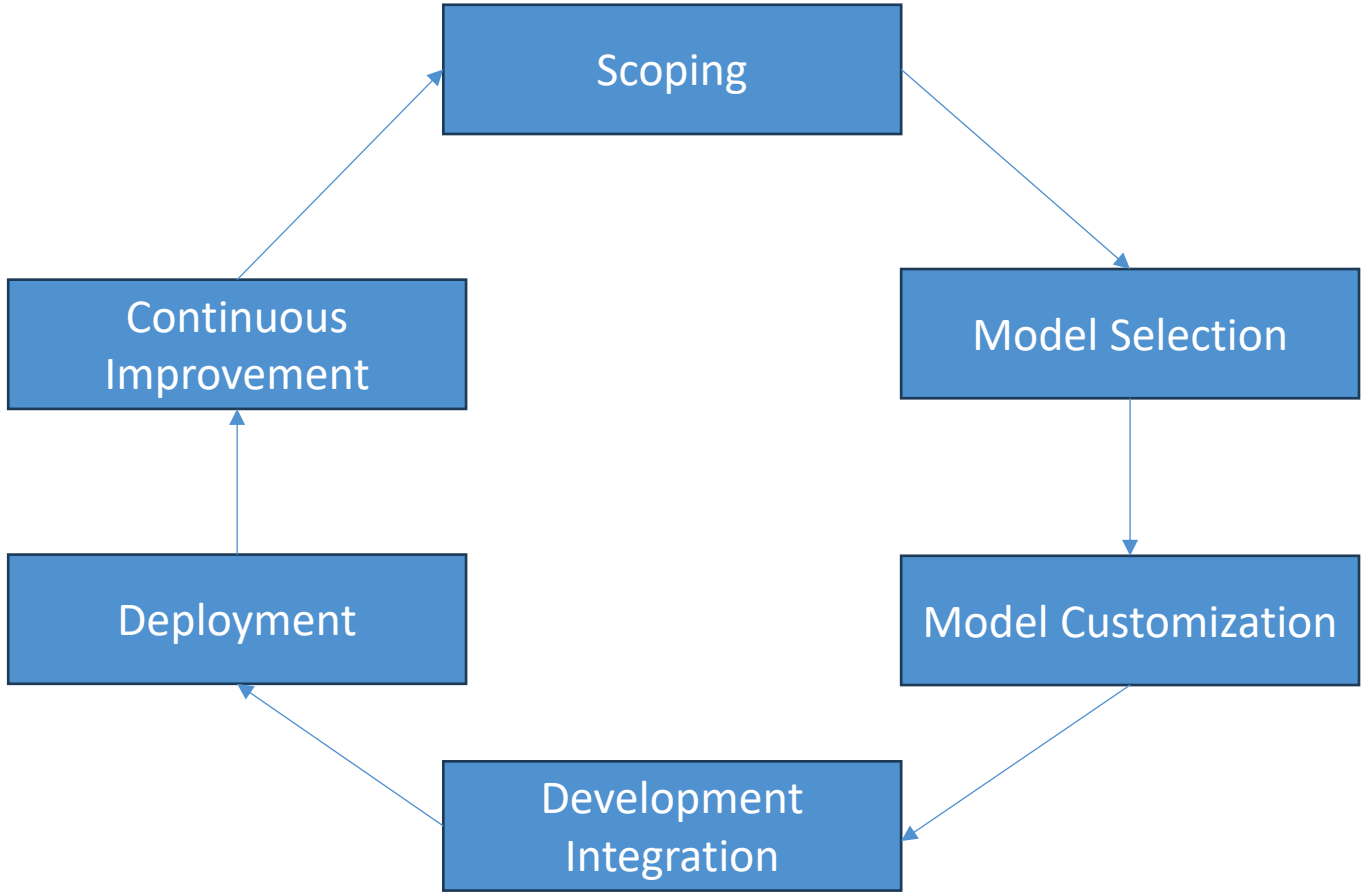
AWS Well-Architected Generative AI Lens: The Generative AI Lifecycle



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Managing Data for Generative AI

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Dealing with Structured Data

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Structured Data: Bedrock API Requests

- Bedrock generally expects request payloads in JSON format
  - Typically, this is the underlying native structure for the models you're working with

```
from botocore.exceptions import ClientError

# Create an Amazon Bedrock Runtime client.
brt = boto3.client("bedrock-runtime")

# Set the model ID, e.g., Amazon Titan Text G1 - Express.
model_id = "amazon.titan-text-express-v1"

# Define the prompt for the model.
prompt = "Describe the purpose of a 'hello world' program in one line."

# Format the request payload using the model's native structure.
native_request = {
    "inputText": prompt,
    "textGenerationConfig": {
        "maxTokenCount": 512,
        "temperature": 0.5,
        "topP": 0.9
    },
}

# Convert the native request to JSON.
request = json.dumps(native_request)

try:
    # Invoke the model with the request.
    response = brt.invoke_model(modelId=model_id, body=request)
```


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Structured Data for SageMaker AI Endpoints

- Models deployed via SageMaker will expect a certain input and output format
- Typically this is JSON for LLMs
  - But for classical ML, it might be csv or something else
- SageMaker will not format your input for you
- Your app and endpoint is responsible for this.

```
body=json.dumps( {
    "max_tokens": max_tokens,
    "system": system_prompt,
    "messages": messages } )

response =
bedrock_runtime.invoke_model(
    body=body,
   modelId=anthropic.claude-3-
haiku-20240307-v1:0")

response_body =
json.loads(response.get('body').re
ad())
```


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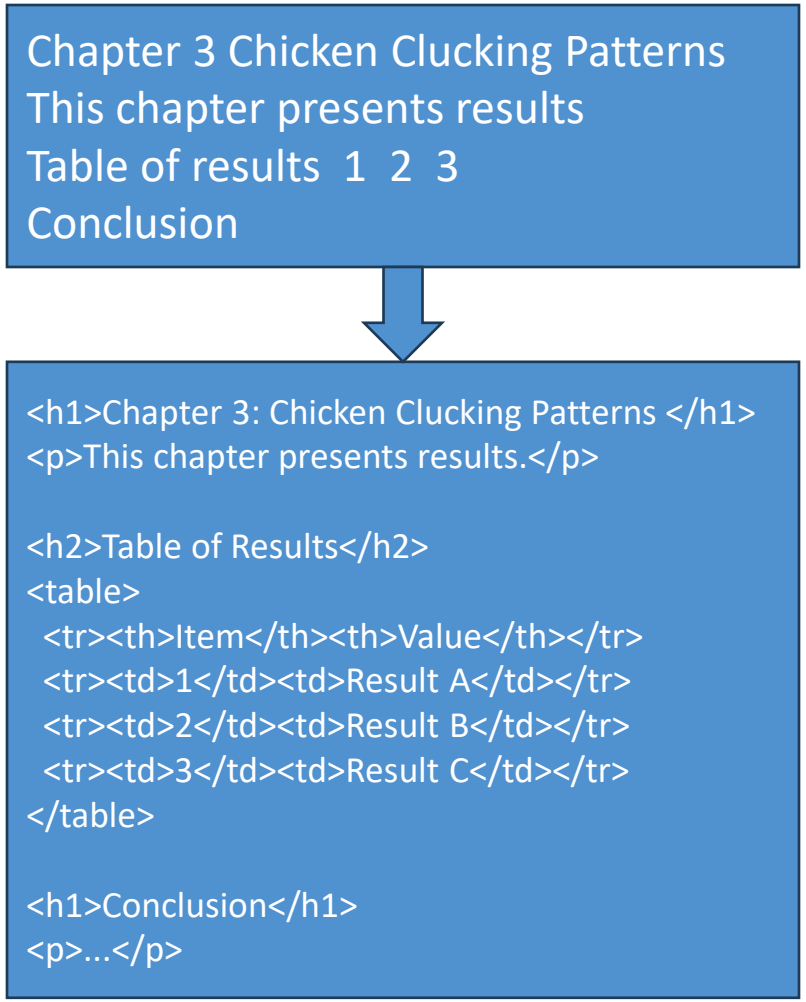
Dealing with Unstructured Text

- Raw text can lose its structure. GenAI likes structure.
- Headings, sections, metadata, tables get lost in the sea of text
- One technique is to convert your unstructured text to HTML (or something) that can preserve that structure
- Then your models can understand the structure and organization of the data better.
- Useful for OCR or PDF text, for example.
- Tools like pandoc, Amazon Textract and Amazon Comprehend can help extract structure from raw text or images.
- An ingestion pipeline performing this conversion might run on top of AWS Glue
- Or, the newer way to do this is with **Bedrock Data Automation (BDA)** (more on this soon.)



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Divider Strings for Segmenting Data

- When chunking data into a vector store, divider strings can be used for better chunking.
- Some ideas on how to do this
  - A Lambda Preprocessor in your Knowledge Base can convert HTML tags to divider strings on the fly
  - AWS Glue ETL could do it as part of a pipeline too

```
def handler(event, context):
    html = event["text"]

    html = html.replace("<h1", "<SECTION_BREAK><h1")
    html = html.replace("<h2", "<SUBSECTION_BREAK><h2")

    return {"text": html}
```

<h1>Executive Summary</h1>
Company revenue increased...

<h2>Key Risks</h2>
Supply-chain instability...

<SECTION_BREAK:Executive Summary>
Company revenue increased...

<SECTION_BREAK:Key Risks>
Supply-chain instability...

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Formatting Data for Conversations

- Bedrock's Converse API requires JSON output that includes:
  - Role
  - Content
- The "role" is who is "talking."
- Here's an example of a conversation history.

```
{
  "messages": [
    {
      "role": "user",
      "content": [{ "text": "You're a chicken expert.
Ready?" }]
    },
    {
      "role": "assistant",
      "content": [{ "text": "Absolutely—let's talk
chickens!" }]
    },
    {
      "role": "user",
      "content": [{ "text": "How fast can a chicken
run?" }]
    }
  ]
}
```


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Amazon Bedrock Data Automation

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What is Bedrock Data Automation (BDA)?

- Extracts structured data from pretty much anything
  - Multi-modal
  - Documents, images, videos, audio
- Useful for:
  - Preparing data for vector stores / KBs
  - Intelligent Document Processing (IDP)
  - Analyzing video
    - Scene summaries
    - ID explicit content
    - Extract text in video
    - Find ads

```
# NEW JERSEY

Motor Vehicle
 Commission

# AUTO DRIVER LICENSE

|     |                                                                |              |
|-----|----------------------------------------------------------------|--------------|
|     | DL M6454 64774 51685                                           | CLASS D      |
|     | DOB 01-01-1968                                                 |              |
|     | ISS 03-19-2019 EXP                                             | 01-01-2023   |
|     | MONTOYA RENEE MARIA 321 GOTHAM AVENUE TRENTON, NJ 08666        |              |
|     | ENC NONE                                                       |              |
|     | RESTR NONE                                                     |              |
|     | SEX F HGT 5'-08" EYES HZL                                      | ORGAN DONOR  |
|     | CM ST201907800000019 CHG                                       | 11.00        |

[SIGNATURE]
```


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BDA Concepts

Standard Output

- If you just throw a file at it, it will guess at what format you want the output
- For documents this is probably JSON. For audio it's a transcript. Etc.

Custom Output

- For documents, audio, or images, you can customize the output
- Specify exactly what you want to extract
- A **Blueprint** specified the fields you want
- There are standard blueprints (like US Driver License) or you can define your own.

You can store your output configurations in a **Project**.

- Kicking off BDA through the API requires calling InvokeDataAutomationAsync with your project.
- A project may include many Blueprints for different document types.

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Quick Demo Walkthrough

Bedrock Data Automation

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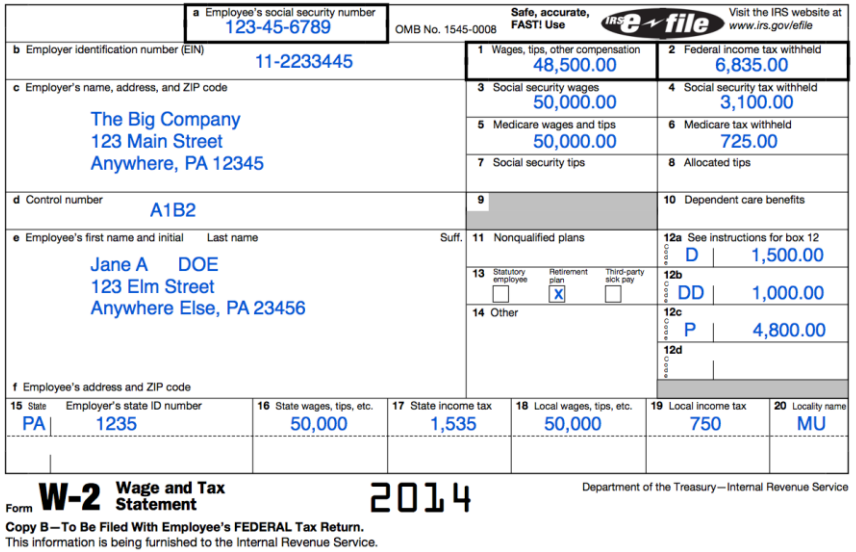
BDA Document Processing

- Accepts PDF, TIFF, JPEG, PNG, DOCX
- Outputs JSON or JSON+files
  - Files might include CSV for tables, overall text extraction, markdown
  - HTML and CSV structured output is also possible
    - Remember this came up earlier in the context of structuring unstructured data for better chunking – this is an easier way to do this
- You can select the granularity of responses
  - Page-level, element-level (default), word-level
  - Bounding boxes and generative summaries can be enabled per level



a Employee's social security number 123-45-6789 | OMB No. 1545-0008 | Safe, accurate, FAST! Use | Visit the IRS website at www.irs.gov/efile

b Employer identification number (EIN) 11-2233445

c Employer's name, address, and ZIP code
The Big Company
123 Main Street
Anywhere, PA 12345

d Control number A1B2

e Employee's first name and initial Last name Suff.
Jane A DOE
123 Elm Street
Anywhere Else, PA 23456

f Employee's address and ZIP code

1 Wages, tips, other compensation	48,500.00
2 Federal income tax withheld	6,835.00
3 Social security wages	50,000.00
4 Social security tax withheld	3,100.00
5 Medicare wages and tips	50,000.00
6 Medicare tax withheld	725.00
7 Social security tips	
8 Allocated tips	
9	
10 Dependent care benefits	
11 Nonqualified plans	
12a See instructions for box 12	D 1,500.00
12b	DD 1,000.00
12c	P 4,800.00
12d	
13 Statutory employee / Retirement plan [X] / Third-party sick pay	
14 Other	

15 State	Employer's state ID number	16 State wages, tips, etc.	17 State income tax	18 Local wages, tips, etc.	19 Local income tax	20 Locality name
PA	1235	50,000	1,535	50,000	750	MU

Form W-2 Wage and Tax Statement 2014 — Department of the Treasury—Internal Revenue Service

Copy B—To Be Filed With Employee's FEDERAL Tax Return.
This information is being furnished to the Internal Revenue Service.

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BDA Image Processing

- Accepts JPEG, PNG
- Generates:
  - Image summary / caption
  - IAB Taxonomy (Interactive Advertising Bureau)
  - Logos found
  - Text in images
  - Content moderation
  - …all in JSON

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BDA Video Processing

- Accepts MP4, MOV, AVI, MKV, WEBM
- Extracts:
  - Full video summary
  - Chapter summaries
    - This implies it can break the video down too!
  - IAB Taxonomy
  - Transcript
  - Text in video
  - Logos found
  - Content Moderation
  - …all in JSON

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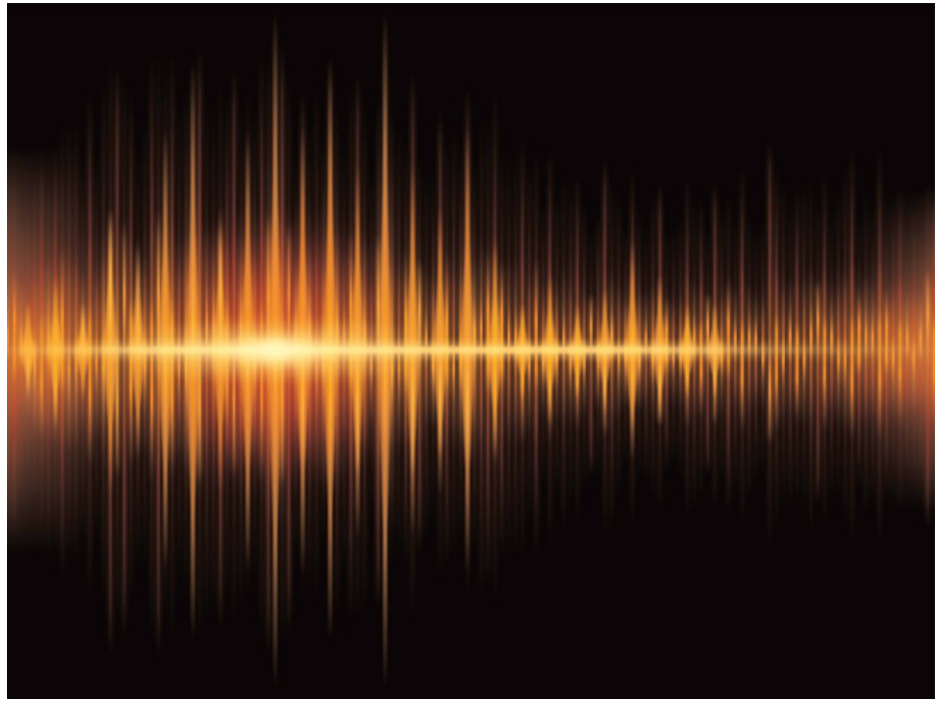
BDA Audio Processing

- Accepts AMR, FLAC, M4A, MP3, Ogg, WAV
  - …and a variety of languages
- Extracts:
  - Summary of the file
  - Full transcript
  - Speaker and Channel labeling in transcript
  - Breaking up into Topics
  - Content moderation
  - …all in JSON



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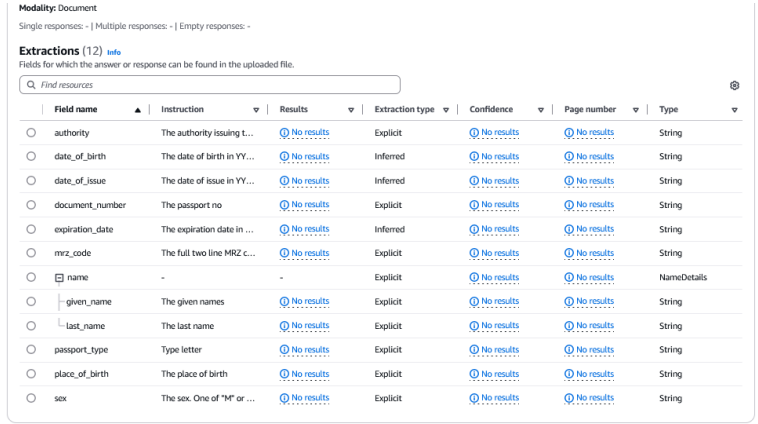
BDA Blueprints for Custom Output

- Basic fields
  - Explicit or implicit (transformed)
- Table fields
- Groups (for organizing fields)
- Custom types
  - Example: “address” consisting of city, street, zip, etc.
- You can create Blueprints just from a prompt as well!
- Again there are many pre-existing blueprints you can use for common document types.



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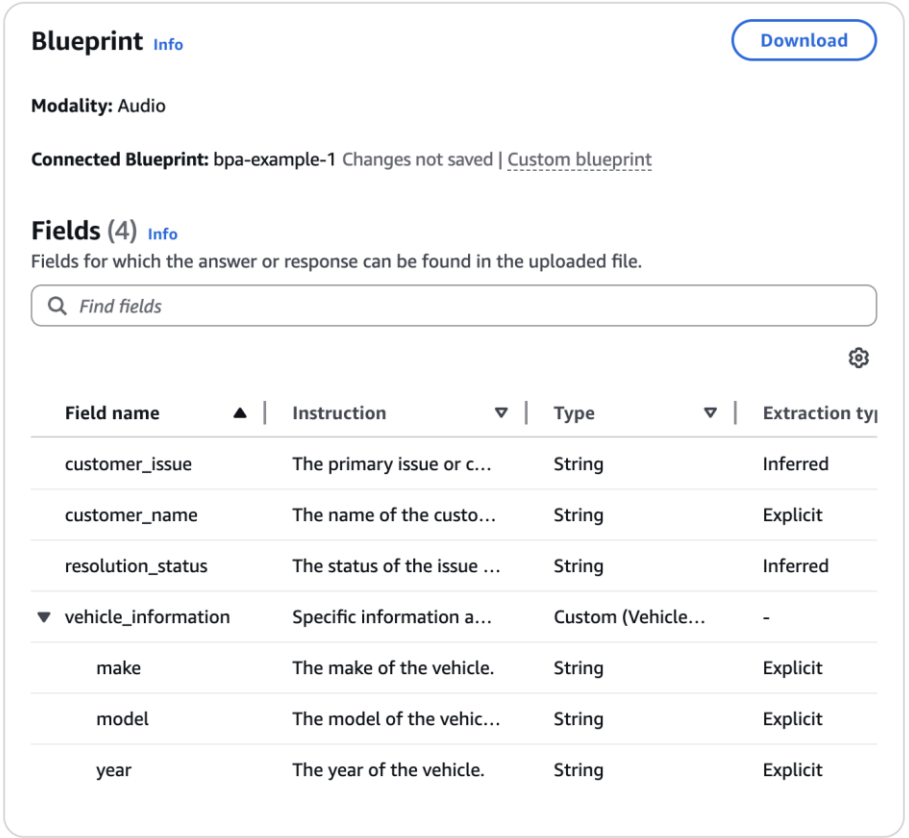
Uses of BDA Blueprints

- Classification
  - The document class and description can be used to automatically classify documents that match this blueprint
- Extraction
  - Allows you extract specific fields, even if it's from tabular data
- Normalization
  - Key Normalization can deal with different names for the same data
  - Value Normalization can ensure data is extracted in a consistent format and in consistent data types
- Transformation
  - You can split and restructure data fields
- Validation
  - Ensure the accuracy of data. Is it within specified ranges? Have the expected size? Etc.



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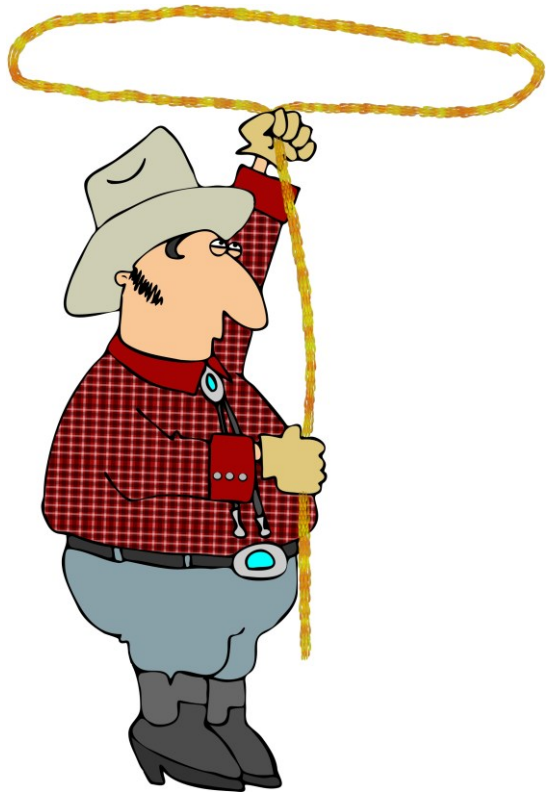
SageMaker Data Wrangler

- Visual interface (in SageMaker Studio) to prepare data for machine learning
- Import data
- Visualize data
- Transform data (300+ transformations to choose from)
  - Or integrate your own custom xforms with pandas, PySpark, PySpark SQL
- “Quick Model” to train your model with your data and measure its results



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Data Wrangler: Notable Capabilities

- Transforming image data
  - Resize, enhance, corrupt images & more
- Balance data
  - Random oversampling, undersampling, SMOTE
- Impute missing data
- Handle outliers
- Dimensionality reduction (PCA)

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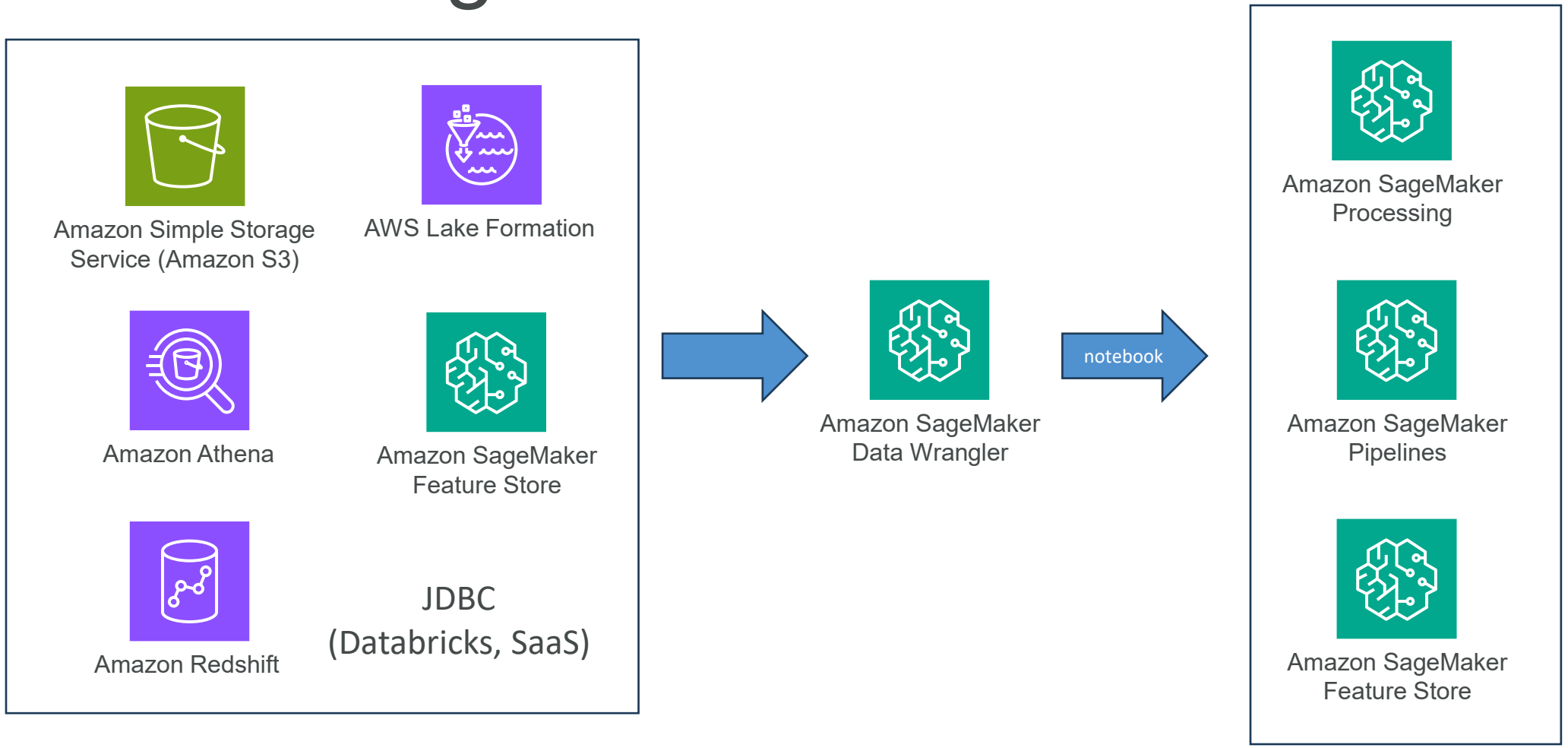
Data Wrangler sources



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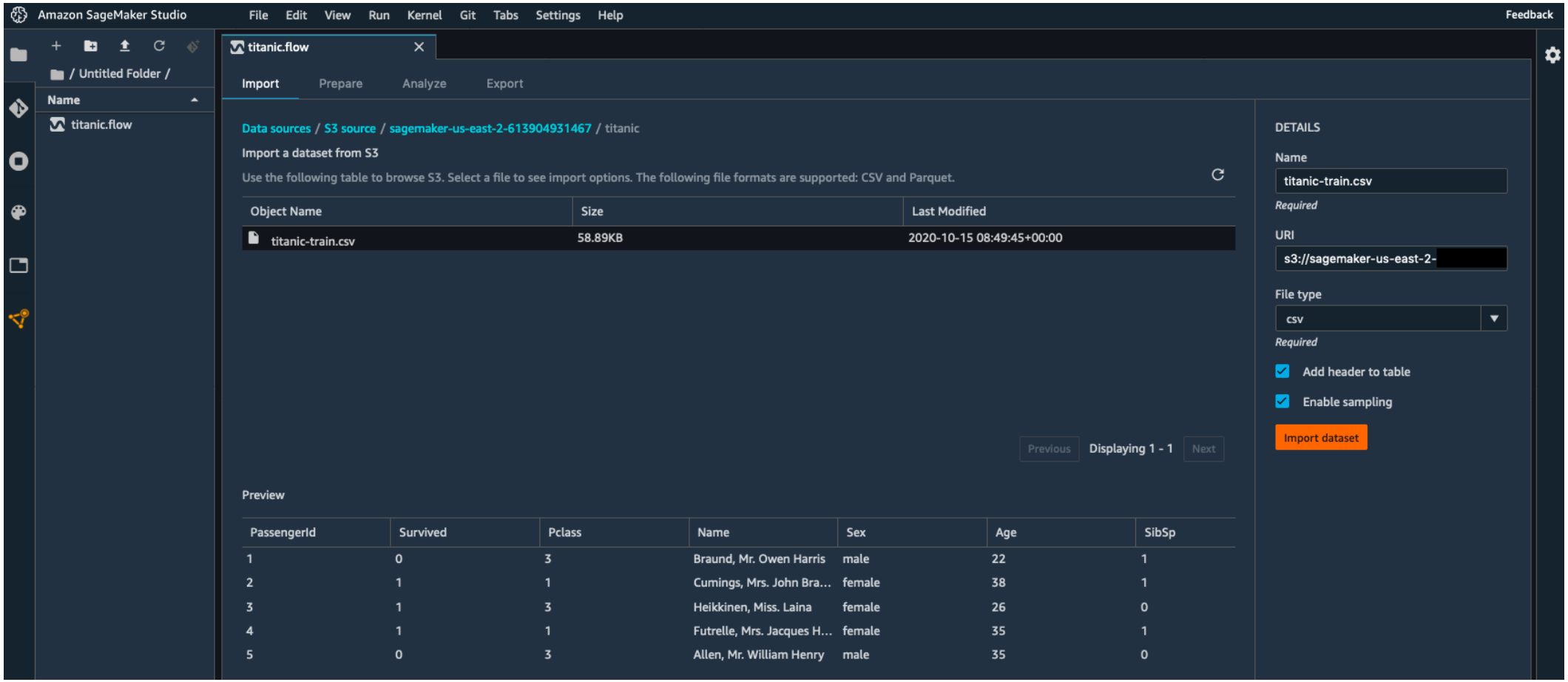
Data Wrangler: Import Data



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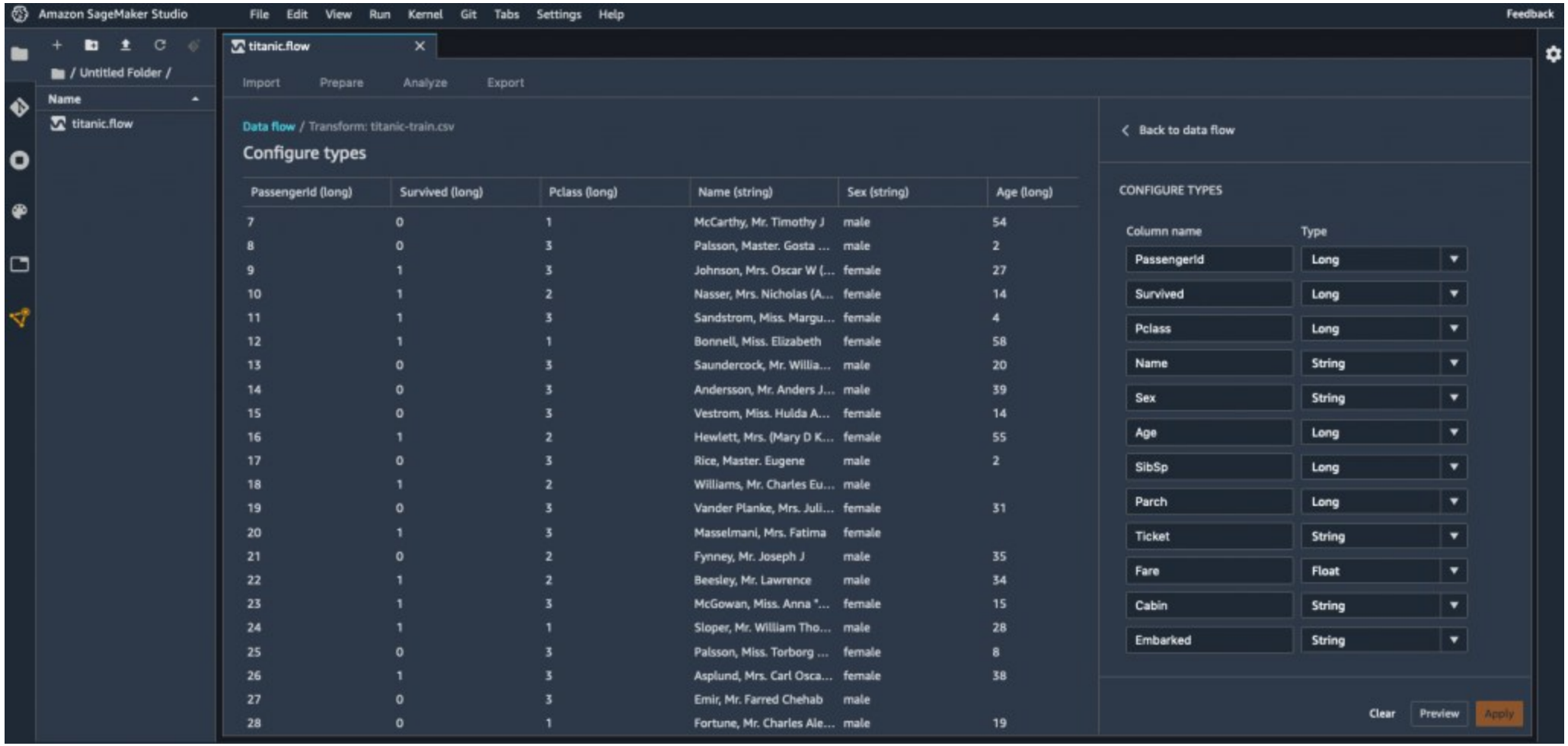
Data Wrangler: Preview Data



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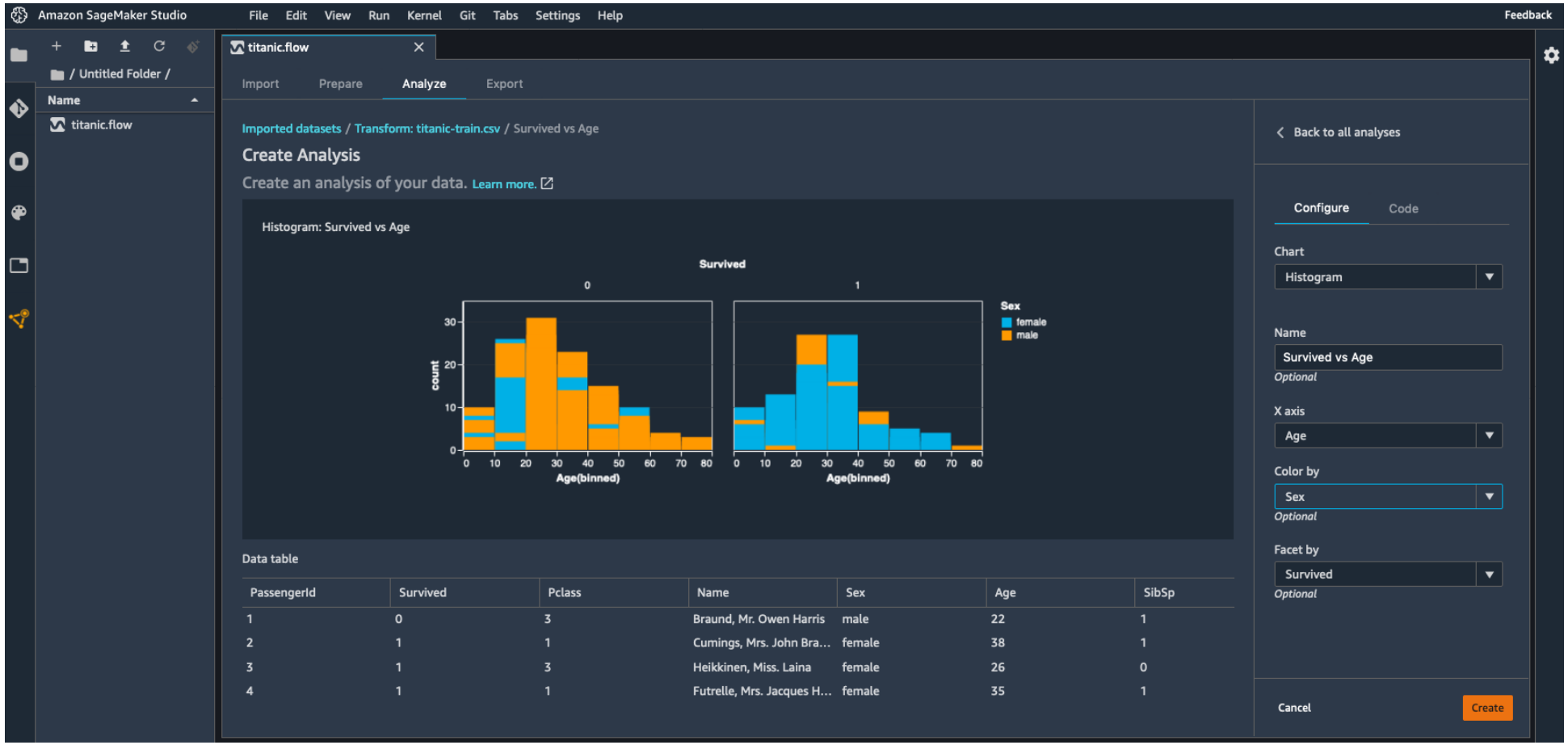
Data Wrangler: Visualize Data



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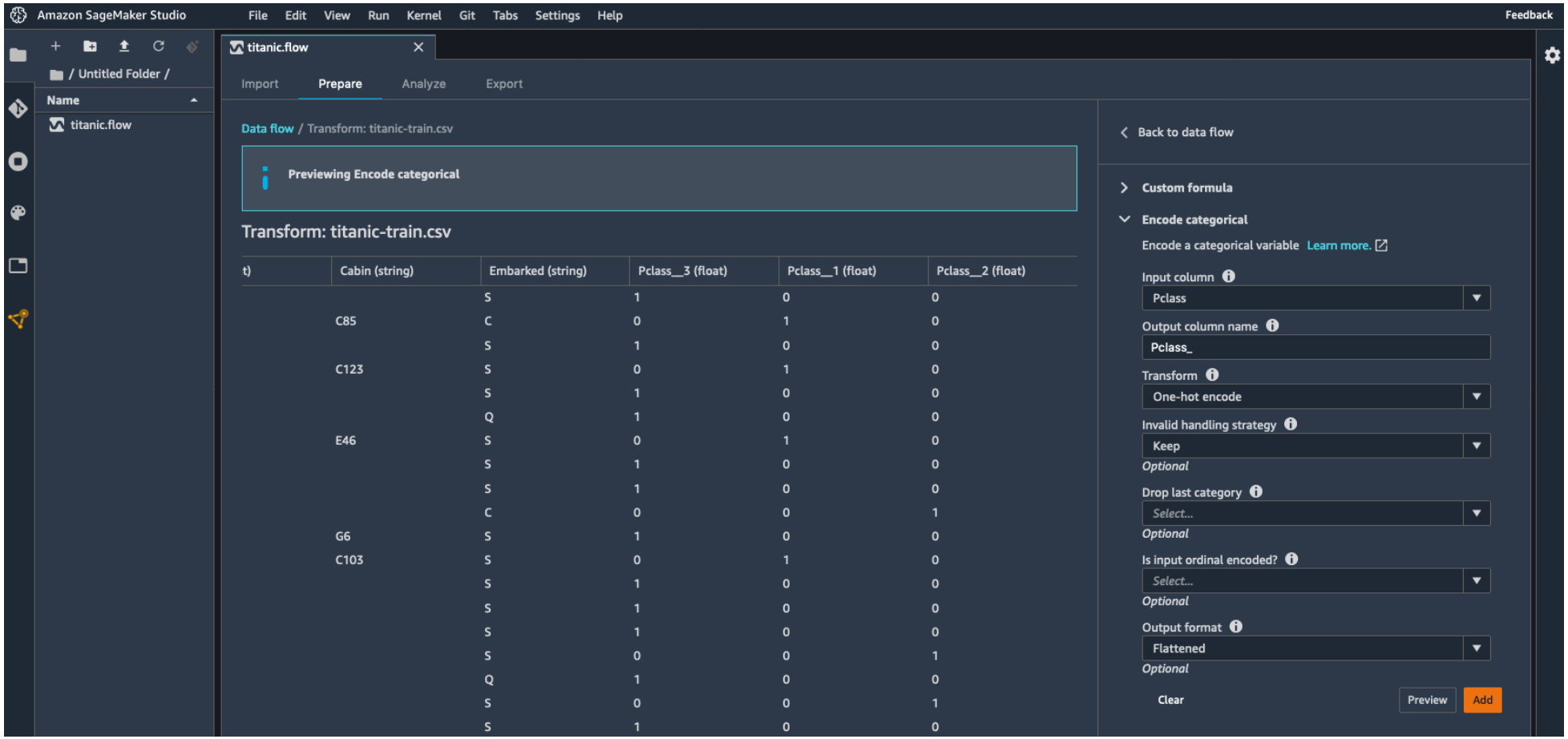
Data Wrangler: Transform Data



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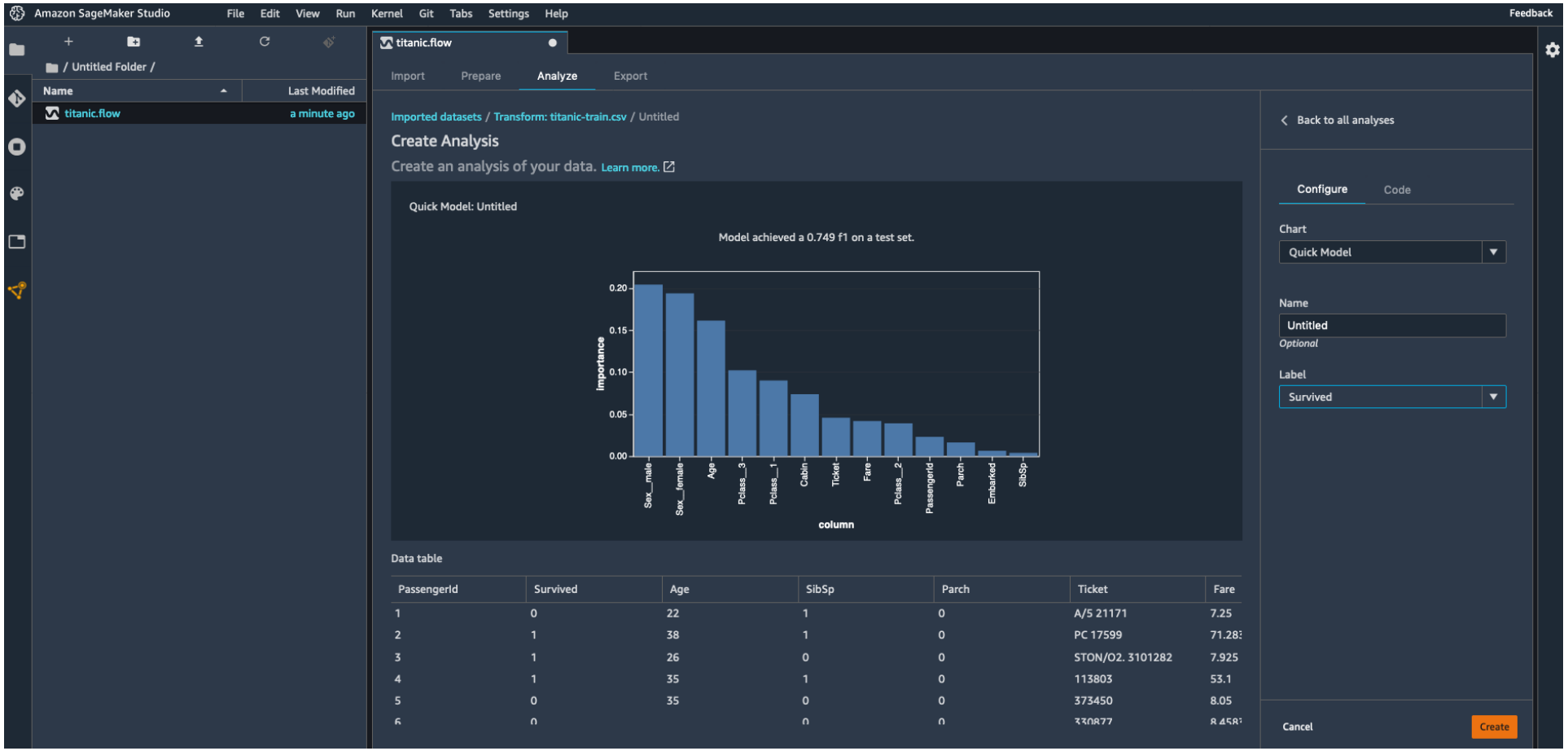
Data Wrangler: Quick Model



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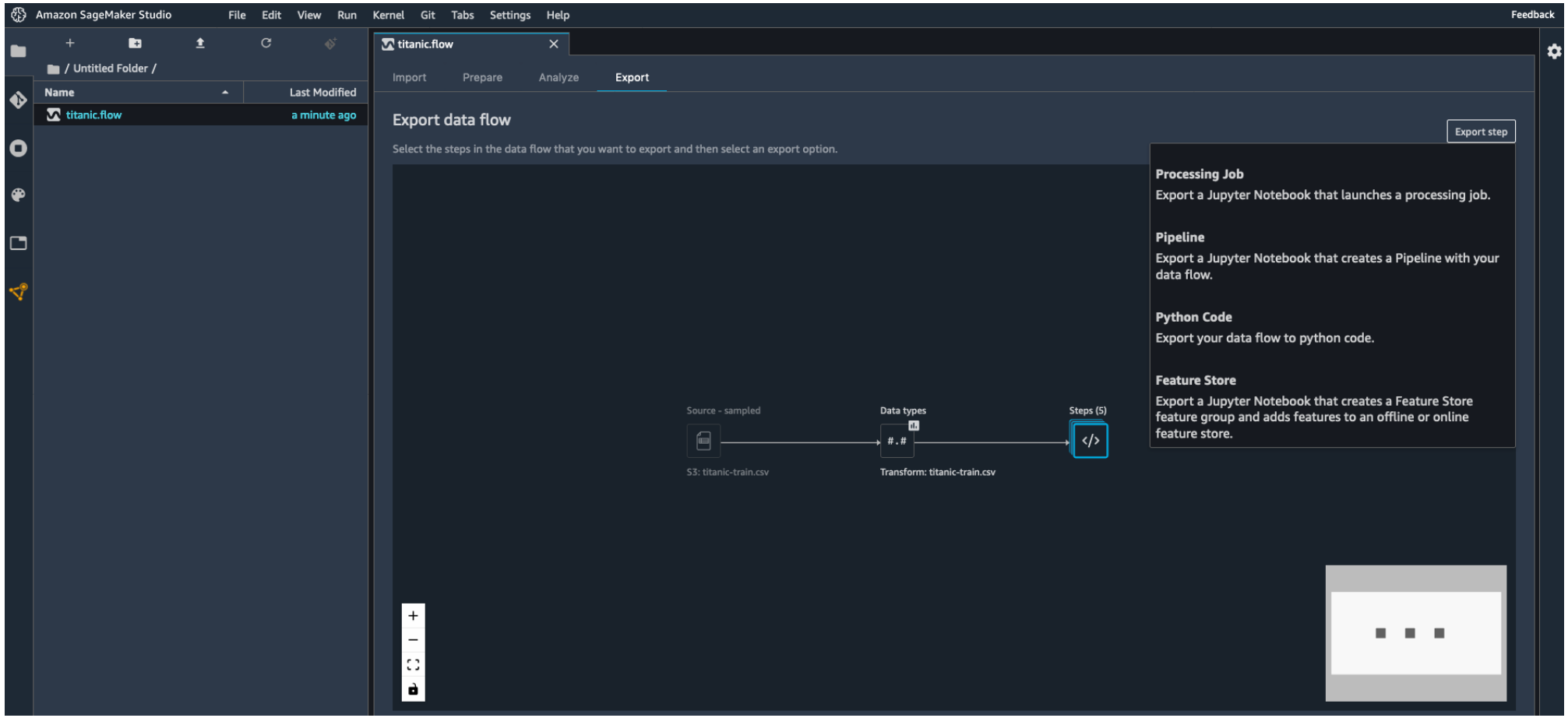
Data Wrangler: Export Data Flow



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Data Wrangler Troubleshooting

- Make sure your Studio user has appropriate IAM roles
- Make sure permissions on your data sources allow Data Wrangler access
  - Add AmazonSageMakerFullAccess policy
- EC2 instance limit
  - If you get “The following instance type is not available…” errors
  - May need to request a quota increase
  - Service Quotas / Amazon SageMaker / Studio KernelGateway Apps running on ml.m5.4xlarge instance

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AWS Glue

Table definitions and ETL

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What is Glue?

- Serverless discovery and definition of table definitions and schema
  - S3 “data lakes”
  - RDS
  - Redshift
  - DynamoDB
  - Most other SQL databases
- Custom ETL jobs
  - Trigger-driven, on a schedule, or on demand
  - Fully managed

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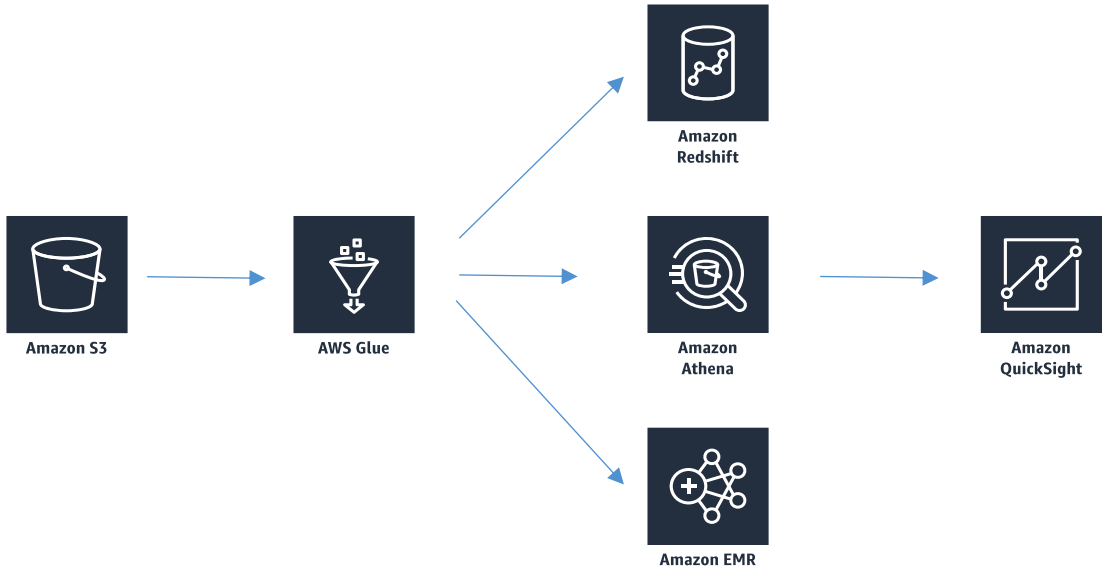
Glue Crawler / Data Catalog



- Glue crawler scans data in S3, creates schema
- Can run periodically
- Populates the Glue Data Catalog
  - Stores only table definition
  - Original data stays in S3
- Once cataloged, you can treat your unstructured data like it's structured
  - Redshift Spectrum
  - Athena
  - EMR
  - Quicksight

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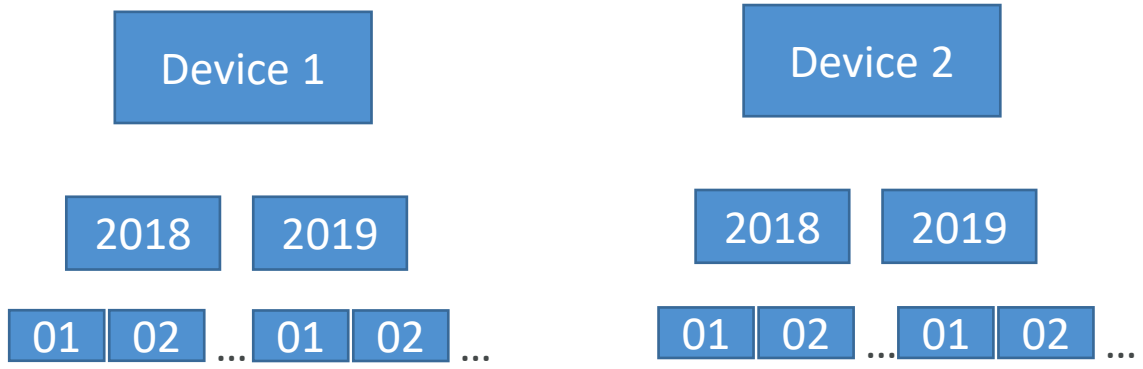
Glue and S3 Partitions

- Glue crawler will extract partitions based on how your S3 data is organized
- Think up front about how you will be querying your data lake in S3
- Example: devices send sensor data every hour
- Do you query primarily by time ranges?
  - If so, organize your buckets as yyyy/mm/dd/device
- Do you query primarily by device?
  - If so, organize your buckets as device/yyyy/mm/dd



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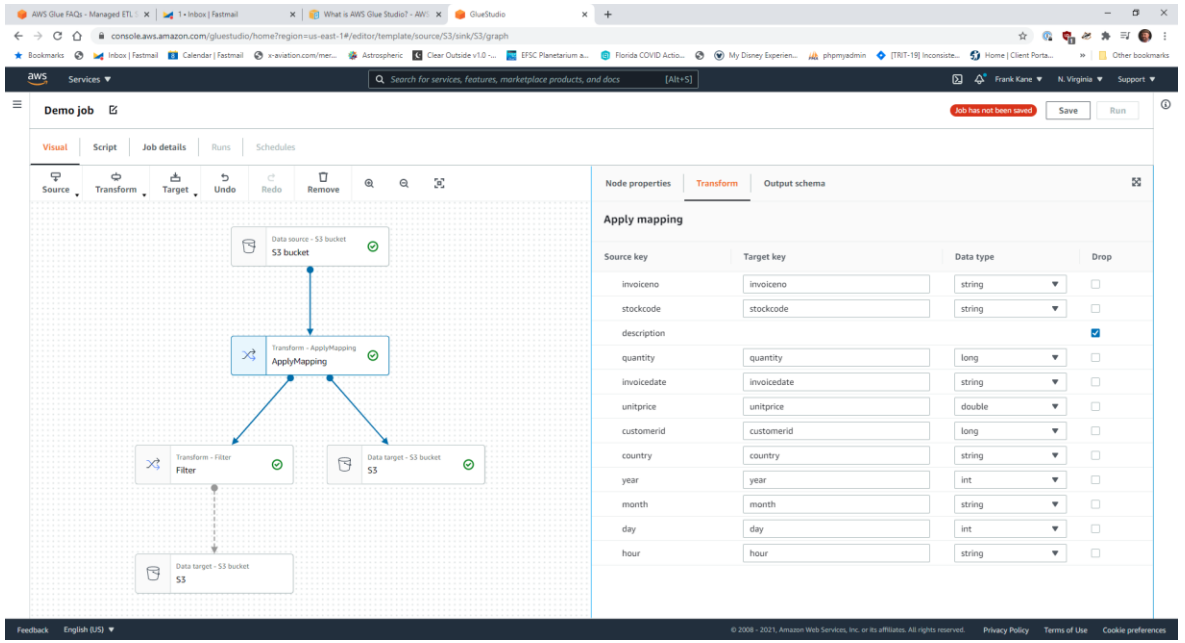
AWS Glue Studio

- Visual interface for ETL workflows
- Visual job editor
  - Create DAGs for complex workflows
  - Sources include S3, Kinesis, Kafka, JDBC
  - Transform / sample / join data
  - Target to S3 or Glue Data Catalog
  - Support partitioning
- Visual job dashboard
  - Overviews, status, run times



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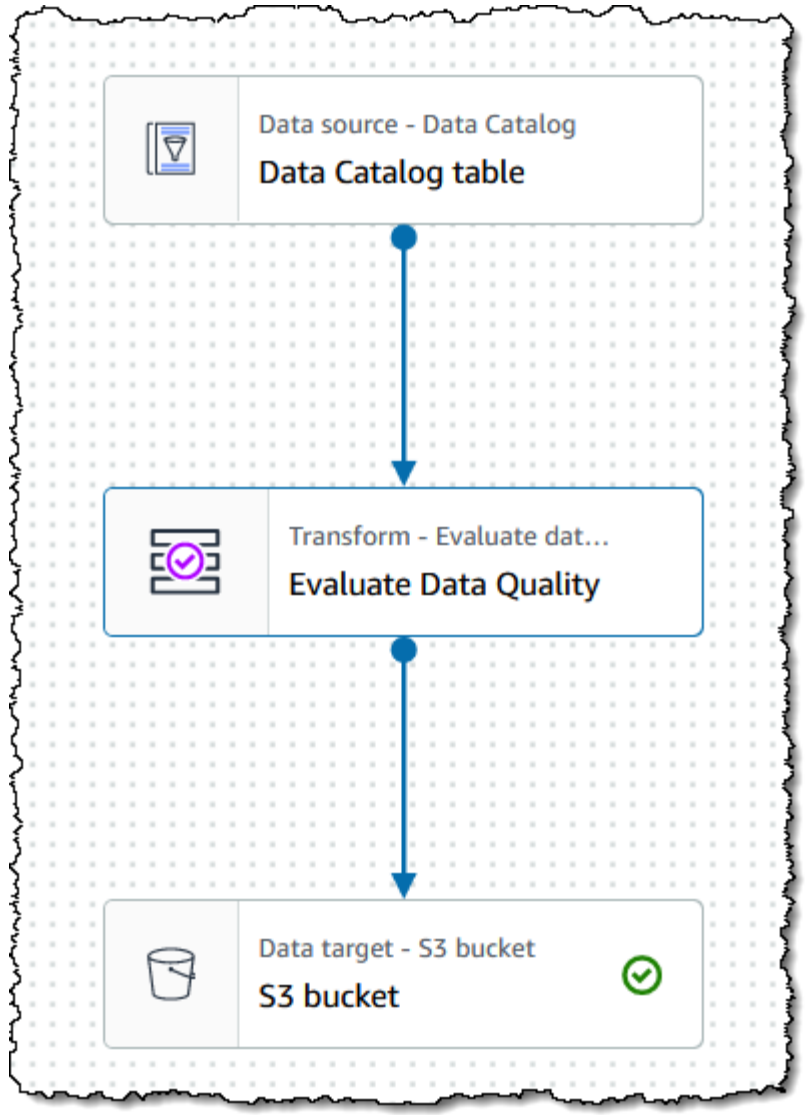
AWS Glue Data Quality

- Data quality rules may be created manually or recommended automatically
- Integrates into Glue jobs
- Uses Data Quality Definition Language (DQDL)
- Results can be used to fail the job, or just be reported to CloudWatch



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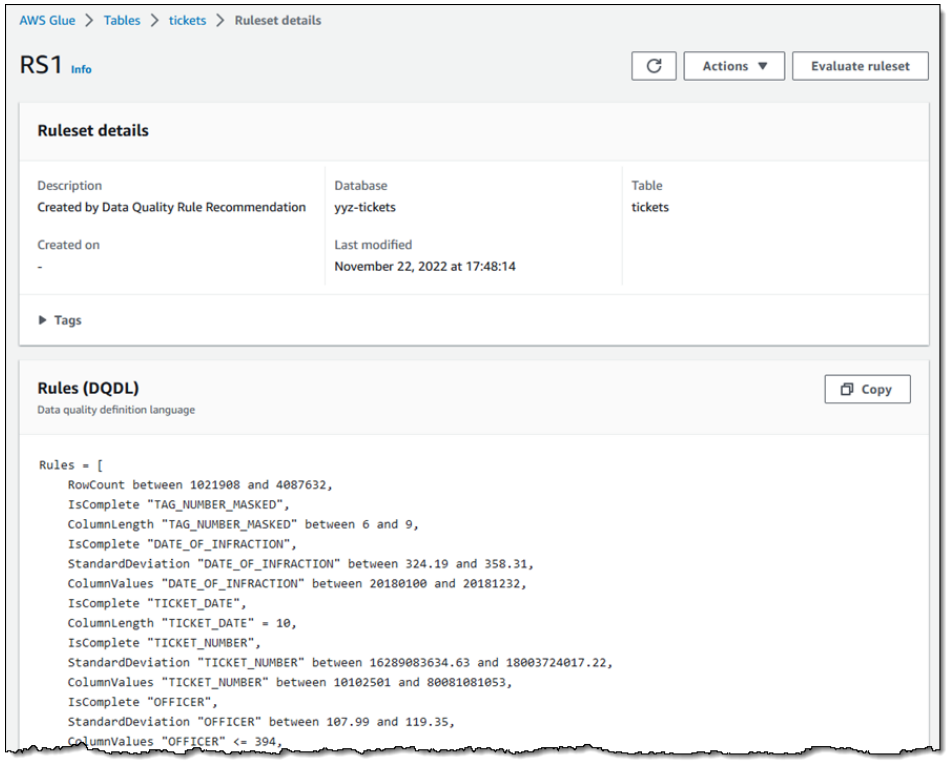
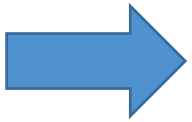
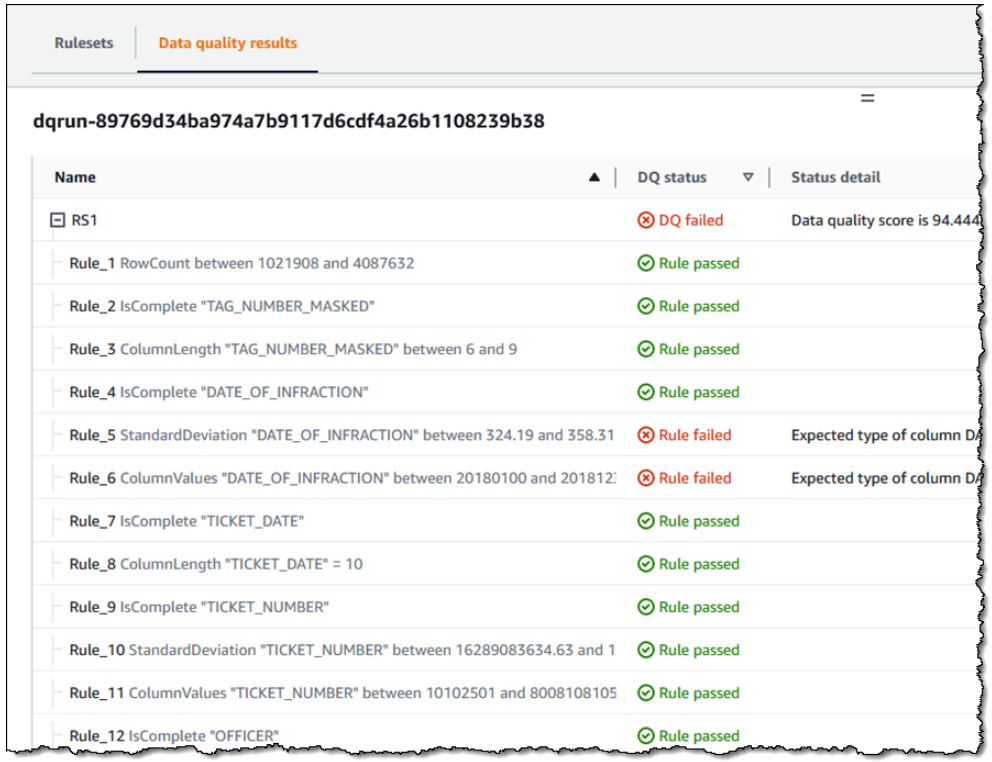
AWS Glue Data Quality







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Amazon CloudWatch Metrics

- CloudWatch provides metrics for *every* services in AWS
- **Metric** is a variable to monitor (CPUUtilization, NetworkIn…)
- Metrics belong to **namespaces**
- **Dimension** is an attribute of a metric (instance id, environment, etc…).
- Up to 30 dimensions per metric
- Metrics have **timestamps**
- Can create CloudWatch dashboards of metrics
- Can create **CloudWatch Custom Metrics** (for the RAM for example)

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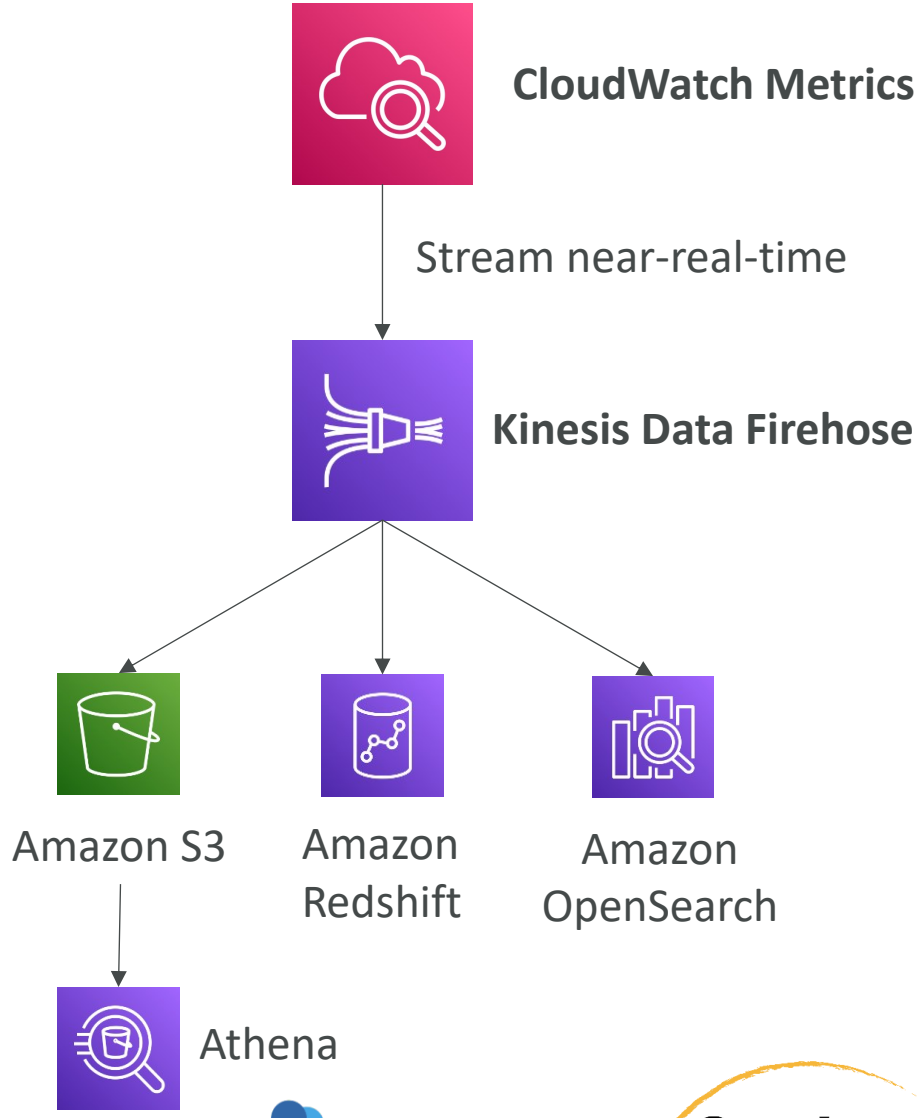
CloudWatch Metric Streams

- Continually stream CloudWatch metrics to a destination of your choice, with **near-real-time delivery** and low latency.
  - Amazon Kinesis Data Firehose (and then its destinations)
  - 3rd party service provider: Datadog, Dynatrace, New Relic, Splunk, Sumo Logic…
- Option to **filter metrics** to only stream a subset of them



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Amazon Transcribe

- Automatically **convert speech to text**
- Uses a **deep learning process** called **automatic speech recognition** (ASR) to convert speech to text quickly and accurately
- **Automatically remove Personally Identifiable Information (PII) using Redaction**
- **Supports Automatic Language Identification for multi-lingual audio**
- Use cases:
  - transcribe customer service calls
  - automate closed captioning and subtitling
  - generate metadata for media assets to create a fully searchable archive

”Hello my name is Stéphane.
I hope you’re enjoying the course!

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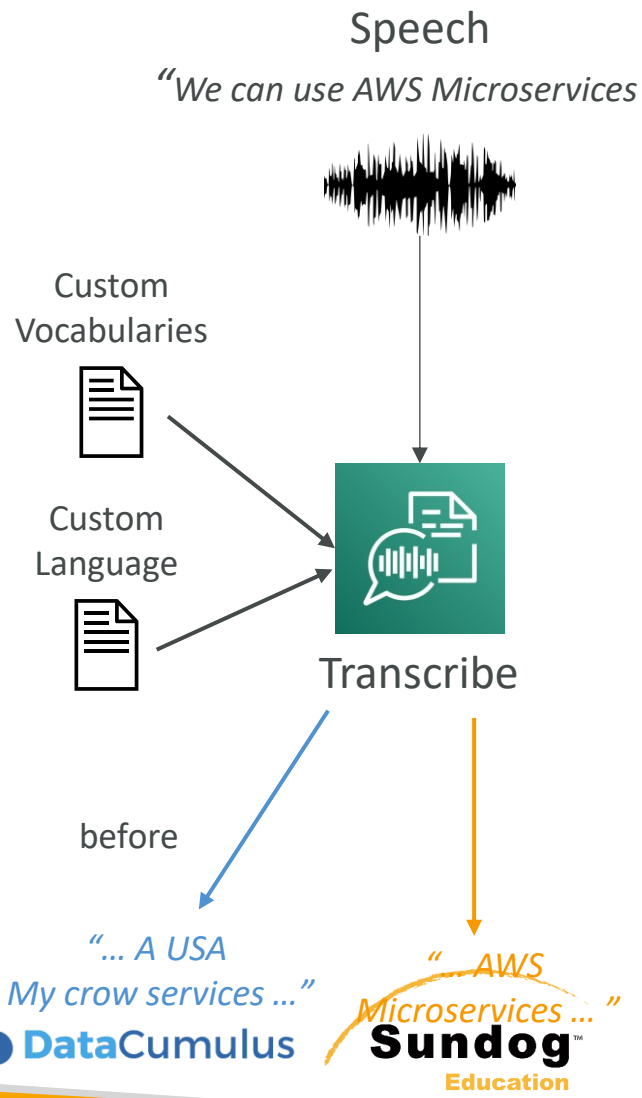
Amazon Transcribe – Improving Accuracy

- Allows Transcribe to capture domain-specific or non-standard terms (e.g., technical words, acronyms, jargon…)
- **Custom Vocabularies (for words)**
  - Add specific words, phrases, domain-specific terms
  - Good for brand names, acronyms…
  - Increase recognition of a new word by providing hints (such as pronunciation..)
- **Custom Language Models (for context)**
  - Train Transcribe model on your own domain-specific text data
  - Good for transcribing large volumes of domain-specific speech
  - Learn the context associated with a given word
- **Note:** use both for the highest transcription accuracy



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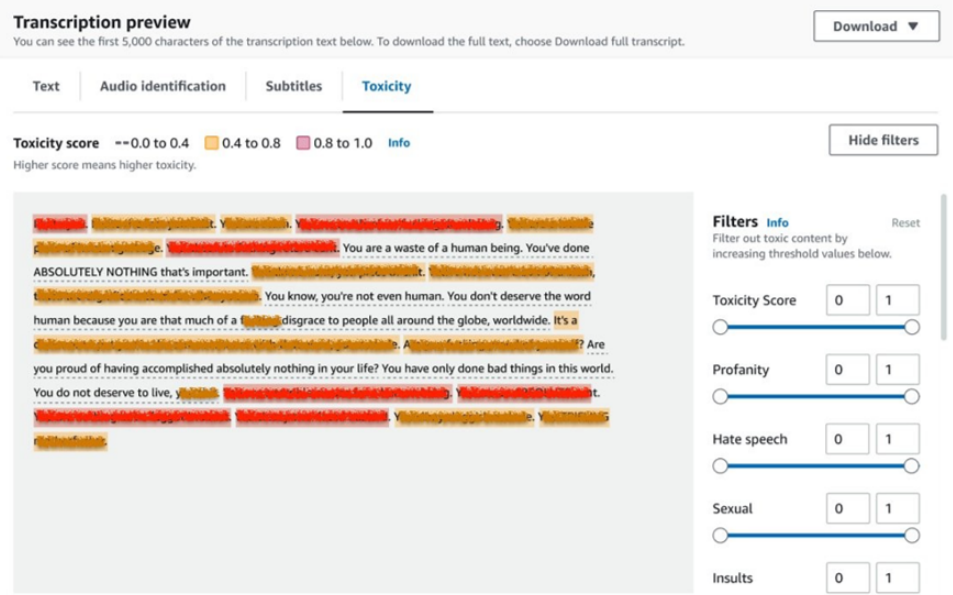
Transcribe – Toxicity Detection

- ML-powered, voice-based toxicity detection capability
- Leverages speech cues: tone and pitch, and text-based cues
- Toxicity categories: sexual harassment, hate speech, threat, abuse, profanity, insult, and graphic….



Toxicity Categories

Profanity: Speech that contains words, phrases, or acronyms that are impolite, vulgar, or offensive.

Hate speech: Speech that criticizes, insults, denounces, or dehumanizes a person or group on the basis of an identity (such as race, ethnicity, gender, religion, sexual orientation, ability, and national origin).

Sexual: Speech that indicates sexual interest, activity, or arousal using direct or indirect references to body parts, physical traits, or sex.

Insults: Speech that includes demeaning, humiliating, mocking, insulting, or belittling language. This type of language is also labeled as bullying.

Violence or threat: Speech that includes threats seeking to inflict pain, injury, or hostility toward a person or group.

Graphic: Speech that uses visually descriptive and unpleasantly vivid imagery. This type of language is often intentionally verbose to amplify a recipient's discomfort.

Harassment or abusive: Speech intended to affect the psychological well-being of the recipient, including demeaning and objectifying terms.

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Amazon Comprehend

- For **Natural Language Processing – NLP**
- Fully managed and serverless service
- Uses machine learning to find insights and relationships in text
  - Language of the text
  - Extracts key phrases, places, people, brands, or events
  - Understands how positive or negative the text is
  - Analyzes text using tokenization and parts of speech
  - Automatically organizes a collection of text files by topic
- Sample use cases:
  - analyze customer interactions (emails) to find what leads to a positive or negative experience
  - Create and groups articles by topics that Comprehend will uncover

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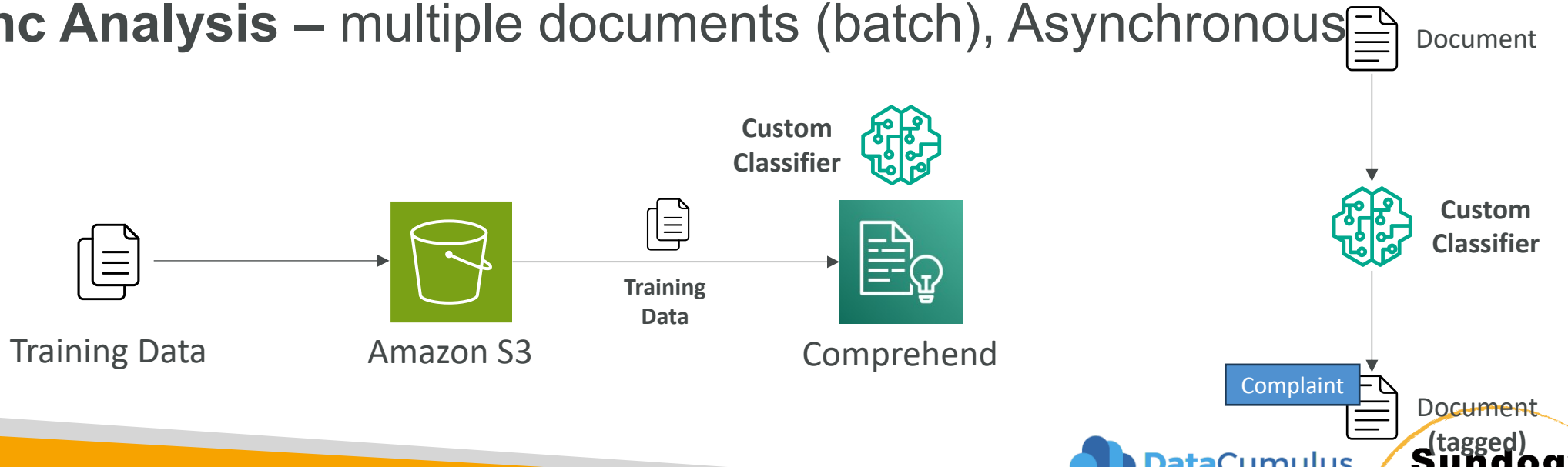
Comprehend – Custom Classification

- Organize documents into categories (classes) that you define
- Example: categorize customer emails so that you can provide guidance based on the type of the customer request
- Supports different document types (text, PDF, Word, images...)
- **Real-time Analysis –** single document, synchronous
- **Async Analysis –** multiple documents (batch), Asynchronous



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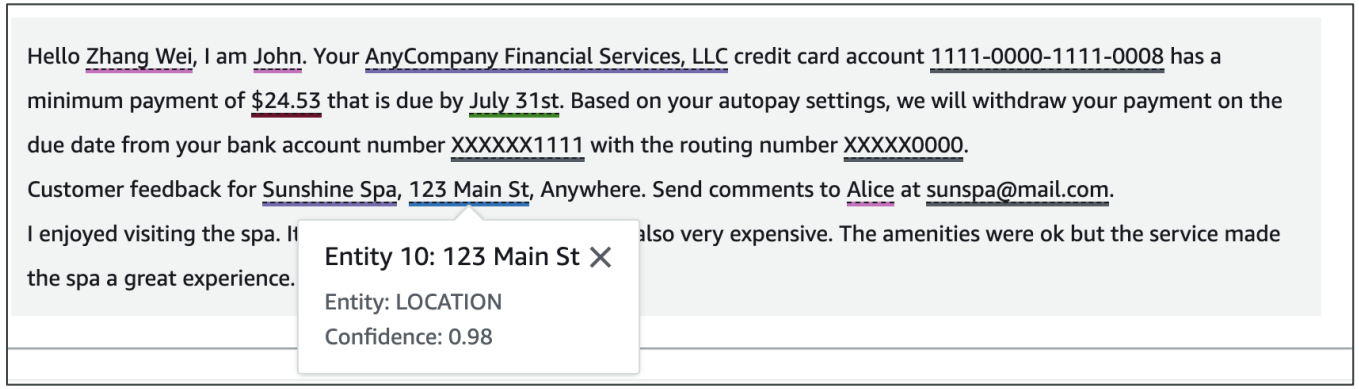
Named Entity Recognition (NER)

- **NER –** Extracts predefined, general-purpose entities like people, places, organizations, dates, and other standard categories, **from text**



Entity	Type
Zhang Wei	Person
John	Person
AnyCompany Financial Services, LLC	Organization
1111-0000-1111-0008	Other
$24.53	Quantity
July 31st	Date
XXXXXX1111	Other
XXXXX0000	Other
Sunshine Spa	Organization
123 Main St	Location

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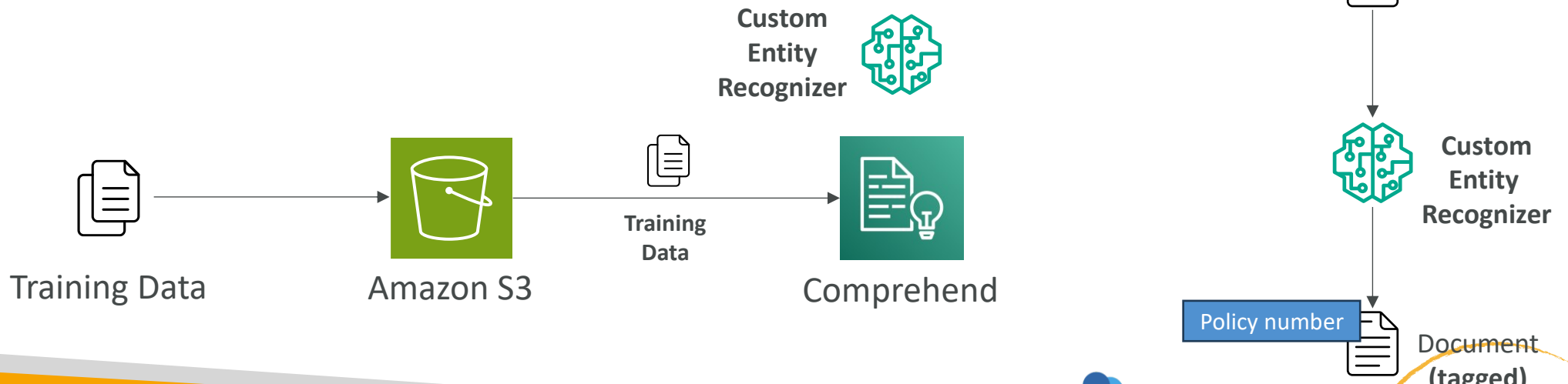
Comprehend – Custom Entity Recognition

- Analyze text for specific terms and noun-based phrases
- Extract terms like policy numbers, or phrases that imply a customer escalation, anything specific to your business
- Train the model with custom data such as a list of the entities and documents that contain them
- Real-time or Async analysis



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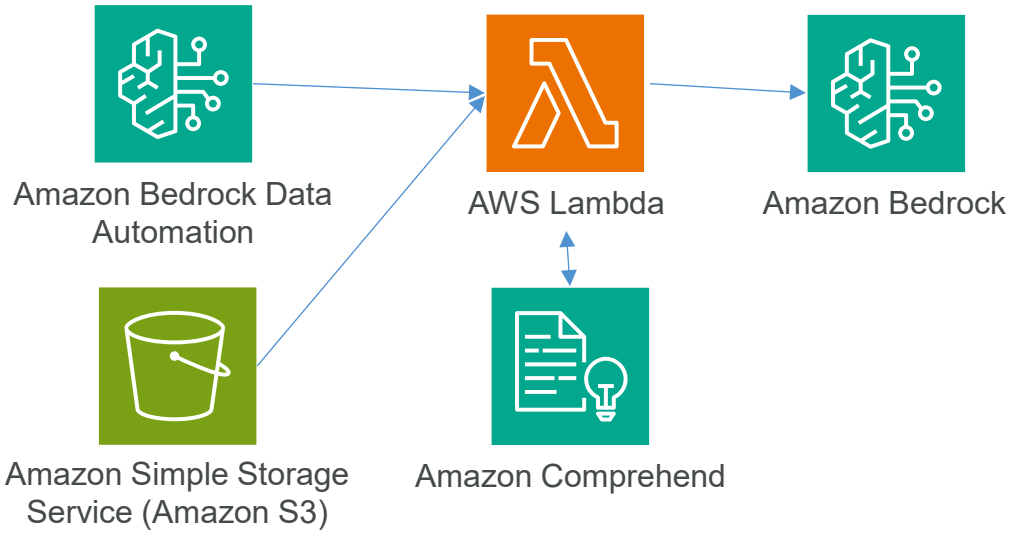
Using Comprehend and Lambda for Data Quality

- A Lambda function can call Amazon Comprehend before data hits Bedrock to…
  - Redact PII
  - Extract entities
  - Detect language
  - Classify the data
- Example applications
  - Clean up transcripts before going into a Knowledge Base
  - Pre-screen user-generated content before feeding it to an agent



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Amazon Comprehend Medical

- A HIPAA-certified version of Comprehend for medical use
- Pre-trained for extracting health data (prescriptions, procedures, diagnoses, protected health information (PHI)) as entities
- Separate API for PHI (DetectPHI instead of DetectEntities)
- Can search for specific medical ontologies
- Individual documents or batch in S3
- Integrates with VPC Endpoints / PrivateLink for security
- Could potentially be used in a post-processing scenario to detect PHI

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Amazon Opensearch Service

Petabyte-scale analysis and reporting

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What is Opensearch?

- A fork of Elasticsearch and Kibana
- A search engine
- An analysis tool
- A visualization tool (Dashboards = Kibana)
- A data pipeline
  - Kinesis replaces Beats & LogStash
- Horizontally scalable

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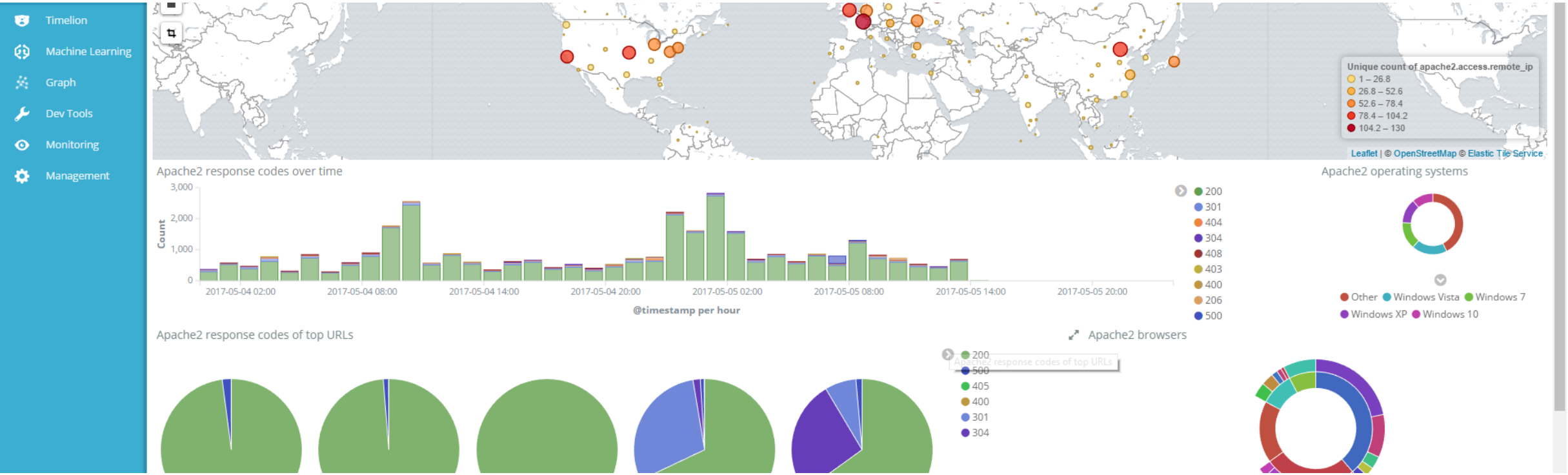
What are Dashboards?



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Opensearch applications

- Full-text search
- Log analytics
- Application monitoring
- Security analytics
- Clickstream analytics

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Opensearch concepts

documents

Documents are the things you're searching for. They can be more than text – any structured JSON data works. Every document has a unique ID, and a type.

types

A type defines the schema and mapping shared by documents that represent the same sort of thing. (A log entry, an encyclopedia article, etc.)

indices

An index powers search into all documents within a collection of types. They contain inverted indices that let you search across everything within them at once.

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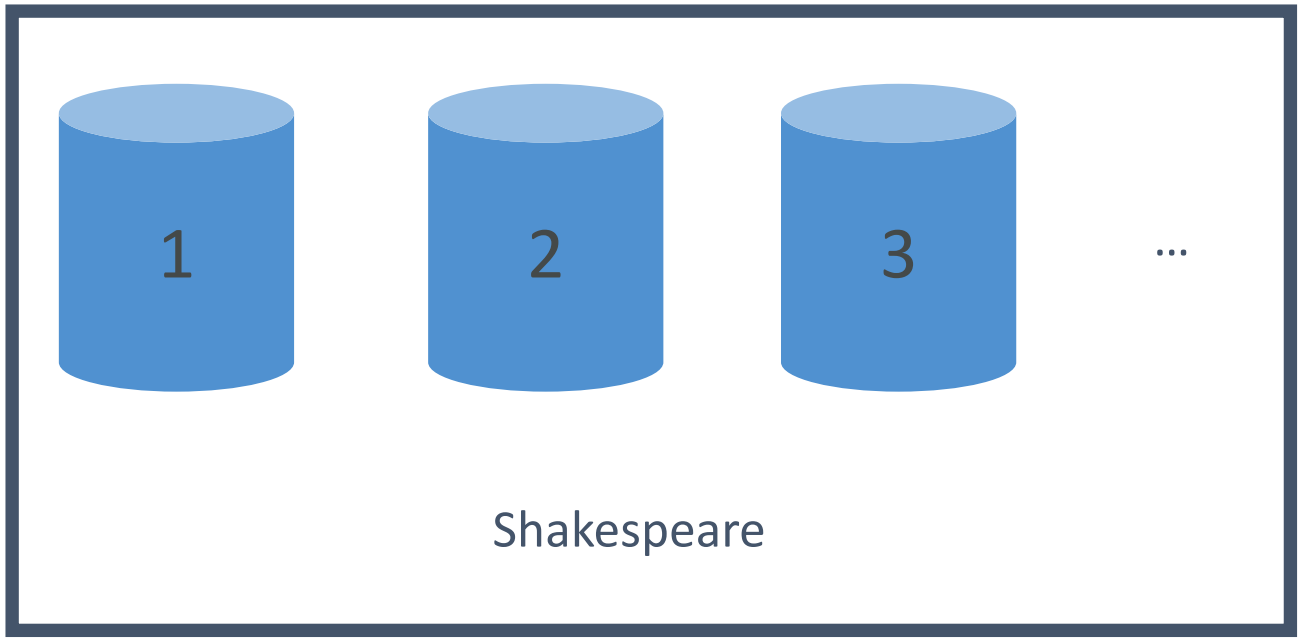
An index is split into shards

Documents are hashed to a particular shard.



Each shard may be on a different node in a cluster.
Every shard is a self-contained Lucene index of its own.

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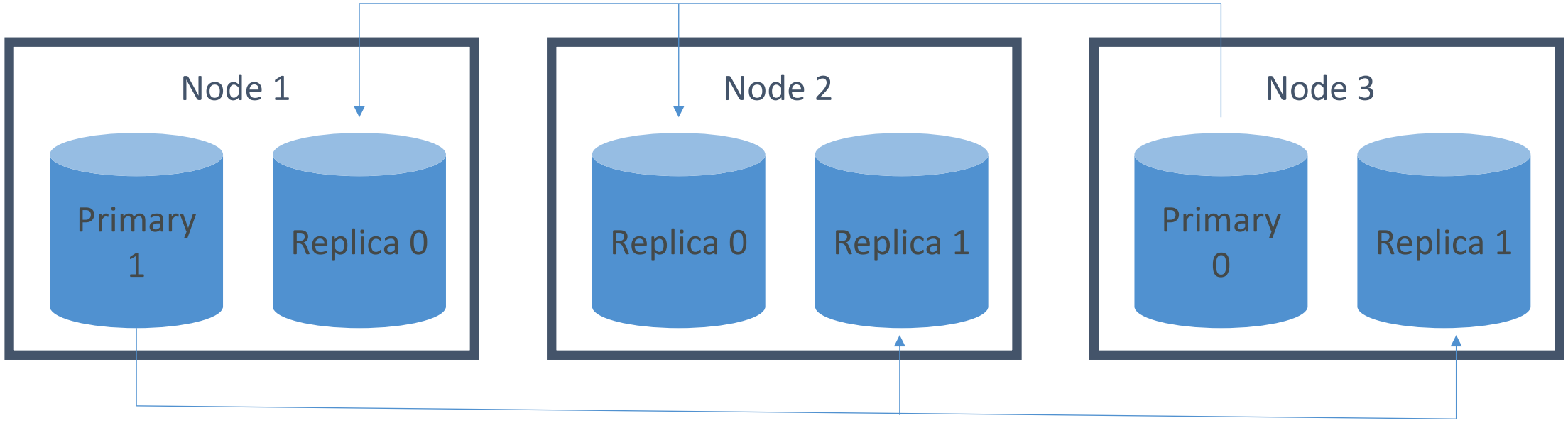
Redundancy

This index has two primary shards and two replicas.
Your application should round-robin requests amongst nodes.



Write requests are routed to the primary shard, then replicated
Read requests are routed to the primary or any replica

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Amazon OpenSearch Service (Managed)

- Fully-managed (but not serverless)
  - There is a separate serverless option now
- Scale up or down without downtime
  - But this isn't automatic
- Pay for what you use
  - Instance-hours, storage, data transfer
- Network isolation
- AWS integration
  - S3 buckets (via Lambda to Kinesis)
  - Kinesis Data Streams
  - DynamoDB Streams
  - CloudWatch / CloudTrail
  - Zone awareness

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Amazon Opensearch options

- Dedicated master node(s)
  - Choice of count and instance types
- “Domains”
- Snapshots to S3
- Zone Awareness

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Cold / warm / ultrawarm / hot storage

- Standard data nodes use “hot” storage
  - Instance stores or EBS volumes / fastest performance
- UltraWarm (warm) storage uses S3 + caching
  - Best for indices with few writes (like log data / immutable data)
  - Slower performance but much lower cost
  - Must have a dedicated master node
- Cold storage
  - Also uses S3
  - Even cheaper
  - For “periodic research or forensic analysis on older data”
  - Must have dedicated master and have UltraWarm enabled too.
  - Not compatible with T2 or T3 instance types on data nodes
  - If using fine-grained access control, must map users to cold_manager role in OpenSearch Dashboards
- Data may be migrated between different storage types

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Index State Management

- Automates index management policies
- Examples
  - Delete old indices after a period of time
  - Move indices into read only state after a period of time
  - Move indices from hot -> UltraWarm -> cold storage over time
  - Reduce replica count over time
  - Automate index snapshots
- ISM policies are run every 30-48 minutes
  - Random jitter to ensure they don't all run at once
- Can even send notifications when done

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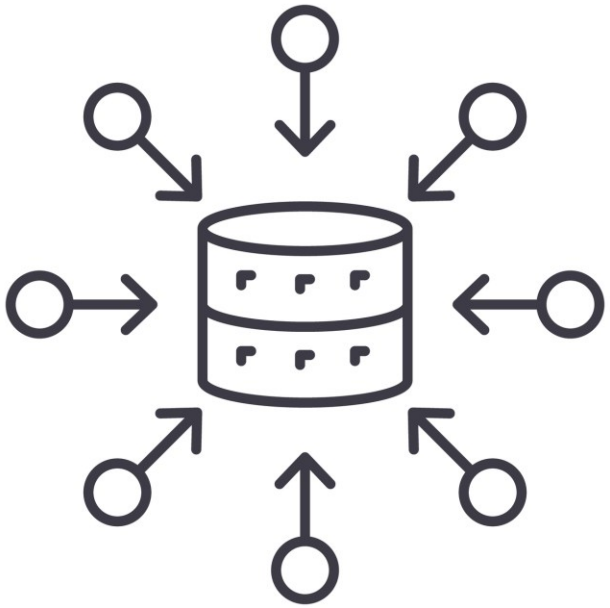
More Index Management

- Index rollups
  - Periodically roll up old data into summarized indices
  - Saves storage costs
  - New index may have fewer fields, coarser time buckets
- Index transforms
  - Like rollups, but purpose is to create a different view to analyze data differently.
  - Groupings and aggregations



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Cross-cluster replication

- Replicate indices / mappings / metadata across domains
- Ensures high availability in an outage
- Replicate data geographically for better latency
- “Follower” index pulls data from “leader” index
- Requires fine-grained access control and node-to-node encryption
- “Remote Reindex” allows copying indices from one cluster to another on demand

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Opensearch Stability

- 3 dedicated master nodes is best
  - Avoids “split brain”
- Don’t run out of disk space
  - Minimum storage requirement is roughly: Source Data * (1 + Number of Replicas) * 1.45
- Choosing the number of shards
  - (source data + room to grow) * (1 + indexing overhead) / desired shard size
  - In rare cases you may need to limit the number of shards per node
    - You usually run out of disk space first.
- Choosing instance types
  - At least 3 nodes
  - Mostly about storage requirements
  - i.e., m6g.large.search, i3.4xlarge.search, i3.16xlarge.search

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Amazon Opensearch Security

- Resource-based policies
- Identity-based policies
- IP-based policies
- Request signing
- VPC
- Cognito

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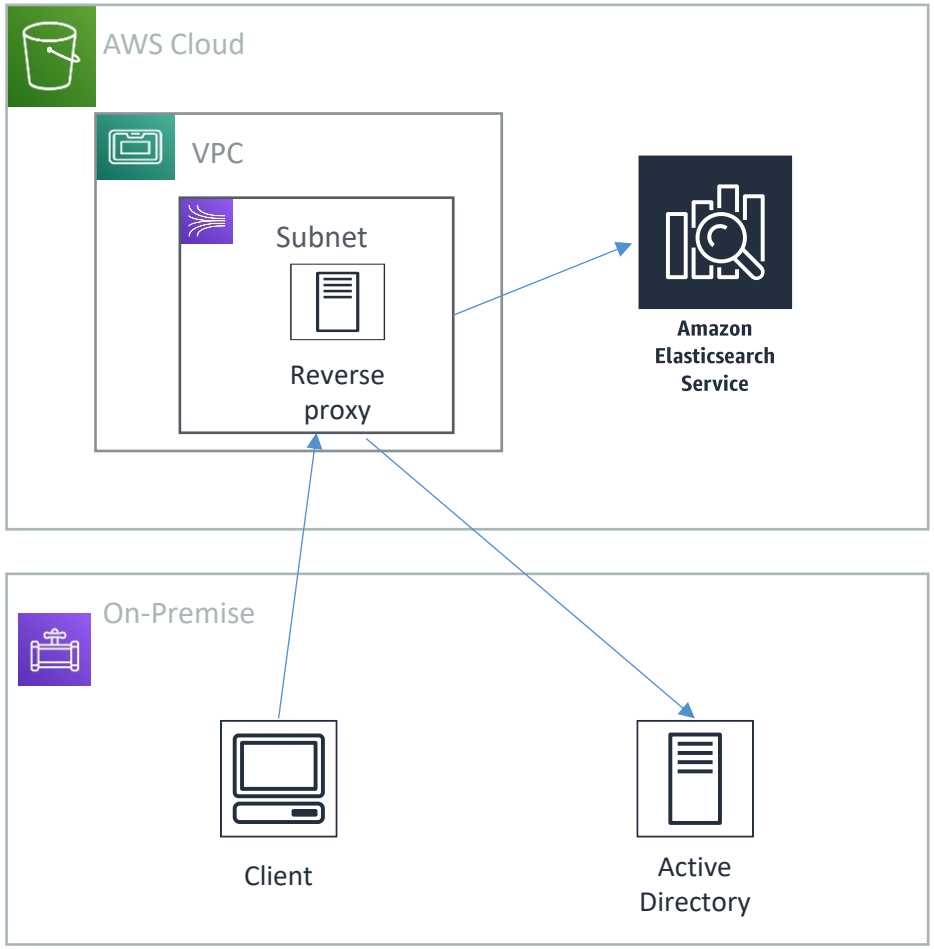
Securing Dashboards

- Cognito
- Getting inside a VPC from outside is hard…
  - Nginx reverse proxy on EC2 forwarding to ES domain
  - SSH tunnel for port 5601
  - VPC Direct Connect
  - VPN



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Amazon Opensearch anti-patterns

- OLTP
  - No transactions
  - RDS or DynamoDB is better
- Ad-hoc data querying
  - Athena is better
- Remember Opensearch is primarily for search & analytics

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Amazon Opensearch performance

- Memory pressure in the JVM can result if:
  - You have unbalanced shard allocations across nodes
  - You have too many shards in a cluster
- Fewer shards can yield better performance if JVMMemoryPressure errors are encountered
  - Delete old or unused indices

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Amazon OpenSearch Serverless

- On-demand autoscaling!
- Works against “collections” instead of provisioned domains
  - May be “search” or “time series” type
- Always encrypted with your KMS key
  - Data access policies
  - Encryption at rest is required
  - May configure security policies across many collections
- Capacity measured in Opensearch Compute Units (OCUs)
  - Can set an upper limit, lower limit is always 2 for indexing, 2 for search

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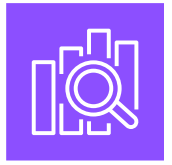
OpenSearch as a Vector Store

- Serverless vector stores
  - Note “serverless” does not mean “scales to zero”
    - Remember to shut down your serverless OpenSearch instances if you’re not using them – or you’ll keep getting billed
  - The exam treats OpenSearch as the primary implementation for backing Bedrock Knowledge Bases
    - But I think this might get subsumed by the newer S3 vectors feature
- Not limited to Bedrock Knowledge Bases
  - Can also integrate with SageMaker AI and HuggingFace or custom models for RAG
- SEMANTIC vs. HYBRID search
  - Semantic refers to vector search
    - As we’ve seen relevancy can be elusive and dependent on your chunking strategy
  - Hybrid **combines vector and keyword search**
    - Requires indexing filterable keyword fields for metadata
  - This is important

Amazon Bedrock



Amazon OpenSearch Service

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OpenSearch Vector Engines

- Top-level engines are Facebook AI Similarity (FAISS,) Non-Metric Space Library (NMSLib) and Apache Lucene.
- Exact Nearest Neighbor search
  - This is slow
- Approximate Nearest Neighbor (ANN) search
  - Heirarchical Navigable Small World ()
    - Fast, high-quality, simple – but uses lots of RAM
  - Inverted File (IVF)
    - Better for really huge datasets, can trade recall for speed & memory
- Tuning
  - M = How many edges per node
    - Higher M -> denser graph, higher recall, but **more memory**
  - ef_construction = Size of dynamic list used for KNN graph
    - Higher -> more accurate graph, **slower indexing speed**
  - ef_search = How thoroughly the graph is explored
    - Higher -> higher recall, but **slower search performance**

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OpenSearch as a Vector Store: Finer Points

- Vector compression
  - Different ANN methods have their own techniques (, IVF)
  - Dense embeddings add up fast
- Binary vectors
  - Bit sequences (stored as bytes)
  - Not embeddings, but used for fast distance search
  - 32x compression compared to float32
- FP16
  - A form of scalar quantization
  - Store embedding dimensions as 16-bit values instead of 32-bit
  - What does

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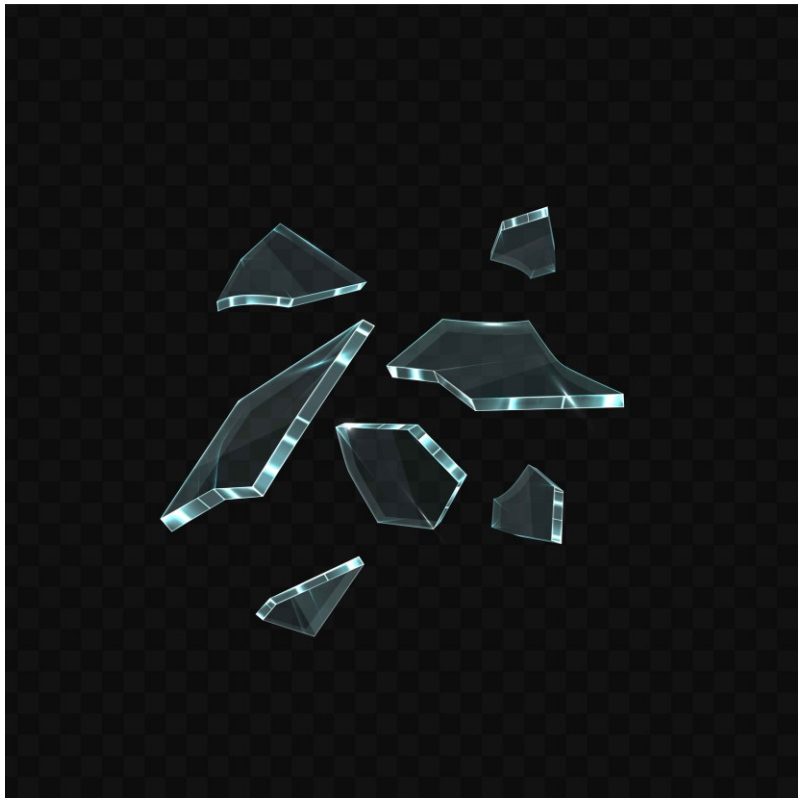
OpenSearch as a Vector Store: Finer Points

- Sharding strategies
  - Balance shard size and CPU ratio
  - Larger shards for semantic search (30-50 GB)
    - Vector search benefits from fewer, larger shards because there is less cross-shard coordination.
  - Smaller for hybrid (10-30 GB)
  - On OpenSearch Serverless, this is automatic
- Multi-index approaches
  - Different document types belong in their own index
  - This way you can tune performance for each corpus
  - You might want different embedding models for different types of data
  - You'll need some sort of query routing to pull this off.



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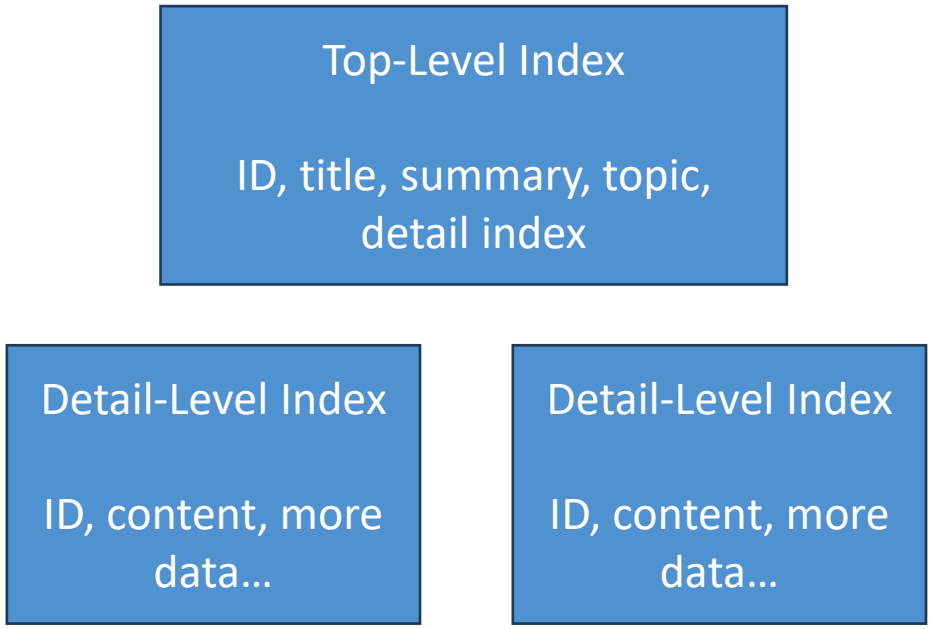
OpenSearch Hierarchical Indices

- A top level index can serve as a general / summary layer
  - This index is small and fast
  - It routes you to an index with more detailed data per domain
  - It's a way to implement multiple indices that are specialized for different domains or data types
    - With their own embedding models, shard sizes, tuning
- You could also use a FM to route to the right index.



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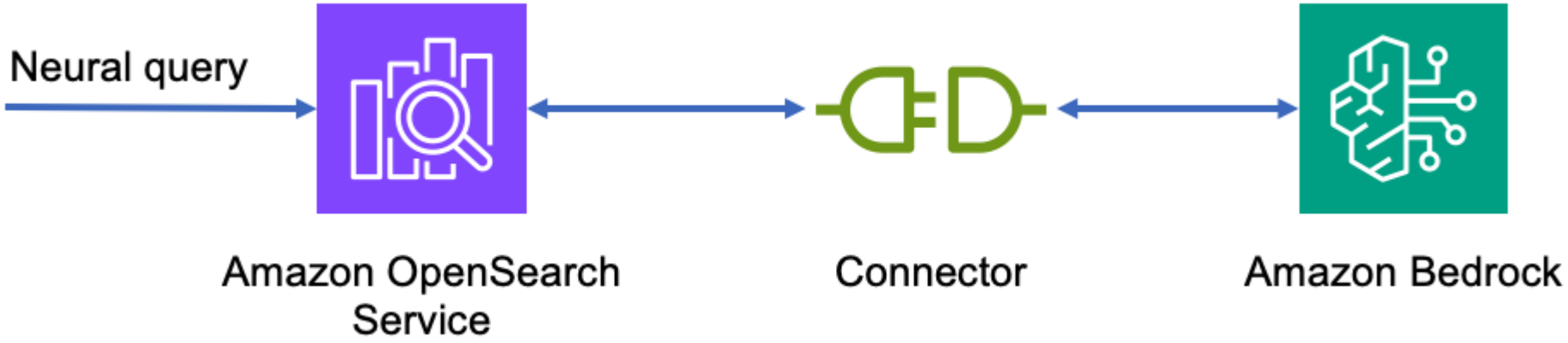
OpenSearch as a Vector Store: Neural Plugin

- Allows you to integrate your embedding model into an OpenSearch ingest pipeline
- “Neural queries” just allow OpenSearch to accept text queries and generate embeddings under the hood for vector queries.
- This is an alternative to using the built-in Bedrock Knowledge Base integration with OpenSearch Serverless
- The Neural plugin allows OpenSearch to call your Bedrock model for embeddings as part of search or ingestion.



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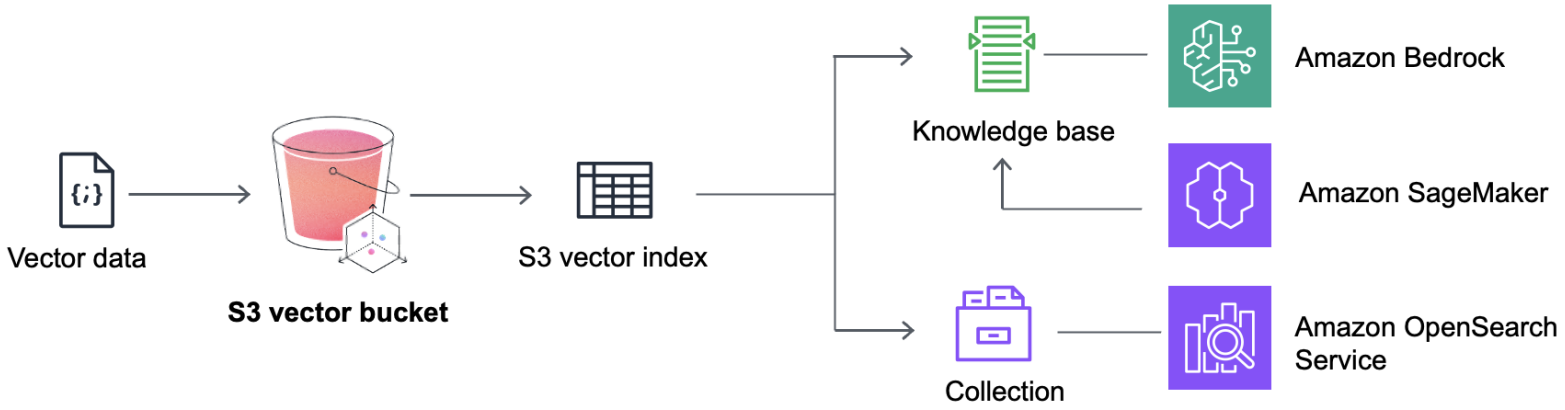
Amazon S3 Vectors

- Up to 90% cheaper (and a lot simpler)
- Create a S3 vector bucket
- Then create a vector index
  - Specify dimensions and distance metric
- Now you can store your vector embeddings with `put_vectors`
  - Include whatever metadata you want
  - Query them with `query_vectors`



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What a S3 Vectors Put Request Looks Like

```
s3vectors.put_vectors(
    vectorBucketName="media-embeddings",
    indexName="movies",
    vectors=[
        {
            "key": "Star Wars",
            "data": {"float32": embeddings[0]},
            "metadata": {"source_text": texts[0], "genre":"scifi"}
        },
        {
            "key": "Jurassic Park",
            "data": {"float32": embeddings[1]},
            "metadata": {"source_text": texts[1], "genre":"scifi"}
        },
        {
            "key": "Finding Nemo",
            "data": {"float32": embeddings[2]},
            "metadata": {"source_text": texts[2], "genre":"family"}
        }
    ]
)
```


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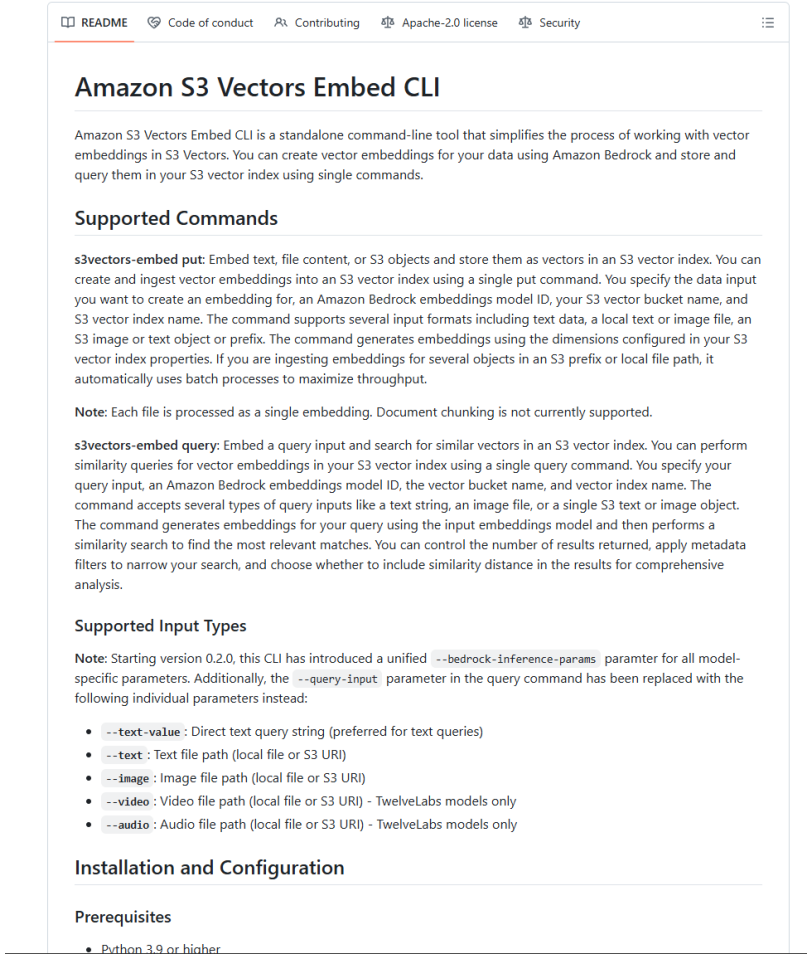
Alternative API: s3vectors-embed-cli

- You can use boto3 or the AWS CLI
- But there's also a dedicated S3 vectors CLI
- Available in GitHub as open source
- `s3vectors-embed put`
  - Converts text or images to embeddings using Bedrock, stores them.
- `s3vectors-embed query`
  - Generates an embedding for your query, queries your vector index.



README Code of conduct Contributing Apache-2.0 license Security

Amazon S3 Vectors Embed CLI

Amazon S3 Vectors Embed CLI is a standalone command-line tool that simplifies the process of working with vector embeddings in S3 Vectors. You can create vector embeddings for your data using Amazon Bedrock and store and query them in your S3 vector index using single commands.

Supported Commands

s3vectors-embed put: Embed text, file content, or S3 objects and store them as vectors in an S3 vector index. You can create and ingest vector embeddings into an S3 vector index using a single put command. You specify the data input you want to create an embedding for, an Amazon Bedrock embeddings model ID, your S3 vector bucket name, and S3 vector index name. The command supports several input formats including text data, a local text or image file, an S3 image or text object or prefix. The command generates embeddings using the dimensions configured in your S3 vector index properties. If you are ingesting embeddings for several objects in an S3 prefix or local file path, it automatically uses batch processes to maximize throughput.

Note: Each file is processed as a single embedding. Document chunking is not currently supported.

s3vectors-embed query: Embed a query input and search for similar vectors in an S3 vector index. You can perform similarity queries for vector embeddings in your S3 vector index using a single query command. You specify your query input, an Amazon Bedrock embeddings model ID, the vector bucket name, and vector index name. The command accepts several types of query inputs like a text string, an image file, or a single S3 text or image object. The command generates embeddings for your query using the input embeddings model and then performs a similarity search to find the most relevant matches. You can control the number of results returned, apply metadata filters to narrow your search, and choose whether to include similarity distance in the results for comprehensive analysis.

Supported Input Types

Note: Starting version 0.2.0, this CLI has introduced a unified `--bedrock-inference-params` paramter for all model-specific parameters. Additionally, the `--query-input` parameter in the query command has been replaced with the following individual parameters instead:

- `--text-value` : Direct text query string (preferred for text queries)
- `--text` : Text file path (local file or S3 URI)
- `--image` : Image file path (local file or S3 URI)
- `--video` : Video file path (local file or S3 URI) - TwelveLabs models only
- `--audio` : Audio file path (local file or S3 URI) - TwelveLabs models only

Installation and Configuration

Prerequisites

- Python 3.9 or higher

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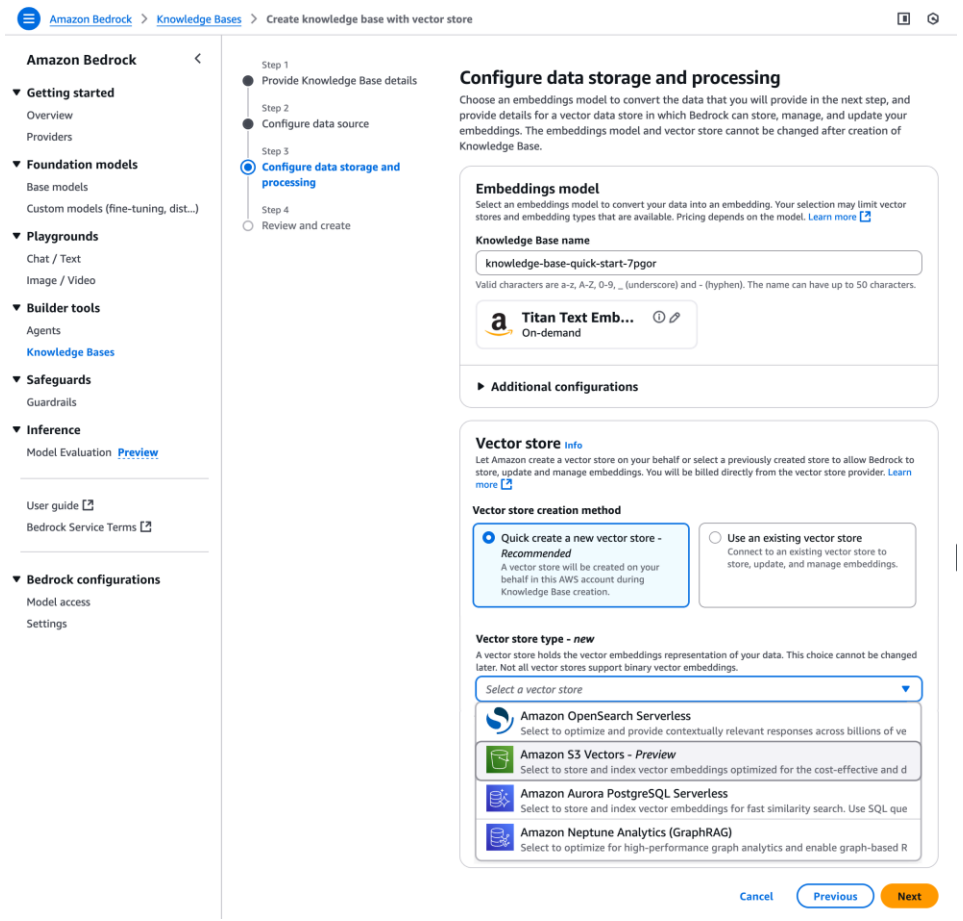
Amazon S3 Vectors

- You can generate your own vectors with an embedding model and store them
- Or it's fully integrated with Bedrock Knowledge Bases
  - It manages it all for you
- Also integrates with Bedrock within SageMaker Unified Studio



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S3 Vectors: More specifics?

- They are strongly consistent
  - Newly added data is immediately available
- The vector data is automatically optimized over time for *price performance*
- But actual performance is only advertised as 100ms-1s
  - **AWS recommends a hybrid approach with OpenSearch for performance-critical queries**
  - They call this a “tiered search strategy” – use S3 for infrequently queried vectors only
  - S3 Vectors has connectivity with OpenSearch for this purpose
    - This COPIES (exports) data from S3 into OpenSearch – you end up paying for both if you’re not careful!
- You can also use OpenSearch with S3 Vectors engine
  - Only works with OpenSearch managed clusters
  - Give you OpenSearch functionality with S3 Vectors as the backend
- Max indices per bucket: 10,000 (2B vectors per index)

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S3 Vectors: Best Practices

- Insert or delete vectors in large batches (500 per API call)
- Use concurrent requests for smaller batches
  - Can get up to 2500 vectors per second
- Have a retry mechanism for queries
  - Only hundreds per second supported per index before you get an 429 error
- Use multiple indexes when possible
  - Improves per-index performance
  - Also useful for data isolation, multi-tenancy
- If a metadata field doesn't need to be filtered, mark it as non-filterable

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Amazon RDS Overview

- RDS stands for Relational Database Service
- It's a managed DB service for DB use SQL as a query language.
- It allows you to create databases in the cloud that are managed by AWS
  - Postgres
  - MySQL
  - MariaDB
  - Oracle
  - Microsoft SQL Server
  - IBM DB2
  - Aurora (AWS Proprietary database)

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Advantage over using RDS versus deploying DB on EC2

- RDS is a managed service:
  - Automated provisioning, OS patching
  - Continuous backups and restore to specific timestamp (Point in Time Restore)!
  - Monitoring dashboards
  - Read replicas for improved read performance
  - Multi AZ setup for DR (Disaster Recovery)
  - Maintenance windows for upgrades
  - Scaling capability (vertical and horizontal)
  - Storage backed by EBS
- BUT you can't SSH into your instances

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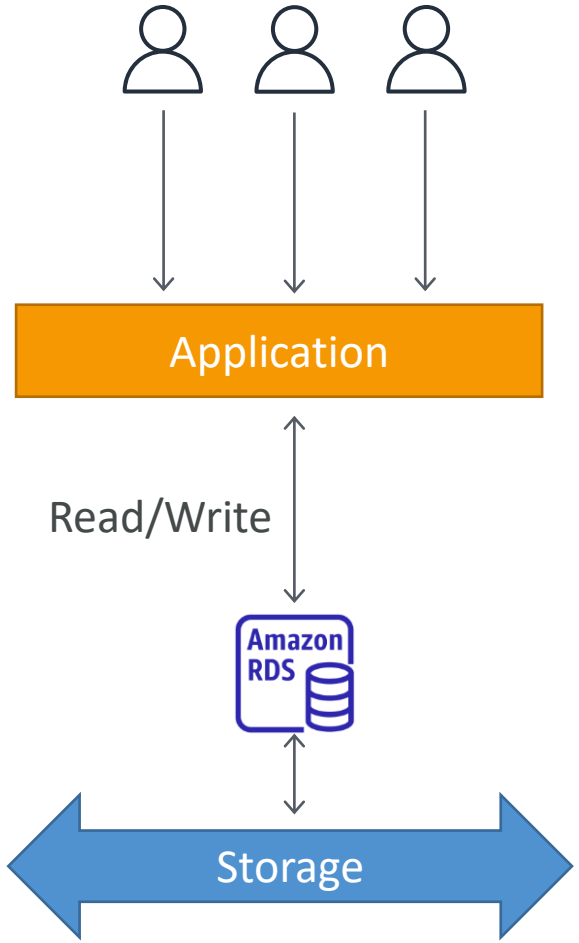
RDS – Storage Auto Scaling

- Helps you increase storage on your RDS DB instance dynamically
- When RDS detects you are running out of free database storage, it scales automatically
- Avoid manually scaling your database storage
- You have to set **Maximum Storage Threshold** (maximum limit for DB storage)
- Automatically modify storage if:
  - Free storage is less than 10% of allocated storage
  - Low-storage lasts at least 5 minutes
  - 6 hours have passed since last modification
- Useful for applications with **unpredictable workloads**
- Supports all RDS database engines



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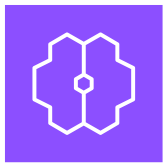
Using RDS with S3 Document Repositories

- RDS can also serve as your vector store
  - RDS SQL Server Vector Store is a thing
- A common pattern is to use RDS for querying structured data, which then points to something like S3 for large unstructured data it refers to.
- So you could do something like this…
  - Your vector store for RAG is backed by RDS
  - The data returned just points you to content in S3



Amazon SageMaker

Amazon Relational Database Service (Amazon RDS)

Amazon Simple Storage Service (Amazon S3)

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Amazon Aurora

- Aurora is a proprietary technology from AWS (not open sourced)
- Postgres and MySQL are both supported as Aurora DB (that means your drivers will work as if Aurora was a Postgres or MySQL database)
- Aurora is “AWS cloud optimized” and claims 5x performance improvement over MySQL on RDS, over 3x the performance of Postgres on RDS
- Aurora storage automatically grows in increments of 10GB, up to 256 TB.
- Aurora can have up to 15 replicas and the replication process is faster than MySQL (sub 10 ms replica lag)
- Failover in Aurora is instantaneous. It’s HA (High Availability) native.
- Aurora costs more than RDS (20% more) – but is more efficient

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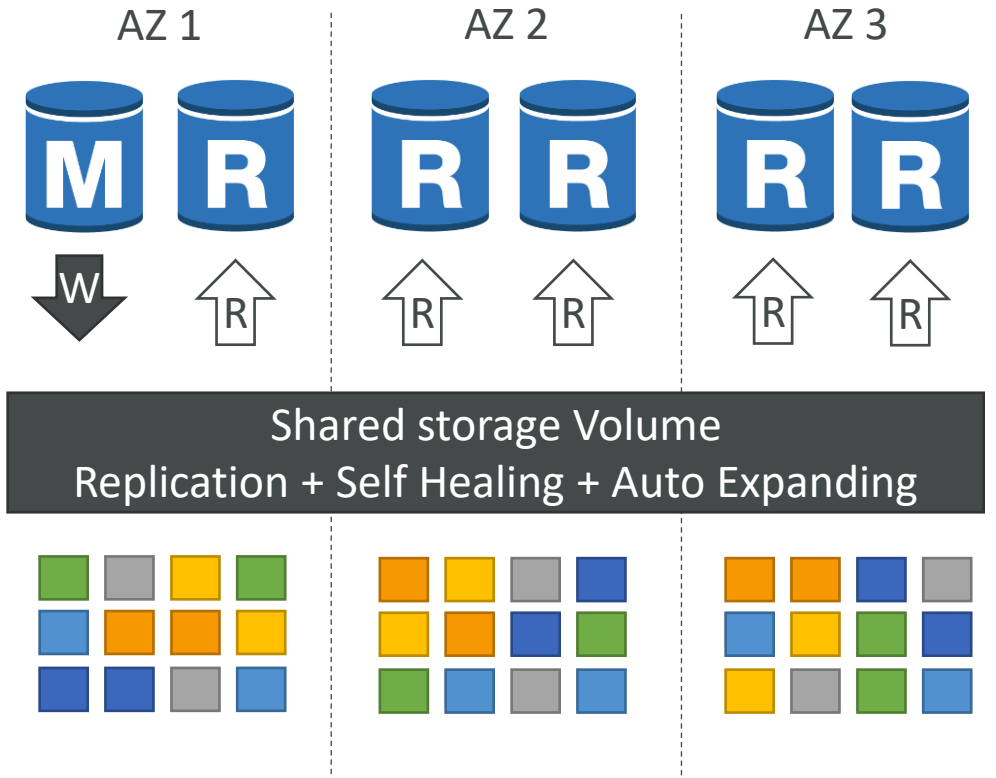
Aurora High Availability and Read Scaling

- 6 copies of your data across 3 AZ:
  - 4 copies out of 6 needed for writes
  - 3 copies out of 6 need for reads
  - Self healing with peer-to-peer replication
  - Storage is striped across 100s of volumes
- One Aurora Instance takes writes (master)
- Automated failover for master in less than 30 seconds
- Master + up to 15 Aurora Read Replicas serve reads
- **Support for Cross Region Replication**



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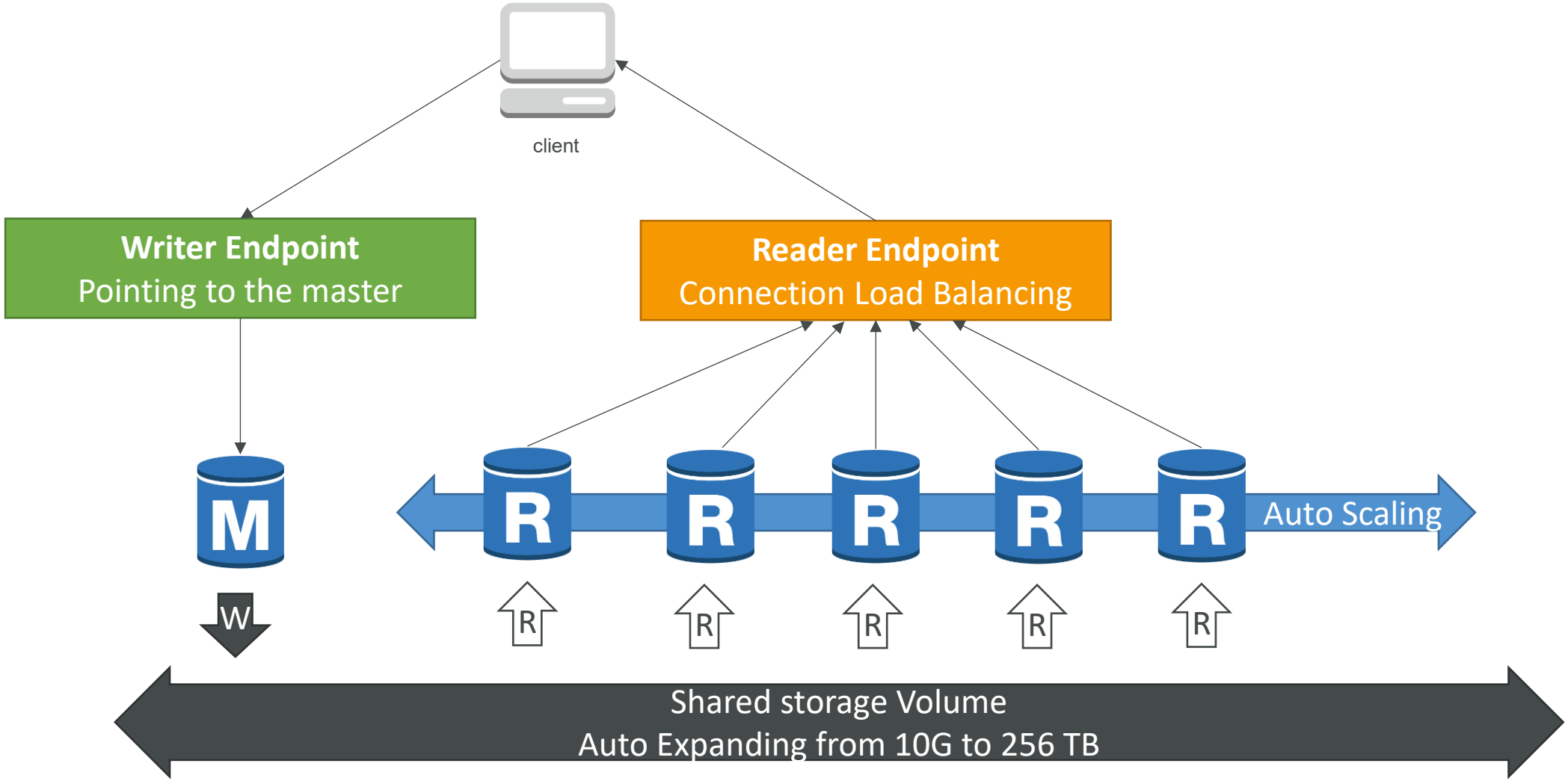
Aurora DB Cluster



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Features of Aurora

- Automatic fail-over
- Backup and Recovery
- Isolation and security
- Industry compliance
- Push-button scaling
- Automated Patching with Zero Downtime
- Advanced Monitoring
- Routine Maintenance
- Backtrack: restore data at any point of time without using backups

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Amazon Aurora as a Vector Store

- **pgvector** extension
  - Creates a new “vector” column type
  - Can compute cosine, L2, or inner product distances
  - Can do similarity search with IVF or
  - Also adds vector operators for SQL
    - This gives it an advantage over OpenSearch if you need complex filtering logic and stuff.
- Allows you to store structured and vector data in the same place
- Can be appropriate for small / medium RAG systems with mostly structured data

Amazon Aurora

Trusted Language Extensions for PostgreSQL

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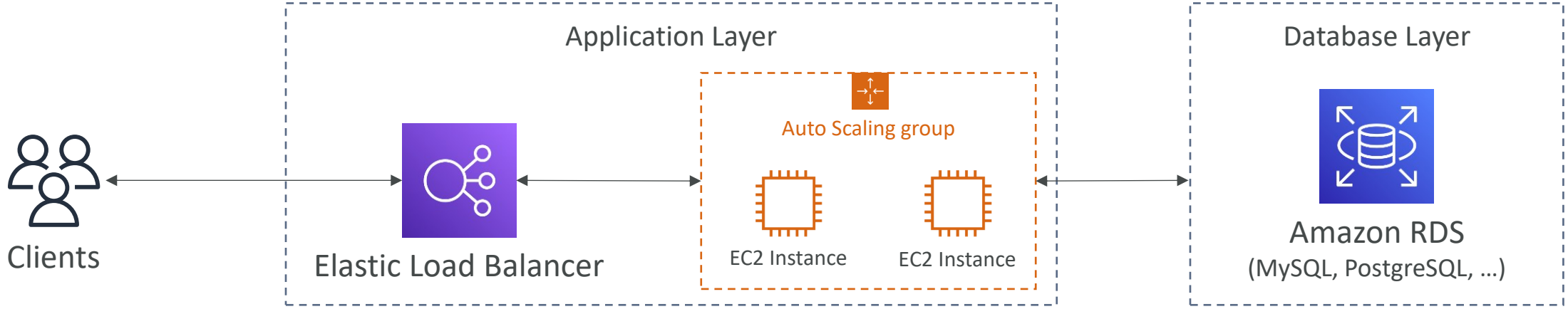
Traditional Architecture



- Traditional applications leverage RDBMS databases
- These databases have the SQL query language
- Strong requirements about how the data should be modeled
- Ability to do query joins, aggregations, complex computations
- Vertical scaling (getting a more powerful CPU / RAM / IO)
- Horizontal scaling (increasing reading capability by adding EC2 / RDS Read Replicas)

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NoSQL databases

- NoSQL databases are non-relational databases and are **distributed**
- NoSQL databases include MongoDB, DynamoDB, …
- NoSQL databases do not support query joins (or just limited support)
- All the data that is needed for a query is present in one row
- NoSQL databases don't perform aggregations such as "SUM", "AVG", …
- **NoSQL databases scale horizontally**

- There's no "right or wrong" for NoSQL vs SQL, they just require to model the data differently and think about user queries differently

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Amazon DynamoDB

- Fully managed, highly available with replication across multiple AZs
- NoSQL database - not a relational database
- Scales to massive workloads, distributed database
- Millions of requests per seconds, trillions of row, 100s of TB of storage
- Fast and consistent in performance (low latency on retrieval)
- Integrated with IAM for security, authorization and administration
- Enables event driven programming with DynamoDB Streams
- Low cost and auto-scaling capabilities
- Standard & Infrequent Access (IA) Table Class

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DynamoDB - Basics

- DynamoDB is made of **Tables**
- Each table has a **Primary Key** (must be decided at creation time)
- Each table can have an infinite number of items (= rows)
- Each item has **attributes** (can be added over time – can be null)
- Maximum size of an item is **400KB**
- Data types supported are:
  - **Scalar Types** – String, Number, Binary, Boolean, Null
  - **Document Types** – List, Map
  - **Set Types** – String Set, Number Set, Binary Set

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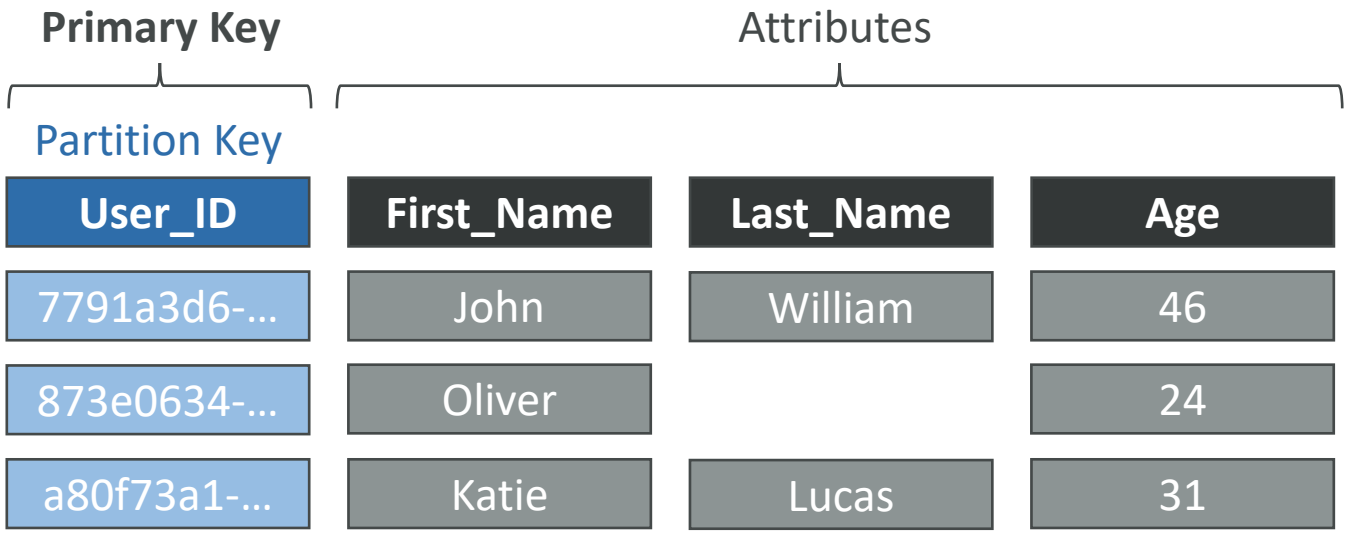
DynamoDB – Primary Keys

- **Option 1: Partition Key (HASH)**
  - Partition key must be unique for each item
  - Partition key must be "diverse" so that the data is distributed
  - Example: "User_ID" for a users table



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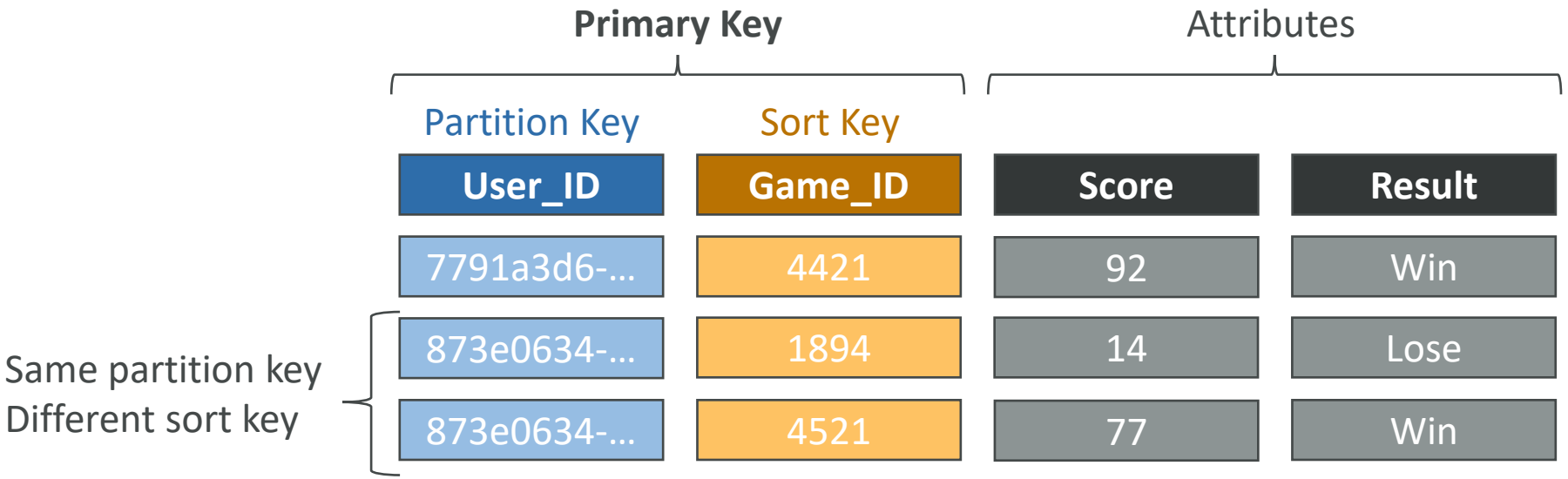
DynamoDB – Primary Keys

- **Option 2: Partition Key + Sort Key (HASH + RANGE)**
  - The combination must be unique for each item
  - Data is grouped by partition key
  - Example: users-games table, “User_ID” for Partition Key and “Game_ID” for Sort Key



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DynamoDB – Partition Keys (Exercise)

- We're building a movie database
- What is the best Partition Key to maximize data distribution?
  - movie_id
  - producer_name
  - leader_actor_name
  - movie_language

- "movie_id" has the highest cardinality so it's a good candidate
- "movie_language" doesn't take many values and may be skewed towards English so it's not a great choice for the Partition Key

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DynamoDB – Read/Write Capacity Modes

- Control how you manage your table's capacity (read/write throughput)

- **Provisioned Mode**
  - You specify the number of reads/writes per second
  - You need to plan capacity beforehand
  - Pay for provisioned read & write capacity units

- **On-Demand Mode (default)**
  - Read/writes automatically scale up/down with your workloads
  - No capacity planning needed
  - Pay for what you use, more expensive ($$$)

- You can switch between different modes once every 24 hours

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R/W Capacity Modes – Provisioned

- Table must have provisioned read and write capacity units
- **Read Capacity Units (RCU) –** throughput for reads
- **Write Capacity Units (WCU) –** throughput for writes
- Option to setup auto-scaling of throughput to meet demand
- Throughput can be exceeded temporarily using **"Burst Capacity"**
- If Burst Capacity has been consumed, you'll get a **"ProvisionedThroughputExceededException"**
- It's then advised to do an **exponential backoff** retry

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DynamoDB – Write Capacity Units (WCU)

- One *Write Capacity Unit (WCU)* represents one write per second for an item up to **1 KB** in size
- If the items are larger than 1 KB, more WCUs are consumed

- **Example 1:** we write 10 items per second, with item size 2 KB
  - We need $10 * (\frac{2\ KB}{1\ KB}) = 20\ WCUs$
- **Example 2:** we write 6 items per second, with item size 4.5 KB
  - We need $6 * (\frac{5\ KB}{1\ KB}) = 30\ WCUs$ (4.5 gets rounded to the upper KB)
- **Example 3:** we write 120 items per minute, with item size 2 KB
  - We need $\left(\frac{120}{60}\right) * (\frac{2\ KB}{1\ KB}) = 4\ WCUs$

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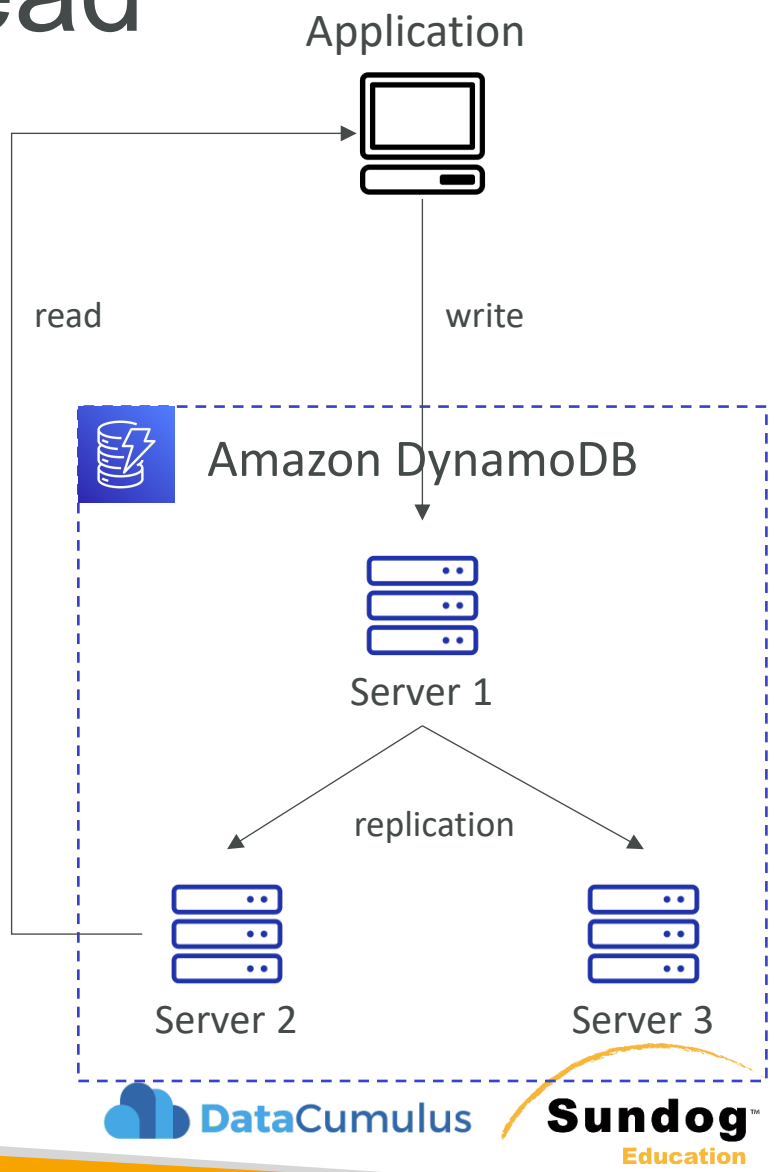
Strongly Consistent Read vs. Eventually Consistent Read

- **Eventually Consistent Read (default)**
  - If we read just after a write, it's possible we'll get some stale data because of replication

- **Strongly Consistent Read**
  - If we read just after a write, we will get the correct data
  - Set **"ConsistentRead"** parameter to **True** in API calls (GetItem, BatchGetItem, Query, Scan)
  - Consumes twice the RCU



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DynamoDB – Read Capacity Units (RCU)

- One *Read Capacity Unit (RCU)* represents **one Strongly Consistent Read** per second, or **two Eventually Consistent Reads** per second, for an item up to **4 KB** in size
- If the items are larger than 4 KB, more RCUs are consumed

- **Example 1:** 10 Strongly Consistent Reads per second, with item size 4 KB
  - We need $10 * \left(\frac{4\ KB}{4\ KB}\right) = 10\ RCUs$
- **Example 2:** 16 Eventually Consistent Reads per second, with item size 12 KB
  - We need $\left(\frac{16}{2}\right) * \left(\frac{12\ KB}{4\ KB}\right) = 24\ RCUs$
- **Example 3:** 10 Strongly Consistent Reads per second, with item size 6 KB
  - We need $10 * \left(\frac{8\ KB}{4\ KB}\right) = 20\ RCUs$ (we must round up 6 KB to 8 KB)

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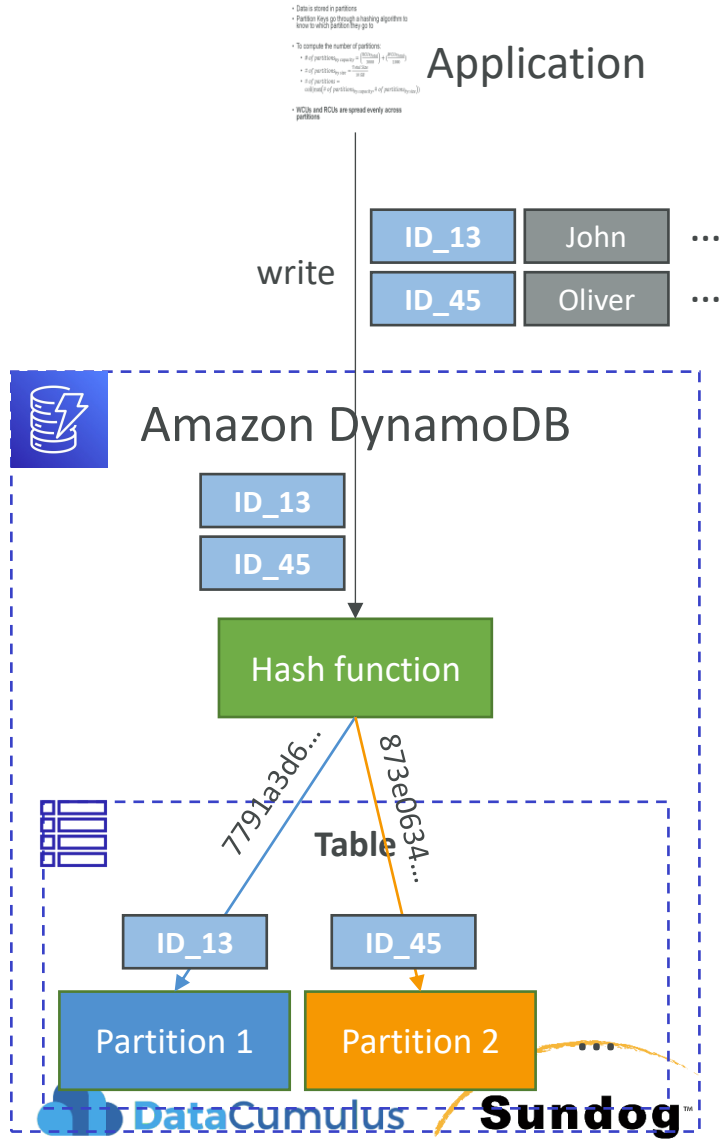
DynamoDB – Partitions Internal

- Data is stored in partitions
- Partition Keys go through a hashing algorithm to know to which partition they go to

- To compute the number of partitions:
  - $\#\ of\ partitions_{by\ capacity} = \left(\frac{RCUs_{Total}}{3000}\right) + \left(\frac{WCUs_{Total}}{1000}\right)$
  - $\#\ of\ partitions_{by\ size} = \frac{Total\ Size}{10\ GB}$
  - $\#\ of\ partitions = \text{ceil}(\max(\#\ of\ partitions_{by\ capacity}, \#\ of\ partitions_{by\ size}))$

- **WCUs and RCUs are spread evenly across partitions**



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DynamoDB – Throttling

- If we exceed provisioned RCUs or WCUs, we get **"ProvisionedThroughputExceededException"**
- Reasons:
  - **Hot Keys** – one partition key is being read too many times (e.g., popular item)
  - **Hot Partitions**
  - **Very large items,** remember RCU and WCU depends on size of items
- Solutions:
  - **Exponential backoff** when exception is encountered (already in SDK)
  - **Distribute partition keys** as much as possible
  - If RCU issue, we can **use DynamoDB Accelerator (DAX)**

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R/W Capacity Modes – On-Demand

- Read/writes automatically scale up/down with your workloads
- No capacity planning needed (WCU / RCU)
- Unlimited WCU & RCU, no throttle, more expensive
- You're charged for reads/writes that you use in terms of **RRU** and **WRU**
- **Read Request Units (RRU) –** throughput for reads (same as RCU)
- **Write Request Units (WRU) –** throughput for writes (same as WCU)
- 2.5x more expensive than provisioned capacity (use with care)
- Use cases: unknown workloads, unpredictable application traffic, …

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DynamoDB – Writing Data

- **PutItem**
  - Creates a new item or fully replace an old item (same Primary Key)
  - Consumes WCUs
- **UpdateItem**
  - Edits an existing item's attributes or adds a new item if it doesn't exist
  - Can be used to implement **Atomic Counters –** a numeric attribute that's unconditionally incremented
- **Conditional Writes**
  - Accept a write/update/delete only if conditions are met, otherwise returns an error
  - Helps with concurrent access to items
  - No performance impact

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DynamoDB – Reading Data

- **GetItem**
  - Read based on Primary key
  - Primary Key can be **HASH** or **HASH+RANGE**
  - Eventually Consistent Read (default)
  - Option to use Strongly Consistent Reads (more RCU - might take longer)
  - **ProjectionExpression** can be specified to retrieve only certain attributes

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DynamoDB – Reading Data (Query)

- **Query** returns items based on:
  - **KeyConditionExpression**
    - Partition Key value (**must be = operator**) – required
    - Sort Key value (=, <, <=, >, >=, Between, Begins with) – optional
  - **FilterExpression**
    - Additional filtering after the Query operation (before data returned to you)
    - Use only with non-key attributes (does not allow HASH or RANGE attributes)
- Returns:
  - The number of items specified in **Limit**
  - Or up to 1 MB of data
- Ability to do pagination on the results
- Can query table, a Local Secondary Index, or a Global Secondary Index

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DynamoDB – Reading Data (Scan)

- **Scan** the entire table and then filter out data (inefficient)
- Returns up to 1 MB of data – use pagination to keep on reading
- Consumes a lot of RCU
- Limit impact using **Limit** or reduce the size of the result and pause
- For faster performance, use **Parallel Scan**
  - Multiple workers scan multiple data segments at the same time
  - Increases the throughput and RCU consumed
  - Limit the impact of parallel scans just like you would for Scans
- Can use **ProjectionExpression** & **FilterExpression** (no changes to RCU)

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DynamoDB – Deleting Data

- **DeleteItem**
  - Delete an individual item
  - Ability to perform a conditional delete

- **DeleteTable**
  - Delete a whole table and all its items
  - Much quicker deletion than calling **DeleteItem** on all items

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DynamoDB – Batch Operations

- Allows you to save in latency by reducing the number of API calls
- Operations are done in parallel for better efficiency
- Part of a batch can fail; in which case we need to try again for the failed items

- **BatchWriteItem**
  - Up to 25 **PutItem** and/or **DeleteItem** in one call
  - Up to 16 MB of data written, up to 400 KB of data per item
  - Can't update items (use **UpdateItem**)
  - **UnprocessedItems** for failed write operations (exponential backoff or add WCU)

- **BatchGetItem**
  - Return items from one or more tables
  - Up to 100 items, up to 16 MB of data
  - Items are retrieved in parallel to minimize latency
  - **UnprocessedKeys** for failed read operations (exponential backoff or add RCU)

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DynamoDB – PartiQL

- SQL-compatible query language for DynamoDB
- Allows you to select, insert, update, and delete data in DynamoDB using SQL
- Run queries across multiple DynamoDB tables
- Run PartiQL queries from:
  - AWS Management Console
  - NoSQL Workbench for DynamoDB
  - DynamoDB APIs
  - AWS CLI
  - AWS SDK

```
SELECT OrderID, Total
FROM Orders
WHERE OrderID IN [1, 2, 3]
ORDER BY OrderID DESC
```


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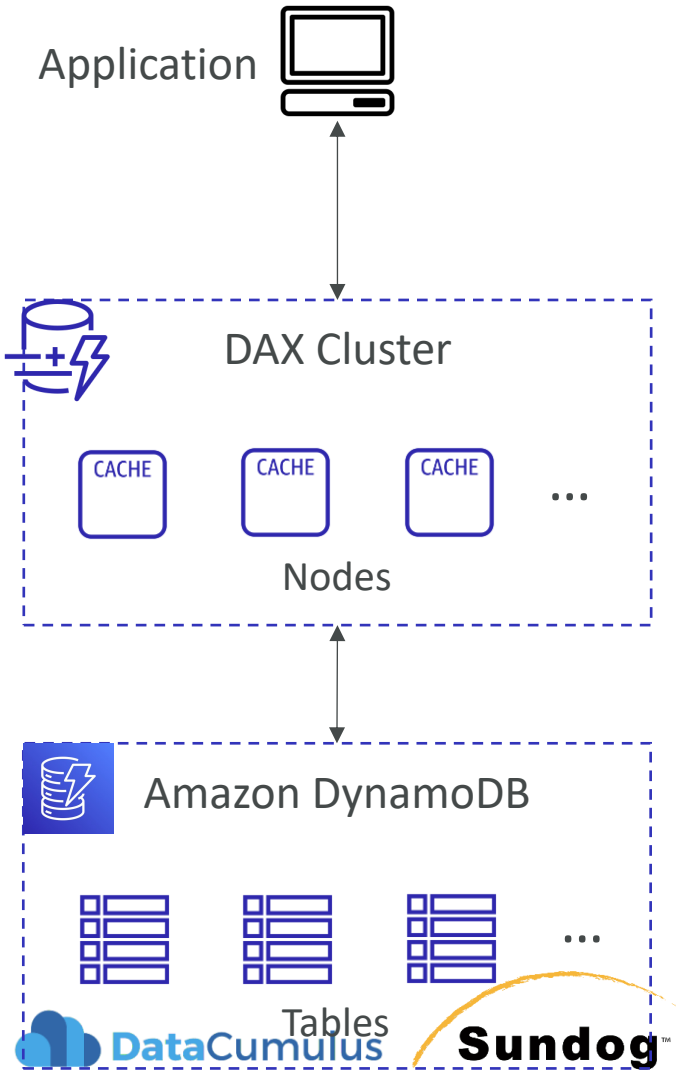
DynamoDB Accelerator (DAX)

- Fully-managed, highly available, seamless in-memory cache for DynamoDB
- Microseconds latency for cached reads & queries
- Doesn't require application logic modification (compatible with existing DynamoDB APIs)
- Solves the "Hot Key" problem (too many reads)
- 5 minutes TTL for cache (default)
- Up to 10 nodes in the cluster
- Multi-AZ (3 nodes minimum recommended for production)
- Secure (Encryption at rest with KMS, VPC, IAM, CloudTrail, …)



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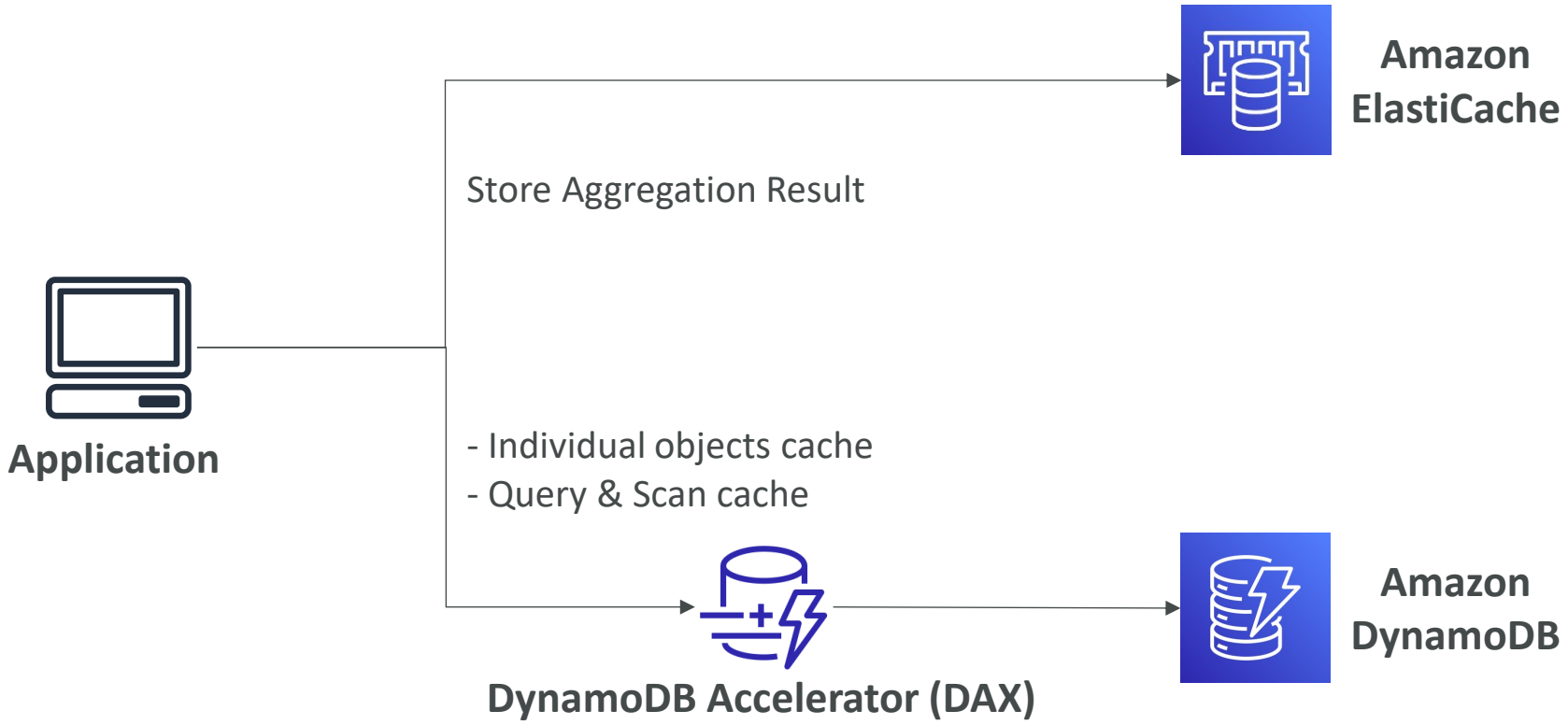
DynamoDB Accelerator (DAX) vs. ElastiCache



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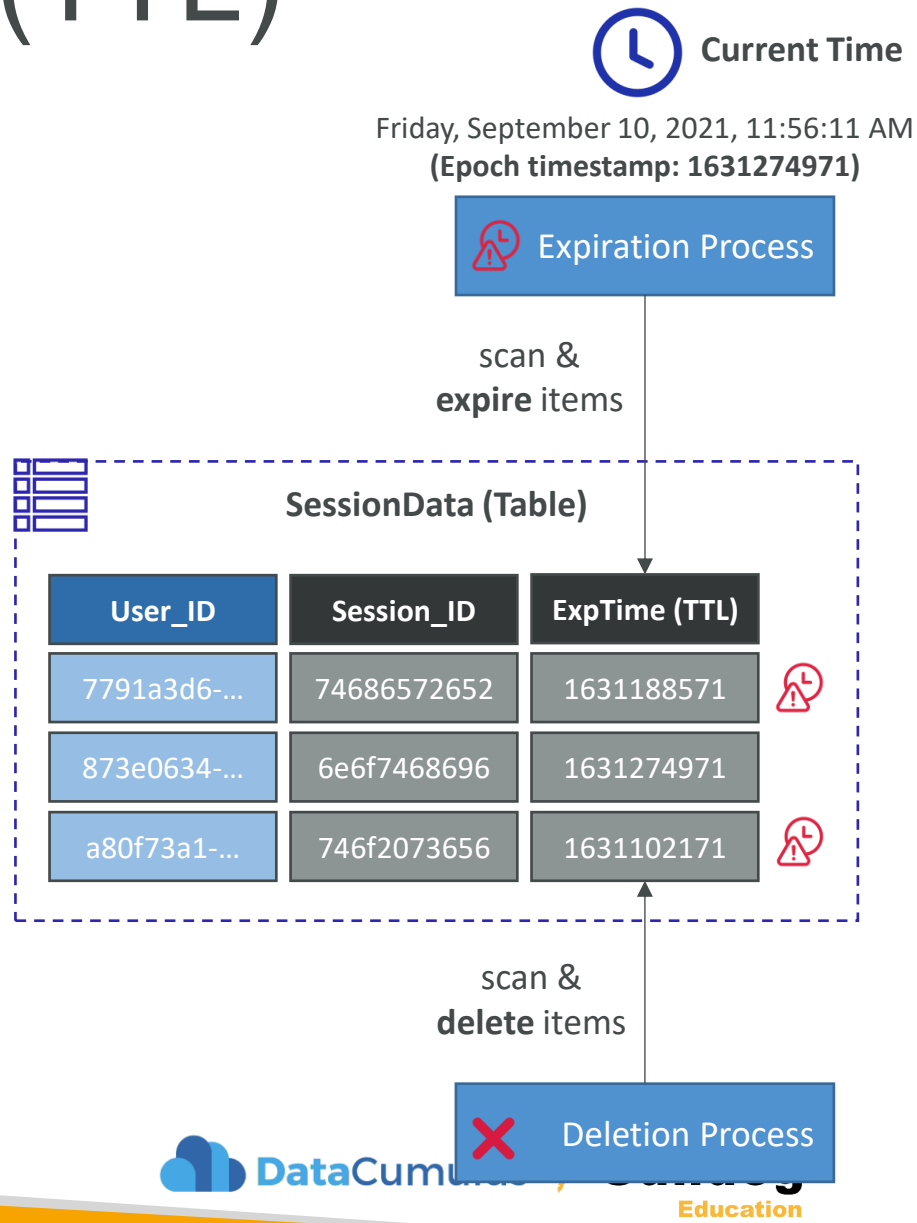
DynamoDB – Time To Live (TTL)

- Automatically delete items after an expiry timestamp
- Doesn't consume any WCUs (i.e., no extra cost)
- The TTL attribute must be a **"Number"** data type with **"Unix Epoch timestamp"** value
- Expired items deleted within few days of expiration
- Expired items, that haven't been deleted, appears in reads/queries/scans (if you don't want them, filter them out)
- Expired items are deleted from both LSIs and GSIs
- A delete operation for each expired item enters the DynamoDB Streams (can help recover expired items)
- Use cases: reduce stored data by keeping only current items, adhere to regulatory obligations, …



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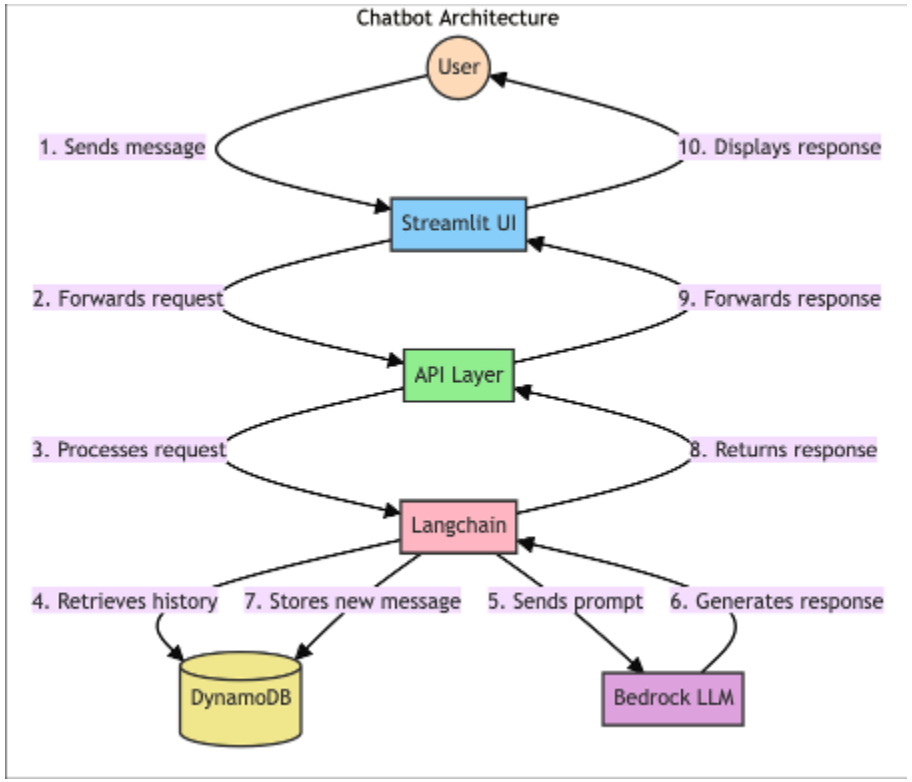
DynamoDB with GenAI

- It's now a vector store, too!
  - Vector indexes enable similarity search on embeddings stored right in your table (more on this next)
- It can serve your model with near real-time data
- It can also serve as a fast way to store and retrieve chat histories
  - "Long term memory" for your AI agents
  - This is important for context awareness in a chatbot



https://aws.amazon.com/blogs/database/build-a-scalable-context-aware-chatbot-with-amazon-dynamodb-amazon-bedrock-and-langchain/

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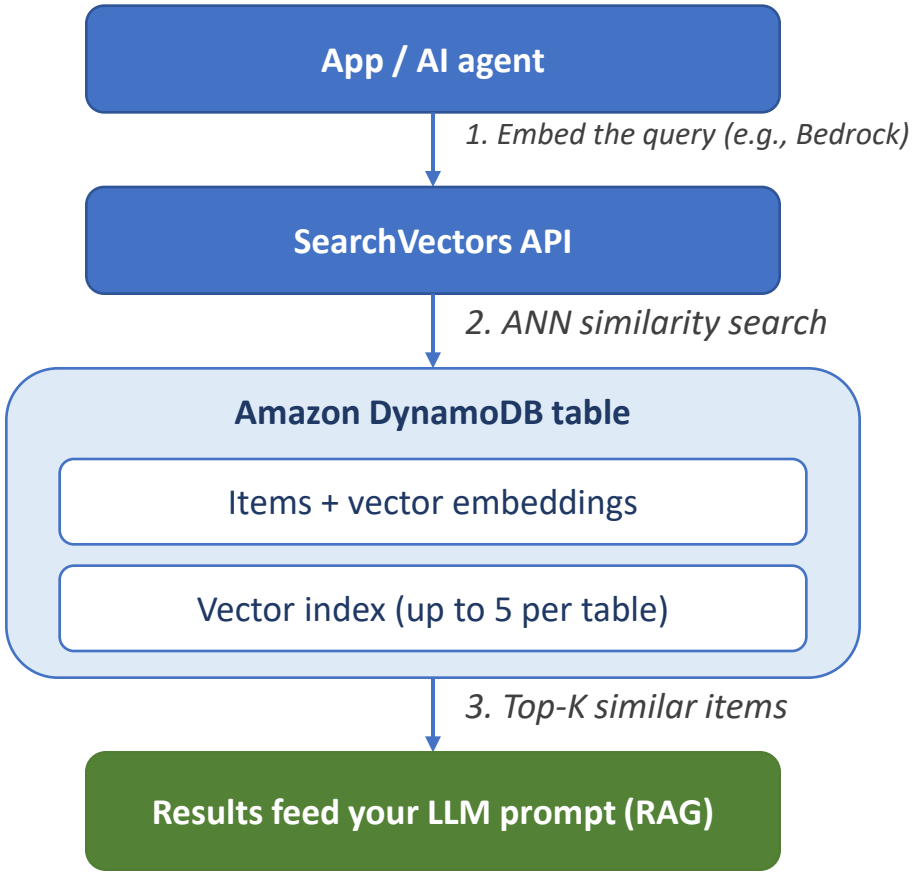
DynamoDB Vector Indexes

- Similarity search on embeddings stored alongside your operational data
  - New SearchVectors API – approximate nearest neighbor (ANN)
  - No separate vector DB, no sync pipeline
- Distance functions: COSINE, EUCLIDEAN, DOT_PRODUCT
  - COSINE is the safe default for text embeddings
- Optional partition key and inline filters to scope and scale searches
- Projections work like global secondary indexes: only projected attributes are returned
- Works with Streams, TTL, global tables, point-in-time recovery
- Use cases: RAG, semantic search, agent memory, recommendations



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Vector Indexes: Limitations, and Why DynamoDB?

- Limitations
  - On-demand capacity mode only; results are approximate (ANN); vectors stored as 32-bit floats
  - Read only via SearchVectors – no Query, Scan, PartiQL, DAX, or fine-grained access control
  - 16 MB response limit with no pagination; inline filters are equality-only
  - Distance function and projection are locked in at index creation
  - Pricing: index writes and searches are billed per byte (1 KB minimum) – on top of normal table charges
- Why choose it over other vector stores?
  - Your operational data (and chat history!) is already in DynamoDB – no replication pipeline to a separate vector DB
  - Serverless, low-latency: agent memory and semantic recall in one table
  - S3 Vectors: cheaper for massive, rarely-queried vector sets – but higher latency
  - OpenSearch: richer search (hybrid, full-text, exact k-NN) – but more to manage and pay for

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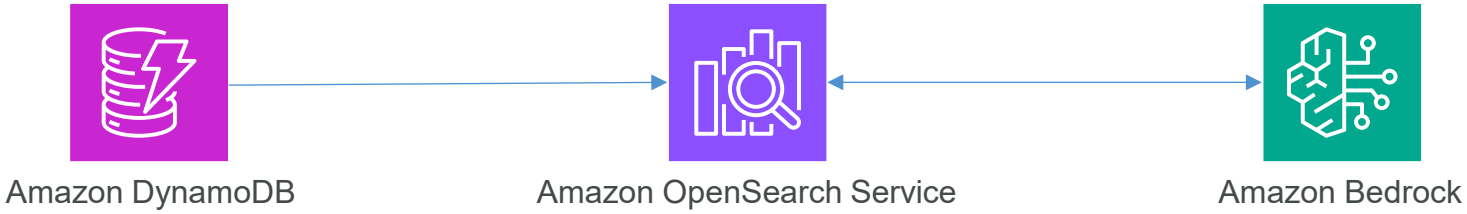
DynamoDB and OpenSearch

- Zero-ETL integration with OpenSearch
- You can create a pipeline from DynamoDB to OpenSearch
- And then use OpenSearch as your vector store for Bedrock
- This is a way to incorporate DynamoDB data into your knowledge bases



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Keeping your Vector Store Up to Date

- We need some way to keep your vector store in sync with new and changed data
  - Incremental updates
  - Real-time change detection
  - Automated synchronization
  - Scheduled refresh pipelines
- EventBridge is your friend
  - More on EventBridge later.

Amazon EventBridge

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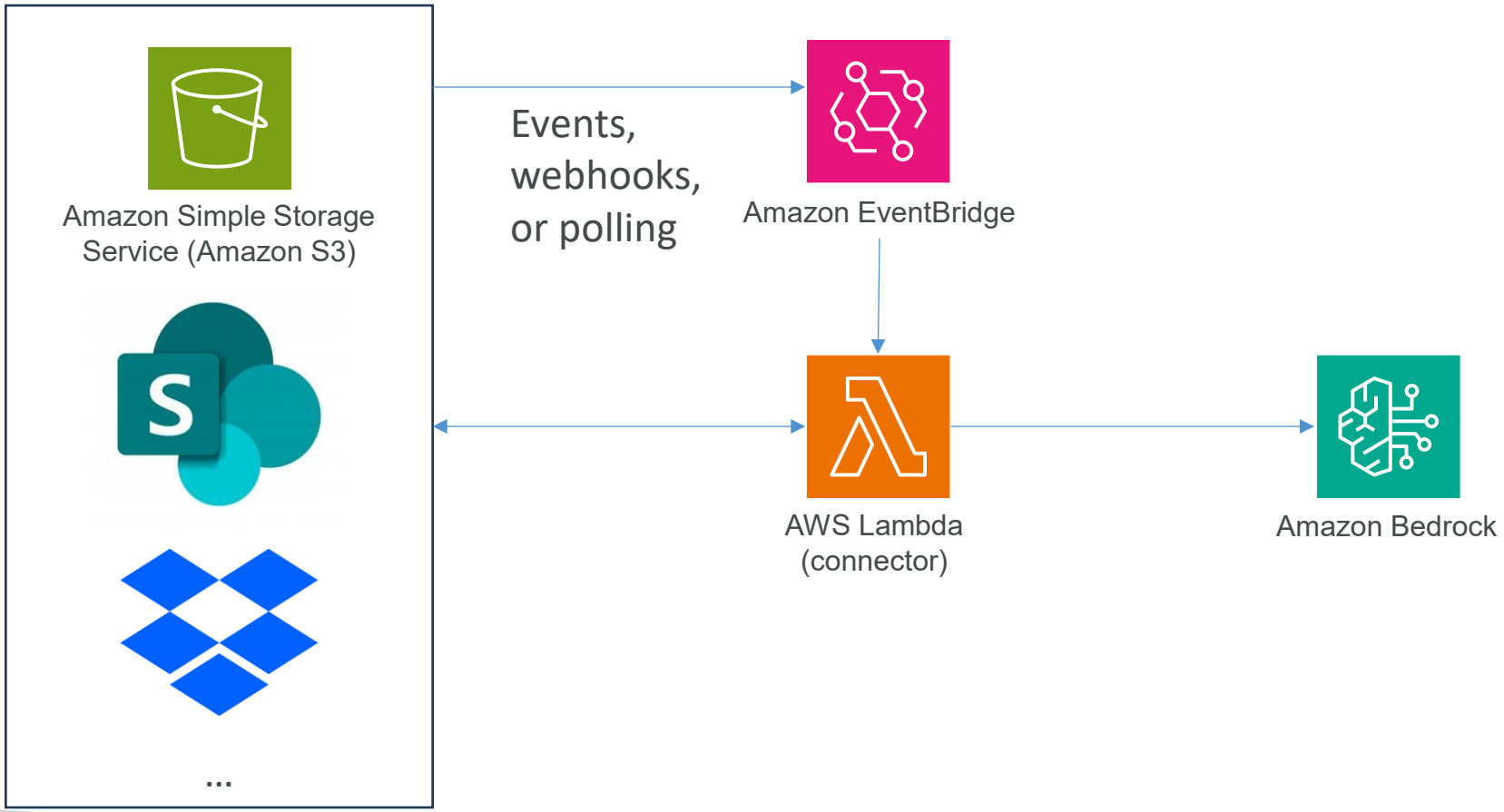
Keeping your Vector Store Up to Date



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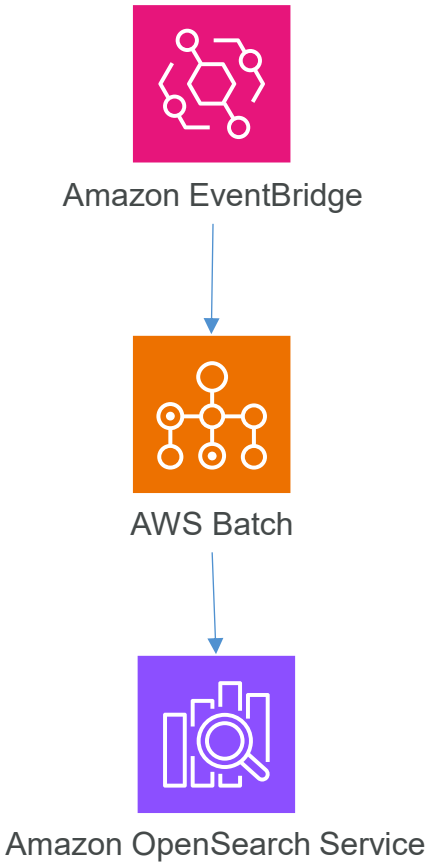
Vector Index Maintenance

- Drift and fragmentation can become issues over time
- You could use EventBridge to schedule a periodic trigger to rebuild your indices
- That could in turn use AWS Batch to queue up a job that
  - Creates new embeddings
  - Creates a new vector DB
  - Validates it
  - Swaps from the old to the new



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Re-ranker Models in Amazon Bedrock

- Attempts to improve relevance of retrieved results for RAG from your KB
- Calculates **relevance of chunks to the query**, orders results accordingly
- There is a Rerank operation in the API you can use to rank whatever documents you want relative to a query
- Or you can specify a reranker model when hitting your Knowledge Base
- Currently you can choose from Amazon or Cohere models
  - In limited regions

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S3 Storage Classes

- Amazon S3 Standard - General Purpose
- Amazon S3 Standard-Infrequent Access (IA)
- Amazon S3 One Zone-Infrequent Access
- Amazon S3 Glacier Instant Retrieval
- Amazon S3 Glacier Flexible Retrieval
- Amazon S3 Glacier Deep Archive
- Amazon S3 Intelligent Tiering

- Can move between classes manually or using S3 Lifecycle configurations

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S3 Durability and Availability

- Durability:
  - High durability (99.999999999%, 11 9's) of objects across multiple AZ
  - If you store 10,000,000 objects with Amazon S3, you can on average expect to incur a loss of a single object once every 10,000 years
  - Same for all storage classes

- Availability:
  - Measures how readily available a service is
  - Varies depending on storage class
  - Example: S3 standard has 99.99% availability = not available 53 minutes a year

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S3 Standard – General Purpose

- 99.99% Availability
- Used for frequently accessed data
- Low latency and high throughput
- Sustain 2 concurrent facility failures

- Use Cases: Big Data analytics, mobile & gaming applications, content distribution…

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S3 Storage Classes – Infrequent Access

- For data that is less frequently accessed, but requires rapid access when needed
- Lower cost than S3 Standard

- **Amazon S3 Standard-Infrequent Access (S3 Standard-IA)**
  - 99.9% Availability
  - Use cases: Disaster Recovery, backups

- **Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA)**
  - High durability (99.999999999%) in a single AZ; data lost when AZ is destroyed
  - 99.5% Availability
  - Use Cases: Storing secondary backup copies of on-premises data, or data you can recreate

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Amazon S3 Glacier Storage Classes

- Low-cost object storage meant for archiving / backup
- Pricing: price for storage + object retrieval cost

- **Amazon S3 Glacier Instant Retrieval**
  - Millisecond retrieval, great for data accessed once a quarter
  - Minimum storage duration of 90 days
- **Amazon S3 Glacier Flexible Retrieval** (formerly Amazon S3 Glacier):
  - Expedited (1 to 5 minutes), Standard (3 to 5 hours), Bulk (5 to 12 hours) – free
  - Minimum storage duration of 90 days
- **Amazon S3 Glacier Deep Archive – for long term storage:**
  - Standard (12 hours), Bulk (48 hours)
  - Minimum storage duration of 180 days

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S3 Intelligent-Tiering

- Small monthly monitoring and auto-tiering fee
- Moves objects automatically between Access Tiers based on usage
- There are no retrieval charges in S3 Intelligent-Tiering

- *Frequent Access tier (automatic):* default tier
- *Infrequent Access tier (automatic)*: objects not accessed for 30 days
- *Archive Instant Access tier (automatic):* objects not accessed for 90 days
- *Archive Access tier (optional):* configurable from 90 days to 700+ days
- *Deep Archive Access tier (optional):* config. from 180 days to 700+ days

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S3 Storage Classes Comparison

	Standard	Intelligent-Tiering	Standard-IA	One Zone-IA	Glacier Instant Retrieval	Glacier Flexible Retrieval	Glacier Deep Archive
Durability	99.999999999% == (11 9's)						
Availability	99.99%	99.9%	99.9%	99.5%	99.9%	99.99%	99.99%
Availability SLA	99.9%	99%	99%	99%	99%	99.9%	99.9%
Availability Zones	>= 3	>= 3	>= 3	1	>= 3	>= 3	>= 3
Min. Storage Duration Charge	None	None	30 Days	30 Days	90 Days	90 Days	180 Days
Min. Billable Object Size	None	None	128 KB	128 KB	128 KB	40 KB	40 KB
Retrieval Fee	None	None	Per GB retrieved	Per GB retrieved	Per GB retrieved	Per GB retrieved	Per GB retrieved

https://aws.amazon.com/s3/storage-classes/

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S3 Storage Classes – Price Comparison Example: us-east-1

	Standard	Intelligent-Tiering	Standard-IA	One Zone-IA	Glacier Instant Retrieval	Glacier Flexible Retrieval	Glacier Deep Archive
Storage Cost (per GB per month)	$0.023	$0.0025 - $0.023	$0.0125	$0.01	$0.004	$0.0036	$0.00099
Retrieval Cost (per 1000 request)	**GET:** $0.0004 **POST:** $0.005	**GET:** $0.0004 **POST:** $0.005	**GET:** $0.001 **POST:** $0.01	**GET:** $0.001 **POST:** $0.01	**GET:** $0.01 **POST:** $0.02	**GET:** $0.0004 **POST:** $0.03 **Expedited:** $10 **Standard:** $0.05 **Bulk:** free	**GET:** $0.0004 **POST:** $0.05 **Standard:** $0.10 **Bulk:** $0.025
Retrieval Time	Instantaneous					**Expedited** (1 – 5 mins) **Standard** (3 – 5 hours) Bulk (5 – 12 hours)	**Standard** (12 hours) **Bulk** (48 hours)
Monitoring Cost (per 1000 objects)		$0.0025					

https://aws.amazon.com/s3/pricing/

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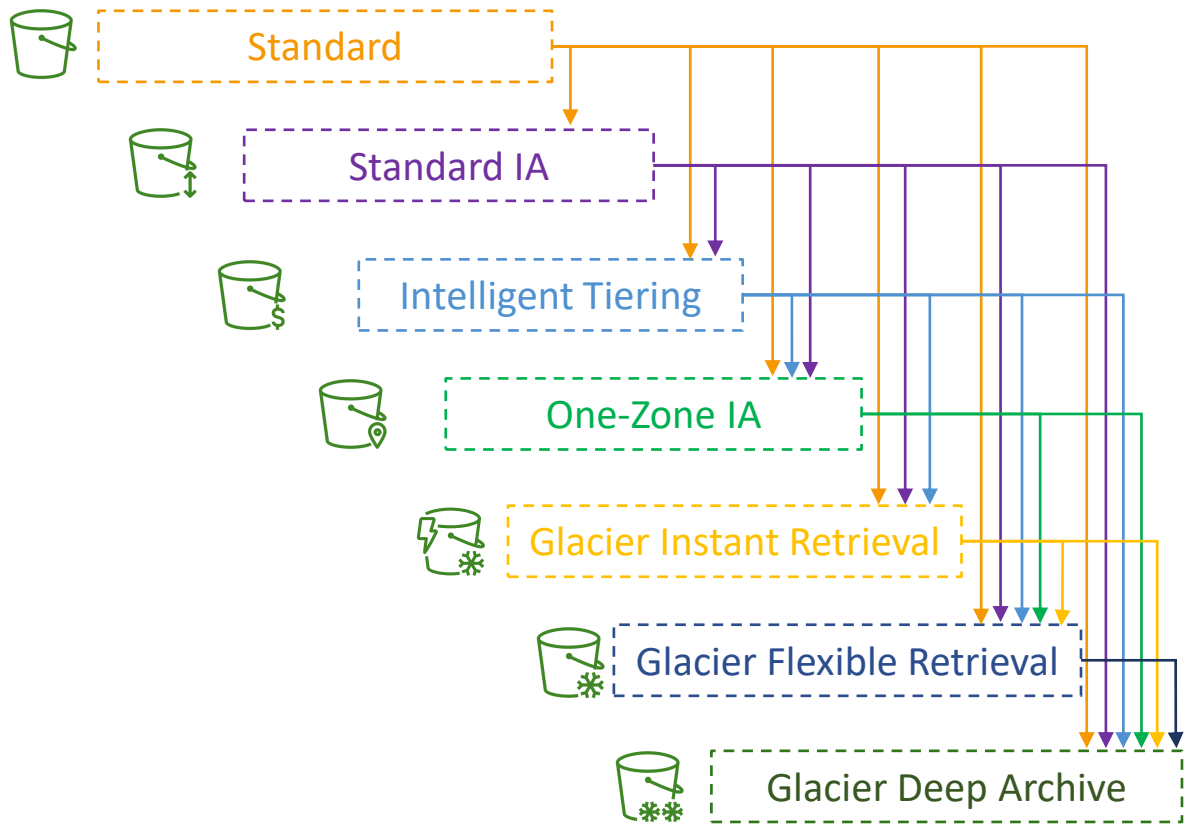
Amazon S3 – Moving between Storage Classes

- You can transition objects between storage classes
- For infrequently accessed object, move them to **Standard IA**
- For archive objects that you don't need fast access to, move them to **Glacier or Glacier Deep Archive**
- Moving objects can be automated using a **Lifecycle Rules**



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Amazon S3 – Lifecycle Rules

- **Transition Actions –** configure objects to transition to another storage class
  - Move objects to Standard IA class 60 days after creation
  - Move to Glacier for archiving after 6 months

- **Expiration actions –** configure objects to expire (delete) after some time
  - Access log files can be set to delete after a 365 days
  - **Can be used to delete old versions of files (if versioning is enabled)**
  - Can be used to delete incomplete Multi-Part uploads

- Rules can be created for a certain prefix (example: *s3://mybucket/mp3/*)*
- Rules can be created for certain objects Tags (example: *Department: Finance*)

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Amazon S3 – Lifecycle Rules (Scenario 1)

- **Your application on EC2 creates images thumbnails after profile photos are uploaded to Amazon S3. These thumbnails can be easily recreated, and only need to be kept for 60 days. The source images should be able to be immediately retrieved for these 60 days, and afterwards, the user can wait up to 6 hours. How would you design this?**

- S3 source images can be on **Standard**, with a lifecycle configuration to transition them to **Glacier** after 60 days
- S3 thumbnails can be on **One-Zone IA**, with a lifecycle configuration to expire them (delete them) after 60 days

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Amazon S3 – Lifecycle Rules (Scenario 2)

- **A rule in your company states that you should be able to recover your deleted S3 objects immediately for 30 days, although this may happen rarely. After this time, and for up to 365 days, deleted objects should be recoverable within 48 hours.**

- E**nable S3 Versioning** in order to have object versions, so that "deleted objects" are in fact hidden by a "delete marker" and can be recovered
- Transition the "noncurrent versions" of the object to **Standard IA**
- Transition afterwards the "noncurrent versions" to **Glacier Deep Archive**

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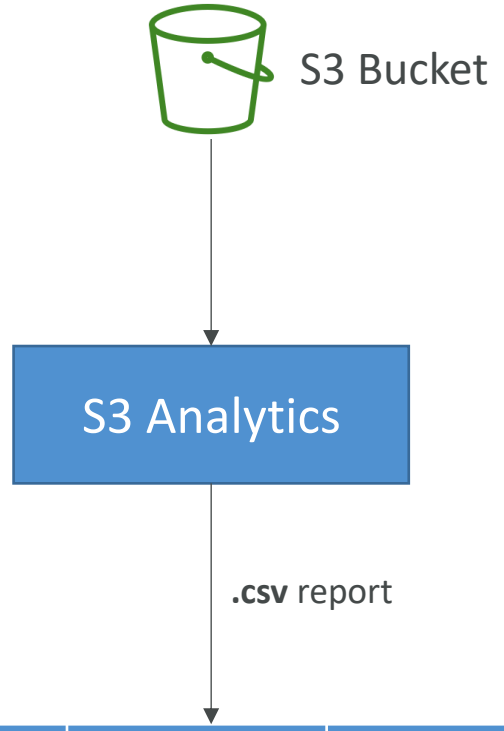
Amazon S3 Analytics – Storage Class Analysis

- Help you decide when to transition objects to the right storage class
- Recommendations for **Standard** and **Standard IA**
  - Does NOT work for One-Zone IA or Glacier
- Report is updated daily
- 24 to 48 hours to start seeing data analysis
- Good first step to put together Lifecycle Rules (or improve them)!



Date	StorageClass	ObjectAge
8/22/2022	STANDARD	000-014
8/25/2022	STANDARD	030-044
9/6/2022	STANDARD	120-149

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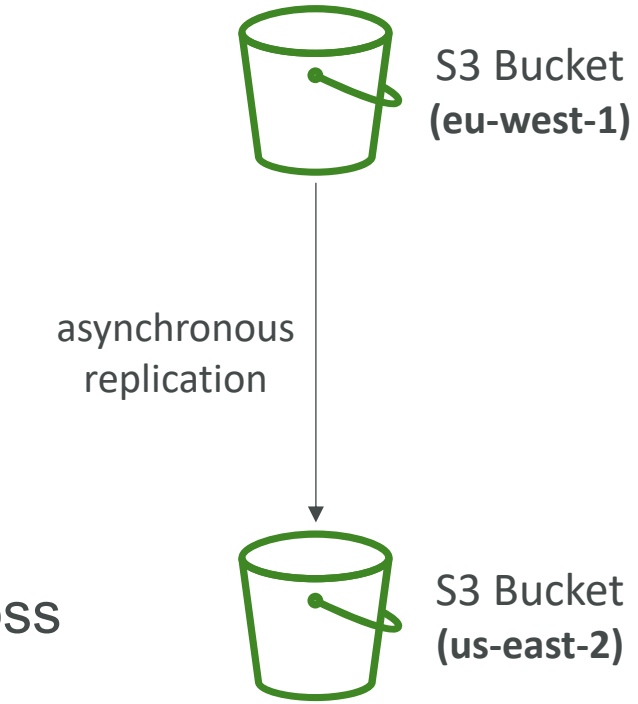
Amazon S3 – Replication (CRR & SRR)

- **Must enable Versioning** in source and destination buckets
- **Cross-Region Replication (CRR)**
- **Same-Region Replication (SRR)**
- Buckets can be in different AWS accounts
- Copying is asynchronous
- Must give proper IAM permissions to S3



- Use cases:
  - **CRR –** compliance, lower latency access, replication across accounts
  - **SRR –** log aggregation, live replication between production and test accounts

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Amazon S3 – Replication (Notes)

- After you enable Replication, only new objects are replicated
- Optionally, you can replicate existing objects using **S3 Batch Replication**
  - Replicates existing objects and objects that failed replication

- For DELETE operations
  - **Can replicate delete markers** from source to target (optional setting)
  - Deletions with a version ID are not replicated (to avoid malicious deletes)

- **There is no "chaining" of replication**
  - If bucket 1 has replication into bucket 2, which has replication into bucket 3
  - Then objects created in bucket 1 are not replicated to bucket 3

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Amazon S3 – Object Encryption

- You can encrypt objects in S3 buckets using one of 4 methods

- **Server-Side Encryption (SSE)**
  - **Server-Side Encryption with Amazon S3-Managed Keys (SSE-S3) – Enabled by Default**
    - Encrypts S3 objects using keys handled, managed, and owned by AWS
  - **Server-Side Encryption with KMS Keys stored in AWS KMS (SSE-KMS)**
    - Leverage AWS Key Management Service (AWS KMS) to manage encryption keys
  - **Server-Side Encryption with Customer-Provided Keys (SSE-C)**
    - When you want to manage your own encryption keys
- **Client-Side Encryption**

- It's important to understand which ones are for which situation for the exam

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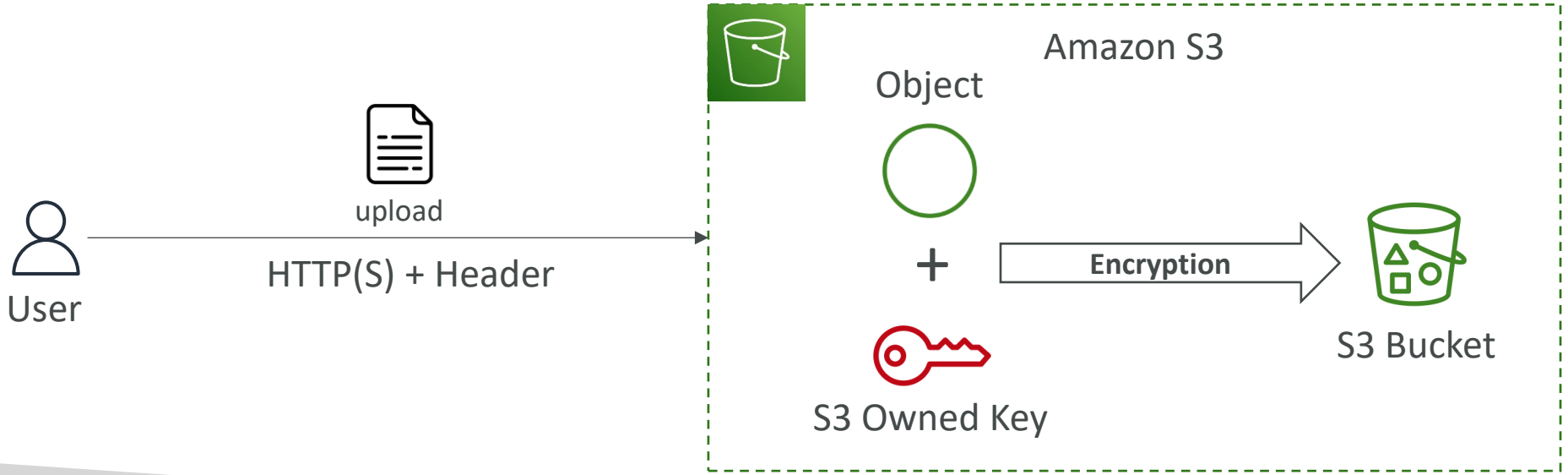
Amazon S3 Encryption – SSE-S3

- Encryption using keys handled, managed, and owned by AWS
- Object is encrypted server-side
- Encryption type is **AES-256**
- Must set header **"x-amz-server-side-encryption": "AES256"**
- **Enabled by default for new buckets & new objects**



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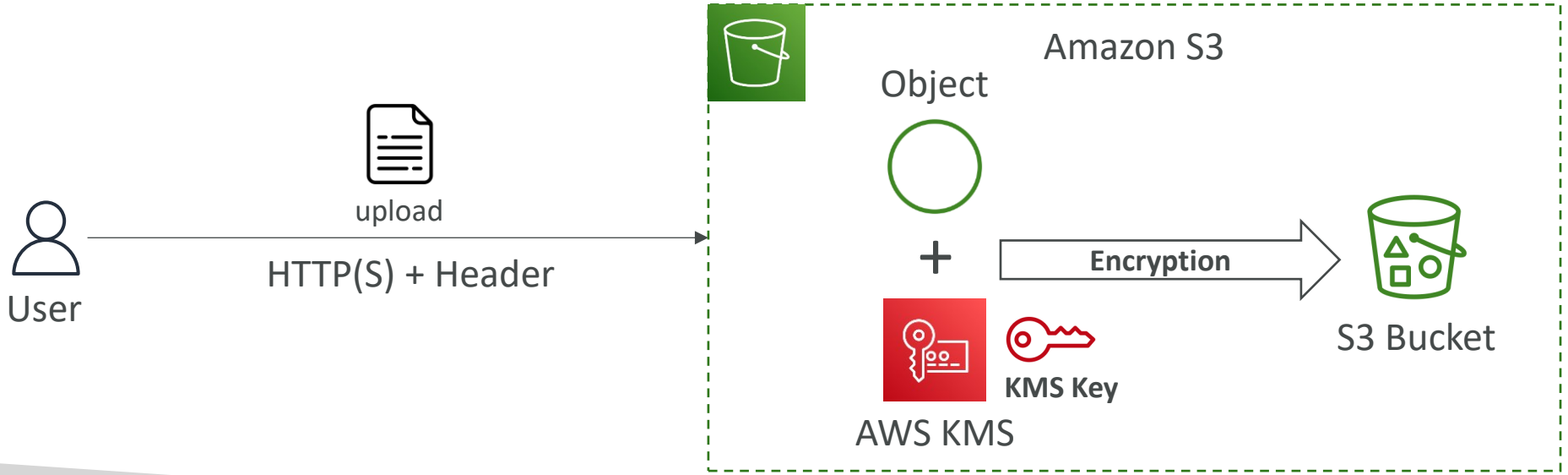
Amazon S3 Encryption – SSE-KMS

- Encryption using keys handled and managed by AWS KMS (Key Management Service)
- KMS advantages: user control + audit key usage using CloudTrail
- Object is encrypted server side
- Must set header **"x-amz-server-side-encryption": "aws:kms"**



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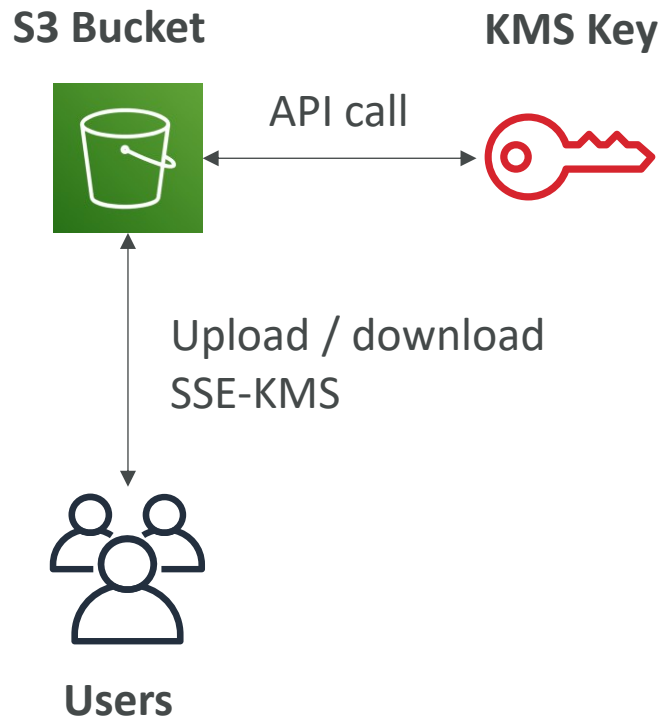
SSE-KMS Limitation

- If you use SSE-KMS, you may be impacted by the KMS limits
- When you upload, it calls the **GenerateDataKey** KMS API
- When you download, it calls the **Decrypt** KMS API
- Count towards the KMS quota per second (5500, 10000, 30000 req/s based on region)
- You can request a quota increase using the Service Quotas Console



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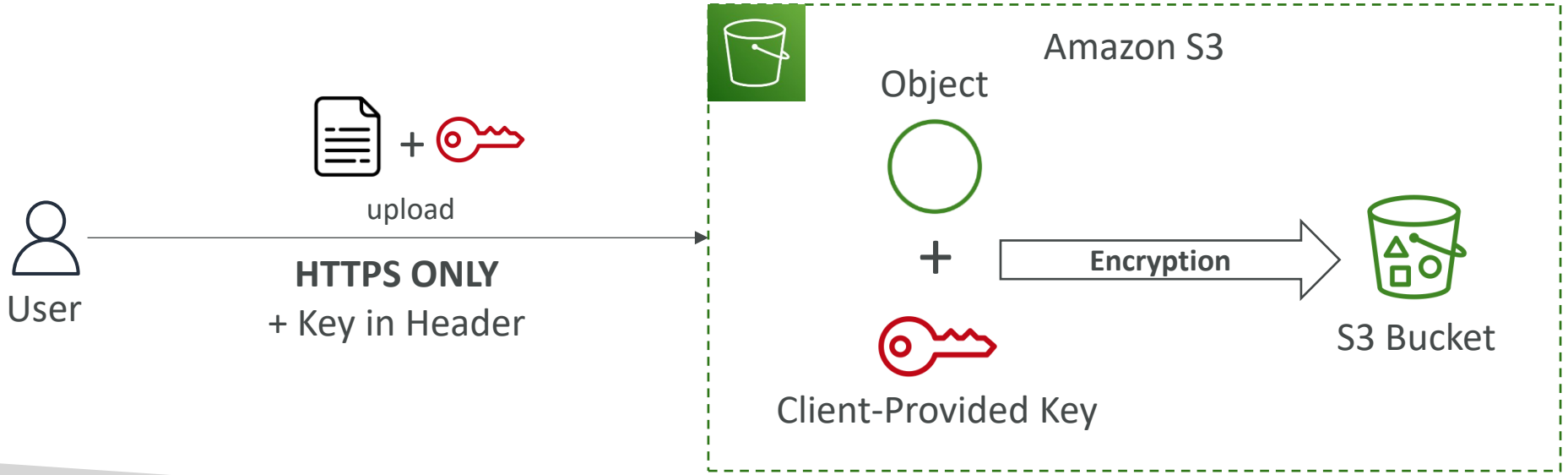
Amazon S3 Encryption – SSE-C

- Server-Side Encryption using keys fully managed by the customer outside of AWS
- Amazon S3 does **NOT** store the encryption key you provide
- **HTTPS must be used**
- Encryption key must provided in HTTP headers, for every HTTP request made



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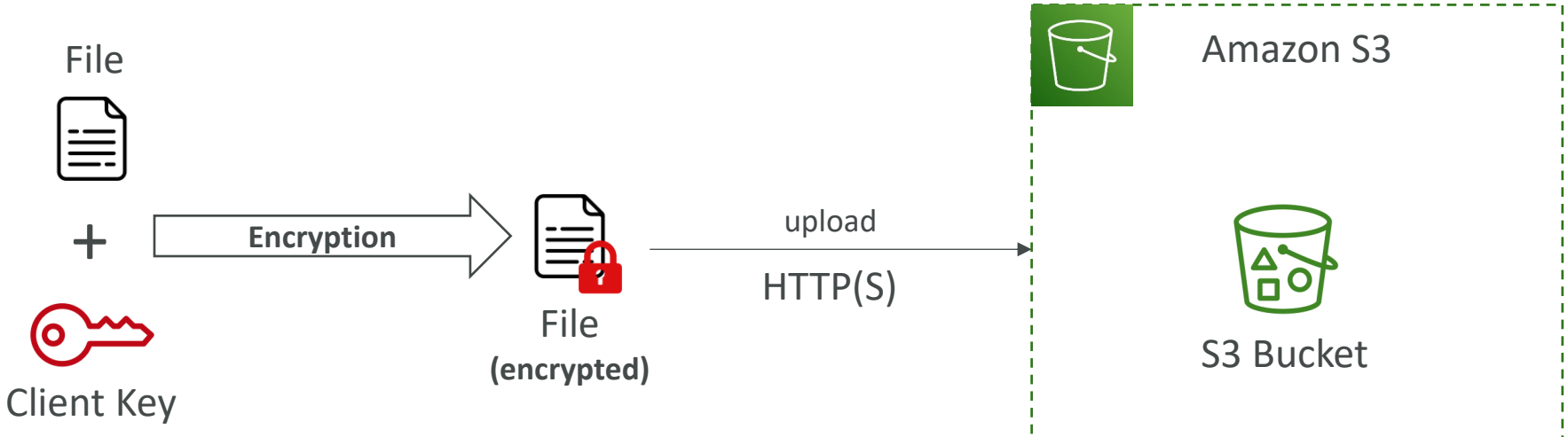
Amazon S3 Encryption – Client-Side Encryption

- Use client libraries such as **Amazon S3 Client-Side Encryption Library**
- Clients must encrypt data themselves before sending to Amazon S3
- Clients must decrypt data themselves when retrieving from Amazon S3
- Customer fully manages the keys and encryption cycle



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Amazon S3 – Encryption in transit (SSL/TLS)

- Encryption in flight is also called SSL/TLS

- Amazon S3 exposes two endpoints:
  - **HTTP Endpoint** – non encrypted
  - **HTTPS Endpoint** – encryption in flight

- **HTTPS is recommended**
- **HTTPS is mandatory for SSE-C**
- Most clients would use the HTTPS endpoint by default

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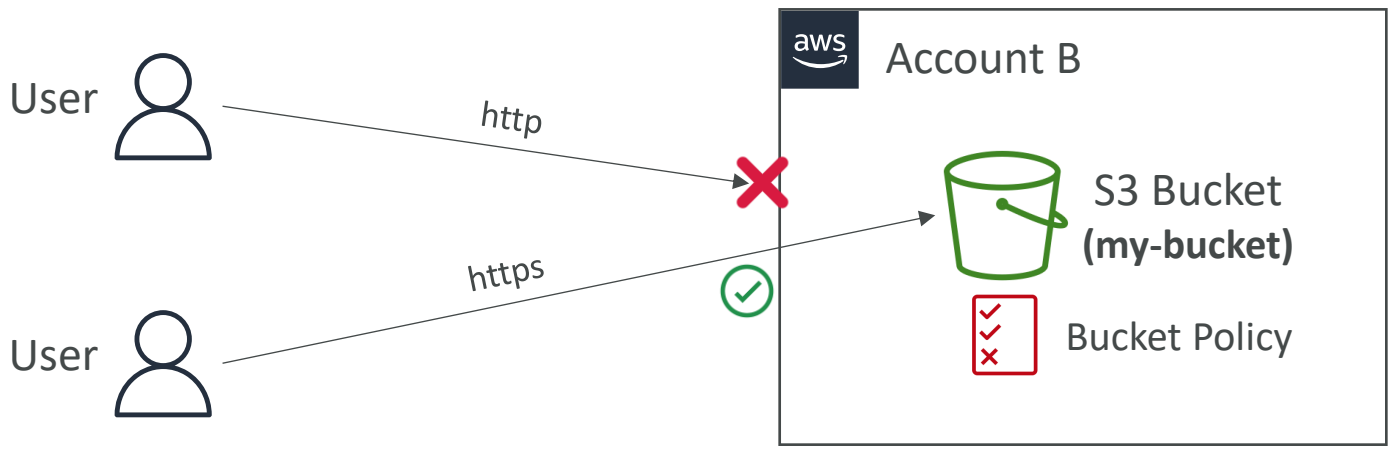
Amazon S3 – Force Encryption in Transit aws:SecureTransport




```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Deny",
            "Principal": "*",
            "Action": "s3:GetObject",
            "Resource": "arn:aws:s3:::my-bucket/*",
            "Condition": {
                "Bool": {
                    "aws:SecureTransport": "false"
                }
            }
        }
    ]
}
```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Amazon S3 – Default Encryption vs. Bucket Policies

- **SSE-S3 encryption is automatically applied to new objects stored in S3 bucket**
- Optionally, you can “force encryption” using a bucket policy and refuse any API call to PUT an S3 object without encryption headers (SSE-KMS or SSE-C)

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Deny",
            "Action": "s3:PutObject",
            "Principal": "*",
            "Resource": "arn:aws:s3:::my-bucket/*",
            "Condition": {
                "StringNotEquals": {
                    "s3:x-amz-server-side-encryption": "aws:kms"
                }
            }
        }
    ]
}
```

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Deny",
            "Action": "s3:PutObject",
            "Principal": "*",
            "Resource": "arn:aws:s3:::my-bucket/*",
            "Condition": {
                "Null": {
                    "s3:x-amz-server-side-encryption-customer-algorithm": "true"
                }
            }
        }
    ]
}
```

- **<u>Note:</u> Bucket Policies are evaluated before “Default Encryption”**

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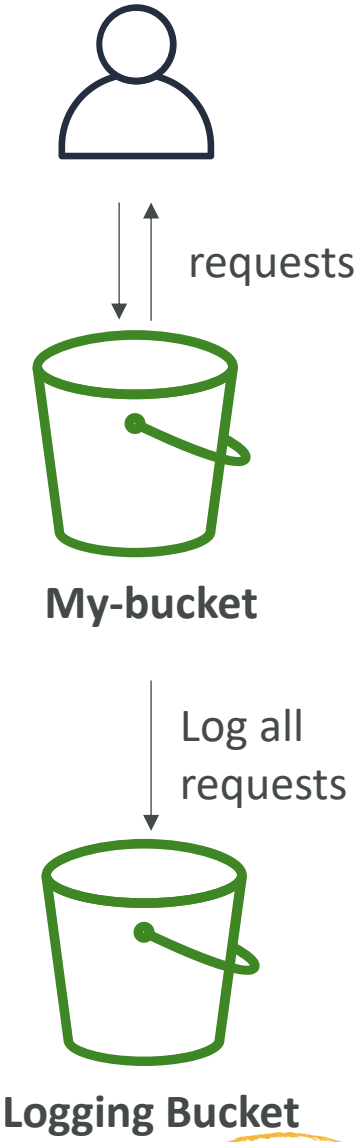
S3 Access Logs

- For audit purpose, you may want to log all access to S3 buckets
- Any request made to S3, from any account, authorized or denied, will be logged into another S3 bucket
- That data can be analyzed using data analysis tools…
- The target logging bucket must be in the same AWS region

- The log format is at: https://docs.aws.amazon.com/AmazonS3/latest/dev/LogFormat.html



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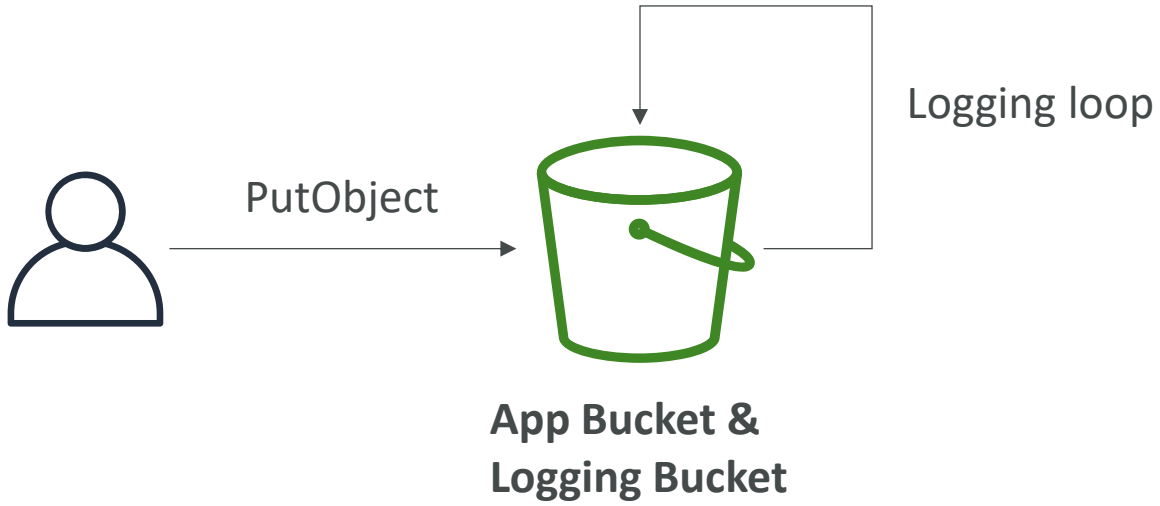
S3 Access Logs: Warning

- Do not set your logging bucket to be the monitored bucket
- It will create a logging loop, and **your bucket will grow exponentially**



Do not try this at home ☺

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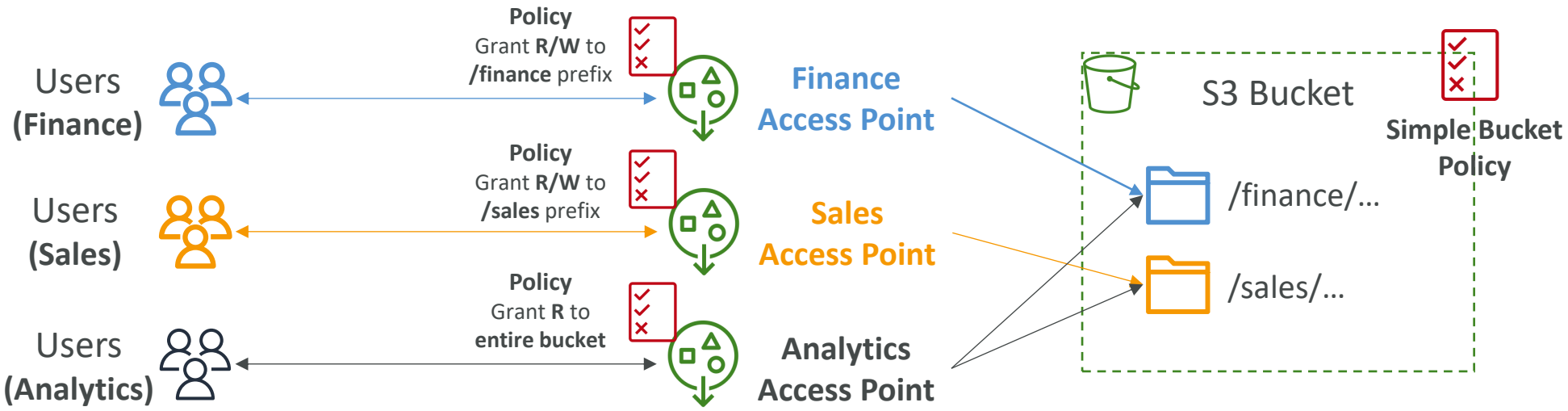
S3 – Access Points



- Access Points simplify security management for S3 Buckets
- Each Access Point has:
  - its own DNS name (Internet Origin or VPC Origin)
  - an access point policy (similar to bucket policy) – manage security at scale

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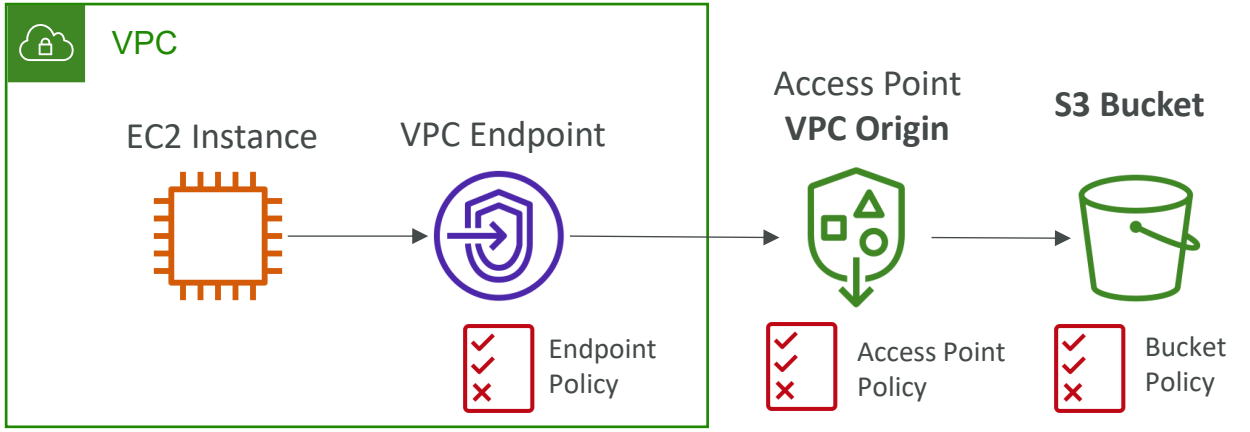
S3 – Access Points – VPC Origin

- We can define the access point to be accessible only from within the VPC
- You must create a VPC Endpoint to access the Access Point (Gateway or Interface Endpoint)
- The VPC Endpoint Policy must allow access to the target bucket and Access Point




```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Principal": "*",
            "Action": [
                "s3:GetObject"
            ],
            "Effect": "Allow",
            "Resource": [
                "arn:aws:s3:::awsexamplebucket1/*",
                "arn:aws:s3:us-west-2:123456789012:accesspoint/example-vpc-ap/object/*"
            ]
        }
    ]
}
```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Agentic AI

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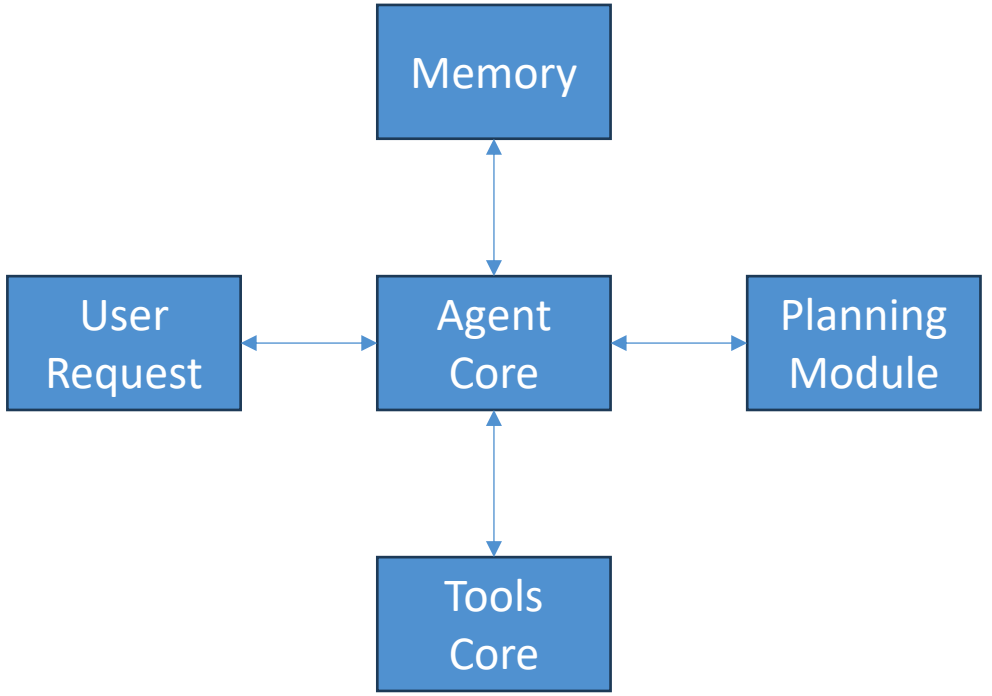
LLM Agents

- Giving tools to your LLM!
- The LLM is given discretion on which tools to use for what purpose
- The agent has a memory, an ability to plan how to answer a request, and tools it can use in the process.
- In practice, the “memory” is just the chat history and external data stores, and the “planning module” is guidance given to the LLM on how to break down a question into sub-questions that the tools might be able to help with.
- Prompts associated with each tool are used to guide it on how to use its tools.



Conceptual diagram of an LLM agent, as described by Nvidia (https://developer.nvidia.com/blog/introduction-to-llm-agents)

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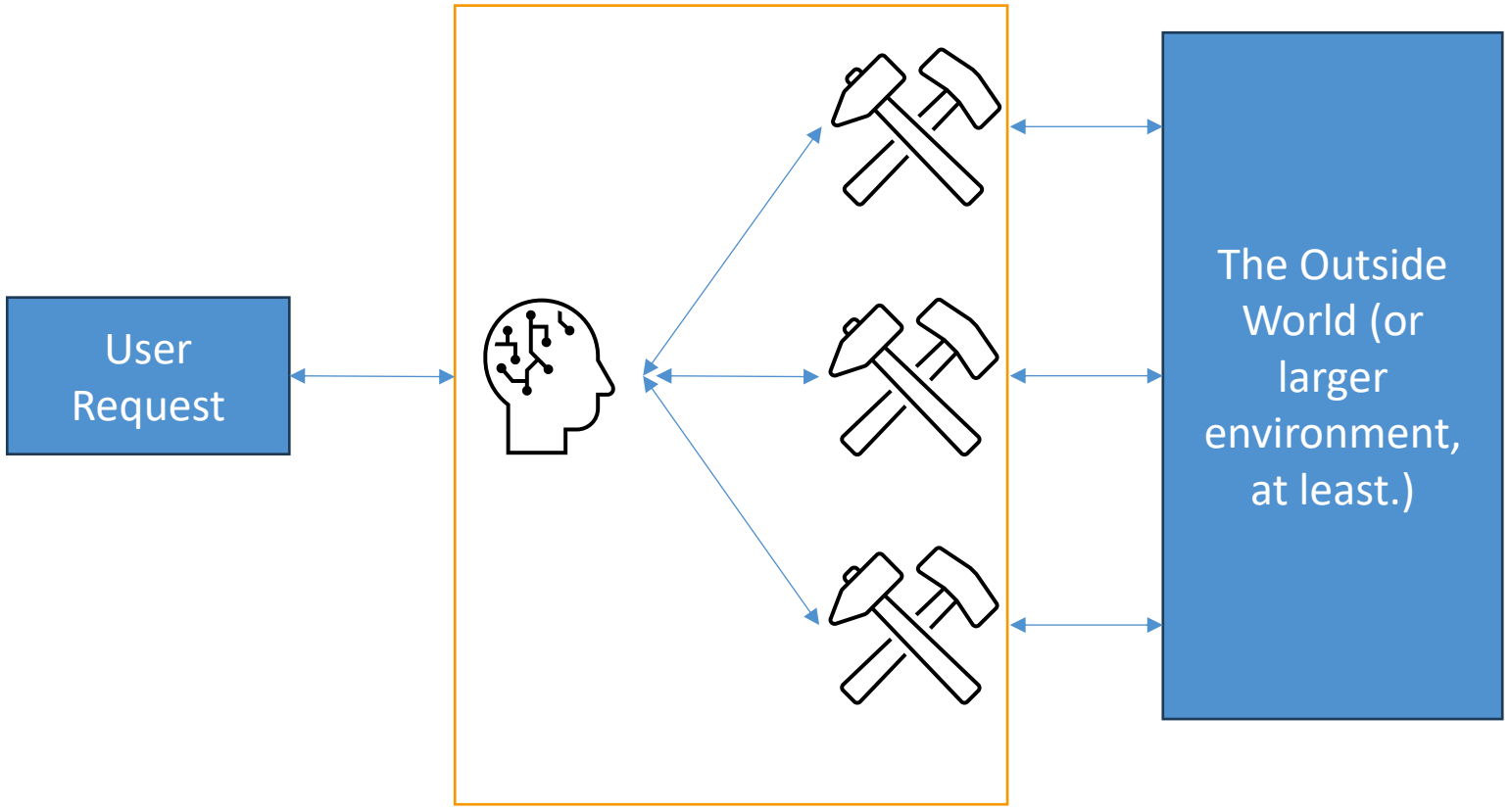
LLM Agents: A More Practical Approach

- "Tools" are just functions provided to a tools API.
  - In Bedrock, this can be a Lambda function.
- Prompts guide the LLM on how to use them.
- Tools may access outside information, retrievers, other Python modules, services, etc.



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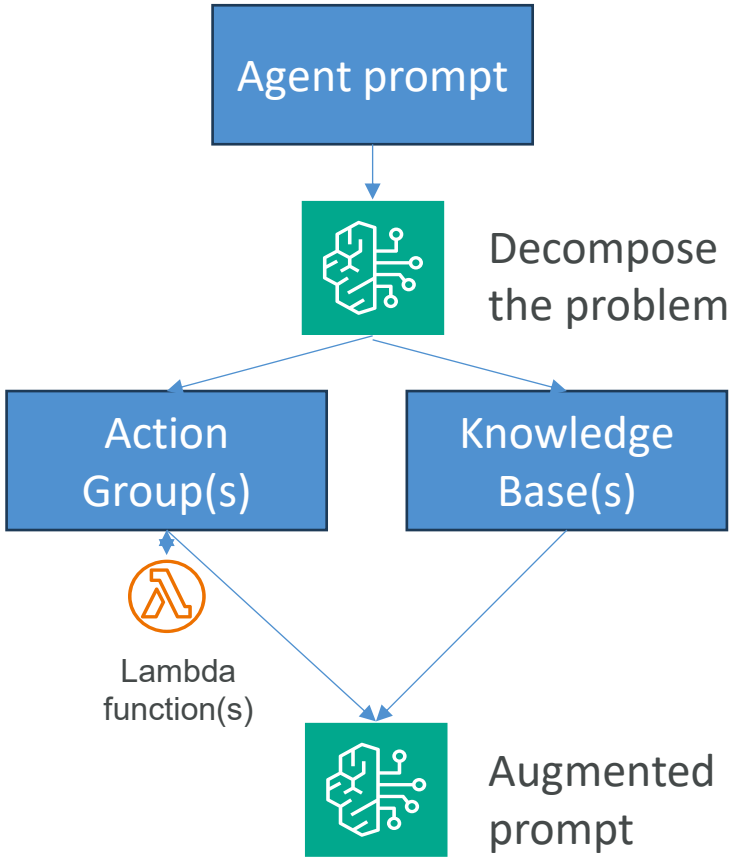
How do Agents Know Which Tools to Use?

- Start with a foundational model to work with
- In Bedrock, “Action Groups” define a tool
  - A prompt informs the FM when to use it
    - “Use this function to determine the current weather in a city”
  - You must define the parameters your tool (Lambda function) expects
    - Define name, description, type, and if it’s required or not
    - The description is actually important and will be used by the LLM to extract the required info from the user prompt
    - You can also allow the FM to go back and ask the user for missing information it needs for a tool
    - This can be done using OpenAI-style schemas, or visually with a table in the Bedrock UI
- Agents may also have knowledge bases associated with them
  - Again, a prompt tells the LLM when it should use it
    - “Use this for answering questions about X”
  - This is “agentic RAG” – RAG is just another tool
- Optional “Code Interpreter” allows the agent to write its own code to answer questions or produce charts.



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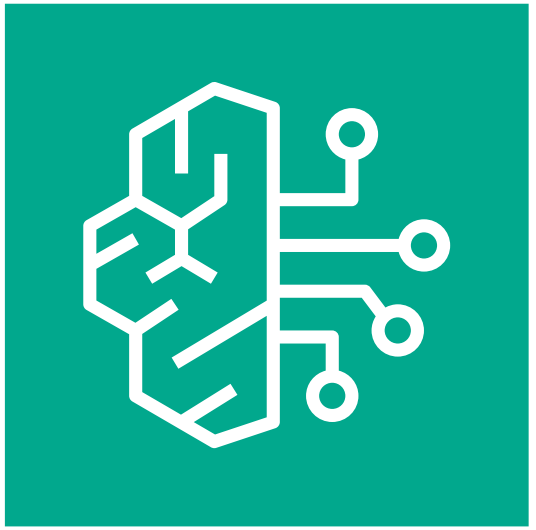
Deploying Bedrock Agents

- ("Bedrock Agents Classic" is being replaced by AgentCore… more on that later.)
- Create an "alias" for an agent
  - This creates a deployed snapshot
- On-Demand throughput (ODT)
  - Allows agent to run at quotas set at the account level
- Provisioned throughput (PT)
  - Allows you to purchase an increased rate and number of tokens for your agent
- Your application can use the **InvokeAgent** request using your alias ID and an **Agents for Amazon Bedrock Runtime Endpoint**.



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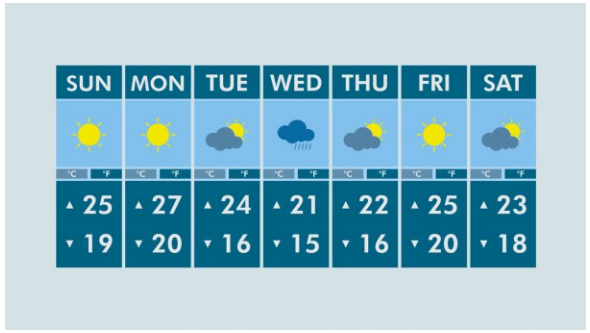
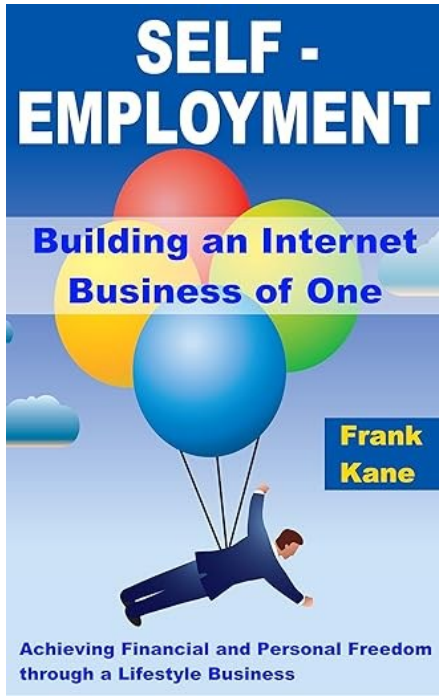
This will make more sense with an example

- Let's tie it all together
  - An agent that uses a knowledge base (my book)
  - And a tool (action group – what's the weather)
  - And guardrails





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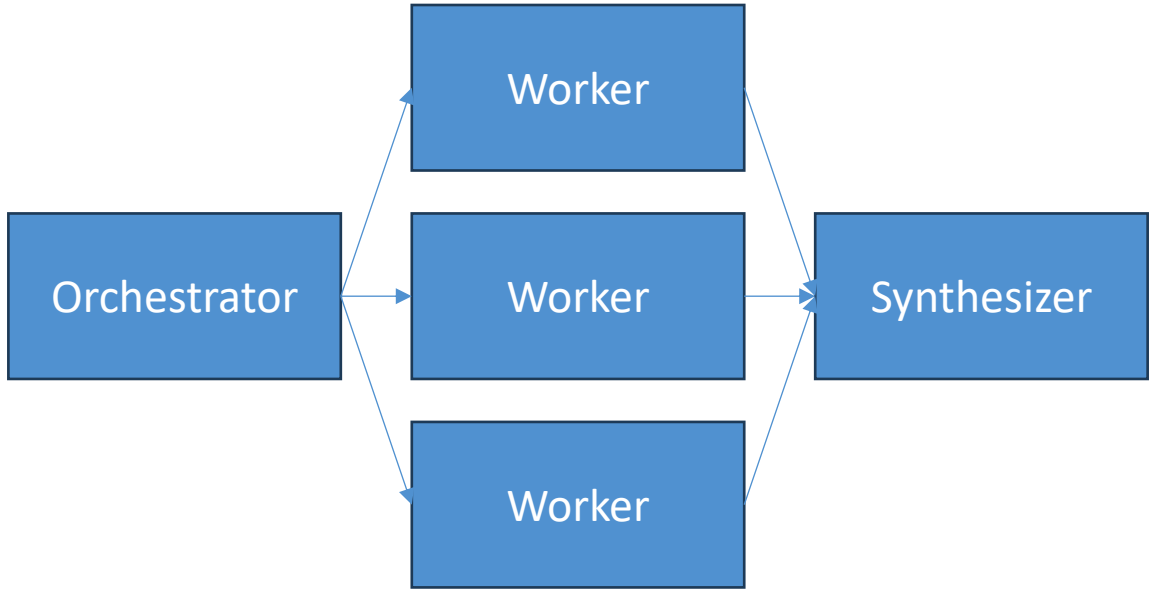
Multi-agent systems: an example

- Similar to just giving tools to a single agent
  - But many tools may be used at once
  - For example, coding agents may need to operate on multiple files in different ways to achieve a larger task
  - Workflows that require complex decision making can benefit from these agents (otherwise, keep it simple with deterministic workflows.)
- The Orchestrator breaks down tasks and delegates them to worker LLMs
- A Synthesizer can then combine their results



For example, this could be a translation tool being asked to translate to multiple languages at once.

Each LLM here may have its own tools and memory.

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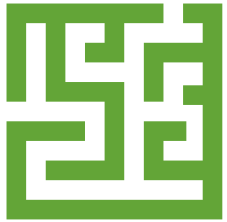
Why use multiple agents (or augmented LLMs)?

Too many tools

If one agent has trouble choosing the appropriate tool, multiple specialized agents with specific tools might work better



Complex logic

Prompts get too complicated with conditional statements, better to define the flow with specialized agents

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Manager (Orchestrator – workers)

- Similar to just giving tools to a single agent
  - But many tools may be used at once
  - For example, coding agents may need to operate on multiple files in different ways to achieve a larger task
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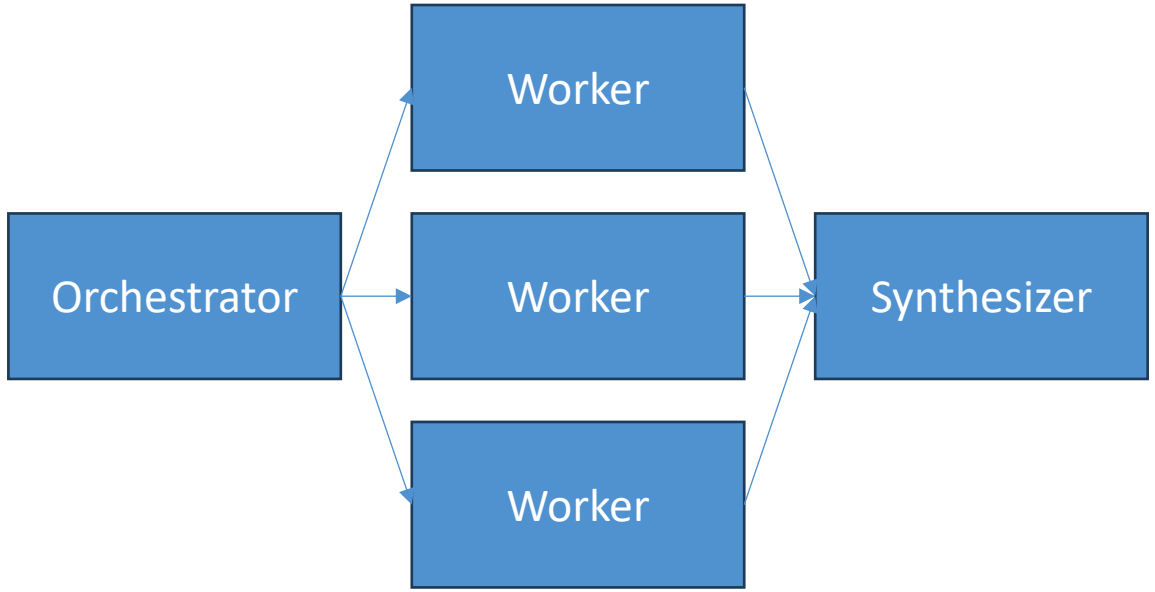

For example, this could be a translation tool being asked to translate to multiple languages at once.

Each LLM here may have its own tools and memory.

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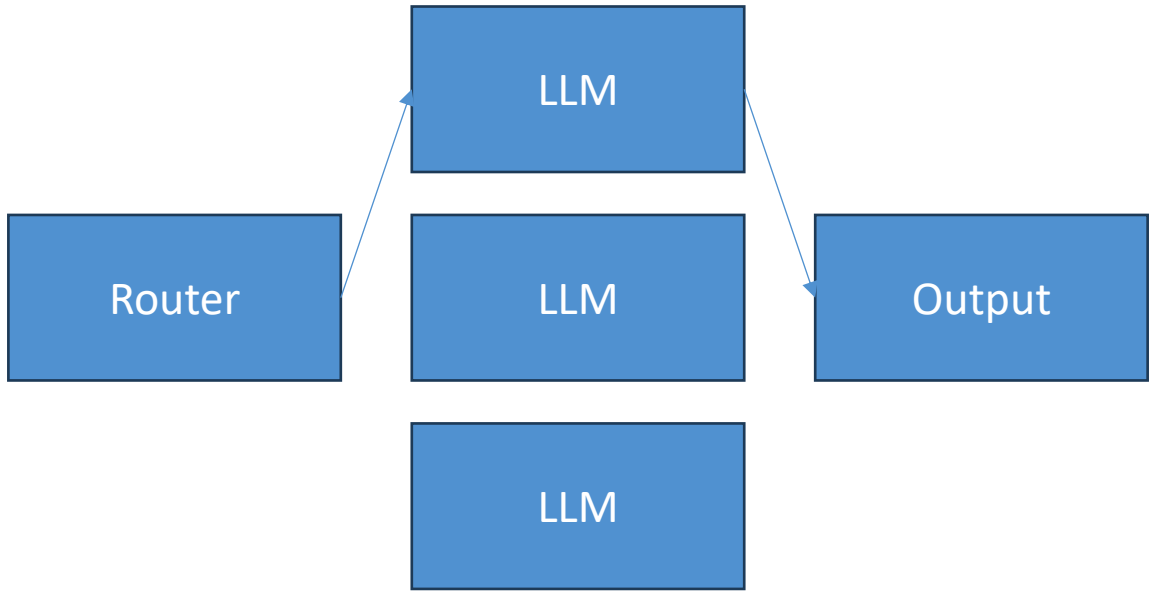
Routing

- A “router” LLM chooses just one of many specialized agents to call in turn
- Use for complex tasks where a single classification is needed
  - i.e., how large of a model is needed for the task
  - What kind of customer service query is being received



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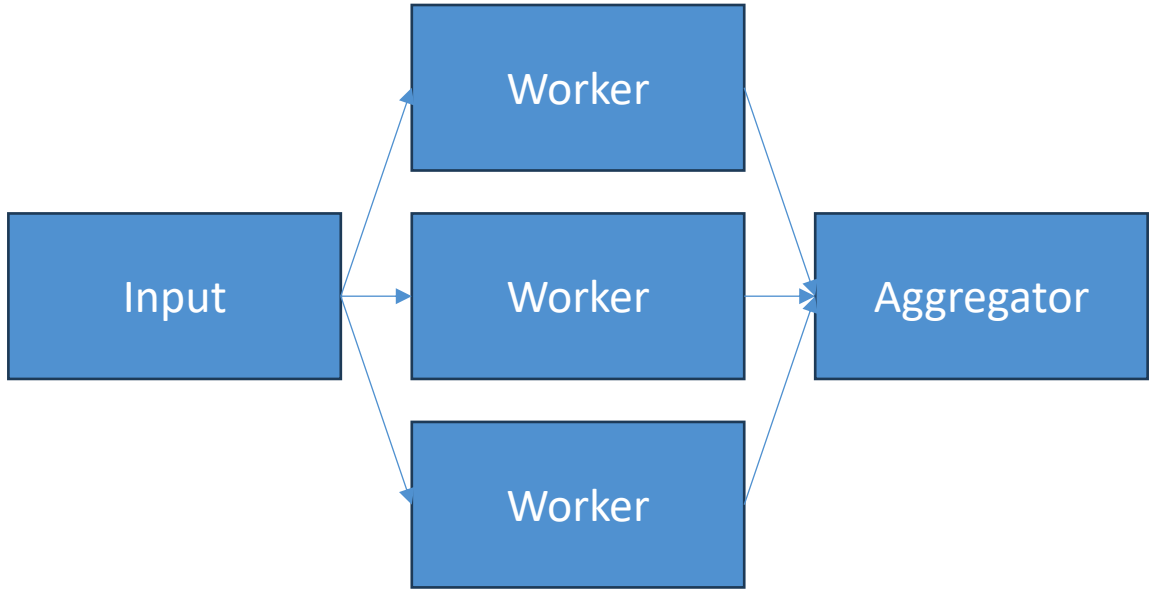
Parallelization

- A lot like the orchestrator
  - But without complex orchestration
- Sectioning: independent subtasks run in parallel
  - Running multiple guardrails at once
  - Running multiple evaluations at once
- Voting
  - Do the same thing with different models or prompts
  - Let them vote on the right result



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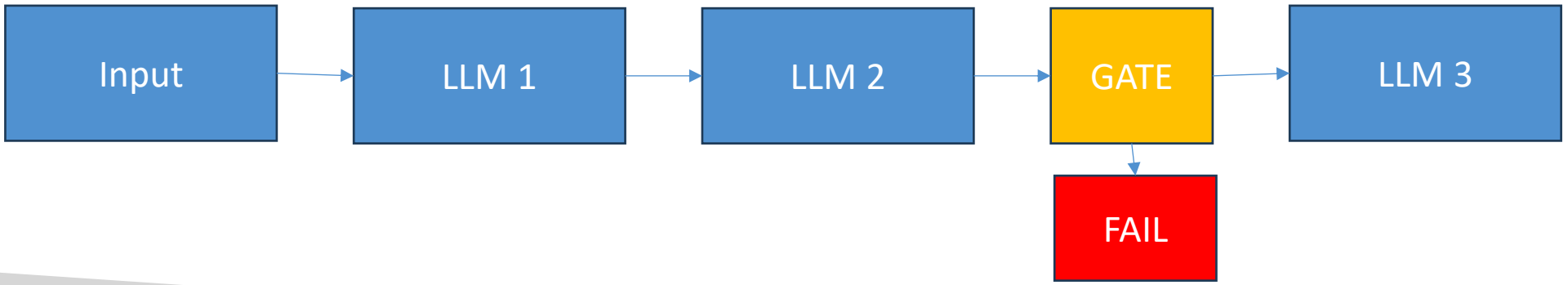
Prompt Chaining

- Use when a task has a discrete sequence of known steps
- Each LLM processes the output of the previous one
- Can add “gates” to keep things on track as you go, or exit early
- Examples
  - Write a document, then translate it to different languages
  - Write an outline, then make slides



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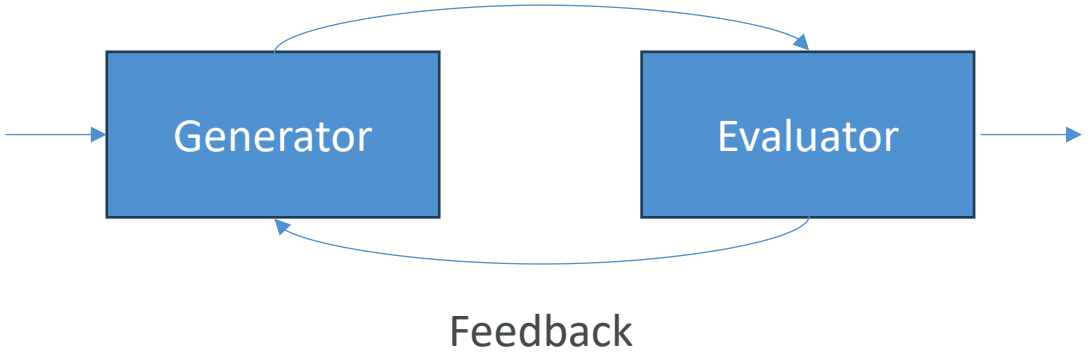
Evaluator - Optimizer

- One LLM generates a response
- Another evaluates that response
- If it's not good enough, provide feedback and try again
- Examples
  - Code reviewing code and incorporating those recommendations
  - Complex search that requires multiple rounds



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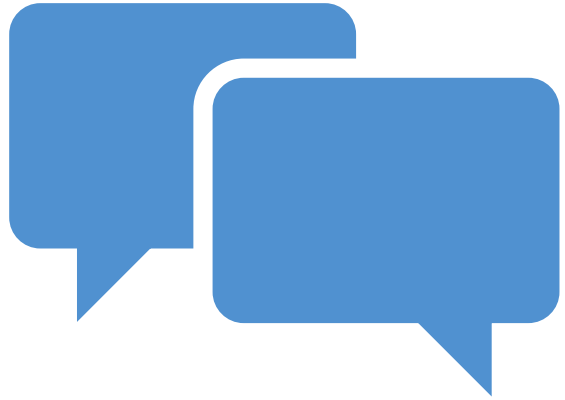
Adding Memory

- Short-term
  - Chat history within a session / immediate context
  - Enables conversations
  - Events within sessions
  - This could be in-memory or in a big distributed cache
    - Elasticache, MemoryDB, even DynamoDB is OK
- Long-term
  - Stores “extracted insights”
  - Summaries of past sessions
  - Preferences (your coding style and favorite tools, for example)
  - Facts you gave it in the past
  - “Memory Records” that store structured information derived from agent interactions
    - “Strategies” for user preferences, semantic facts, session summaries
- This all needs to be stored somewhere!
  - Long-term: DynamoDB, SQLLite, RDS, Aurora, etc.
  - AgentCore Memory (we’re getting there)
  - Mem0 (Strands uses this)
  - You could frame Knowledge Bases as a form of long-term memory too.



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Strands Agents

- Another Agents Python SDK (like OpenAI Agents SDK, CrewAI, LangGraph, Google ADK)
  - Released by Amazon as open source
- You can create specialized agents and multi-agent systems
  - Complex task decomposition
  - Collaborative problem solving
- Of course it has tighter AWS integration
  - Bedrock models, Lambda, Step Functions, and more
- But it's not tied to AWS
  - You can use it with OpenAI and others if you want to
- Multimodal support (text, speech, images, video)
- MCP (Model Context Protocol) support

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Strands Built-in Tools (just a few)

- AWS Services / boto3
- RAG and memory in Bedrock Knowledge Bases
- Mem0 agent memory and personalization
- Run Python code
- http (call anything external)
- Run shell commands
- Manipulate files
- Coordinate agent swarms
- Speak (with Amazon Polly)

…or your own custom tools, from arbitrary Python code

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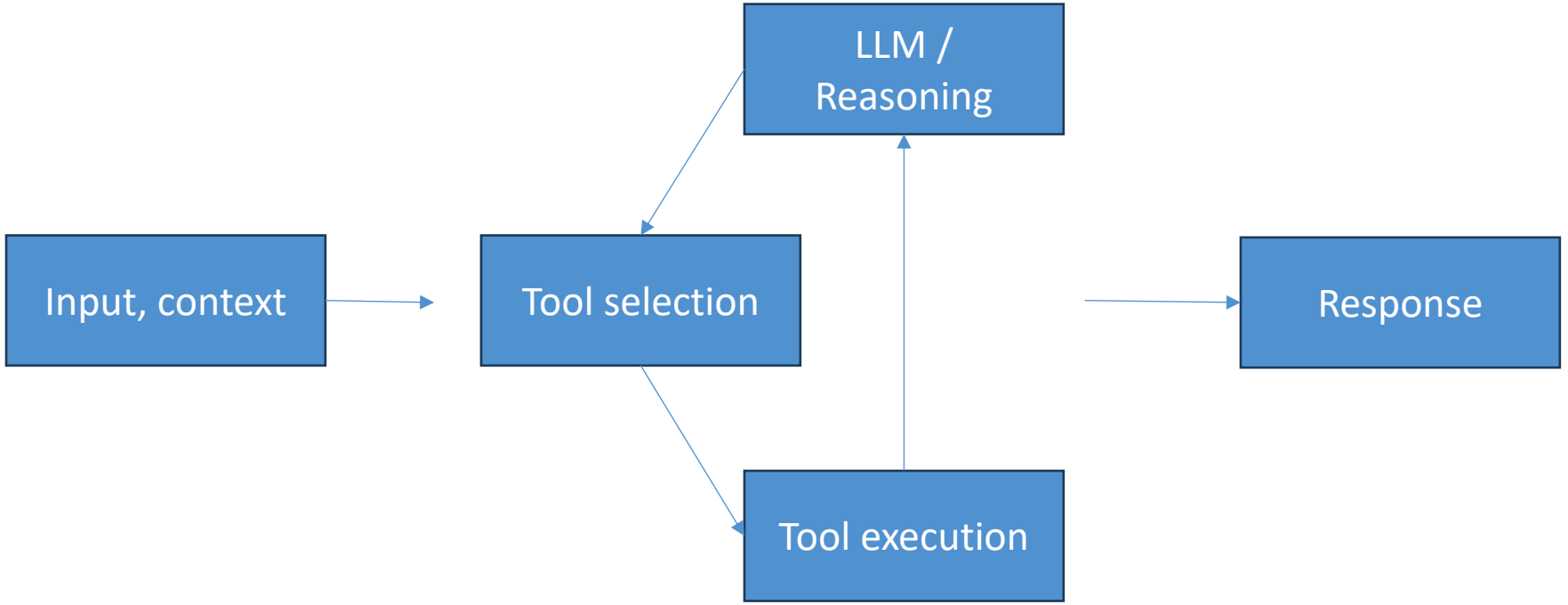
The Strands Agent Loop



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Making it real – the Strands quickstart

This example illustrates a mix of custom and built-in tools:

```
from strands import Agent, tool
from strands_tools import calculator, current_time, python_repl

# Define a custom tool as a Python function using the @tool decorator
@tool
def letter_counter(word: str, letter: str) -> int:
if not isinstance(word, str) or not isinstance(letter, str):
      return 0

   if len(letter) != 1:
      raise ValueError("The 'letter' parameter must be a single character")

   return word.lower().count(letter.lower())

# Create an agent with tools from the strands-tools example tools package
# as well as our custom letter_counter tool
agent = Agent(tools=[calculator, current_time, python_repl, letter_counter])

# Ask the agent a question that uses the available tools
message = """
I have 4 requests:

1. What is the time right now?
2. Calculate 3111696 / 74088
3. Tell me how many letter R's are in the word "strawberry" 🍓
4. Output a script that does what we just spoke about!
  Use your python tools to confirm that the script works before outputting it
"""
agent(message)
```

Source: https://strandsagents.com/0.1.x/documentation/docs/user-guide/quickstart/

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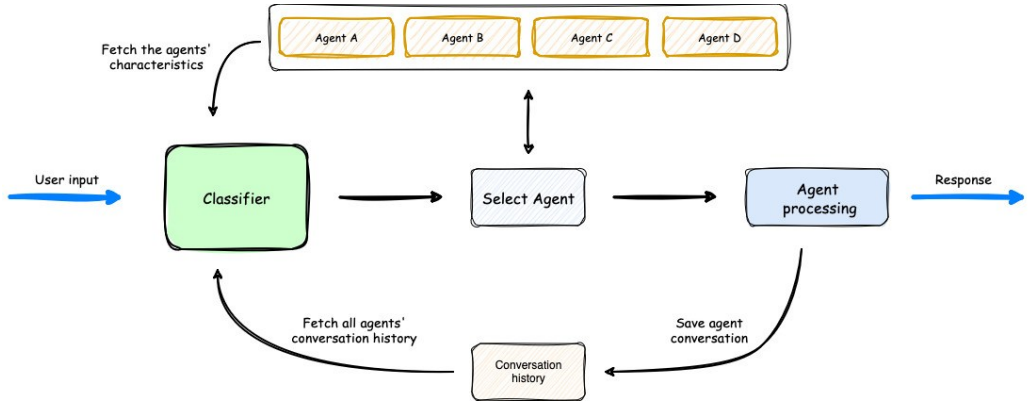
AWS Agent Squad

- Another open-sourced framework for multi-agent workflows
  - https://github.com/awslabs/agent-squad
- Intelligent intent classification
  - Routes queries to the right agent
- Python and TypeScript
- Shares contexts across agents
- Many pre-built agents and classifiers
- Integrates with Bedrock Agents
- Can extend Bedrock Flows
  - Conversation memory, orchestration
- Supervisor Agent
  - Coordinates multiple specialized agents (like Strands!)
- Agent Squad is focused on routing, Strands on tool use in a single agent loop



Source: awslabs/agent-squad repo

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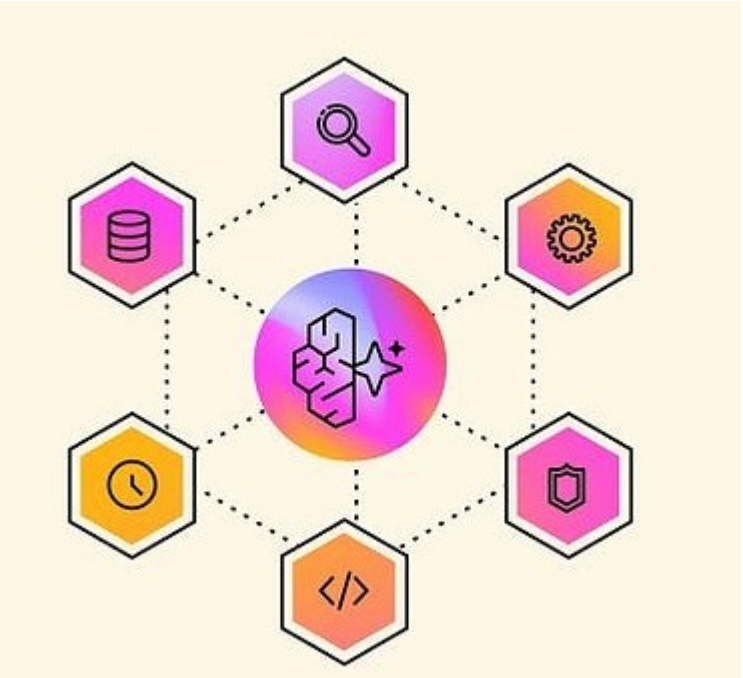
Amazon Bedrock AgentCore

- Hands deployment and operation of AI agents at scale
  - No fussing with Docker, ECR, ECS, etc
  - Serverless
- And it works with ANY agent framework
  - You aren't tied to Bedrock agents
  - You aren't even tied to AWS's Strands framework
  - OpenAI Agent SDK, LangGraph / LangChain, CrewAI, Google ADK, whatever
  - (although… Strands does get better support…)
- Includes a "agentcore CLI" to make deployment of agents to AWS super easy
- Includes several tools and capabilities you can include in your agents
  - These can be contained in a "harness" without writing code
- Integrates with Bedrock Knowledge Bases



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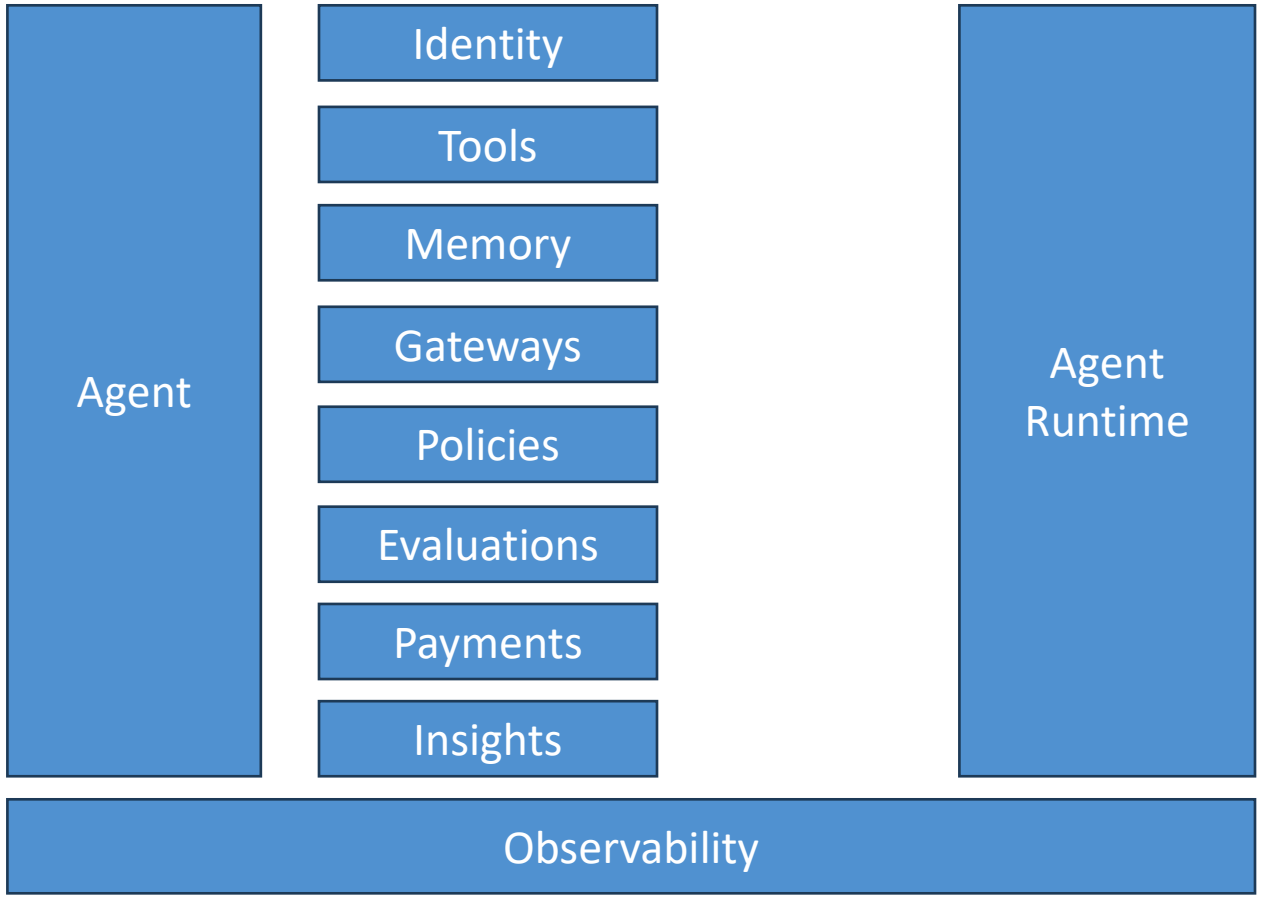
AgentCore Capabilities



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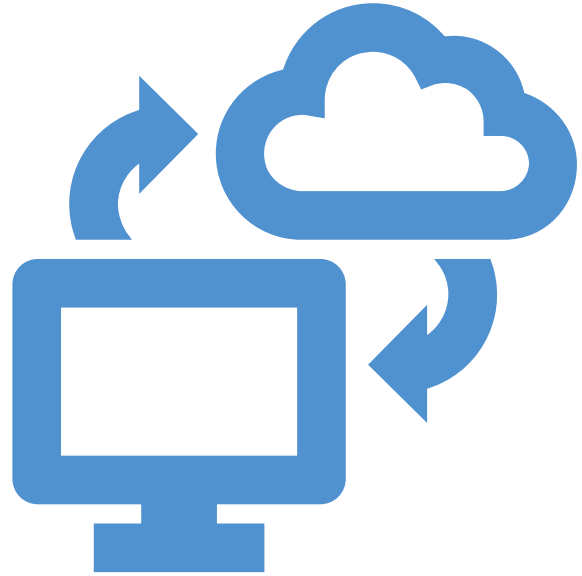
AgentCore Agent Runtime

- Serverless endpoints
- Deploy your agent to ECR, enhanced with AgentCore capabilities
  - Agentcore CLI manages it all for you, using CodeBuild
  - But you can build your own Docker containers if you want
- Can have multiple endpoints
- Observability dashboard lets you track usage and performance
  - Through “GenAI Observability” feature in CloudWatch



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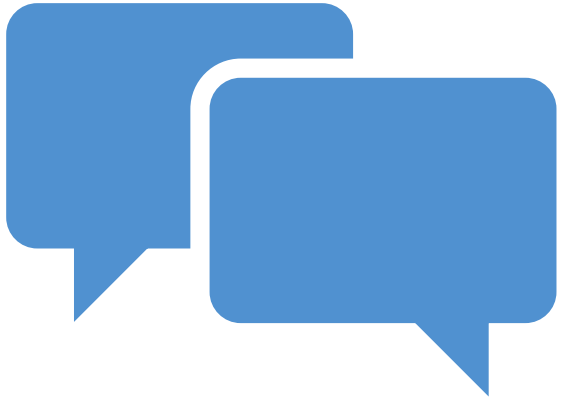
Adding AgentCore Memory

- Short-term
  - Chat history within a session / immediate context
  - Enables conversations
  - API centered around Session objects that contain Events
- Long-term
  - Stores “extracted insights”
  - Summaries of past sessions
  - Preferences (your coding style and favorite tools, for example)
  - Facts you gave it in the past
  - API involves “Memory Records” that store structured information derived from agent interactions
    - “Strategies” for user preferences, semantic facts, session summaries
- This all needs to be stored somewhere!
  - The OpenAI Agents SDK gives you a SQLLite implementation
  - But maybe you need something that scales better, and is serverless
  - Enter AgentCore Memory



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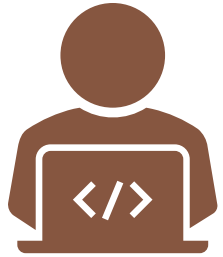
AgentCore Built-In Tools

Browser Tool

Allows control of your browser to interact with the web



Code Interpreter

Lets you run code (in an isolated container)

Python, JavaScript, TypeScript

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Importing Bedrock Agents

- Bedrock has its own system for endpoints…
  - But Bedrock Agents is deprecated for new customers.
  - AWS recommends migration to AgentCore
- Building agents in Bedrock is super easy
- Importing is just a matter of running `agentcore create --type import`
  - Specify your agent-id, agent-alias-id, region
  - This generates Strands code (or LangChain / LangGraph) in an output directory
  - From there you can test or deploy it like any other AgentCore agent

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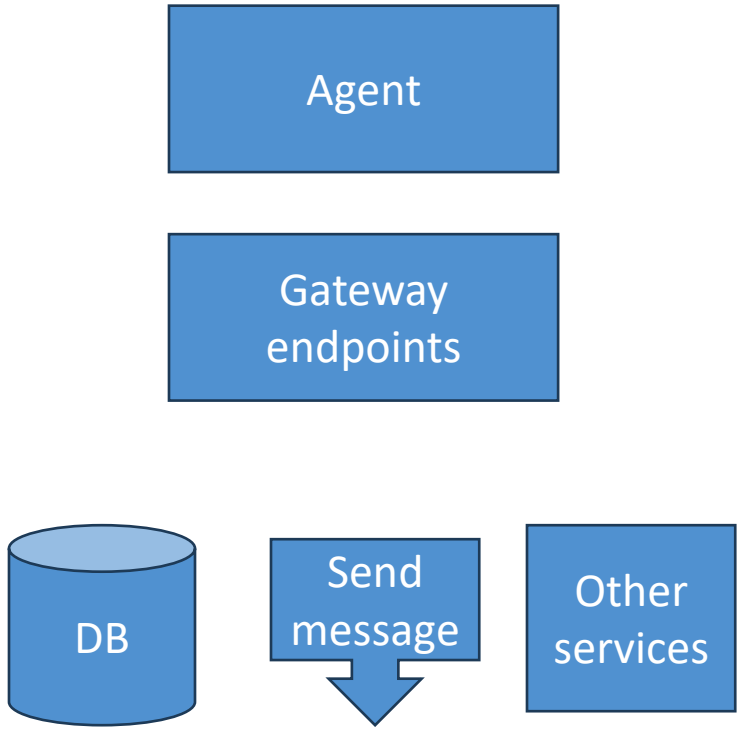
AgentCore Gateway

- Addresses the problem of using external tools at scale
- Convert APIs, Lambda functions, or other services into MCP tools
  - Targets can be OpenAPI (REST), Smithy models (that's an AWS thing), or Lambda
- Agents can then access them through Gateway endpoints
  - Manages OAuth security / credentials
- Semantic tool selection



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AgentCore Identity

- This is different from OAuth identity for users and connecting to services we talked about earlier
- This is about your agent's identity / identities
- Secure access to external tools and AWS services
- Central repository for all of your agent identities
  - Similar to a Cognito user pool
- Secure credential storage
- OAuth 2.0 support
  - Built-in support for Google, GitHub, Slack, Salesforce, Atlassian
- There is a lot of depth to this, but you probably don't need it right away
  - Refer to https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/identity.html

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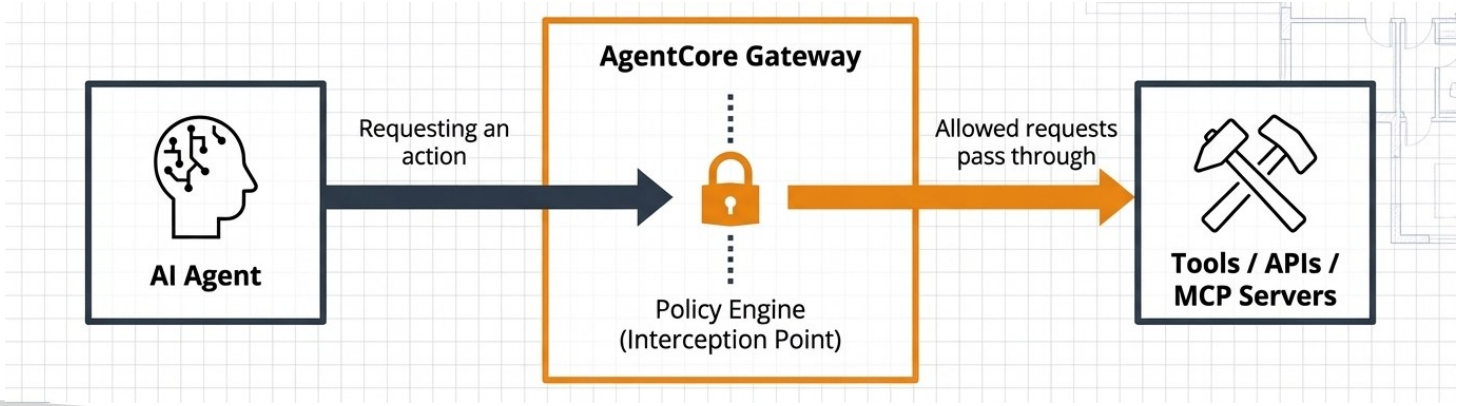
Policy in AgentCore

- Integrates with AgentCore Gateways
- Intercepts tool calls, evaluates against your defined policies
- Defined in the Cedar language
  - You can also construct policies using NLP or via a form
- Very fine-grained control – who can do what under what conditions
- Evaluated outside of the agent's code

```
permit (
  principal == Role::"refund-agent",
  action == Action::"process_refund",
  resource
) when { context.amount < 200 };
```



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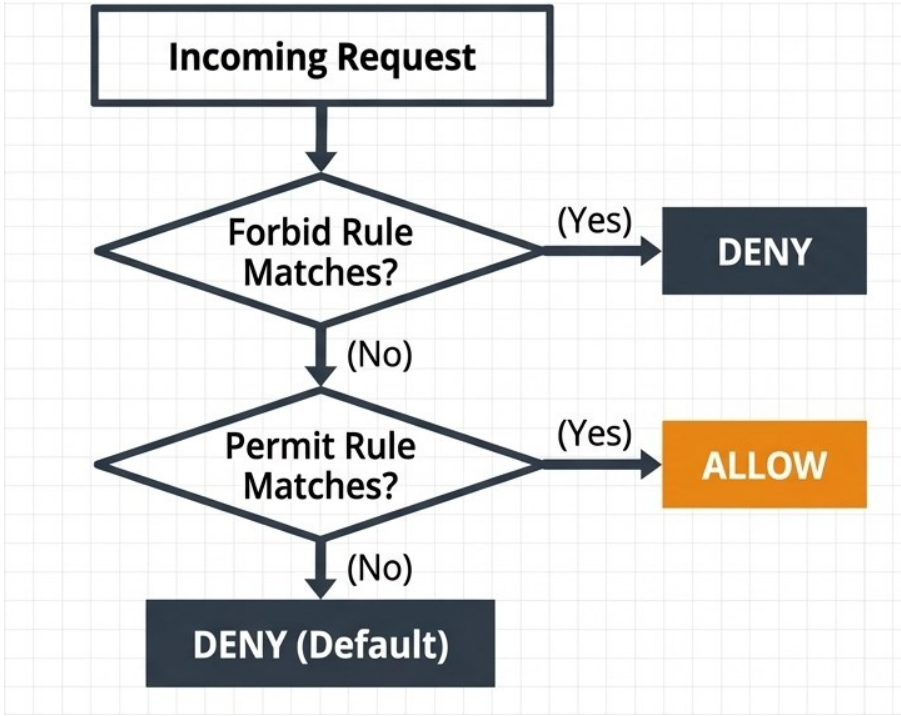
Policy Enforcement

- Deny by default
  - All actions denied unless explicitly permitted
- Contextual validation
  - Validate full context, OAuth sub-claims, specific tool arguments
- Dual modes
  - Enforce mode to actively block actions
  - Log-only mode to test and validate policies



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AgentCore Evaluations

- Assess how well your agent:
  - Performs tasks
  - Handles edge cases
  - Maintains consistency
- Integrates with:
  - Strands
  - LangGraph
  - OpenTelemetry
  - OpenInference
- Visualize results in CloudWatch
  - AgentCore Observability Insights
  - The usual CloudWatch alerts / alarms
- Cross-region inference
  - Automatically selects optimal region for processing
  - Data remains resident to where the request originated
  - But may be transmitted and processed – encrypted – to other regions.

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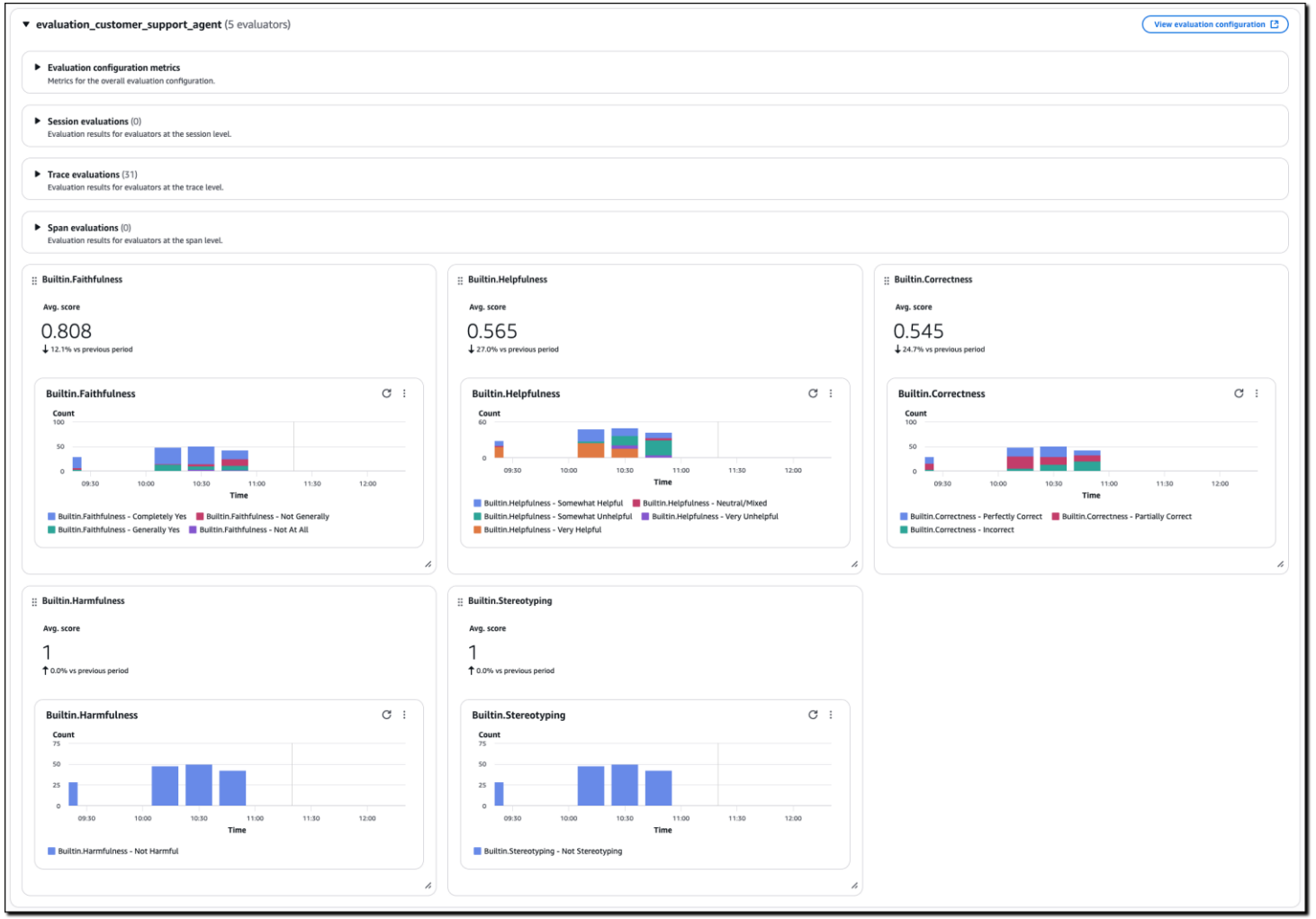

Sample CloudWatch Visualization

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Some Built-In Evaluations (prompt templates)

- Correctness (is it factually correct)
- Helpfulness (from the user's standpoint)
- Conciseness
- Instruction following
- Faithfulness (is response supported by context/sources)
- Response relevance (to user query)
- Coherence (logical structure)
- Refusal (is it evasive)
- Goal success rate
- Tool selection and parameter accuracy
- Harmfulness, Stereotyping
- Evaluation may be limited by filters and sampling rate you specify

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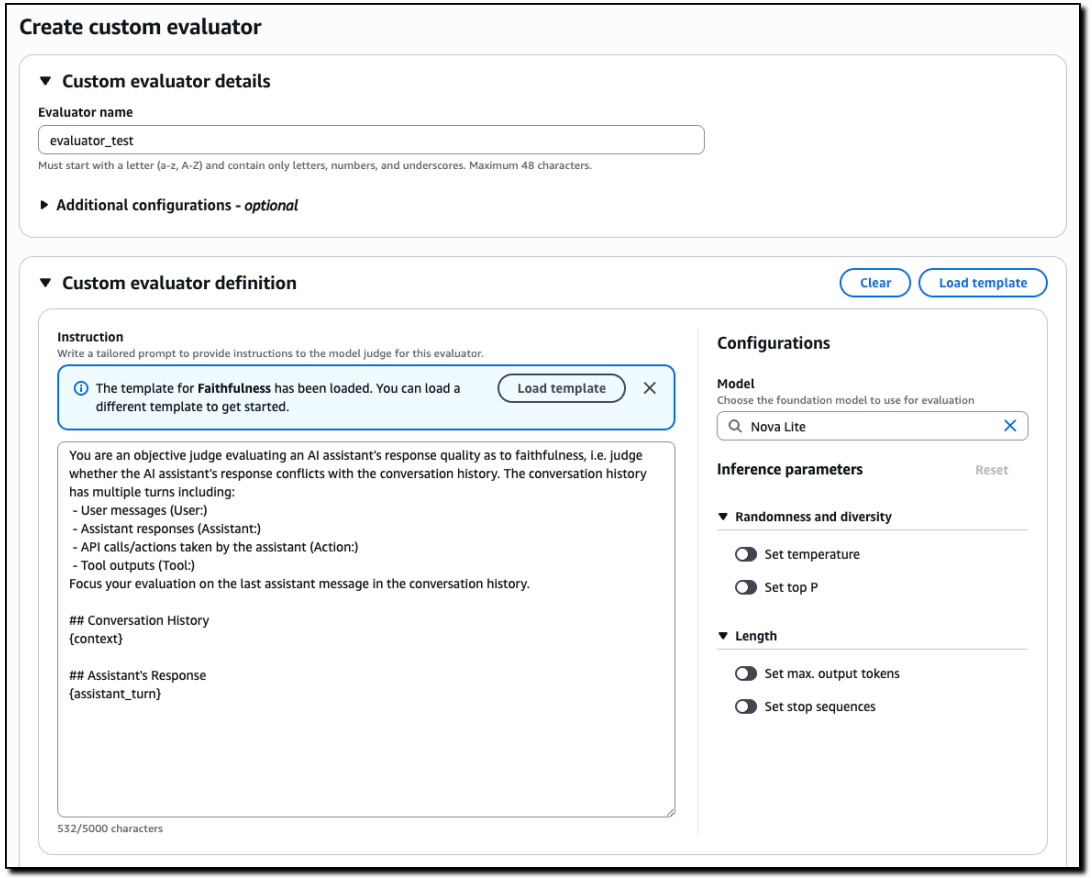
Custom Evaluators

- Specify a model to use as a judge
  - And configure it
- And define it with a prompt
  - Maybe start with a built-in one
- Define scales
  - Maps ranges of results to labels
- Define evaluation level
  - Session, trace, tool call



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Lab: Strands Agents, AgentCore, Agent Squad

- If you want to follow along
  - You'll need a Python environment
  - AWS CLI installed and configured
    - Preferably with administrator permissions
  - A Bedrock API Key
    - Set in AWS_BEARER_TOKEN_BEDROCK env variable
  - Agentcore CLI dependencies:
    - Node.js 20+
    - uv
    - git
- Warning: Agent Squad is a bit fragile ATM
  - And this is all changing quickly. You might want to just watch; the concepts are what's important.
- Remember to clean up when you're done
  - agentcore remove all / agentcore deploy
  - Double check the Bedrock Agentcore console, EKS, ECR, Codebuild

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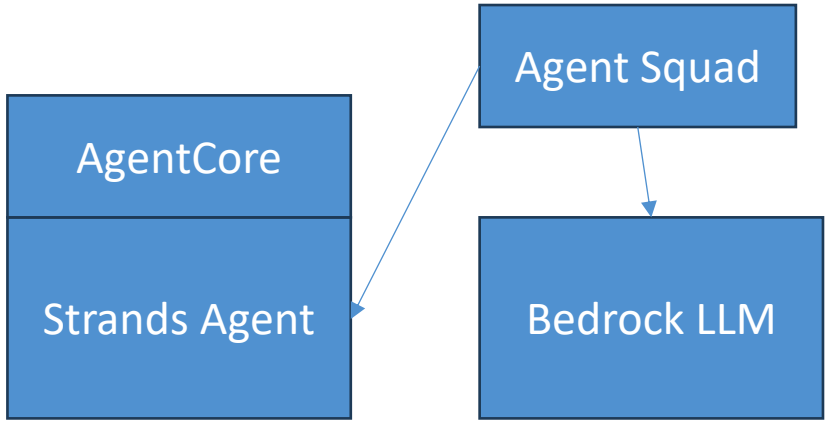
Lab: Strands Agents, AgentCore, Agent Squad

- What we'll build:
  - Strands agent with tools
    - Calculator, clock, and custom "count keywords" tool
  - Strands agent deployed through AgentCore
  - Strands agent orchestrated with Agent Squad
    - Routes appropriate queries to Strands, everything else to a generic LLM



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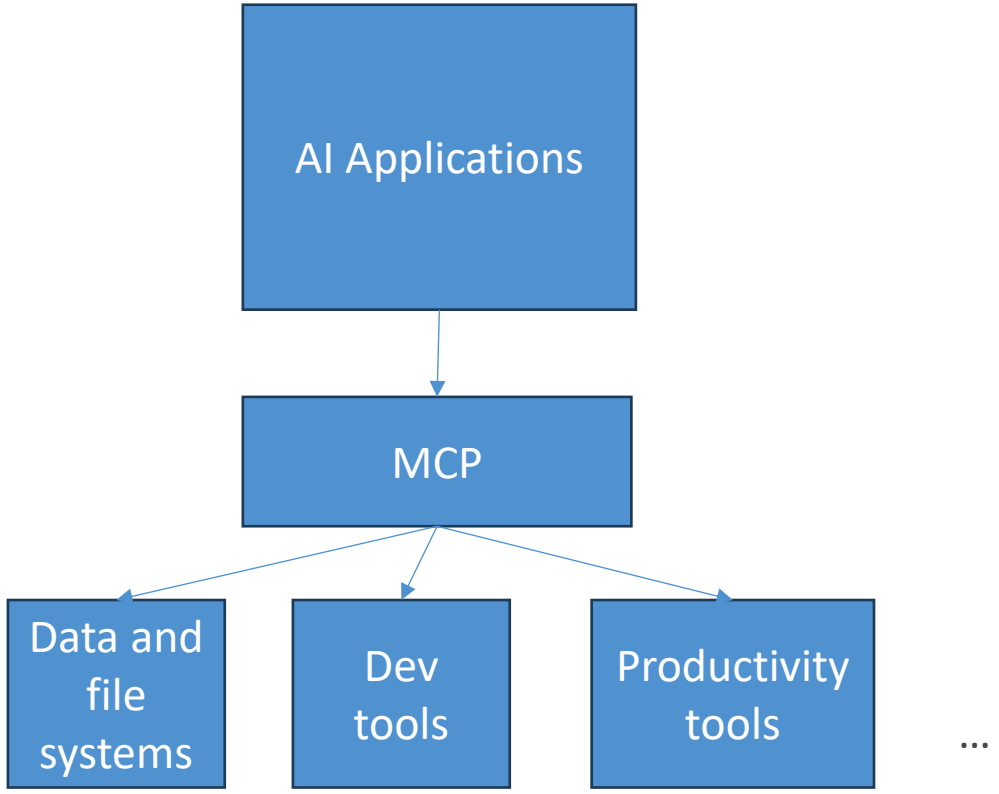
Model Context Protocol (MCP)

- Standardized interface for agent-tool interactions
  - “Like a USB-C port for AI Applications” – Anthropic
  - Technically the data layer is JSON-RPC 2.0
  - Transport is stdio or HTTP streaming
- Access external tools with consistent interaction patterns
  - Standardizes how apps provide context to LLMs
- Pretty much any agent SDK can call external MCP servers (including Strands)
- Servers may expose *tools*, *resources*, and *prompts*
- Some popular MCP servers:
  - GitHub, Atlassian, PostgreSQL, Slack, Google Maps… even Udemy!



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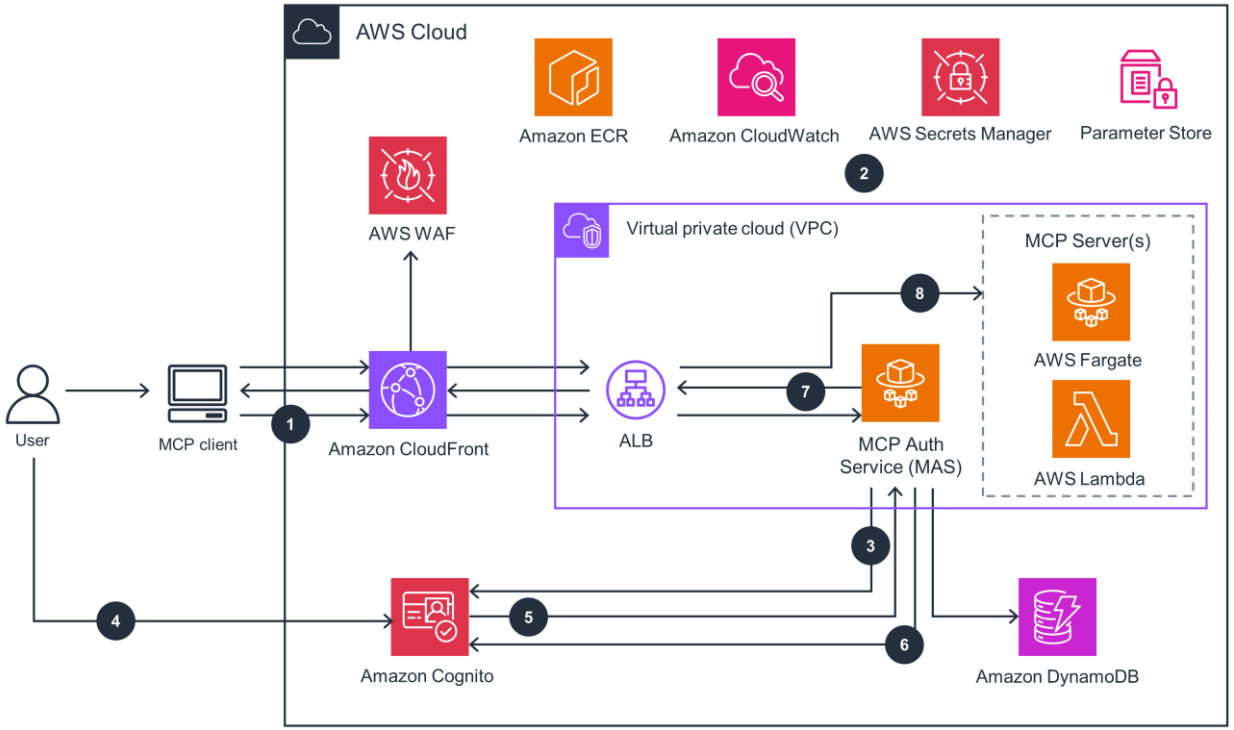
Deploying your Own MCP Server

- For stateless, lightweight access:
  - AWS Lambda
- For complex tools:
  - Amazon ECS
    - Or Fargate
- API Gateway
  - Can also expose MCP endpoints
- Use MCP client libraries
  - Abstract away the complexities of the protocol, don't re-invent the wheel!
- AWS's official MCP architecture guidance is on the right here
  - But it's different from what the exam guide and SkillBuilder talk about :/
- You can also deploy MCP servers with AgentCore



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OpenAPI and GenAI

- Originally Swagger – a spec for defining interfaces for web services
- Can also be used to define interfaces between FMs and tools
  - Helps with reliable tool operations
- Standardize function definitions
  - Define input parameters, expected outputs, error conditions per tool
- May be used **with Bedrock action groups**
  - Upload OpenAPI schema to S3
    - Or edit it in the console
  - The action group can then refer to it to guide its tool use

```
openapi: 3.1.0
info:
  title: Calculator Tool
  version: 1.0.0

paths:
  /add:
    get:
      operationId: add
      summary: Add two numbers
      parameters:
        - name: a
          in: query
          required: true
          schema: { type: number }
        - name: b
          in: query
          required: true
          schema: { type: number }
      responses:
        "200":
          description: Sum
          content:
            application/json:
              schema:
                type: object
                properties:
                  result:
                    type: number
```


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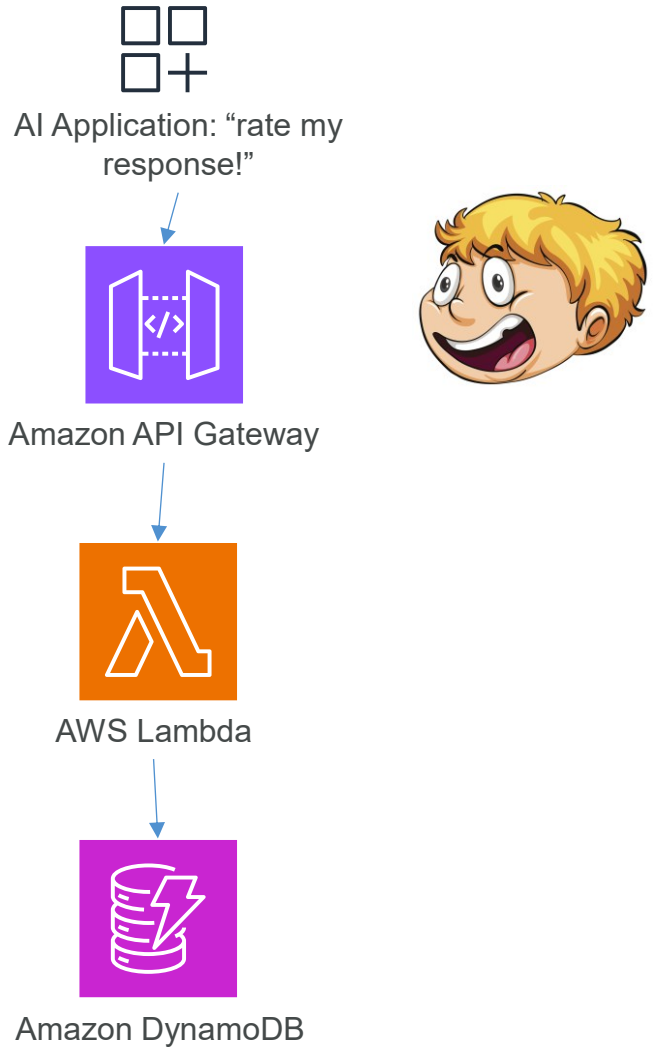
Humans in the Loop

- Can be low-tech
  - “Human augmentation pattern” – AI prepares drafts, humans refine
- AI systems can implement escalation criteria
  - Route complex or uncertain cases to human experts
  - Based on confidence scores or risk assessments
- Human feedback enables continuous improvement
  - Front feedback collection with API Gateway to just ask humans if they like the output
  - Store that feedback in DynamoDB
    - With indexing to allow later analysis
    - Use to measure which model / variant is preferred



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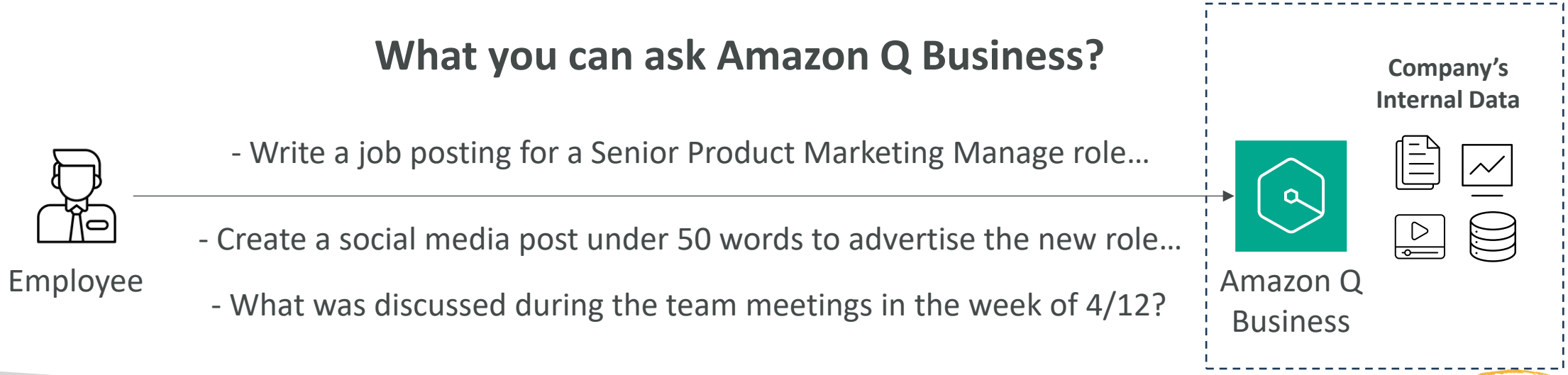
Amazon Q Business

- Fully managed Gen-AI assistant for your employees
- Based on your company's knowledge and data
  - Answer questions, provide summaries, generate content, automate tasks
  - Perform routine actions (e.g., submit time-off requests, send meeting invites)
- Built on Amazon Bedrock (but you can't choose the underlying FM)



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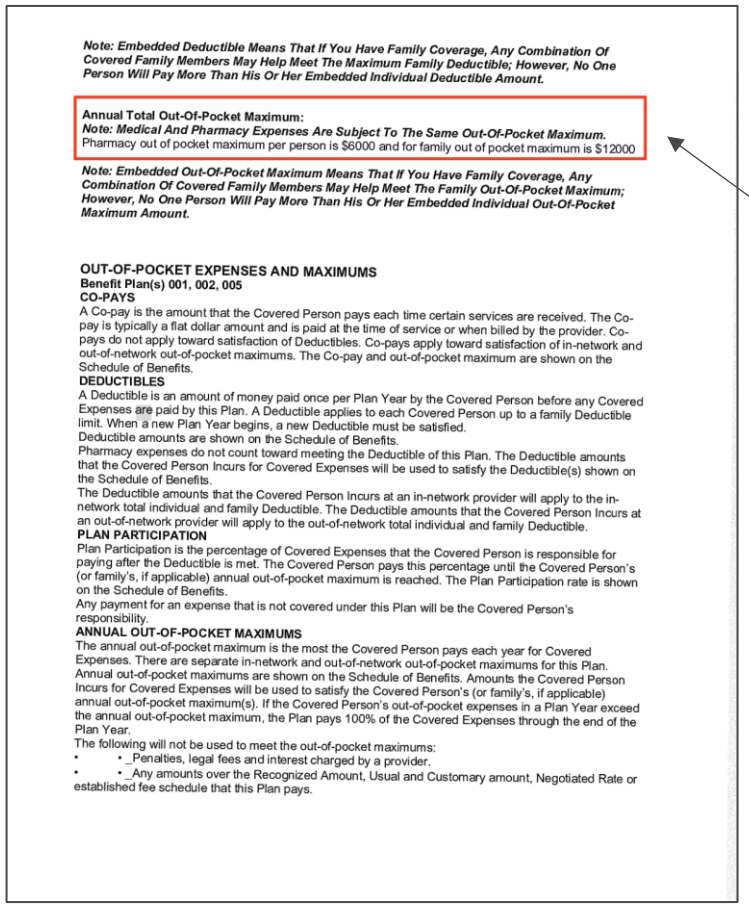
Amazon Q Business Example



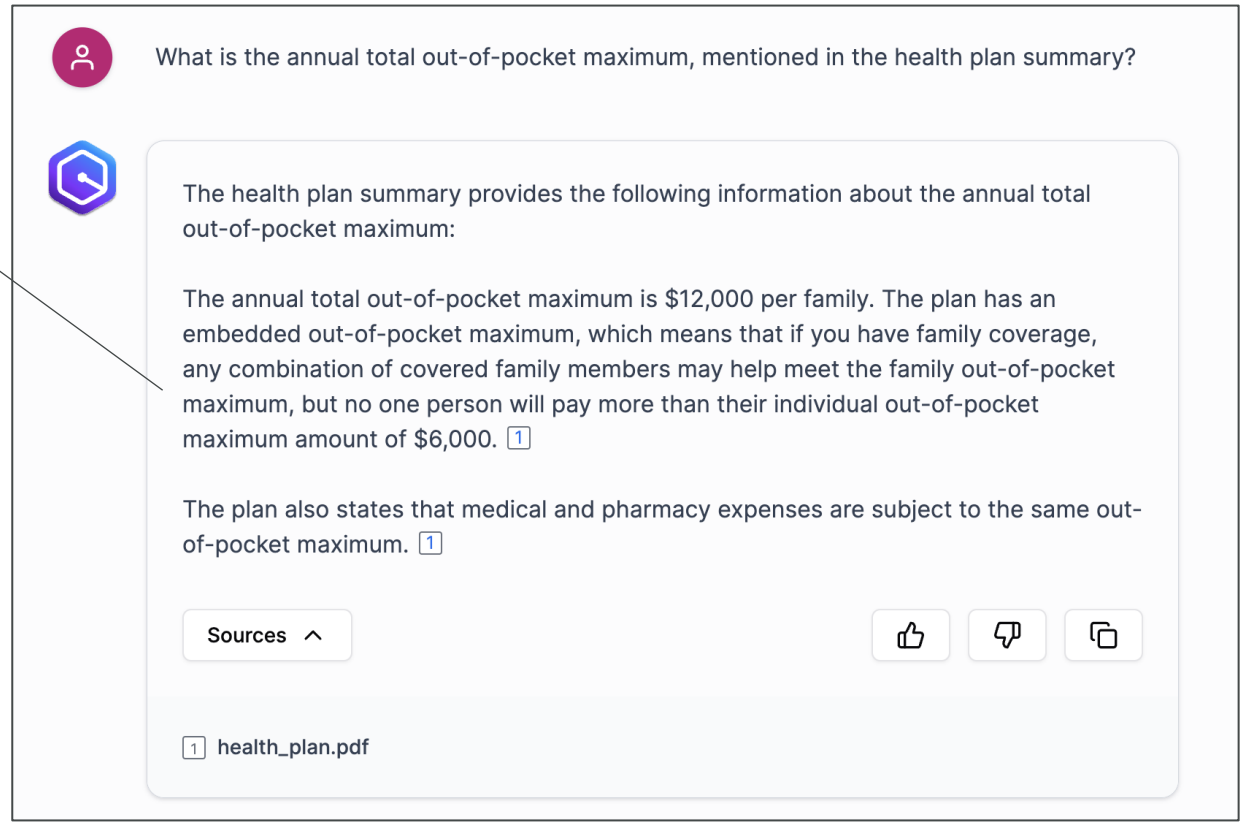
Note: Embedded Deductible Means That If You Have Family Coverage, Any Combination Of Covered Family Members May Help Meet The Maximum Family Deductible; However, No One Person Will Pay More Than His Or Her Embedded Individual Deductible Amount.

Annual Total Out-Of-Pocket Maximum:
Note: Medical And Pharmacy Expenses Are Subject To The Same Out-Of-Pocket Maximum.
Pharmacy out of pocket maximum per person is $6000 and for family out of pocket maximum is $12000

Note: Embedded Out-Of-Pocket Maximum Means That If You Have Family Coverage, Any Combination Of Covered Family Members May Help Meet The Family Out-Of-Pocket Maximum; However, No One Person Will Pay More Than His Or Her Embedded Individual Out-Of-Pocket Maximum Amount.

OUT-OF-POCKET EXPENSES AND MAXIMUMS
Benefit Plan(s) 001, 002, 005
CO-PAYS
A Co-pay is the amount that the Covered Person pays each time certain services are received. The Co-pay is typically a flat dollar amount and is paid at the time of service or when billed by the provider. Co-pays do not apply toward satisfaction of Deductibles. Co-pays apply toward satisfaction of in-network and out-of-network out-of-pocket maximums. The Co-pay and out-of-pocket maximum are shown on the Schedule of Benefits.
DEDUCTIBLES
A Deductible is an amount of money paid once per Plan Year by the Covered Person before any Covered Expenses are paid by this Plan. A Deductible applies to each Covered Person up to a family Deductible limit. When a new Plan Year begins, a new Deductible must be satisfied.
Deductible amounts are shown on the Schedule of Benefits.
Pharmacy expenses do not count toward meeting the Deductible of this Plan. The Deductible amounts that the Covered Person Incurs for Covered Expenses will be used to satisfy the Deductible(s) shown on the Schedule of Benefits.
The Deductible amounts that the Covered Person Incurs at an in-network provider will apply to the in-network total individual and family Deductible. The Deductible amounts that the Covered Person Incurs at an out-of-network provider will apply to the out-of-network total individual and family Deductible.
PLAN PARTICIPATION
Plan Participation is the percentage of Covered Expenses that the Covered Person is responsible for paying after the Deductible is met. The Covered Person pays this percentage until the Covered Person's (or family's, if applicable) annual out-of-pocket maximum is reached. The Plan Participation rate is shown on the Schedule of Benefits.
Any payment for an expense that is not covered under this Plan will be the Covered Person's responsibility.
ANNUAL OUT-OF-POCKET MAXIMUMS
The annual out-of-pocket maximum is the most the Covered Person pays each year for Covered Expenses. There are separate in-network and out-of-network out-of-pocket maximums for this Plan.
Annual out-of-pocket maximums are shown on the Schedule of Benefits. Amounts the Covered Person Incurs for Covered Expenses will be used to satisfy the Covered Person's (or family's, if applicable) annual out-of-pocket maximum(s). If the Covered Person's out-of-pocket expenses in a Plan Year exceed the annual out-of-pocket maximum, the Plan pays 100% of the Covered Expenses through the end of the Plan Year.
The following will not be used to meet the out-of-pocket maximums:
- _Penalties, legal fees and interest charged by a provider.
- _Any amounts over the Recognized Amount, Usual and Customary amount, Negotiated Rate or established fee schedule that this Plan pays.

Company Document



https://aws.amazon.com/blogs/machine-learning/improve-productivity-when-processing-scanned-pdfs-using-amazon-q-business/

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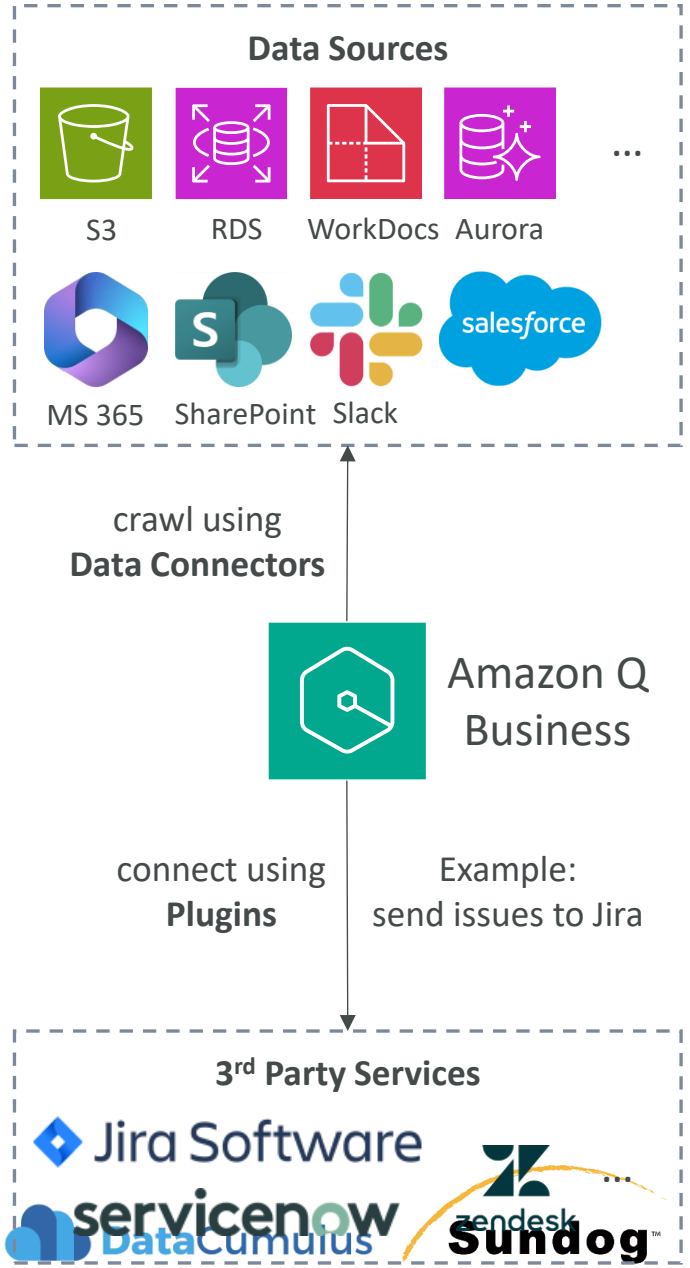
Amazon Q Business

- **Data Connectors (fully managed RAG)** – connects to 40+ popular enterprise data sources
  - Amazon S3, RDS, Aurora, WorkDocs…
  - Microsoft 365, Salesforce, GDrive, Gmail, Slack, Sharepoint…
- **Plugins** – allows you to interact with 3rd party services
  - Jira, ServiceNow, Zendesk, Salesforce…
  - **Custom Plugins** – connects to any 3rd party application using APIs



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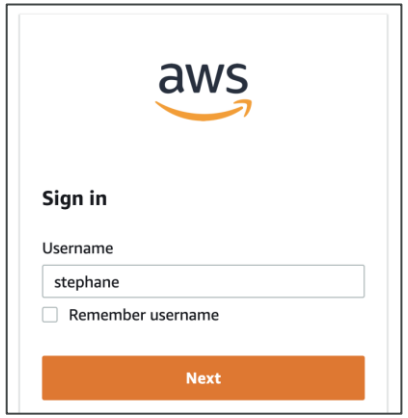
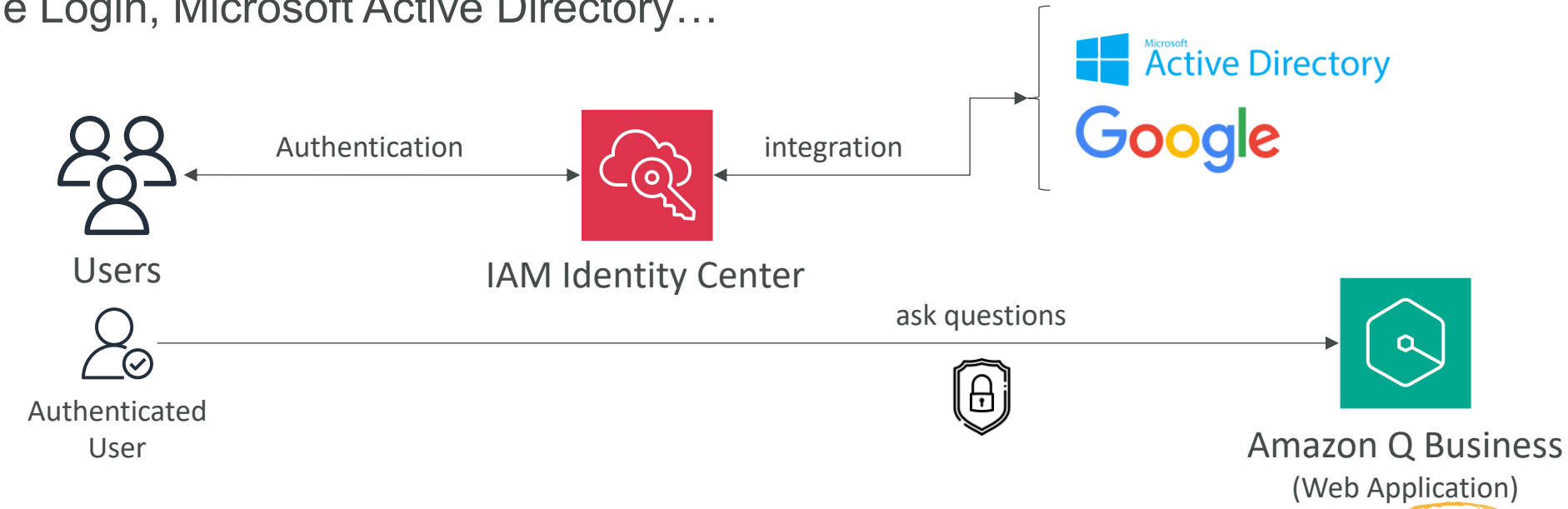
Amazon Q Business + IAM Identity Center

- Users can be authenticated through **IAM Identity Center**
- Users receive responses generated only from the documents they have access to
- IAM Identity Center can be configured with external Identity Providers
  - IdP: Google Login, Microsoft Active Directory…





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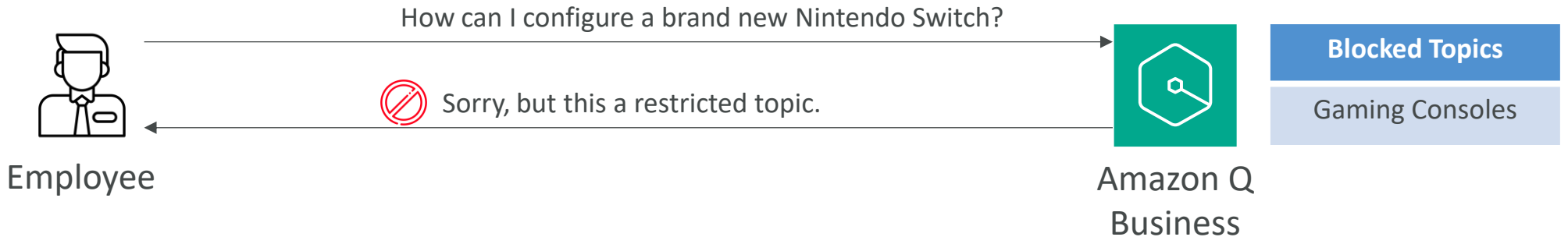
Amazon Q Business – Admin Controls

- Controls and customize responses to your organizational needs
- Admin controls == Guardrails
- Block specific words or topics
- Respond only with internal information (vs using external knowledge)
- Global controls & topic-level controls (more granular rules)



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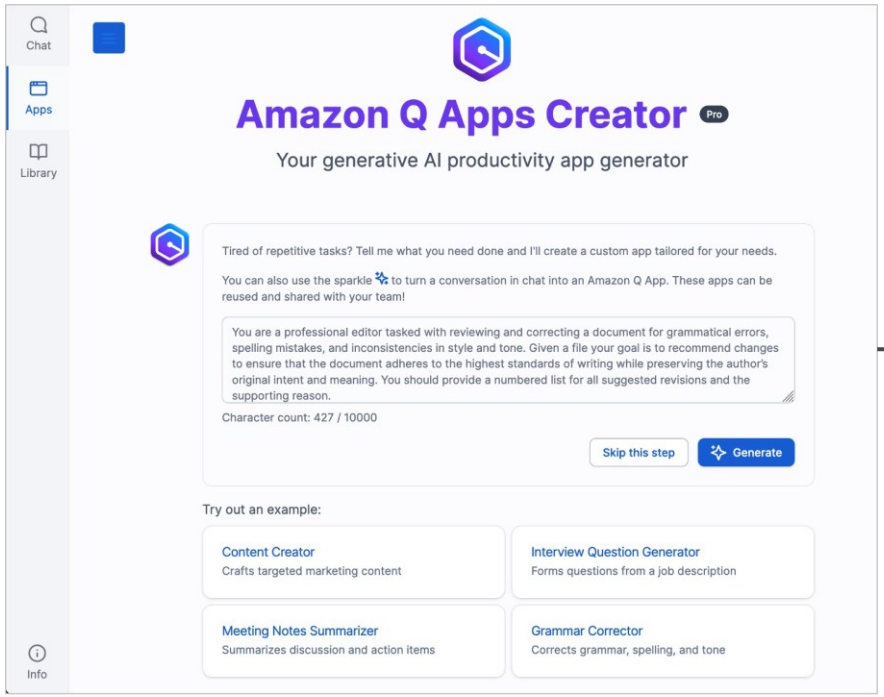
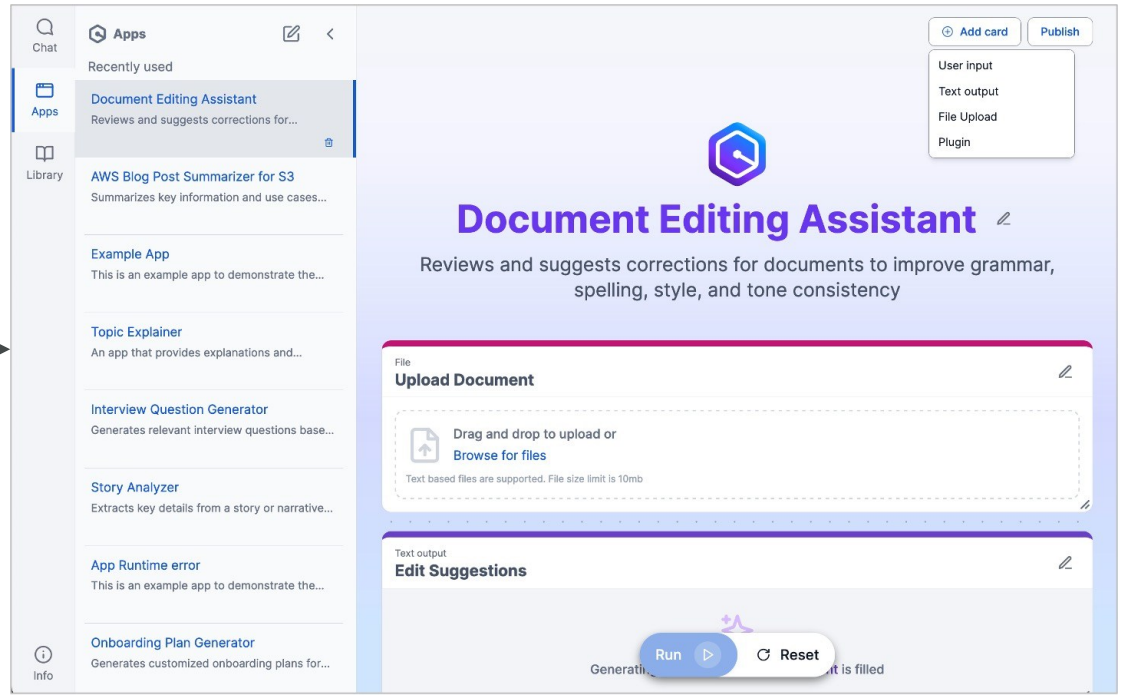
Amazon Q Apps (Q Business)

- Create Gen AI-powered apps without coding by using natural language
- Leverages your company's internal data
- Possibility to leverage plugins (Jira, etc…)





https://aws.amazon.com/blogs/aws/amazon-q-business-now-generally-available-helps-boost-workforce-productivity-with-generative-ai/

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Amazon Quick

Agentic AI for business users

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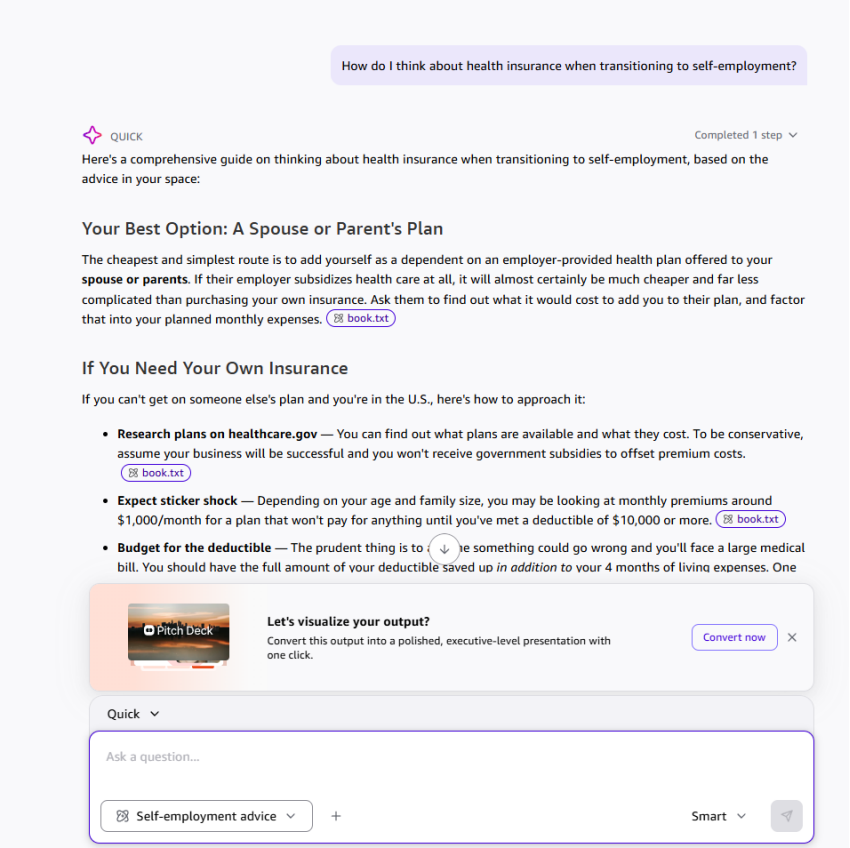
Amazon Quick – What It Is

- AI assistant for work: chat, research, BI, automation and no-code apps in one place
- Evolved from QuickSight, which continues as Quick Sight inside Quick; existing APIs and integrations keep working
- Fully managed – no infrastructure, no models to host, no ML expertise
- Grounded in your systems through knowledge bases, connectors and structured data
- Open standards – MCP and OpenAPI connect anything
- Use Quick from the web, desktop app, or embedded in Chrome, Slack, Teams and Microsoft 365



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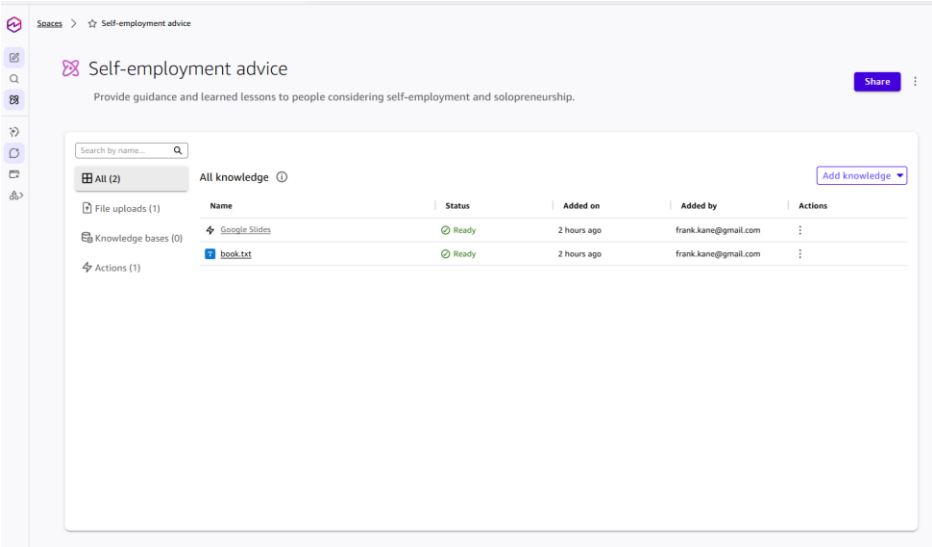
The Problems Amazon Quick Solves

- Knowledge scattered across S3, SharePoint, Confluence, Drive, Slack, Jira and CRMs
- Generic chatbots that can't see enterprise data – confident answers, no citations
- BI locked behind analysts; business users wait days for a number
- Research and synthesis done by hand across dozens of tabs
- Insight without action – the tool gives you the answer, but a person still has to open Jira or Salesforce and make the change
- Shadow AI: staff pasting confidential data into unapproved tools



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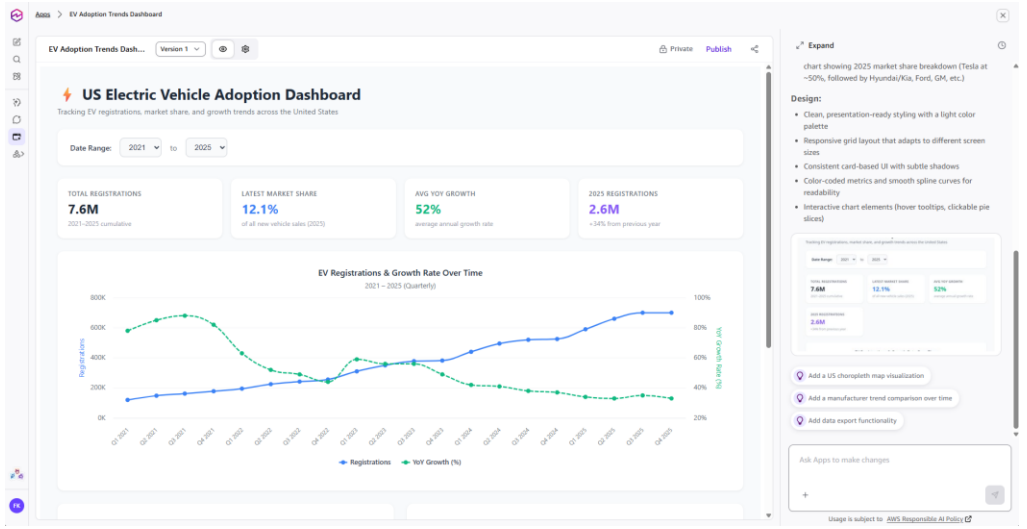
Why Quick vs. ChatGPT, Claude or Gemini on AWS

- Identity you already run – IAM, IAM Identity Center, SAML 2.0, MFA, RBAC and SSO
- Permission-aware answers – agents return only what the user may already see
- Private data paths – VPC, Direct Connect and Site-to-Site VPN to on-premises sources
- Native analytics over Redshift, Athena, S3 and SPICE, not a file upload
- Your data stays in your control and is not used to train someone else's models
- One AWS agreement, bill, Region choice and compliance posture



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How Quick Works – The Building Blocks

- Every request flows through the same five layers – chat interprets it, an agent plans it, a space supplies the context, integrations reach your systems, and a feature does the work.

Chat	Agents	Spaces	Integrations	Features
the front door for every request	instructions, knowledge, tools	files, dashboards, datasets, actions	knowledge bases, connectors, extensions	Sight, Flows, Automate, Research, Apps

Delivered through the web app, desktop app, Chrome, Slack, Microsoft Teams and Microsoft 365

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Chat – The Primary Interface

- One natural-language front door: answer from data, generate content, run a flow, start research or take an action in a connected app.

Scope the answer

All data and apps
spaces, dashboards, datasets, topics, knowledge bases, actions and model knowledge

General knowledge
the underlying model only, no enterprise resources

Specific data and apps
choose the exact resources to answer from

In the conversation

Agent picker	Chat with datasets	Web search	Upload files (20 per chat)
Invoke actions	Run flows	Source citations	Memory and personalisation

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Chat Agents

- An agent is instructions plus knowledge sources plus tools – preconfigured agents stay inside their assigned resources, unopinionated ones let chat decide the boundary.

Instructions
what the agent does and how it behaves

Knowledge sources
spaces, dashboards, topics, knowledge bases

Tools
action connectors it may invoke

Chat agent – answers are always scoped to what the invoking user may already see

System chat agent – “My assistant”
Available to everyone. A base planner with no data of its own; it resolves the user's resources at chat time.

Custom chat agents
Built for a domain with a persona and toolset, then shared. Admins control who may create them.

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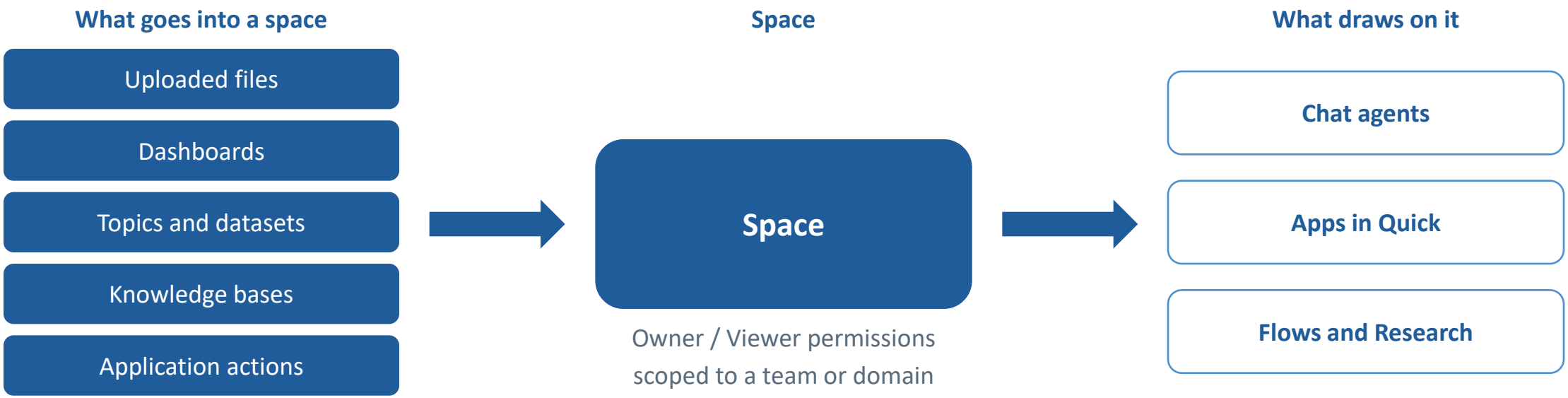
Spaces – Shared Context for Teams

- A space is a collection of data and Quick resources scoped to a team or domain, so agents answer only from that context
- Owners share, link and unlink resources; viewers ask questions and download files. Sharing a space does not by itself grant access to the underlying assets.



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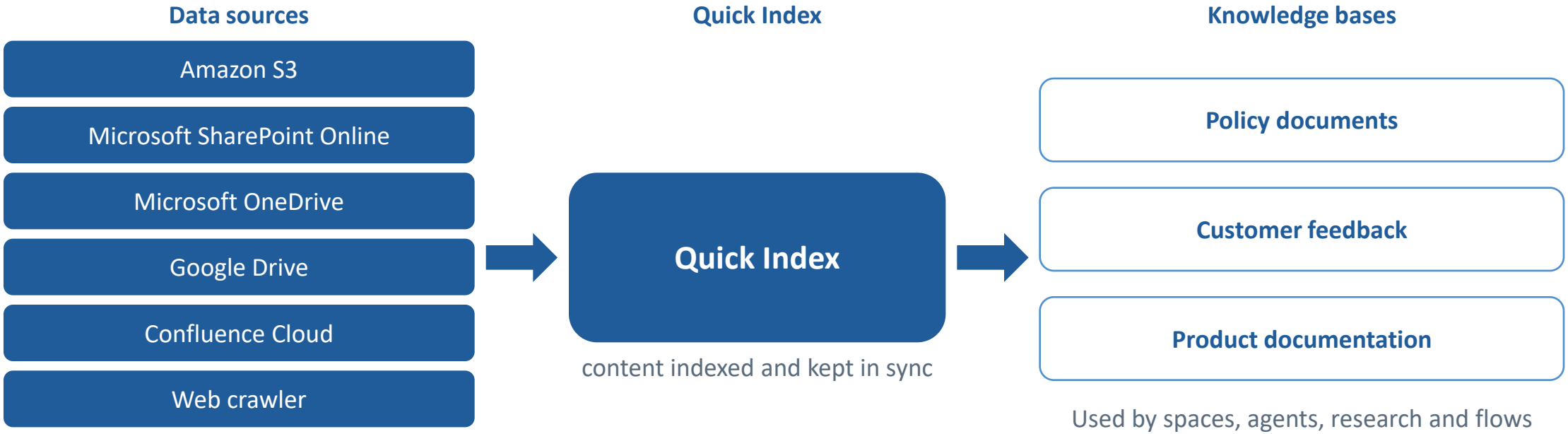
Knowledge and the Quick Index

- An indexed collection of content from your data sources, optimised for AI retrieval, semantic search and cited answers
- Many knowledge bases share one Quick Index; access control sits at the knowledge base level. Bring your own Q Business index or Bedrock knowledge base. Pooled storage: 25 GB Professional, 50 GB Enterprise.



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Connectors and Integrations

- Around 50 supported applications, in three integration types. Most support actions only – the knowledge base list is much shorter.

Action connectors	Data access integrations	Knowledge bases
Real-time calls into external services – read data, update records, trigger processes Slack, Teams, Jira, Salesforce, ServiceNow, GitHub, Notion, HubSpot, QuickBooks, Box, Asana	Connect to an external application to reach its data The foundation for knowledge bases – not used directly for analysis	Selected content, indexed for AI retrieval and question answering Amazon S3, SharePoint Online, OneDrive, Google Drive, Confluence Cloud, Web crawler

Build your own with MCP servers, OpenAPI specifications or REST APIs • Extensions embed Quick in Chrome, Slack, Teams and Microsoft 365

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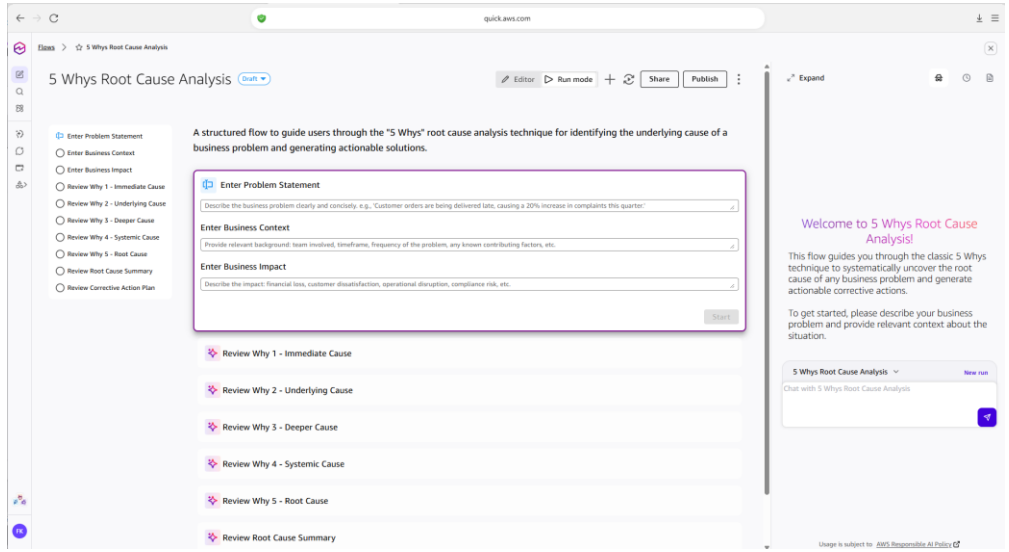
Quick Flows – Automate Routine Tasks

- A flow is a reusable sequence of steps you define once and run whenever you need it
- Build one three ways: generate it from a chat conversation, describe it in natural language, or assemble it in the visual editor
- Run it guided step-by-step, or conversationally to refine outputs and ask follow-ups
- Publish to an admin-managed library to share across the organisation
- No technical skills required – aimed at any business user



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Flow Step Types – The Palette

- Steps come from five categories.

User input

Text • Files

Gathers what the flow needs from the person running it

AI responses

Chat agent • Research • Web search • General knowledge • UI agent • Create image

Six ways to have AI produce something

Data insights

Spaces and knowledge bases • Dashboards and topics

Pulls grounded answers from your own data

Actions

Application actions

Reads or writes in connected apps such as Jira or Salesforce

Flow logic

Reasoning groups

Applies run instructions across one or more steps

Mix and match

A flow uses as many or as few of these as it needs, in whatever order the task requires.

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Quick Flows vs. Quick Automate

- Two sibling automation capabilities, not tiers of one product – Automate is a separate builder, not Flows with the limits raised
- Flows lets any business user automate routine tasks; Automate lets power users orchestrate end-to-end business processes at enterprise scale

	Quick Flows	Quick Automate
Who it's for	Individual users – personal and team productivity	Centralised and enterprise-wide; runs independently of any one user
Scope and duration	Shorter tasks: generating reports, drafting content, responding to tickets	Long-running, high-volume: loan applications, invoice processing, vendor compliance
Customisation	A simple set of no-code building blocks	Control flow logic, case management, error handling, custom code, human-in-the-loop
Availability	All plans	Enterprise only – not available on Free or Plus

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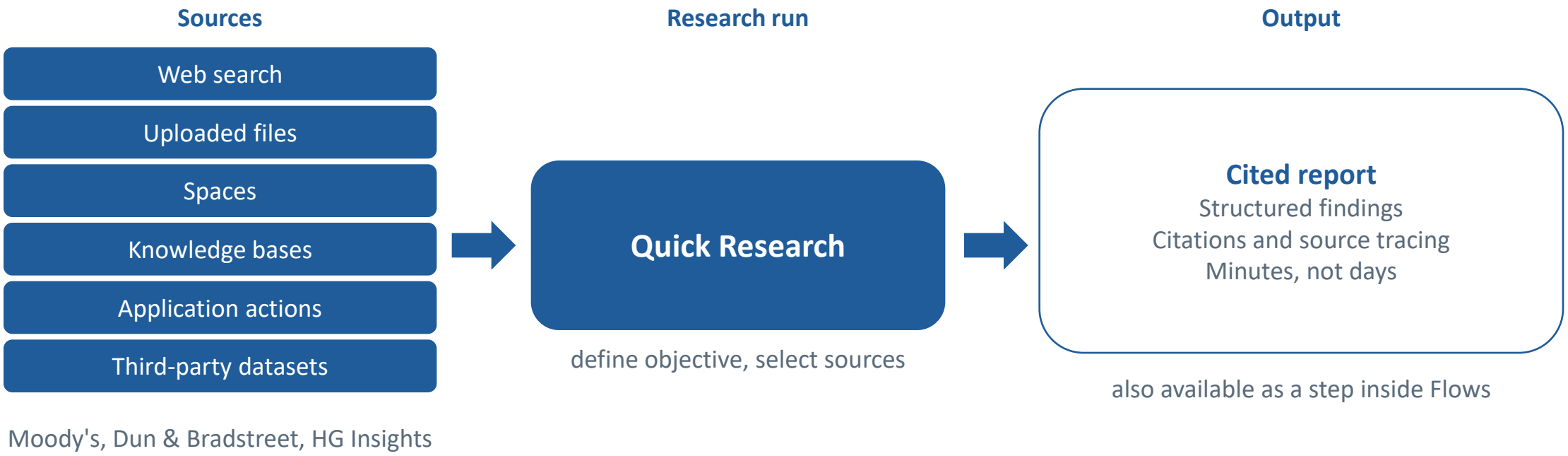
Research Mode

- Quick Research runs deep, multi-source investigations – market and competitor analysis, literature reviews, regulatory research, customer insights – and returns a structured, cited report.



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Apps in Quick – No-Code Web Applications

- Describe the app in natural language and the agent writes it live while you watch.

Build an app in five conversational steps

What you get

Conversational authoring	Live Quick Sight embeds	Action connectors	Built-in AI inference
Spaces integration	Persistent data storage	SSO and enterprise auth	One-click publishing

Intake forms, status trackers, calculators, document viewers – shared internally or published as a public site

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Security and Governance

- Quick runs on the identity, network and compliance controls your security team already approved.

Identity	IAM, IAM Identity Center, SAML 2.0 federation, Active Directory, MFA, RBAC and SSO
Network	Private VPC connections via elastic network interfaces, Direct Connect or Site-to-Site VPN, plus IP and VPC access restrictions
Encryption	TLS 1.2 minimum in transit; SPICE block-level encryption at rest; KMS customer-managed keys, configured before resources are created
Data controls	Row-level security, namespaces for multi-tenancy, domain allow-listing, mobile device settings, data sovereignty controls
Assurance	Guardrails and toxicity filters, CloudTrail audit logging, existing AWS compliance programmes; your data is not used to train models

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Pricing – Four Plans

- 30-day trial. Free and Plus need no AWS account.

	Free	Plus	Professional	Enterprise
Price / user / month	$0	$20 (annual)	$20	$40
Sign-up	Email or social	Email or social	AWS account	AWS account
Custom chat agents	–	Yes	Yes	Yes
Shared spaces and knowledge bases	–	Yes	Yes	Yes
Create dashboards and datasets	–	–	–	Yes
Quick Automate	–	–	–	Yes
Pooled index storage	~1 GB	10 GB	25 GB	50 GB

Free: single user, ~1 GB index storage, no published agent-hour entitlement and no way to buy more. The plan itself doesn't expire – the 30-day Plus trial does.
Professional and Enterprise: $250 / account / month infrastructure fee. Agent hours beyond 4 (Pro) or 8 (Enterprise) per month cost $3 per agent hour, $6 per research hour. Index storage overage $5 / GB / month.

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Limitations and Things to Watch

- Worth knowing before you commit to a rollout.

Region coverage
AWS GovCloud (US) supports only classic Quick Sight BI – no AI features

Space capacity
10,000 files under 10 GB; 500 MB per PDF, Word or PowerPoint file, 50 MB for other types

Sharing
Sharing a space does not grant access to its linked assets, and deleted assets fail silently

Connectors
Most integrations support actions only; the knowledge base source list is much shorter

Space structure
Spaces cannot nest; at most 20 resources of each type per space

Chat uploads
Regular chat conversations are limited to 20 uploaded files

Access control
Per knowledge base by default; document-level ACLs are available for S3, SharePoint, Confluence and Google Drive

Plan gating
Quick Automate, dashboard authoring and asset certification are Enterprise-only; agent hours and index overages bill separately

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Operational Efficiency and Optimization

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Token Efficiency

- Token Estimation and Tracking
  - Bedrock has a **CountTokens** API
    - Returns token count for a given request without actually running it
    - Different models behave differently
    - Costs nothing
  - Estimate costs prior to inference
  - Optimize prompts to fit within token limits
- CloudWatch tracks this too – also on the output side
  - InputTokenCount
  - outputTokenCount

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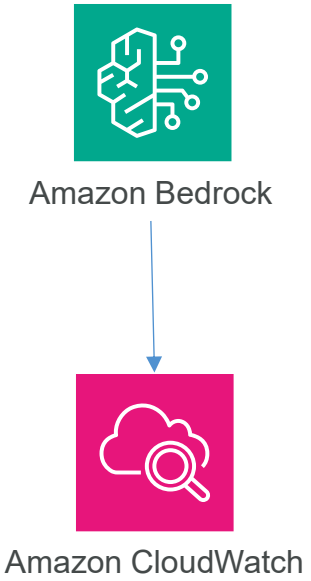
Other Stuff CloudWatch Monitors

- You can use these for additional optimization opportunities
- Count, time for requests
- TTFT Time to First Token (in streaming)
- Throttles / server errors / client errors on invocation
- Model latency, invocation count, throttles
- And again – input / output token counts



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More Token Efficiency Techniques

- Context Window Optimization / Context Pruning
  - Chunking RAG data, limiting # of chunks retrieved
  - Filtering chunks by metadata to toss irrelevant stuff
  - Summarize (or even toss) older parts of conversation history
- Response Size Controls / Response Limiting
  - Use maxTokens
  - Bake desired length in the prompt (“respond in 50 words or less”)
  - Use few-shot examples to illustrate the desired verbosity
  - Use JSON output to force a given format / length
- Prompt Compression
  - Use a small model to summarize large chats, documents before sending to larger model
  - Use Knowledge Bases instead of complete documents in the prompt

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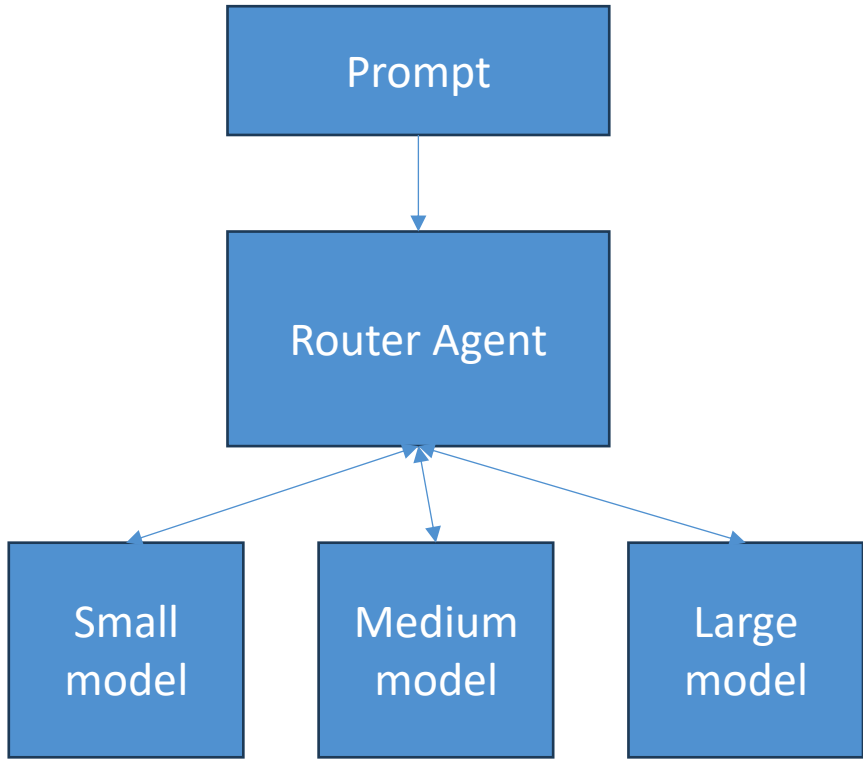
Cost-Effective Model Selection

- Cost / Capability Tradeoff
  - Do you really need the largest, most expensive model?
  - If the “smarts” are in RAG and tool usage, maybe a smaller model will do just fine
  - Smaller models may also play a role in pre-processing
    - Summarization, compression, classification, chunking – doesn’t need PhD-level smarts
- Tiered usage based on query complexity – Dynamic Routing
  - AKA **Intelligent Prompt Routing**
    - (This is now a feature in Bedrock as well)
  - Route to different models based on the complexity of the query
    - (This is how ChatGPT works under the hood…)
  - Many ways to route
    - A conditional Flow; Lambda; Agent Squad; Strands
    - This is a great application of agents – with a tool to score “complexity”



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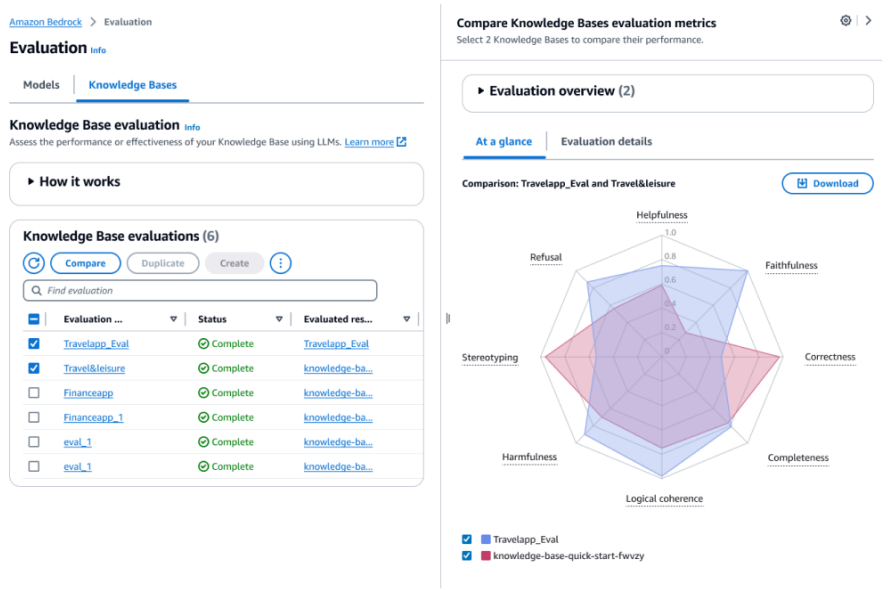
Measuring Price : Performance Ratio

- You can use tools like **Amazon Bedrock Evaluations** to measure Foundation Model performance against your prompt dataset
  - This can use human or LLM judges, or just predefined metrics
  - The **compare** feature in Evaluations lets you visualize the performance of different models with your prompts
- Pair that with token counting to estimate costs (or track actual costs)
- Now you can make informed tradeoffs…



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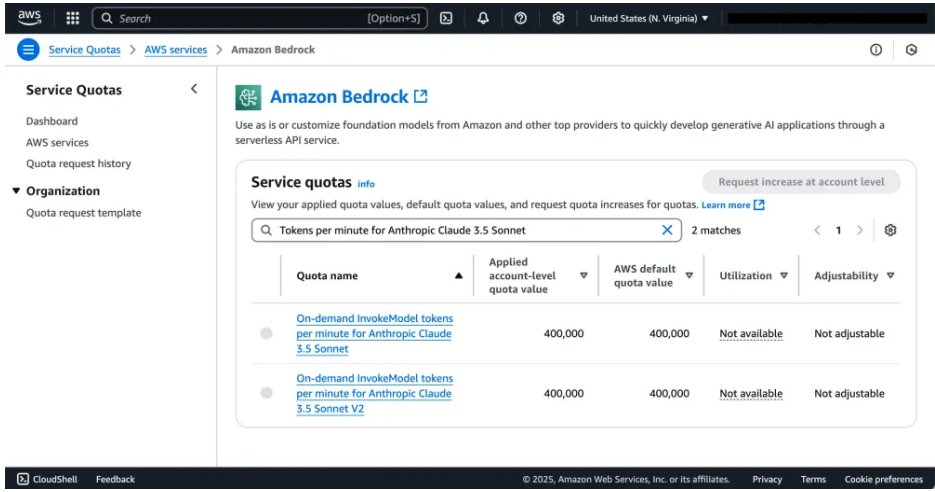
Maximizing Resource Utilization

- Batching strategies
  - Batch embeddings up for vector stores
  - Bedrock Batch Inference
    - Submit many prompts together in S3, get responses in S3
- Capacity planning with Bedrock
  - Work from desired tokens per minute (TPM), requests per minute (RPM)
  - **AWS Service Quotas** has a Bedrock tool (seen here)
  - Request a quota increase if necessary, or provisioned throughput
  - AWS **CloudFormation** templates can help with capacity planning
- Tensor parallelism
  - Shards LLM weights across GPUs
  - Better memory utilization per container



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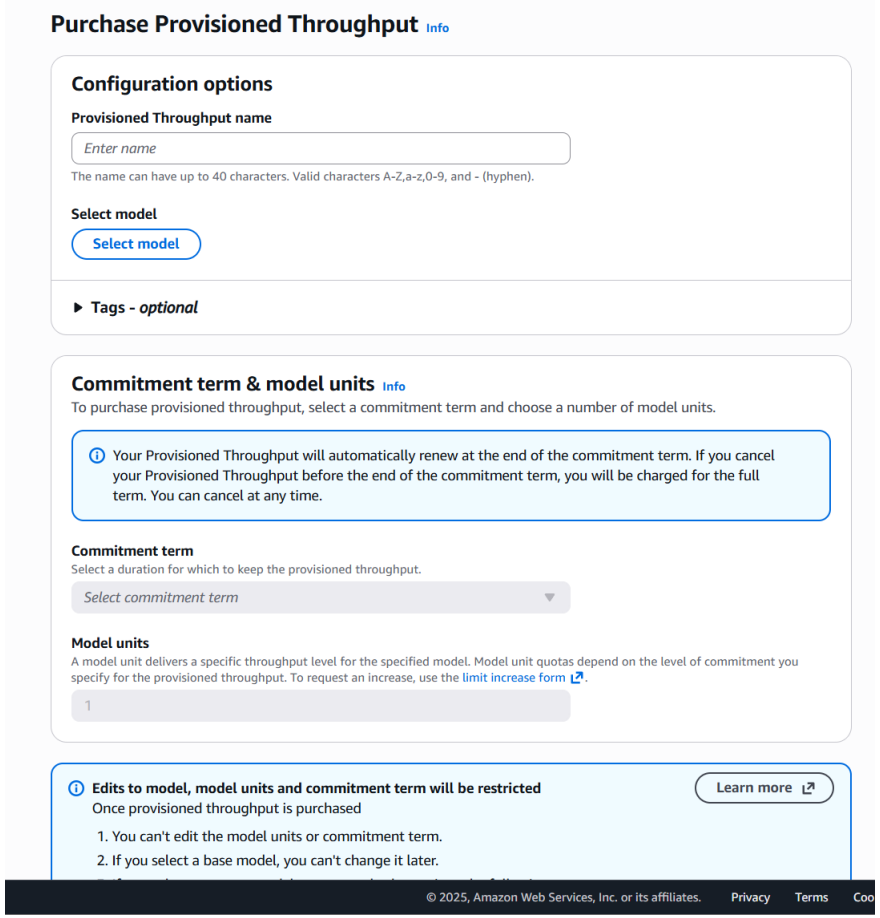
Maximizing Resource Utilization

- Provisioned throughput
  - Bedrock allows you to provision by **Tokens** or by **Model Units** (MU's) which corresponds to token throughput
  - You must provision for customized models
  - Useful for consistent performance with high volume
  - You provision for a given model, and must **provide the provisioned model ARN as the modelID** when running inference on it
- Utilization monitoring
  - CloudWatch is your friend, with dashboards
  - AWS Cost Explorer to attribute model costs to business functions
- Auto-scaling
  - Serverless solutions like Lambda, Bedrock, OpenSearch serverless, AgentCore



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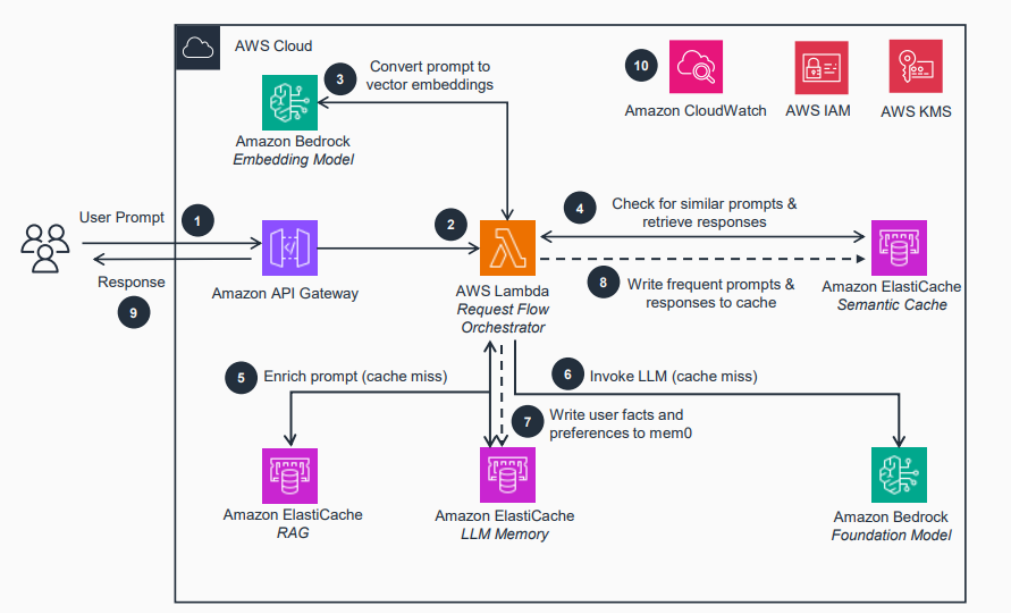
Intelligent Caching Systems: Semantic Caching

- Cache embeddings of prompts & responses instead of the prompt text
  - Store in an in-memory vector store (ElastiCache for Valkey, MemoryDB)
  - You *could* use OpenSearch too
  - For new prompts, create an embedding, find nearest neighbors in cache, if similarity score > threshold, return cached response
- Tune the similarity threshold carefully
  - Balance cache hits with relevant responses
- Make sure the overhead does not outweigh the benefits
- Can dramatically improve latency for some applications.



AWS reference architecture

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Intelligent Caching Systems: Prompt Caching

- Improves latency for supported models
  - Built into Bedrock
- You cache a *prompt prefix*
  - Static content, like instructions, and few-shot examples
  - Like a system prompt
  - Place dynamic content at the end
  - A *cache checkpoint* separates the two
- No need to tokenize the prefix again
- You might also cache an uploaded document that is queried repeatedly
- Cached content is discounted per token
  - But writes may be more expensive
- Cache reads & writes are monitored in CloudWatch
- Prompt caching is an option in Bedrock Prompt Management as well

```
instructions = ( "I will provide you with a
document, followed by a question about its
content. " "Your task is to analyze the
document, extract relevant information, and
provide " "a comprehensive answer to the
question. Please follow these detailed
instructions:" "\n\n1. Identifying Relevant
Quotes:" "\n - Carefully read through the
entire document." "\n - Identify sections of
the text that are directly relevant to
answering the question." "\n - Select quotes
that provide key information, context, or
support for the answer." "\n - Quotes should
be concise and to the point, typically no more
than 2-3 sentences each." "\n - Choose a
diverse range of quotes if multiple aspects of
the question need to be addressed." "\n - …
```


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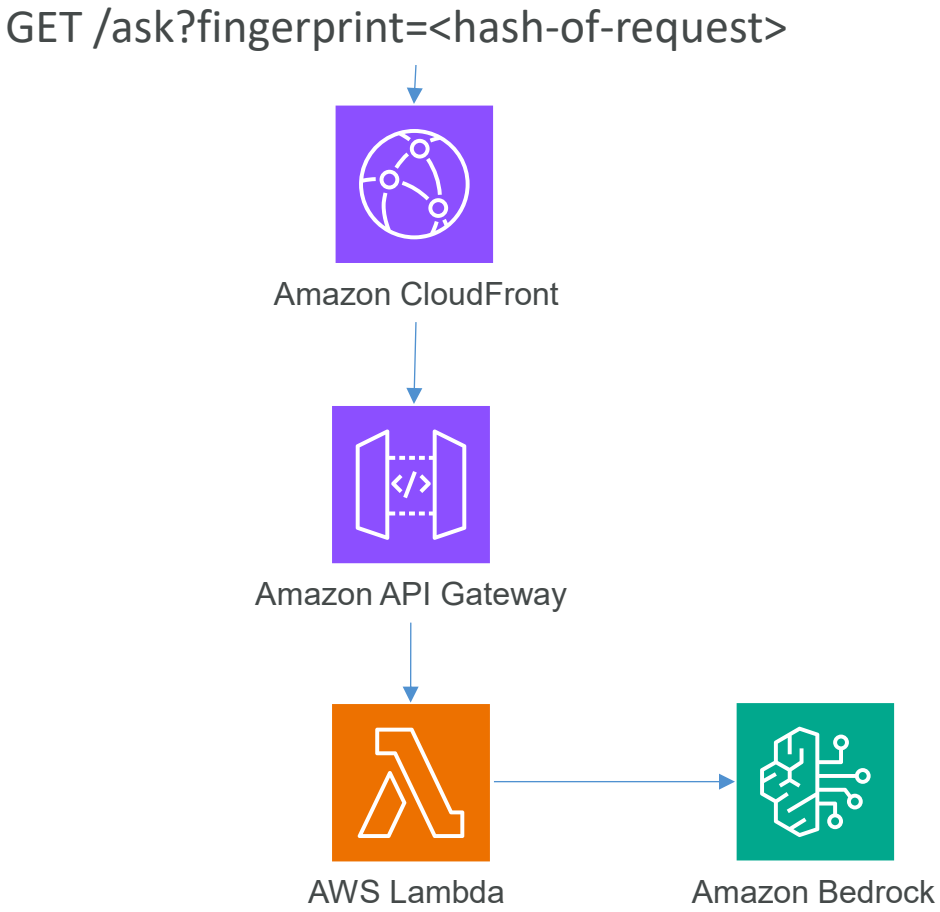
Intelligent Caching Systems: Edge Caching

- Use CloudFront when you can
  - Reduces latency and backend requests
- But wait, isn't GenAI dynamic? What can you cache?
  - Well, some prompts / responses are common and can be cached
  - If you have a deterministic request hash, you can bake that fingerprint into a GET request, which CloudFront could cache
  - The results can also be fingerprinted to identify when it's appropriate (based on model configuration values)
  - Might want an aggressive TTL if things change often
- There's more to an app than the LLM piece
  - Cache that too



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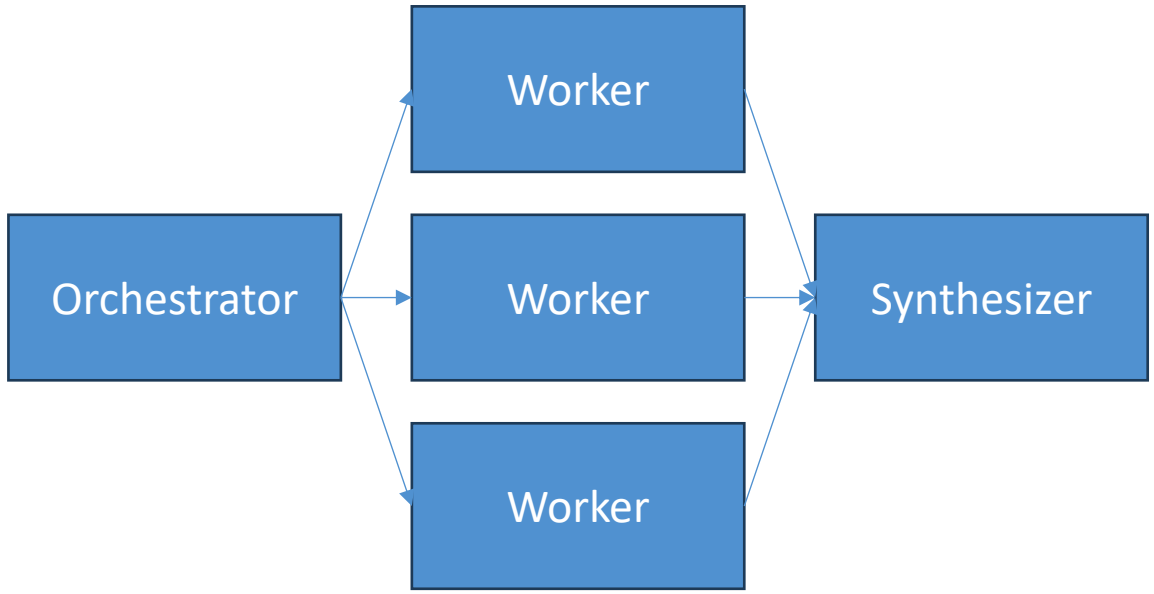
Building Responsive AI Systems

- Use parallel requests for complex workflows
  - Multi-agent workflows in parallel
  - Step Functions can also do this
- Cache / pre-compute predictable queries
- Response streaming
- Latency-optimized inference for FMs
  - A feature of Bedrock
    - Specify `performanceConfig={'latency': 'optimized'}` in your API calls
  - Optimizes
    - Time to First Token (TTFT)
    - Output Tokens per Second (OTPS)
    - End to End Latency (E2E)



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Building Responsive AI Systems

- Use Amazon Bedrock Intelligent Prompt Routing
  - Direct requests to different models based on complexity
- Keep your prompts concise
  - Put the important stuff first in case it gets truncated
- Context pruning
  - Ditch RAG results that aren't valuable or are redundant
- Limit response sizes
- Break up complex tasks

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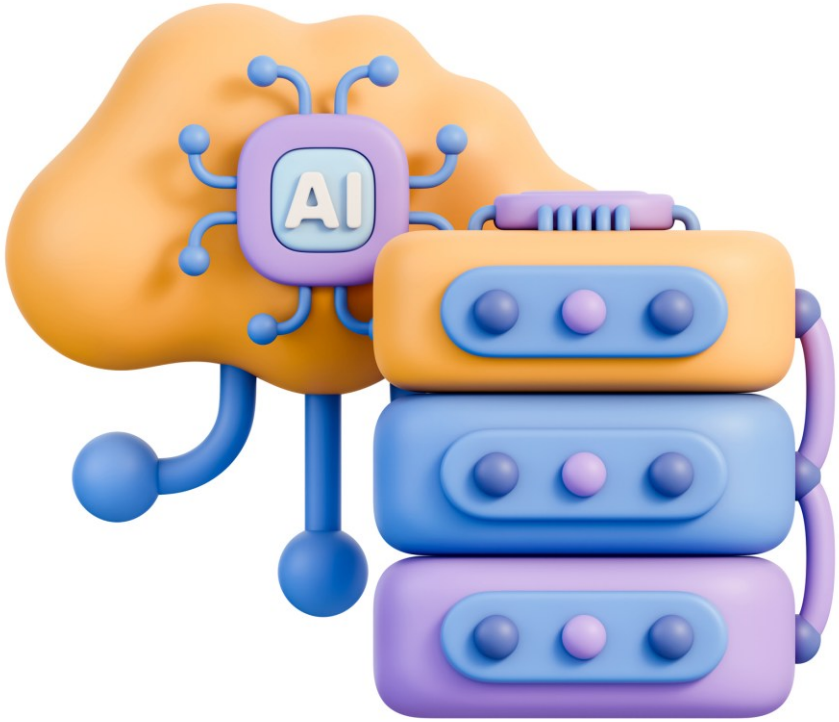
Optimizing Retrieval Performance

- We covered a lot of these details when talking about OpenSearch vector stores
- Optimize your indices
- Hybrid search improves relevancy
  - You can also have your own custom hybrid scoring functions
- Query pre-processing
  - Normalize query to the corpus in terms of style
  - Break up multi-part questions – don't try to look up two different things at once
  - Filter out irrelevant stuff
  - Reduce ambiguity
  - Keyword extraction for hybrid search



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Optimizing for Specific Use Cases

- Different models have different parameters you can tune
- Use A/B testing to evaluate changes
  - Bedrock Evaluations
  - CloudWatch Evidently
- Some common parameters
  - Temperature – Amount of randomness in the response (0=not random, 1 = “creative”)
  - Top_p – “Nucleus sampling” – Probability threshold for token candidates (specify this OR temperature)
  - Top_k – How many token options to sample from

Evidently

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Optimizing Foundation Model System Performance

- API call profiling
  - Find patterns in your requests
  - This might identify opportunities for caching, batching, RAG improvements
- Use structured input and output when possible
  - JSON, XML templates
  - Improves efficiency and accuracy
- Chain of Thought instruction patterns
  - “Reasoning” – forcing the FM to show the process step by step produces more accurate conclusions for complex reasoning tasks
- Feedback loops capturing user satisfaction & behavior
  - Use this to identify effective prompts

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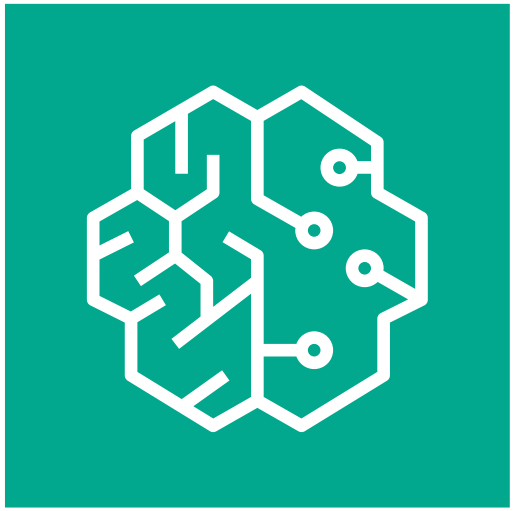
Optimizing Foundation Model System Performance: SageMaker stuff

- SageMaker AI can deploy models up to 500GB
  - Adjust container health check and download timeout quotas to allow enough time to download them and their resources!
- 3rd party model parallelization supported
  - Triton, FasterTransformer, DeepSpeed
- Instance type guidance
  - Large models: ml.p4d.24xlarge (GPU)
  - Small models (ie, named entity recognition): ml.c5.9xlarge (CPU)



Amazon SageMaker AI

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Optimizing Foundation Model System Performance

- UltraServers (Trn2, P6e-GB200)
  - Connects EC2 instances hosting your AI/ML workloads
  - Low latency, high bandwidth accelerator interconnects
- Use Lambda endpoint lifecycle management
  - Can automatically initialize endpoints
  - Download model artifacts from S3
  - …on demand, when processing is triggered

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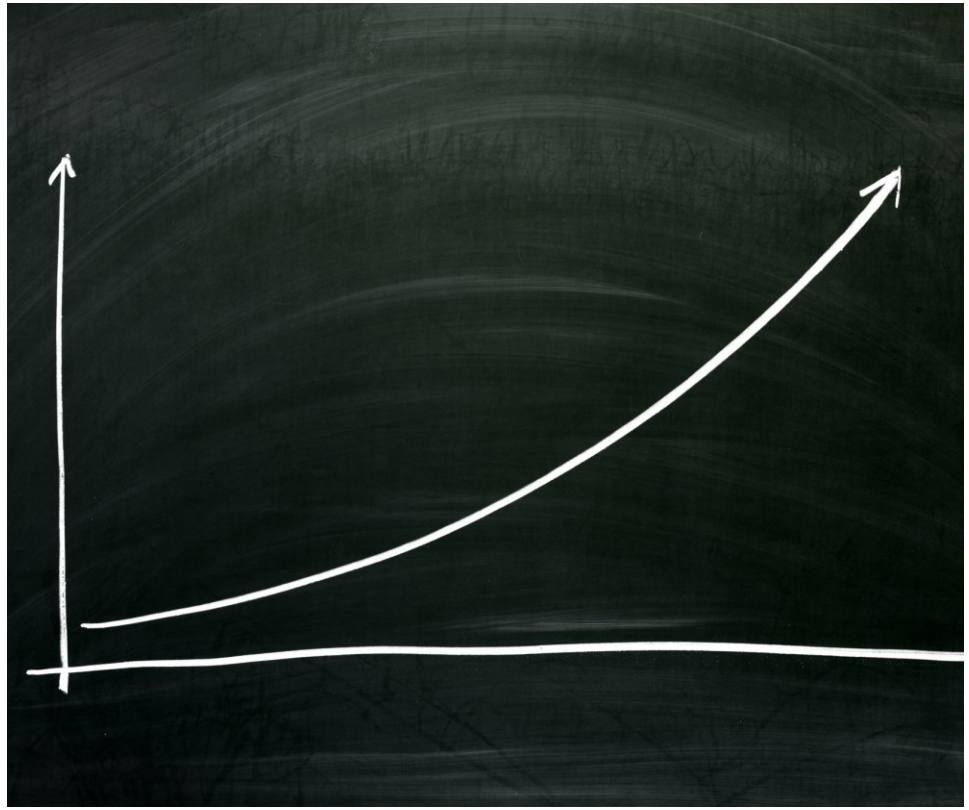
Exponential Backoff

- How to retry failed API calls in a controlled manner
  - Don’t flood the poor broken service
- Custom retry policies in AWS SDK clients
  - Start at 100ms
  - Backoff factor 2
  - Max retry count of 3-5 attempts
  - Jitter +- 100ms to prevent synchronized retries across clients
- Oddly specific guidance in SkillBuilder…



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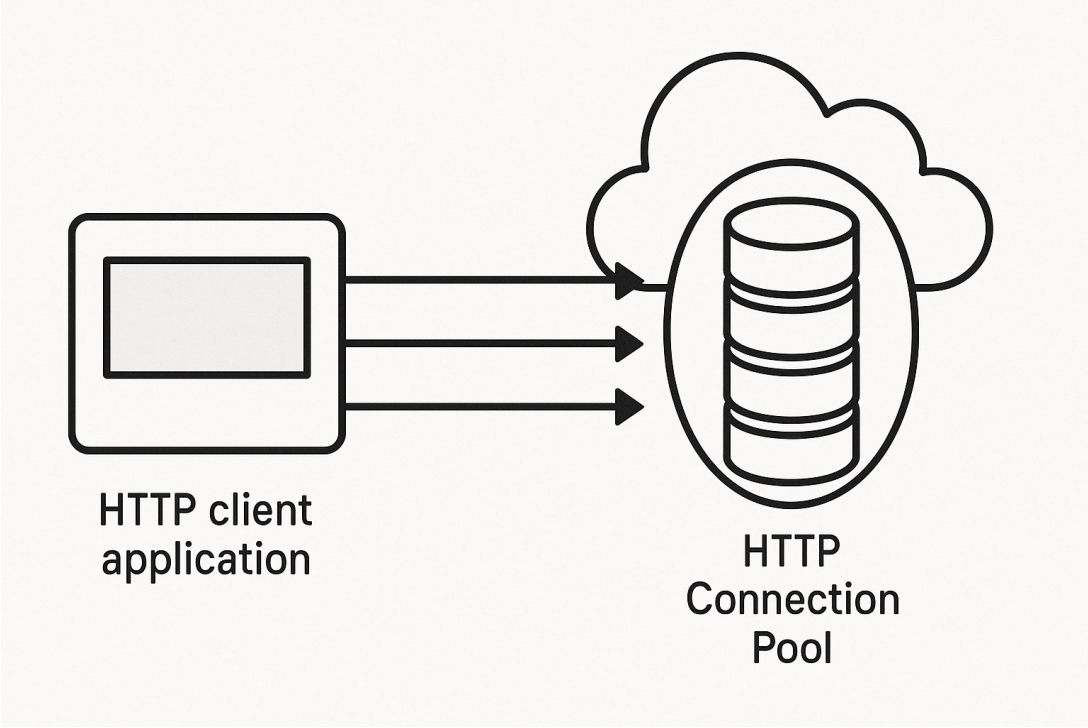
Connection Pooling

- For HTTP clients
- Instead of establishing a new connection for every request, maintain a pool of open connections that are used all the time.
- 10-20 connections per instance
- Connection TTL 60-300 seconds
- Balances resource utilization with connection re-use efficiency
- More oddly specific guidance in SkillBuilder…



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Bedrock Cross-Region Inference

- This comes up a lot… many features require it, many recommend it strongly
- Distributes workloads across AWS regions during service interruptions
  - …or restrictions from quotas or peak usage in specific regions
- Things like Amazon Organizations SCP's (service control policies) can block regions on you and prevent it from working :/
  - Make sure SCP allows all regions you are enabling, or leave the region unspecified
- Inference profiles
  - Specific geography: selects optimal region within your geography
  - Global: selects optimal commercial AWS region
  - Doesn't work with provisioned throughput

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Bedrock Cross-Region Inference

- Geographic cross-region inference
  - Useful if you have data residency requirements
  - Standard pricing
- Global cross-region inference
  - Maximized throughput
  - 10% cost savings
- Pricing based on where you called it from
- You don’t have to enable these regions in your account
- Data is encrypted in transit, and logged in CloudTrail

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Managing Models with SageMaker AI

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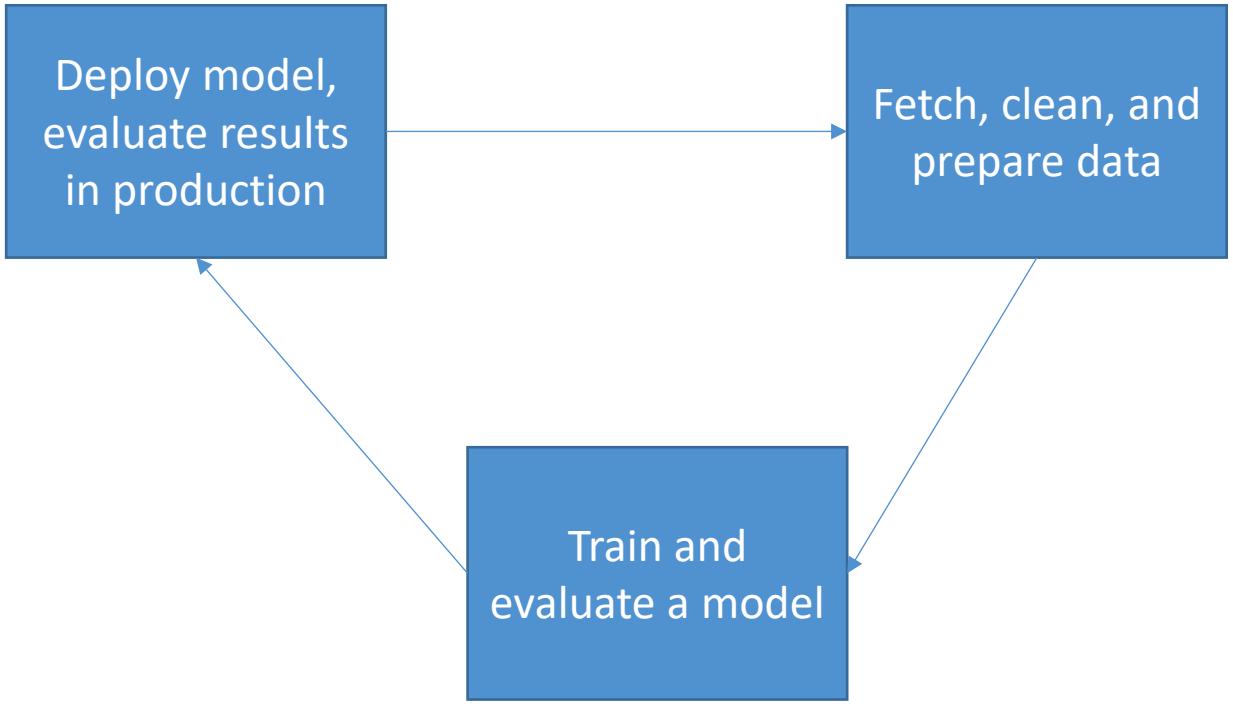
SageMaker is built to handle the entire machine learning workflow.



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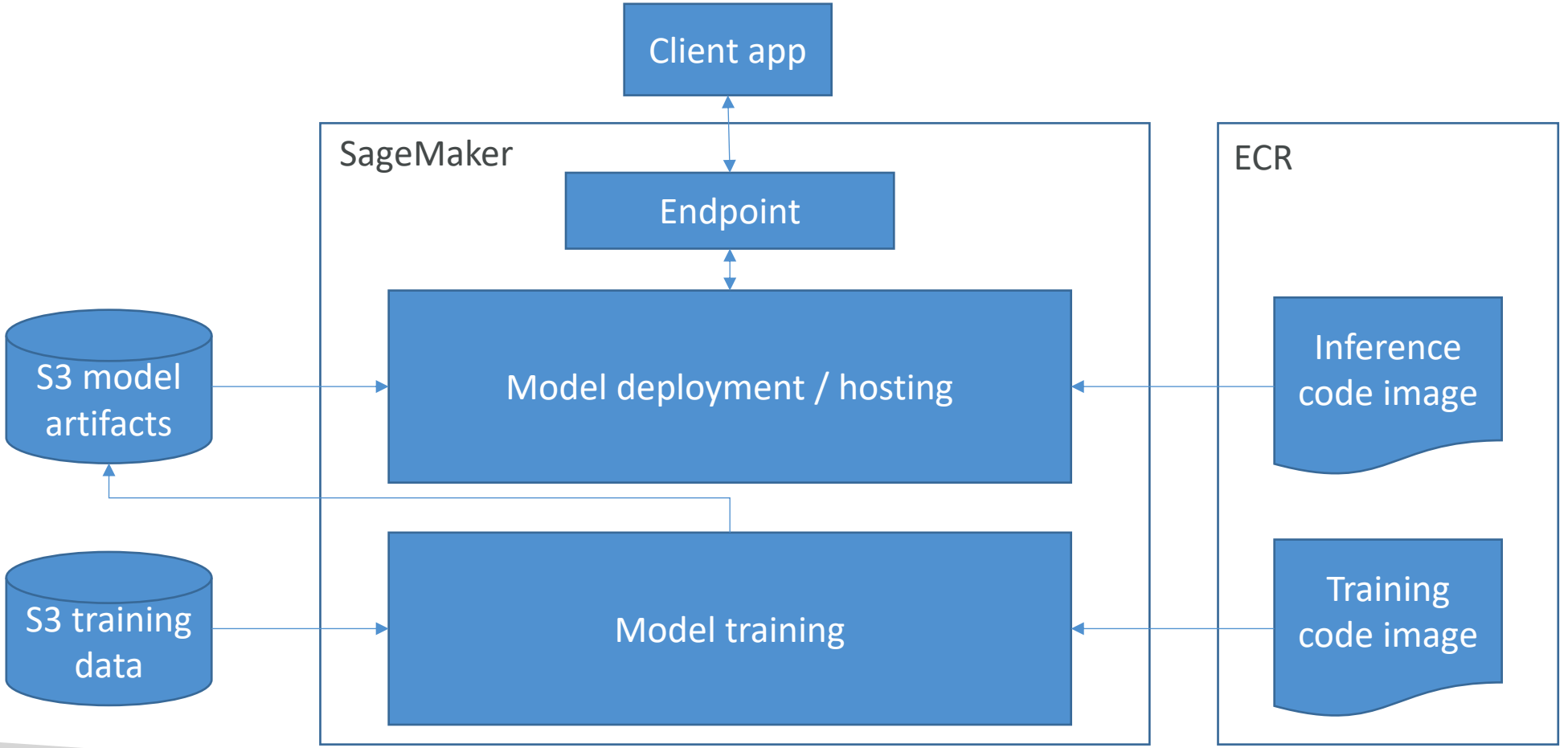
SageMaker Training & Deployment



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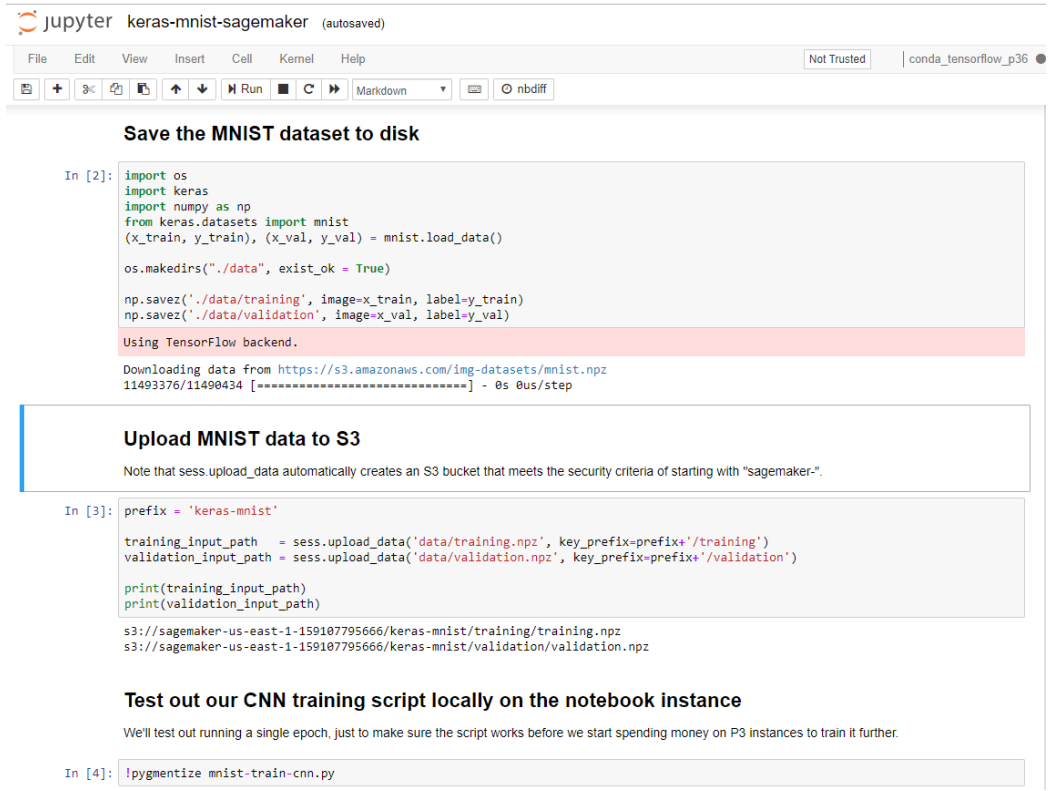
SageMaker Notebooks can direct the process

- Notebook Instances on EC2 are spun up from the console
  - S3 data access
  - Scikit_learn, Spark, Tensorflow
  - Wide variety of built-in models
  - Ability to spin up training instances
  - Abilty to deploy trained models for making predictions at scale



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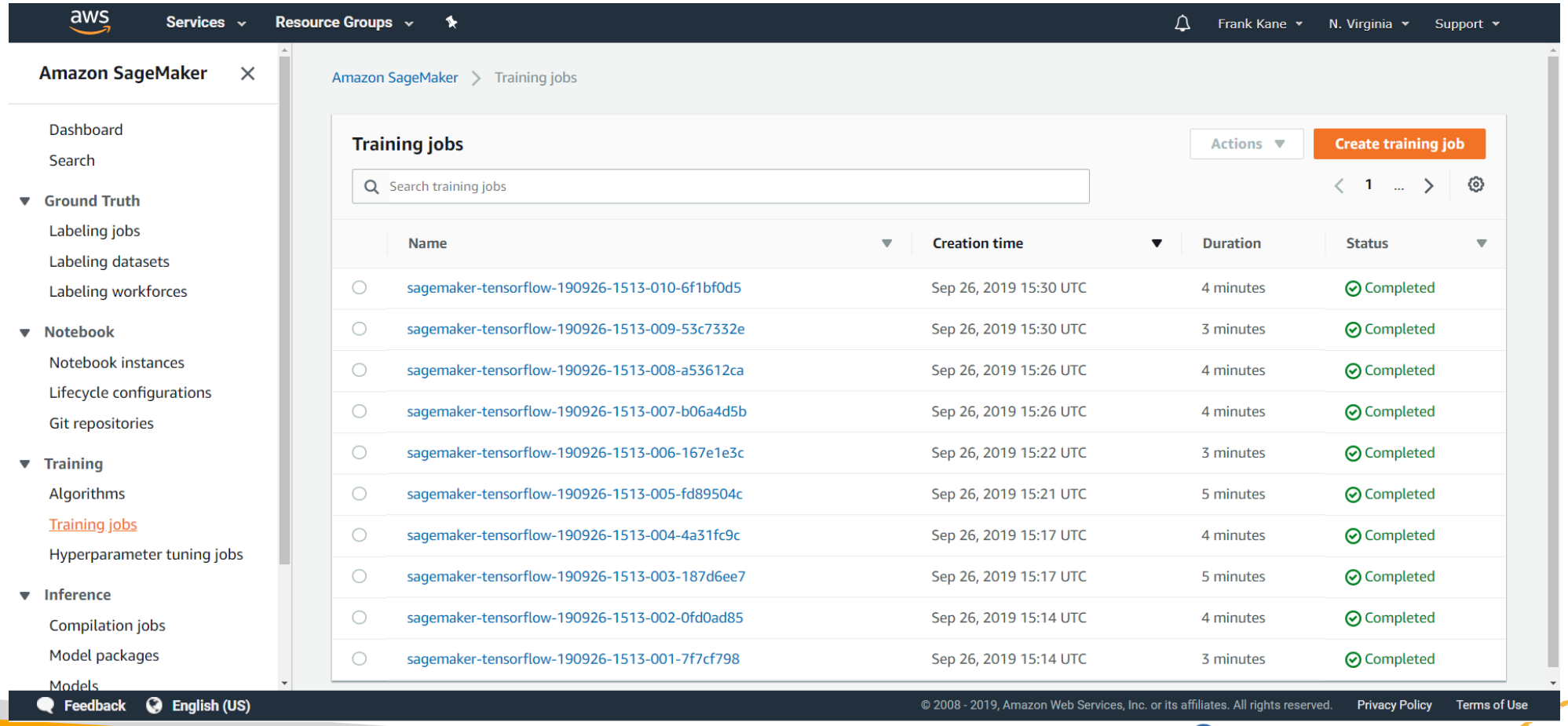
So can the SageMaker console



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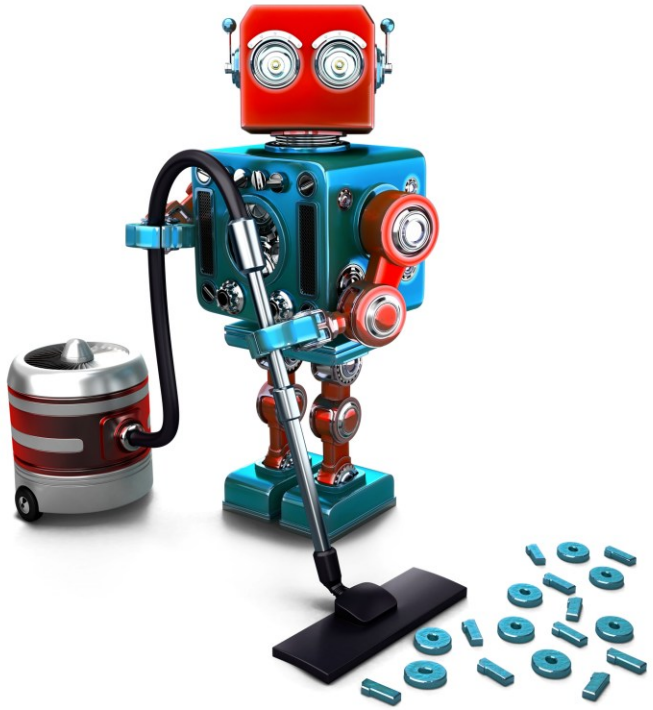
Data prep on SageMaker

- Data usually comes from S3
  - Ideal format varies with algorithm – often it is RecordIO / Protobuf
- Can also ingest from Athena, EMR, Redshift, and Amazon Keyspaces DB
- Apache Spark integrates with SageMaker
- Scikit_learn, numpy, pandas all at your disposal within a notebook



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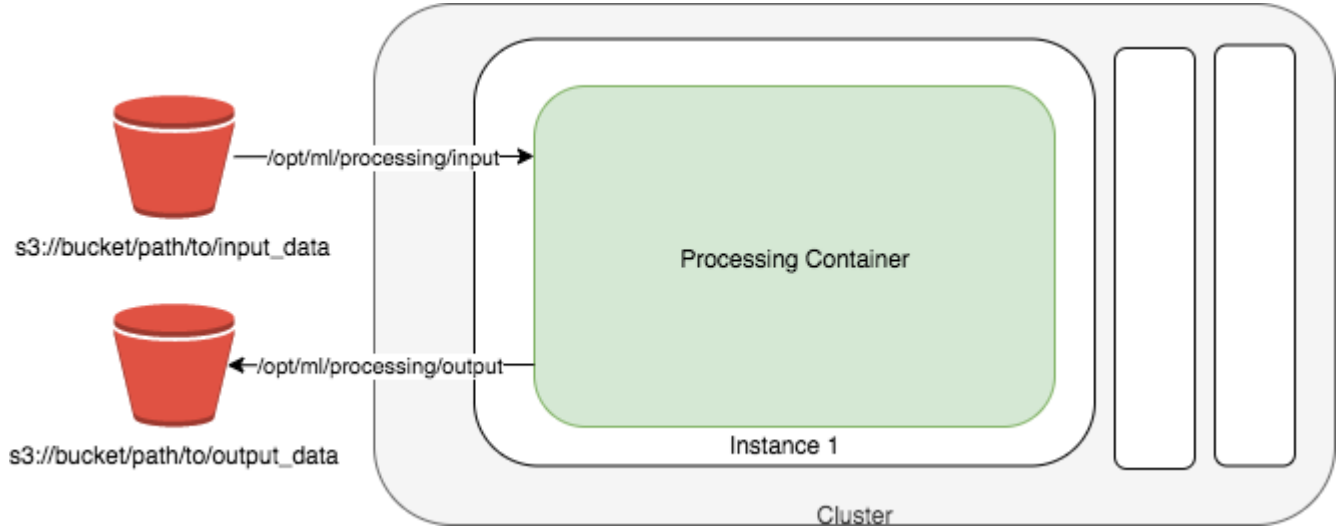
SageMaker Processing

- Processing jobs
  - Copy data from S3
  - Spin up a processing container
    - SageMaker built-in or user provided
  - Output processed data to S3



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Deploying Trained Models

- Save your trained model to S3
- Can deploy two ways:
  - Persistent endpoint for making individual predictions on demand
  - SageMaker Batch Transform to get predictions for an entire dataset
- Lots of cool options
  - Inference Pipelines for more complex processing
  - SageMaker Neo for deploying to edge devices
  - Elastic Inference for accelerating deep learning models
  - Automatic scaling (increase # of endpoints as needed)
  - Shadow Testing evaluates new models against currently deployed model to catch errors

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Deployment Safeguards

- Deployment Guardrails
  - For asynchronous or real-time inference endpoints
  - Controls shifting traffic to new models
    - “Blue/Green Deployments”
      - All at once: shift everything, monitor, terminate blue fleet
      - Canary: shift a small portion of traffic and monitor
      - Linear: Shift traffic in linearly spaced steps
  - Auto-rollbacks
- Shadow Tests
  - Compare performance of shadow variant to production
  - You monitor in SageMaker console and decide when to promote it

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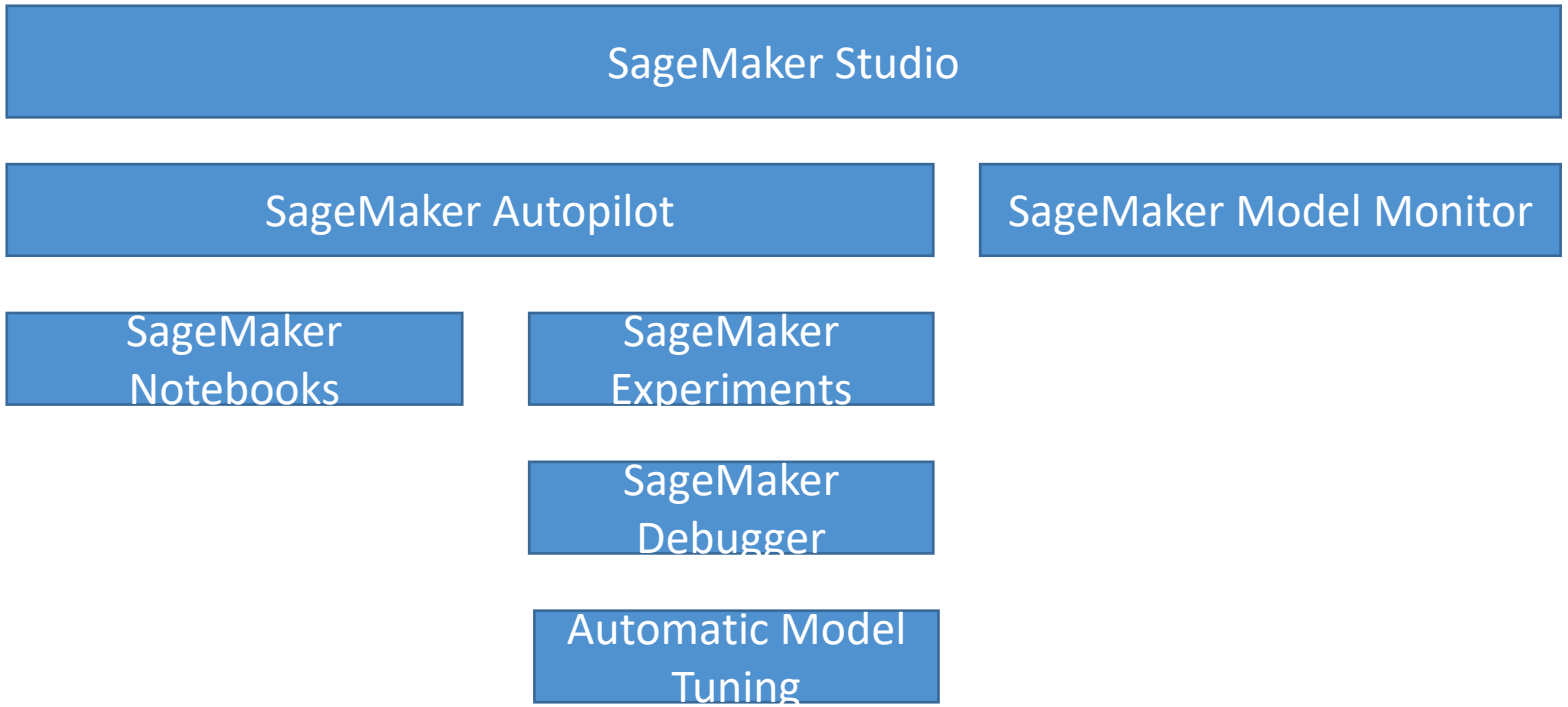
Putting them together



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SageMaker: More Features

- SageMaker JumpStart
  - One-click models and algorithms from model zoos
  - Over 150 open source models in NLP, object detections, image classification, etc.
- SageMaker Data Wrangler
  - Import / transform / analyze / export data within SageMaker Studio
- SageMaker Feature Store
  - Find, discover, and share features in Studio
  - Online (low latency) or offline (for training or batch inference) modes
  - Features organized into Feature Groups
- SageMaker Edge Manager
  - Software agent for edge devices
  - Model optimized with SageMaker Neo
  - Collects and samples data for monitoring, labeling, retraining
- Asynchronous Inference endpoints

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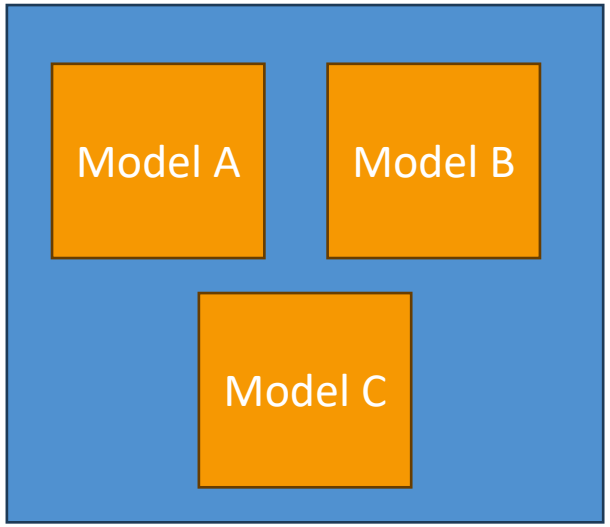
Optimizing FM Deployments

- SageMaker AI offers single and multi-model endpoints
  - More generally, multi-container endpoints
  - Each endpoint supports deployment guardrails, VPC, network isolation
- You can train/tune a model in SageMaker AI, and deploy through Bedrock
  - You use Bedrock Custom Model Import for this
  - Now your inference is serverless ☺
- SageMaker AI Inference Components
  - Each model gets its own scaling policy



SageMaker multi-model endpoint

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Optimizing FM Deployments

- Cross-region inference profiles
  - For endpoints on Bedrock
- EC2 Auto Scaling Groups
  - Load balancing in front of SageMaker AI endpoints
- Available model servers with SageMaker AI
  - TorchServe
  - DJL Serving (Deep Java Library)
    - Deep Learning Containers
    - DJL was created by Amazon, so more likely to show up on exam
  - Triton Inference Server

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Optimizing FM Deployments

- Use asynchronous inference if latency isn't important
  - SageMaker AI async endpoints
  - Your own queue with SNS / SQS
- Model compression
  - Quantization (of model weights)
  - Pruning
  - Knowledge distillation
    - A smaller model is trained from a larger model
- Avoid premature optimization
  - Measure your performance, costs, resource utilization
  - Don't solve problems that don't exist!

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SageMaker Ground Truth

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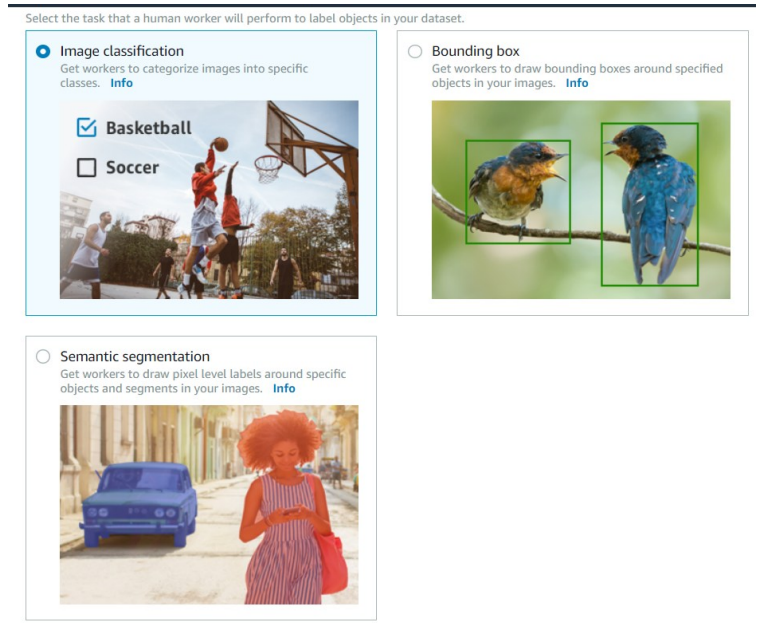
What is Ground Truth?

- Sometimes you don't have training data at all, and it needs to be generated by humans first.
- Example: training an image classification model. Somebody needs to tag a bunch of images with what they are images of before training a neural network
- Ground Truth manages humans who will label your data for training purposes



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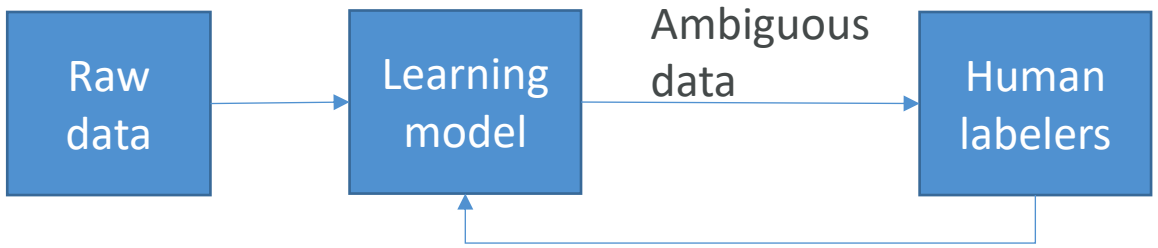
But it's more than that

- Ground Truth creates its own model as images are labeled by people
- As this model learns, only images the model isn't sure about are sent to human labelers
- This can reduce the cost of labeling jobs by 70%



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Who are these human labelers?

- Mechanical Turk
- Your own internal team
- Professional labeling companies

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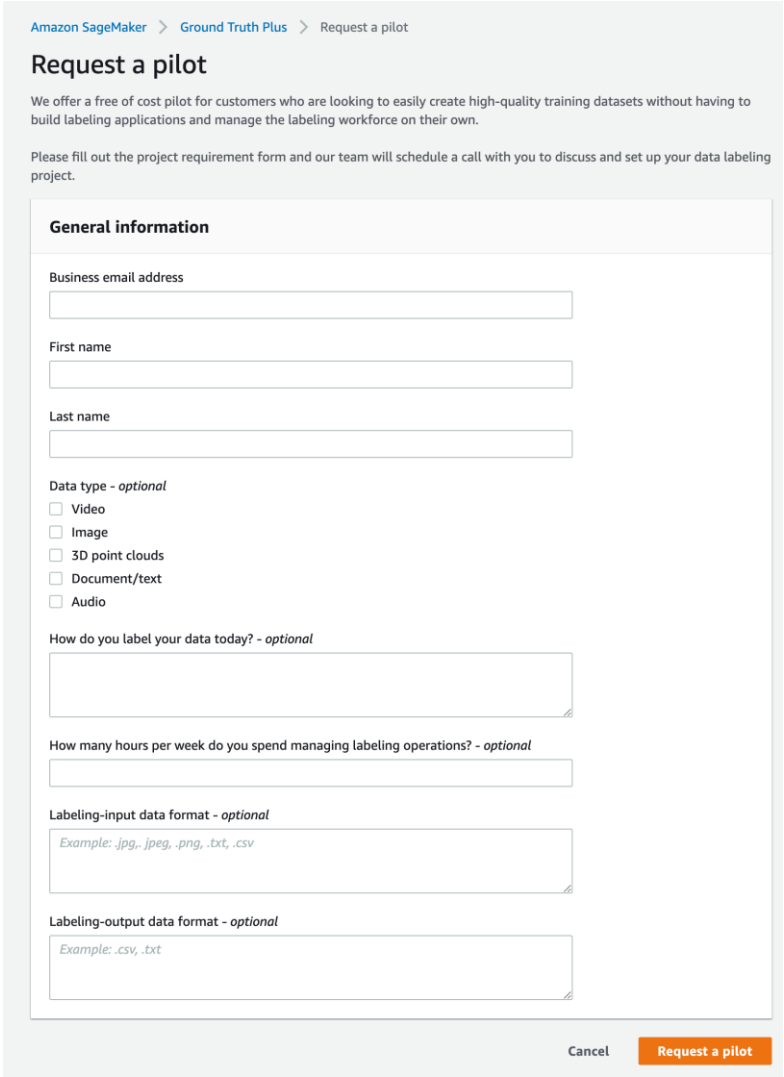
Ground Truth Plus

- Turnkey solution
- “Our team of AWS Experts” manages the workflow and team of labelers
  - You fill out an intake form
  - They contact you and discuss pricing
- You track progress via the Ground Truth Plus Project Portal
- Get labeled data from S3 when done



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Other ways to generate training labels

- Rekognition
  - AWS service for image recognition
  - Automatically classify images
- Comprehend
  - AWS service for text analysis and topic modeling
  - Automatically classify text by topics, sentiment
- Any pre-trained model or unsupervised technique that may be helpful

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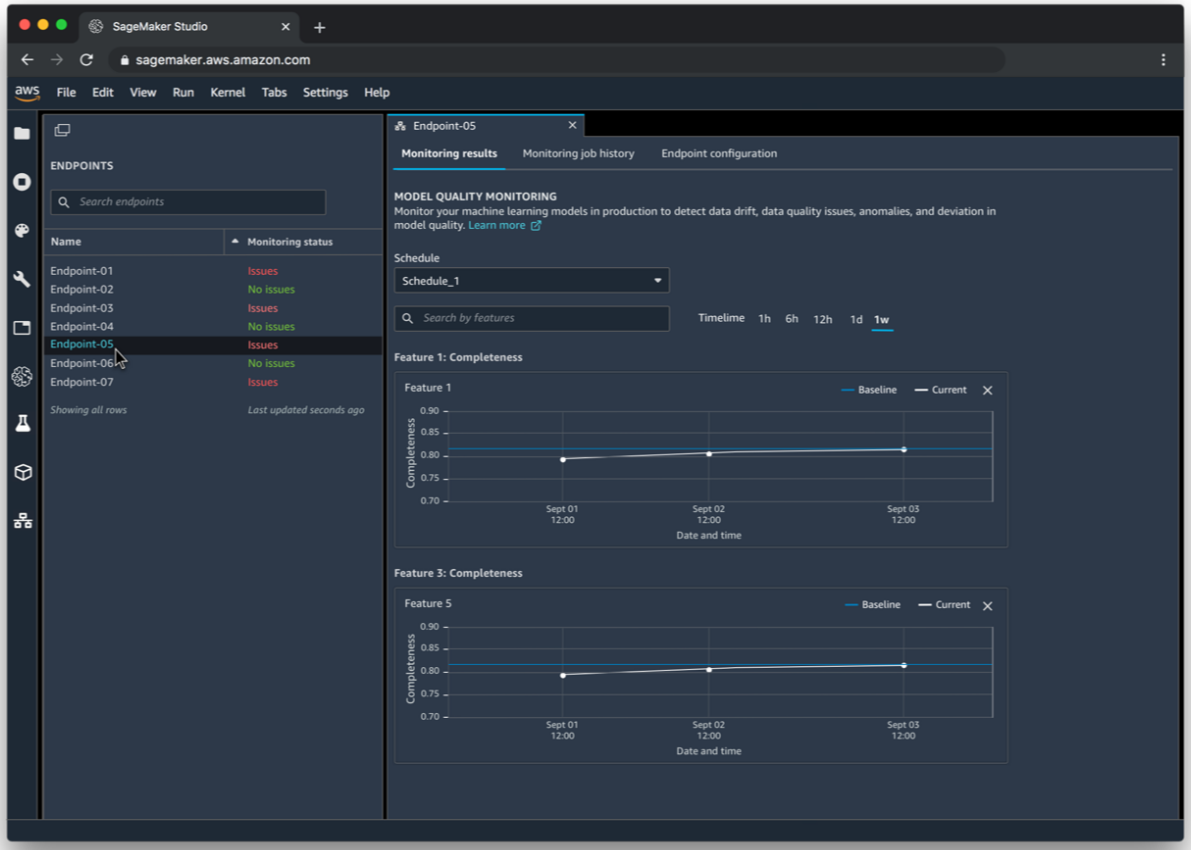
SageMaker Model Monitor

- Get alerts on quality deviations on your deployed models (via CloudWatch)
- Visualize data drift
  - Example: loan model starts giving people more credit due to drifting or missing input features
- Detect anomalies & outliers
- Detect new features
- No code needed



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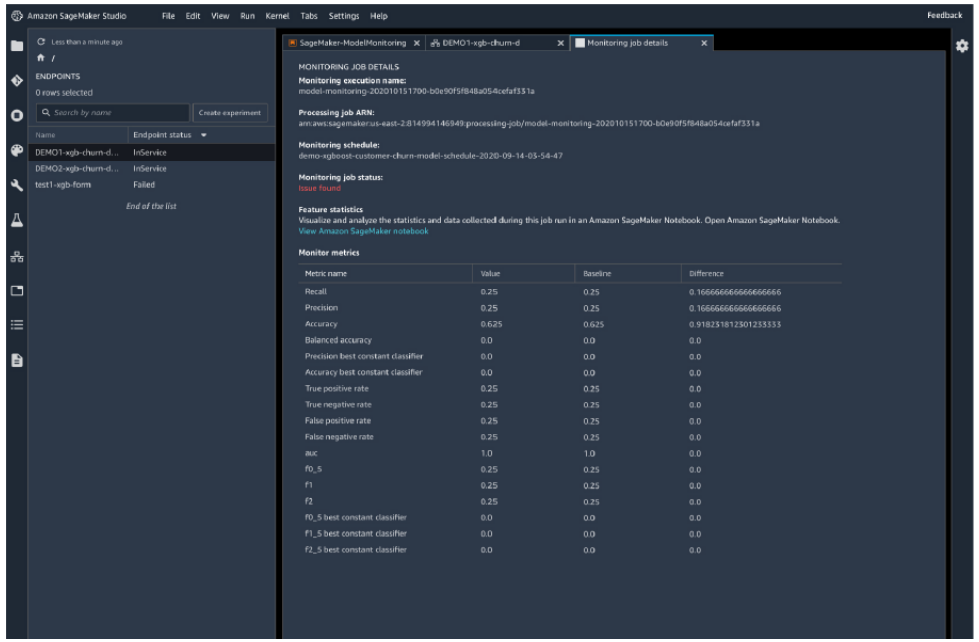
SageMaker Model Monitor + Clarify

- Integrates with SageMaker Clarify
  - SageMaker Clarify detects potential bias
  - i.e., imbalances across different groups / ages / income brackets
  - With ModelMonitor, you can monitor for bias and be alerted to new potential bias via CloudWatch
  - SageMaker Clarify also helps explain model behavior
    - Understand which features contribute the most to your predictions



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Pre-training Bias Metrics in Clarify

- Class Imbalance (CI)
  - One facet (demographic group) has fewer training values than another
- Difference in Proportions of Labels (DPL)
  - Imbalance of positive outcomes between facet values
- Kullback-Leibler Divergence (KL), Jensen-Shannon Divergence(JS)
  - How much outcome distributions of facets diverge
- Lp-norm (LP)
  - P-norm difference between distributions of outcomes from facets
- Total Variation Distance (TVD)
  - L1-norm difference between distributions of outcomes from facets
- Kolmogorov-Smirnov (KS)
  - Maximum divergence between outcomes in distributions from facets
- Conditional Demographic Disparity (CDD)
  - Disparity of outcomes between facets as a whole, and by subgroups

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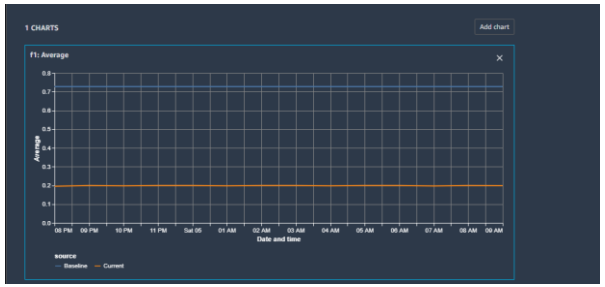
SageMaker Model Monitor

- Data is stored in S3 and secured
- Monitoring jobs are scheduled via a Monitoring Schedule
- Metrics are emitted to CloudWatch
  - CloudWatch notifications can be used to trigger alarms
  - You'd then take corrective action (retrain the model, audit the data)
- Integrates with Tensorboard, QuickSight, Tableau
  - Or just visualize within SageMaker Studio



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SageMaker Model Monitor

- Monitoring Types:
  - Drift in data quality
    - Relative to a baseline you create
    - “Quality” is just statistical properties of the features
  - Drift in model quality (accuracy, etc)
    - Works the same way with a model quality baseline
    - Can integrate with Ground Truth labels
  - Bias drift
  - Feature attribution drift
    - Based on Normalized Discounted Cumulative Gain (NDCG) score
    - This compares feature ranking of training vs. live data

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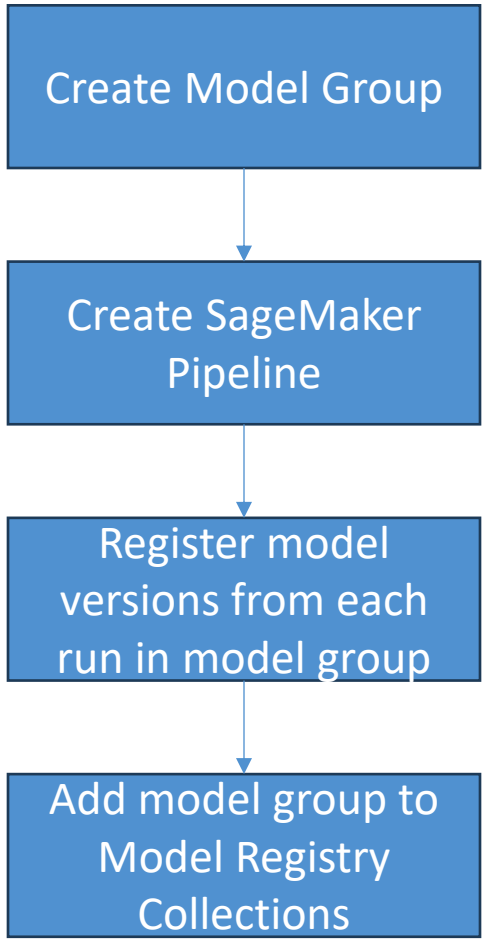
SageMaker Model Registry

- Catalog your models, manage model versions
- Associate metadata with models
- Manage approval status of a model
- Deploy models to production
- Automate deployment with CI/CD
- Share models
- Integrate with SageMaker Model



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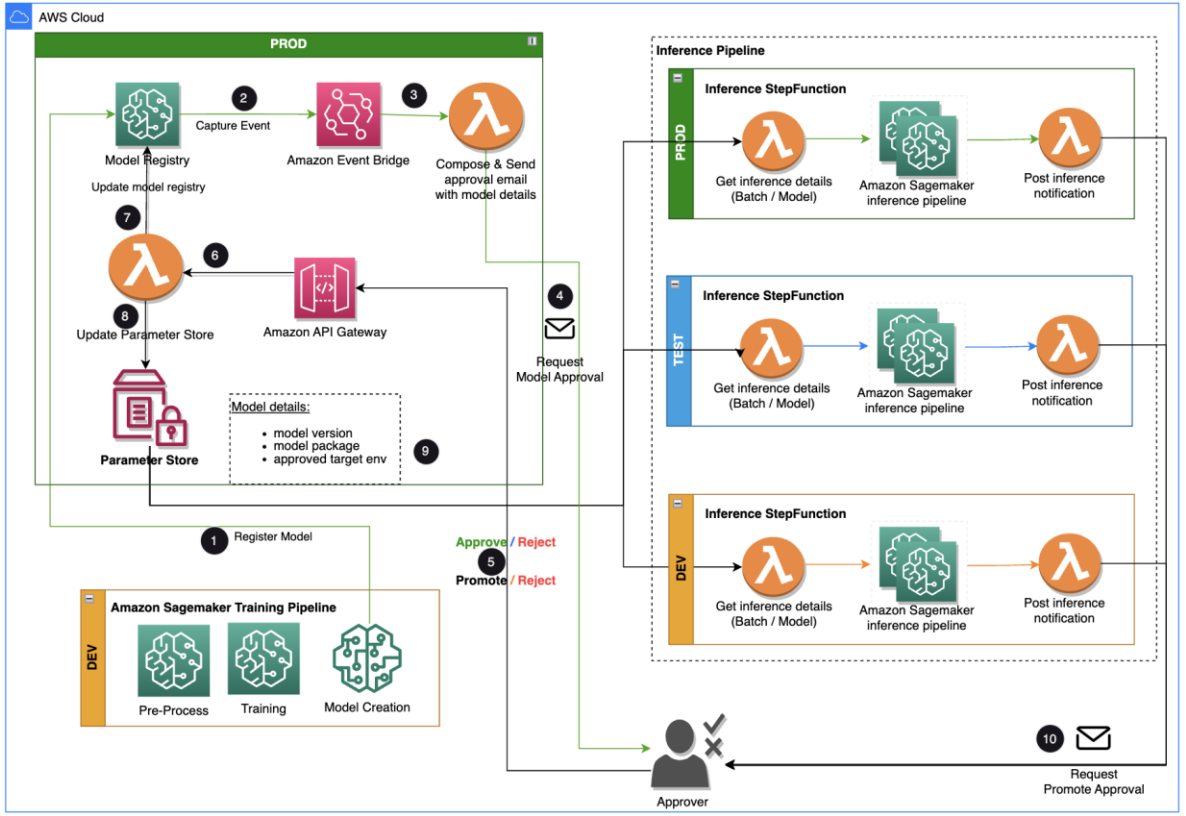
Example workflow for approving and promoting models with Model Registry



https://aws.amazon.com/blogs/machine-learning/build-an-amazon-sagemaker-model-registry-approval-and-promotion-workflow-with-human-intervention/

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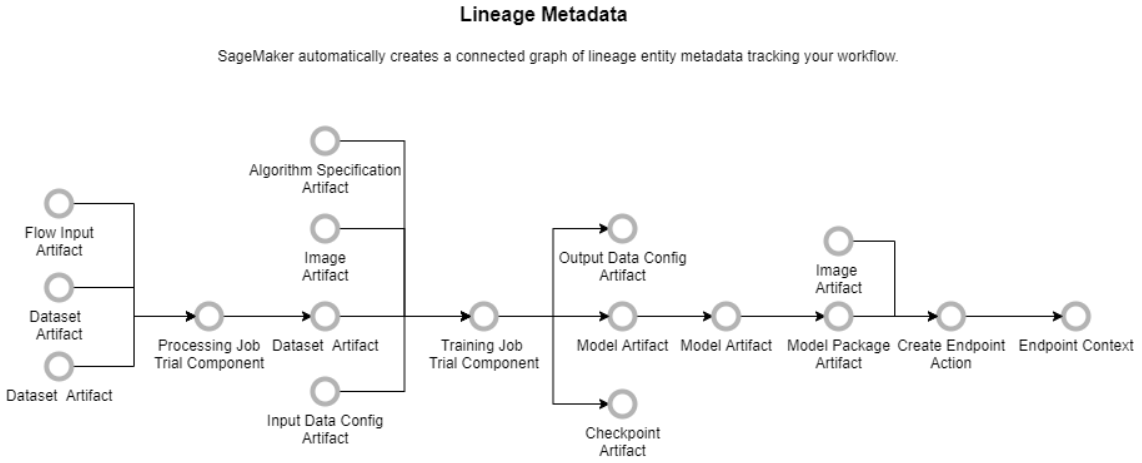
SageMaker ML Lineage Tracking

- Creates & stores your ML workflow (MLOps)
- Keep a running history of your models
- Tracking for auditing and compliance
- Automatically or manually-created tracking entities
- Integrates with AWS Resource Access Manager for cross-account lineage
- Sample SageMaker-created lineage graph:



AWS

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Lineage Tracking Entities

- Trial component (processing jobs, training jobs, transform jobs)
- Trial (a model composed of trial components)
- Experiment (a group of Trials for a given use case)
- Context (logical grouping of entities)
- Action (workflow step, model deployment)
- Artifact (Object or data, such as an S3 bucket or an image in ECR)
- Association (connects entities together) – has optional AssociationType:
  - ContributedTo
  - AssociatedWith
  - DerivedFrom
  - Produced
  - SameAs

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Querying Lineage Entities

- Use the LineageQuery API from Python
  - Part of the Amazon SageMaker SDK for Python
- Do things like find all models / endpoints / etc. that use a given artifact
- Produce a visualization
  - Requires external Visualizer helper class

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Cross-Account Lineage Tracking

- Can trace across organizational boundaries
- Use SageMaker AddAssociation API
  - Plus required IAM roles & permissions

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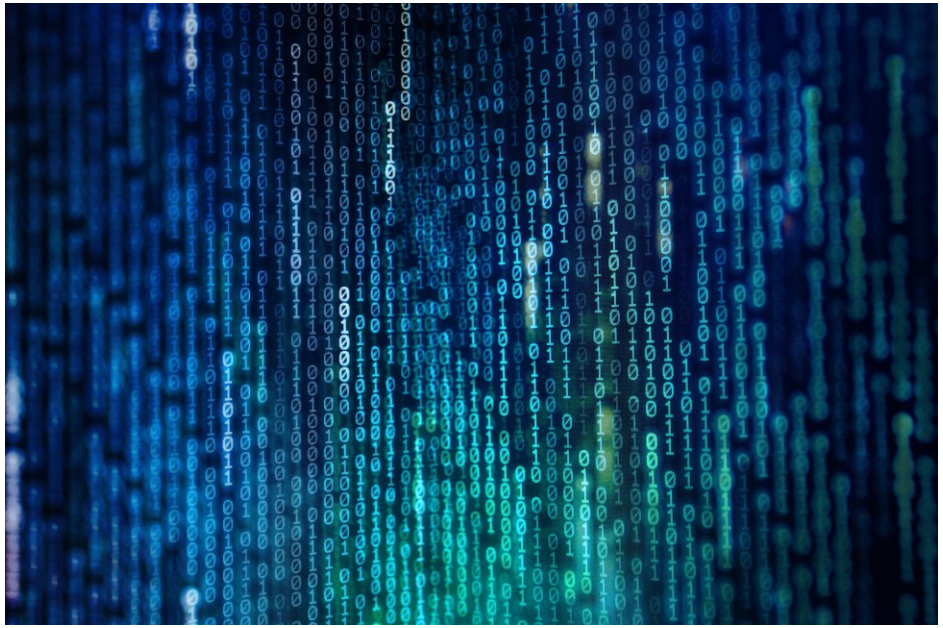
SageMaker Neo

- Train once, run anywhere
  - Edge devices
    - ARM, Intel, Nvidia processors
    - Embedded in whatever – your car?
- Optimizes code for specific devices
  - Tensorflow, MXNet, PyTorch, ONNX, XGBoost, DarkNet, Keras
- Consists of a **compiler** and a **runtime**



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Neo + AWS IoT Greengrass

- Neo-compiled models can be deployed to an HTTPS endpoint
  - Hosted on C5, M5, M4, P3, or P2 instances
  - Must be same instance type used for compilation
- OR! You can deploy to IoT Greengrass
  - This is how you get the model to an actual edge device
  - Inference at the edge with local data, using model trained in the cloud
  - Uses Lambda inference applications

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SageMaker Unified Studio

- Single interface for data, analytics, AI, and ML services
  - This includes Bedrock, Q, and Quicksight
  - Also notebooks, and all the stuff that was in the old SageMaker Studio
- Single interface for build / deploy / execute / monitor
- Built for teams
  - Administrators manage users/groups and what they access
  - A *domain* connects assets, users, and projects
- Can connect from Visual Studio Code

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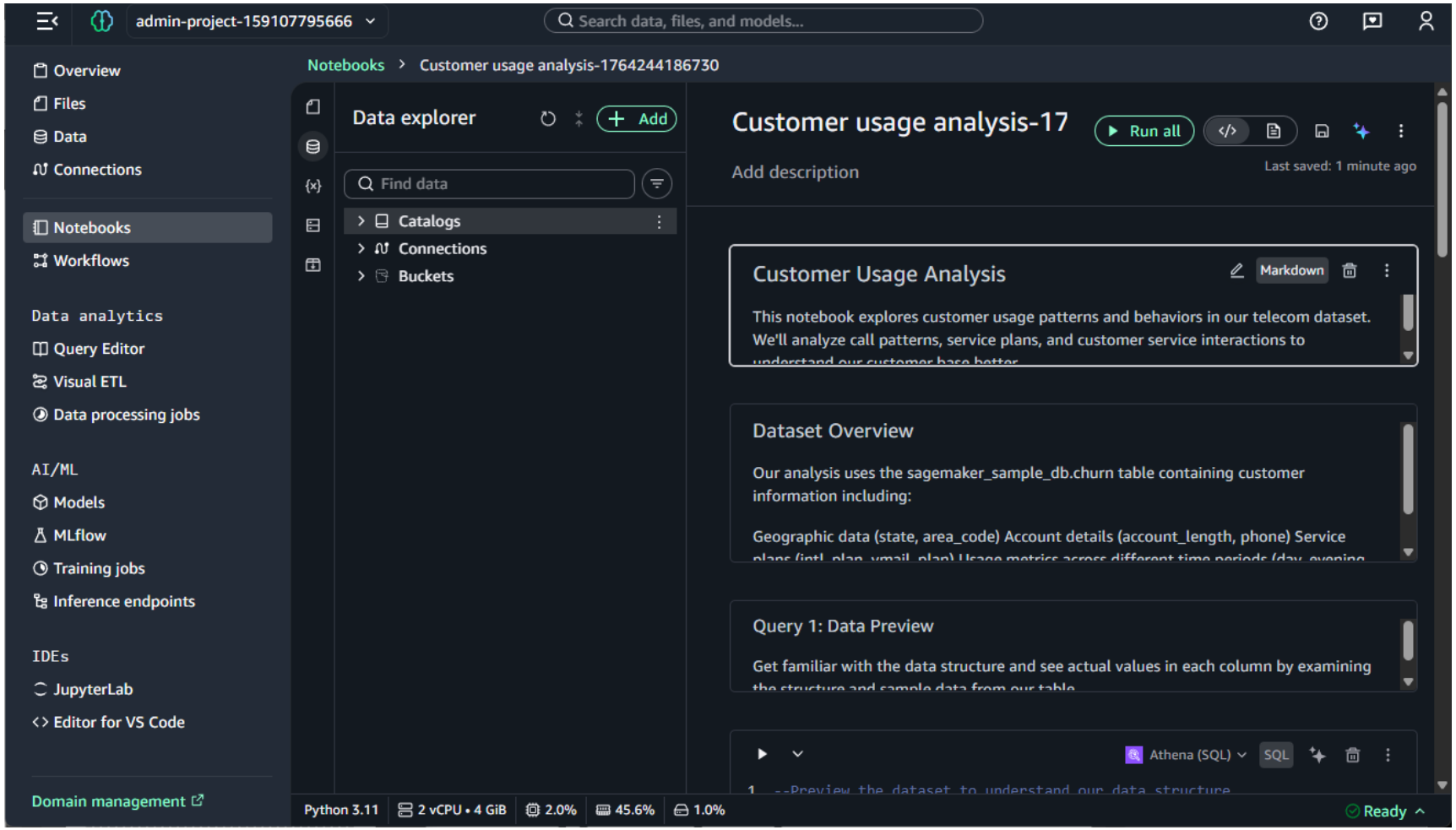
SageMaker Unified Studio



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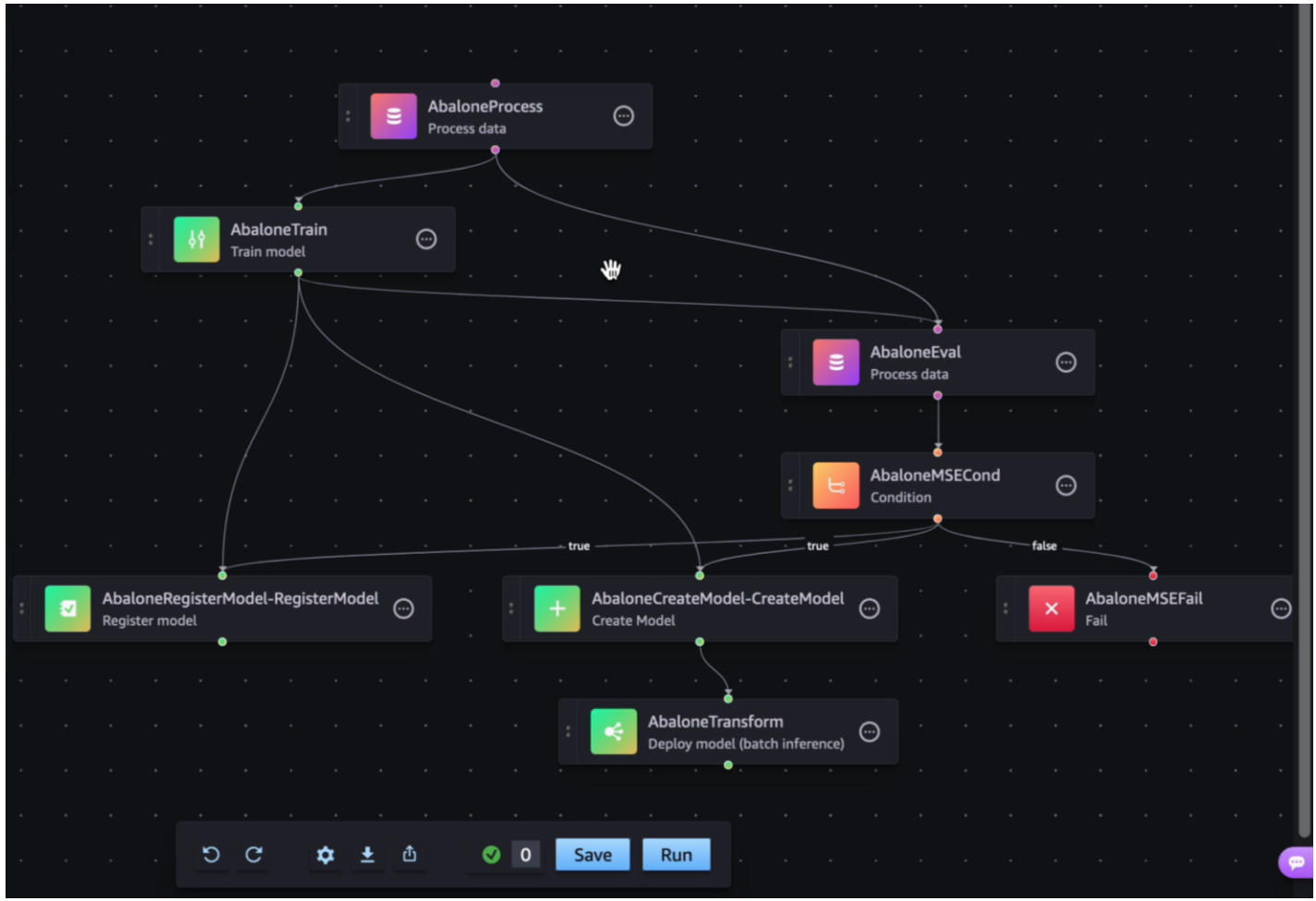
SageMaker Pipelines

Directed Acyclic Graph (DAG) for ML Workflows

Can define visually in pipeline designer, or via JSON

Integrated with SageMaker Unified Studio and SageMaker AI



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More Tools for Building AI Applications

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AWS Lambda

Serverless data processing

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What is Lambda?

- A way to run code snippets “in the cloud”
  - Serverless
  - Continuous scaling
- Often used to process data as it’s moved around

AWS Lambda

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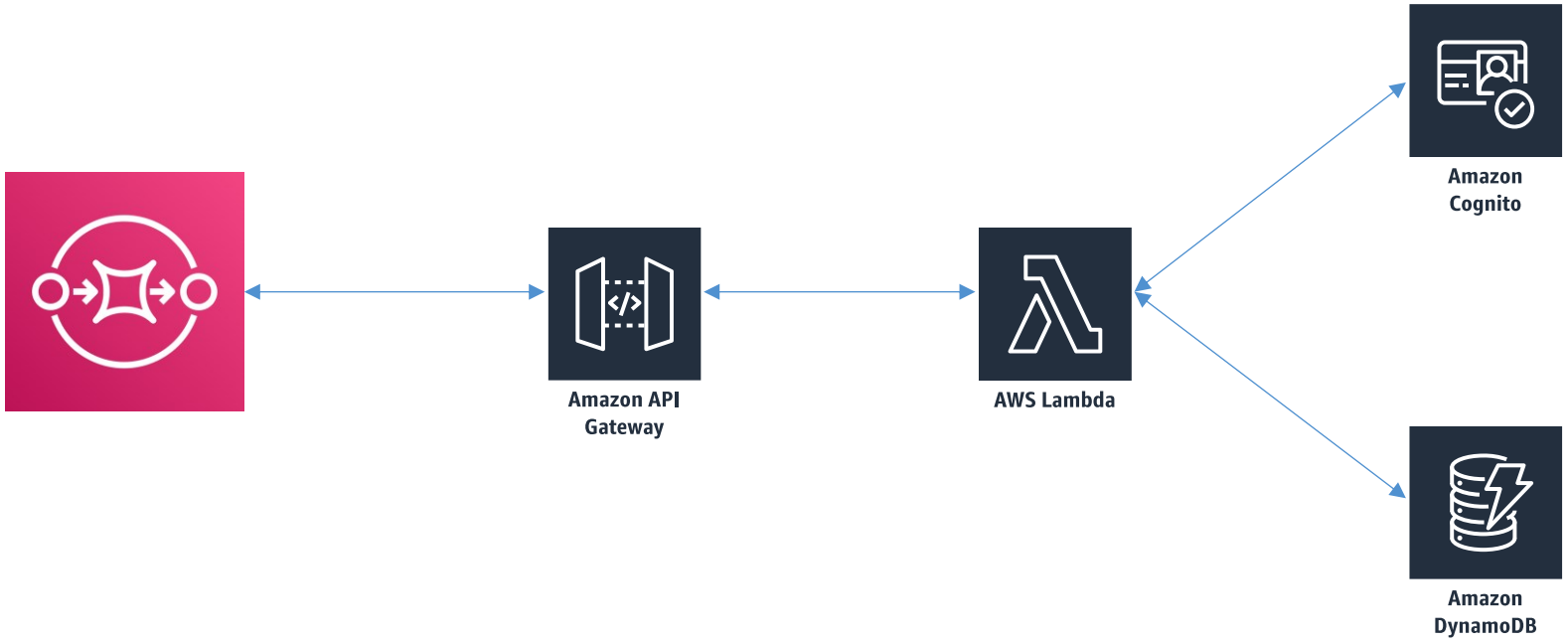
Example: Serverless Website



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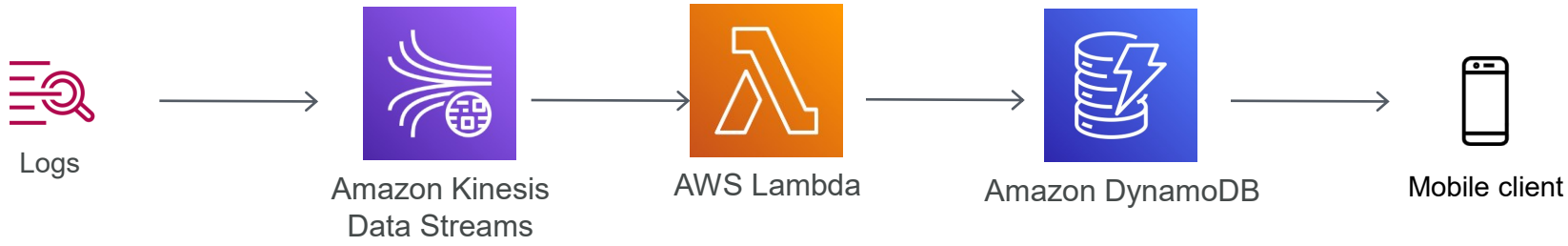
Example:
Order history app



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Example:
Transaction rate alarm

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Why not just run a server?

- Server management (patches, monitoring, hardware failures, etc.)
- Servers can be cheap, but scaling gets expensive really fast
- You don’t pay for processing time you don’t use
- Easier to split up development between front-end and back-end

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Main uses of Lambda

- Real-time file processing
- Real-time stream processing
- ETL
- Cron replacement
- Process AWS events

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Supported languages

- Node.js
- Python
- Java
- C#
- Go
- Powershell
- Ruby

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Lambda triggers

Amazon S3

Amazon Simple Email Service

Amazon Kinesis Data Firehose

Amazon Kinesis Data Streams

Amazon DynamoDB

Amazon SNS

Amazon SQS

AWS Config

AWS IoT Button

Amazon Lex

Amazon CloudWatch

AWS CloudFormation

Amazon API Gateway

Amazon CloudFront

Amazon Cognito

AWS CodeCommit

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Lambda and Amazon Opensearch Service

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Lambda and Data Pipeline

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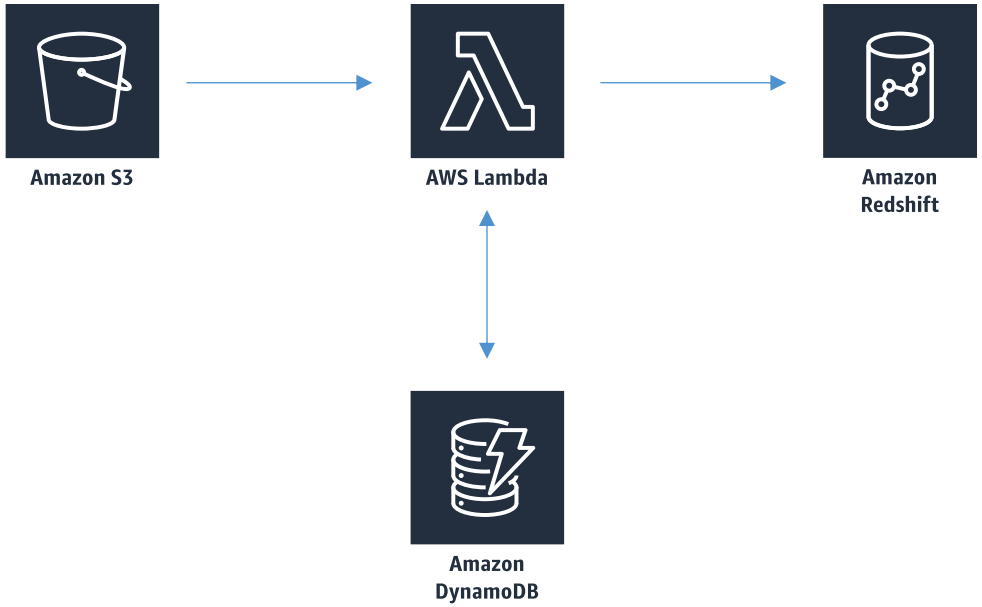
Lambda and Redshift



Best practice for loading data into Redshift is the COPY command

But, you can use Lambda if you need to respond to new data that shows up at any time

Can use DynamoDB to keep track of what's been loaded

Lambda can batch up new data and load them with COPY

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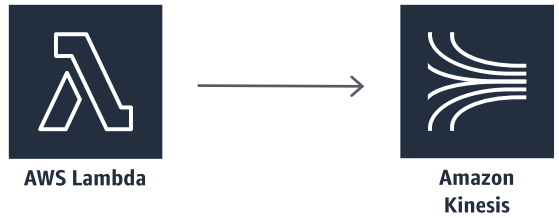
Lambda + Kinesis

- Your Lambda code receives an event with a **batch** of stream records
  - You specify a batch size when setting up the trigger (up to 10,000 records)
  - Too large a batch size can cause timeouts!
  - Batches may also be split beyond Lambda's payload limit (6 MB)
- Lambda will retry the batch until it succeeds or the data expires
  - This can stall the shard if you don't handle errors properly
  - Use more shards to ensure processing isn't totally held up by errors
- Lambda processes shard data synchronously



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Using Lambda with Bedrock / GenAI

- Connecting agents with tools
  - Parameter validation, error handling
- On-demand FM invocation
  - Without provisioning capacity
- Receiving webhooks for integrations
- Handle JSON events from API Gateway
- Custom aggregation logic
  - From multiple models (weighted averaging?), or models that vote

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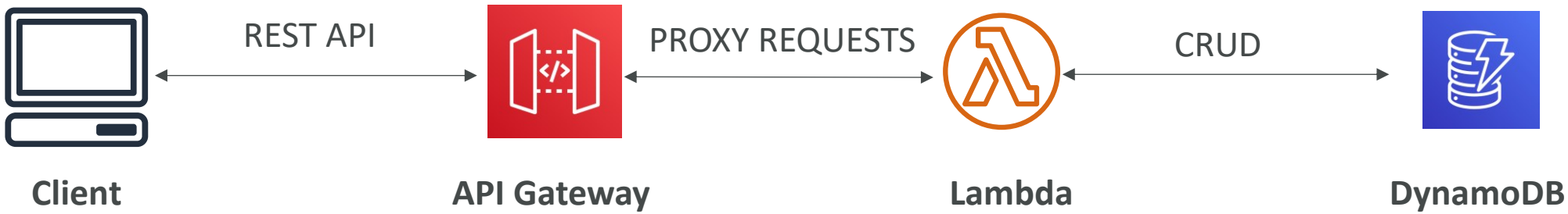
Example: Building a Serverless API



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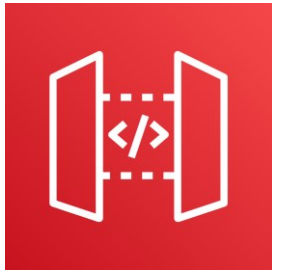
AWS API Gateway



- AWS Lambda + API Gateway: No infrastructure to manage
- Support for the WebSocket Protocol
- Handle API versioning (v1, v2…)
- Handle different environments (dev, test, prod…)
- Handle security (Authentication and Authorization)
- Create API keys, handle request throttling
- Swagger / Open API import to quickly define APIs
- Transform and validate requests and responses
- Generate SDK and API specifications
- Cache API responses

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API Gateway – Integrations High Level

- **Lambda Function**
  - Invoke Lambda function
  - Easy way to expose REST API backed by AWS Lambda
- **HTTP**
  - Expose HTTP endpoints in the backend
  - Example: internal HTTP API on premise, Application Load Balancer…
  - Why? Add rate limiting, caching, user authentications, API keys, etc…
- **AWS Service**
  - Expose any AWS API through the API Gateway
  - Example: start an AWS Step Function workflow, post a message to SQS
  - Why? Add authentication, deploy publicly, rate control…

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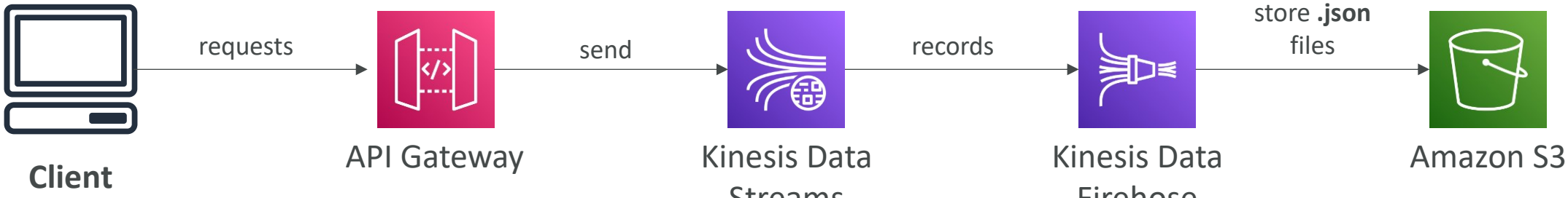
API Gateway – AWS Service Integration Kinesis Data Streams example



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API Gateway - Endpoint Types

- **Edge-Optimized (default):** For global clients
  - Requests are routed through the CloudFront Edge locations (improves latency)
  - The API Gateway still lives in only one region
- **Regional:**
  - For clients within the same region
  - Could manually combine with CloudFront (more control over the caching strategies and the distribution)
- **Private:**
  - Can only be accessed from your VPC using an interface VPC endpoint (ENI)
  - Use a resource policy to define access

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API Gateway – Security

- **User Authentication through**
  - IAM Roles (useful for internal applications)
  - Cognito (identity for external users – example mobile users)
  - Custom Authorizer (your own logic)

- **Custom Domain Name HTTPS** security through integration with AWS Certificate Manager (ACM)
  - If using Edge-Optimized endpoint, then the certificate must be in **us-east-1**
  - If using Regional endpoint, the certificate must be in the API Gateway region
  - Must setup CNAME or A-alias record in Route 53

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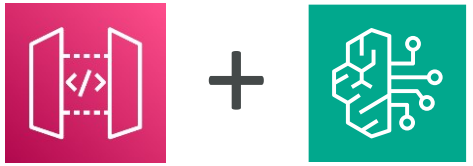
API Gateway and GenAI

- Use as a front for feedback collection
- Use as a front for your models
- API Gateway usage plans
  - Base on your model capacity and client requirements
  - Throttling (~10-50 requests per second)
  - Burst capacity (~2-3X)
- Response filtering
- Retry strategies when models time out
- Use for token limit management
  - Request validator + JSON schema definition can enforce constraints
- Implement routing logic with request transformations
- Validate required fields



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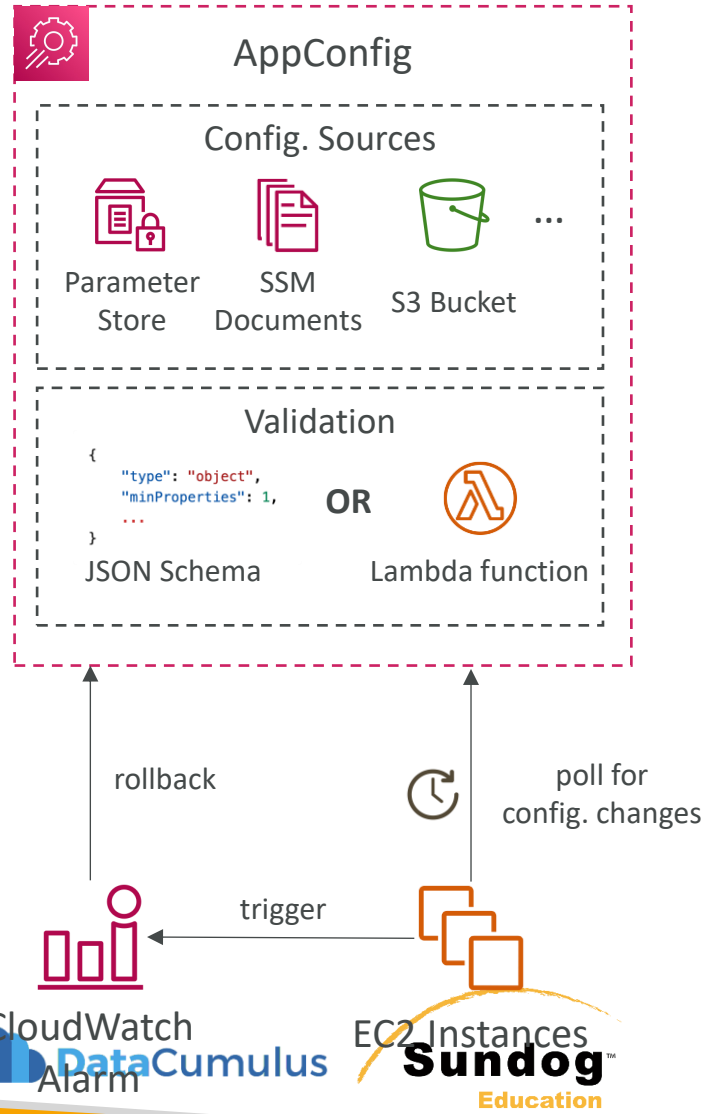
AWS AppConfig

- Configure, validate, and deploy dynamic configurations
- Deploy dynamic configuration changes to your applications independently of any code deployments
  - You don't need to restart the application
- Feature flags, application tuning, allow/block listing…
- Use with apps on EC2 instances, Lambda, ECS, EKS…
- Gradually deploy the configuration changes and rollback if issues occur
- Validate configuration changes before deployment using:
  - **JSON Schema** (syntactic check) or
  - **Lambda Function –** run code to perform validation (semantic check)



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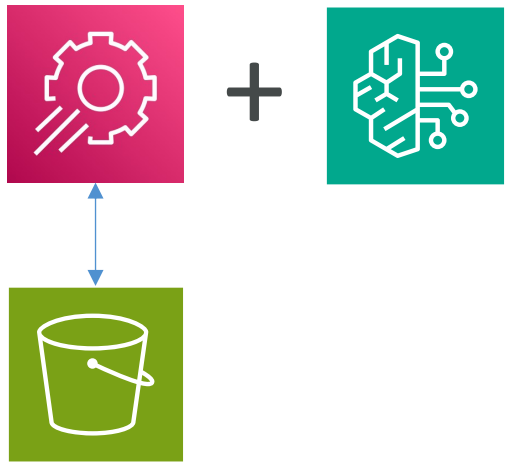
AppConfig and GenAI

- This is a possible solution for dynamic FM selection
- No code changes required to switch models
  - Use feature flags and configuration profiles
- Configuration may be stored in S3
  - AppConfig is integrated with S3
- Can be used as a tool for rolling out new models, A/B testing, rollbacks
  - Of course SageMaker does all this too.
  - Bedrock Evaluations and CloudWatch Evidently can also manage A/B tests



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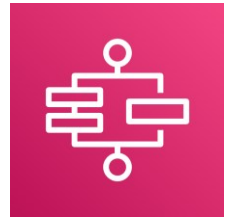
AWS Step Functions



- **Use to design workflows**
- Easy visualizations
- Advanced Error Handling and Retry mechanism outside the code
- Audit of the history of workflows
- Ability to “Wait” for an arbitrary amount of time
- Max execution time of a State Machine is 1 year

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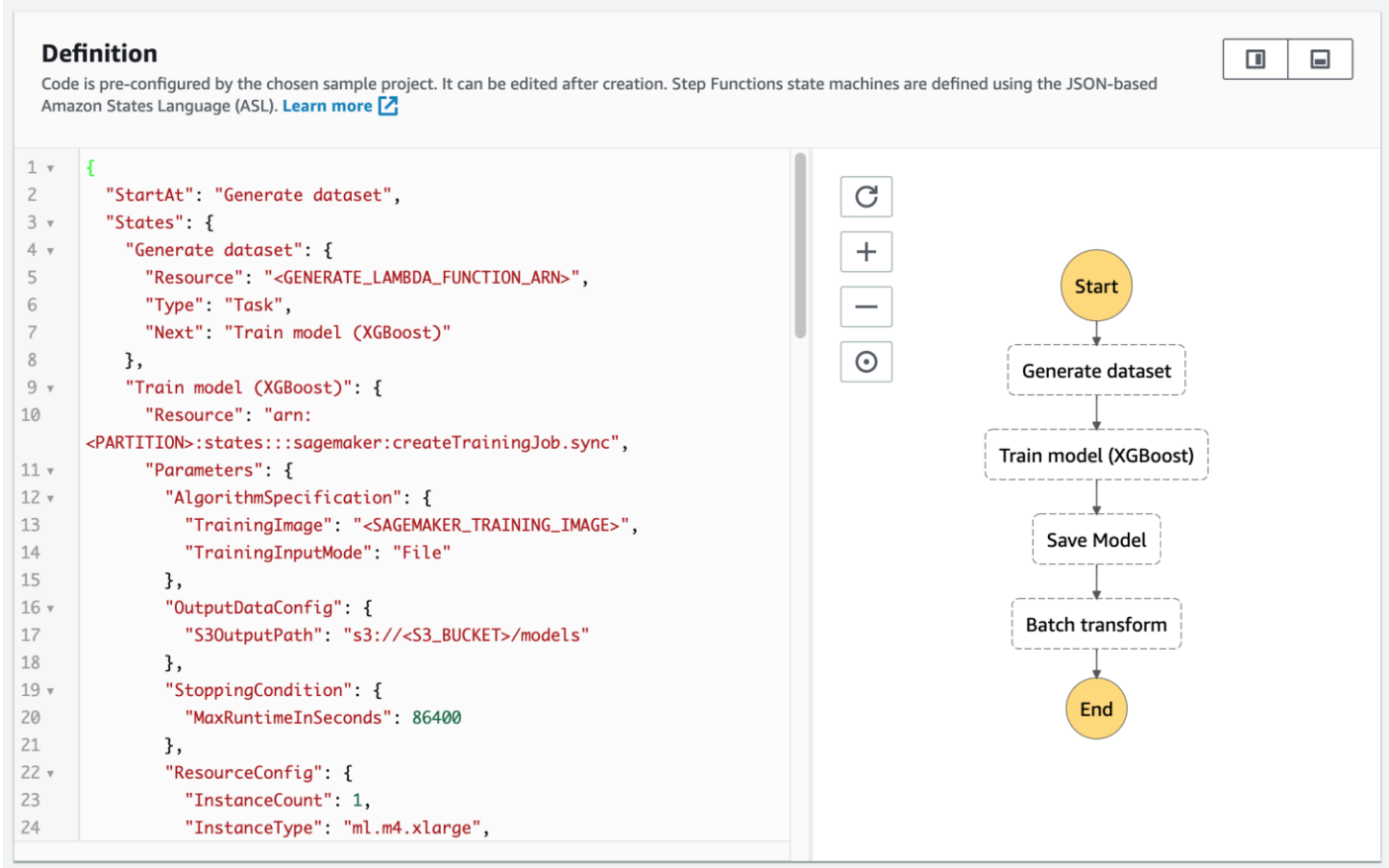
Step Functions – Examples
Train a Machine Learning Model



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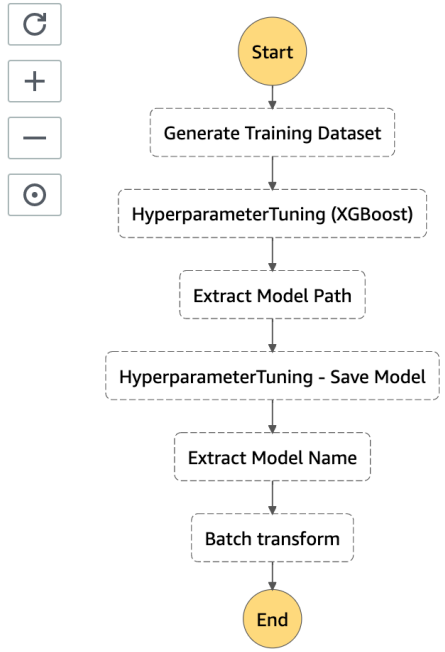
Step Functions – Examples
Tune a Machine Learning Model

Definition

Code is pre-configured by the chosen sample project. It can be edited after creation. Step Functions state machines are defined using the JSON-based Amazon States Language (ASL). Learn more

```
{
    "StartAt": "Generate Training Dataset",
    "States": {
        "Generate Training Dataset": {
            "Resource": "<GENERATE_LAMBDA_FUNCTION_ARN>",
            "Type": "Task",
            "Next": "HyperparameterTuning (XGBoost)"
        },
        "HyperparameterTuning (XGBoost)": {
            "Resource": "arn:
<PARTITION>:states:::sagemaker:createHyperParameterTuningJob.sync",
            "Parameters": {
                "HyperParameterTuningJobName.$": "
<JOB_NAME_FROM_LAMBDA>",
                "HyperParameterTuningJobConfig": {
                    "Strategy": "Bayesian",
                    "HyperParameterTuningJobObjective": {
                        "Type": "Minimize",
                        "MetricName": "validation:rmse"
                    },
                    "ResourceLimits": {
                        "MaxNumberOfTrainingJobs": 2,
                        "MaxParallelTrainingJobs": 2
                    },
                    "ParameterRanges": {
```



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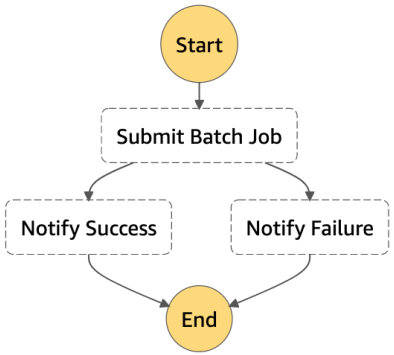
Step Functions – Examples
Manage a Batch Job

Definition

Code is pre-configured by the chosen sample project. It can be edited after creation. Step Functions state machines are defined using the JSON-based Amazon States Language (ASL). Learn more

```
{
  "Comment": "An example of the Amazon States Language for notification on an AWS Batch job completion",
  "StartAt": "Submit Batch Job",
  "TimeoutSeconds": 3600,
  "States": {
    "Submit Batch Job": {
      "Type": "Task",
      "Resource": "arn:<PARTITION>:states:::batch:submitJob.sync",
      "Parameters": {
        "JobName": "BatchJobNotification",
        "JobQueue": "<BATCH_QUEUE_ARN>",
        "JobDefinition": "<BATCH_JOB_DEFINITION_ARN>"
      },
      "Next": "Notify Success",
      "Catch": [
        {
          "ErrorEquals": [ "States.ALL" ],
          "Next": "Notify Failure"
        }
      ]
    },
    "Notify Success": {
      "Type": "Task",
      "Resource": "arn:<PARTITION>:states:::sns:publish",
```



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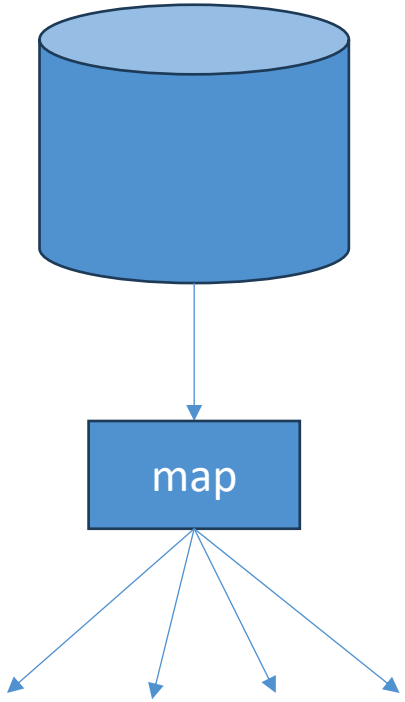
AWS Step Functions

- Your workflow is called a *state machine*
- Each step in a workflow is a *state*
- Types of states
  - **Task**: Does something with Lambda, other AWS services, or third party API's
  - **Choice**: Adds conditional logic via Choice Rules (ie, comparisons)
  - **Wait**: Delays state machine for a specified time
  - **Parallel**: Add separate branches of execution
  - **Map**: Run a set of steps for each item in a dataset, in parallel
    - **This is most relevant to data engineering! Works with JSON, S3 objects, CSV files**
  - Also **Pass**, **Succeed**, **Fail**



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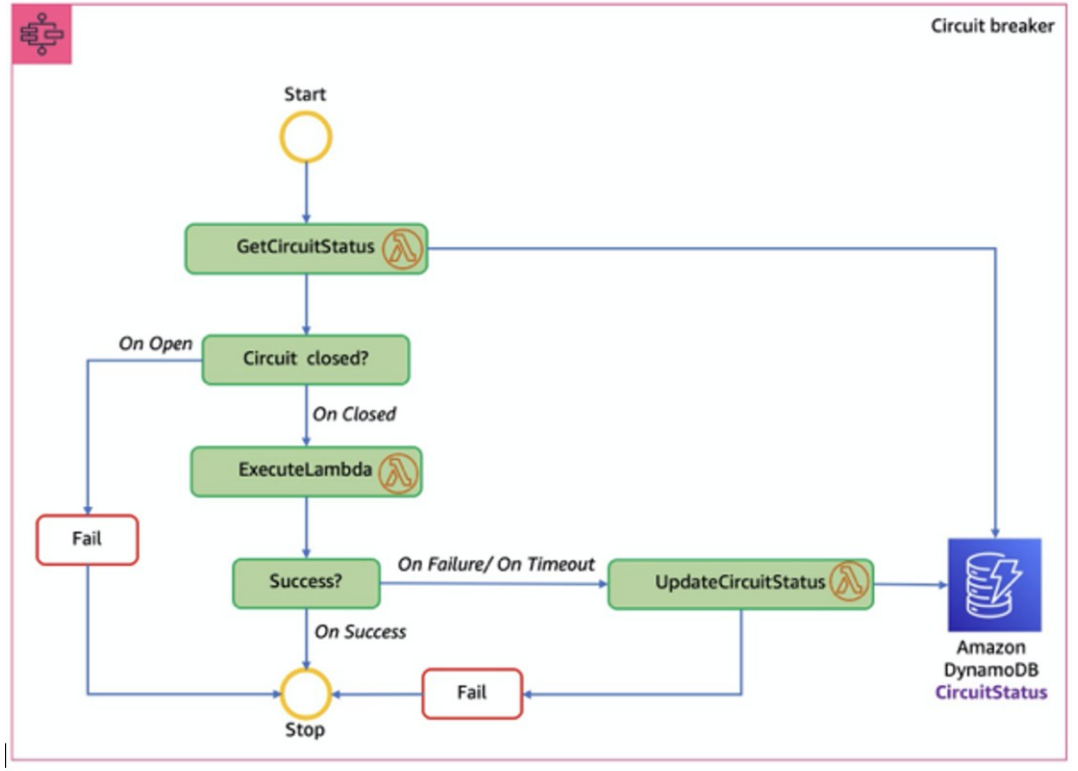
Step Functions: Circuit Breakers

- Circuit Breaker is a pattern
  - Prevents calling a service that is timing out and failing
  - Detects when it's functional again
  - Can be implemented with Step Functions, Lambda, and DynamoDB
- This can be used to safeguard your AI workflows
  - Route requests to fallback models if your model is down
  - Or enter a degraded service mode



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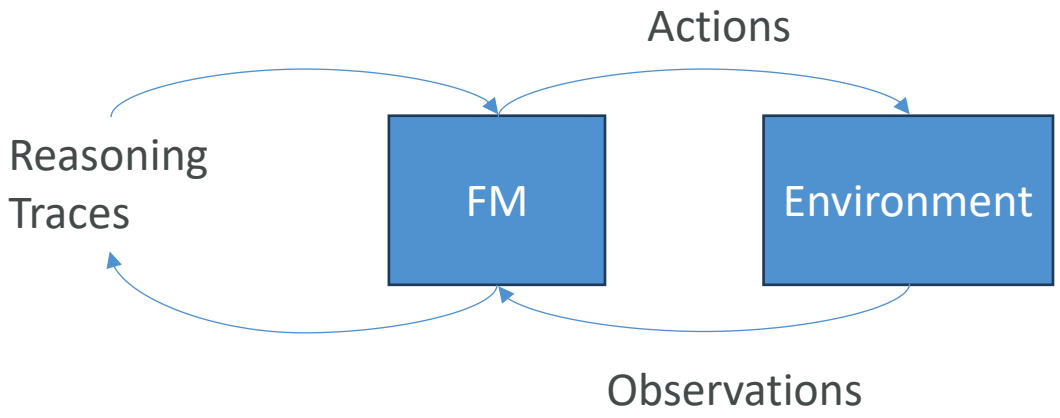
More Step Functions + GenAI

- ReAct patterns (Reasoning and Acting)
  - Step functions orchestrate steps in a structured workflow
  - Chain of thought reasoning
  - Dynamic routing to specialized FMs
- Orchestrate model review and approval process



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Even More Step Functions Stuff

- There is a 256KB limit between data passed between steps
  - You might squirrel data away in DynamoDB or S3 if this is a problem
- Step Functions is integrated with Bedrock
  - InvokeModel, CreateModelCustomizationJob both supported
  - You can chain multiple FM calls together using Step Functions
  - Note that InvokeModel can also include guardrails you specify
    - You can chain those too.

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AWS CodePipeline

- Visual Workflow to orchestrate your CICD
- **Source** – CodeCommit, ECR, S3, Bitbucket, GitHub
- **Build** – CodeBuild, Jenkins, CloudBees, TeamCity
- **Test** – CodeBuild, AWS Device Farm, 3rd party tools, …
- **Deploy** – CodeDeploy, Elastic Beanstalk, CloudFormation, ECS, S3, …
- **Invoke** – Lambda, Step Functions
- Consists of stages:
  - Each stage can have sequential actions and/or parallel actions
  - Example: Build ➔ Test ➔ Deploy ➔ Load Testing ➔ …
  - Manual approval can be defined at any stage

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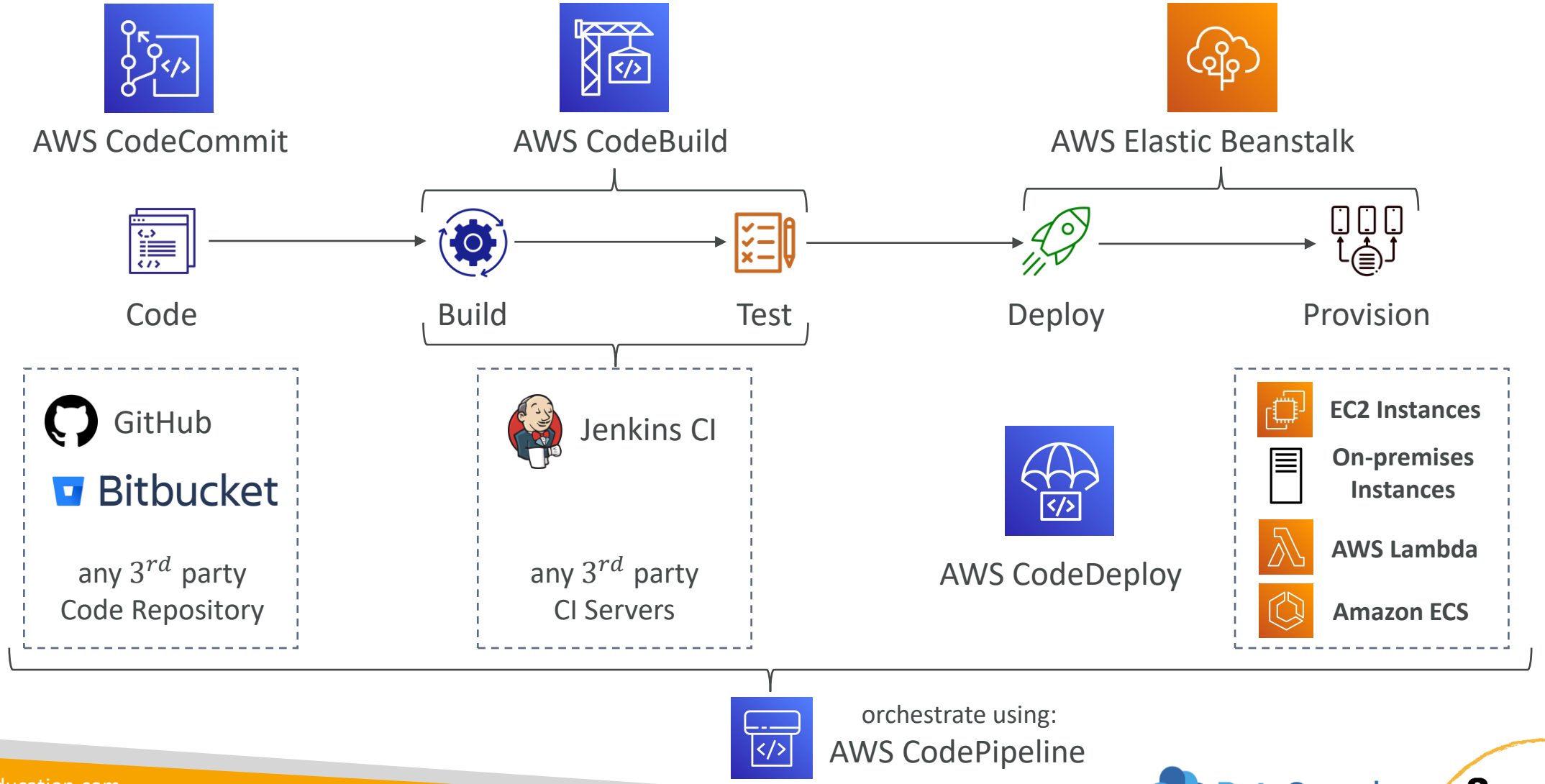
Technology Stack for CICD



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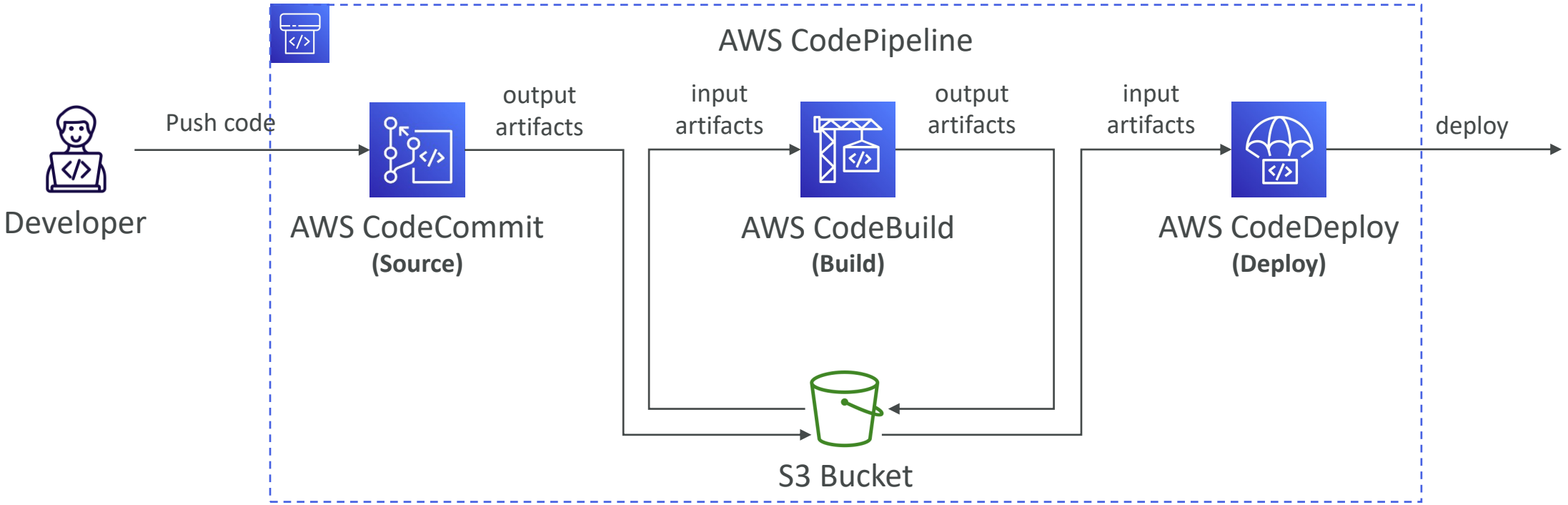
CodePipeline – Artifacts

- Each pipeline stage can create **artifacts**
- Artifacts stored in an S3 bucket and passed on to the next stage



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CodePipeline – Troubleshooting

- For CodePipeline Pipeline/Action/Stage Execution State Changes
- Use **CloudWatch Events (Amazon EventBridge).** Example:
  - You can create events for failed pipelines
  - You can create events for cancelled stages

- If CodePipeline fails a stage, your pipeline stops, and you can get information in the console
- If pipeline can't perform an action, make sure the "IAM Service Role" attached does have enough IAM permissions (IAM Policy)
- AWS CloudTrail can be used to audit AWS API calls

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AWS CodeBuild

- A fully managed continuous integration (CI) service
- Continuous scaling (no servers to manage or provision – no build queue)
- Compile source code, run tests, produce software packages, …
- Alternative to other build tools (e.g., Jenkins)
- Charged per minute for compute resources (time it takes to complete the builds)
- Leverages Docker under the hood for reproducible builds
- Use prepackaged Docker images or create your own custom Docker image
- Security:
  - Integration with KMS for encryption of build artifacts
  - IAM for CodeBuild permissions, and VPC for network security
  - AWS CloudTrail for API calls logging

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AWS CodeBuild

- **Source** – CodeCommit, S3, Bitbucket, GitHub
- **Build instructions**: Code file **buildspec.yml** or insert manually in Console
- **Output logs** can be stored in Amazon S3 & CloudWatch Logs
- Use CloudWatch Metrics to monitor build statistics
- Use EventBridge to detect failed builds and trigger notifications
- Use CloudWatch Alarms to notify if you need “thresholds” for failures

- **Build Projects can be defined within CodePipeline or CodeBuild**

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CodeBuild – Supported Environments

- Java
- Ruby
- Python
- Go
- Node.js
- Android
- .NET Core
- PHP
- Docker – extend any environment you like

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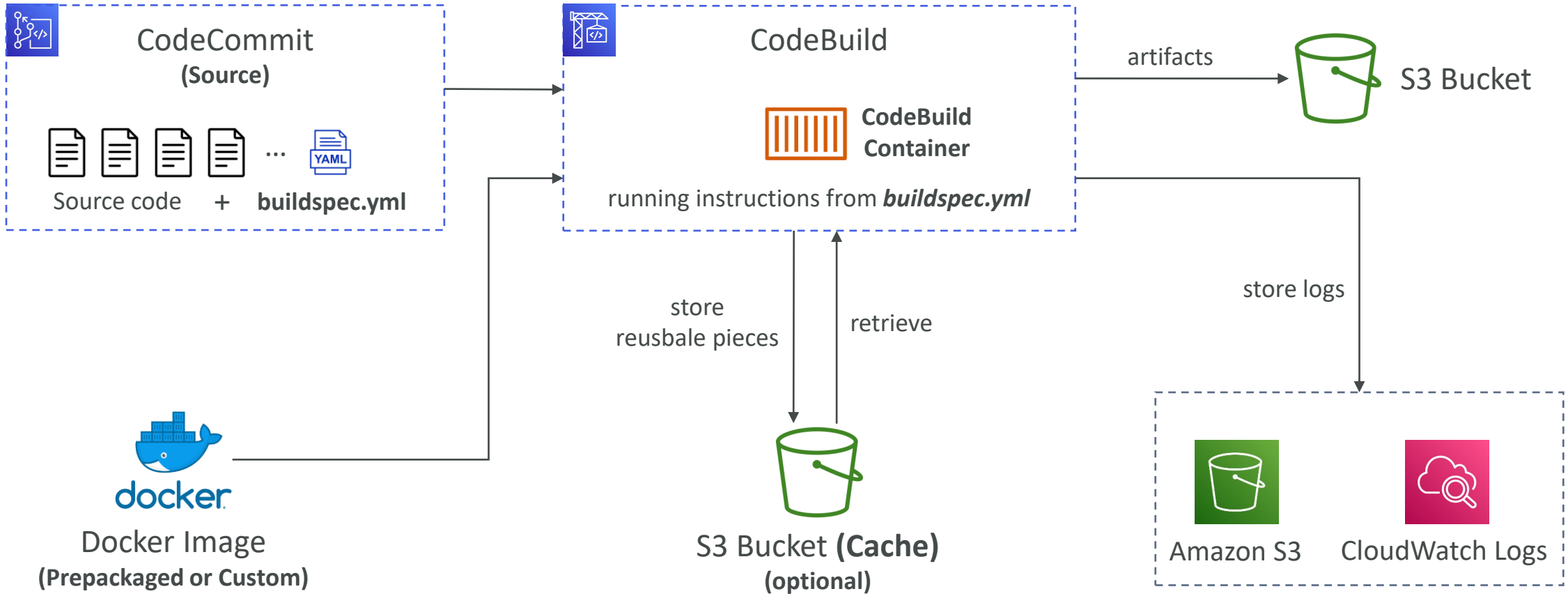
CodeBuild – How it Works



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CodeBuild – buildspec.yml

- **buildspec.yml** file must be at the **root** of your code
- **env** – define environment variables
  - **variables** – plaintext variables
  - **parameter-store** – variables stored in SSM Parameter Store
  - **secrets-manager** – variables stored in AWS Secrets Manager
- **phases** – specify commands to run:
  - **install** – install dependencies you may need for your build
  - **pre_build** – final commands to execute before build
  - **Build – actual build commands**
  - **post_build** – finishing touches (e.g., zip output)
- **artifacts** – what to upload to S3 (encrypted with KMS)
- **cache** – files to cache (usually dependencies) to S3 for future build speedup

```yaml
version: 0.2

env:
  variables:
    JAVA_HOME: "/usr/lib/jvm/java-8-openjdk-amd64"
  parameter-store:
    LOGIN_PASSWORD: /CodeBuild/dockerLoginPassword

phases:
  install:
    commands:
      - echo "Entered the install phase..."
      - apt-get update -y
      - apt-get install -y maven
  pre_build:
    commands:
      - echo "Entered the pre_build phase..."
      - docker login -u User -p $LOGIN_PASSWORD
  build:
    commands:
      - echo "Entered the build phase..."
      - echo "Build started on `date`"
      - mvn install
  post_build:
    commands:
      - echo "Entered the post_build phase..."
      - echo "Build completed on `date`"

artifacts:
  files:
    - target/messageUtil-1.0.jar

cache:
  paths:
    - "/root/.m2/**/*"
```


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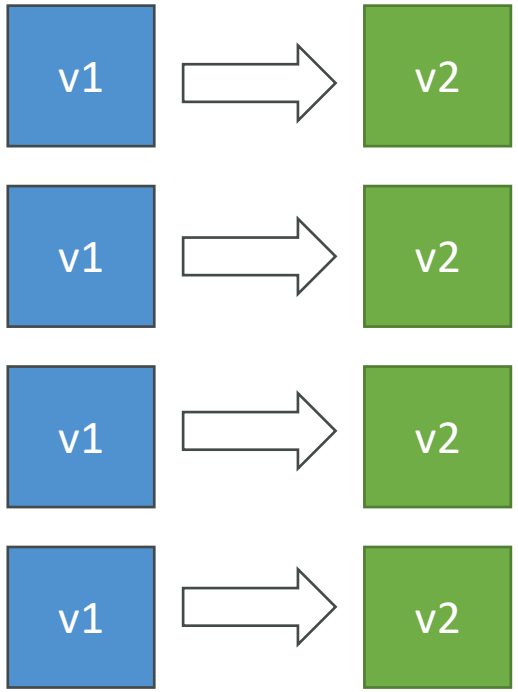
AWS CodeDeploy

- Deployment service that automates application deployment
- Deploy new applications versions to EC2 Instances, On-premises servers, Lambda functions, ECS Services
- Automated Rollback capability in case of failed deployments, or trigger CloudWatch Alarm
- Gradual deployment control
- A file named **appspec.yml** defines how the deployment happens



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CodeDeploy – EC2/On-premises Platform

- Can deploy to EC2 Instances & on-premises servers
- Perform in-place deployments or blue/green deployments
- Must run the **CodeDeploy Agent** on the target instances
- Define deployment speed
  - AllAtOnce: most downtime
  - HalfAtATime: reduced capacity by 50%
  - OneAtATime: slowest, lowest availability impact
  - Custom: define your %

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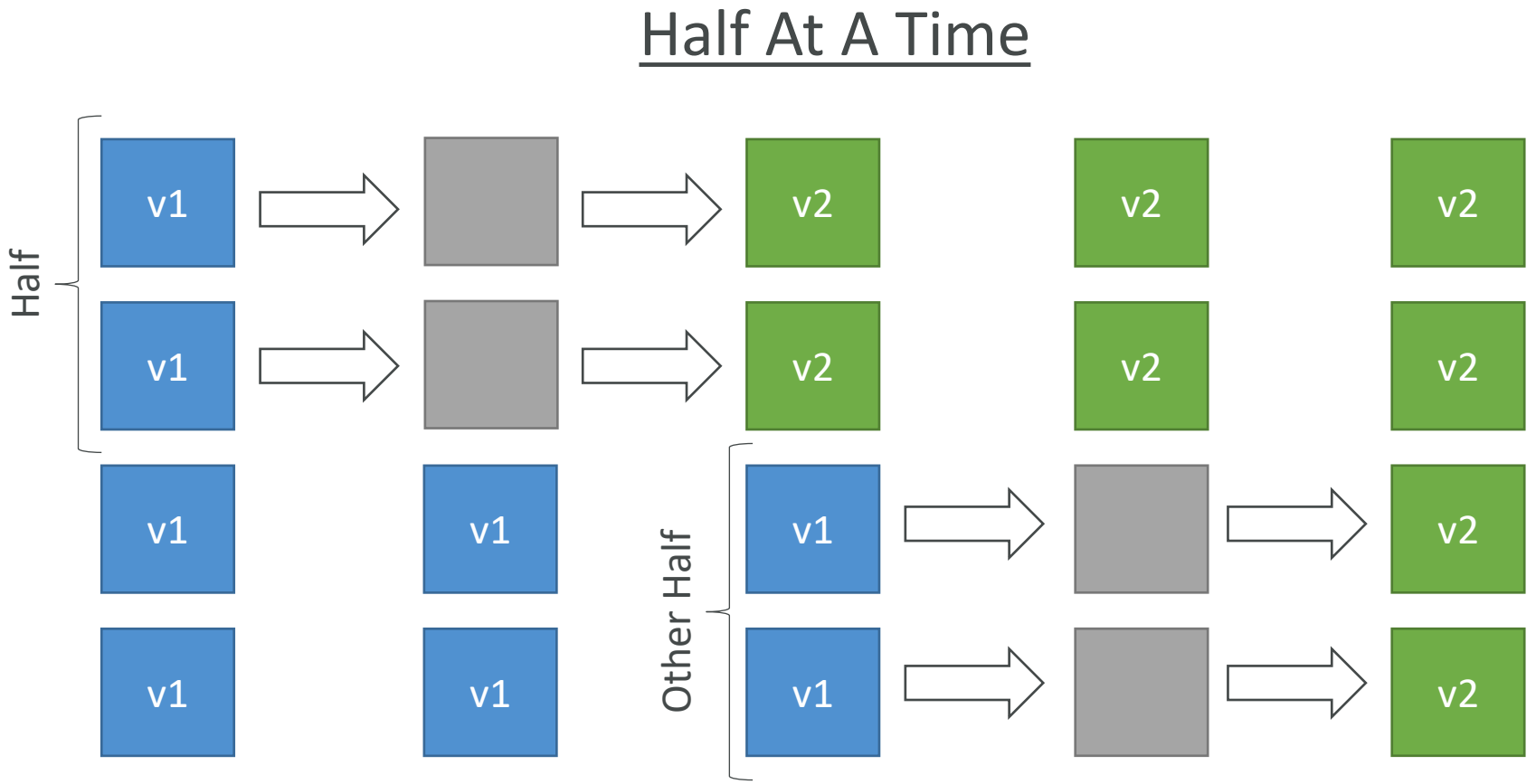
CodeDeploy – In-Place Deployment



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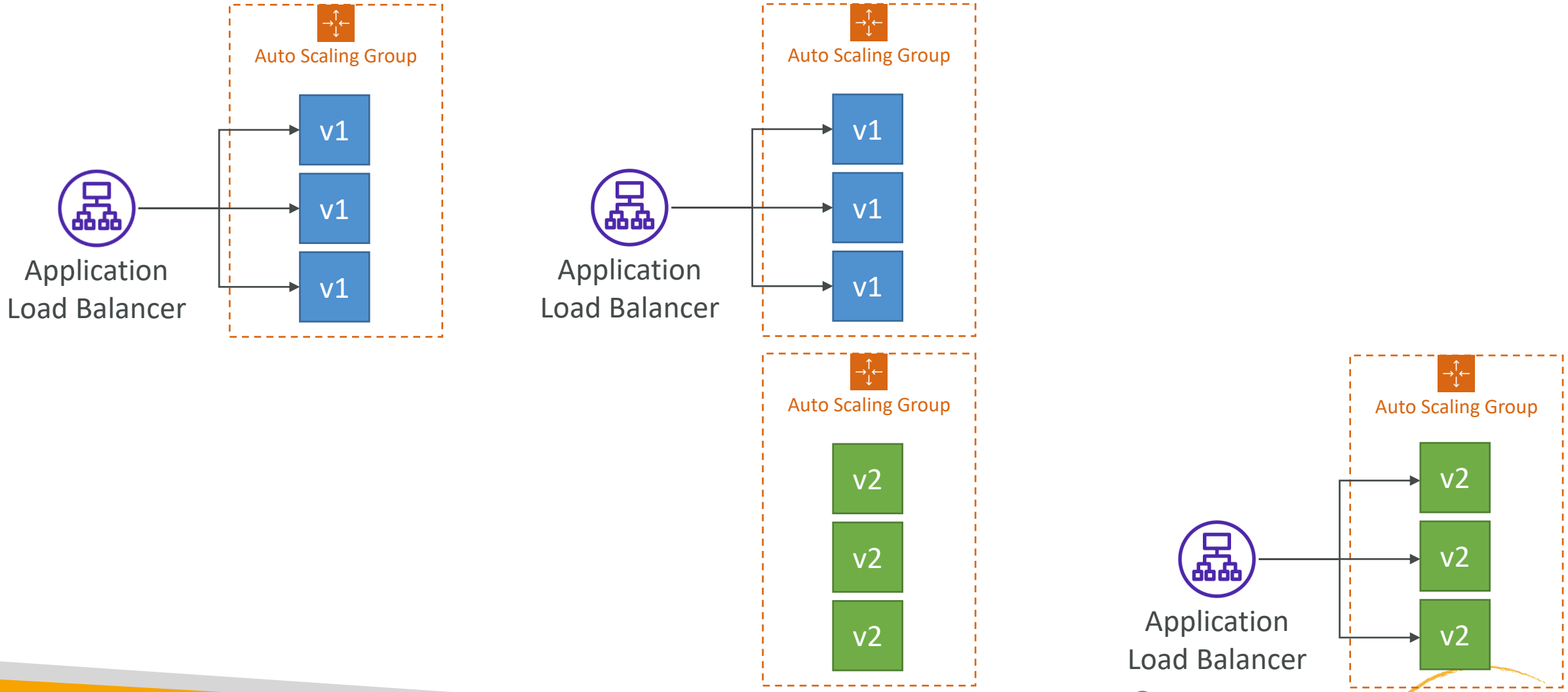
CodeDeploy – Blue-Green Deployment



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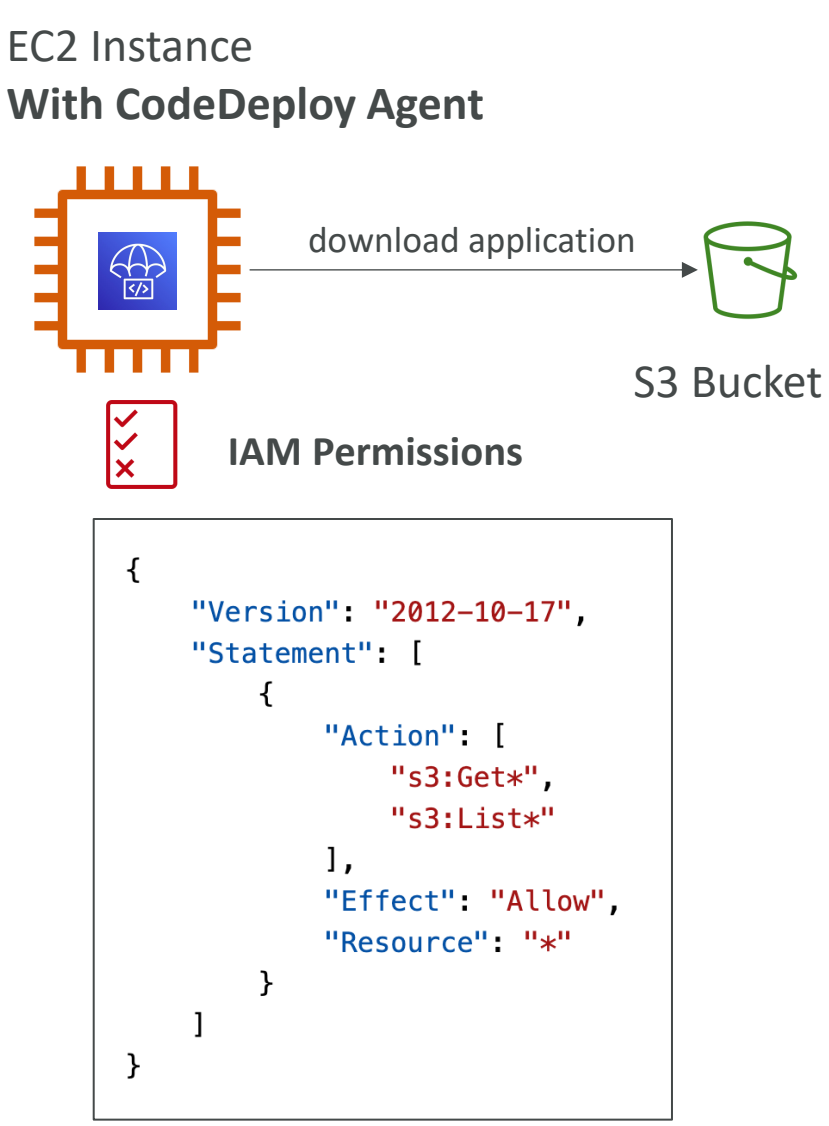
CodeDeploy Agent

- The CodeDeploy Agent must be running on the EC2 instances as a pre-requisites
- It can be installed and updated automatically if you're using Systems Manager
- The EC2 Instances must have sufficient permissions to access Amazon S3 to get deployment bundles



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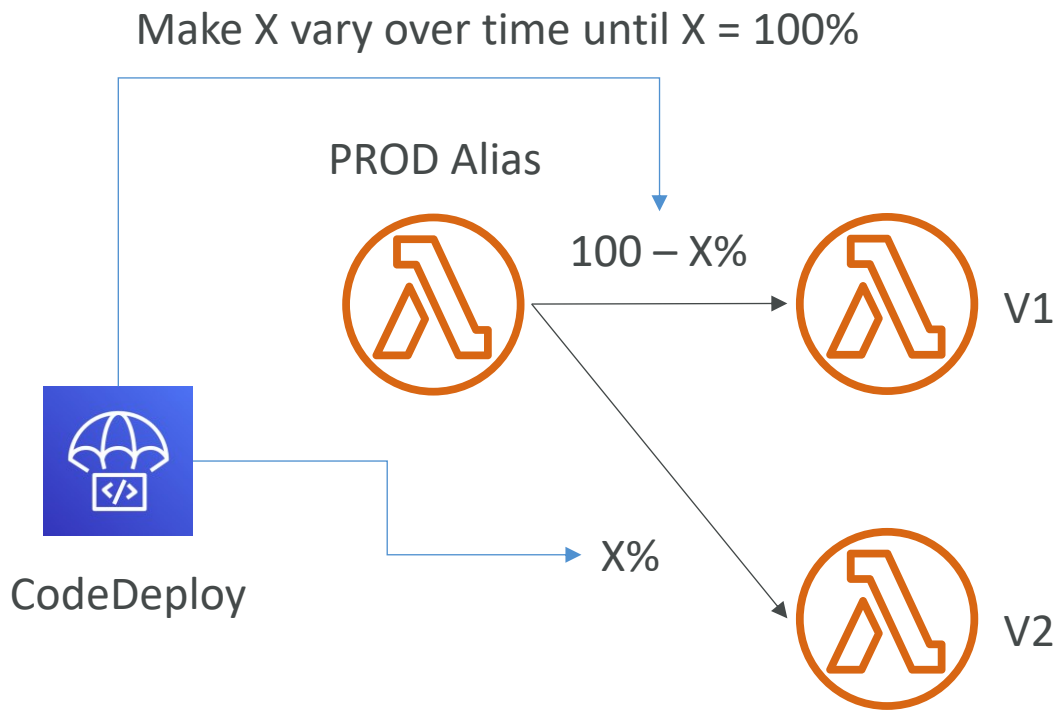
CodeDeploy – Lambda Platform

- **CodeDeploy** can help you automate traffic shift for Lambda aliases
- Feature is integrated within the SAM framework
- **Linear:** grow traffic every N minutes until 100%
  - LambdaLinear10PercentEvery3Minutes
  - LambdaLinear10PercentEvery10Minutes
- **Canary:** try X percent then 100%
  - LambdaCanary10Percent5Minutes
  - LambdaCanary10Percent30Minutes
- **AllAtOnce:** immediate



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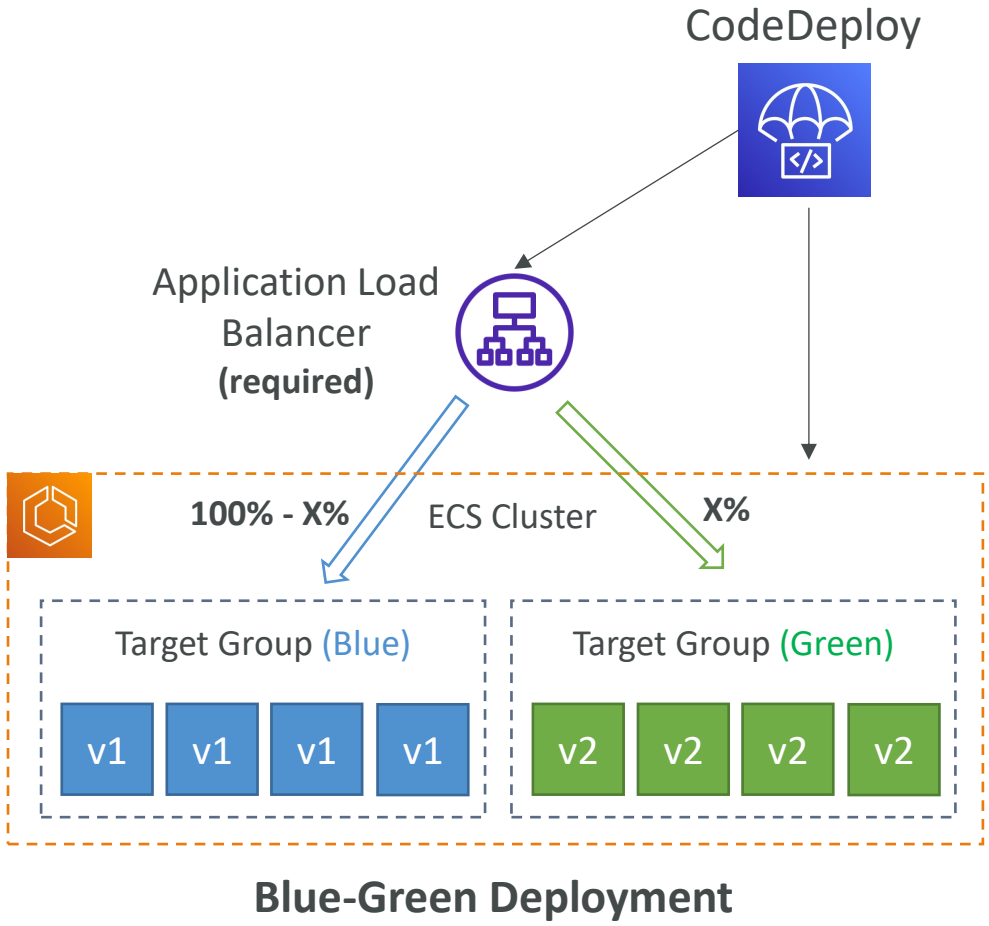
CodeDeploy – ECS Platform

- **CodeDeploy** can help you automate the deployment of a new ECS Task Definition
- Only Blue/Green Deployments
- **Linear:** grow traffic every N minutes until 100%
  - ECSLinear10PercentEvery3Minutes
  - ECSLinear10PercentEvery10Minutes
- **Canary:** try X percent then 100%
  - ECSCanary10Percent5Minutes
  - ECSCanary10Percent30Minutes
- **AllAtOnce:** immediate



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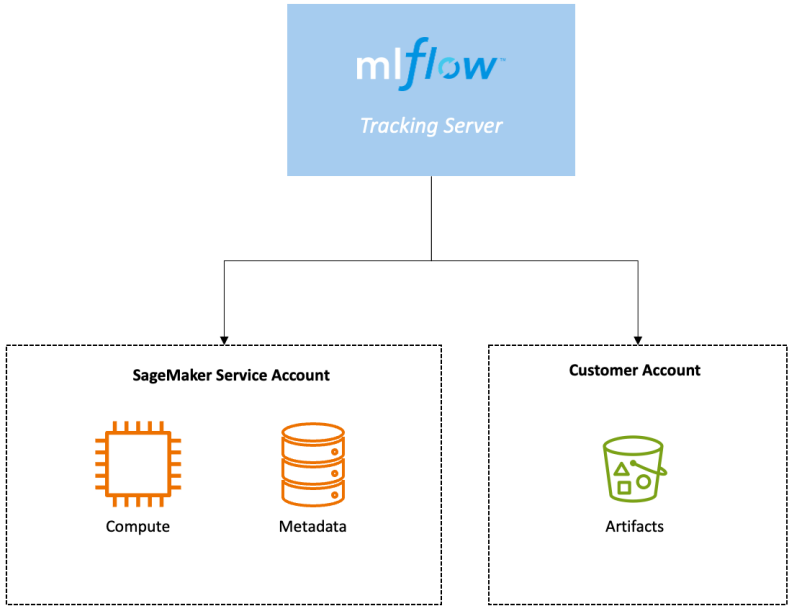
MLFlow

- Open-source platform for ML / GenAI workflows
- Offers observability, evaluations, tracking, tracing, AI gateway
- Model management and deployment
- MLFlow is integrated into Amazon SageMaker AI
  - Managed service
  - Integrates with model development, management, deployment, and tracking
  - Integration points:
    - SageMaker Studio
    - SageMaker Model Registry
    - SageMaker AI Inference
    - IAM
    - CloudTrail
    - EventBridge



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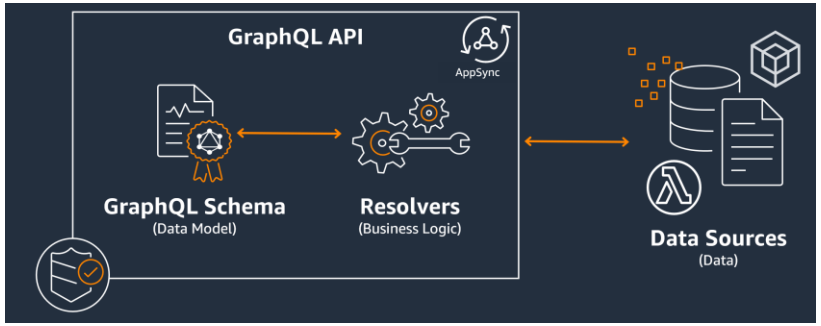
AWS AppSync GraphQL and GenAI

- Connects apps and services to data and events
  - GraphQL and Pub/Sub API's
    - GraphQL is what it sounds like – Graph query language
    - Can retrieve data from different sources and return it as a graph
    - Developed by Facebook
  - Serverless
  - JavaScript / TypeScript support for business logic
- GenAI applications
  - Integrate Lambda with AppSync resolvers for real-time FM inference
  - Transform GraphQL requests to model inputs using Apache Velocity Template Language (VTL) mapping
    - VTL is just what AppSync uses under the hood to translate GraphQL requests to your data source

AWS AppSync



AWS documentation

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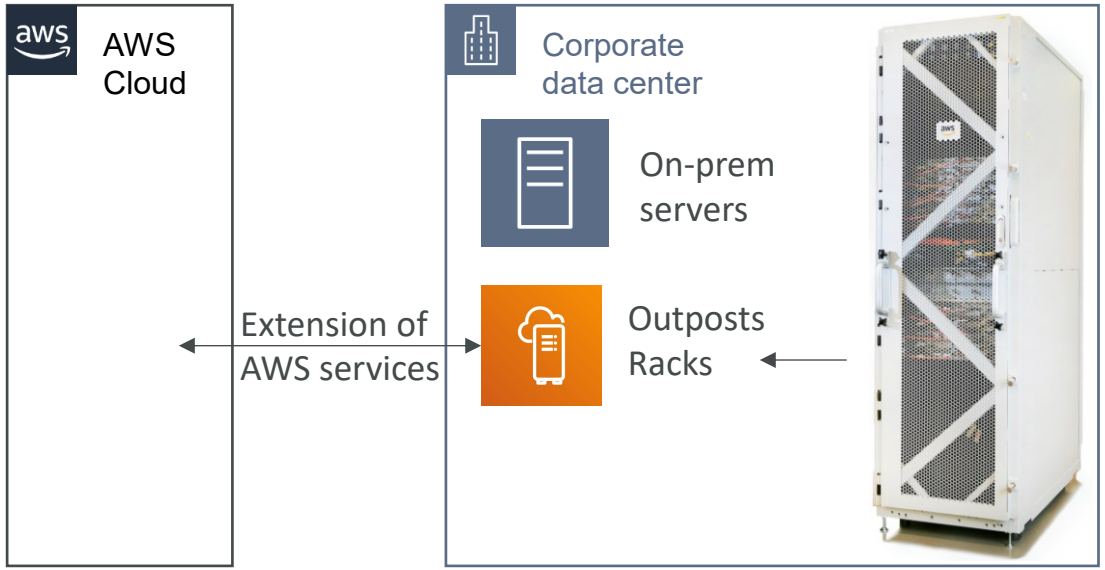
AWS Outposts

- **Hybrid Cloud**: businesses that keep an on-premises infrastructure alongside a cloud infrastructure
- Therefore, two ways of dealing with IT systems:
  - One for the AWS cloud (using the AWS console, CLI, and AWS APIs)
  - One for their on-premises infrastructure
- **AWS Outposts are "server racks"** that offers the same AWS infrastructure, services, APIs & tools to build your own applications on-premises just as in the cloud
- **AWS will setup and manage "Outposts Racks"** within your on-premises infrastructure and you can start leveraging AWS services on-premises
- You are responsible for the Outposts Rack physical security



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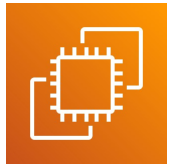
AWS Outposts

- Benefits:
  - Low-latency access to on-premises systems
  - Local data processing
  - Data residency
  - Easier migration from on-premises to the cloud
  - Fully managed service
- Some services that work on Outposts:


Amazon EC2

Amazon EBS

Amazon S3

Amazon EKS

Amazon ECS

Amazon RDS

Amazon EMR

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AWS Outposts and GenAI

- Relevant for data compliance across jurisdictions
  - Different countries have different AI laws
- Data privacy
  - On-premise data integration
- Need sufficient compute and storage for your FM inference
  - Based on your on-prem data volume and requirements
  - Local caching can minimize data movement

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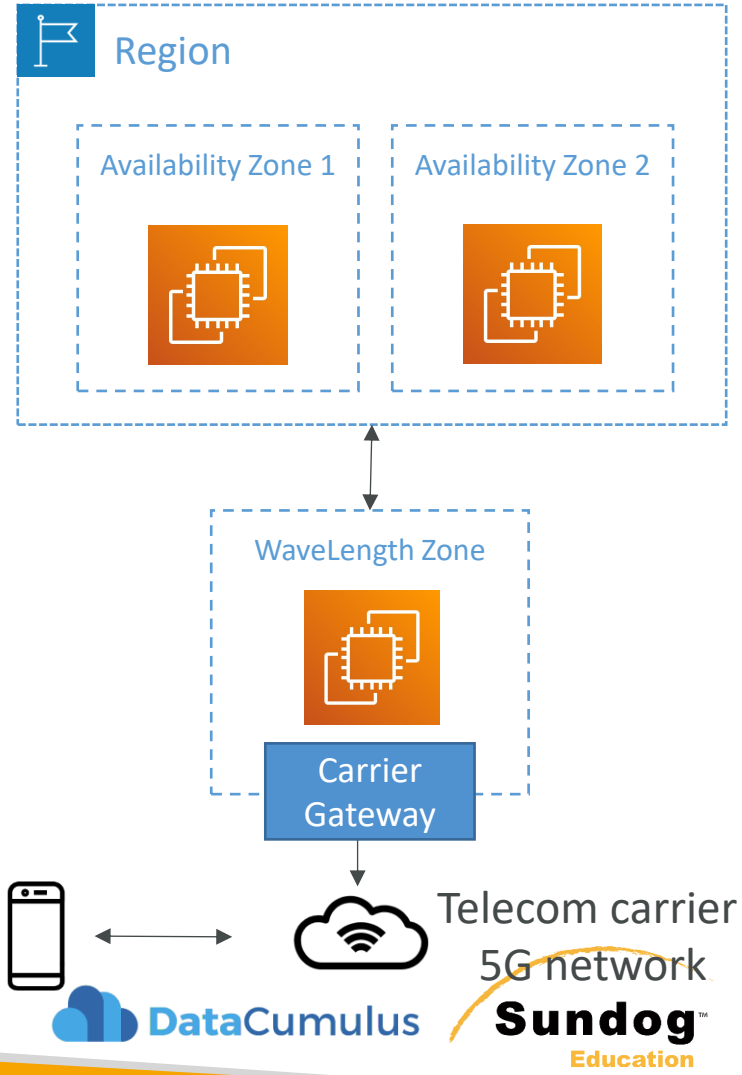
AWS WaveLength

- **WaveLength Zones** are infrastructure deployments embedded within the telecommunications providers' datacenters at the edge of the 5G networks
- Brings AWS services to the edge of the 5G networks
- Example: EC2, EBS, VPC…
- Ultra-low latency applications through 5G networks
- Traffic doesn't leave the Communication Service Provider's (CSP) network
- High-bandwidth and secure connection to the parent AWS Region
- No additional charges or service agreements
- Use cases: Smart Cities, ML-assisted diagnostics, Connected Vehicles, Interactive Live Video Streams, AR/VR, Real-time Gaming, …



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AWS WaveLength and GenAI

- Edge deployments
- Secure routing between cloud and on-prem
- Mobile foundation model apps
- Distribute traffic between Wavelength Zones and parent regions based on workload
  - Low-latency, lighter work at the edge
  - Heavier in the Region

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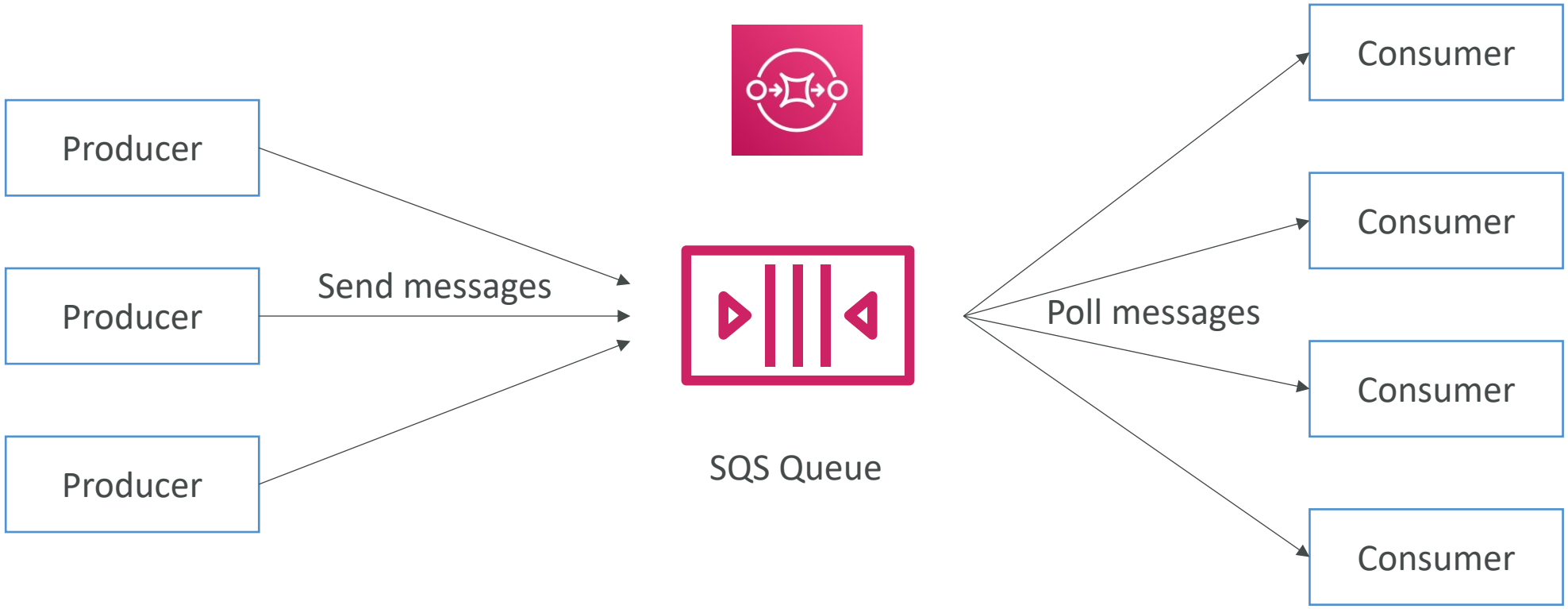
Amazon SQS
What's a queue?



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Amazon SQS – Standard Queue

- Oldest offering (over 10 years old)
- Fully managed service, used to **decouple applications**

- Attributes:
  - Unlimited throughput, unlimited number of messages in queue
  - Default retention of messages: 4 days, maximum of 14 days
  - Low latency (<10 ms on publish and receive)
  - Limitation of 1,024 KB per message sent

- Can have duplicate messages (at least once delivery, occasionally)
- Can have out of order messages (best effort ordering)

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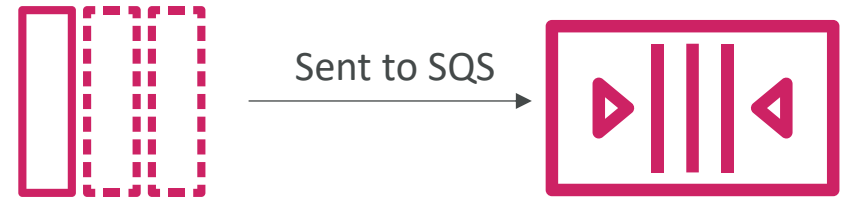
SQS – Producing Messages

- Produced to SQS using the SDK (SendMessage API)
- The message is **persisted** in SQS until a consumer deletes it
- Message retention: default 4 days, up to 14 days

- Example: send an order to be processed
  - Order id
  - Customer id
  - Any attributes you want



Message
Up to 1024 KB

- SQS standard: unlimited throughput

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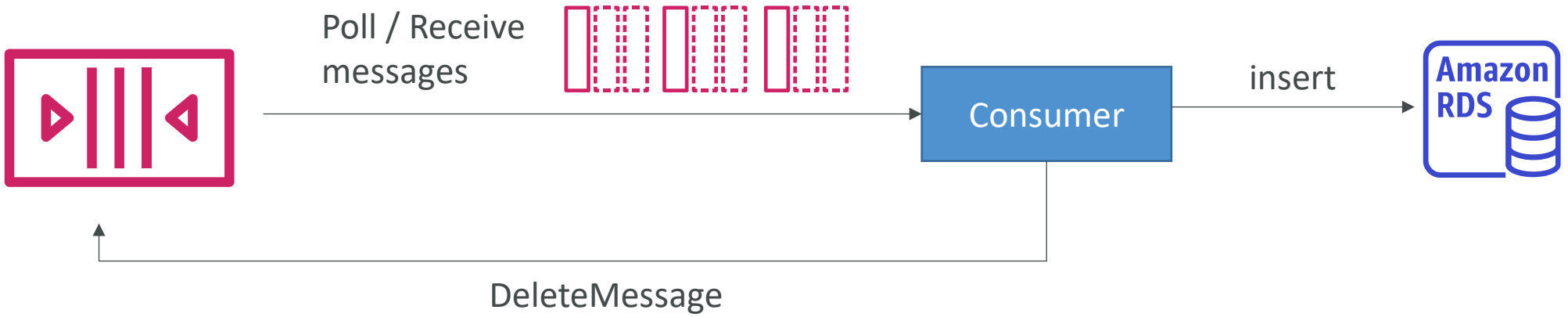
SQS – Consuming Messages

- Consumers (running on EC2 instances, servers, or AWS Lambda)…
- Poll SQS for messages (receive up to 10 messages at a time)
- Process the messages (example: insert the message into an RDS database)
- Delete the messages using the DeleteMessage API



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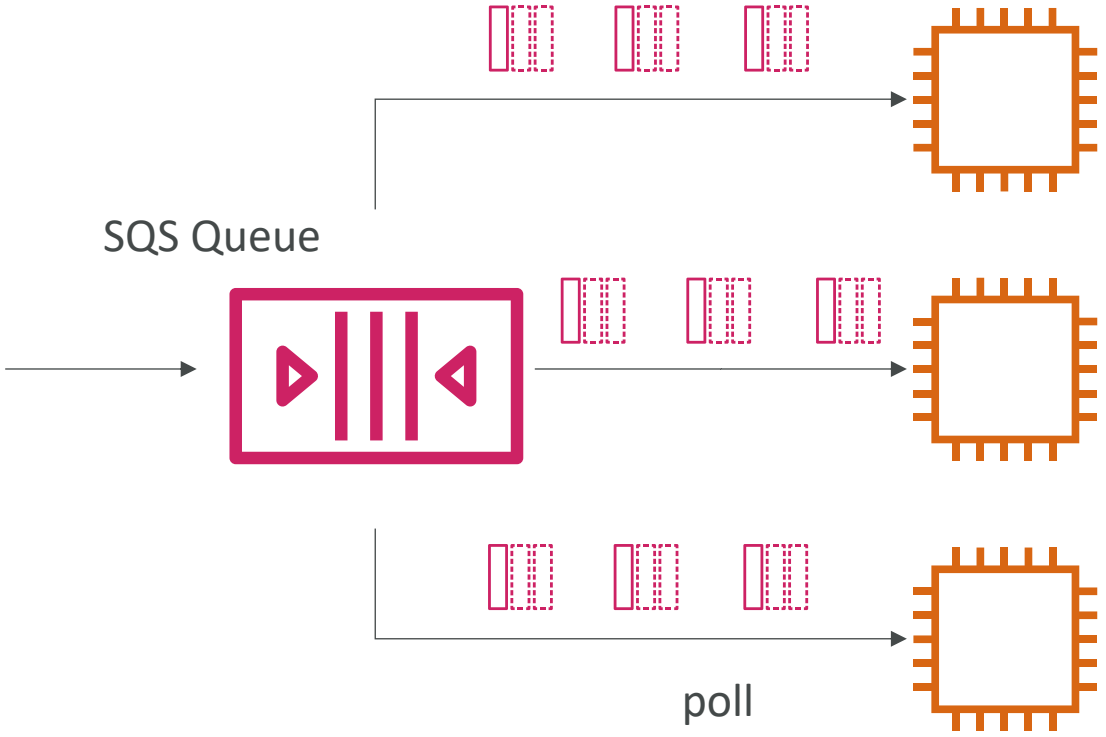
SQS – Multiple EC2 Instances Consumers



- Consumers receive and process messages in parallel
- At least once delivery
- Best-effort message ordering
- Consumers delete messages after processing them
- We can scale consumers horizontally to improve throughput of processing

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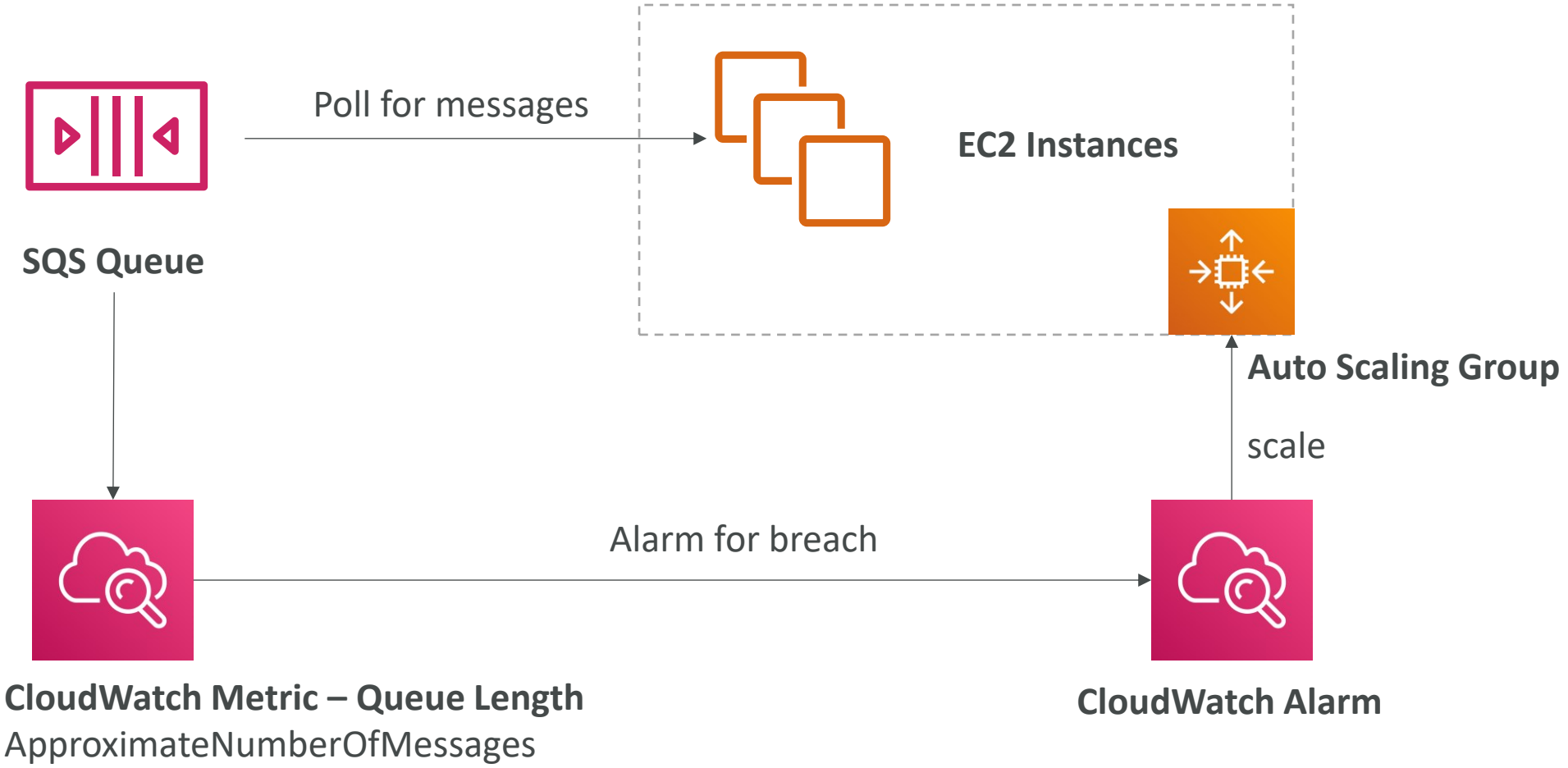
SQS with Auto Scaling Group (ASG)



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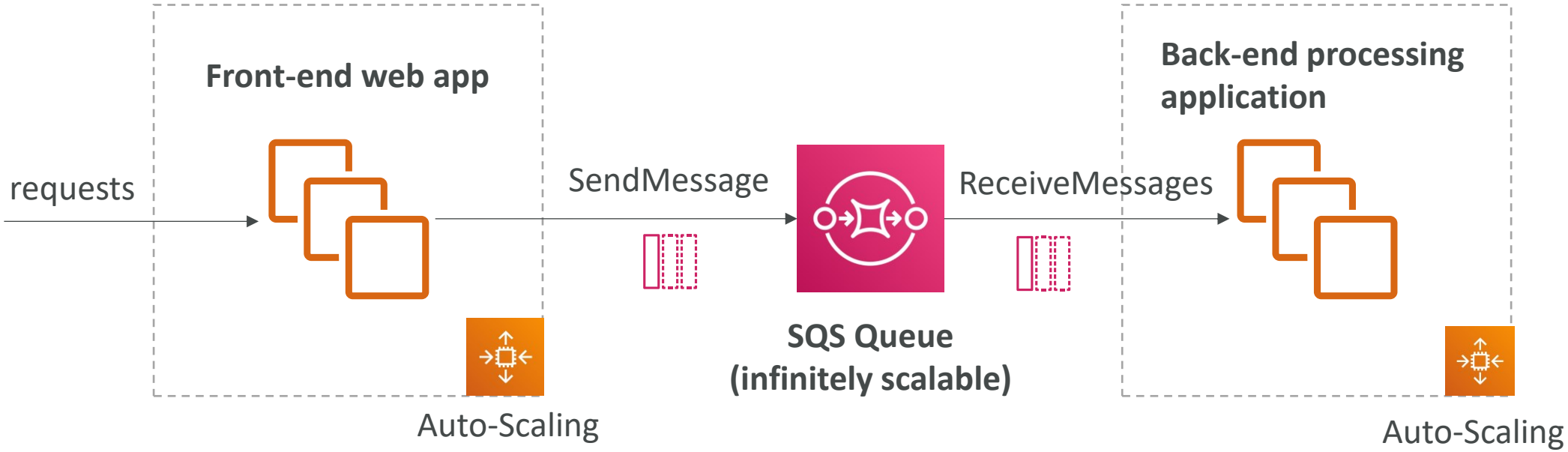
SQS to **decouple** between application tiers



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Amazon SQS - Security

- **Encryption:**
  - In-flight encryption using HTTPS API
  - At-rest encryption using KMS keys
  - Client-side encryption if the client wants to perform encryption/decryption itself
- **Access Controls:** IAM policies to regulate access to the SQS API
- **SQS Access Policies** (similar to S3 bucket policies)
  - Useful for cross-account access to SQS queues
  - Useful for allowing other services (SNS, S3…) to write to an SQS queue

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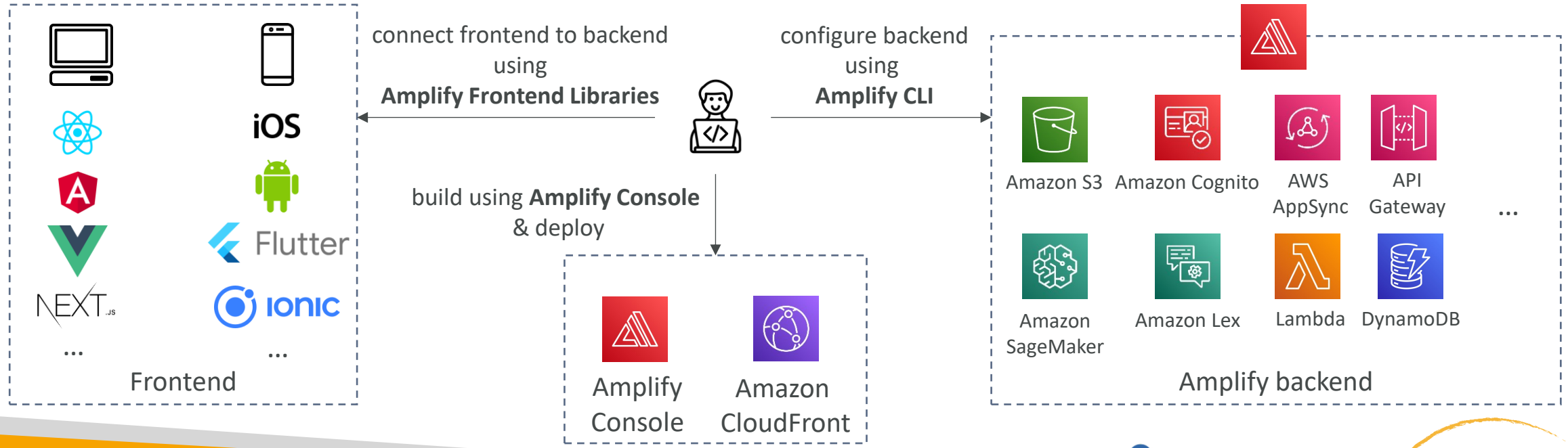
AWS Amplify - web and mobile applications

- A set of tools and services that helps you develop and deploy scalable full stack web and mobile applications
- Authentication, Storage, API (REST, GraphQL), CI/CD, PubSub, Analytics, AI/ML Predictions, Monitoring, …
- Connect your source code from GitHub, AWS CodeCommit, Bitbucket, GitLab, or upload directly



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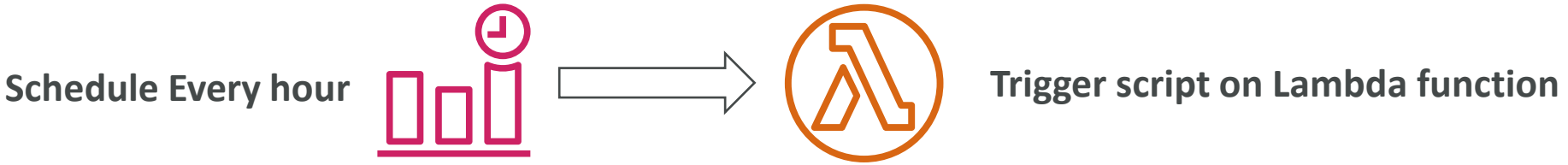
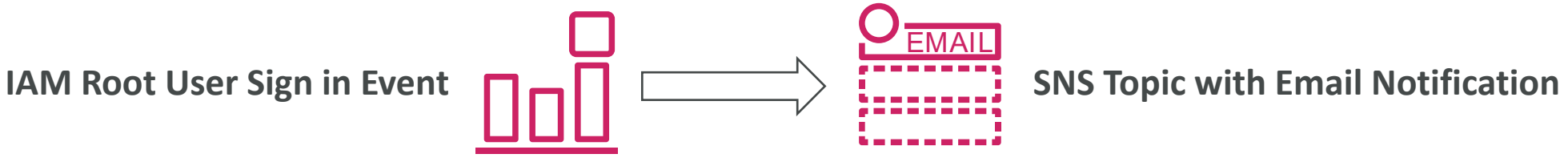
Amazon EventBridge (formerly CloudWatch Events)

- Schedule: Cron jobs (scheduled scripts)



- Event Pattern: Event rules to react to a service doing something



- Trigger Lambda functions, send SQS/SNS messages…

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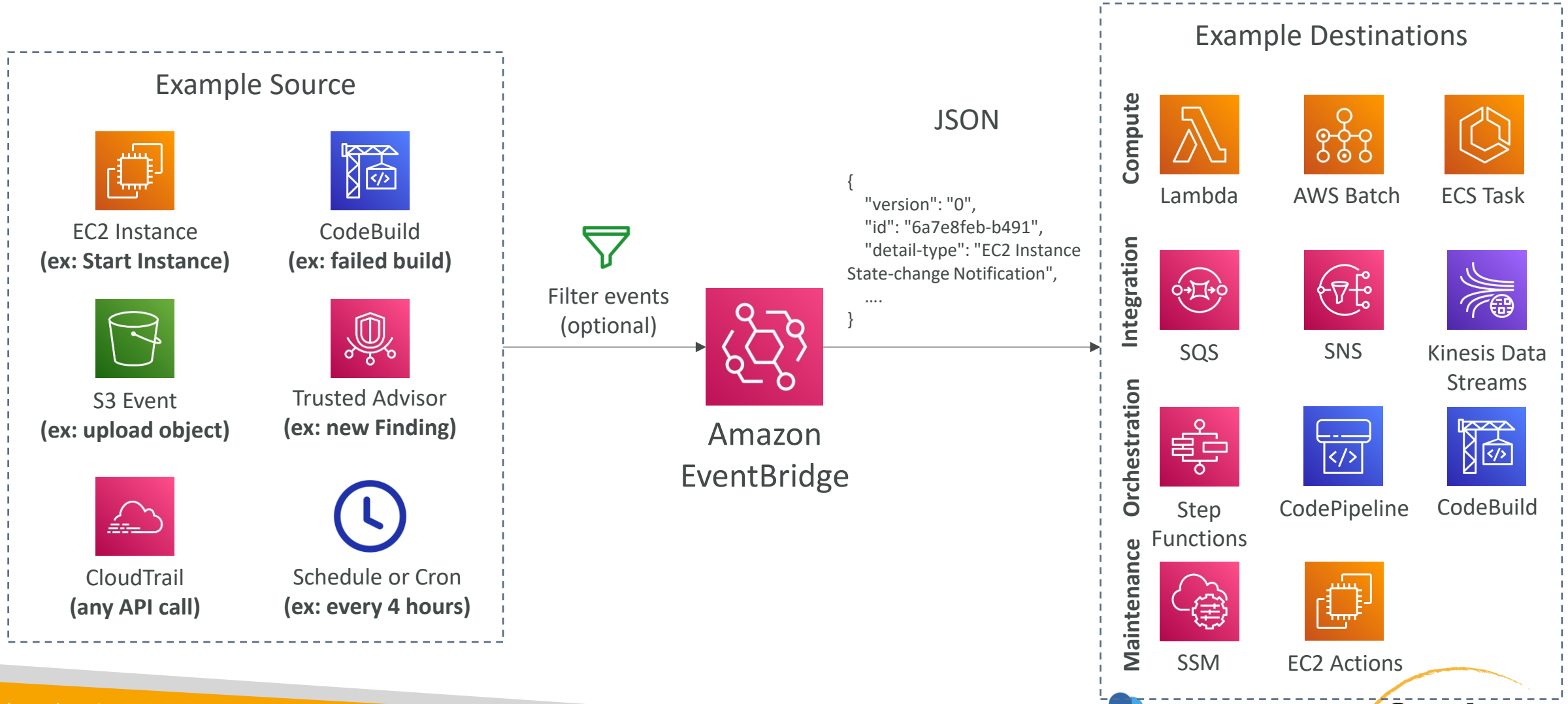
Amazon EventBridge Rules



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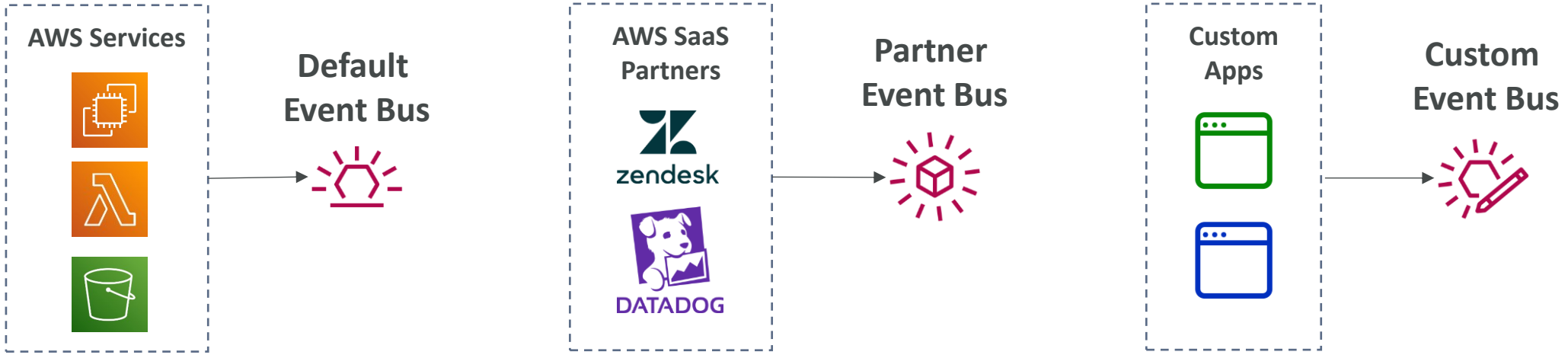
Amazon EventBridge



- Event buses can be accessed by other AWS accounts using Resource-based Policies

- You can **archive events** (all/filter) sent to an event bus (indefinitely or set period)
- Ability to **replay archived events**

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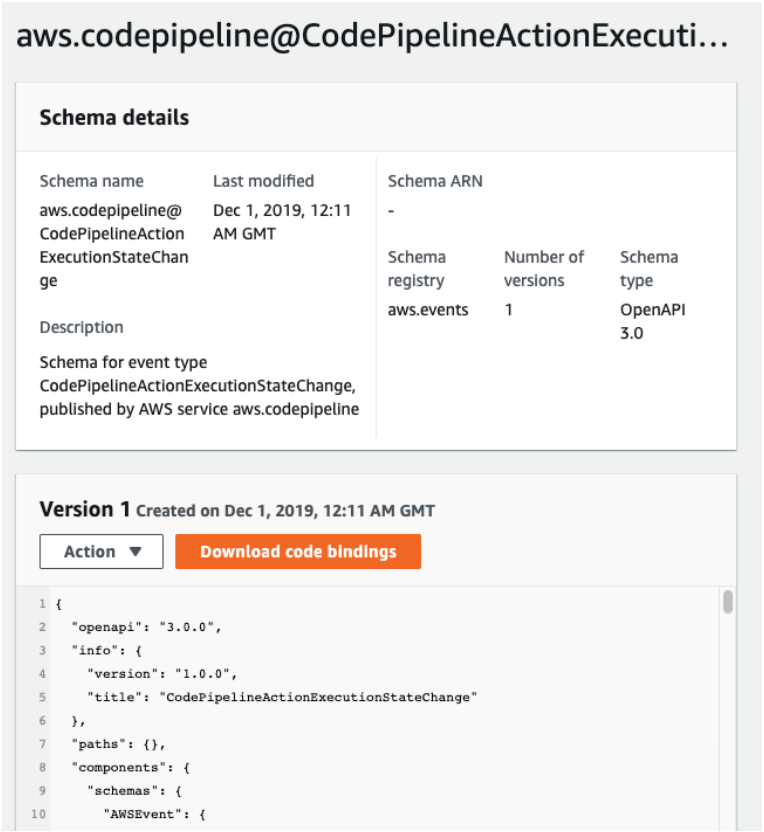
Amazon EventBridge – Schema Registry

- EventBridge can analyze the events in your bus and infer the **schema**
- The **Schema Registry** allows you to generate code for your application, that will know in advance how data is structured in the event bus
- Schema can be versioned



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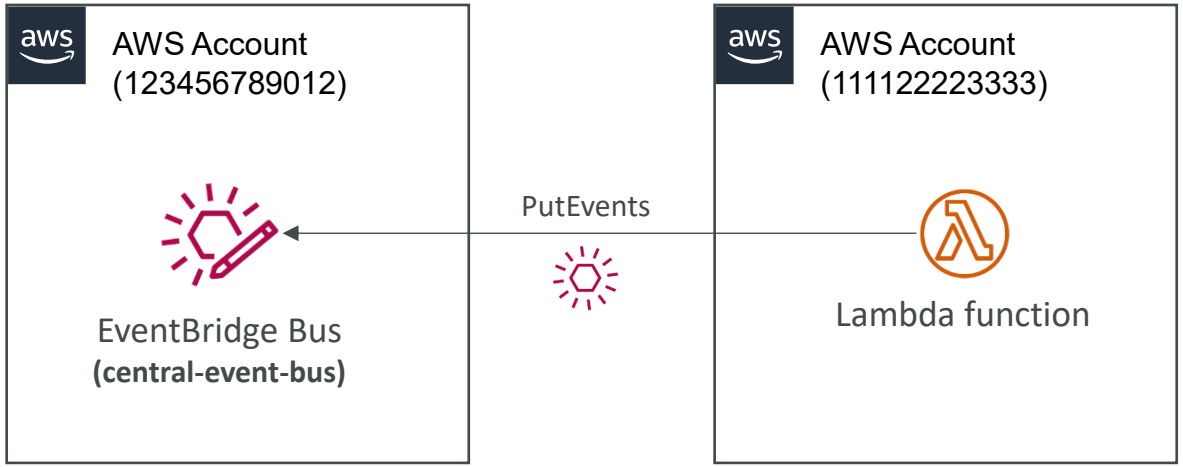
Amazon EventBridge – Resource-based Policy

- Manage permissions for a specific Event Bus
- Example: allow/deny events from another AWS account or AWS region
- Use case: aggregate all events from your AWS Organization in a single AWS account or AWS region

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "events:PutEvents",
            "Principal": { "AWS": "111122223333" },
            "Resource": "arn:aws:events:us-east-1:123456789012:
event-bus/central-event-bus"
        }
    ]
}
```

Allow **events** from another AWS account



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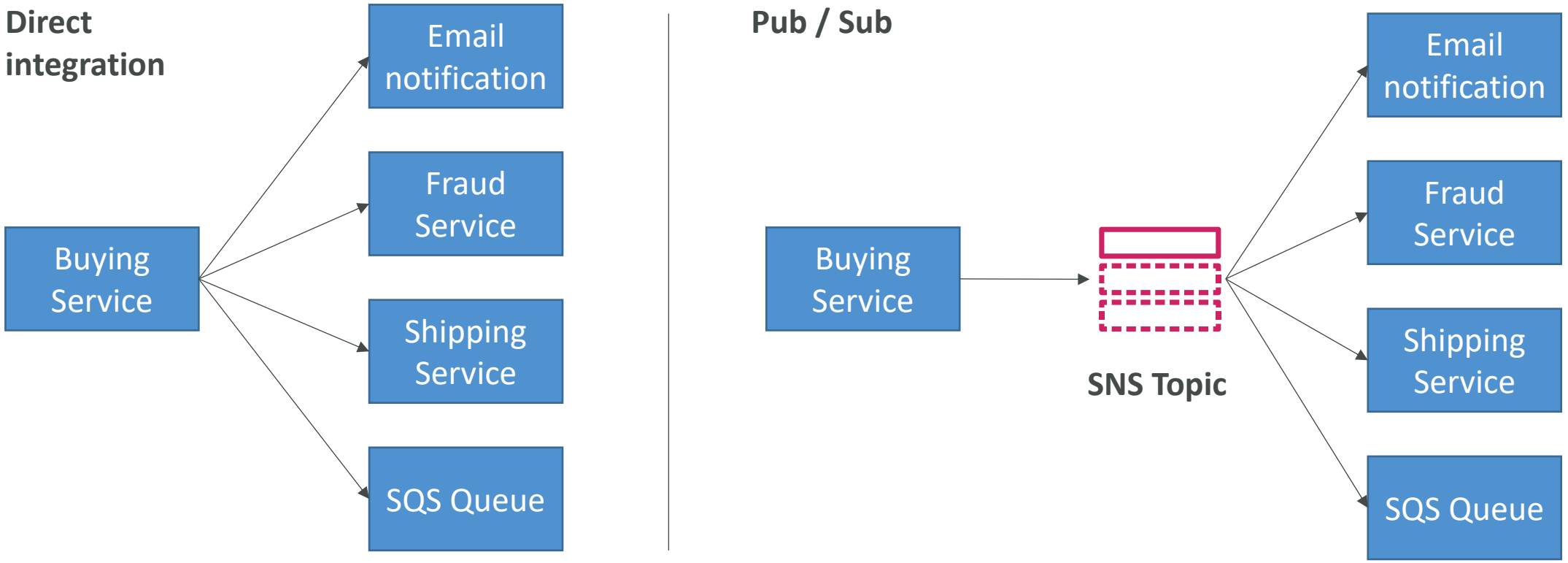
Amazon SNS

- What if you want to send one message to many receivers?



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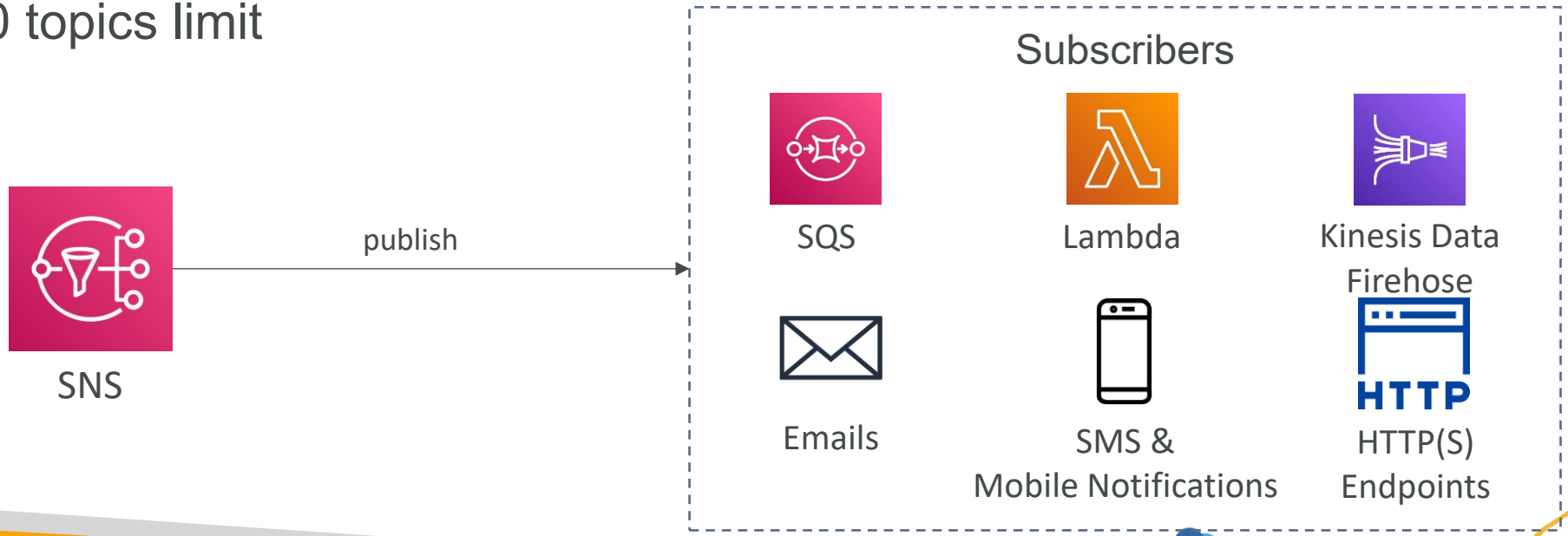
Amazon SNS

- The “event producer” only sends message to one SNS topic
- As many “event receivers” (subscriptions) as we want to listen to the SNS topic notifications
- Each subscriber to the topic will get all the messages (note: new feature to filter messages)
- Up to 12,500,000 subscriptions per topic
- 100,000 topics limit



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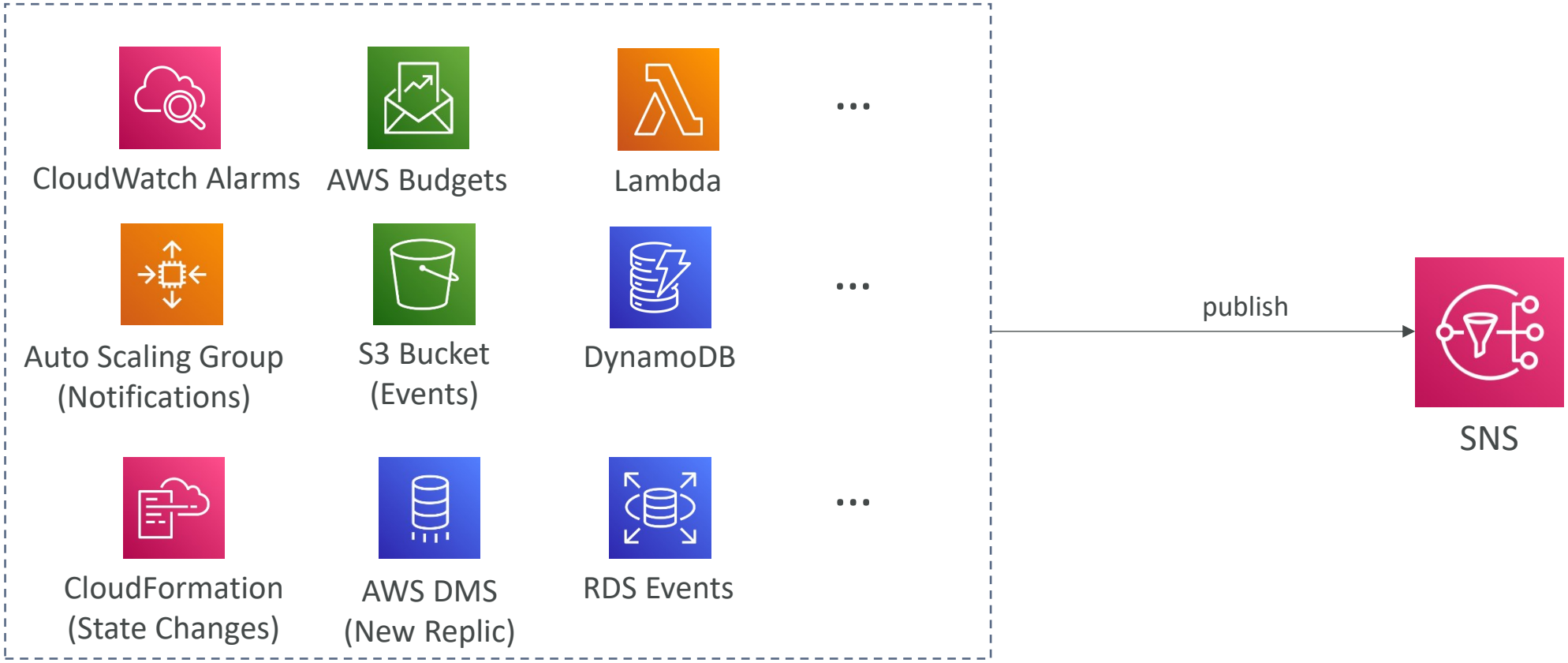
SNS integrates with a lot of AWS services

- Many AWS services can send data directly to SNS for notifications



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Amazon SNS – How to publish

- Topic Publish (using the SDK)
  - Create a topic
  - Create a subscription (or many)
  - Publish to the topic

- Direct Publish (for mobile apps SDK)
  - Create a platform application
  - Create a platform endpoint
  - Publish to the platform endpoint
  - Works with Google GCM, Apple APNS, Amazon ADM…

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Amazon SNS – Security

- **Encryption:**
  - In-flight encryption using HTTPS API
  - At-rest encryption using KMS keys
  - Client-side encryption if the client wants to perform encryption/decryption itself
- **Access Controls:** IAM policies to regulate access to the SNS API
- **SNS Access Policies** (similar to S3 bucket policies)
  - Useful for cross-account access to SNS topics
  - Useful for allowing other services (S3…) to write to an SNS topic

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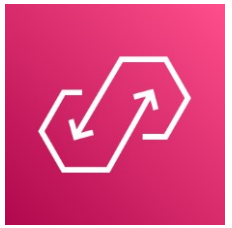
Amazon AppFlow



- Fully managed integration service that enables you to securely transfer data between **Software-as-a-Service (SaaS) applications and AWS**
- Sources: **Salesforce**, SAP, Zendesk, Slack, and ServiceNow
- Destinations: AWS services like **Amazon S3, Amazon Redshift** or non-AWS such as SnowFlake and Salesforce
- Frequency: on a schedule, in response to events, or on demand
- Data transformation capabilities like filtering and validation
- Encrypted over the public internet or privately over AWS PrivateLink
- Don’t spend time writing the integrations and leverage APIs immediately

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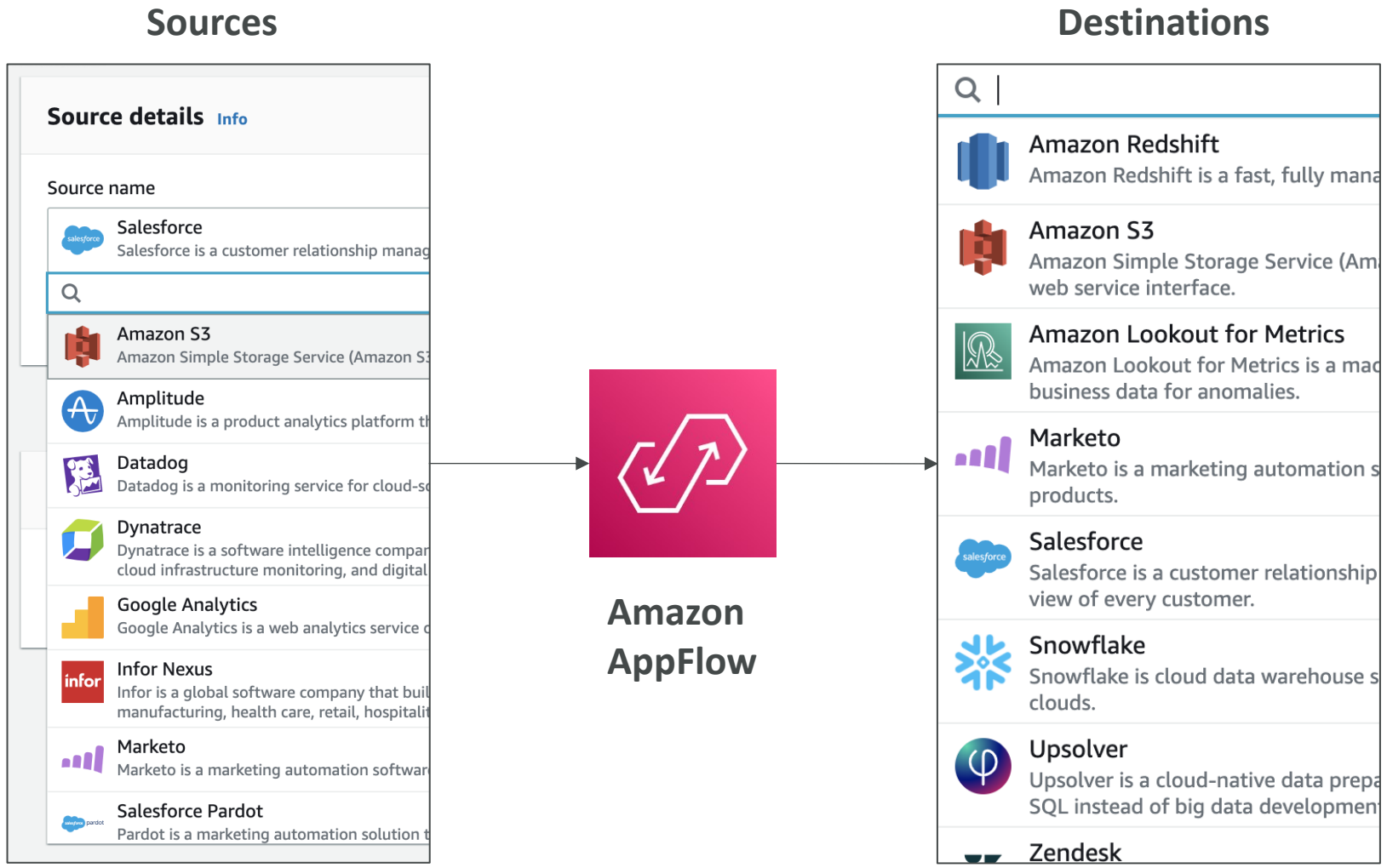
Amazon AppFlow



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Governance and QA

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Bedrock Agents Classic Tracing

- Every Agent response includes a trace
  - Shows its “reasoning” process
  - What knowledge bases did it hit, what did it get back
  - How did it use action groups
  - Error details
- Different trace types:
  - PreProcessing – categorizing user input
  - Orchestration – Calling action groups and knowledge bases
  - PostProcessing – Collect final response
  - CustomOrchestration – Determining order of actions
  - RoutingClassifier – How it classified and routed the request
  - Failure – Why a step failed
  - Guardrail – What your guardrails did


```
{
  "orchestrationTrace": {
    "observation": {
      "actionGroupInvocationOutput": {
        "metadata": {
          "clientRequestId": "75f5e9fc-6981-4887-a793-
77bbd3bd83af",
          "endTime": "2025-06-19T01:56:04.191408Z",
          "startTime": "2025-06-19T01:56:03.929317Z",
          "totalTimeMs": 262
        },
        "text": "The function get_time was called
successfully. The current time is 01:56:04"
      },
      "traceId": "1776395b-435a-4c3a-b66c-4ab947f9dc6c-
0",
      "type": "ACTION_GROUP"
    }
  }
}
```


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Foundation model evaluation criteria

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Human Evaluation

- The complex and non-deterministic nature of generative AI makes it hard to measure.
- Humans are often your best source of feedback
  - UX: How does it feel to use this model?
  - Contextual relevancy / sensitivity
  - Creativity & flexibility
    - Handling unexpected or complex problems

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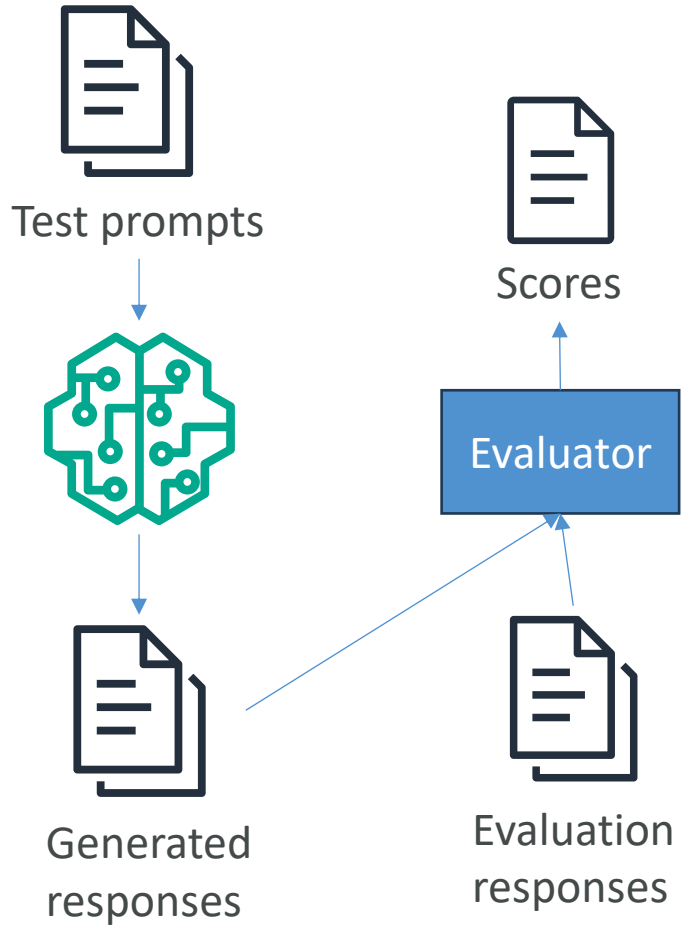
Evaluation against Benchmark Datasets

- Generally a set of sample questions (prompts) and answers generated by SME's
  - Might test performance for a specific application
  - Might test performance for specific kinds of problems (like reasoning ability)
- Can be useful as initial metrics before getting human feedback
- Often used for FM "leaderboards"
- Things you can measure
  - Accuracy
    - How close was the generated result to the "ideal" one?
  - Speed / Efficiency
    - How quickly was the result generated?
  - Scalability
    - How does it hold up to more data? More users? Longer prompts? More context?
  - Context retrieval
    - We're getting there… but if the system retrieves external data, how relevant is that data?
- Only as good as your test and evaluation data… these metrics can be flawed and incomplete



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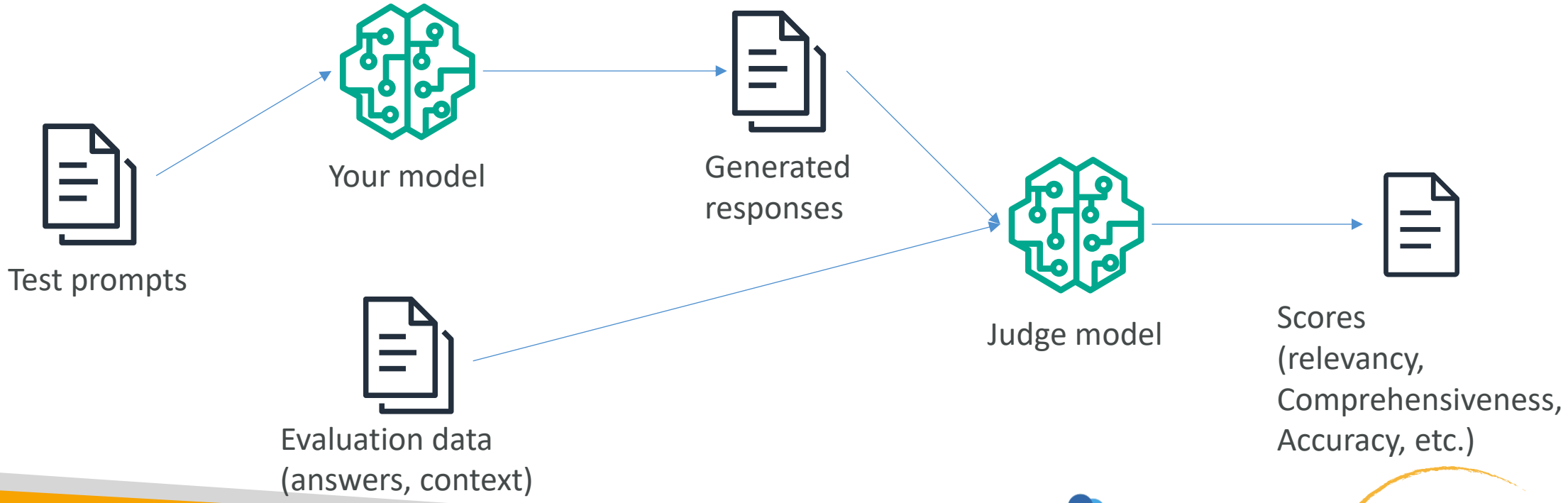
Using another model as a “judge”

- A “trusted” model is used to evaluate responses from your test prompts
  - This assumes you have a “trusted” model (who watches the watchers?)
    - If your model and the judge model share some of the same problems, that could be a problem.
  - Perhaps better for testing “small” language models designed for efficiency



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Hybrid approaches

- Why not both?
  - Comparing human and benchmark-based approaches can reveal the limitations of both

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ROUGE

- Set of metrics for:
  - Text summarization
  - Machine translation
- Counts number of overlapping “units” between computer-generated output and evaluation ground-truth output.
  - Words, n-grams, sentence fragments
- ROUGE-N
  - Overlap on n-grams (how many words at a time are compared)
  - ROUGE-1: unigrams (one word,) ROUGE-2: bigrams (two words)
- ROUGE-L
  - Uses longest common subsequence between generated and reference text
  - Good at evaluating coherence and order of narrative

Certified AI Practitioner
Unigrams: **Certified**, **AI**, **Practitioner**
Bigrams: **Certified AI**, **AI Practitioner**

Output: Certified ML Practitioner
ROUGE-1: 2/3
ROUGE-2: 0

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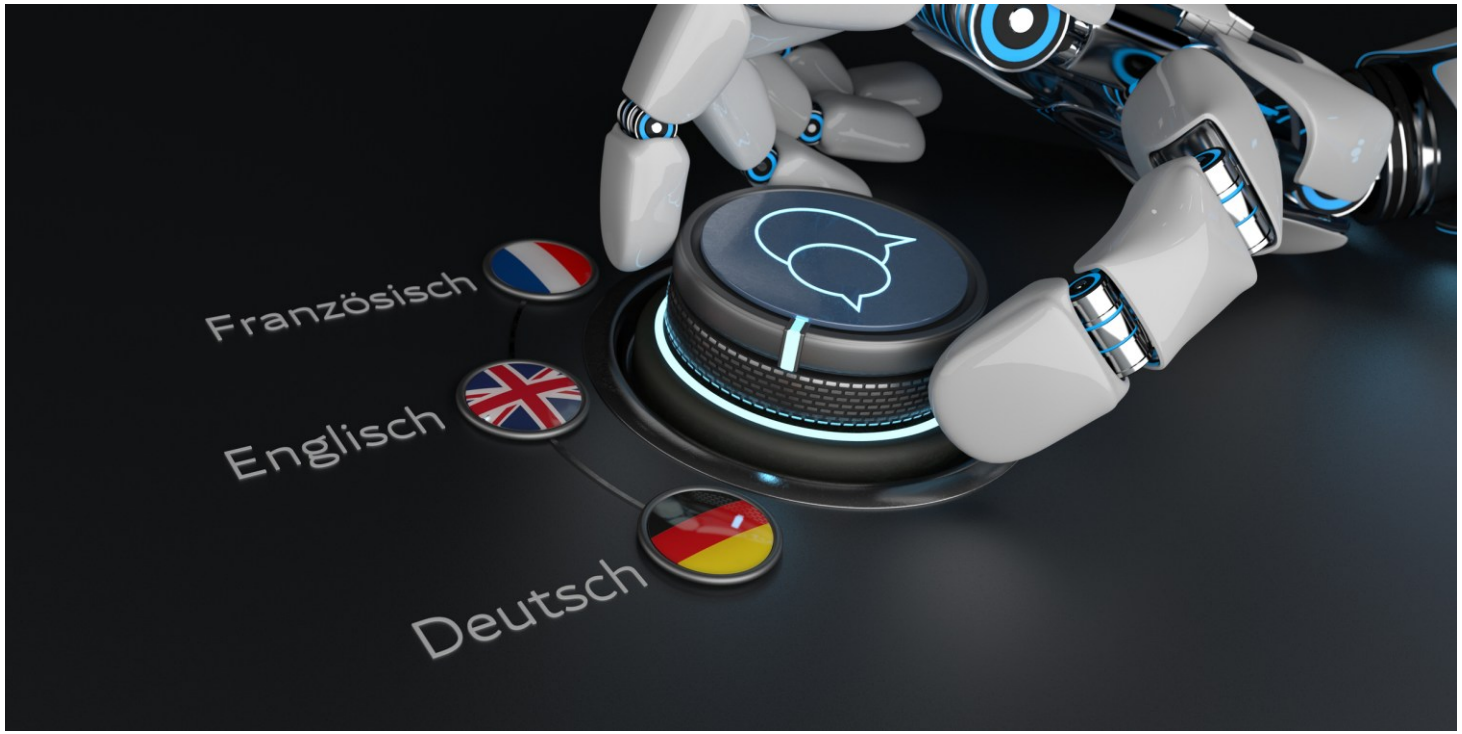

BLEU

- Used for machine translation
- Compares machine translation to human translation
- Measure PRECISION of N-grams
  - As opposed to ROUGE, which is recall
  - Precision compares to everything, recall just to “relevant” results (overlapping results in the case of ROUGE)
  - Checks how many words / phrases appear in the reference translation
  - Works at the sentence level
- Brevity penalty
  - Overly short translations are penalized
- Limited use in assessing fluency and grammar

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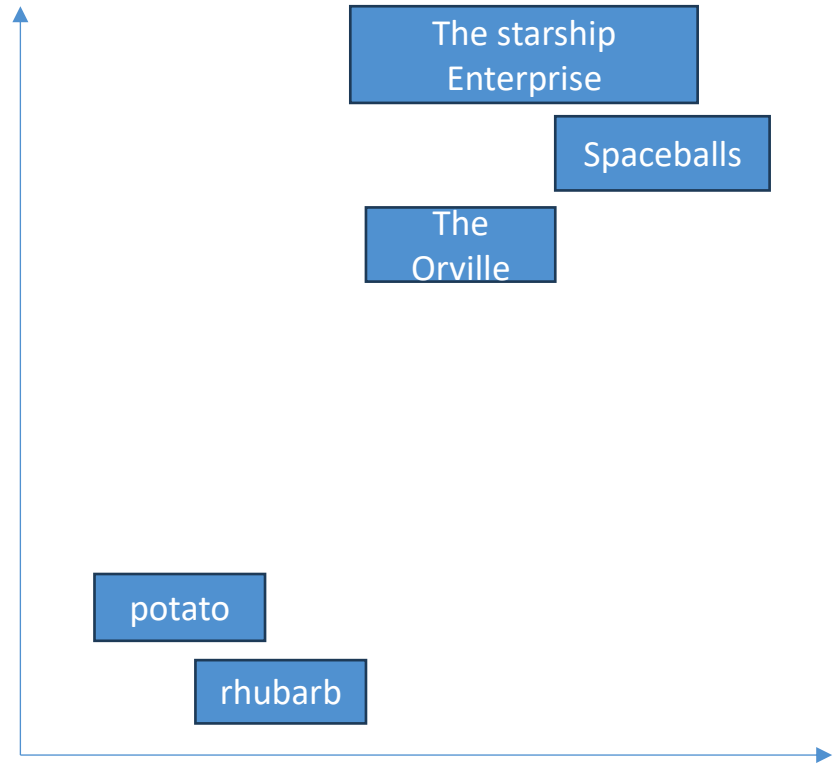
BERTscore

- LLMs rely on “embeddings” that store the meaning of a word as a vector
  - I am oversimplifying, but that’s the idea.
- BERTscore uses these embedding vectors to compare the semantic similarity between a model’s output and the ideal output
  - BERT is a language model that predated GPT
- Less sensitive to synonyms and paraphrasing that don’t really change the meaning of what’s being generated



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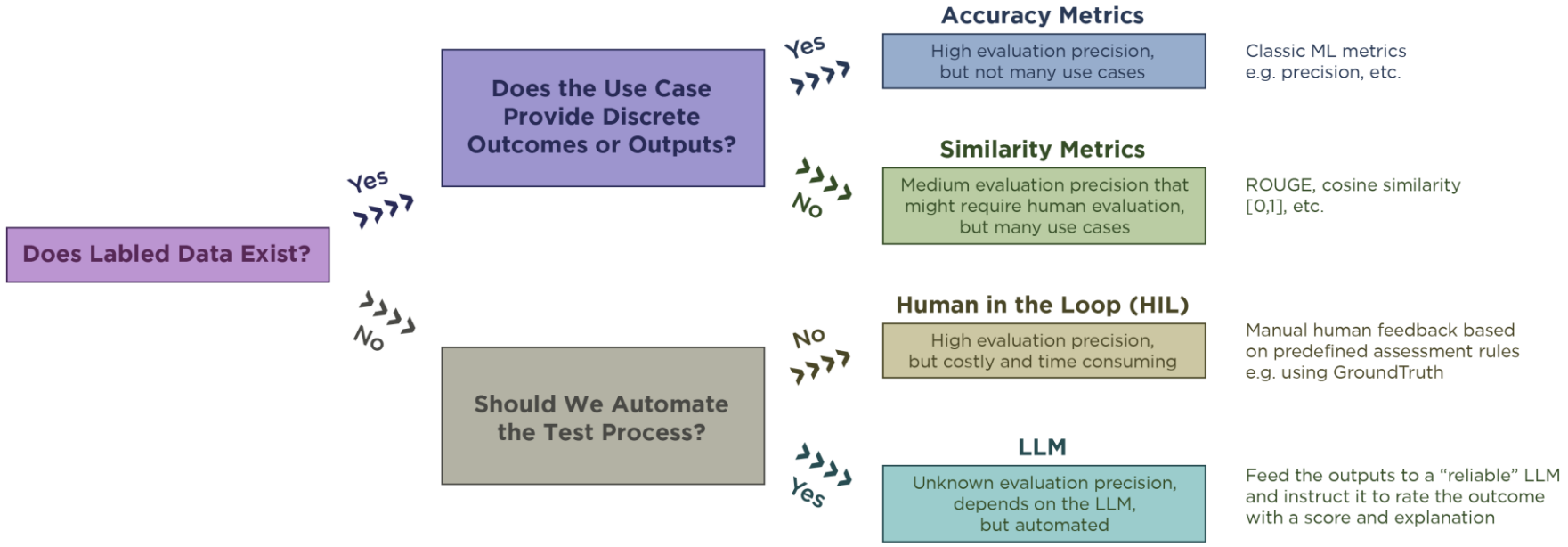
Testing and evaluation: choosing a strategy



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Bedrock Model Evaluations

- Automatic model evaluations
  - Tons of built-in task types, datasets, and metrics (seen here)
  - Text generation, summarization, Q&A, classification
- Human-based
  - Humans compare two models
    - Given your prompt dataset
  - Set up a work team in Cognito, SageMaker Ground Truth, or Amazon Augmented AI

Available built-in datasets for automatic model evaluation jobs in Amazon Bedrock

Task type	Metric	Built-in datasets	Computed metric
General text generation	Accuracy	TREX	Real world knowledge (RWK) score
	Robustness	BOLD	Word error rate
		TREX	
		WikiText2	
	Toxicity	RealToxicityPrompts	Toxicity
		BOLD	
Text summarization	Accuracy	Gigaword	BERTScore
	Toxicity	Gigaword	Toxicity
	Robustness	Gigaword	BERTScore and deltaBERTScore
Question and answer	Accuracy	BoolQ	NLP-F1
		NaturalQuestions	
		TriviaQA	
	Robustness	BoolQ	F1 and deltaF1
		NaturalQuestions	
		TriviaQA	
	Toxicity	BoolQ	Toxicity
		NaturalQuestions	
		TriviaQA	
Text classification	Accuracy	Women's Ecommerce Clothing Reviews	Accuracy (Binary accuracy from classification_accuracy_score)
	Robustness	Women's Ecommerce Clothing Reviews	classification_accuracy_score and delta_classification_accuracy_score

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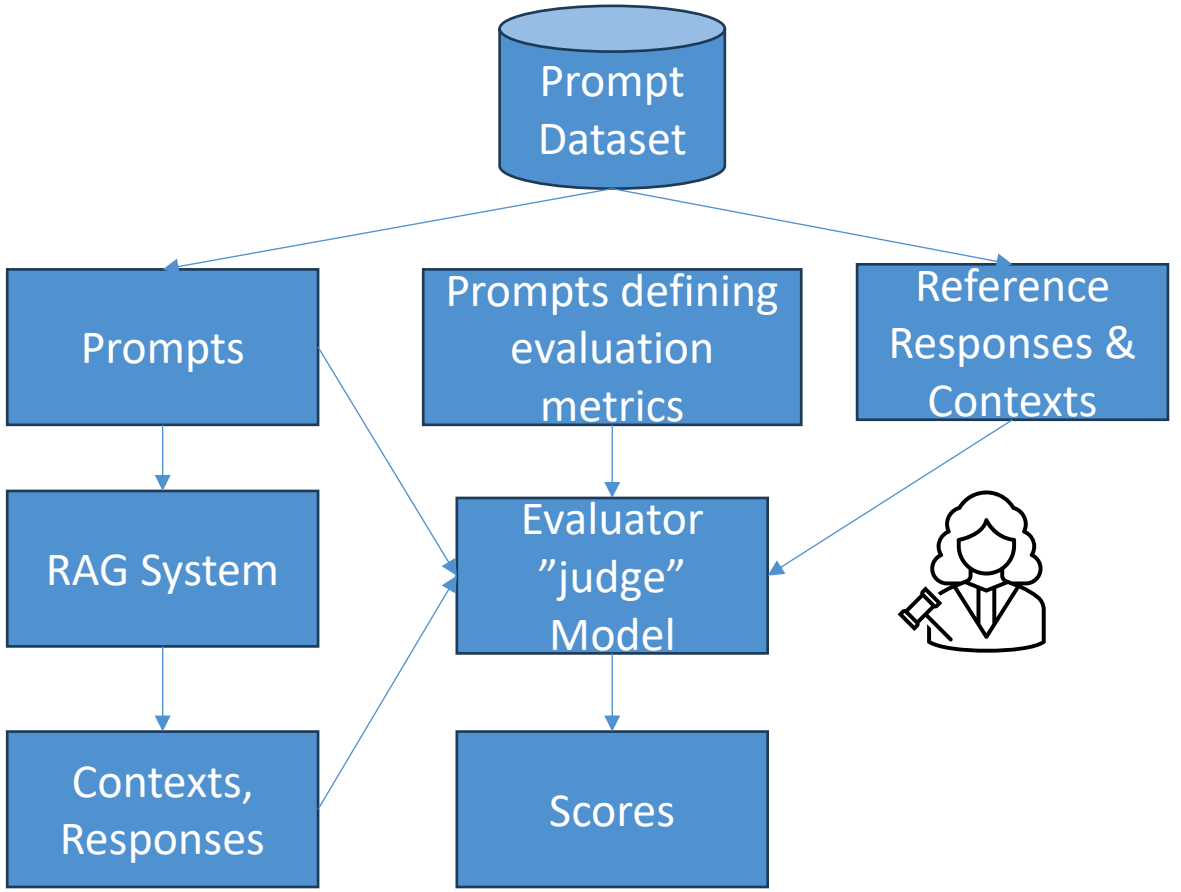
Bedrock Model Evaluations

- LLM as a judge
  - Evaluator model and generator model
  - You provide a prompt dataset
  - Many metrics available – we talked about this before
- RAG evaluation jobs
  - Retrieve only – measures retrieval from your RAG source
    - Relevance and coverage
  - Retrieve and generate – Measures the response as well
    - Correctness, completeness, helpfulness, logical coherence, etc.
  - Provide a prompt dataset in S3 in JSON line format
  - You can add your own metrics (based on prompts for the judge model)



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Considerations for Deployment Validation

- Synthetic user workflows
  - Unit testing often doesn't cut it, because GenAI is non-deterministic
  - Simulate end to end usage of your application
    - Before and after deployments
  - Some useful tools
    - Cloudwatch synthetic monitoring (canaries,) Step Functions, EventBridge, Lambda
    - Store results in S3 / Athena, analyze in QuickSight
- AI-Specific output validation
  - Measure hallucination rate, semantic drift, faithfulness, compliance, etc.
  - Bedrock Model Evaluations does this
  - You could roll your own with Lambda + Step Functions
- Checking response consistency
  - Best you can do is make sure style is consistent, answers are similar for a test prompt dataset
  - Test that variability is within an accepted range

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Responsible AI: Core Dimensions

- Fairness
- Explainability
- Privacy and Security
- Safety
- Controllability
- Veracity and Robustness
- Governance
- Transparency

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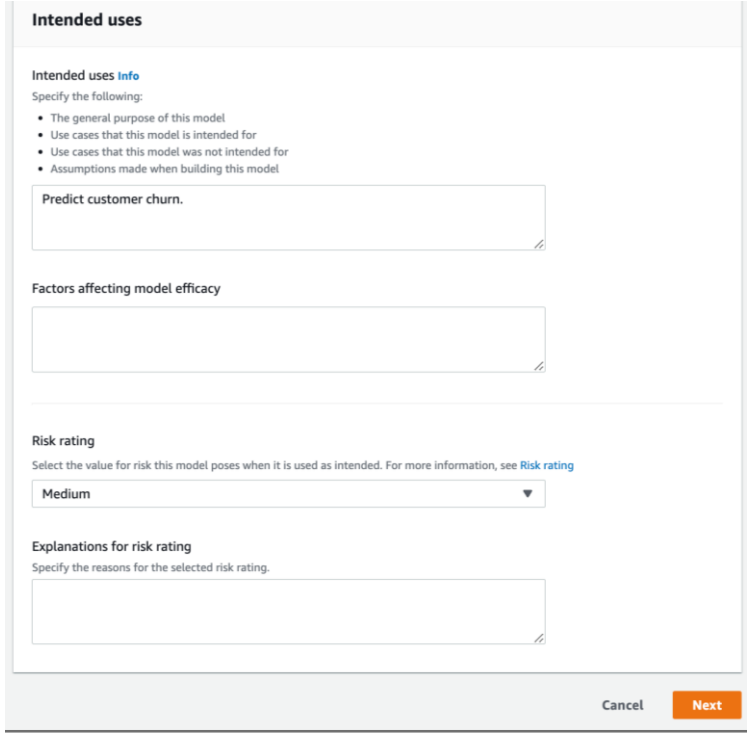
AWS Tools for Responsible AI

- Amazon Bedrock
  - Model evaluation tools
- SageMaker Clarify
  - Bias detection
  - Model evaluation
  - Explainability
- SageMaker Model Monitor
  - Get alerts for inaccurate responses
- Amazon Augmented AI
  - Insert humans in the loop to help correct results
- SageMaker ML Governance
  - SageMaker Role Manager
  - Model
  - Model Dashboard



Portion of model card creation UI

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CloudWatch Logs

- **Log groups:** arbitrary name, usually representing an application
- **Log stream:** instances within application / log files / containers
- Can define log expiration policies (never expire, 1 day to 10 years…)
- **CloudWatch Logs can send logs to:**
  - Amazon S3 (exports)
  - Kinesis Data Streams
  - Kinesis Data Firehose
  - AWS Lambda
  - OpenSearch
- Logs are encrypted by default
- Can setup KMS-based encryption with your own keys

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CloudWatch Logs - Sources

- SDK, CloudWatch Logs Agent, CloudWatch Unified Agent
- Elastic Beanstalk: collection of logs from application
- ECS: collection from containers
- AWS Lambda: collection from function logs
- VPC Flow Logs: VPC specific logs
- API Gateway
- CloudTrail based on filter
- Route53: Log DNS queries

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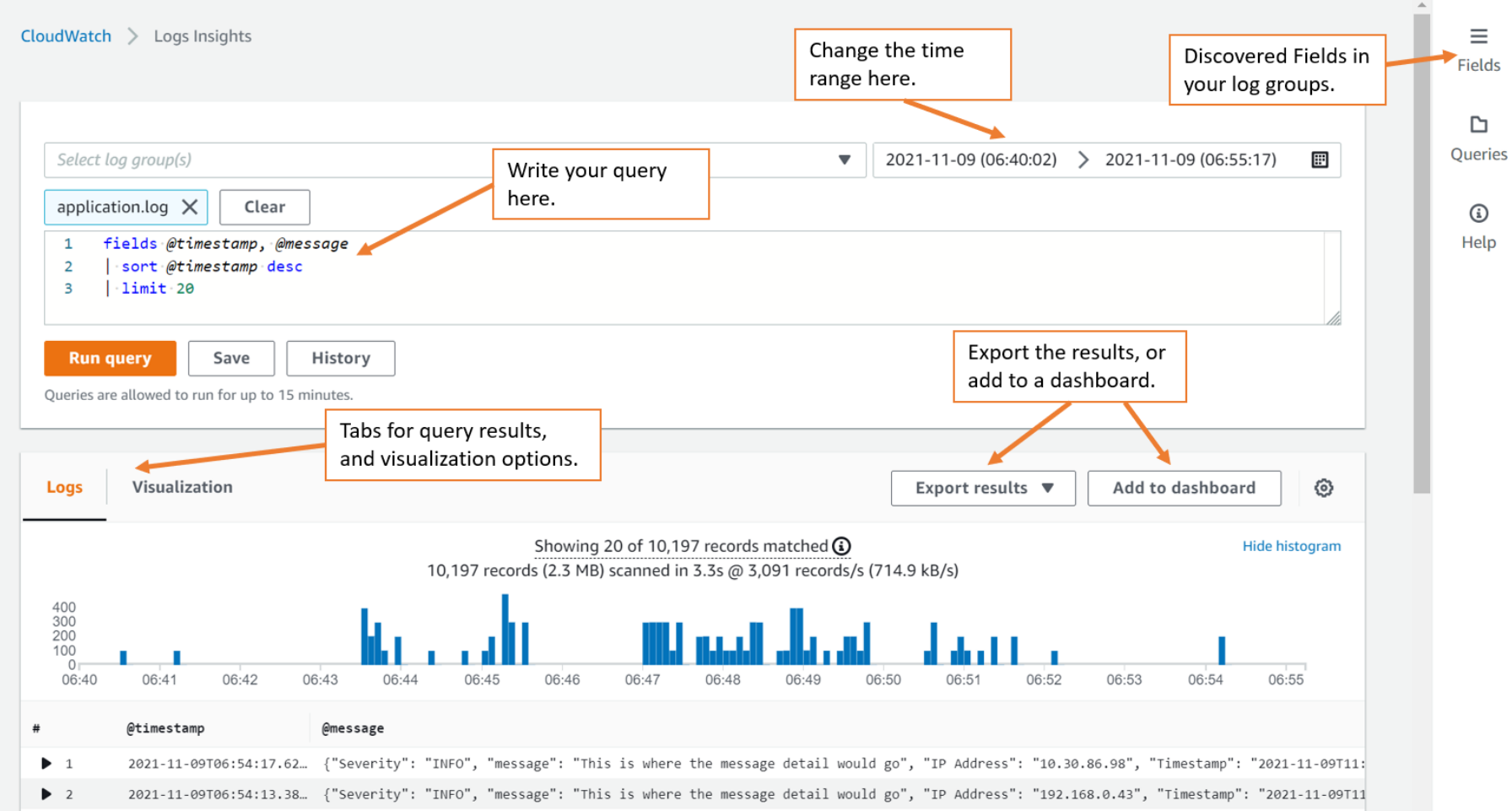
CloudWatch Logs Insights



https://mng.workshop.aws/operations-2022/detect/cwlogs.html

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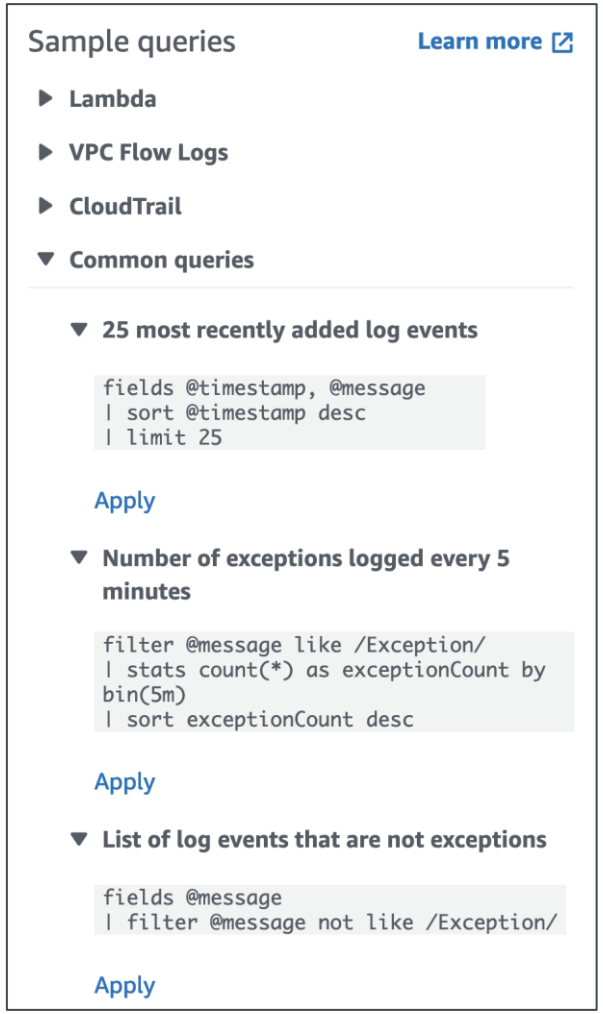
CloudWatch Logs Insights

- Search and analyze log data stored in CloudWatch Logs
- Example: find a specific IP inside a log, count occurrences of “ERROR” in your logs…
- Provides a purpose-built query language
  - Automatically discovers fields from AWS services and JSON log events
  - Fetch desired event fields, filter based on conditions, calculate aggregate statistics, sort events, limit number of events…
  - Can save queries and add them to CloudWatch Dashboards
- Can query multiple Log Groups in different AWS accounts
- It’s a query engine, not a real-time engine



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CloudWatch Logs – S3 Export

CloudWatch Logs Amazon S3

- Log data can take **up to 12 hours** to become available for export
- The API call is **CreateExportTask**

- Not near-real time or real-time… use Logs Subscriptions instead

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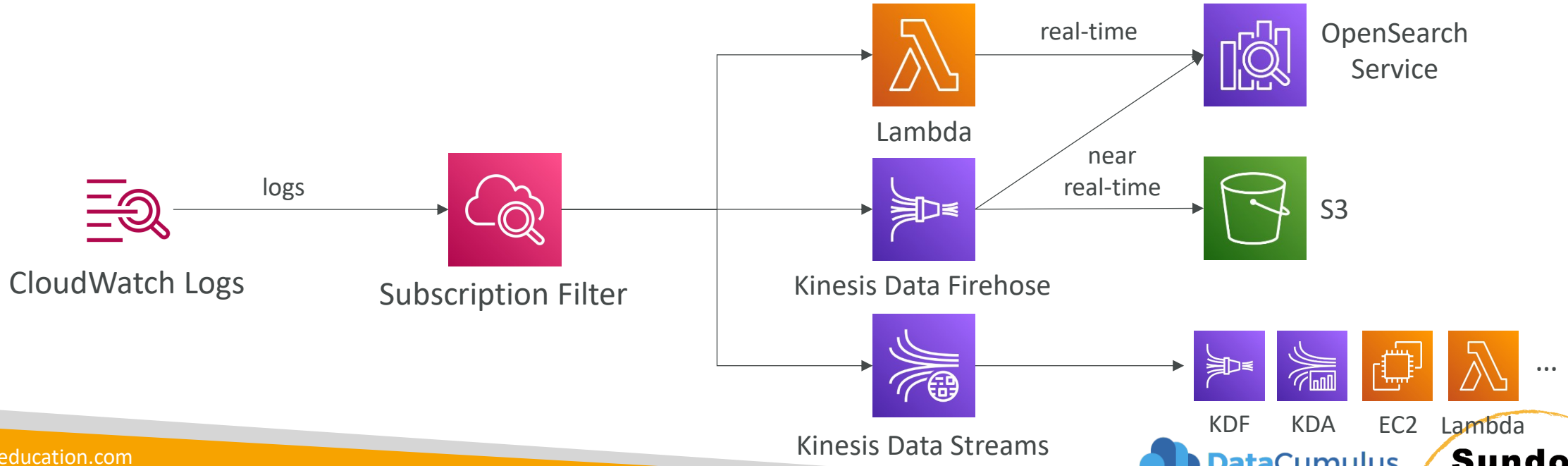
CloudWatch Logs Subscriptions

- Get a real-time log events from CloudWatch Logs for processing and analysis
- Send to Kinesis Data Streams, Kinesis Data Firehose, or Lambda
- **Subscription Filter –** filter which logs are events delivered to your destination



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CloudWatch Alarms

- Alarms are used to trigger notifications for any metric
- Various options (sampling, %, max, min, etc…)
- Alarm States:
  - OK
  - INSUFFICIENT_DATA
  - ALARM
- Period:
  - Length of time in seconds to evaluate the metric
  - High resolution custom metrics: 10 sec, 30 sec or multiples of 60 sec

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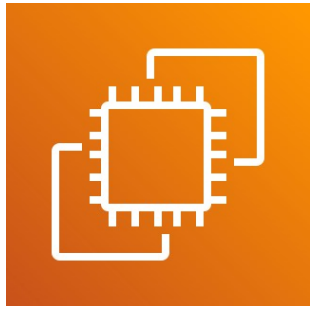
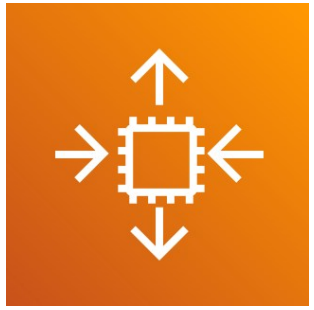
CloudWatch Alarm Targets

- Stop, Terminate, Reboot, or Recover an EC2 Instance
- Trigger Auto Scaling Action
- Send notification to SNS (from which you can do pretty much anything)



Amazon EC2



EC2 Auto Scaling

Amazon SNS

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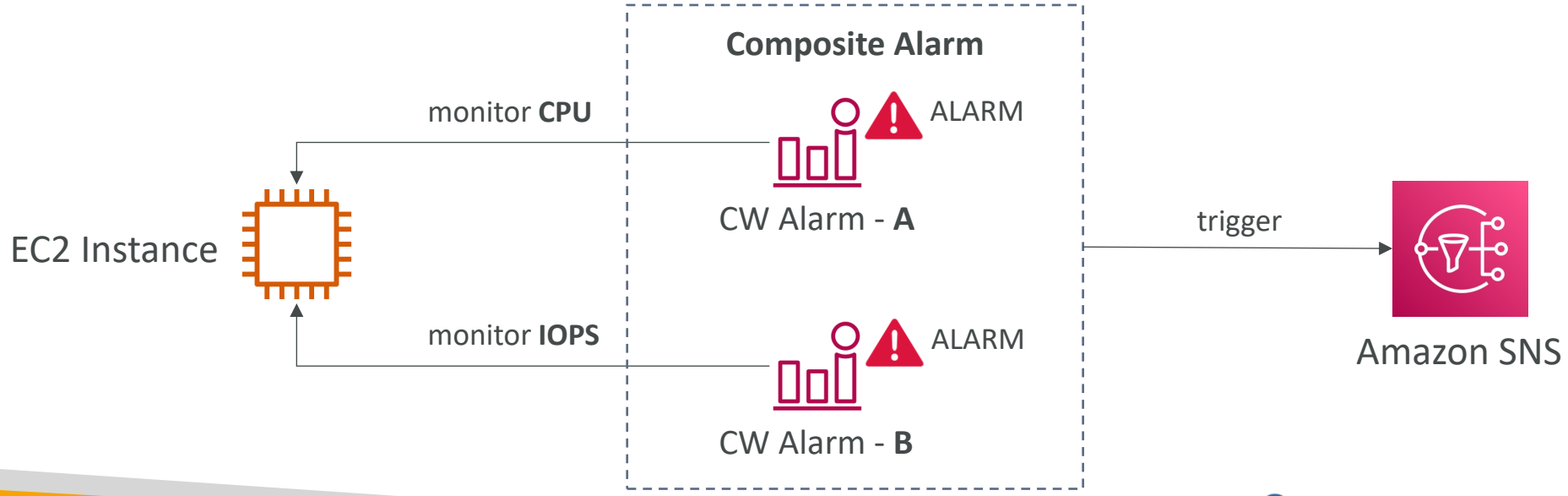
CloudWatch Alarms – Composite Alarms

- CloudWatch Alarms are on a single metric
- **Composite Alarms are monitoring the states of multiple other alarms**
- AND and OR conditions
- Helpful to reduce “alarm noise” by creating complex composite alarms



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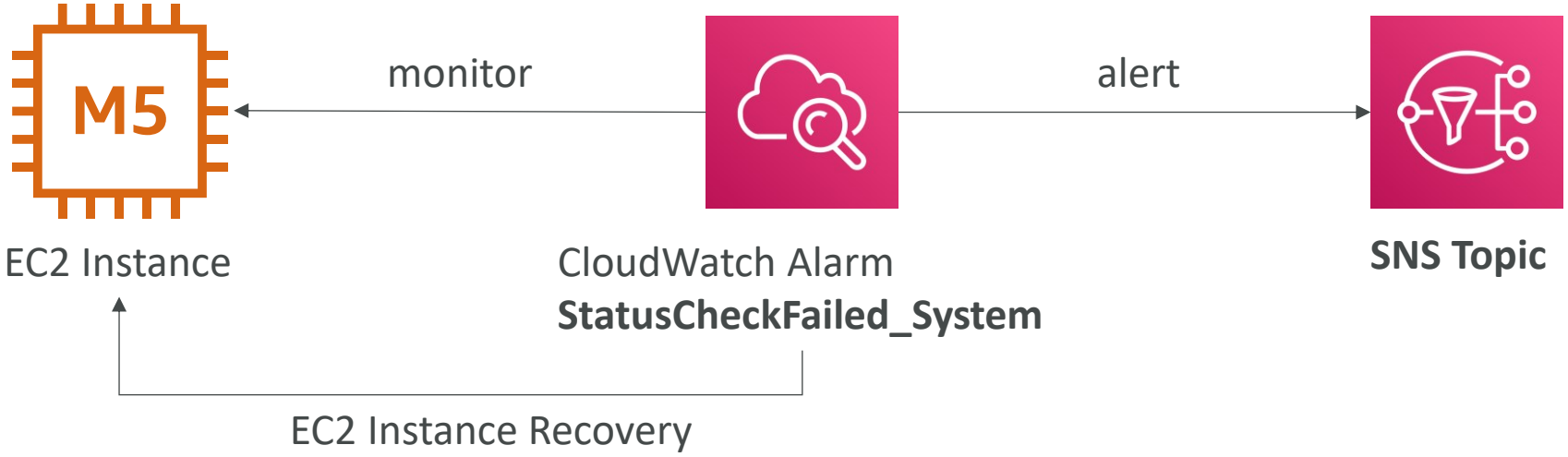
EC2 Instance Recovery

- Status Check:
  - Instance status = check the EC2 VM
  - System status = check the underlying hardware
  - Attached EBS status = check attached EBS volumes



- **Recovery:** Same Private, Public, Elastic IP, metadata, placement group

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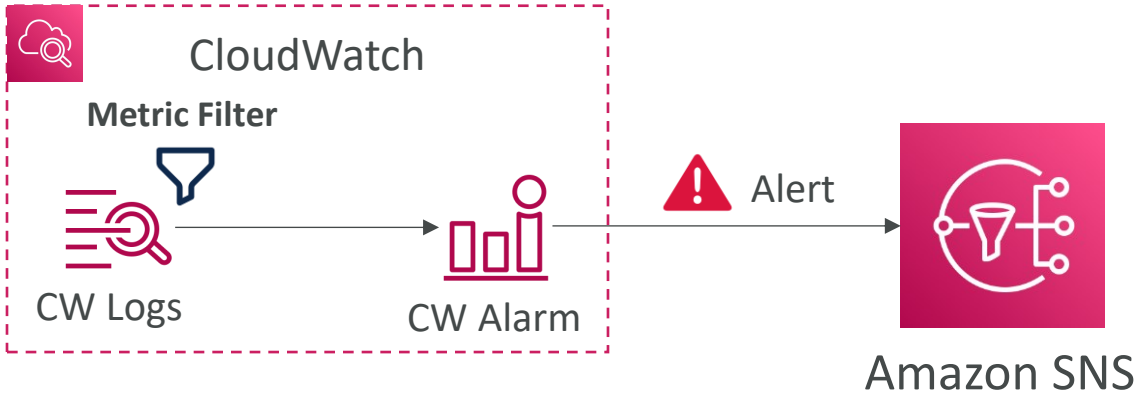
CloudWatch Alarm: good to know

- Alarms can be created based on CloudWatch Logs Metrics Filters



- To test alarms and notifications, set the alarm state to Alarm using CLI

aws cloudwatch set-alarm-state --alarm-name "myalarm" --state-value ALARM --state-reason "testing purposes"

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CloudWatch and GenAI

Testing prompt regression

CloudWatch logs

- Prompt inputs and model responses
- Foundational to monitoring and troubleshooting

Monitor KPIs

- Prompt effectiveness / response quality
- Latency
- Error rates

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Other things to monitor

- Foundation model interaction tracing
- Business impact metrics
- Prompt effectiveness
- Hallucination rates
- Anomaly detection
  - Token burst patterns
  - Response drift
- Bedrock model invocation logs
- Cost anomaly detection

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CloudWatch Real User Monitoring (RUM)

- Mostly for testing mobile apps (iOS or Android)
  - Measures page load times, errors, app launch times, etc.
  - From a real user session
- Integrates with Application Signals
  - View results in X-Ray traces
- Relevant for measuring end to end performance of your mobile GenAI apps

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AWS CloudTrail

- **Provides governance, compliance and audit for your AWS Account**
- CloudTrail is enabled by default!
- Get **an history of events / API calls made within your AWS Account** by:
  - Console
  - SDK
  - CLI
  - AWS Services
- Can put logs from CloudTrail into CloudWatch Logs or S3
- **A trail can be applied to All Regions (default) or a single Region.**
- If a resource is deleted in AWS, investigate CloudTrail first!

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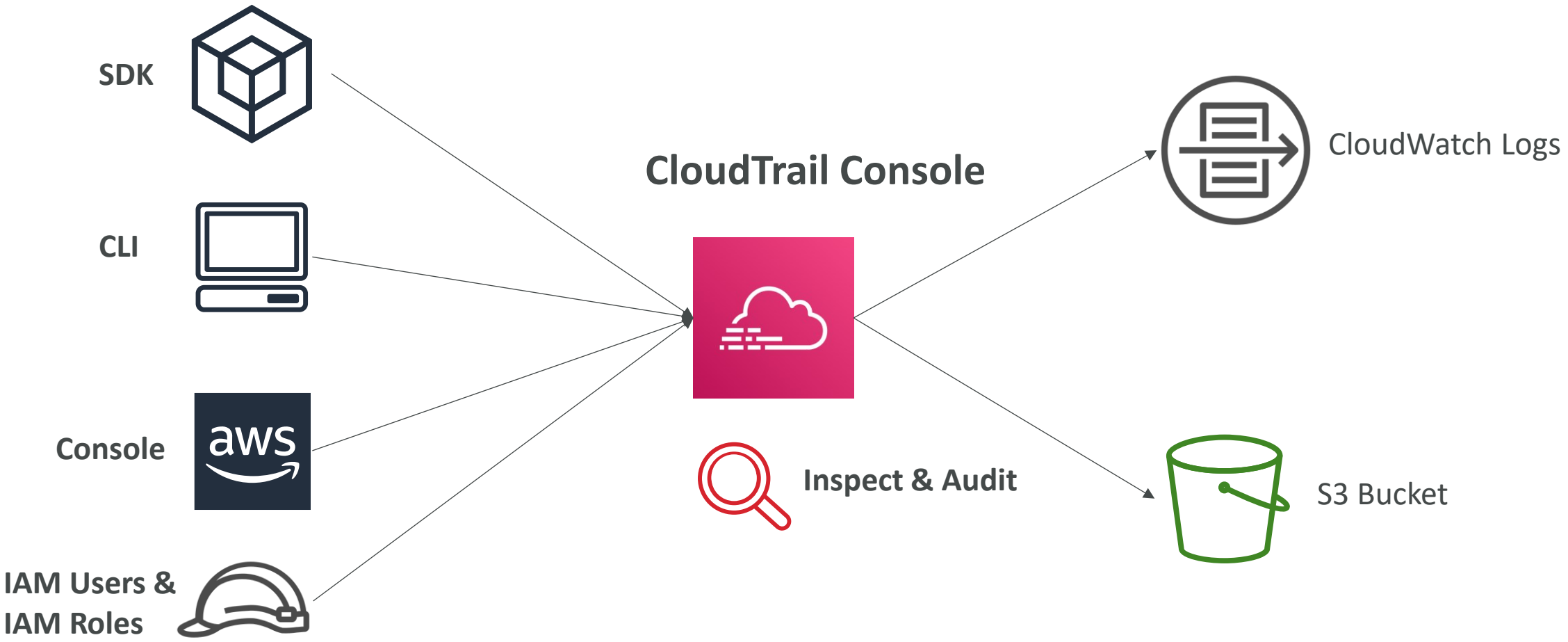
CloudTrail Diagram



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CloudTrail Events

- **Management Events:**
  - Operations that are performed on resources in your AWS account
  - Examples:
    - Configuring security (IAM AttachRolePolicy)
    - Configuring rules for routing data (Amazon EC2 CreateSubnet)
    - Setting up logging (AWS CloudTrail CreateTrail)
  - **By default, trails are configured to log management events.**
  - Can separate **Read Events** (that don't modify resources) from **Write Events** (that may modify resources)

- **Data Events:**
  - **By default, data events are not logged (because high volume operations)**
  - Amazon S3 object-level activity (ex: GetObject, DeleteObject, PutObject): can separate Read and Write Events
  - AWS Lambda function execution activity (the Invoke API)

- **CloudTrail Insights Events:**
  - See next slide ☺

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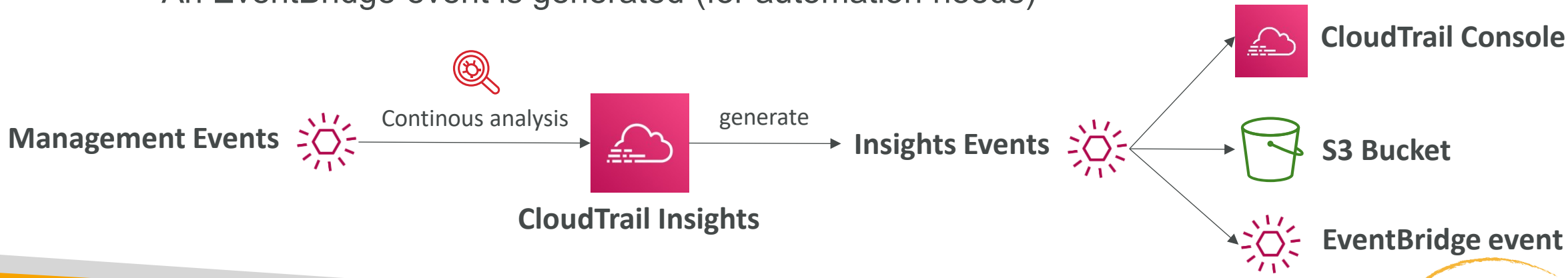
CloudTrail Insights

- Enable **CloudTrail Insights to detect unusual activity** in your account:
  - inaccurate resource provisioning
  - hitting service limits
  - Bursts of AWS IAM actions
  - Gaps in periodic maintenance activity
- CloudTrail Insights analyzes normal management events to create a baseline
- And then **continuously analyzes write events to detect unusual patterns**
  - Anomalies appear in the CloudTrail console
  - Event is sent to Amazon S3
  - An EventBridge event is generated (for automation needs)



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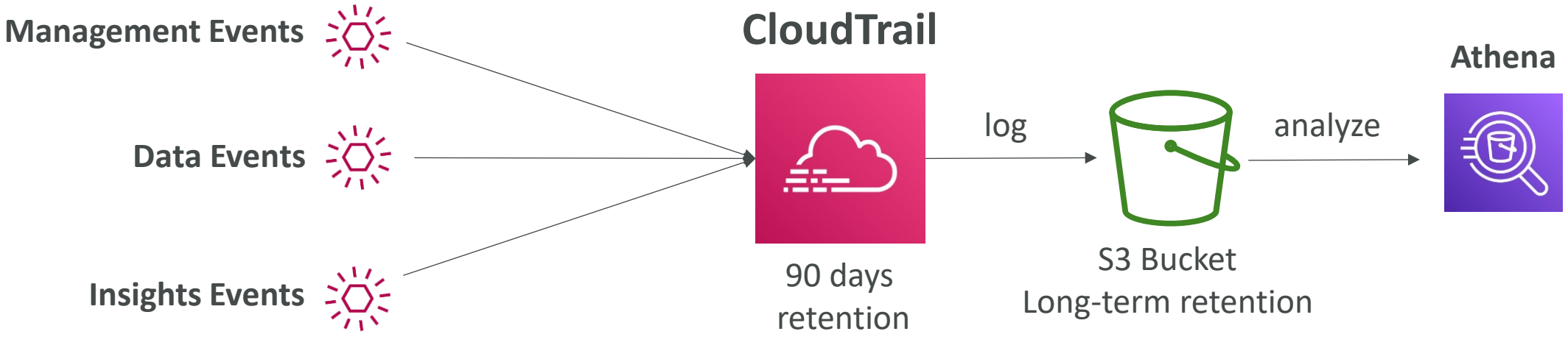
CloudTrail Events Retention

- Events are stored for 90 days in CloudTrail
- To keep events beyond this period, log them to S3 and use Athena



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CloudTrail and GenAI

- CloudTrail can track all API calls to Amazon Bedrock
  - Audit trails of which prompt resources were used
  - When and by who
- This is often a compliance requirement
  - Yes, this stuff gets logged

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AWS X-Ray

- Debugging in Production, the good old way:
  - Test locally
  - Add log statements everywhere
  - Re-deploy in production
- Log formats differ across applications using CloudWatch and analytics is hard.
- Debugging: monolith “easy”, distributed services “hard”
- No common views of your entire architecture!

- Enter… AWS X-Ray!

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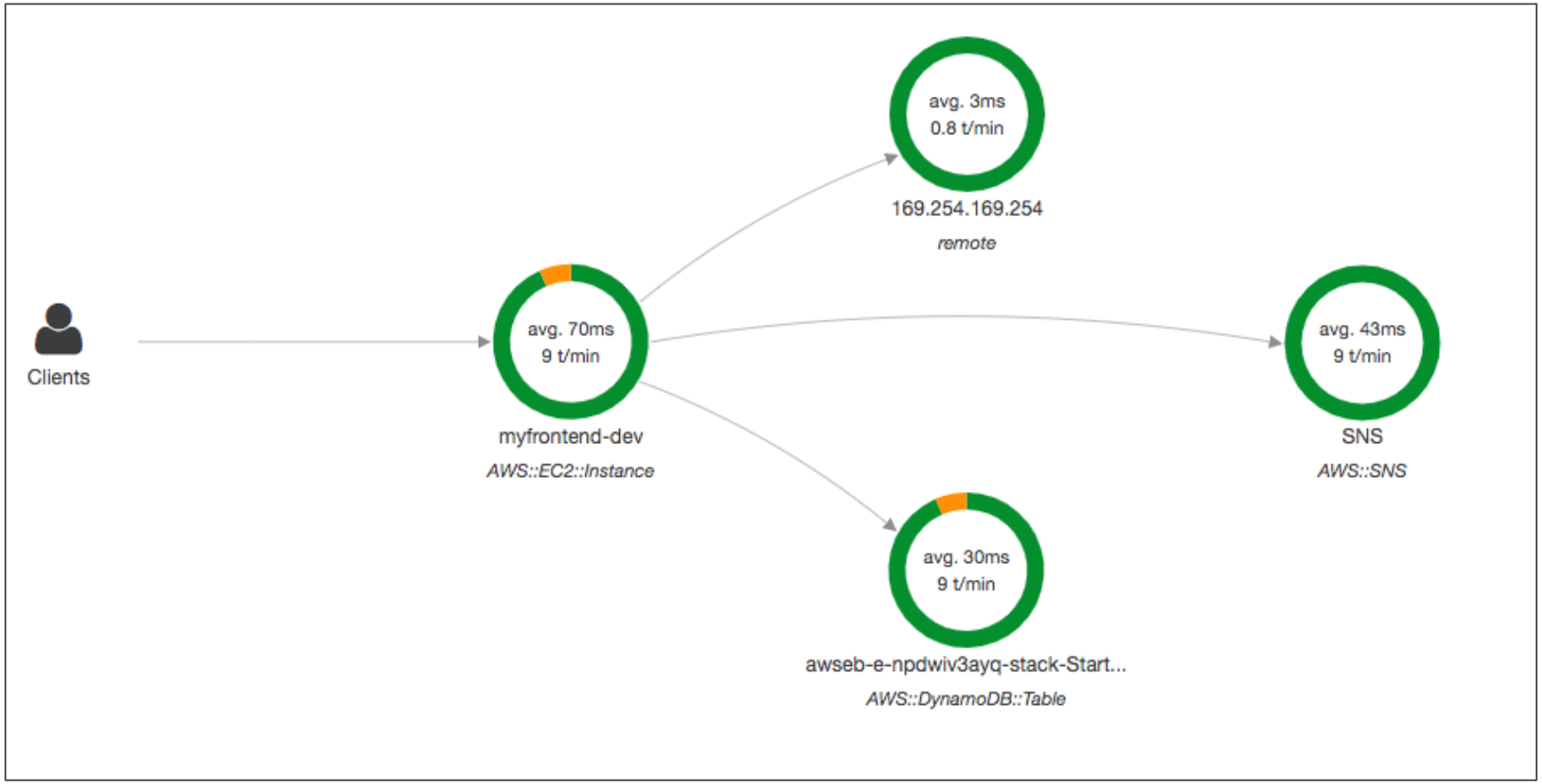
AWS X-Ray
Visual analysis of our applications



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AWS X-Ray advantages

- Troubleshooting performance (bottlenecks)
- Understand dependencies in a microservice architecture
- Pinpoint service issues
- Review request behavior
- Find errors and exceptions
- Are we meeting time SLA?
- Where I am throttled?
- Identify users that are impacted

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X-Ray compatibility

- AWS Lambda
- Elastic Beanstalk
- ECS
- ELB
- API Gateway
- EC2 Instances or any application server (even on premise)

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AWS X-Ray Leverages Tracing

- Tracing is an end to end way to following a “request”
- Each component dealing with the request adds its own “trace”
- Tracing is made of segments (+ sub segments)
- Annotations can be added to traces to provide extra-information
- Ability to trace:
  - Every request
  - Sample request (as a % for example or a rate per minute)
- X-Ray Security:
  - IAM for authorization
  - KMS for encryption at rest

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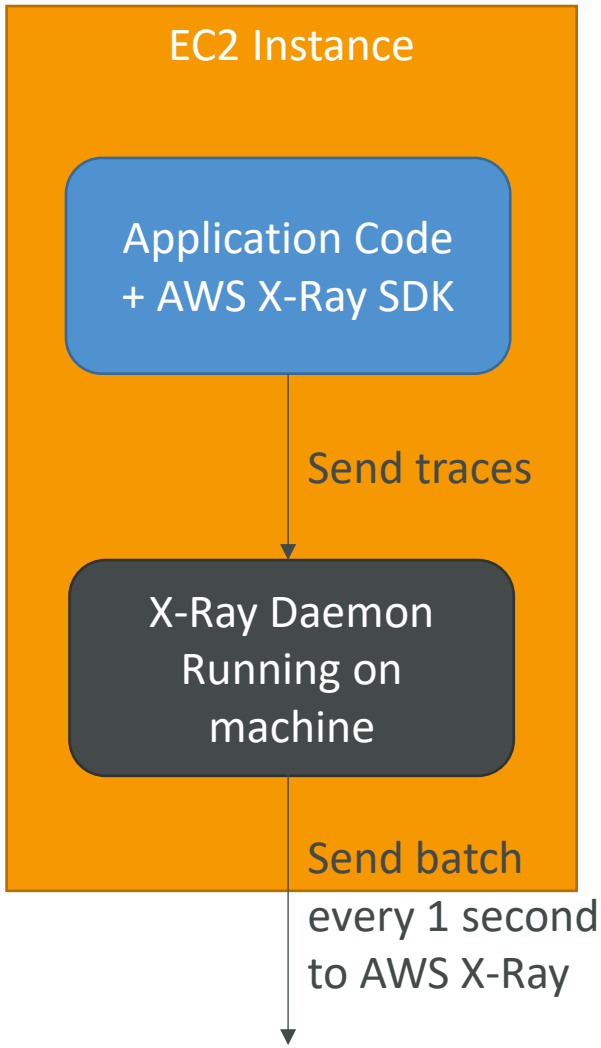
AWS X-Ray
How to enable it?

I) Your code (Java, Python, Go, Node.js, .NET) must import the AWS X-Ray SDK

- Very little code modification needed
- The application SDK will then capture:
  - Calls to AWS services
  - HTTP / HTTPS requests
  - Database Calls (MySQL, PostgreSQL, DynamoDB)
  - Queue calls (SQS)

2) Install the X-Ray daemon or enable X-Ray AWS Integration

- X-Ray daemon works as a low level UDP packet interceptor (Linux / Windows / Mac…)
- AWS Lambda / other AWS services already run the X-Ray daemon for you
- Each application must have the IAM rights to write data to X-Ray



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The X-Ray magic

- X-Ray service collects data from all the different services
- Service map is computed from all the segments and traces
- X-Ray is graphical, so even non technical people can help troubleshoot

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AWS X-Ray Troubleshooting

- If X-Ray is not working on EC2
  - Ensure the EC2 IAM Role has the proper permissions
  - Ensure the EC2 instance is running the X-Ray Daemon

- To enable on AWS Lambda:
  - Ensure it has an IAM execution role with proper policy (AWSX-RayWriteOnlyAccess)
  - Ensure that X-Ray is imported in the code
  - Enable **Lambda X-Ray Active Tracing**

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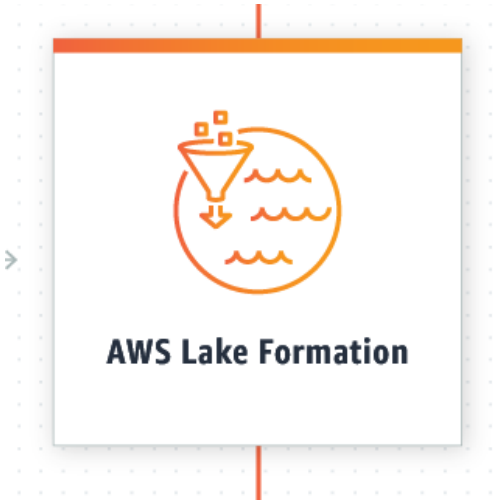
AWS Lake Formation

- “Makes it easy to set up a secure data lake in days”
- Loading data & monitoring data flows
- Setting up partitions
- Encryption & managing keys
- Defining transformation jobs & monitoring them
- Access control
- Auditing
- Built on top of Glue



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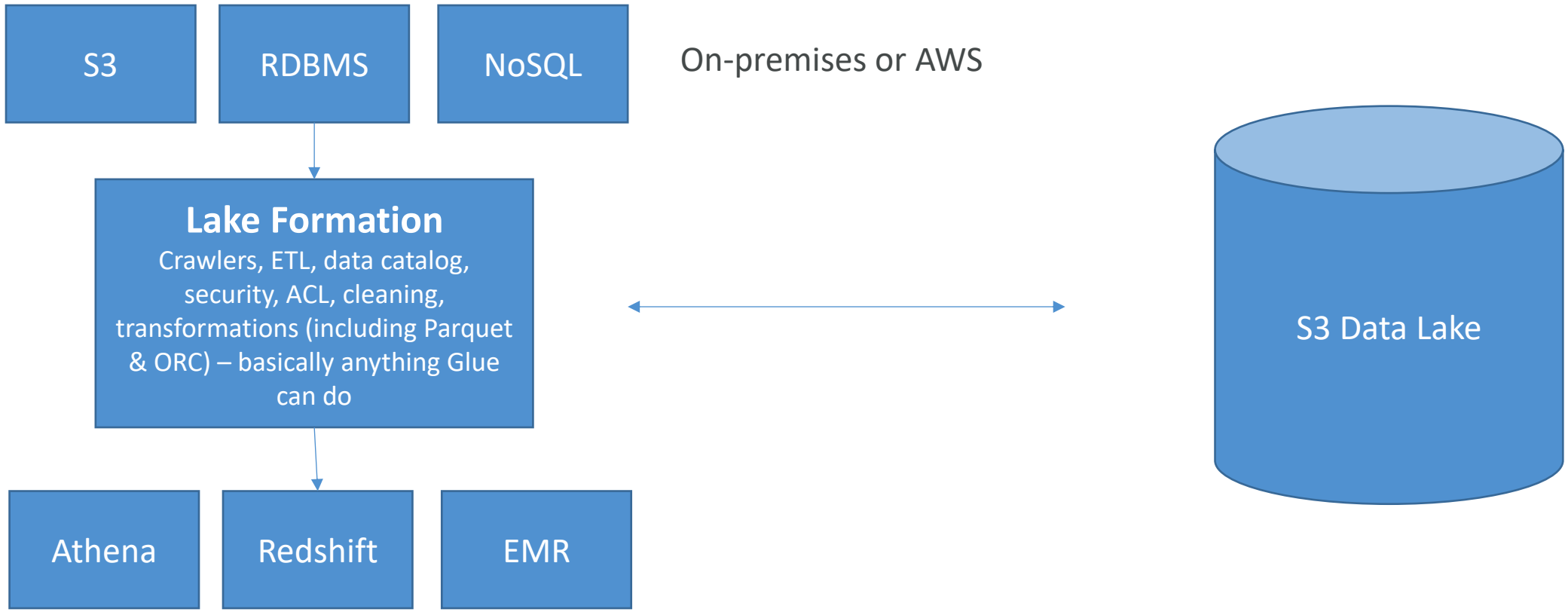
AWS Lake Formation



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AWS Lake Formation: Pricing

- No cost for Lake Formation itself
- But underlying services incur charges
  - Glue
  - S3
  - EMR
  - Athena
  - Redshift

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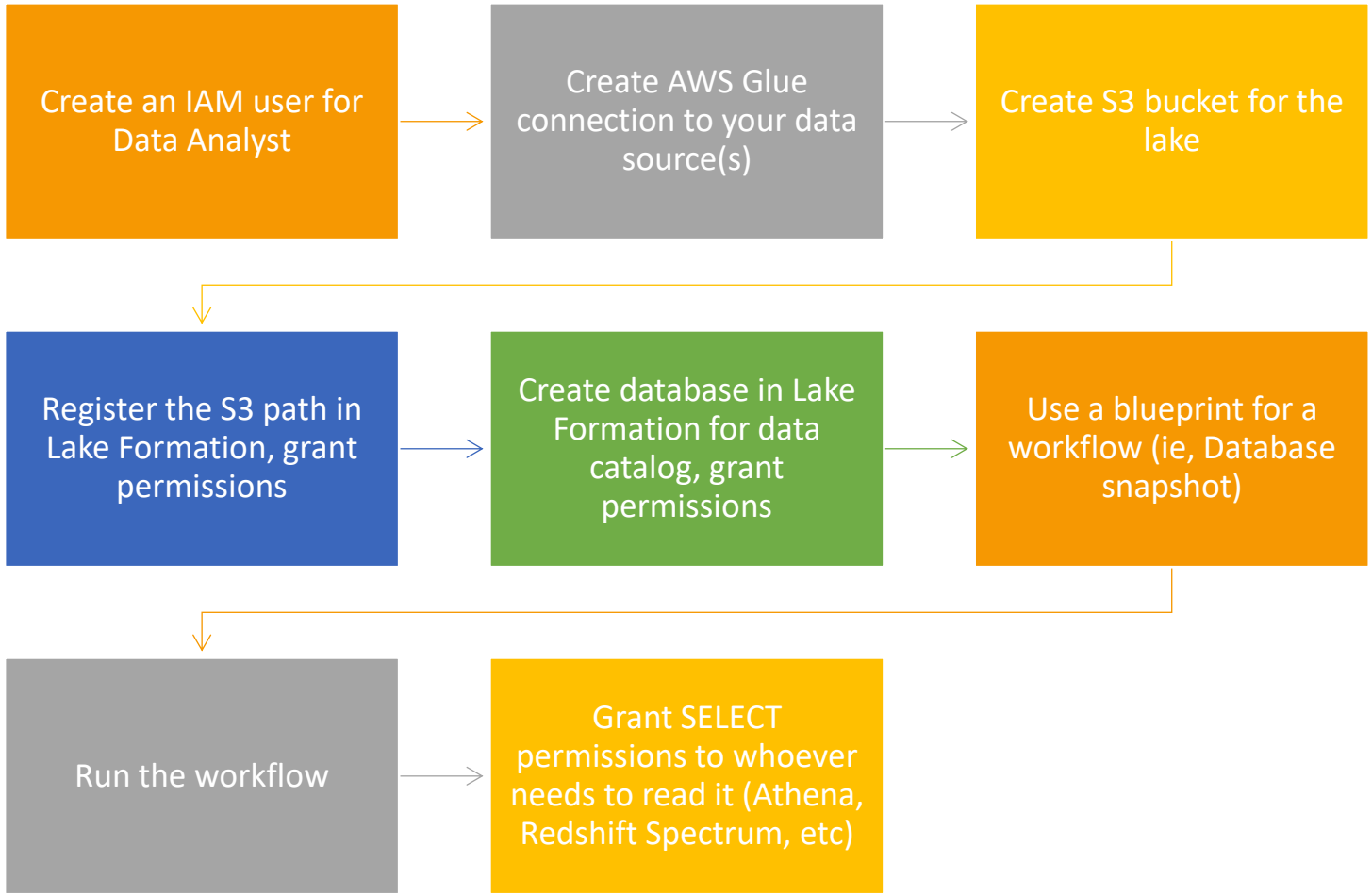
AWS Lake Formation: Building a Data Lake



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AWS Lake Formation: The Finer Points

- Cross-account Lake Formation permission
  - Recipient must be set up as a data lake administrator
  - Can use AWS Resource Access Manager for accounts external to your organization
  - IAM permissions for cross-account access
- Lake Formation does not support manifests in Athena or Redshift queries
- IAM permissions on the KMS encryption key are needed for encrypted data catalogs in Lake Formation
- IAM permissions needed to create blueprints and workflows

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AWS Lake Formation: Governed Tables and Security

- Now supports “Governed Tables” that support ACID transactions across multiple tables
  - New type of S3 table
  - Can’t change choice of governed afterwards
  - Works with streaming data too (Kinesis)
  - Can query with Athena
- Storage Optimization with Automatic Compaction
- Granular Access Control with Row and Cell-Level Security
  - Both for governed and S3 tables
- Above features incur additional charges based on usage

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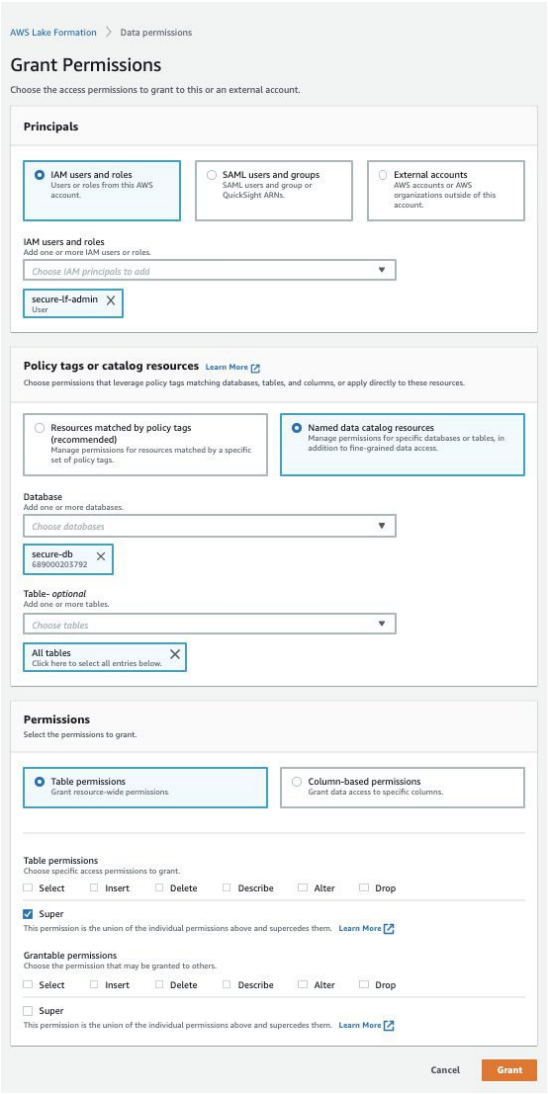
Data Permissions in Lake Formation

- Can tie to IAM users/roles, SAML, or external AWS accounts
- Can use policy tags on databases, tables, or columns
- Can select specific permissions for tables or columns



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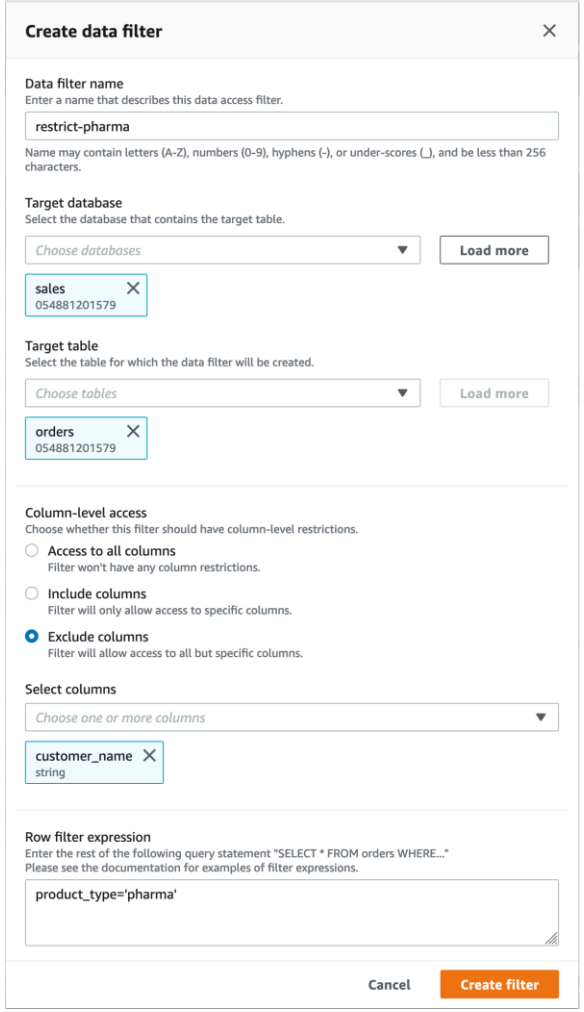
Data Filters in Lake Formation

- Column, row, or cell-level security
- Apply when granting SELECT permission on tables
- “All columns” + row filter = row-level security
- “All rows” + specific columns = column-level security
- Specific columns + specific rows = cell-level security
- Create filters via the console (seen here) or via `CreateDataCellsFilter` API.



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Security, Identity, and Compliance

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Principle of Least Privilege

- Grant only the permissions required to perform a task
- Start with broad permissions while developing
- But lock it down once you have a better idea of the exact services and operations a workload requires
- Can use IAM Access Analyzer to generate least-privilege policies based on access activity

```
{
   "Version": "2012-10-17",
   "Statement": [
      {
         "Sid": "AllowListBucket",
         "Effect": "Allow",
         "Action": "s3:ListBucket",
         "Resource": "arn:aws:s3:::my-specific-bucket",
         "Condition": {
            "StringLike": {
               "s3:prefix": "data/reports/*"
            }
         }
      },
      {
         "Sid": "AllowReadCSVFiles",
         "Effect": "Allow",
         "Action": [
            "s3:GetObject",
            "s3:GetObjectVersion"
         ],
         "Resource": "arn:aws:s3:::my-specific-bucket/data/reports/*.csv"
      }
   ]
}
```


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Data Masking and Anonymization

- Dealing with PII or other sensitive data
- Masking obfuscates data
  - For example, masking all but the last 4 digits of a credit card or social security number
  - Masking passwords
  - Supported in Glue DataBrew and Redshift

```
--create a masking policy that fully masks the credit card number
CREATE MASKING POLICY mask_credit_card_full WITH (credit_card
VARCHAR(256)) USING ('000000XXXX0000'::TEXT);
```


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Data Masking and Anonymization

- Anonymization techniques
  - Replace with random
  - Shuffle
  - Encrypt (deterministic or probabilistic)
  - Hashing
- Or just delete it or don't import it in the first place.

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Key Salting

- Salting involves appending or prepending a random value, known as a "salt," to a piece of data (often a password) before hashing it.
- Prevents pre-computed "rainbow table" attacks where adversaries use pre-generated hashes of commonly used passwords to find matches.
- Ensures that the same piece of data (like two identical passwords) doesn't produce the same hash across different instances due to the unique salt.
- Use strong, cryptographically secure random values for salts
- Rotate salts periodically
- Each user should have a unique salt
- Salt and hash passwords before storing

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Key Salting: Example

Username	Salt value	String to be hashed	Hashed value = SHA256 (Password + Salt value)
user1	D;%yL9TS:5PalS/d	**password123**D;%yL9TS:5PalS/d	9C9B913EB1B6254F4737CE947EFD16F16E916F9D6EE5C1102A2002E48D4C88BD
user2	)<,-<U(jLezy4j>*	**password123**)<,-<U(jLezy4j>*	6058B4EB46BD6487298B59440EC8E70EAE482239FF2B4E7CA69950DFBD5532F2

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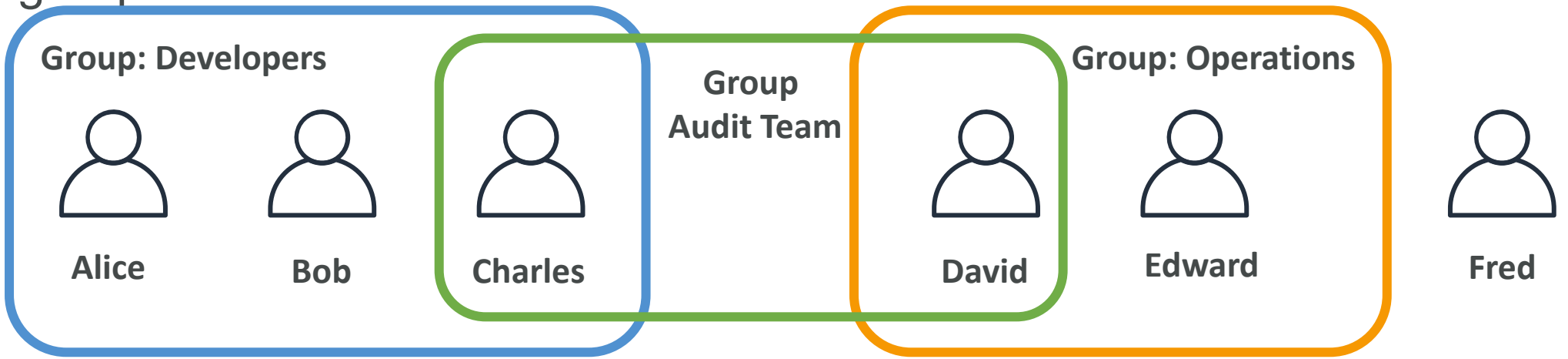
IAM: Users & Groups

- IAM = Identity and Access Management, **Global** service
- **Root account** created by default, shouldn't be used or shared
- **Users** are people within your organization, and can be grouped
- **Groups** only contain users, not other groups
- Users don't have to belong to a group, and user can belong to multiple groups



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IAM: Permissions

- **Users or Groups** can be assigned JSON documents called policies
- These policies define the **permissions** of the users
- In AWS you apply the **least privilege principle**: don't give more permissions than a user needs

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "ec2:Describe*",
            "Resource": "*"
        },
        {
            "Effect": "Allow",
            "Action": "elasticloadbalancing:Describe*",
            "Resource": "*"
        },
        {
            "Effect": "Allow",
            "Action": [
                "cloudwatch:ListMetrics",
                "cloudwatch:GetMetricStatistics",
                "cloudwatch:Describe*"
            ],
            "Resource": "*"
        }
    ]
}
```


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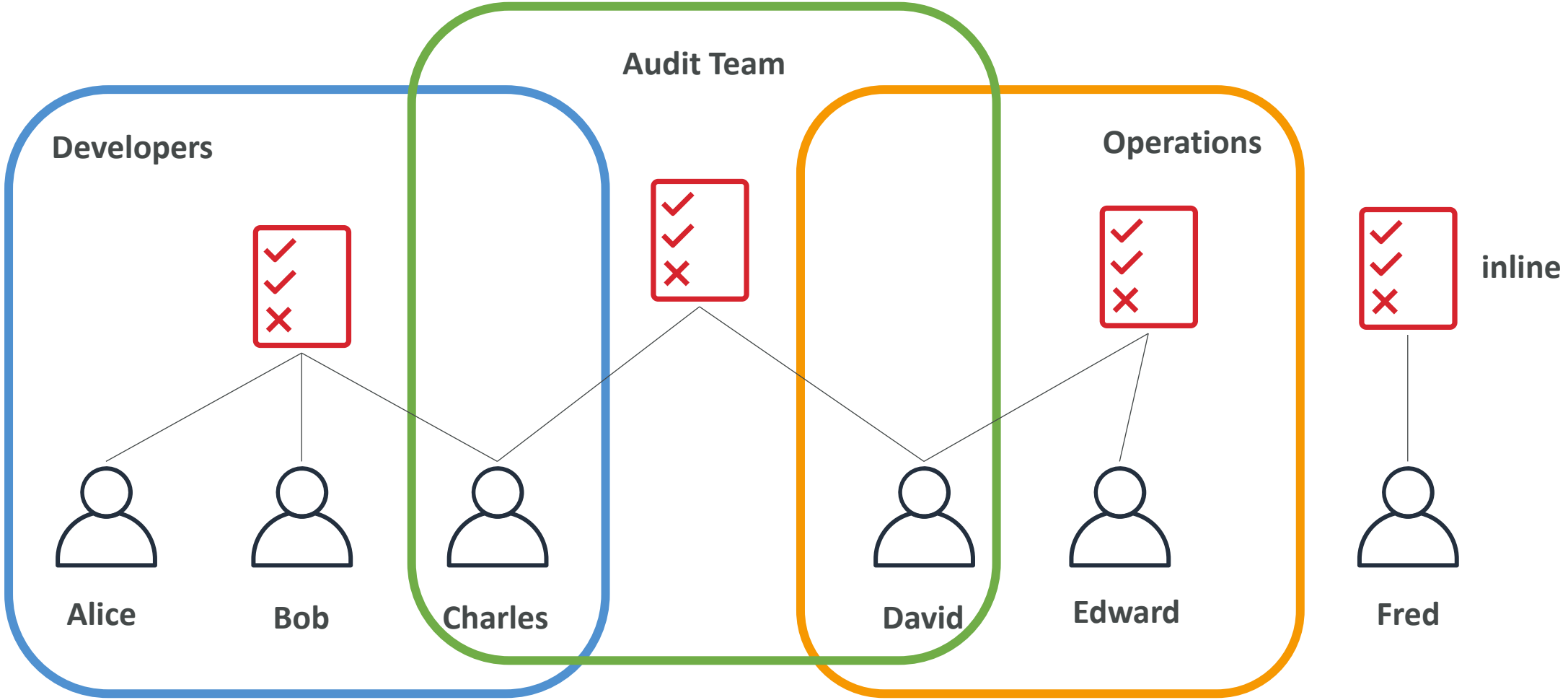
IAM Policies inheritance



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IAM Policies Structure

- Consists of
  - **Version:** policy language version, always include “2012-10-17”
  - **Id:** an identifier for the policy (optional)
  - **Statement:** one or more individual statements (required)
- Statements consists of
  - **Sid:** an identifier for the statement (optional)
  - **Effect:** whether the statement allows or denies access (Allow, Deny)
  - **Principal:** account/user/role to which this policy applied to
  - **Action:** list of actions this policy allows or denies
  - **Resource:** list of resources to which the actions applied to
  - **Condition:** conditions for when this policy is in effect (optional)

```
{
    "Version": "2012-10-17",
    "Id": "S3-Account-Permissions",
    "Statement": [
        {
            "Sid": "1",
            "Effect": "Allow",
            "Principal": {
                "AWS": ["arn:aws:iam::123456789012:root"]
            },
            "Action": [
                "s3:GetObject",
                "s3:PutObject"
            ],
            "Resource": ["arn:aws:s3:::mybucket/*"]
        }
    ]
}
```


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IAM – Password Policy

- Strong passwords = higher security for your account
- In AWS, you can setup a password policy:
  - Set a minimum password length
  - Require specific character types:
    - including uppercase letters
    - lowercase letters
    - numbers
    - non-alphanumeric characters
  - Allow all IAM users to change their own passwords
  - Require users to change their password after some time (password expiration)
  - Prevent password re-use

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Multi Factor Authentication - MFA

- Users have access to your account and can possibly change configurations or delete resources in your AWS account
- **You want to protect your Root Accounts and IAM users**
- MFA = password *you know* + security device *you own*

- **Main benefit of MFA:**
  if a password is stolen or hacked, the account is not compromised

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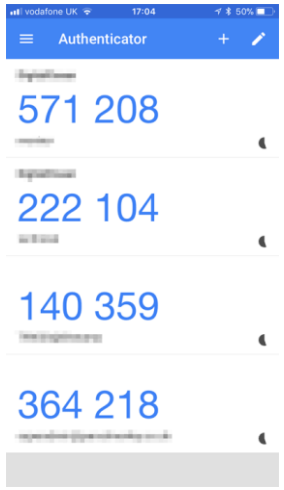
MFA devices options in AWS

Virtual MFA device



Google Authenticator
(phone only)

Authy
(phone only)

Support for multiple tokens on a single device.

Universal 2nd Factor (U2F) Security Key

YubiKey by Yubico (3rd party)

Support for multiple root and IAM users using a single security key

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MFA devices options in AWS

Hardware Key Fob MFA Device

Provided by Gemalto (3rd party)

Hardware Key Fob MFA Device for AWS GovCloud (US)

Provided by SurePassID (3rd party)

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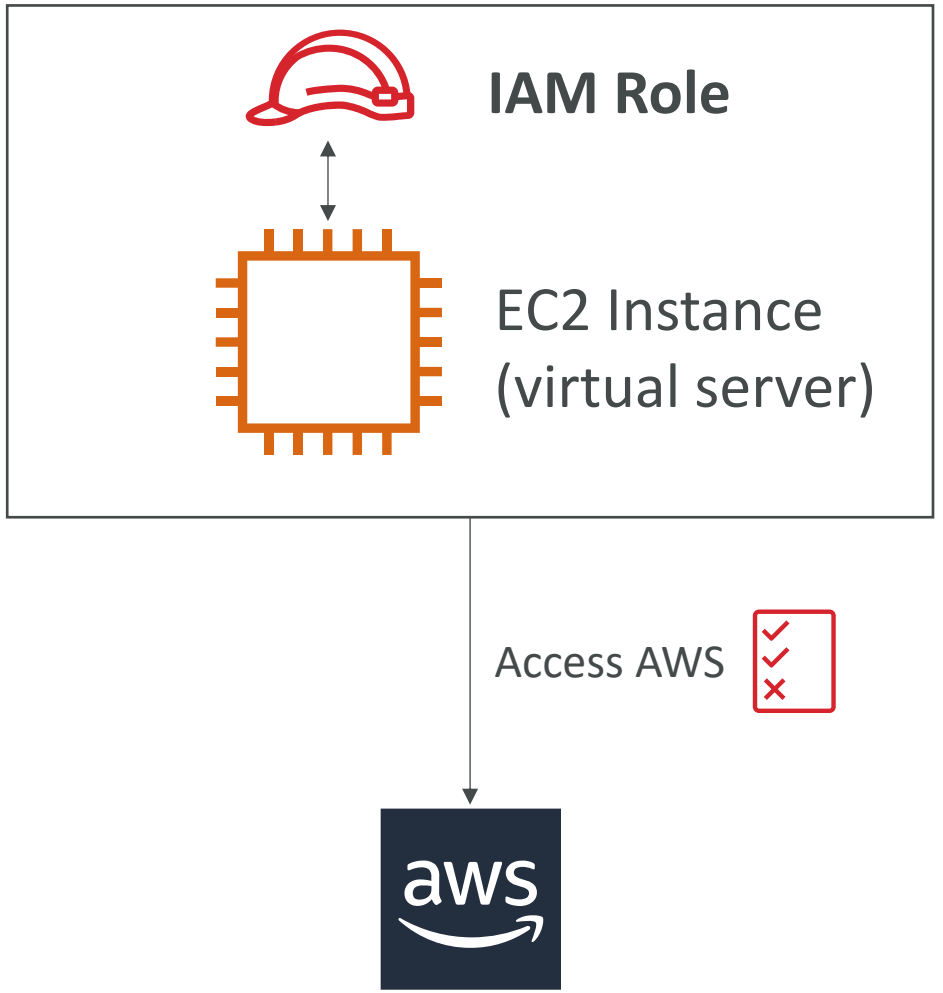
IAM Roles for Services

- Some AWS service will need to perform actions on your behalf
- To do so, we will assign **permissions** to AWS services with **IAM Roles**
- Common roles:
  - EC2 Instance Roles
  - Lambda Function Roles
  - Roles for CloudFormation



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AWS IAM Identity Center (successor to AWS Single Sign-On)

- One login (single sign-on) for all your
  - **AWS accounts in AWS Organizations**
  - Business cloud applications (e.g., Salesforce, Box, Microsoft 365, …)
  - SAML2.0-enabled applications
  - EC2 Windows Instances

- Identity providers
  - Built-in identity store in IAM Identity Center
  - 3rd party: Active Directory (AD), OneLogin, Okta…

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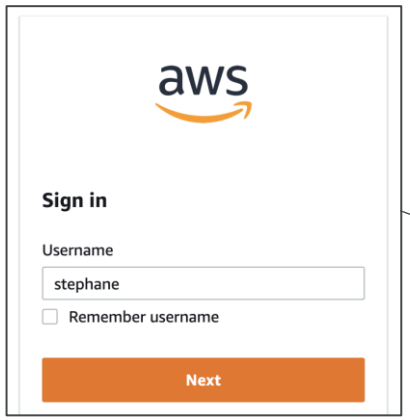
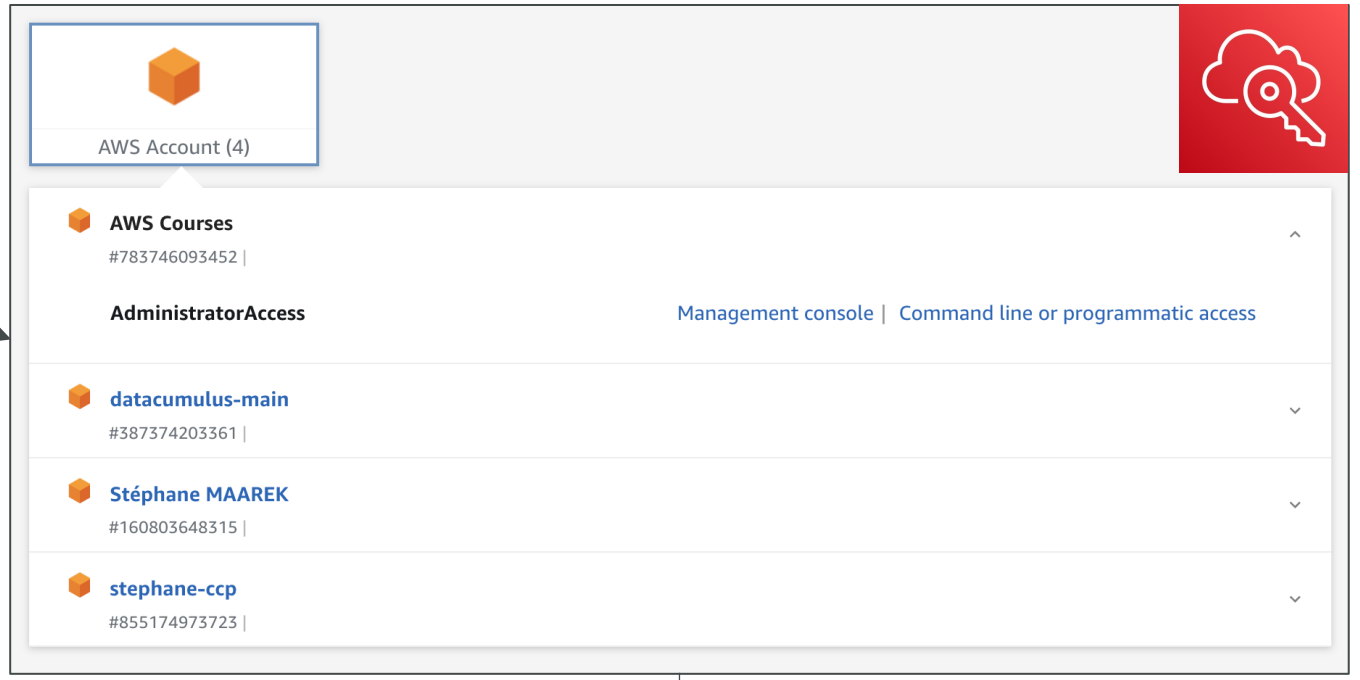
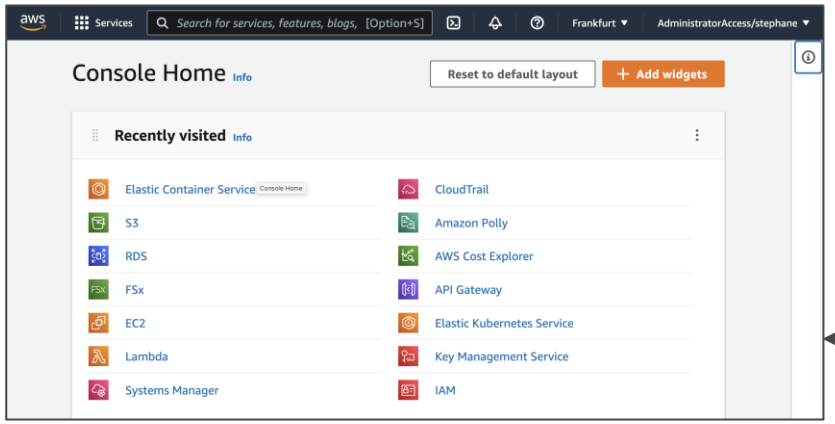
AWS IAM Identity Center – Login Flow







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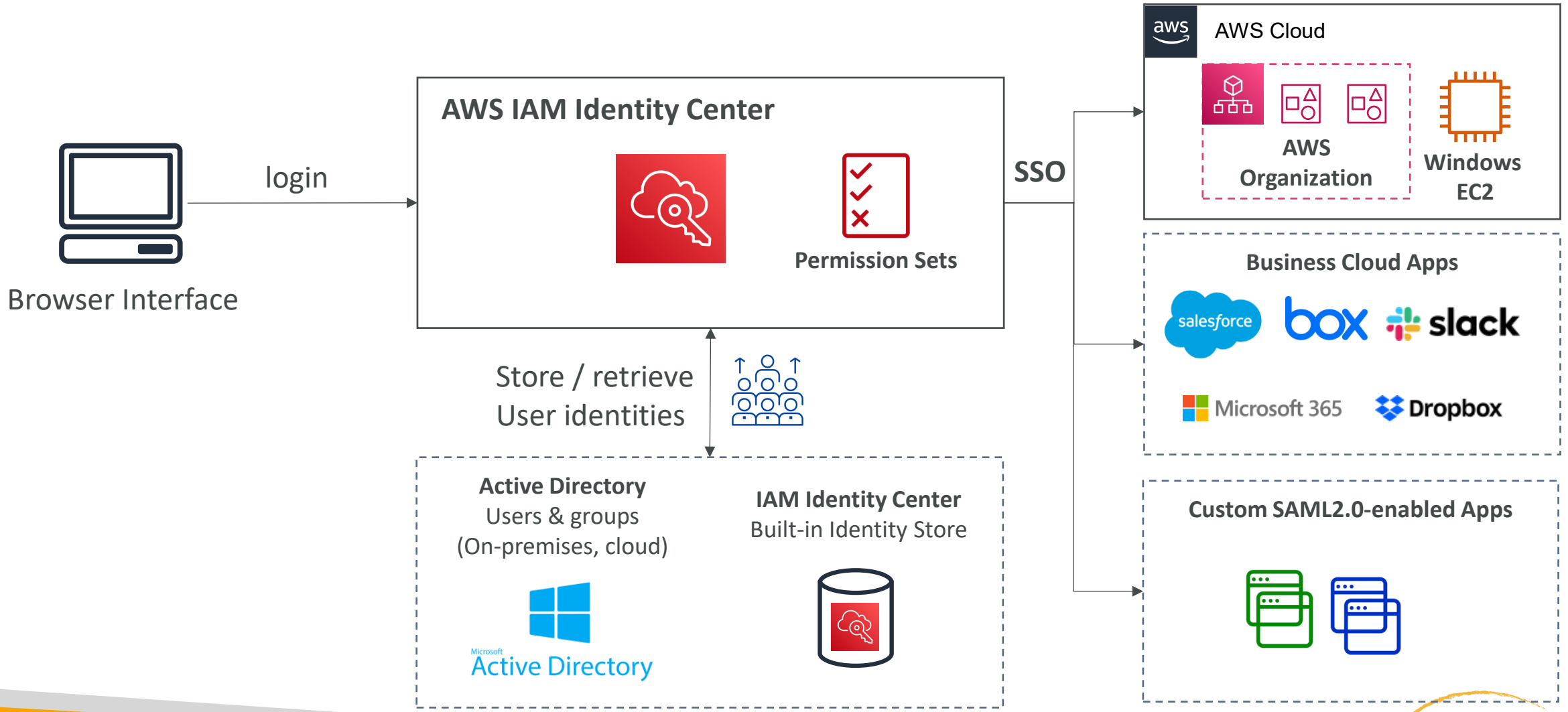
AWS IAM Identity Center



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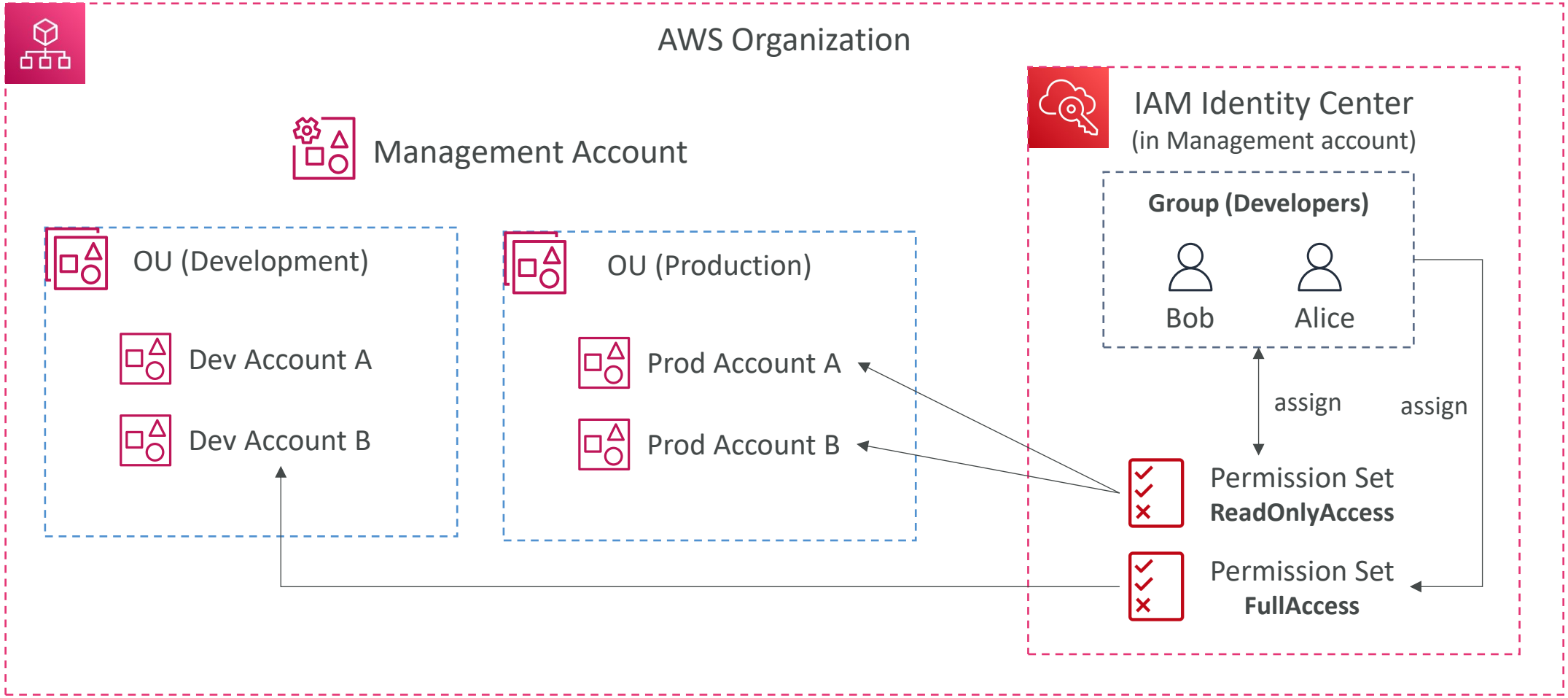
IAM Identity Center



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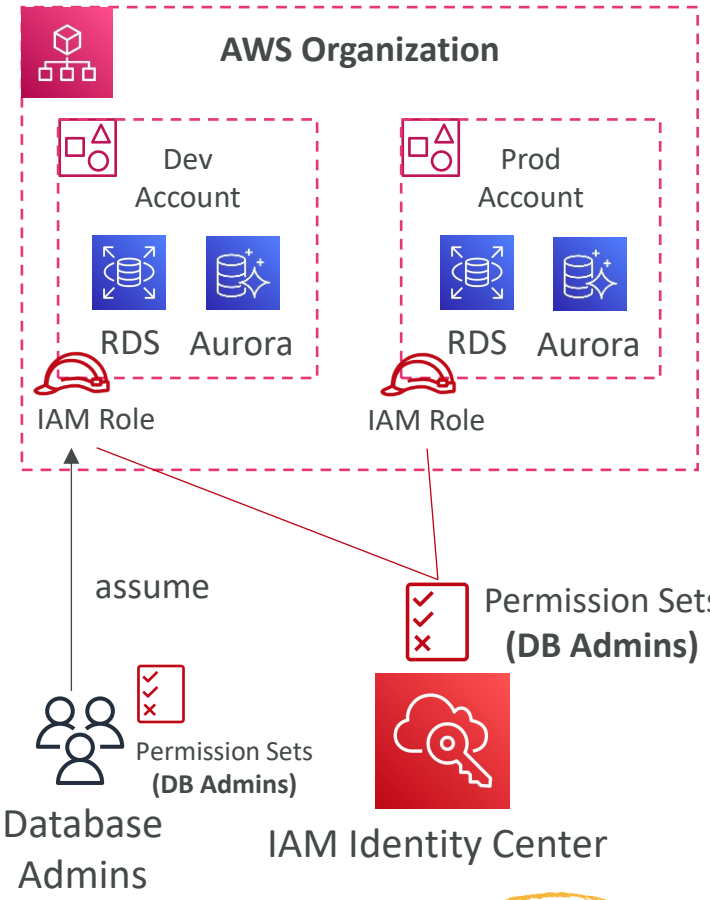
AWS IAM Identity Center
Fine-grained Permissions and Assignments

- **Multi-Account Permissions**
  - Manage access across AWS accounts in your AWS Organization
  - Permission Sets – a collection of one or more IAM Policies assigned to users and groups to define AWS access

- **Application Assignments**
  - SSO access to many SAML 2.0 business applications (Salesforce, Box, Microsoft 365, …)
  - Provide required URLs, certificates, and metadata

- **Attribute-Based Access Control (ABAC)**
  - Fine-grained permissions based on users' attributes stored in IAM Identity Center Identity Store
  - Example: cost center, title, locale, …
  - Use case: Define permissions once, then modify AWS access by changing the attributes



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AWS Control Tower

- Easy way to **set up and govern a secure and compliant multi-account AWS environment** based on best practices
- AWS Control Tower uses AWS Organizations to create accounts

- Benefits:
  - Automate the set up of your environment in a few clicks
  - Automate ongoing policy management using guardrails
  - Detect policy violations and remediate them
  - Monitor compliance through an interactive dashboard

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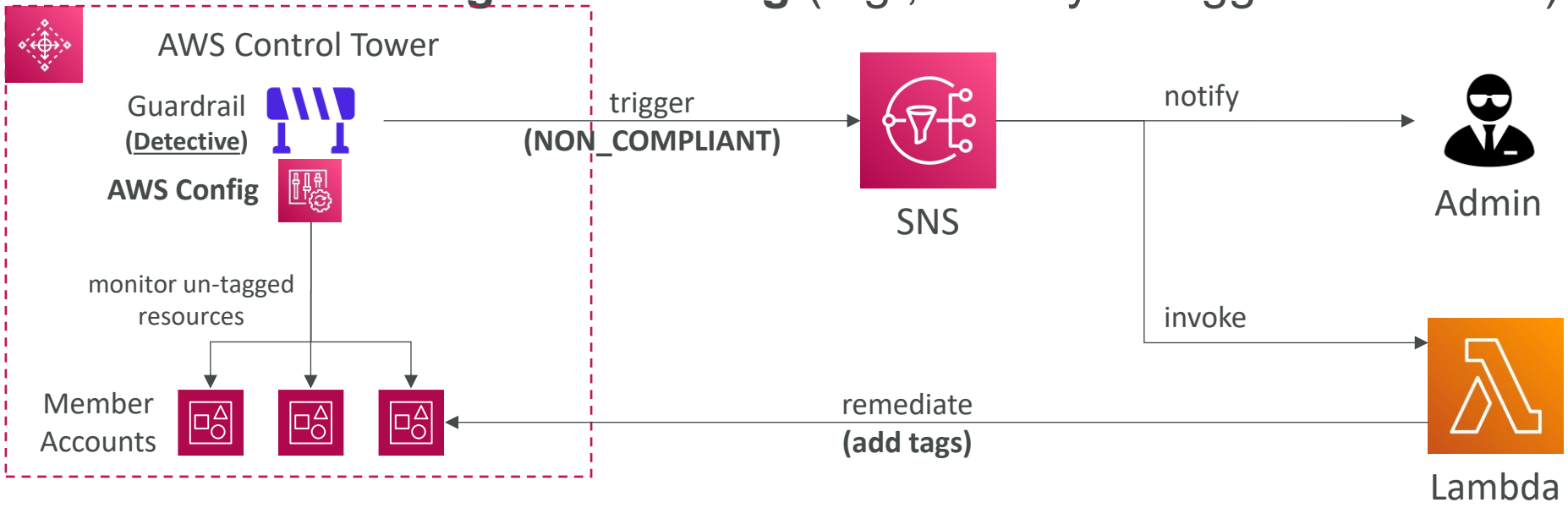
AWS Control Tower – Guardrails

- Provides ongoing governance for your Control Tower environment (AWS Accounts)
- **Preventive Guardrail – using SCPs** (e.g., Restrict Regions across all your accounts)
- **Detective Guardrail – using AWS Config** (e.g., identify untagged resources)



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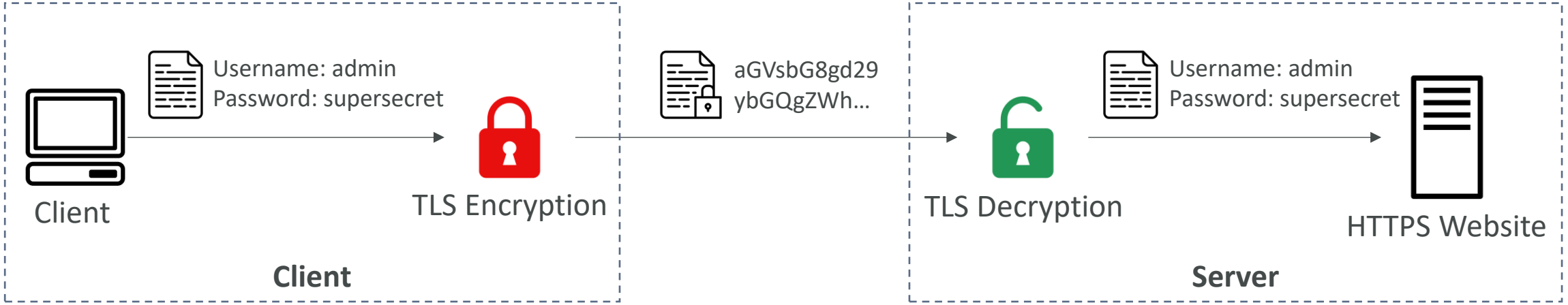
Why encryption?
Encryption in flight (TLS / SSL)

- Data is encrypted before sending and decrypted after receiving
- TLS certificates help with encryption (HTTPS)
- Encryption in flight ensures no MITM (man in the middle attack) can happen



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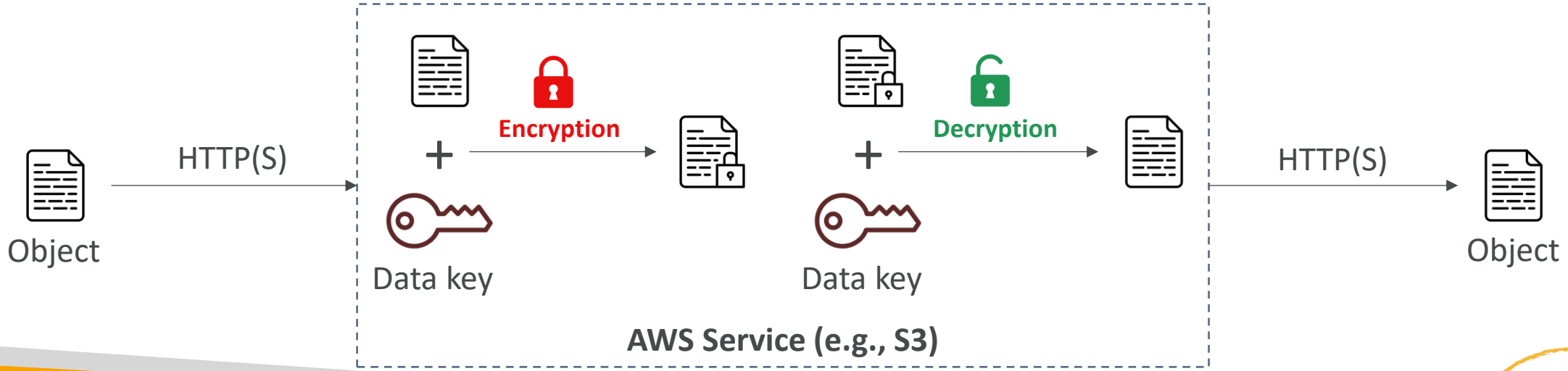
Why encryption?
Server-side encryption at rest

- Data is encrypted after being received by the server
- Data is decrypted before being sent
- It is stored in an encrypted form thanks to a key (usually a data key)
- The encryption / decryption keys must be managed somewhere, and the server must have access to it



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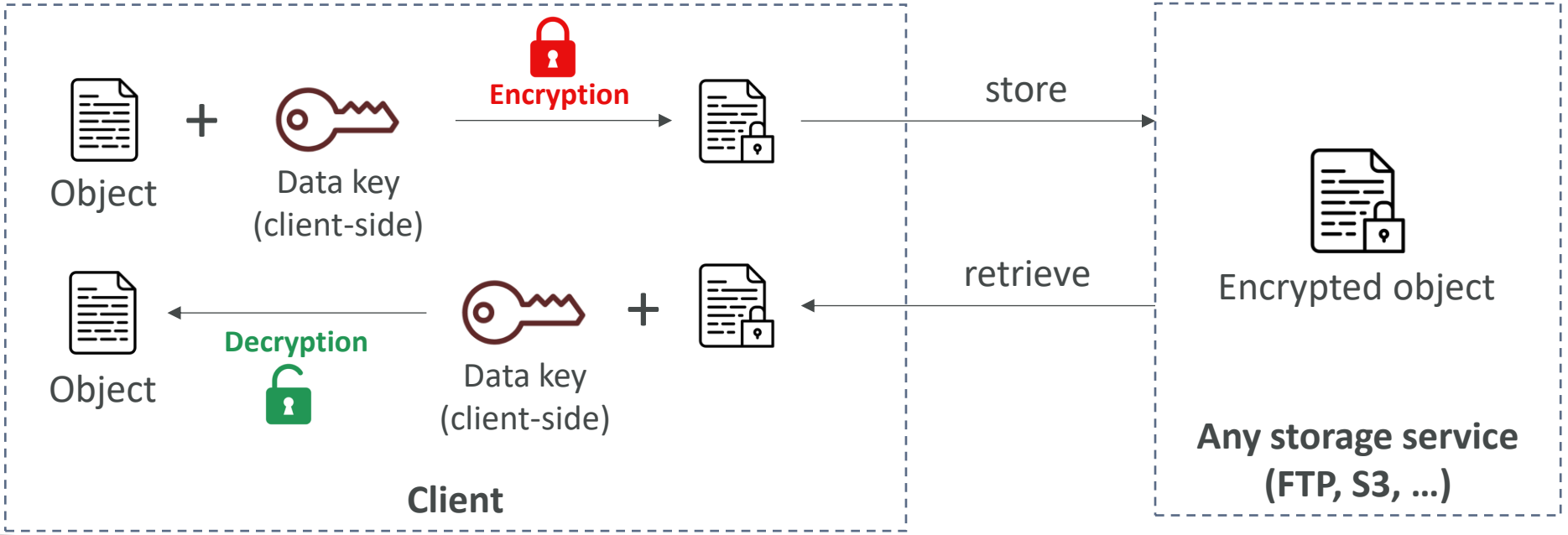
Why encryption?
Client-side encryption

- Data is encrypted by the client and never decrypted by the server
- Data will be decrypted by a receiving client
- The server should not be able to decrypt the data
- Could leverage Envelope Encryption



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AWS KMS (Key Management Service)

- Anytime you hear “encryption” for an AWS service, it’s most likely KMS
- AWS manages encryption keys for us
- Fully integrated with IAM for authorization
- Easy way to control access to your data
- Able to audit KMS Key usage using CloudTrail
- Seamlessly integrated into most AWS services (EBS, S3, RDS, SSM…)
- **Never ever store your secrets in plaintext, especially in your code!**
  - KMS Key Encryption also available through API calls (SDK, CLI)
  - Encrypted secrets can be stored in the code / environment variables

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KMS Keys Types

- **KMS Keys is the new name of KMS Customer *Master* Key**
- **Symmetric (AES-256 keys)**
  - Single encryption key that is used to Encrypt and Decrypt
  - AWS services that are integrated with KMS use Symmetric CMKs
  - You never get access to the KMS Key unencrypted (must call KMS API to use)
- **Asymmetric (RSA & ECC key pairs)**
  - Public (Encrypt) and Private Key (Decrypt) pair
  - Used for Encrypt/Decrypt, or Sign/Verify operations
  - The public key is downloadable, but you can't access the Private Key unencrypted
  - Use case: encryption outside of AWS by users who can't call the KMS API

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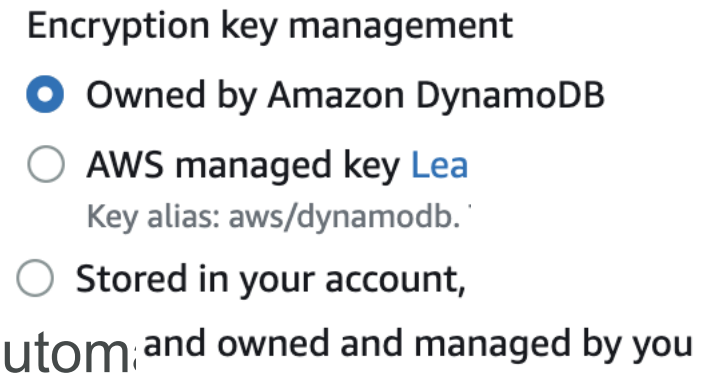
AWS KMS (Key Management Service)

- Types of KMS Keys:
  - AWS Owned Keys (free): SSE-S3, SSE-SQS, SSE-DDB (default key)
  - AWS Managed Key: **free** (aws/*service-name*, example: aws/rds or aws/ebs)
  - Customer managed keys created in KMS: **$1 / month**
  - Customer managed keys imported: **$1 / month**
  - + pay for API call to KMS ($0.03 / 10000 calls)



- Automatic Key rotation:
  - AWS-managed KMS Key: automatic every 1 year
  - Customer-managed KMS Key: (must be enabled) autom
  - Imported KMS Key: only manual rotation possible using alias

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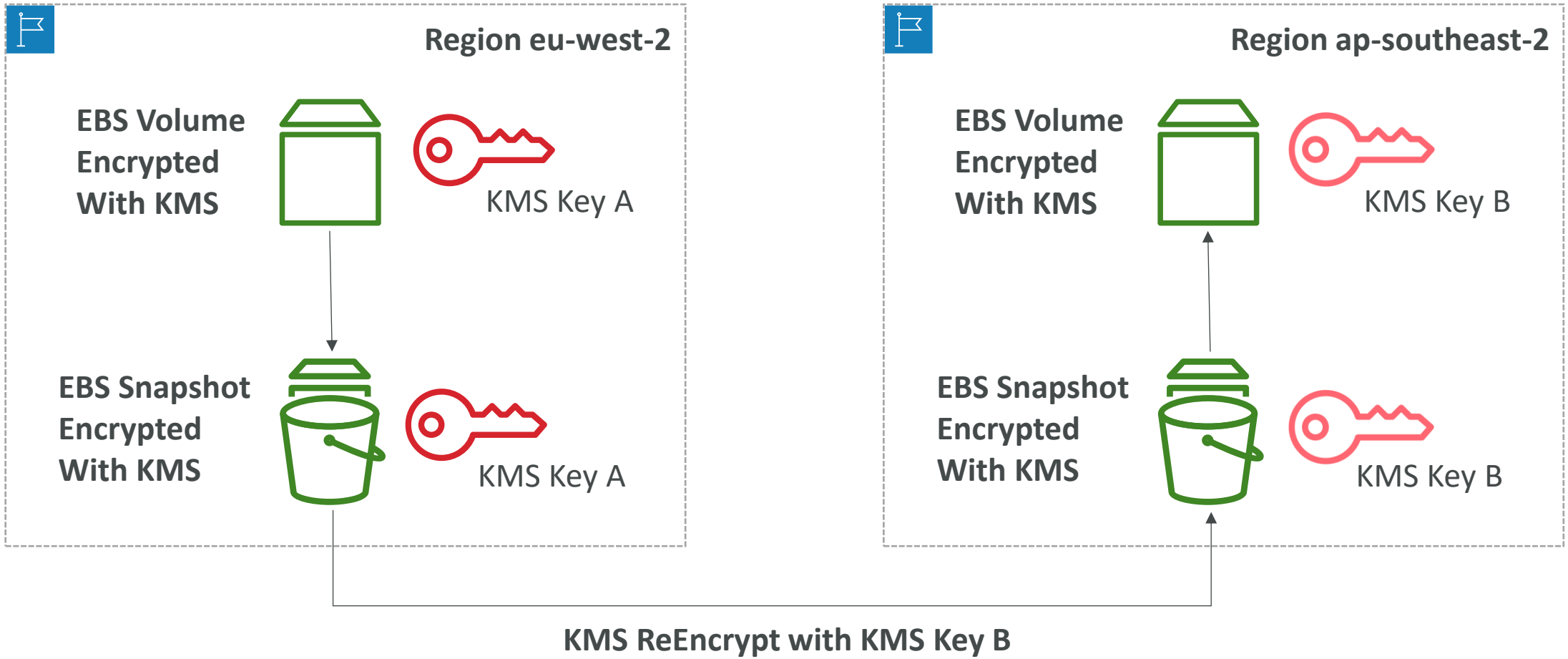
Copying Snapshots across regions



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AWS Macie

- Amazon Macie is a fully managed data security and data privacy service that uses **machine learning and pattern matching to discover and protect your sensitive data in AWS.**
- Macie helps identify and alert you to **sensitive data, such as personally identifiable information (PII)**

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AWS Secrets Manager

- Newer service, meant for storing secrets
- Capability to force **rotation of secrets** every X days
- Automate generation of secrets on rotation (uses Lambda)
- Integration with **Amazon RDS** (MySQL, PostgreSQL, Aurora)
- Secrets are encrypted using KMS

- Mostly meant for RDS integration

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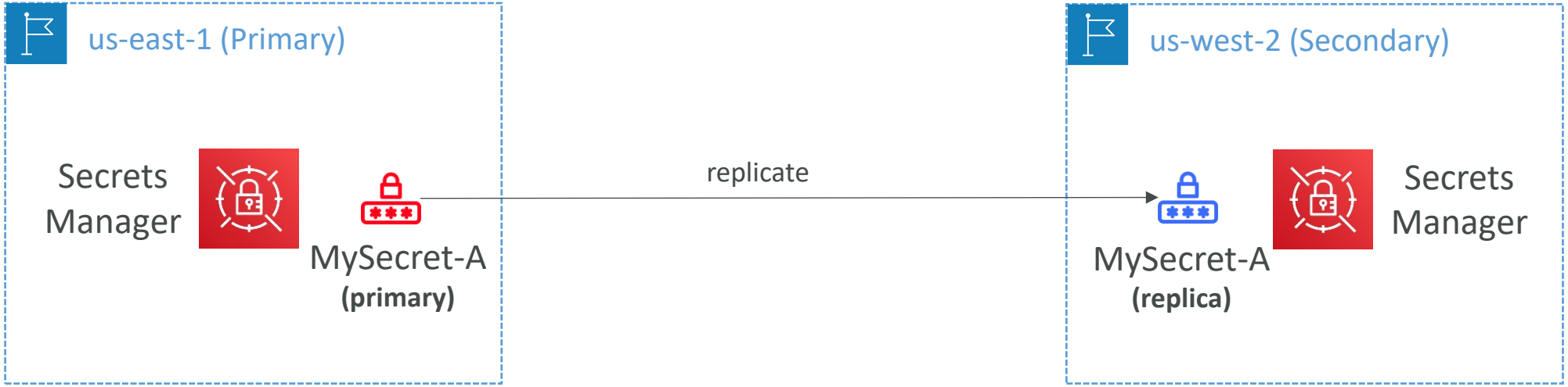
AWS Secrets Manager – Multi-Region Secrets

- Replicate Secrets across multiple AWS Regions
- Secrets Manager keeps read replicas in sync with the primary Secret
- Ability to promote a read replica Secret to a standalone Secret
- Use cases: multi-region apps, disaster recovery strategies, multi-region DB…



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Amazon Cognito

- Give users an identity to interact with our web or mobile application
- **Cognito User Pools:**
  - Sign in functionality for app users
  - Integrate with API Gateway & Application Load Balancer
- **Cognito Identity Pools (Federated Identity):**
  - Provide AWS credentials to users so they can access AWS resources directly
  - Integrate with Cognito User Pools as an identity provider
- **Cognito vs IAM:** “hundreds of users”, ”mobile users”, “authenticate with SAML”

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Cognito User Pools (CUP) – User Features

- **Create a serverless database of user for your web & mobile apps**
- Simple login: Username (or email) / password combination
- Password reset
- Email & Phone Number Verification
- Multi-factor authentication (MFA)
- Federated Identities: users from Facebook, Google, SAML…

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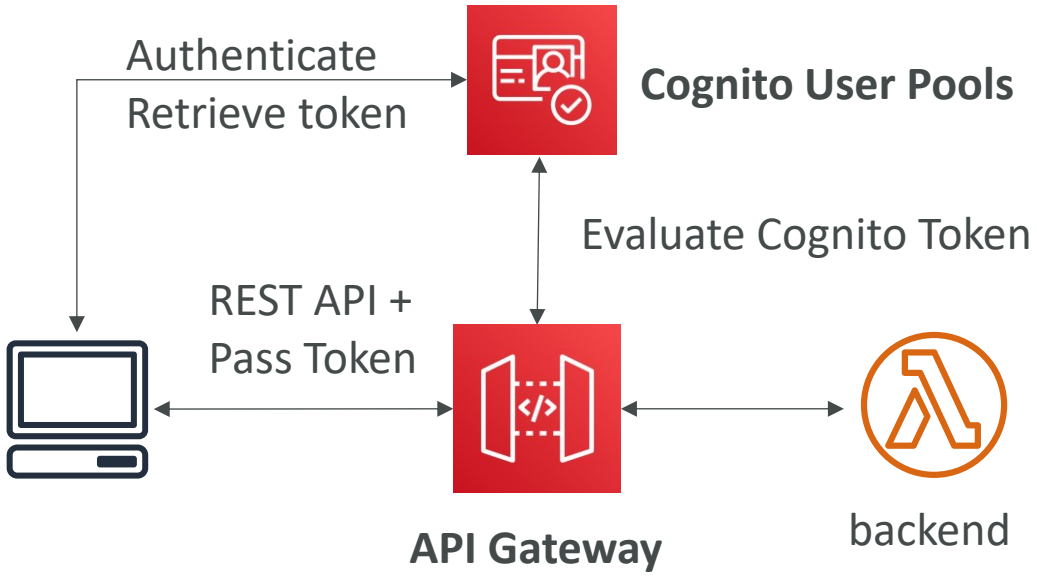
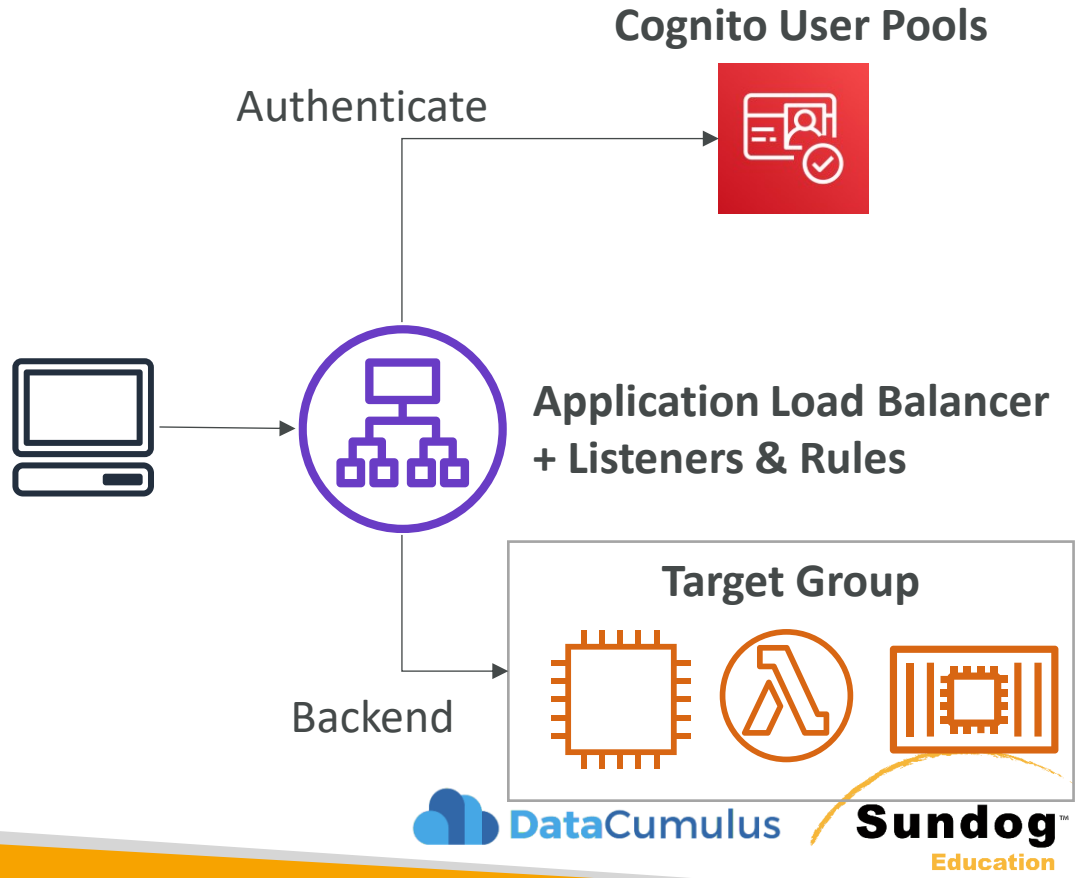
Cognito User Pools (CUP) - Integrations

- CUP integrates with **API Gateway** and **Application Load Balancer**





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Cognito Identity Pools (Federated Identities)

- **Get identities for "users" so they obtain temporary AWS credentials**
- Users source can be Cognito User Pools, 3rd party logins, etc…

- **Users can then access AWS services directly or through API Gateway**
- The IAM policies applied to the credentials are defined in Cognito
- They can be customized based on the user_id for fine grained control
- **Default IAM roles** for authenticated and guest users

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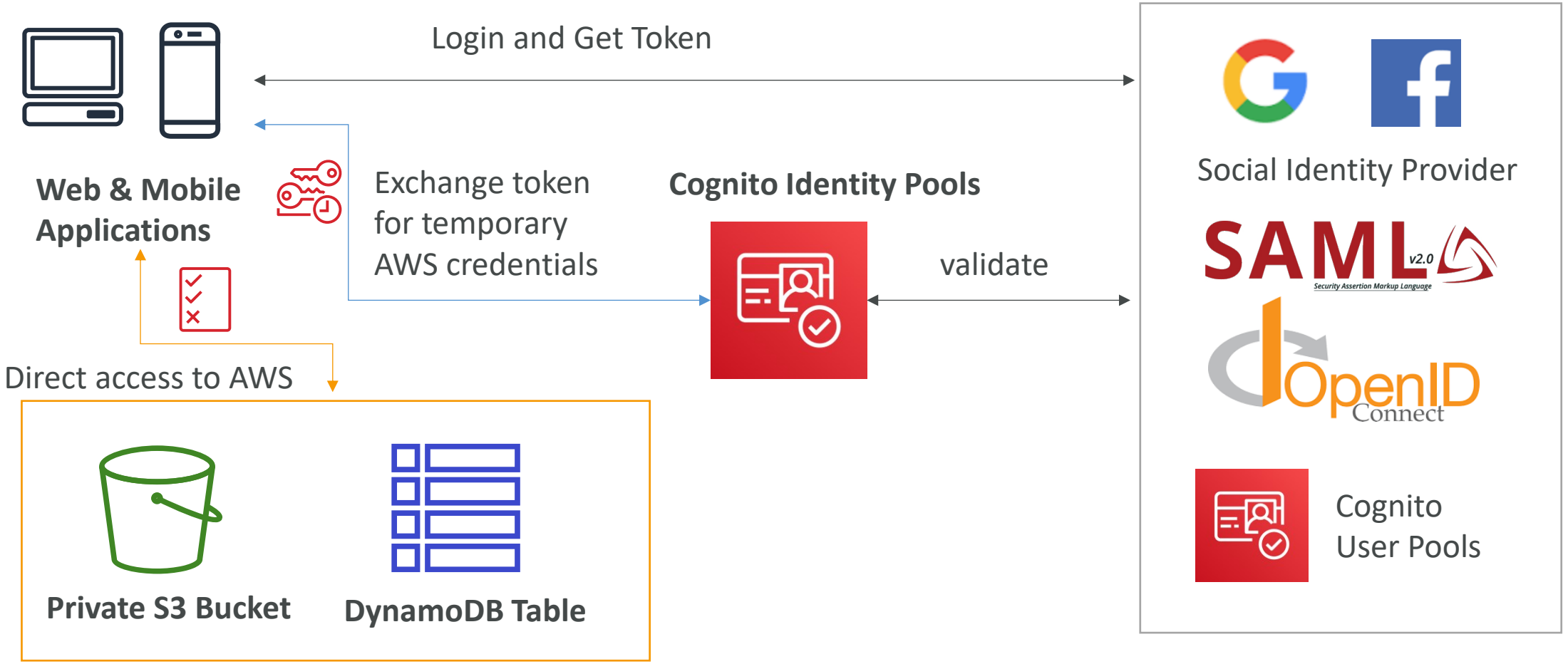
Cognito Identity Pools – Diagram



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Cognito Identity Pools
Row Level Security in DynamoDB

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "dynamodb:GetItem", "dynamodb:BatchGetItem", "dynamodb:Query",
                "dynamodb:PutItem", "dynamodb:UpdateItem", "dynamodb:DeleteItem",
                "dynamodb:BatchWriteItem"
            ],
            "Resource": [
                "arn:aws:dynamodb:us-west-2:123456789012:table/MyTable"
            ],
            "Condition": {
                "ForAllValues:StringEquals": {
                    "dynamodb:LeadingKeys": [
                        "${cognito-identity.amazonaws.com:sub}"
                    ]
                }
            }
        }
    ]
}
```


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AWS WAF – Web Application Firewall

- Protects your web applications from common web exploits (Layer 7)
- **Layer 7 is HTTP** (vs Layer 4 is TCP/UDP)

- Deploy on
  - **Application Load Balancer**
  - **API Gateway**
  - **CloudFront**
  - **AppSync GraphQL API**
  - **Cognito User Pool**

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AWS WAF – Web Application Firewall

- Define Web ACL (Web Access Control List) Rules:
  - **IP Set: up to 10,000 IP addresses** – use multiple Rules for more IPs
  - HTTP headers, HTTP body, or URI strings Protects from common attack - **SQL injection** and **Cross-Site Scripting (XSS)**
  - Size constraints, **geo-match (block countries)**
  - **Rate-based rules** (to count occurrences of events) **– for DDoS protection**

- Web ACL are Regional except for CloudFront
- A rule group is **a reusable set of rules that you can add to a web ACL**

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VPC – Crash Course

- VPC is something you should know in depth for the AWS Certified Solutions Architect Associate & AWS Certified SysOps Administrator

- **At the AWS Certified Developer Level, you should know about:**
  - VPC, Subnets, Internet Gateways & NAT Gateways
  - Security Groups, Network ACL (NACL), VPC Flow Logs
  - VPC Peering, VPC Endpoints
  - Site to Site VPN & Direct Connect

- I will just give you an overview, less than1 or 2 questions at your exam.
- Later in the course, I will be highlighting when VPC concepts are helpful

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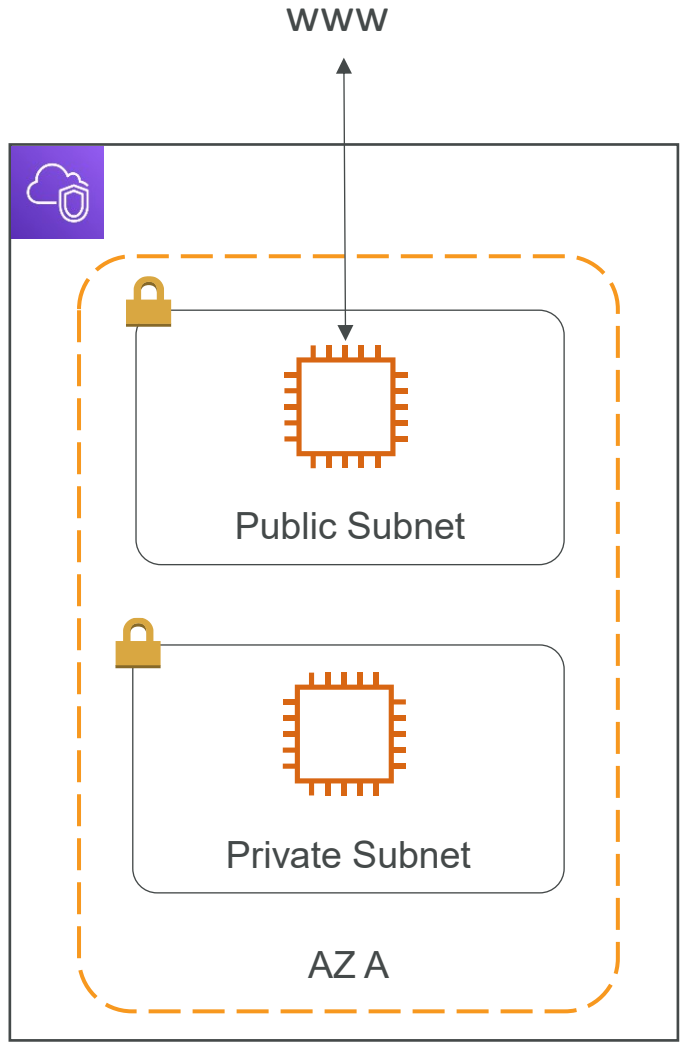
VPC & Subnets Primer

- **VPC**: private network to deploy your resources (regional resource)
- **Subnets** allow you to partition your network inside your VPC (Availability Zone resource)
- A **public subnet** is a subnet that is accessible from the internet
- A **private subnet** is a subnet that is not accessible from the internet
- To define access to the internet and between subnets, we use **Route Tables**.



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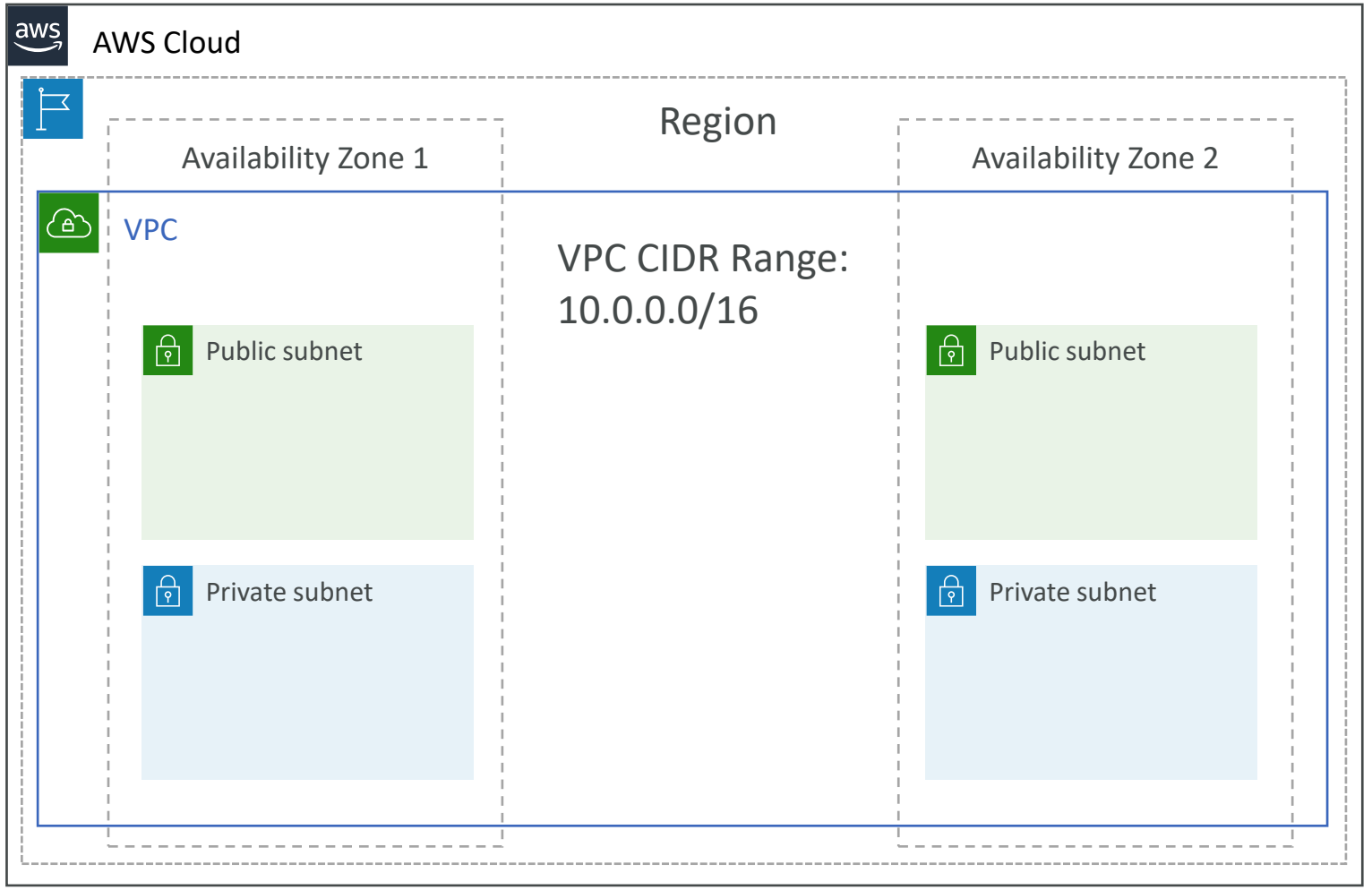
VPC Diagram



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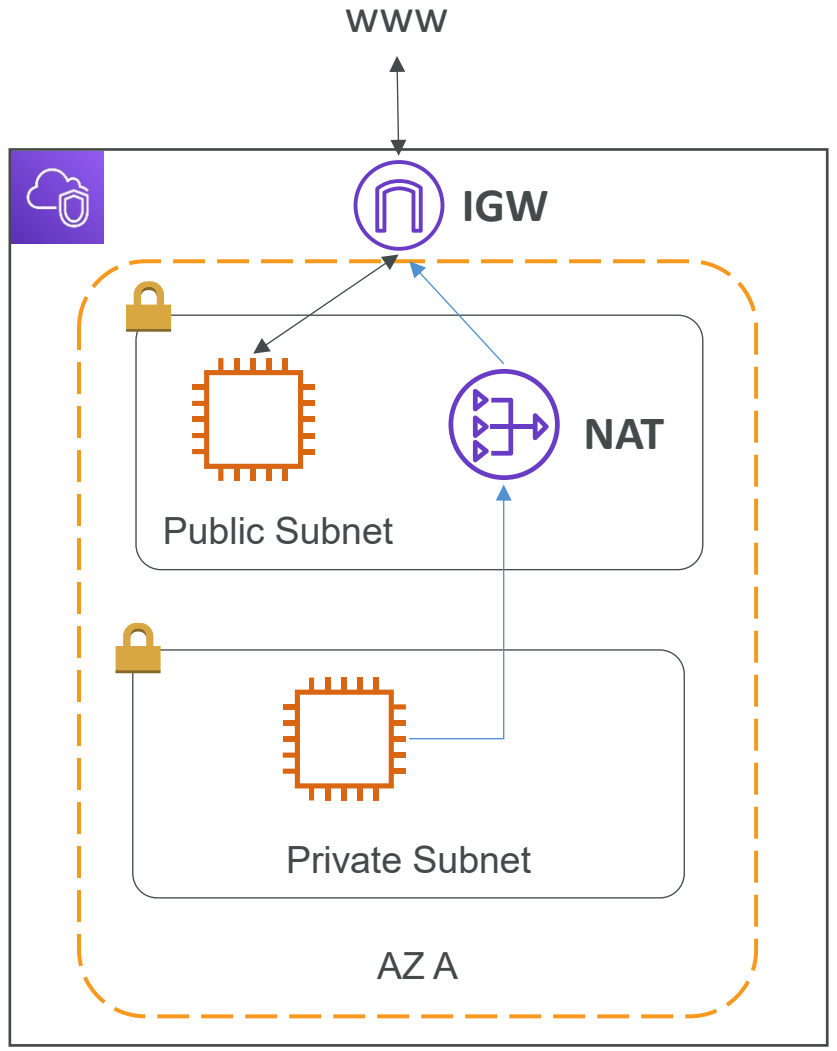
Internet Gateway & NAT Gateways

- **Internet Gateways** helps our VPC instances connect with the internet
- Public Subnets have a route to the internet gateway.

- **NAT Gateways** (AWS-managed) & **NAT Instances** (self-managed) allow your instances in your **Private Subnets** to access the internet while remaining private



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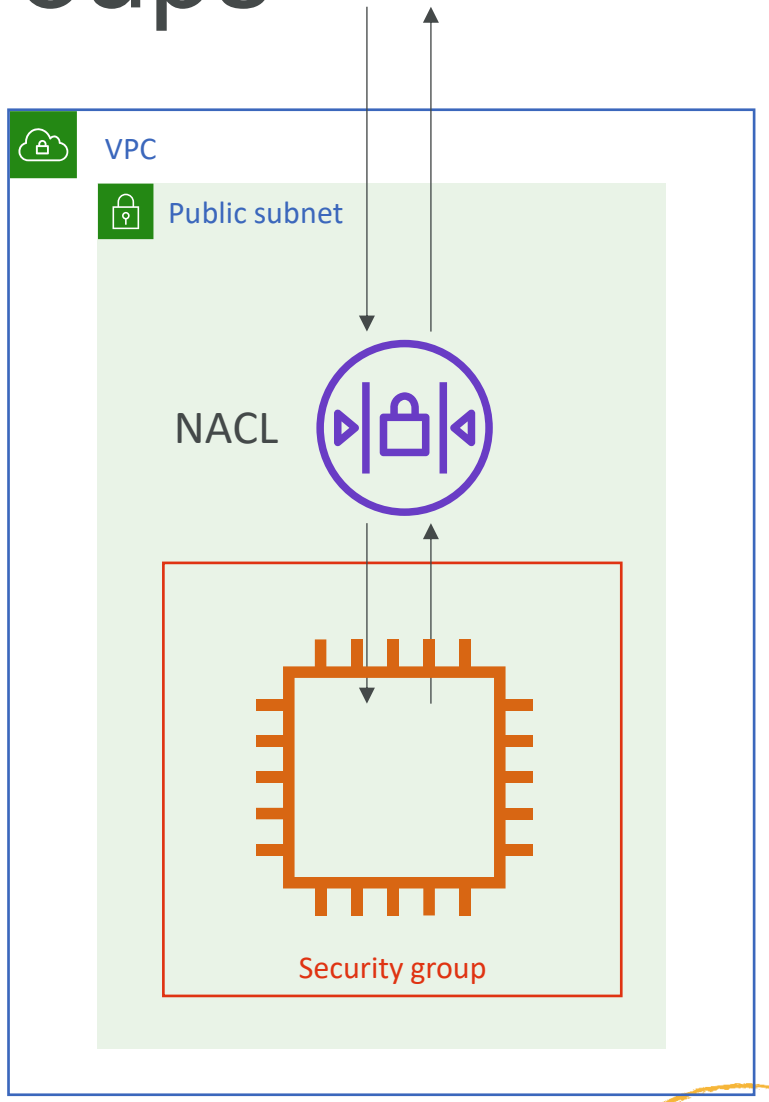
Network ACL & Security Groups

- **NACL** (Network ACL)
  - A firewall which controls traffic from and to subnet
  - Can have ALLOW and DENY rules
  - Are attached at the **Subnet** level
  - Rules only include IP addresses
- **Security Groups**
  - A firewall that controls traffic to and from **an ENI / an EC2 Instance**
  - Can have only ALLOW rules
  - Rules include IP addresses and other security groups



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Network ACLs vs Security Groups

Security Group	Network ACL
Operates at the instance level	Operates at the subnet level
Supports allow rules only	Supports allow rules and deny rules
Is stateful: Return traffic is automatically allowed, regardless of any rules	Is stateless: Return traffic must be explicitly allowed by rules
We evaluate all rules before deciding whether to allow traffic	We process rules in number order when deciding whether to allow traffic
Applies to an instance only if someone specifies the security group when launching the instance, or associates the security group with the instance later on	Automatically applies to all instances in the subnets it's associated with (therefore, you don't have to rely on users to specify the security group)

https://docs.aws.amazon.com/vpc/latest/userguide/VPC_Security.html#VPC_Security_Comparison

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VPC Flow Logs

- Capture information about IP traffic going into your interfaces:
  - **VPC** Flow Logs
  - **Subnet** Flow Logs
  - **Elastic Network Interface** Flow Logs
- Helps to monitor & troubleshoot connectivity issues. Example:
  - Subnets to internet
  - Subnets to subnets
  - Internet to subnets
- Captures network information from AWS managed interfaces too: Elastic Load Balancers, ElastiCache, RDS, Aurora, etc…
- VPC Flow logs data can go to S3, CloudWatch Logs, and Kinesis Data Firehose

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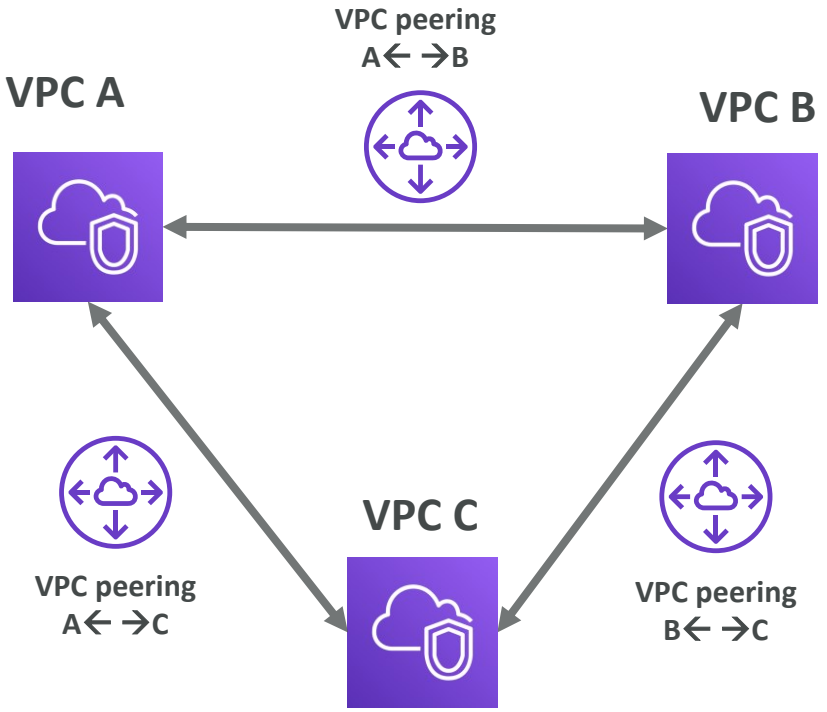
VPC Peering

- Connect two VPC, privately using AWS' network
- Make them behave as if they were in the same network
- Must not have overlapping CIDR (IP address range)
- VPC Peering connection is **not transitive** (must be established for each VPC that need to communicate with one another)



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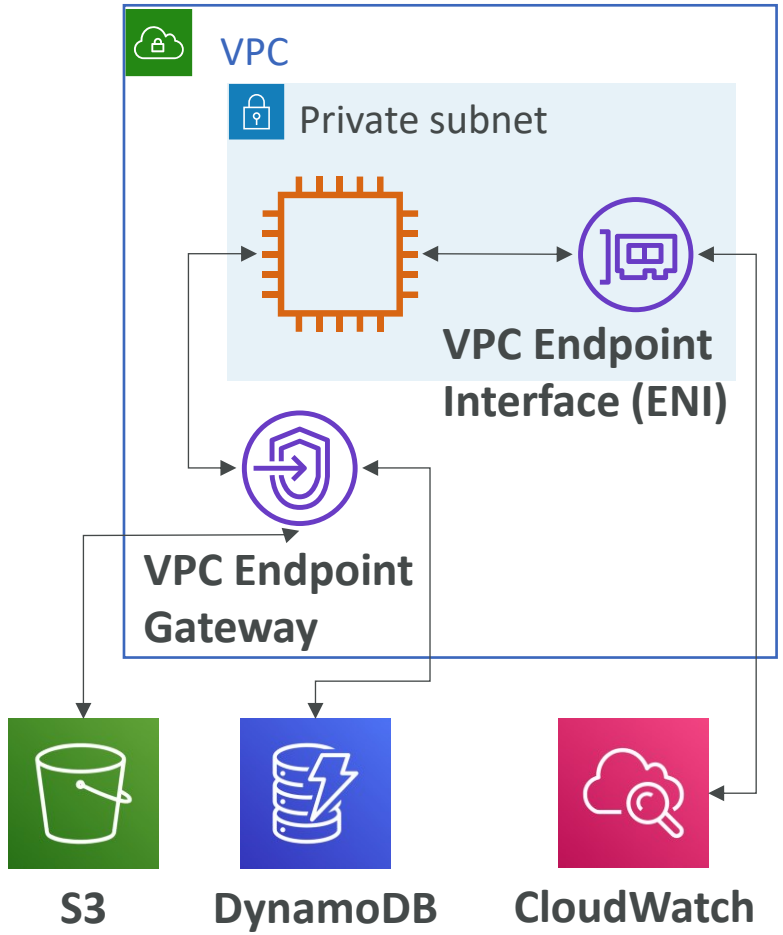
VPC Endpoints

- Endpoints allow you to connect to AWS Services **using a private network** instead of the public www network
- This gives you enhanced security and lower latency to access AWS services

- VPC Endpoint Gateway: S3 & DynamoDB
- VPC Endpoint Interface: most services (including S3 & DynamoDB)

- Only used within your VPC



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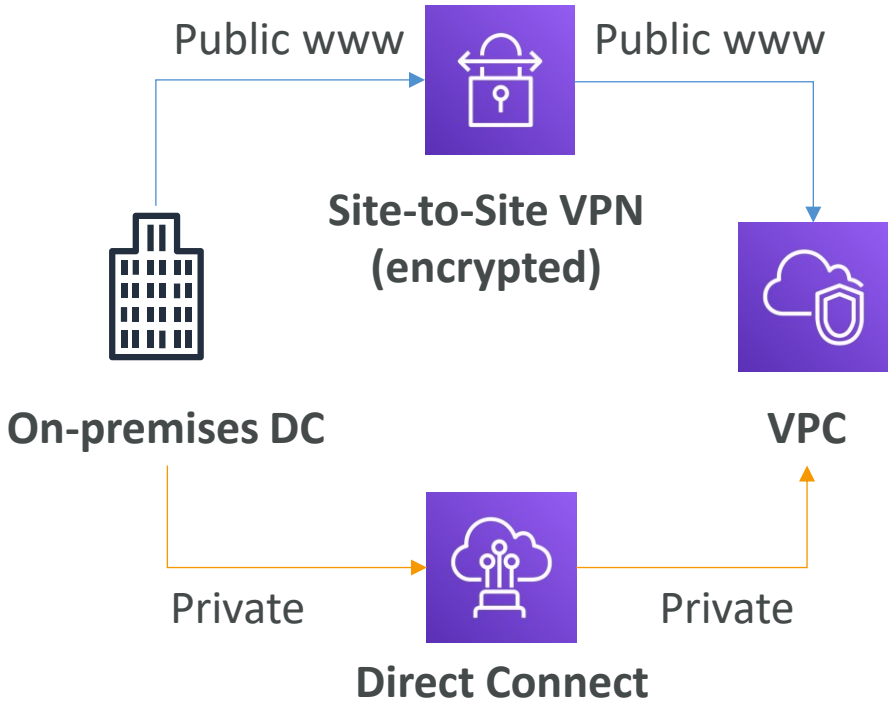
Site to Site VPN & Direct Connect

- **Site to Site VPN**
  - Connect an on-premises VPN to AWS
  - The connection is automatically encrypted
  - Goes over the public internet
- **Direct Connect (DX)**
  - Establish a physical connection between on-premises and AWS
  - The connection is private, secure and fast
  - Goes over a private network
  - Takes at least a month to establish



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VPC Closing Comments

- **VPC:** Virtual Private Cloud
- **Subnets:** Tied to an AZ, network partition of the VPC
- **Internet Gateway:** at the VPC level, provide Internet Access
- **NAT Gateway / Instances:** give internet access to private subnets
- **NACL:** Stateless, subnet rules for inbound and outbound
- **Security Groups:** Stateful, operate at the EC2 instance level or ENI
- **VPC Peering:** Connect two VPC with non overlapping IP ranges, non transitive
- **VPC Endpoints:** Provide private access to AWS Services within VPC
- **VPC Flow Logs:** network traffic logs
- **Site to Site VPN:** VPN over public internet between on-premises DC and AWS
- **Direct Connect:** direct private connection to a AWS

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VPC note – AWS Certified Developer

- Don’t stress if you didn’t understand everything in that section
- I will be highlighting in the course the specific VPC features we need

- Feel free to revisit that section after you’re done in the course !

- Moving on ☺

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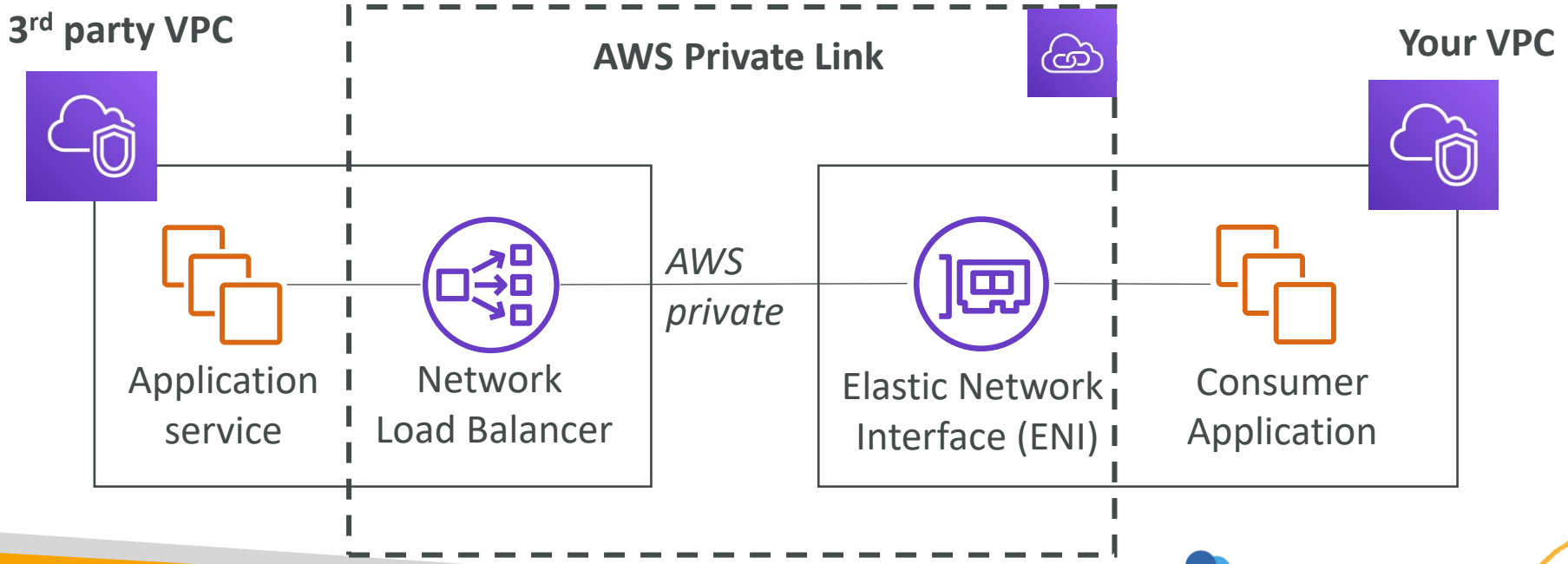
AWS PrivateLink (VPC Endpoint Services)

- Most secure & scalable way to expose a service to 1000s of VPCs
- Does not require VPC peering, internet gateway, NAT, route tables…
- Requires a network load balancer (Service VPC) and ENI (Customer VPC)



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Other Services You Should Know: Analytics

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What is Athena?

- Interactive query service for S3 (SQL)
  - No need to load data, it stays in S3
- Presto under the hood
- Serverless!
- Supports many data formats
  - CSV, TSV (human readable)
  - JSON (human readable)
  - ORC (columnar, splittable)
  - Parquet (columnar, splittable)
  - Avro (splittable)
  - Snappy, Zlib, LZO, Gzip compression
- Unstructured, semi-structured, or structured

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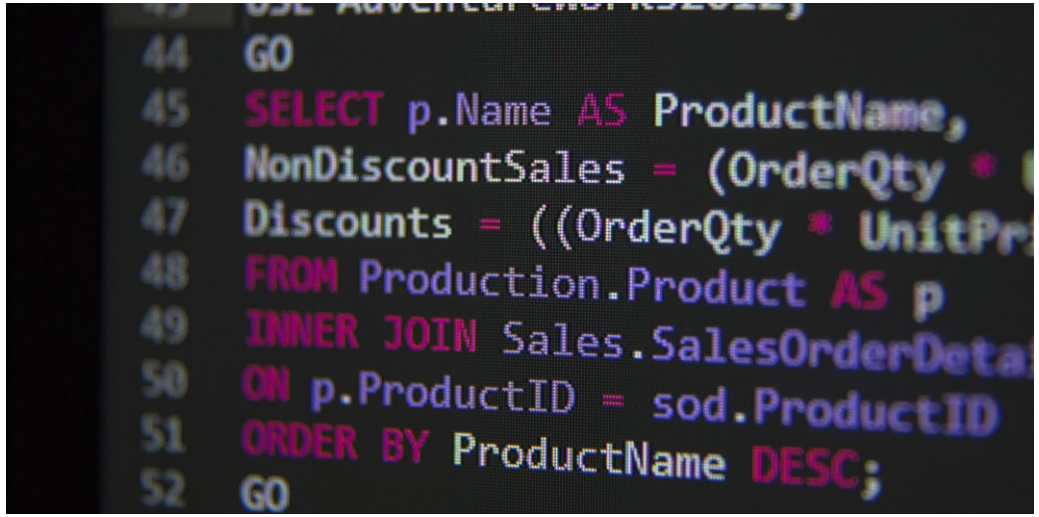
Some examples

- Ad-hoc queries of web logs
- Querying staging data before loading to Redshift
- Analyze CloudTrail / CloudFront / VPC / ELB etc logs in S3
- Integration with Jupyter, Zeppelin, RStudio notebooks
- Integration with QuickSight
- Integration via ODBC / JDBC with other visualization tools



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What is EMR?

- Elastic MapReduce
- Managed Hadoop framework on EC2 instances
- Includes Spark, HBase, Presto, Flink, Hive & more
- EMR Notebooks
- Several integration points with AWS

Amazon EMR

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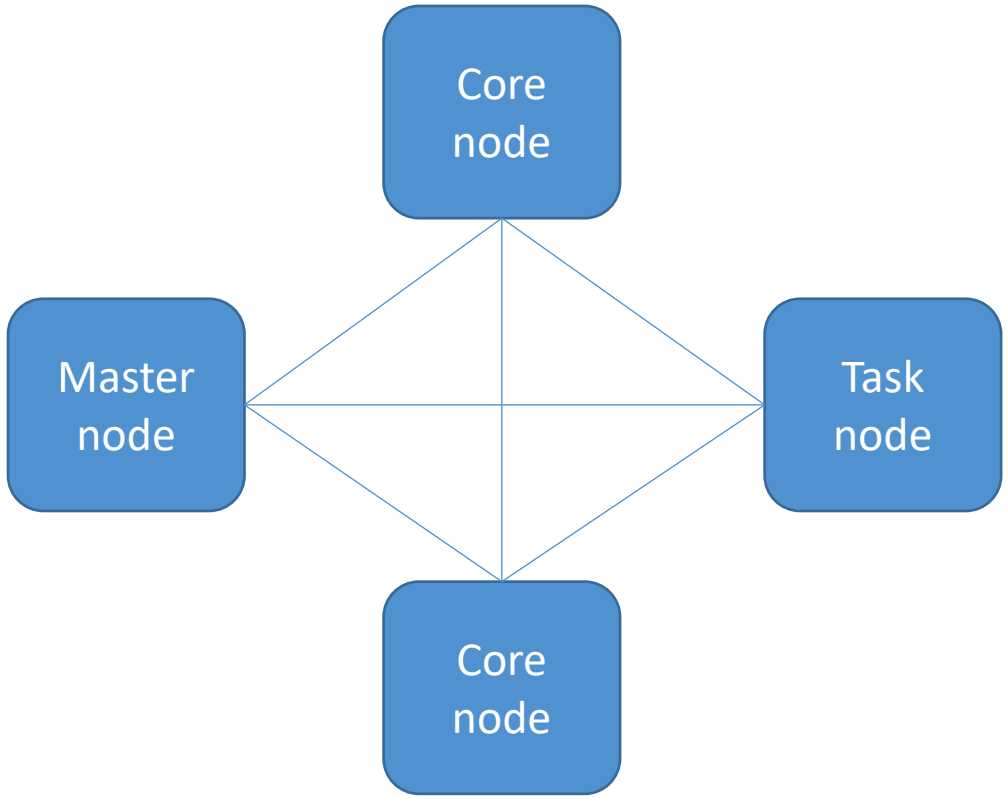
An EMR Cluster



- **Master node**: manages the cluster
  - Tracks status of tasks, monitors cluster health
  - Single EC2 instance (it can be a single node cluster even)
  - AKA “leader node”
- **Core node**: Hosts HDFS data and runs tasks
  - Can be scaled up & down, but with some risk
  - Multi-node clusters have at least one
- **Task node**: Runs tasks, does not host data
  - Optional
  - No risk of data loss when removing
  - Good use of **spot instances**

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EMR Usage

- Transient vs Long-Running Clusters
  - Transient clusters terminate once all steps are complete
    - Loading data, processing, storing – then shut down
    - Saves money
  - Long-running clusters must be manually terminated
    - Basically a data warehouse with periodic processing on large datasets
    - Can spin up task nodes using Spot instances for temporary capacity
    - Can use reserved instances on long-running clusters to save $
    - Termination protection on by default, auto-termination off

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EMR Usage

- Frameworks and applications are specified at cluster launch
- Connect directly to master to run jobs directly
- Or, submit ordered steps via the console
  - Process data in S3 or HDFS
  - Output data to S3 or somewhere
  - Once defined, steps can be invoked via the console

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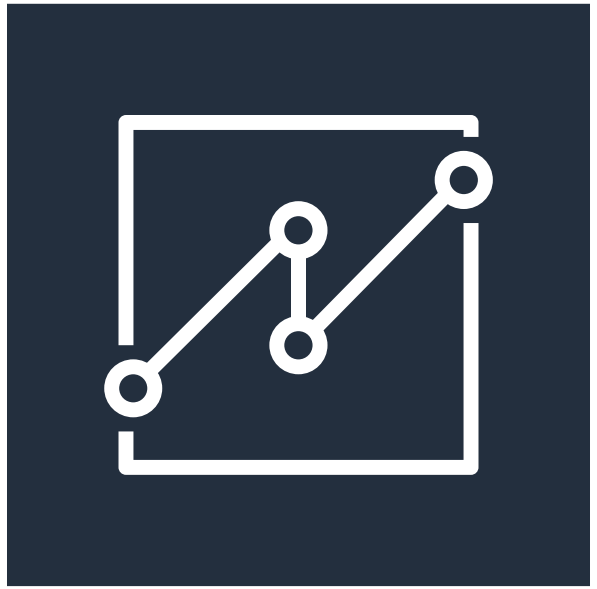
What is QuickSight?

- Fast, easy, cloud-powered business analytics service
- Allows all employees in an organization to:
  - Build visualizations
  - Build paginated reports
  - Perform ad-hoc analysis
  - Get alerts on detected anomalies
  - Quickly get business insights from data
  - Anytime, on any device (browsers, mobile)
- Serverless



Amazon QuickSight

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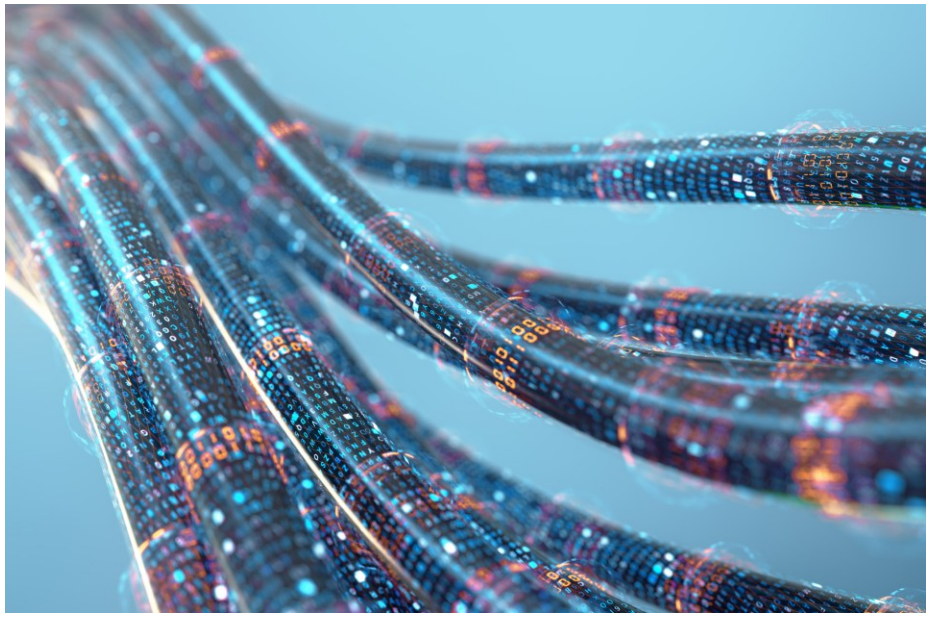
QuickSight Data Sources

- Redshift
- Aurora / RDS
- Athena
- OpenSearch
- IoT Analytics
- EC2-hosted databases
- Files (S3 or on-premises)
  - Excel
  - CSV, TSV
  - Common or extended log format
- Data preparation allows limited ETL



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SPICE

- Data sets are imported into SPICE
  - Super-fast, Parallel, In-memory Calculation Engine
  - Uses columnar storage, in-memory, machine code generation
  - Accelerates interactive queries on large datasets
- Each user gets 10GB of SPICE
- Highly available / durable
- Scales to hundreds of thousands of users
- Can accelerate large queries that would time out in direct query mode (hitting Athena directly)
  - But if it takes more than 30 minutes to import your data into SPICE it will still time out

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QuickSight Use Cases

- Interactive ad-hoc exploration / visualization of data
- Dashboards and KPIs
- Analyze / visualize data from:
  - Logs in S3
  - On-premise databases
  - AWS (RDS, Redshift, Athena, S3)
  - SaaS applications, such as Salesforce
  - Any JDBC/ODBC data source

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QuickSight Anti-Patterns

- ETL
  - Use Glue instead, although QuickSight can do some transformations

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QuickSight Security

- Multi-factor authentication on your account
- VPC connectivity
  - Add QuickSight's IP address range to your database security groups
- Row-level security
  - New for 2021: Column-level security too (CLS) – Enterprise edition only
- Private VPC access
  - Elastic Network Interface, AWS Direct Connect

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QuickSight Security

- Resource access
  - Must ensure QuickSight is authorized to use Athena / S3 / your S3 buckets
  - This can be managed within the QuickSight console (Manage Quicksight / Security & Permissions)\
- Data access
  - Can create IAM policies to restrict what data in S3 given QuickSight users can access

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Quicksight + Redshift: Security

- By default Quicksight can only access data stored IN THE SAME REGION as the one Quicksight is running within
- So if Quicksight is running in one region, and Redshift in another, that's a problem
- A VPC configured to work across AWS regions won't work!
- Solution: **create a new security group with an inbound rule authorizing access from the IP range of QuickSight servers in that region**
  - Those ranges are documented at https://docs.aws.amazon.com/quicksight/latest/user/regions.html

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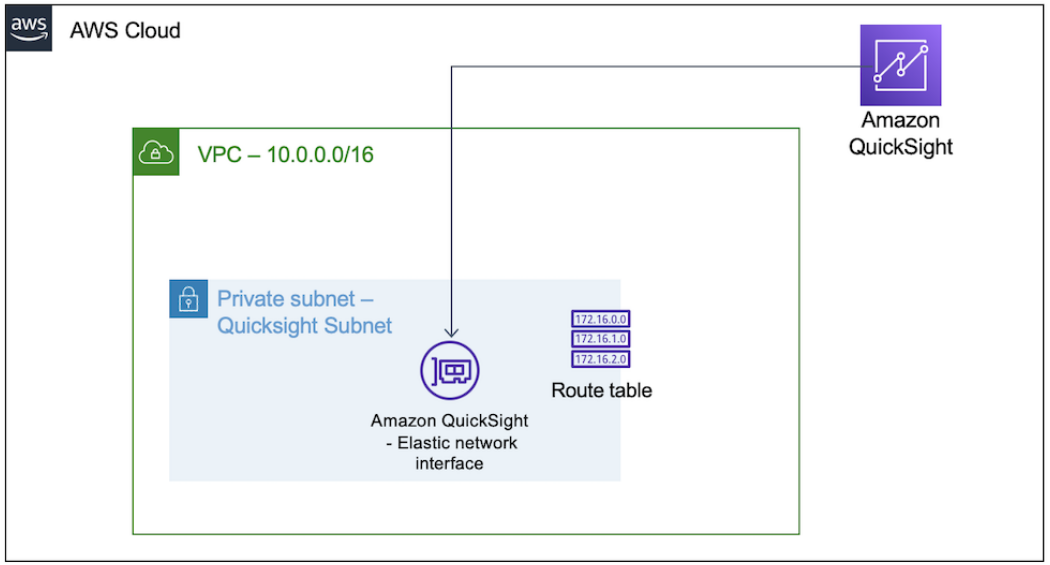
Quicksight / Redshift / RDS cross-region

- Other ways to do it (if you have Enterprise Edition)
- Create a private subnet in a VPC
- Use Elastic Network Interface to put Quicksight in the subnet
  - This is the part that requires Enterprise Edition
- This can also enable cross-account access



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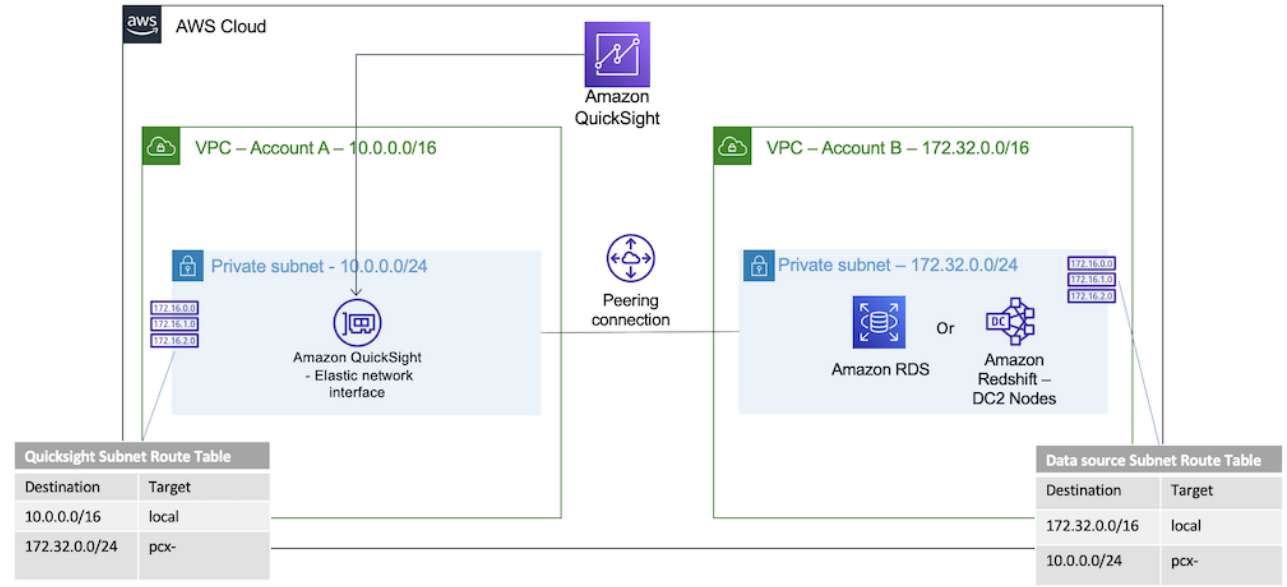
Quicksight / Redshift / RDS cross-region

- Now you can create a **peering connection** between your private subnet and another private subnet containing your data
  - This also works for cross-account access



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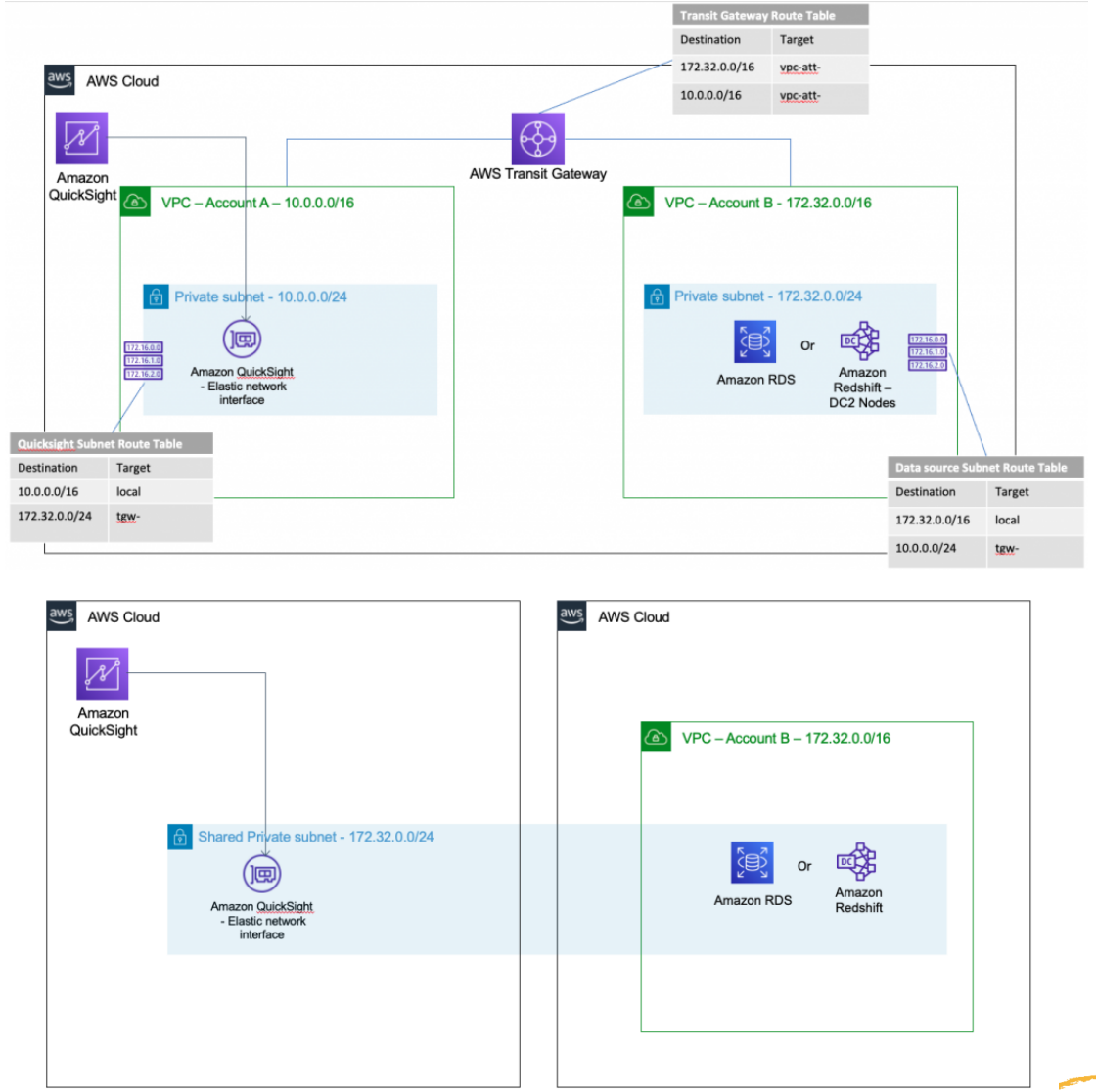
Quicksight / Redshift / RDS cross-account

- Can use an AWS Transit Gateway to connect your subnets
  - But this must be in the same org & region
  - Or you can peer Transit Gateways in different regions
- OR use AWS PrivateLink to connect them
- OR use VPC sharing to connect them



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QuickSight User Management

- Users defined via IAM, or email signup
- Active Directory connector with QuickSight Enterprise Edition
  - All keys are managed by AWS; you CANNOT use customer-provided keys
  - Enterprise edition only!
  - Can tweak security access using IAM if needed

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Amazon Kinesis Data Streams

- Collect and store streaming data in **real-time**

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Kinesis Data Streams

- Retention between up to 365 days
- Ability to reprocess (replay) data by consumers
- Data can’t be deleted from Kinesis (until it expires)
- Data up to 1MB (typical use case is lot of “small” **real-time data**)
- Data ordering guarantee for data with the same “Partition ID”
- At-rest KMS encryption, in-flight HTTPS encryption
- Kinesis Producer Library (KPL) to write an optimized producer application
- Kinesis Client Library (KCL) to write an optimized consumer application

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Kinesis Data Streams – Capacity Modes

- **Provisioned mode:**
  - Choose number of shards
  - Each shard gets 1MB/s in (or 1000 records per second)
  - Each shard gets 2MB/s out
  - Scale manually to increase or decrease the number of shards
  - You pay per shard provisioned per hour

- **On-demand mode:**
  - No need to provision or manage the capacity
  - Default capacity provisioned (4 MB/s in or 4000 records per second)
  - Scales automatically based on observed throughput peak during the last 30 days
  - Pay per stream per hour & data in/out per GB

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Amazon Managed Streaming for Apache Kafka (Amazon MSK)

- Alternative to Amazon Kinesis
- Fully managed Apache Kafka on AWS
  - Allow you to create, update, delete clusters
  - MSK creates & manages Kafka brokers nodes & Zookeeper nodes for you
  - Deploy the MSK cluster in your VPC, multi-AZ (up to 3 for HA)
  - Automatic recovery from common Apache Kafka failures
  - Data is stored on EBS volumes **for as long as you want**
- **MSK Serverless**
  - Run Apache Kafka on MSK without managing the capacity
  - MSK automatically provisions resources and scales compute & storage

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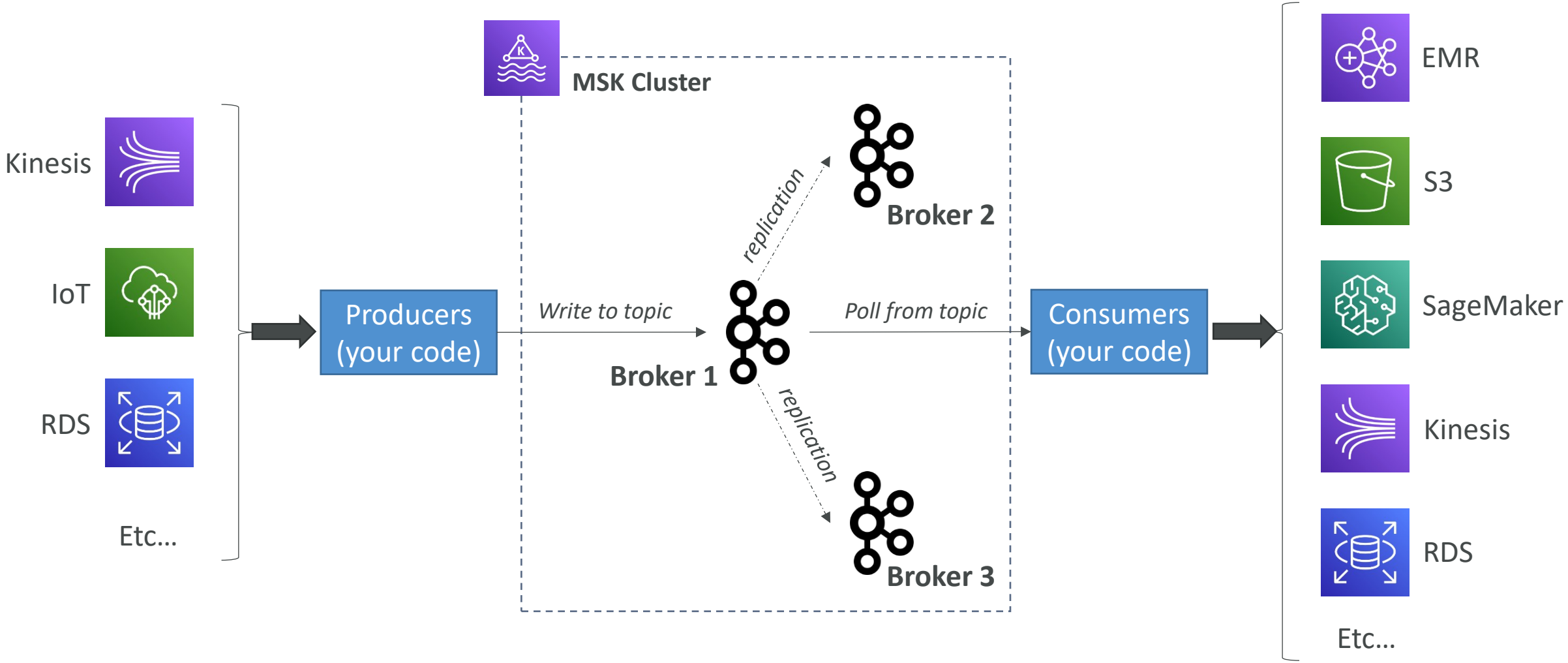
Apache Kafka at a high level



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Kinesis Data Streams vs. Amazon MSK

Kinesis Data Streams

- 1 MB message size limit
- Data Streams with Shards
- Shard Splitting & Merging
- TLS In-flight encryption
- KMS at-rest encryption

Amazon MSK

- 1MB default, configure for higher (ex: 10MB)
- Kafka Topics with Partitions
- Can only add partitions to a topic
- PLAINTEXT or TLS In-flight Encryption
- KMS at-rest encryption

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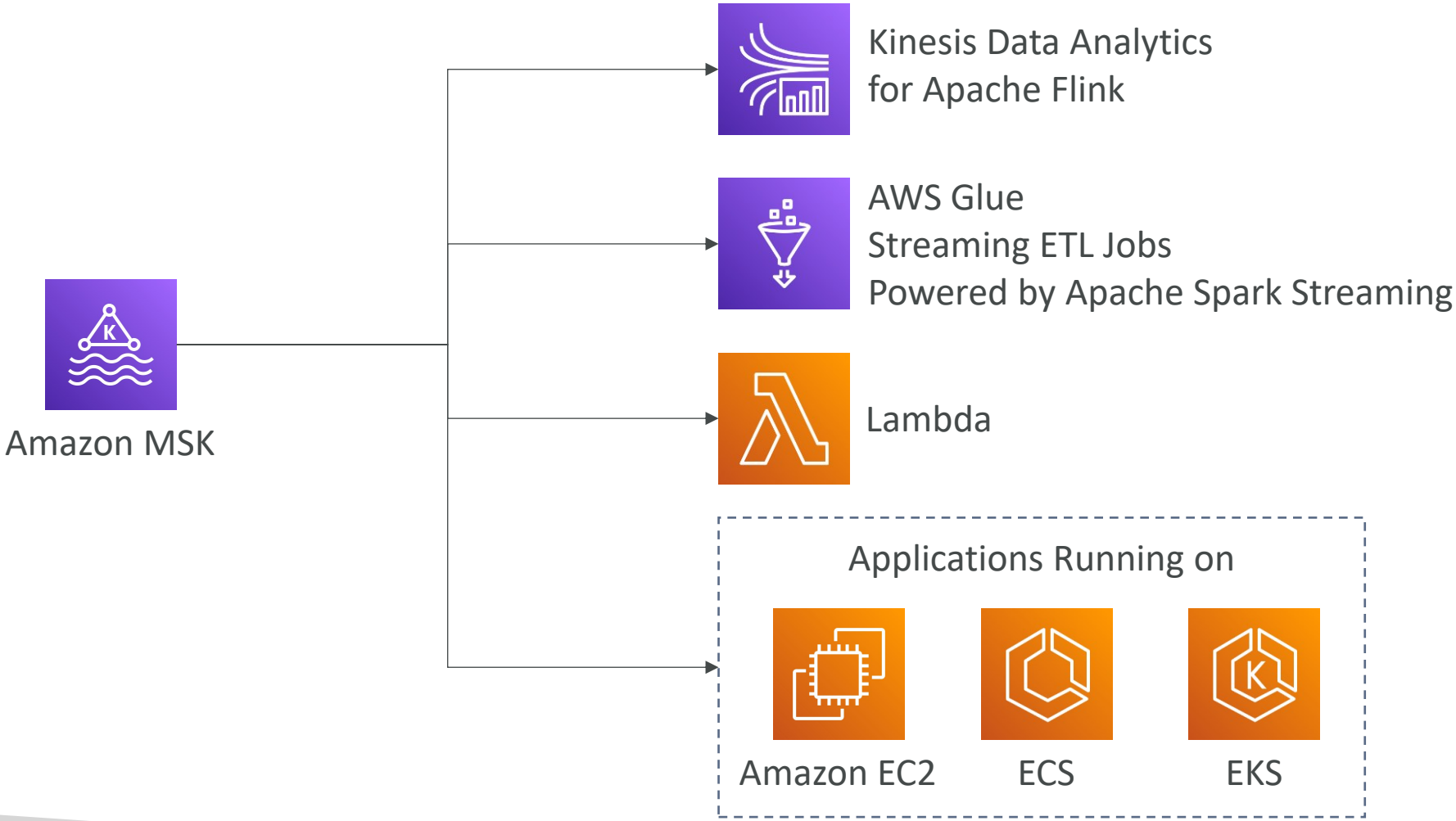
Amazon MSK Consumers



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Other Services You Should Know

Compute / Containers / Customer Engagement

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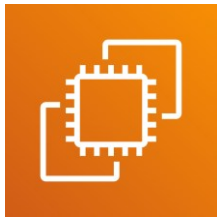
Amazon EC2



- EC2 is one of the most popular of AWS' offering
- EC2 = Elastic Compute Cloud = Infrastructure as a Service
- It mainly consists in the capability of :
  - Renting virtual machines (EC2)
  - Storing data on virtual drives (EBS)
  - Distributing load across machines (ELB)
  - Scaling the services using an auto-scaling group (ASG)

- Knowing EC2 is fundamental to understand how the Cloud works

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EC2 sizing & configuration options

- Operating System (**OS**): Linux, Windows or Mac OS
- How much compute power & cores (**CPU**)
- How much random-access memory (**RAM**)
- How much storage space:
  - Network-attached (**EBS & EFS**)
  - hardware (**EC2 Instance Store**)
- Network card: speed of the card, Public IP address
- Firewall rules: **security group**
- Bootstrap script (configure at first launch): EC2 User Data

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EC2 User Data

- It is possible to bootstrap our instances using an EC2 User data script.
- bootstrapping means launching commands when a machine starts
- That script is only run once at the instance first start
- EC2 user data is used to automate boot tasks such as:
  - Installing updates
  - Installing software
  - Downloading common files from the internet
  - Anything you can think of
- The EC2 User Data Script runs with the root user

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Hands-On:
Launching an EC2 Instance running Linux

- We'll be launching our first virtual server using the AWS Console
- We'll get a first high-level approach to the various parameters
- We'll see that our web server is launched using EC2 user data
- We'll learn how to start / stop / terminate our instance.

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Docker Containers Management on AWS

- **Amazon Elastic Container Service (Amazon ECS**
  - Amazon's own container platform

Amazon ECS

- **Amazon Elastic Kubernetes Service (Amazon EK**
  - Amazon's managed Kubernetes (open source)

Amazon EKS

- **AWS Fargate**
  - Amazon's own Serverless container platform
  - Works with ECS and with EKS

AWS Fargate

- **Amazon ECR:**
  - Store container images

Amazon ECR

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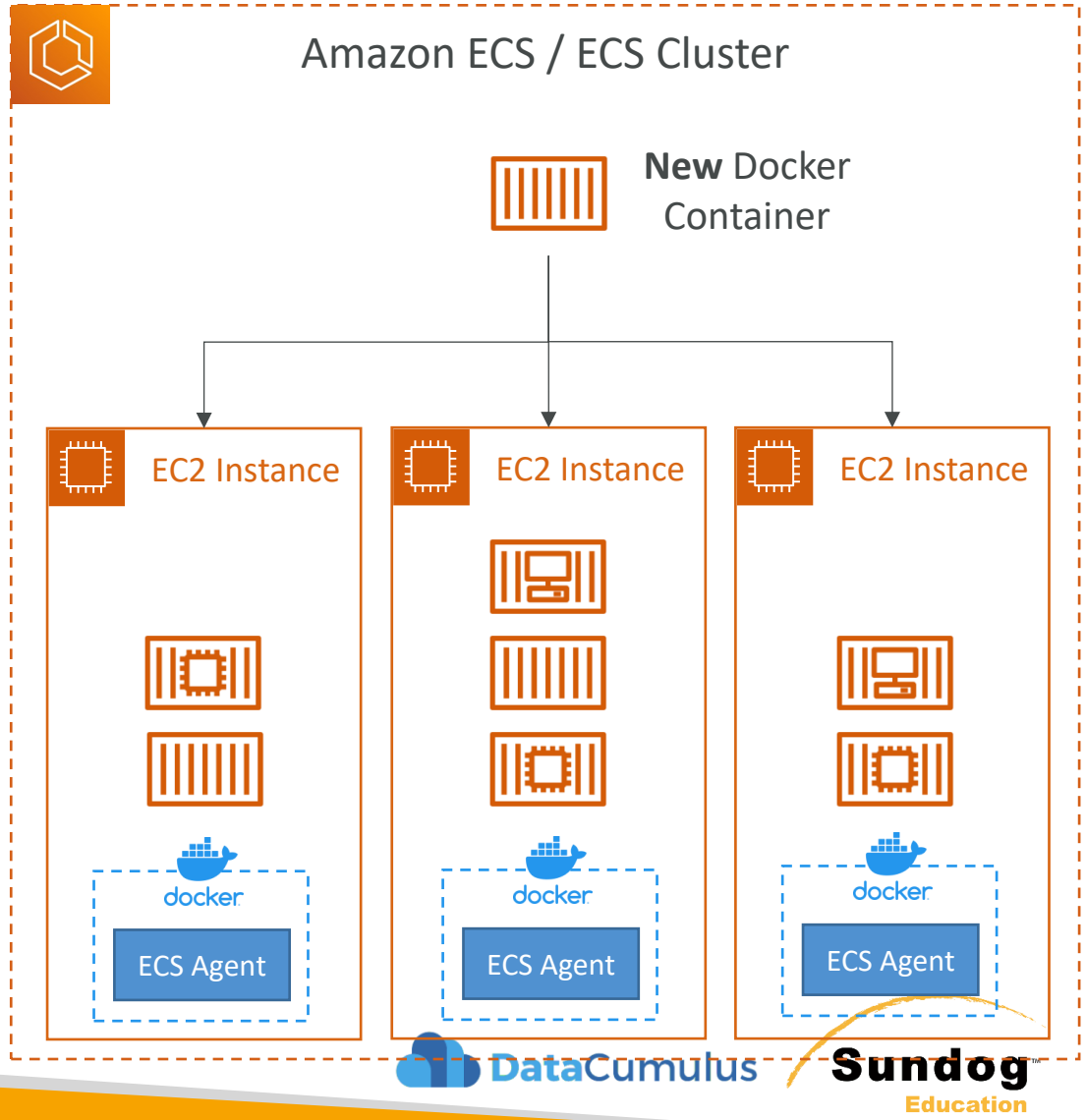
Amazon ECS - EC2 Launch Type

- ECS = Elastic Container Service
- Launch Docker containers on AWS = Launch **ECS Tasks** on ECS Clusters
- **EC2 Launch Type: you must provision & maintain the infrastructure (the EC2 instances)**
- Each EC2 Instance must run the ECS Agent to register in the ECS Cluster
- AWS takes care of starting / stopping containers



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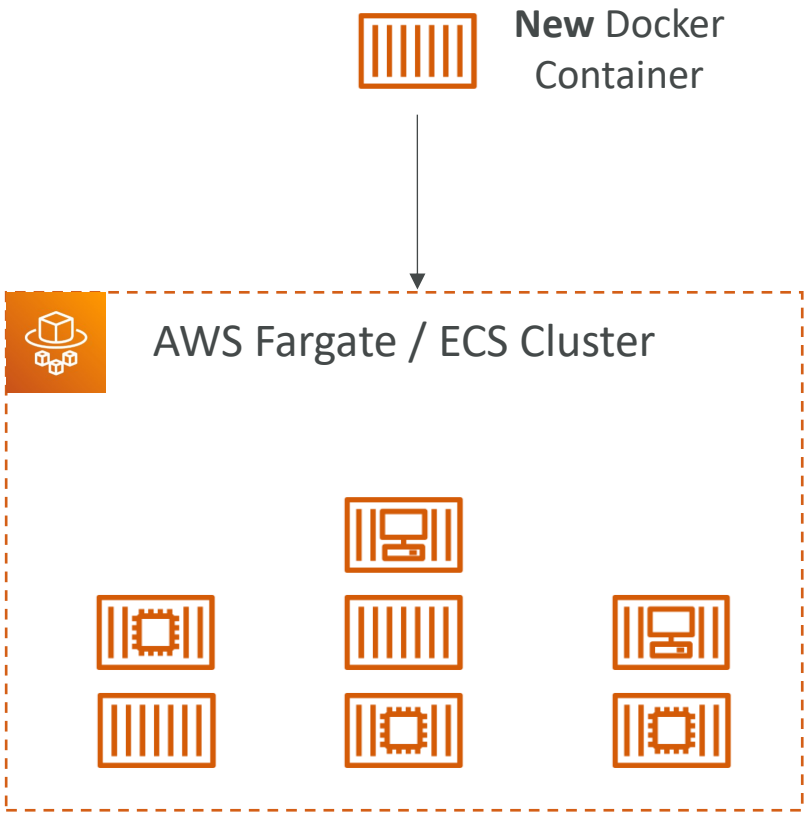
Amazon ECS – Fargate Launch Type

- Launch Docker containers on AWS
- **You do not provision the infrastructure (no EC2 instances to manage)**
- **It's all Serverless!**
- You just create task definitions
- AWS just runs ECS Tasks for you based on the CPU / RAM you need
- To scale, just increase the number of tasks. Simple - no more EC2 instances



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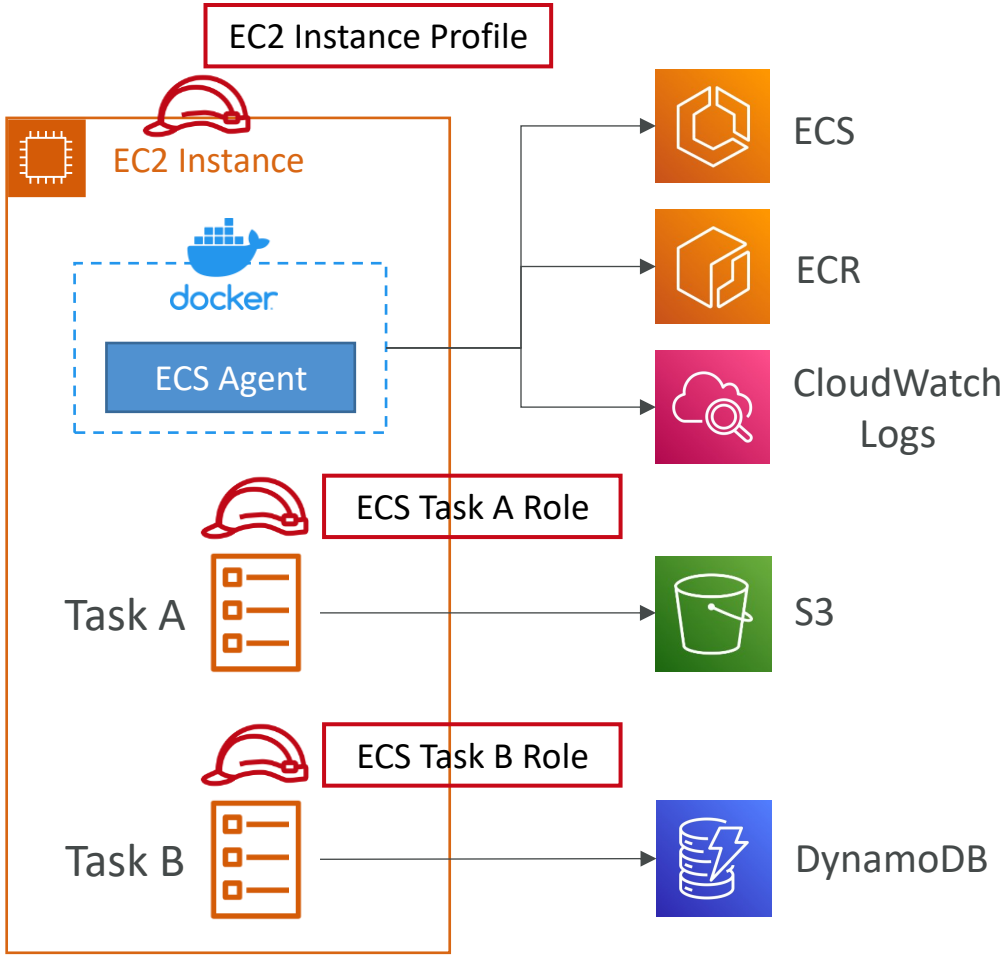
Amazon ECS – IAM Roles for ECS

- **EC2 Instance Profile (EC2 Launch Type only):**
  - Used by the ECS agent
  - Makes API calls to ECS service
  - Send container logs to CloudWatch Logs
  - Pull Docker image from ECR
  - Reference sensitive data in Secrets Manager or SSM Parameter Store
- **ECS Task Role:**
  - Allows each task to have a specific role
  - Use different roles for the different ECS Services you run
  - Task Role is defined in the **task definition**



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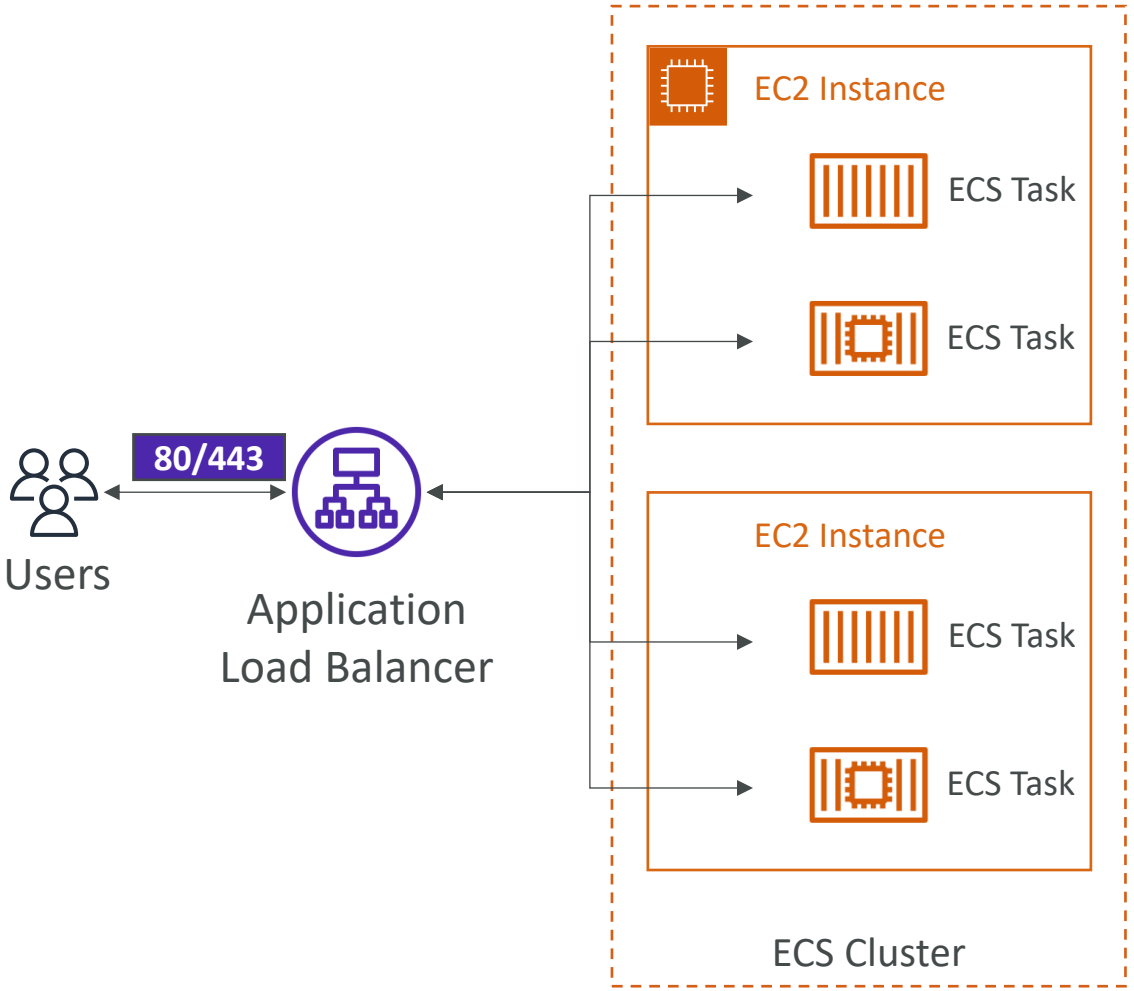
Amazon ECS – Load Balancer Integrations

- **Application Load Balancer** supported and works for most use cases
- **Network Load Balancer** recommended only for high throughput / high performance use cases, or to pair it with AWS Private Link
- **Classic Load Balancer** supported but not recommended (no advanced features – no Fargate)



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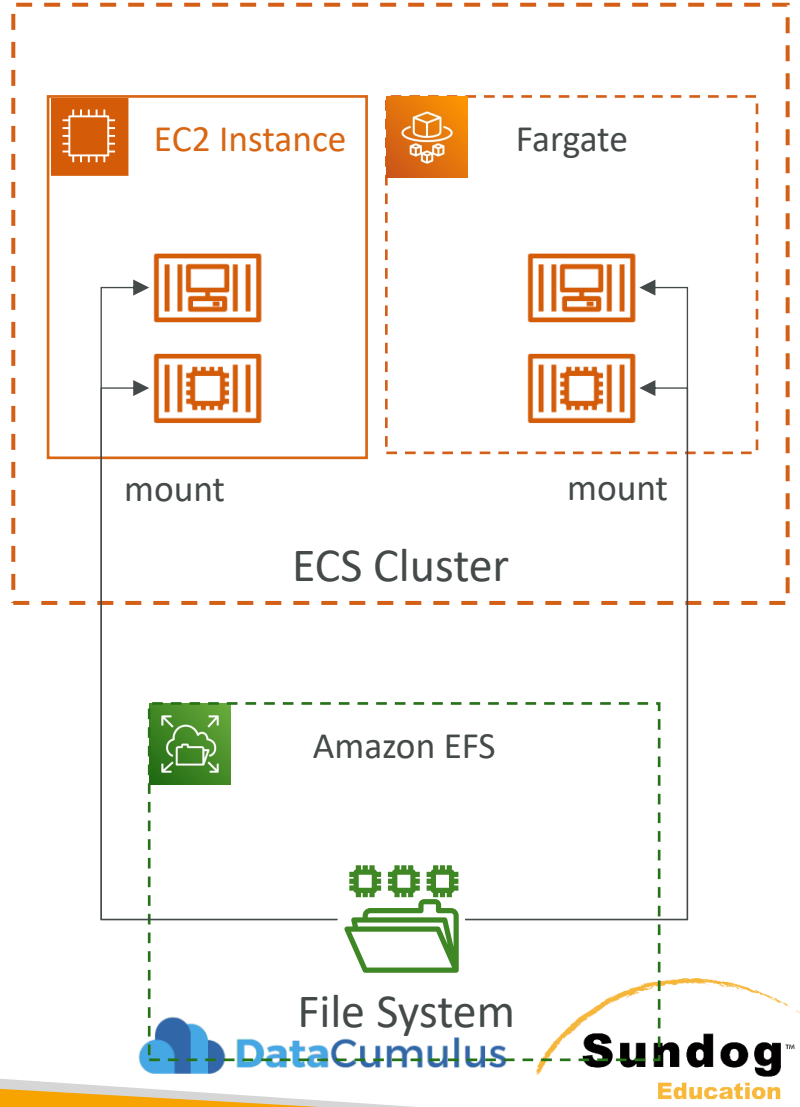
Amazon ECS – Data Volumes (EFS)

- Mount EFS file systems onto ECS tasks
- Works for both **EC2** and **Fargate** launch types
- Tasks running in any AZ will share the same data in the EFS file system
- **Fargate + EFS = Serverless**

- Use cases: persistent multi-AZ shared storage for your containers

- Note:
  - Amazon S3 cannot be mounted as a file system



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Amazon EKS Overview

- Amazon EKS = Amazon Elastic **Kubernetes** Service
- It is a way to launch **managed Kubernetes clusters on AWS**
- Kubernetes is an **open-source system** for automatic deployment, scaling and management of containerized (usually Docker) application
- It's an alternative to ECS, similar goal but different API
- EKS supports **EC2** if you want to deploy worker nodes or **Fargate** to deploy serverless containers
- **Use case:** if your company is already using Kubernetes on-premises or in another cloud, and wants to migrate to AWS using Kubernetes
- **Kubernetes is cloud-agnostic** (can be used in any cloud – Azure, GCP…)
- For multiple regions, deploy one EKS cluster per region
- Collect logs and metrics using **CloudWatch Container Insights**

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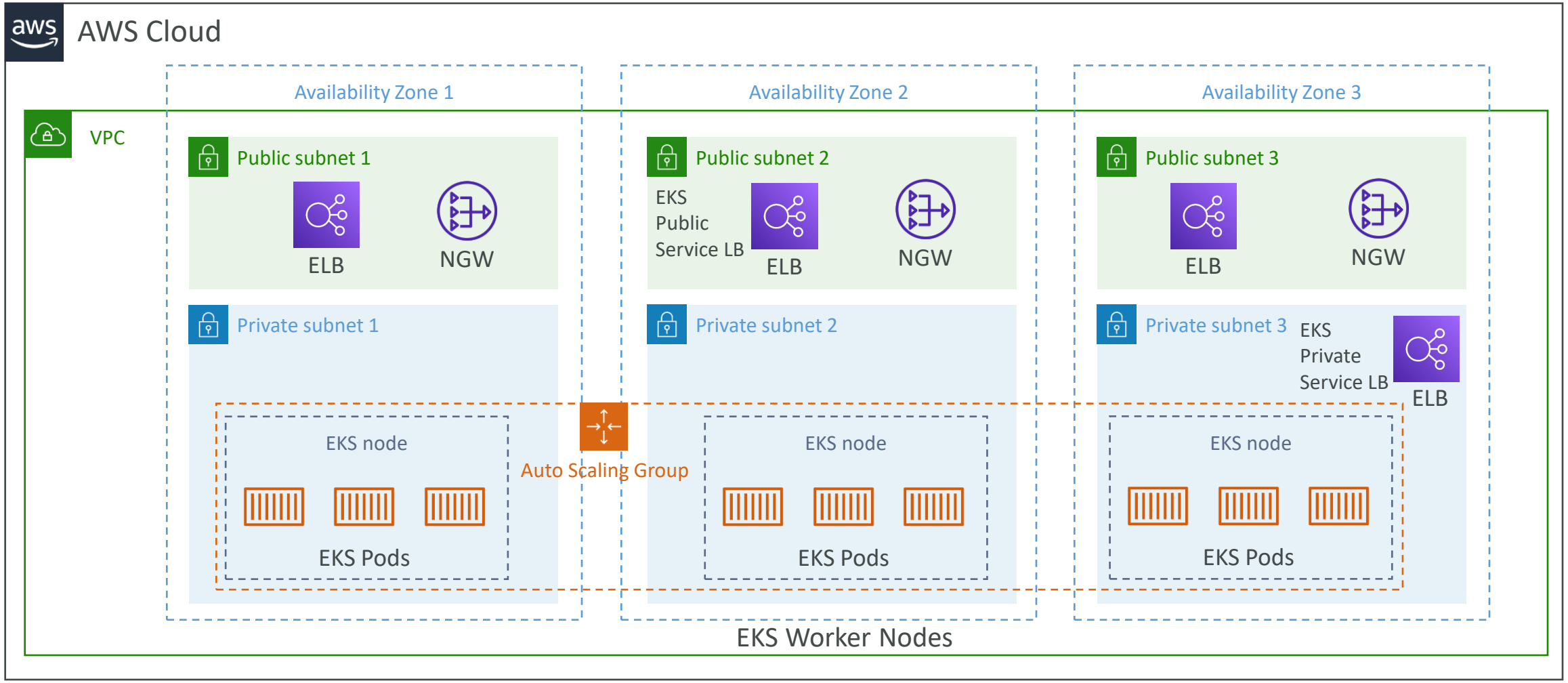
Amazon EKS - Diagram



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Amazon EKS – Node Types

- **Managed Node Groups**
  - Creates and manages Nodes (EC2 instances) for you
  - Nodes are part of an ASG managed by EKS
  - Supports On-Demand or Spot Instances
- **Self-Managed Nodes**
  - Nodes created by you and registered to the EKS cluster and managed by an ASG
  - You can use prebuilt AMI - Amazon EKS Optimized AMI
  - Supports On-Demand or Spot Instances
- **AWS Fargate**
  - No maintenance required; no nodes managed

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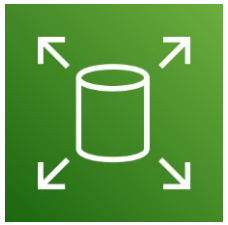
Amazon EKS – Data Volumes

- Need to specify **StorageClass** manifest on your EKS cluster
- Leverages a **Container Storage Interface (CSI)** compliant driver

- Support for…
- Amazon EBS
- Amazon EFS (works with Fargate)
- Amazon FSx for Lustre
- Amazon FSx for NetApp ONTAP



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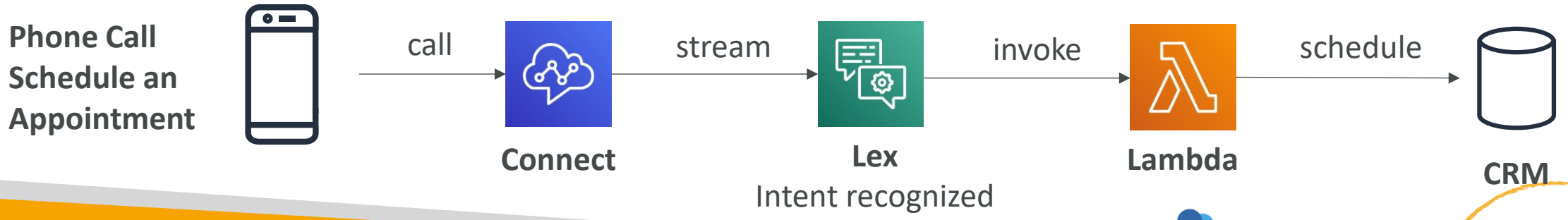
Amazon Lex & Connect

- **Amazon Lex:** (same technology that powers Alexa)
  - Automatic Speech Recognition (ASR) to convert speech to text
  - Natural Language Understanding to recognize the intent of text, callers
  - Helps build chatbots, call center bots
- **Amazon Connect:**
  - Receive calls, create contact flows, cloud-based **virtual contact center**
  - Can integrate with other CRM systems or AWS
  - No upfront payments, 80% cheaper than traditional contact center solutions



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Other Services You Should Know: Database

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DocumentDB

- Aurora is an “AWS-implementation” of PostgreSQL / MySQL …
- **DocumentDB is the same for MongoDB (which is a NoSQL database)**

- MongoDB is used to store, query, and index JSON data
- Similar “deployment concepts” as Aurora
- Fully Managed, highly available with replication across 3 AZ
- DocumentDB storage automatically grows in increments of 10GB

- Automatically scales to workloads with millions of requests per seconds

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Amazon ElastiCache Overview

- The same way RDS is to get managed Relational Databases…
- ElastiCache is to get managed Redis or Memcached
- Caches are in-memory databases with really high performance, low latency
- Helps reduce load off of databases for read intensive workloads
- Helps make your application stateless
- AWS takes care of OS maintenance / patching, optimizations, setup, configuration, monitoring, failure recovery and backups
- **Using ElastiCache involves heavy application code changes**

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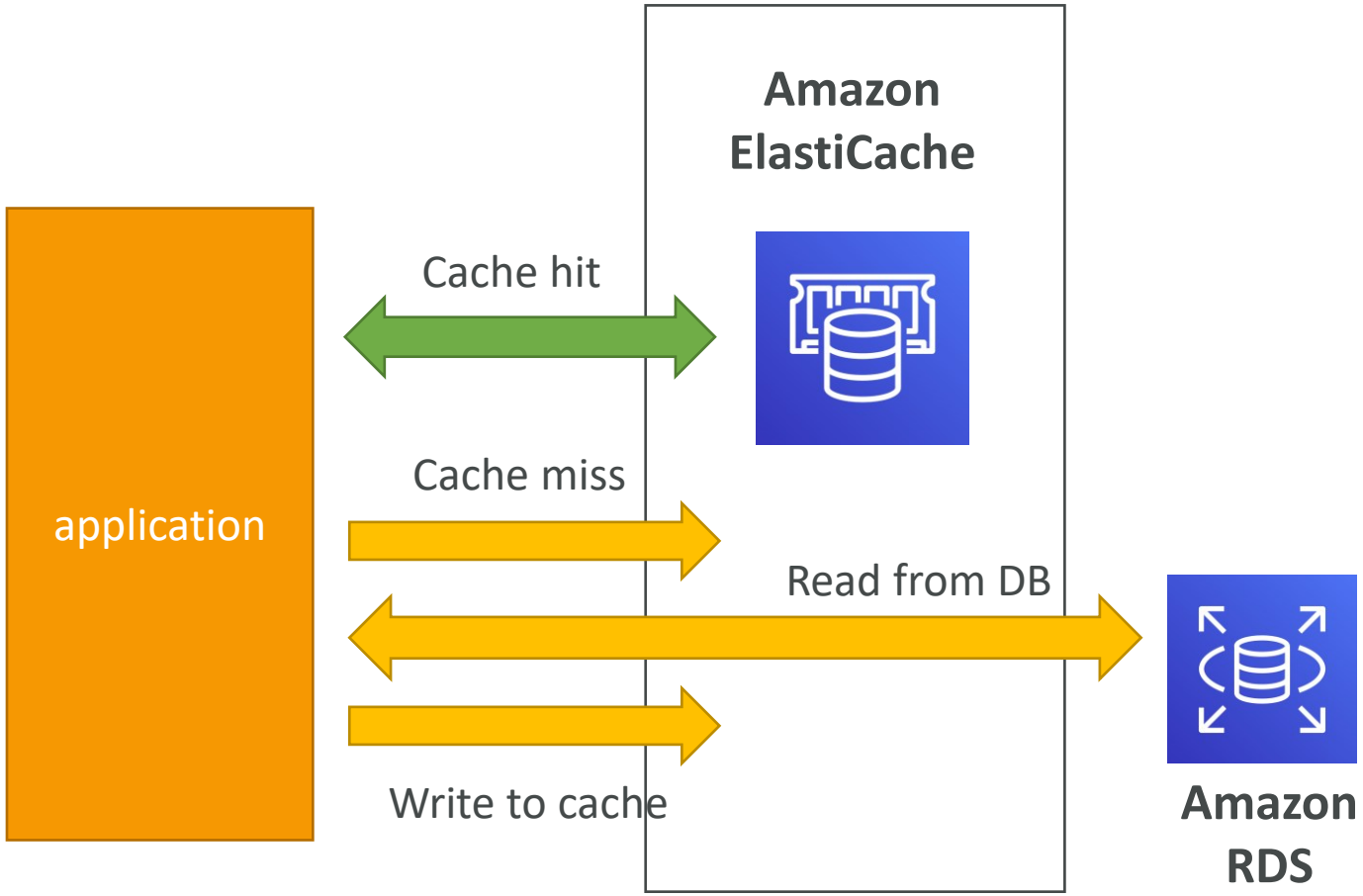
ElastiCache Solution Architecture - DB Cache

- Applications queries ElastiCache, if not available, get from RDS and store in ElastiCache.
- Helps relieve load in RDS
- Cache must have an invalidation strategy to make sure only the most current data is used in there.



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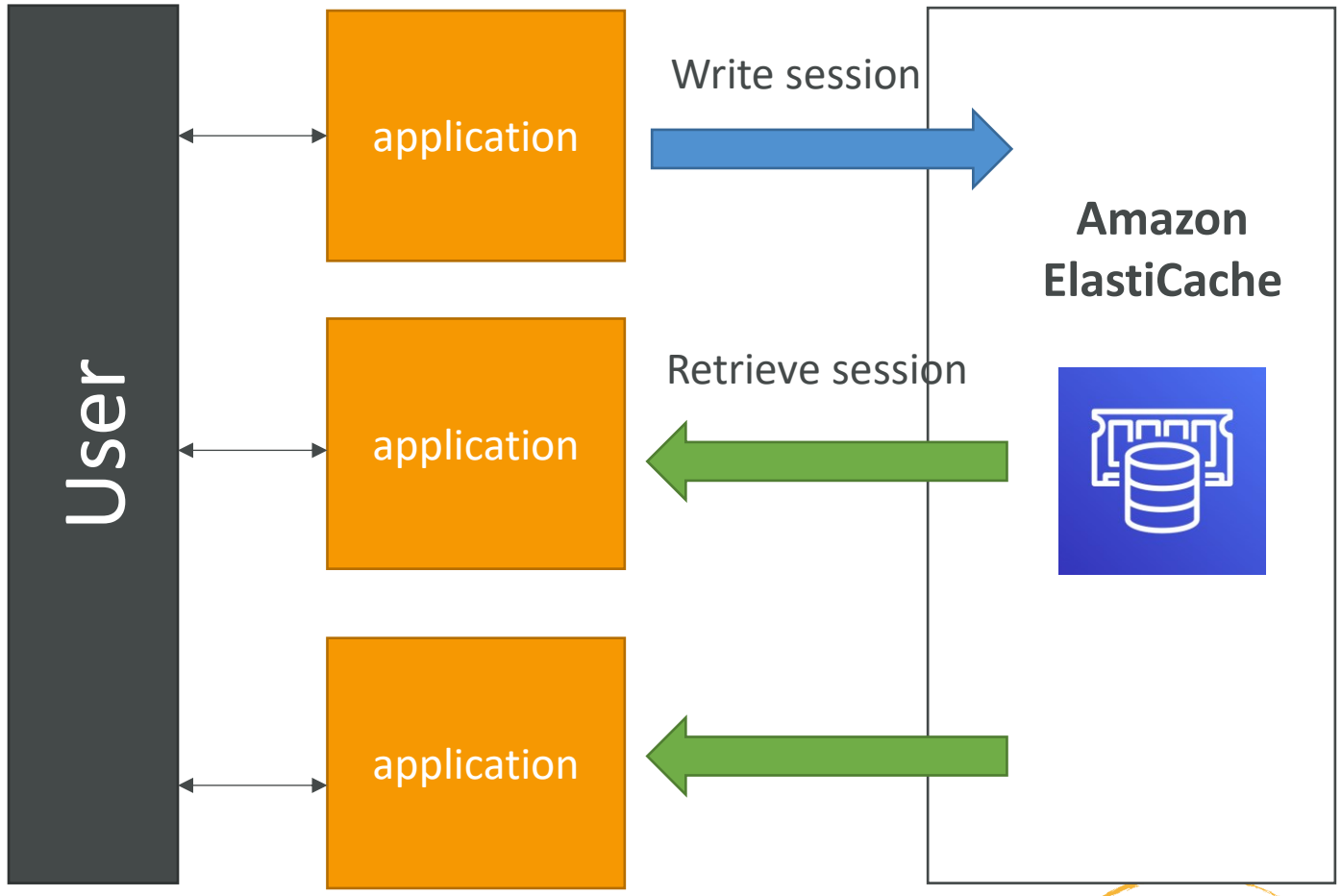
ElastiCache
Solution Architecture – User Session Store

- User logs into any of the application
- The application writes the session data into ElastiCache
- The user hits another instance of our application
- The instance retrieves the data and the user is already logged in



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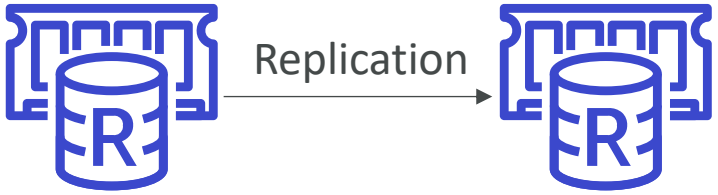
ElastiCache – Redis vs Memcached

REDIS

- **Multi AZ** with Auto-Failover
- **Read Replicas** to scale reads and have **high availability**
- Data Durability using AOF persistence
- **Backup and restore features**
- Supports Sets and Sorted Sets

MEMCACHED

- Multi-node for partitioning of data (sharding)
- **No high availability (replication)**
- **Non persistent**
- **Backup and restore (Serverless)**
- Multi-threaded architecture



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ElastiCache for Valkey

- Full compatibility with Valkey (open source)
  - Basically an alternative to Redis OSS
  - It's actually a fork of Redis
- ElastiCache for Valkey supports vector search!
  - Obviously important for GenAI / RAG / semantic caching
  - Supports billions of high-dimensional embedding vectors
  - Microsecond latency
  - Recall > 99%
  - Appropriate for high performance & scalability requirements

Valkey

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Amazon MemoryDB

- Also supports Valkey and Redis OSS vector search
- All in memory (super fast latency and throughput)
- Multi-AZ
- Vector index algorithms
  - Flat: brute force linear processing
  - : just like Opensearch, gives an approximation for faster execution

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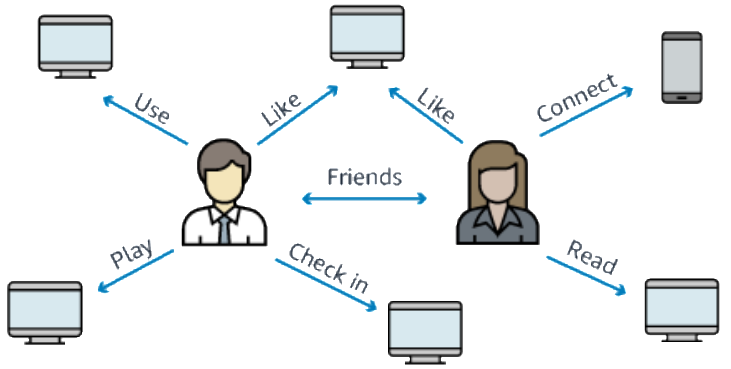
Amazon Neptune

- Fully managed **graph** database
- A popular **graph dataset** would be a **social network**
  - Users have friends
  - Posts have comments
  - Comments have likes from users
  - Users share and like posts…
- Highly available across 3 AZ, with up to 15 read replicas
- Build and run applications working with highly connected datasets – optimized for these complex and hard queries
- Can store up to billions of relations and query the graph with milliseconds latency
- Highly available with replications across multiple AZs
- Great for knowledge graphs (Wikipedia), fraud detection, recommendation engines, social networking



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Neptune and GenAI

- Neptune Analytics
  - Analytics engine built on top of Neptune database
  - vectors.topKByEmbedding
    - Query vector databases with explicit vector values
    - Can filter by vertex attributes
    - Returns top nodes and their scores

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Other Services You Should Know: Developer Tools

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AWS Cloud Development Kit (CDK)

- Define your cloud infrastructure using a familiar language:
  - JavaScript/TypeScript, Python, Java, and .NET
- Contains high level components called **constructs**
- The code is “compiled” into a CloudFormation template (JSON/YAML)
- **You can therefore deploy infrastructure and application runtime code together**
  - Great for Lambda functions
  - Great for Docker containers in ECS / EKS

```
export class MyEcsConstructStack extends core.Stack {
  constructor(scope: core.App, id: string, props?: core.StackProps)
    super(scope, id, props);

    const vpc = new ec2.Vpc(this, "MyVpc", {
      maxAzs: 3 // Default is all AZs in region
    });

    const cluster = new ecs.Cluster(this, "MyCluster", {
      vpc: vpc
    });

    // Create a load-balanced Fargate service and make it public
    new ecs_patterns.ApplicationLoadBalancedFargateService(this, "My
      cluster: cluster, // Required
      cpu: 512, // Default is 256
      desiredCount: 6, // Default is 1
      taskImageOptions: { image: ecs.ContainerImage.fromRegistry("am
      memoryLimitMiB: 2048, // Default is 512
      publicLoadBalancer: true // Default is false
    });
  }
}
```


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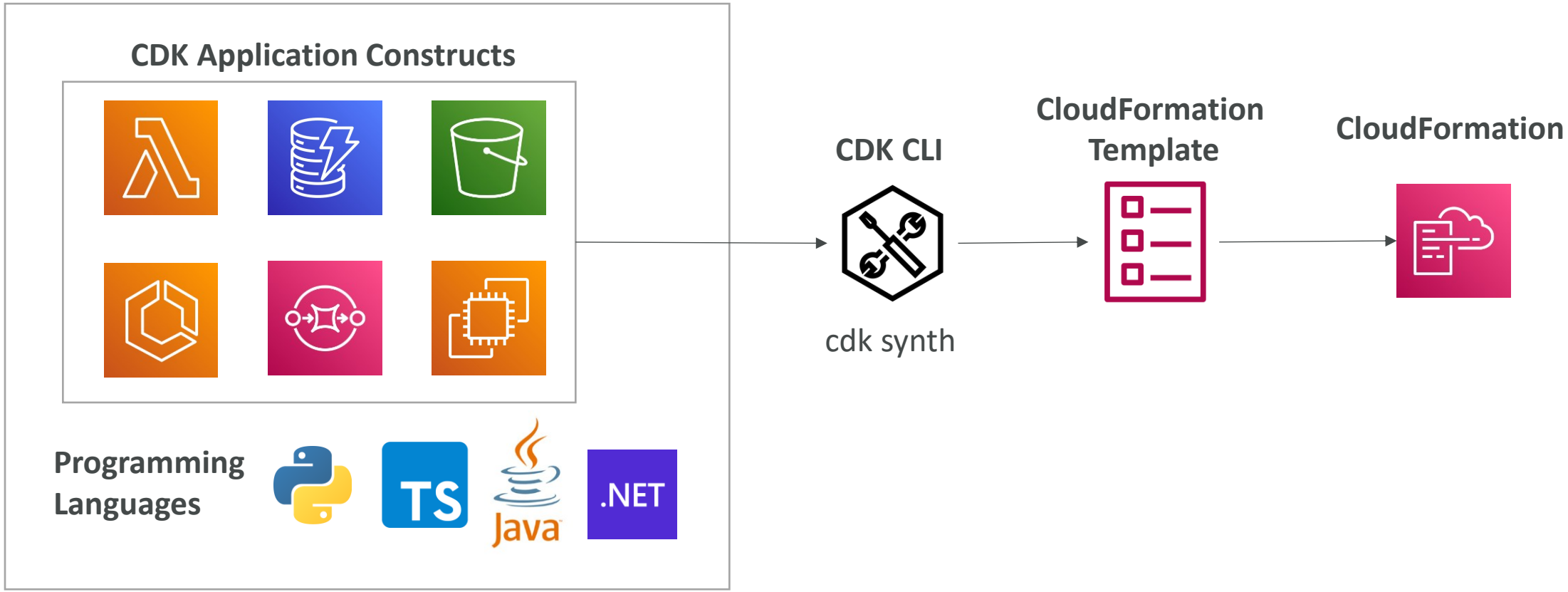
CDK in a diagram



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CDK vs SAM

- SAM:
  - Serverless focused
  - Write your template declaratively in JSON or YAML
  - Great for quickly getting started with Lambda
  - Leverages CloudFormation

- CDK:
  - All AWS services
  - Write infra in a programming language JavaScript/TypeScript, Python, Java, and .NET
  - Leverages CloudFormation

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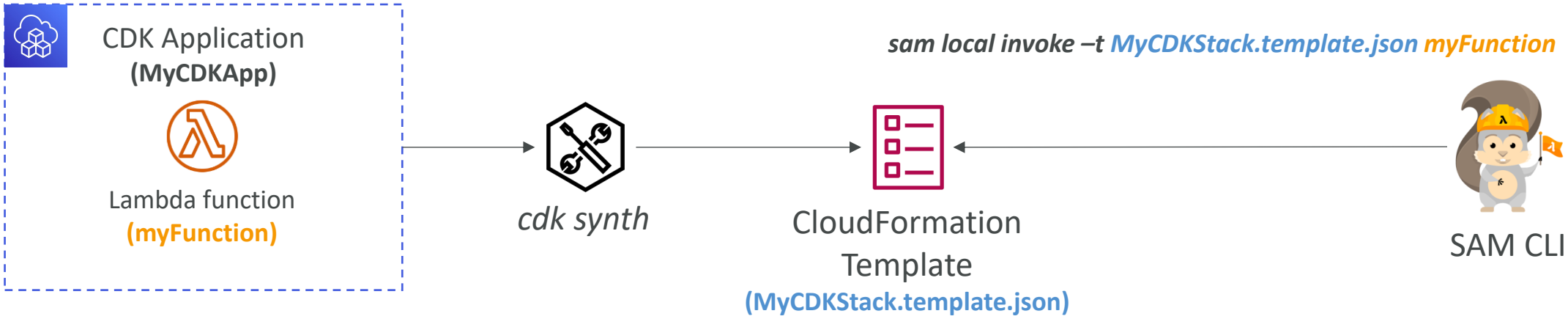
CDK + SAM

- You can use SAM CLI to locally test your CDK apps
- **You must first run *cdk synth***



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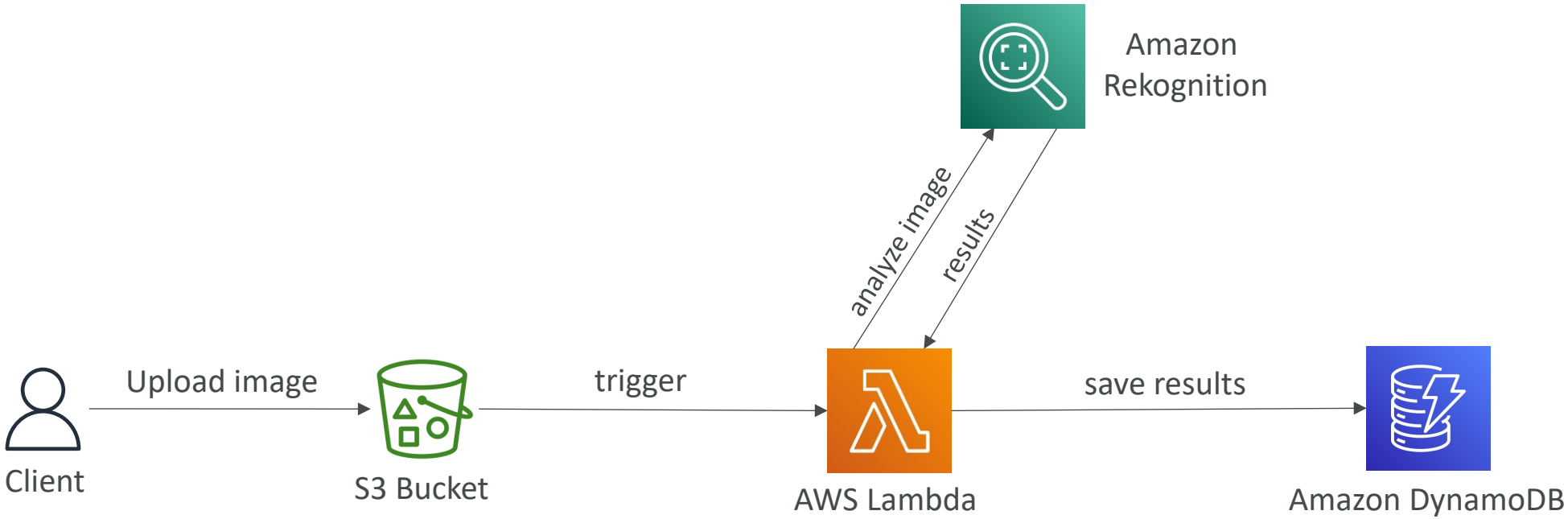
CDK Hands-On



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How can users access AWS ?

- To access AWS, you have three options:
  - **AWS Management Console** (protected by password + MFA)
  - **AWS Command Line Interface (CLI):** protected by access keys
  - **AWS Software Developer Kit (SDK)** - for code: protected by access keys
- Access Keys are generated through the AWS Console
- Users manage their own access keys
- **Access Keys are secret, just like a password. Don't share them**
- Access Key ID ~= username
- Secret Access Key ~= password

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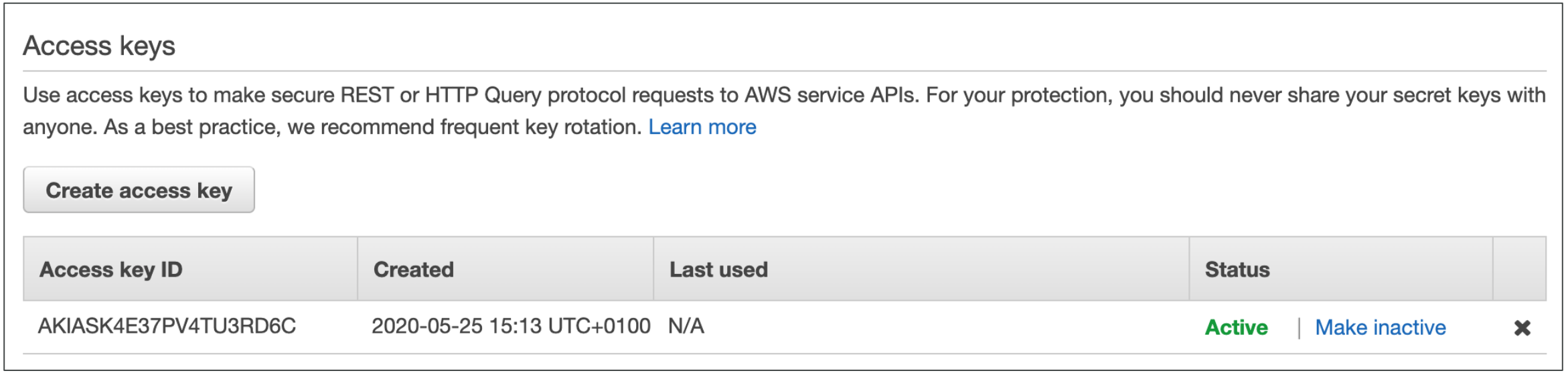
Example (Fake) Access Keys



Access keys

Use access keys to make secure REST or HTTP Query protocol requests to AWS service APIs. For your protection, you should never share your secret keys with anyone. As a best practice, we recommend frequent key rotation. Learn more

Create access key

Access key ID	Created	Last used	Status	
AKIASK4E37PV4TU3RD6C	2020-05-25 15:13 UTC+0100	N/A	Active | Make inactive	✖

- Access key ID: AKIASK4E37PV4983d6C
- Secret Access Key: AZPN3zojWozWCndIjhB0Unh8239a1bzbzO5fqqkZq
- **Remember: don't share your access keys**

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What's the AWS CLI?

- A tool that enables you to interact with AWS services using commands in your command-line shell
- Direct access to the public APIs of AWS services
- You can develop scripts to manage your resources
- It's open-source https://github.com/aws/aws-cli
- Alternative to using AWS Management Console

```
➜  ~ aws s3 cp myfile.txt s3://ccp-mybucket/myfile.txt
upload: ./myfile.txt to s3://ccp-mybucket/myfile.txt
➜  ~ aws s3 ls s3://ccp-mybucket
2021-05-14 03:22:52          0 myfile.txt
➜  ~
```


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What's the AWS SDK?

- AWS Software Development Kit (AWS SDK)
- Language-specific APIs (set of libraries)
- Enables you to access and manage AWS services programmatically
- Embedded within your application
- Supports
  - SDKs (JavaScript, Python, PHP, .NET, Ruby, Java, Go, Node.js, C++)
  - Mobile SDKs (Android, iOS, …)
  - IoT Device SDKs (Embedded C, Arduino, …)
- Example: AWS CLI is built on AWS SDK for Python

Your Application

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What is CloudFormation

- CloudFormation is a declarative way of outlining your AWS Infrastructure, for any resources (most of them are supported).
- For example, within a CloudFormation template, you say:
  - I want a security group
  - I want two EC2 instances using this security group
  - I want an S3 bucket
  - I want a load balancer (ELB) in front of these machines
- Then CloudFormation creates those for you, in the **right order**, with the **exact configuration** that you specify

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Benefits of AWS CloudFormation (1/2)

- **Infrastructure as code**
  - No resources are manually created, which is excellent for control
  - Changes to the infrastructure are reviewed through code

- Cost
  - Each resources within the stack is tagged with an identifier so you can easily see how much a stack costs you
  - You can estimate the costs of your resources using the CloudFormation template
  - Savings strategy: In Dev, you could automation deletion of templates at 5 PM and recreated at 8 AM, safely

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Benefits of AWS CloudFormation (2/2)

- Productivity
  - Ability to destroy and re-create an infrastructure on the cloud on the fly
  - Automated generation of Diagram for your templates!
  - Declarative programming (no need to figure out ordering and orchestration)
- Don’t re-invent the wheel
  - Leverage existing templates on the web!
  - Leverage the documentation
- **Supports (almost) all AWS resources:**
  - Everything we’ll see in this course is supported
  - You can use “custom resources” for resources that are not supported

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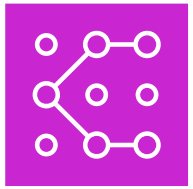
CloudFormation + Infrastructure Composer

- Example: WordPress CloudFormation Stack

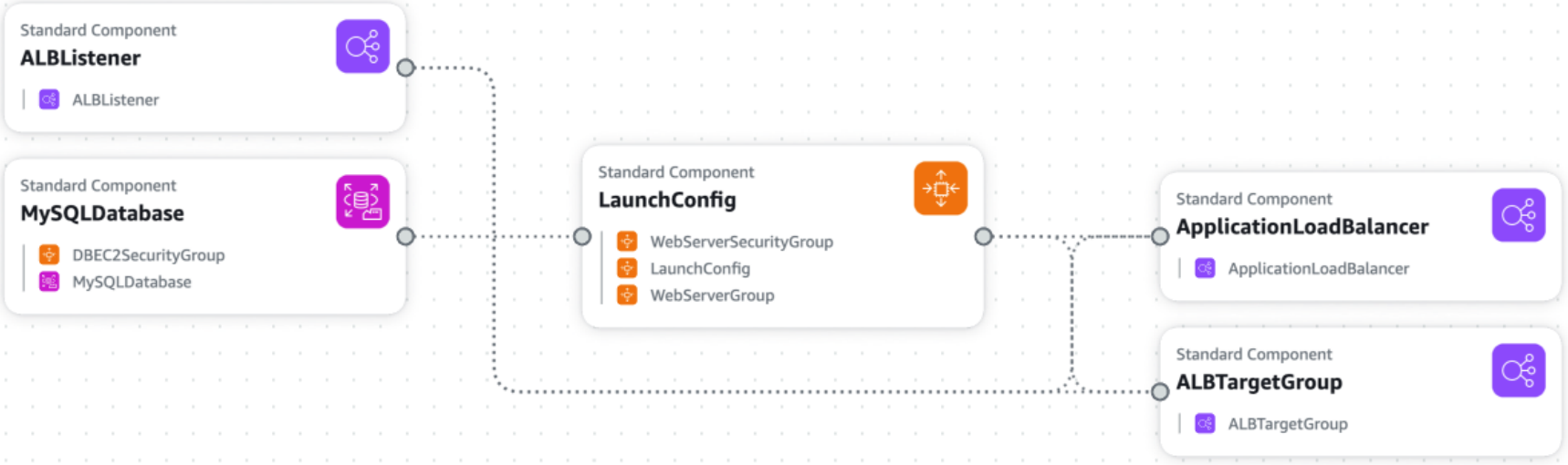
CloudFormation

+



Infrastructure Composer

- We can see all the **resources**
- We can see the **relations** between the components



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AWS CodeArtifact

- Software packages depend on each other to be built (also called code dependencies), and new ones are created
- Storing and retrieving these dependencies is called **artifact management**
- Traditionally you need to setup your own artifact management system
- **CodeArtifact** is a secure, scalable, and cost-effective **artifact management** for software development
- Works with common dependency management tools such as Maven, Gradle, npm, yarn, twine, pip, and NuGet
- **Developers and CodeBuild can then retrieve dependencies straight from CodeArtifact**

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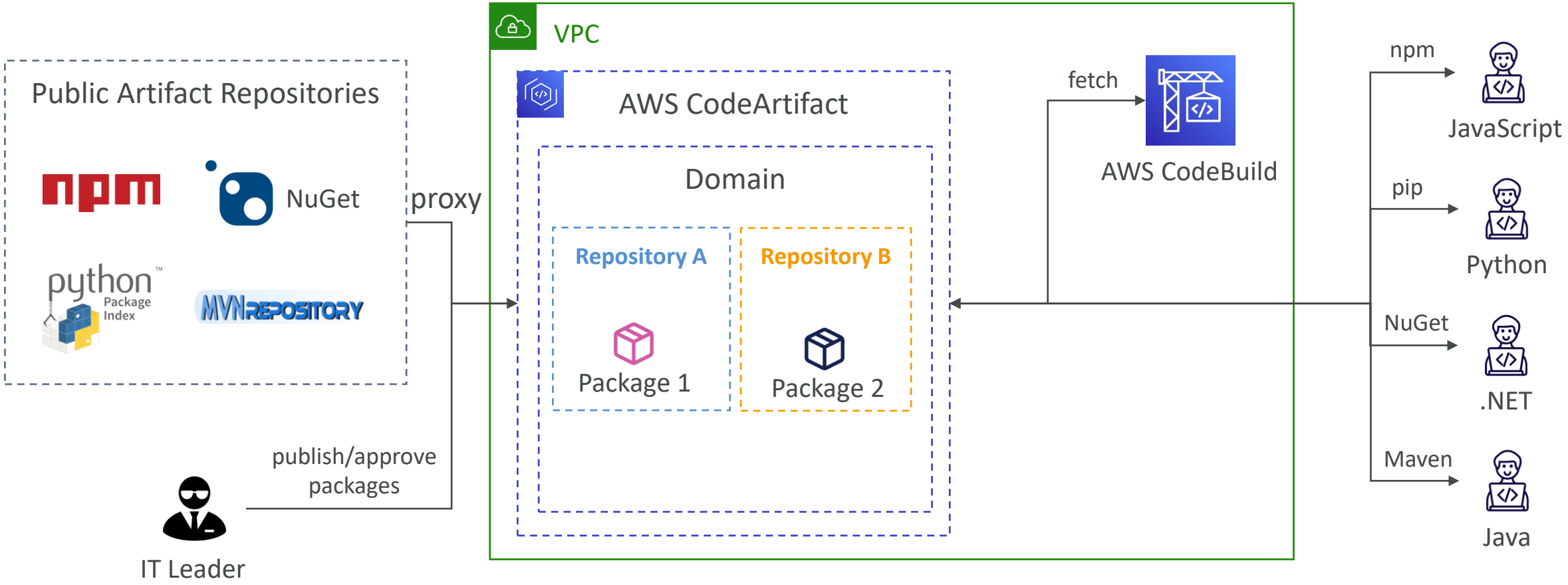
AWS CodeArtifact



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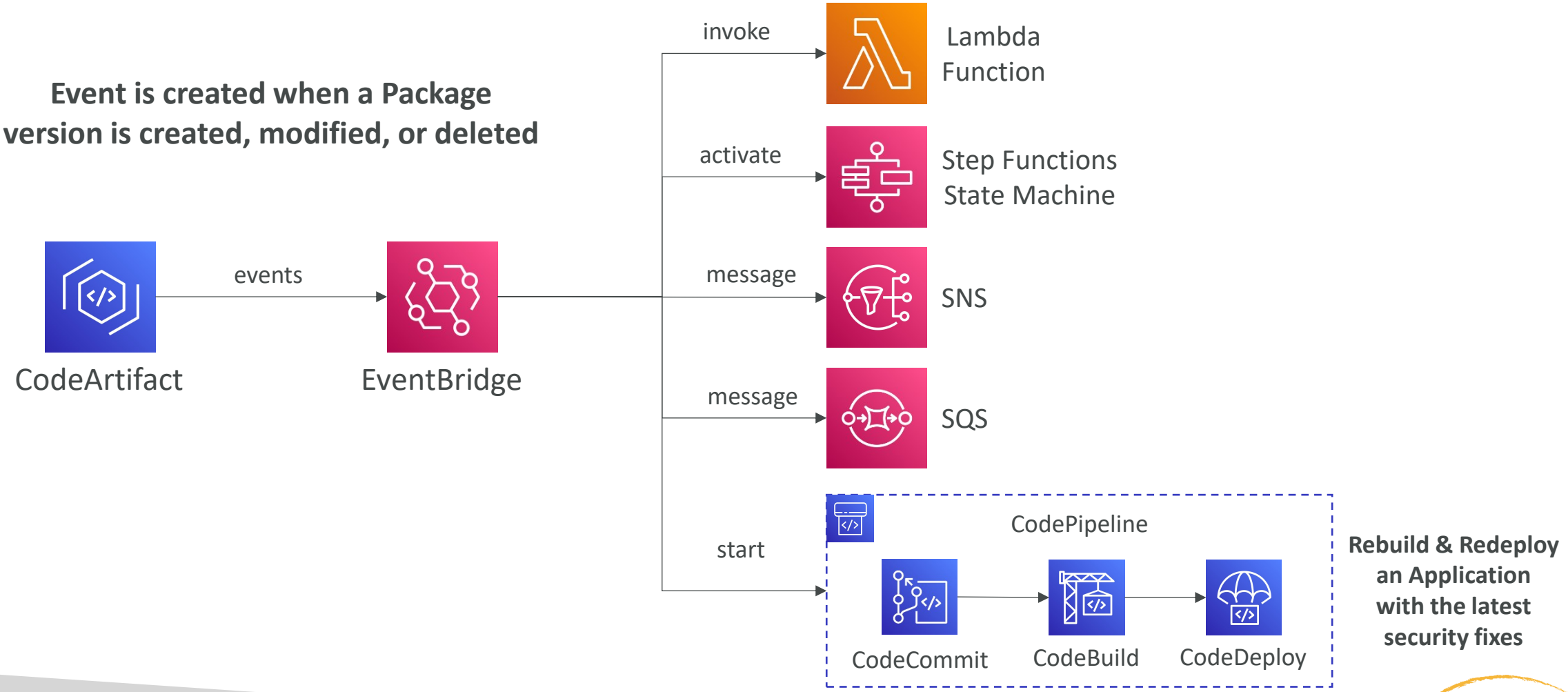
CodeArtifact – EventBridge Integration

Event is created when a Package version is created, modified, or deleted



Rebuild & Redeploy an Application with the latest security fixes

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CodeArtifact – Resource Policy

- Can be used to authorize another account to access CodeArtifact
- A given principal can either read all the packages in a repository or none of them

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "codeartifact:DescribePackageVersion",
                "codeartifact:DescribeRepository",
                "codeartifact:GetPackageVersionReadme",
                "codeartifact:GetRepositoryEndpoint",
                "codeartifact:ListPackages",
                "codeartifact:ListPackageVersions",
                "codeartifact:ListPackageVersionAssets",
                "codeartifact:ListPackageVersionDependencies",
                "codeartifact:ReadFromRepository"
            ],
            "Principal": {
                "AWS": [
                    "arn:aws:iam::123456789012:root",
                    "arn:aws:iam::222333344555:user/bob"
                ]
            },
            "Resource": "*"
        }
    ]
}
```

Repository Resource Policy

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Other Services You Should Know: Machine Learning

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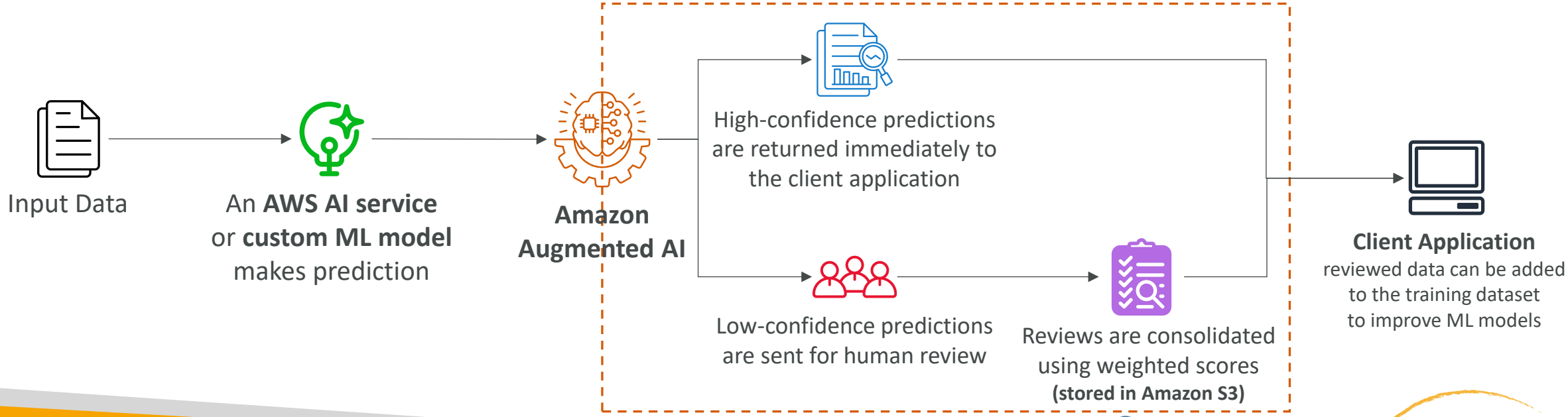
Amazon Augmented AI (A2I)

- Human oversight of Machine Learning predictions in production
  - Can be your own employees, over 500,000 contractors from AWS, or AWS Mechanical Turk
  - Some vendors are pre-screened for confidentiality requirements
- The ML model can be built on AWS or elsewhere (SageMaker, Rekognition…)



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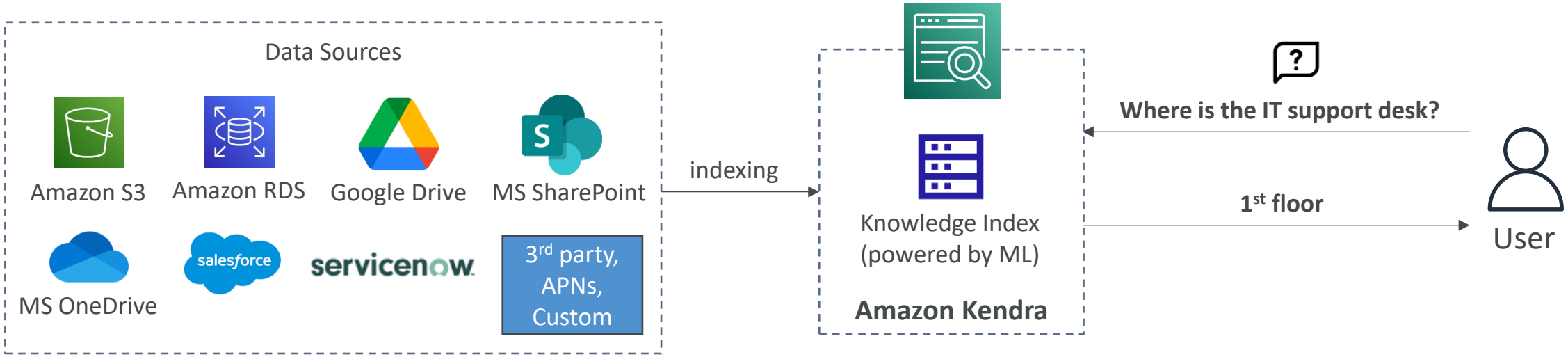
Amazon Kendra

- Fully managed **document search service** powered by Machine Learning
- Extract answers from within a document (text, pdf, HTML, PowerPoint, MS Word, FAQs…)
- Natural language search capabilities
- Learn from user interactions/feedback to promote preferred results **(Incremental Learning)**
- Ability to manually fine-tune search results (importance of data, freshness, custom, …)



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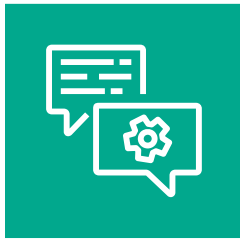
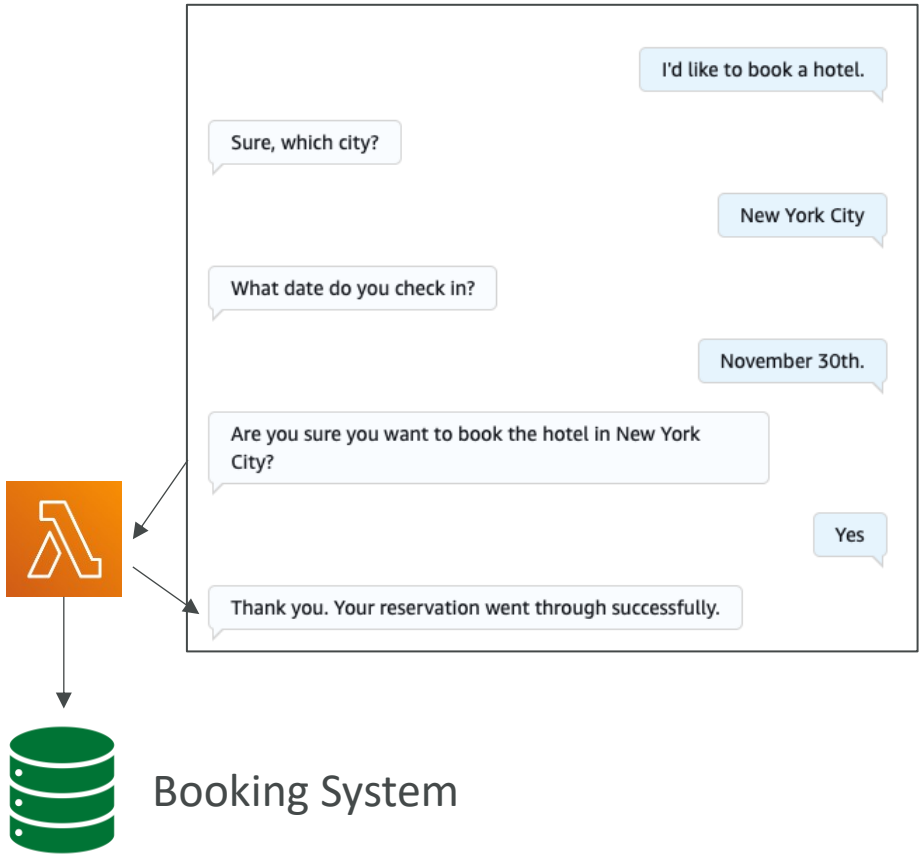
Amazon Lex



- Build chatbots quickly for your applications using voice and text
- Example: a chatbot that allows your customers to order pizzas or book a hotel
- Supports multiple languages
- Integration with AWS Lambda, Connect, Comprehend, Kendra
- The bot automatically understands the user intent to invoke the correct Lambda function to “fulfill the intent”
- The bot will ask for ”Slots” (input parameters) if necessary



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Amazon Rekognition

- Find **objects, people, text, scenes** in **images and videos** using ML
- **Facial analysis** and **facial search** to do user verification, people counting
- Create a database of “familiar faces” or compare against celebrities
- Use cases:
  - Labeling
  - Content Moderation
  - Text Detection
  - Face Detection and Analysis (gender, age range, emotions…)
  - Face Search and Verification
  - Celebrity Recognition
  - Pathing (ex: for sports game analysis)

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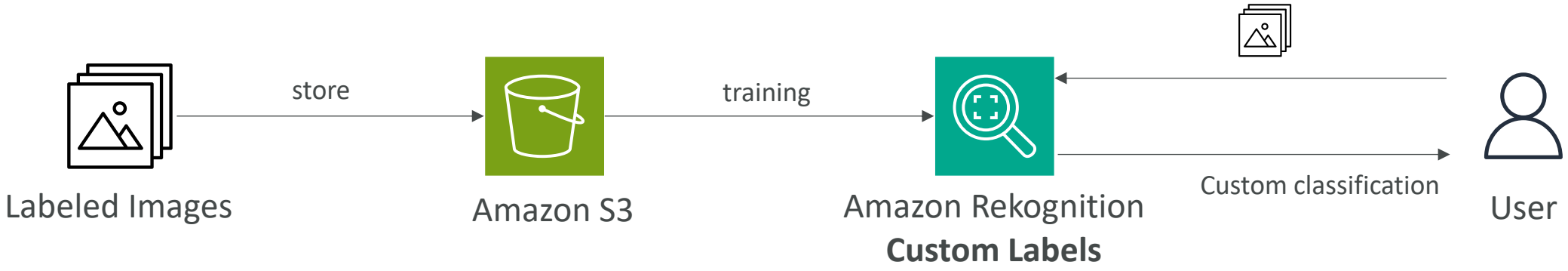
Amazon Rekognition – Custom Labels

- Examples: find your logo in social media posts, identify your products on stores shelves (National Football League – NFL – uses it to find their logo in pictures)
- **Label** your training images and upload them to Amazon Rekognition
- Only needs a few hundred images or less
- Amazon Rekognition creates a custom model on your images set
- New subsequent images will be categorized the custom way you have defined



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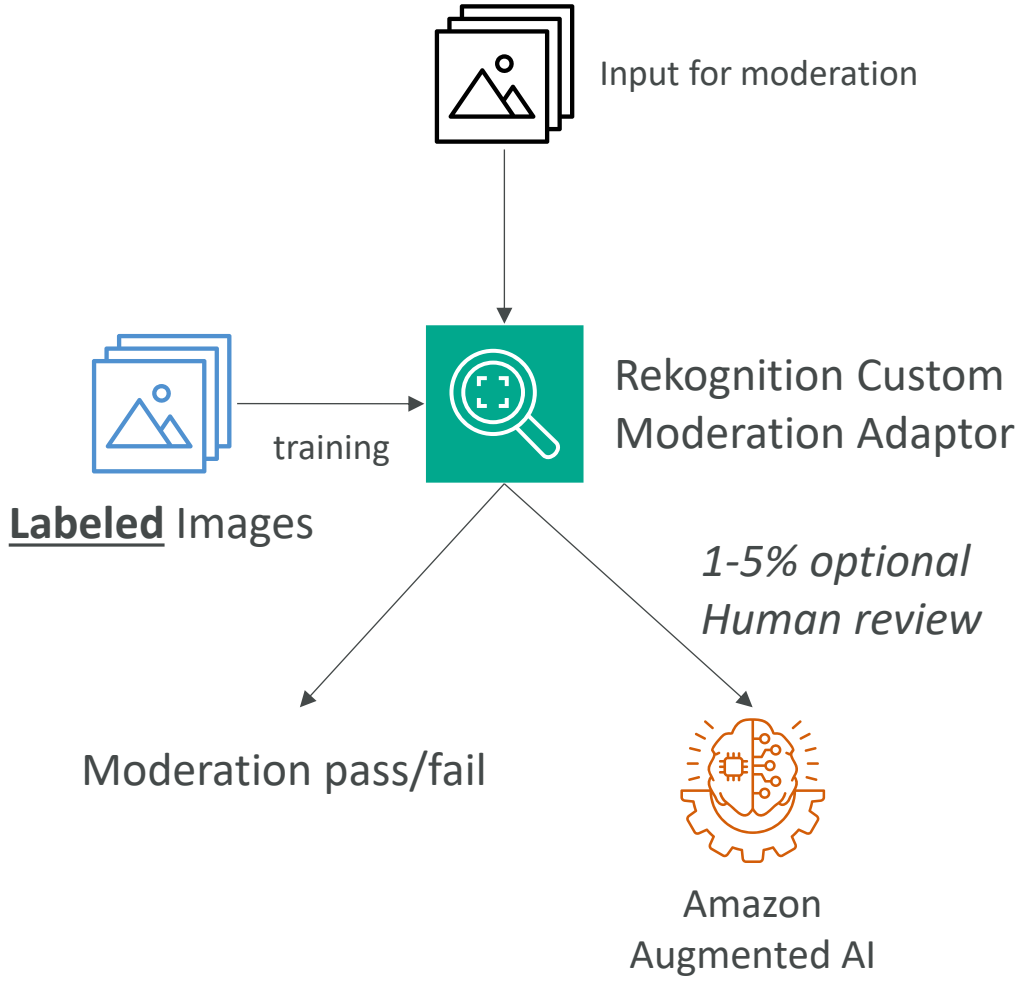
Amazon Rekognition – Content Moderation

- Automatically detect inappropriate, unwanted, or offensive content
- Example: filter out harmful images in social media, broadcast media, advertising…
- Bring down human review to 1-5% of total content volume
- Integrated with Amazon Augmented AI (Amazon A2I) for human review
- **Custom Moderation Adaptors**
  - Extends Rekognition capabilities by providing your own **labeled** set of images
  - Enhances the accuracy of Content Moderation or create a specific use case of Moderation



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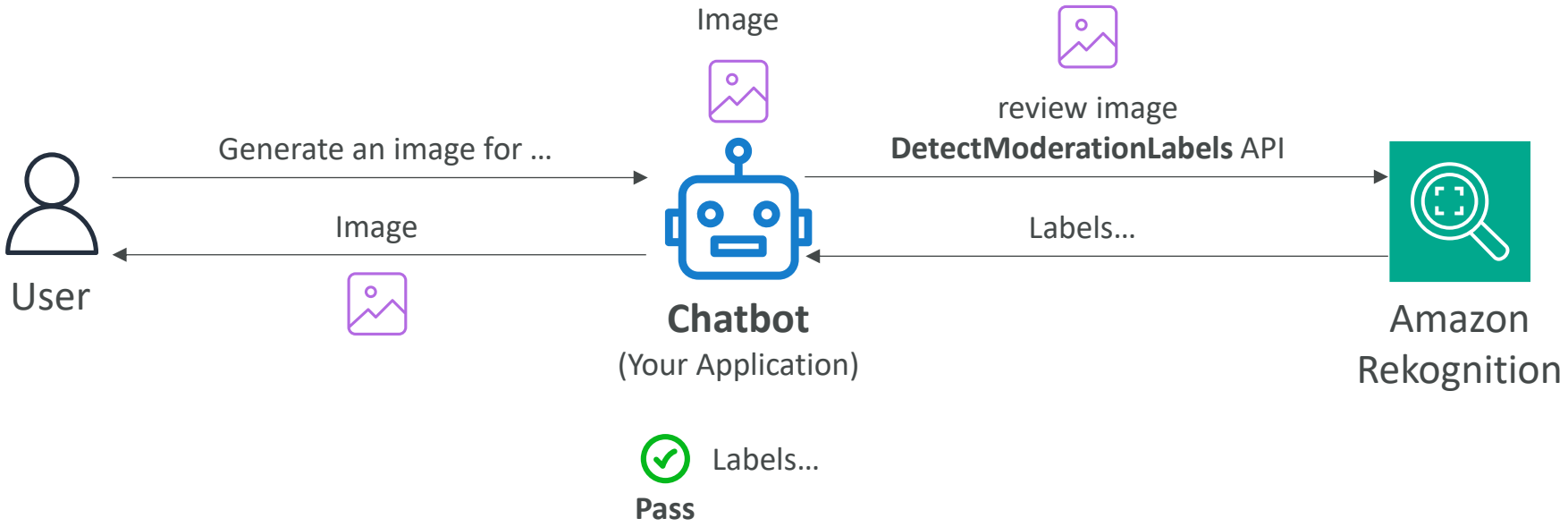
Content Moderation API – Diagram



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Amazon Textract

- Automatically extracts text, handwriting, and data from any scanned documents using AI and ML

analyze

Amazon Textract

result

```
{
   “Document ID”: “123456789-005”,
   “Name”: “”,
   “SEX”: “F”,
   “DOB”: “23.05.1997”,
   …
}
```

- Extract data from forms and tables
- Read and process any type of document (PDFs, images, …)
- Use cases:
  - Financial Services (e.g., invoices, financial reports)
  - Healthcare (e.g., medical records, insurance claims)
  - Public Sector (e.g., tax forms, ID documents, passports)

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Amazon Transcribe

- Automatically **convert speech to text**
- Uses a **deep learning process** called **automatic speech recognition** (ASR) to convert speech to text quickly and accurately
- **Automatically remove Personally Identifiable Information (PII) using Redaction**
- **Supports Automatic Language Identification for multi-lingual audio**
- Use cases:
  - transcribe customer service calls
  - automate closed captioning and subtitling
  - generate metadata for media assets to create a fully searchable archive

”Hello my name is Stéphane.
I hope you’re enjoying the course!

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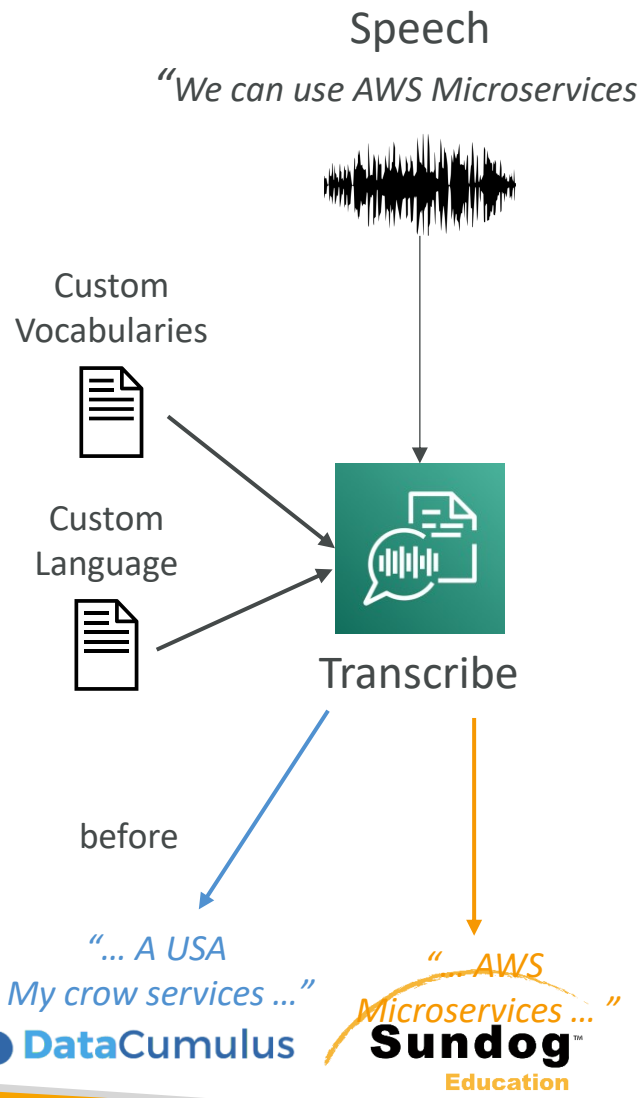
Amazon Transcribe – Improving Accuracy

- Allows Transcribe to capture domain-specific or non-standard terms (e.g., technical words, acronyms, jargon…)
- **Custom Vocabularies (for words)**
  - Add specific words, phrases, domain-specific terms
  - Good for brand names, acronyms…
  - Increase recognition of a new word by providing hints (such as pronunciation..)
- **Custom Language Models (for context)**
  - Train Transcribe model on your own domain-specific text data
  - Good for transcribing large volumes of domain-specific speech
  - Learn the context associated with a given word
- **Note:** use both for the highest transcription accuracy



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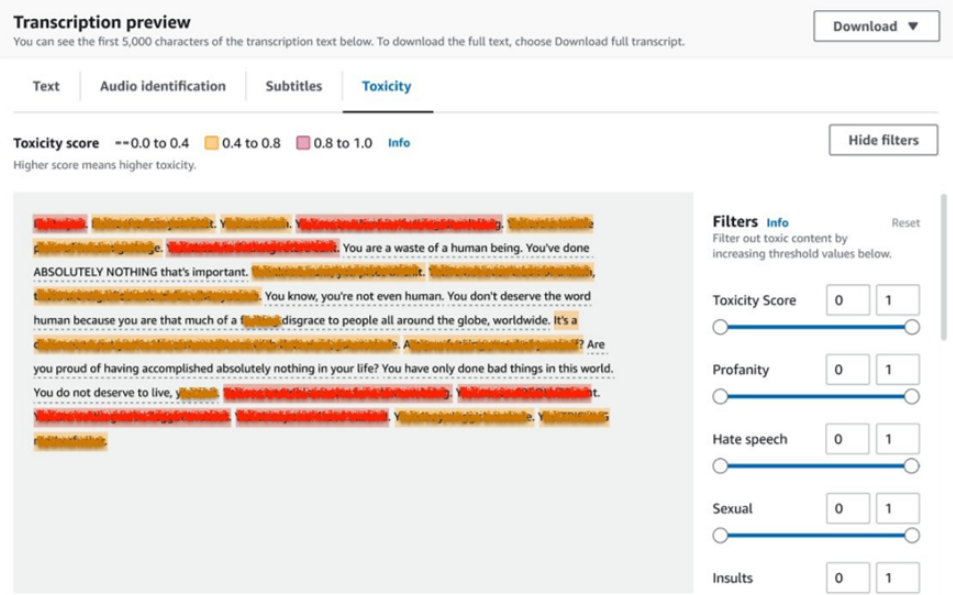
Transcribe – Toxicity Detection

- ML-powered, voice-based toxicity detection capability
- Leverages speech cues: tone and pitch, and text-based cues
- Toxicity categories: sexual harassment, hate speech, threat, abuse, profanity, insult, and graphic….



Toxicity Categories

Profanity: Speech that contains words, phrases, or acronyms that are impolite, vulgar, or offensive.

Hate speech: Speech that criticizes, insults, denounces, or dehumanizes a person or group on the basis of an identity (such as race, ethnicity, gender, religion, sexual orientation, ability, and national origin).

Sexual: Speech that indicates sexual interest, activity, or arousal using direct or indirect references to body parts, physical traits, or sex.

Insults: Speech that includes demeaning, humiliating, mocking, insulting, or belittling language. This type of language is also labeled as bullying.

Violence or threat: Speech that includes threats seeking to inflict pain, injury, or hostility toward a person or group.

Graphic: Speech that uses visually descriptive and unpleasantly vivid imagery. This type of language is often intentionally verbose to amplify a recipient's discomfort.

Harassment or abusive: Speech intended to affect the psychological well-being of the recipient, including demeaning and objectifying terms.

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Other Services You Should Know: Management and Governance

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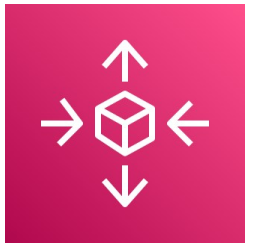
AWS Auto Scaling



- Backbone service of auto scaling for scalable resources in AWS:

- **Amazon EC2 Auto Scaling groups**: Launch or terminate EC2 instances
- **Amazon EC2 Spot Fleet requests**: Launch or terminate instances from a Spot Fleet request, or automatically replace instances that get interrupted for price or capacity reasons.
- **Amazon ECS**: Adjust the ECS service desired count up or down
- **Amazon DynamoDB (table or global secondary index)**: WCU & RCU
- **Amazon Aurora**: Dynamic Read Replicas Auto Scaling

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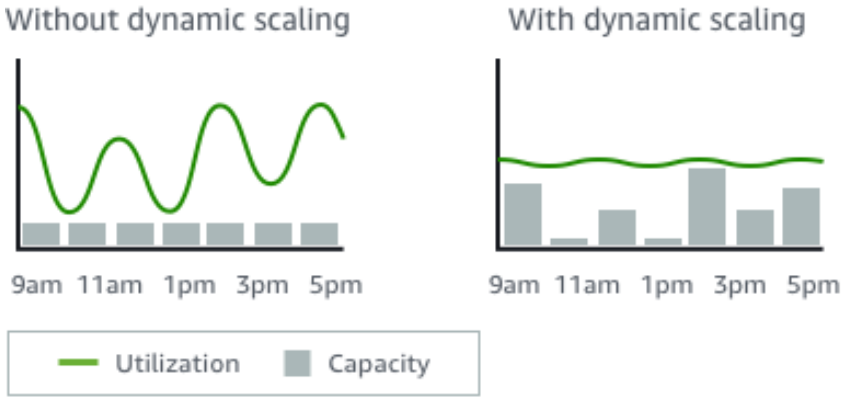
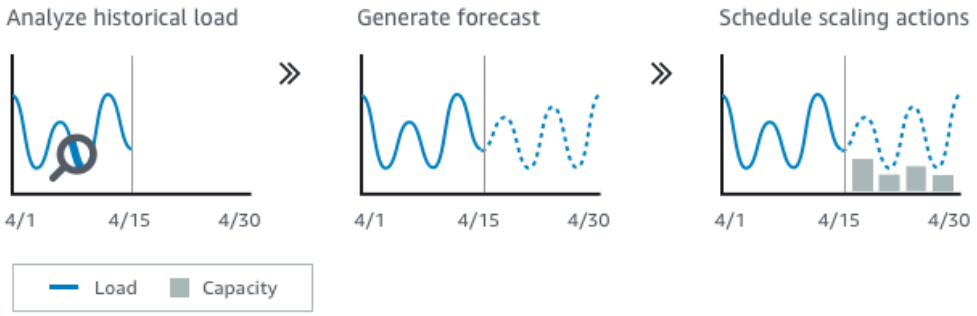
AWS Auto Scaling – Scaling Plans

- **Dynamic scaling:** creates a target tracking scaling policy
  - Optimize for availability => 40% of resource utilization
  - Balance availability and cost => 50% of resource utilization
  - Optimize for cost => 70% of resource utilization
  - Custom => choose own metric and target value
  - Options: Disable scale-in, cooldown period, warmup time (for ASG)

- **Predictive scaling:** continuously forecast load and schedule scaling ahead





https://docs.aws.amazon.com/autoscaling/plans/userguide/how-it-works.html

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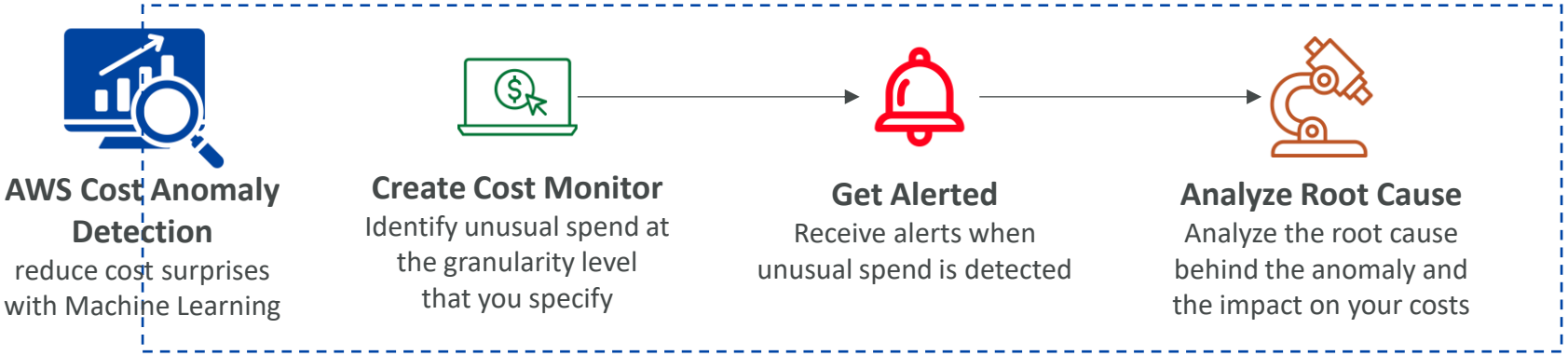
AWS Cost Anomaly Detection

- **Continuously monitor your cost and usage using ML to detect unusual spends**
- It learns your unique, historic spend patterns to detect one-time cost spike and/or continuous cost increases (you don't need to define thresholds)
- Monitor AWS services, member accounts, cost allocation tags, or cost categories
- Sends you the anomaly detection report with root-cause analysis
- Get notified with individual alerts or daily/weekly summary (using SNS)



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Cost Explorer

- Visualize, understand, and manage your AWS costs and usage over time
- Create custom reports that analyze cost and usage data.
- Analyze your data at a high level: total costs and usage across all accounts
- Or Monthly, hourly, resource level granularity
- Choose an optimal **Savings Plan** (to lower prices on your bill)
- **Forecast usage up to 18 months based on previous usage**

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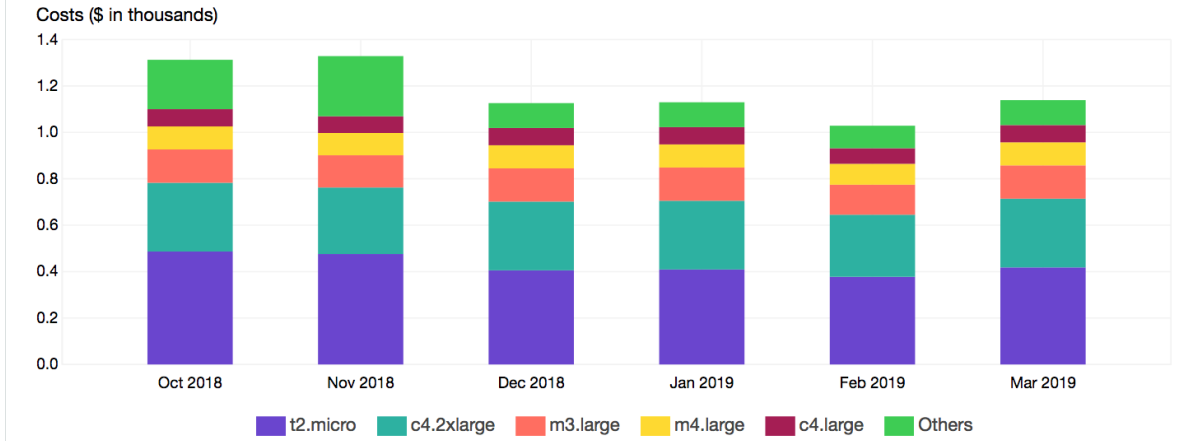
Cost Explorer – Monthly Cost by AWS Service

Monthly costs by service

Last 6 Months | Monthly | Stack

Group by: Instance Type | Service | Linked Account | Region | Usage Type | Tag | API Operation | Availability Zone | More



Download CSV

Instance Type	Oct 1, 2018	Nov 1, 2018	Dec 1, 2018	Jan 1, 2019
Total cost ($)	1,312.71	1,328.54	1,125.99	1,129.65
t2.micro ($)	486.75	475.89	405.63	409.27
c4.2xlarge ($)	296.11	286.56	296.11	296.11

FILTERS CLEAR ALL

Service: Include only — EC2-Instances ✖ 1

Linked Account: Include all

Region: Include all

Instance Type: Include all

Usage Type: Include all

Usage Type Group: Include all

Tag: Include All

API Operation: Include all

Charge Type: Include all

More filters

ADVANCED OPTIONS

Show costs as

Unblended costs

Include costs related to

Show only untagged resources

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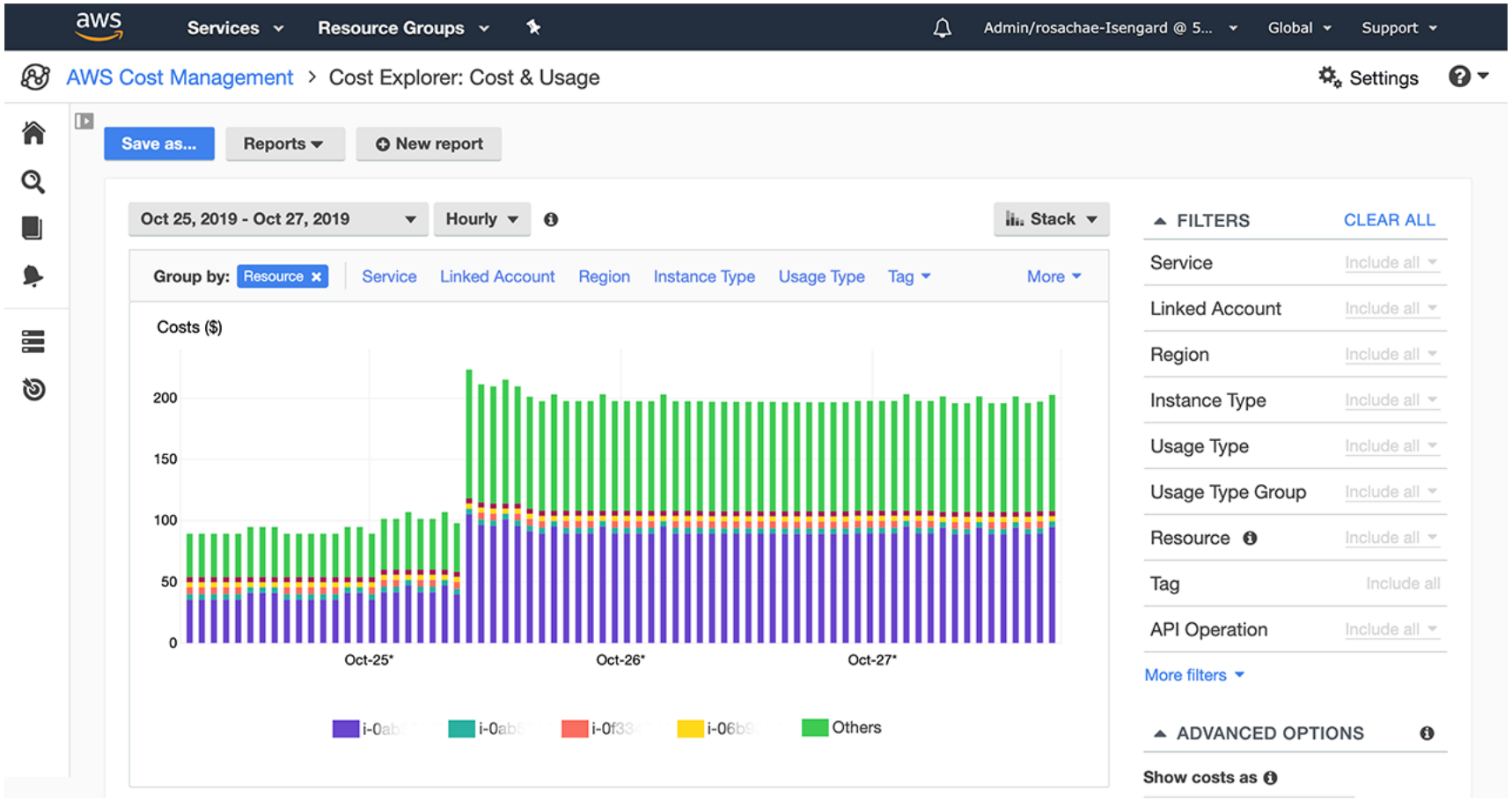
DataCumulus

Sundog Education

Cost Explorer– Hourly & Resource Level



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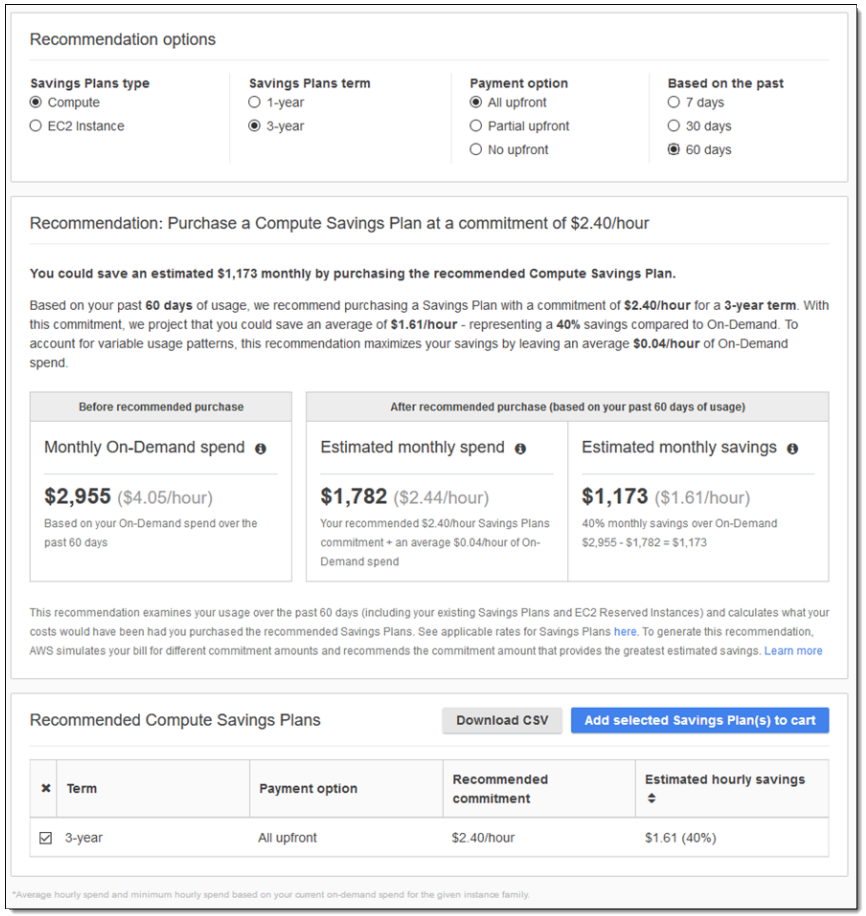
Cost Explorer – Savings Plan Alternative to Reserved Instances



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Cost Explorer – Forecast Usage

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Amazon Managed Grafana

- Grafana is a popular open-source platform used to monitor, visualize, and alert on metrics and logs.
- Integrated with IAM Identity Center (formerly AWS SSO) and/or SAML for user management and permissions
- Compatible with Grafana plugins and alerts
- Fully managed, scales automatically
- Encrypted at rest and in transit, can use KMS.

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Amazon Managed Grafana

- Integrated with many AWS data sources
  - CloudWatch, OpenSearch, Timestream, Athena, Redshift, X-Ray
  - Amazon Managed Service for Prometheus (AMP)
- Also with everything else Grafana integrates with
  - GitHub, Google, Azure, MySQL, Redis, JSON, OpenTelemetry, much more.

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AWS Systems Manager (SSM)

- Helps you manage your **EC2 and On-Premises** systems at scale
- Another **Hybrid** AWS service
- Get operational insights about the state of your infrastructure
- Suite of 10+ products
- Most important features are:
  - **Patching automation for enhanced compliance**
  - **Run commands across an entire fleet of servers**
  - Store parameter configuration with the SSM Parameter Store
- Works for Linux, Windows, MacOS, and Raspberry Pi OS (Raspbian)

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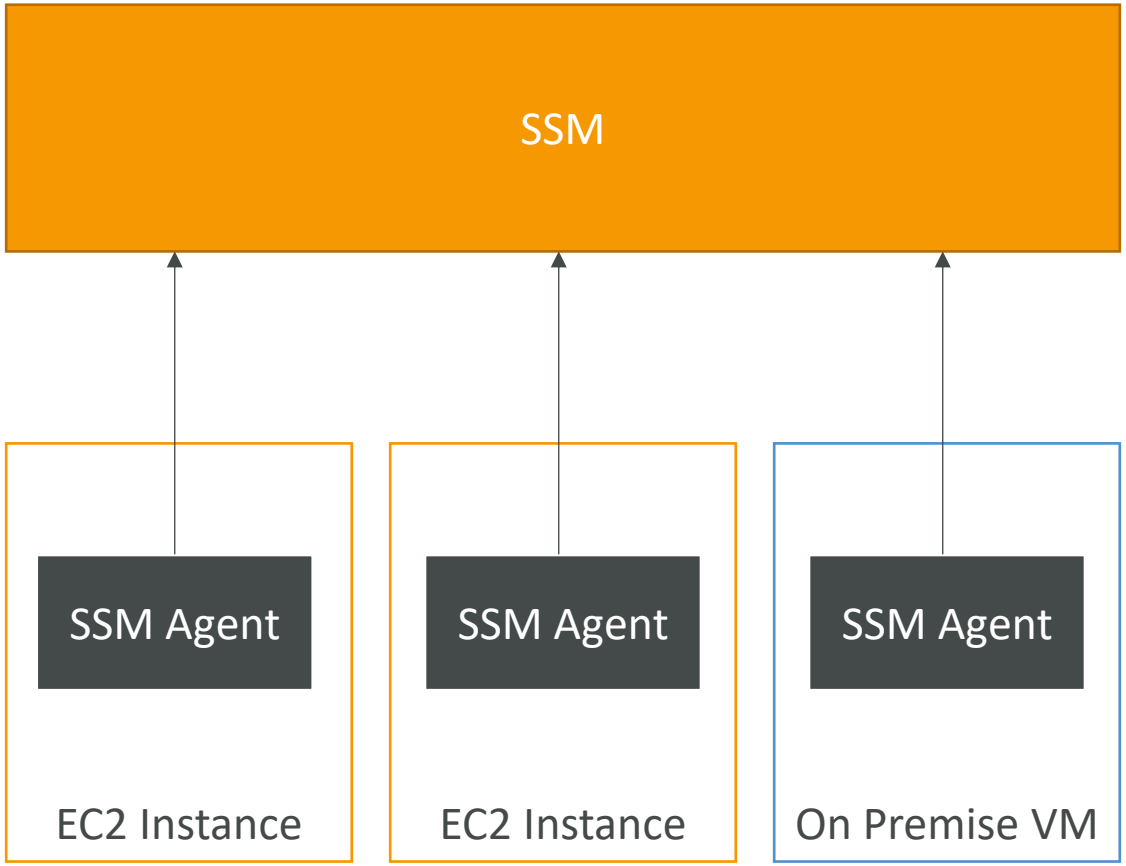
How Systems Manager works

- We need to install the SSM agent onto the systems we control
- Installed by default on Amazon Linux AMI & some Ubuntu AMI
- If an instance can't be controlled with SSM, it's probably an issue with the SSM agent!
- Thanks to the SSM agent, we can **run commands**, **patch** & **configure** our servers



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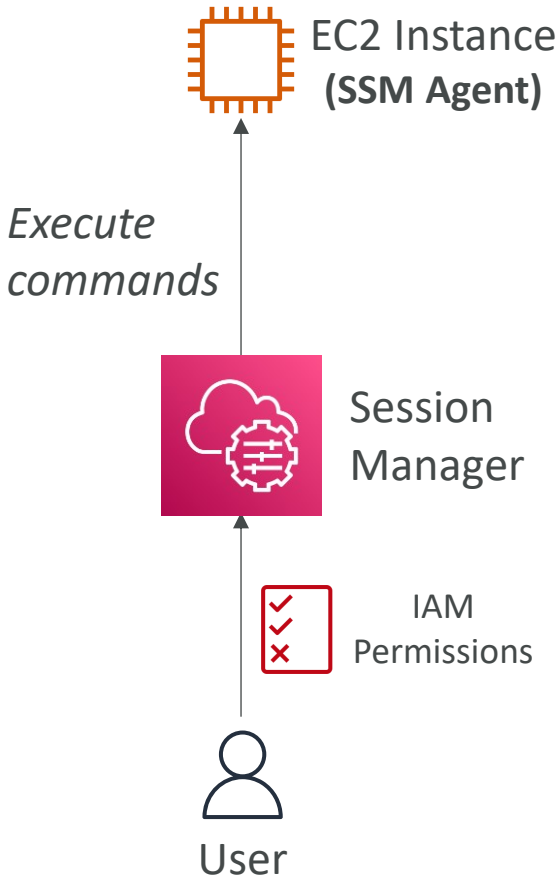
Systems Manager – SSM Session Manager

- Allows you to start a secure shell on your EC2 and on-premises servers

- **No SSH access, bastion hosts, or SSH keys needed**
- **No port 22 needed (better security)**
- Supports Linux, macOS, and Windows
- Send session log data to S3 or CloudWatch Logs



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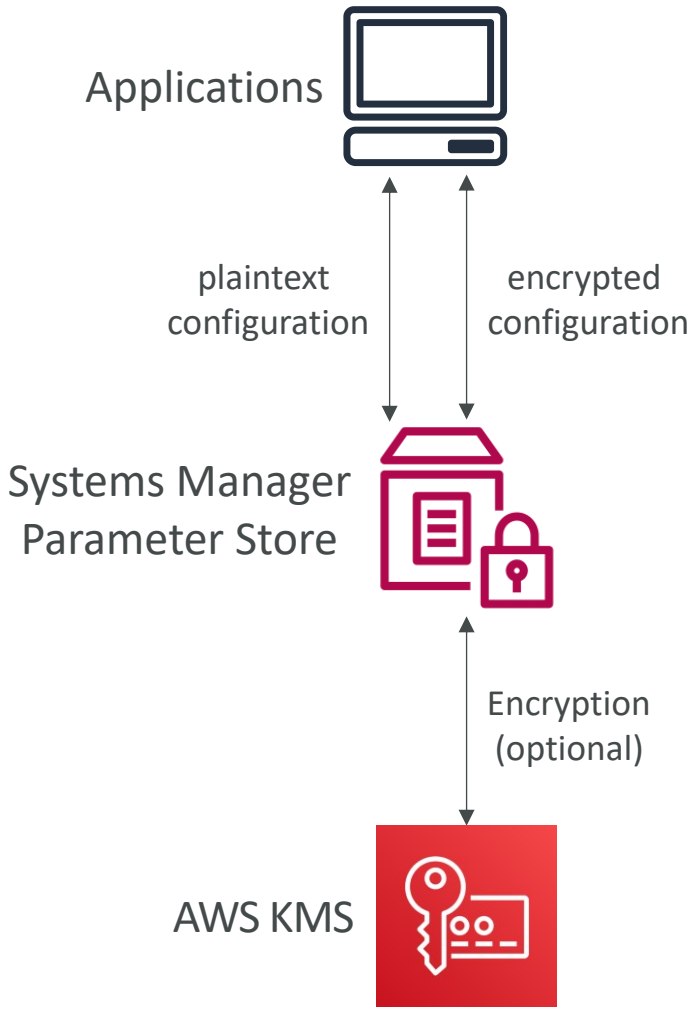
Systems Manager Parameter Store

- Secure storage for configuration and secrets
- API Keys, passwords, configurations…
- Serverless, scalable, durable, easy SDK
- Control access permissions using IAM
- Version tracking & encryption (optional)



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Other Services You Should Know: Migration and Data Transfer

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AWS DataSync

- Move large amount of data to and from
  - On-premises / other cloud to AWS (NFS, SMB, HDFS, S3 API…) – **needs agent**
  - AWS to AWS (different storage services) – no agent needed
- Can synchronize to:
  - Amazon S3 (any storage classes – including Glacier)
  - Amazon EFS
  - Amazon FSx (Windows, Lustre, NetApp, OpenZFS...)
- Replication tasks can be scheduled hourly, daily, weekly
- **File permissions and metadata are preserved** (NFS POSIX, SMB…)
- One agent task can use 10 Gbps, can setup a bandwidth limit

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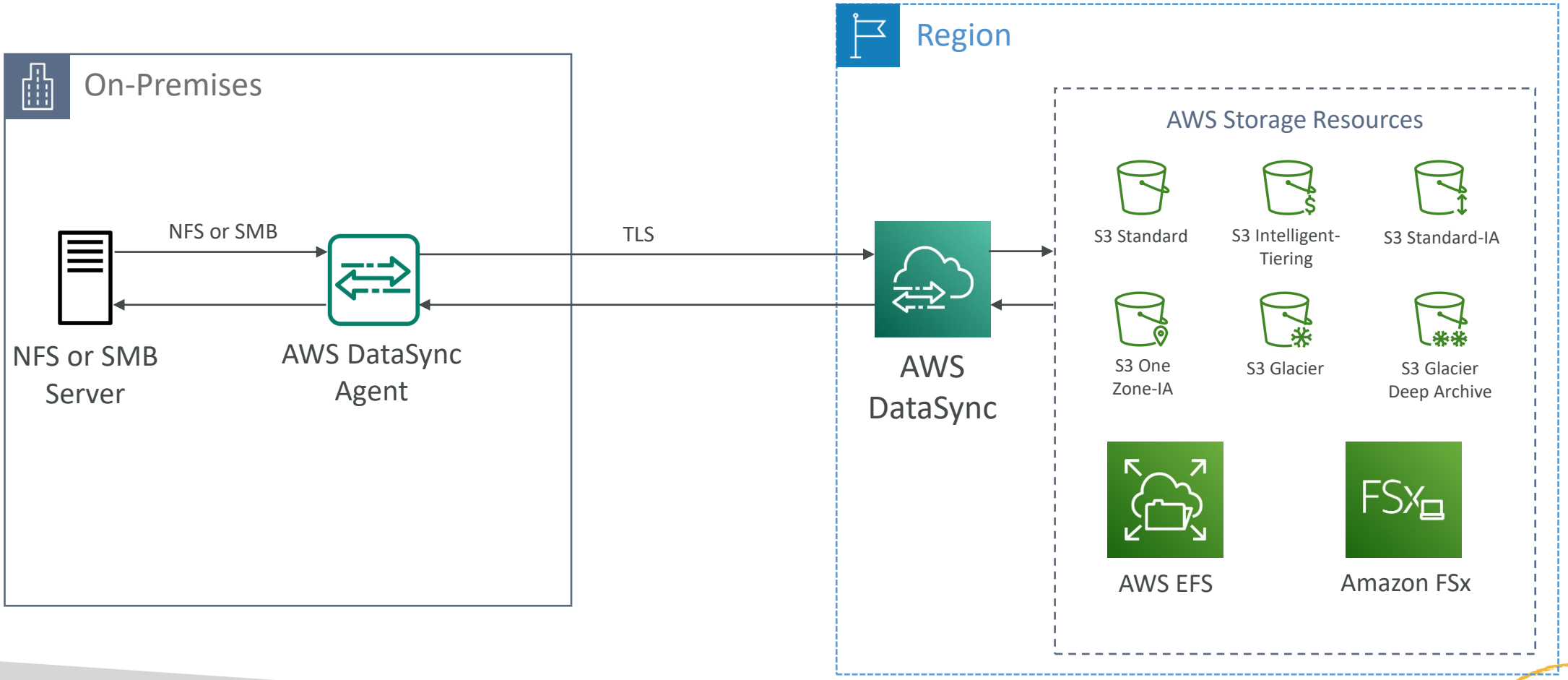
AWS DataSync
NFS / SMB to AWS (S3, EFS, FSx...)



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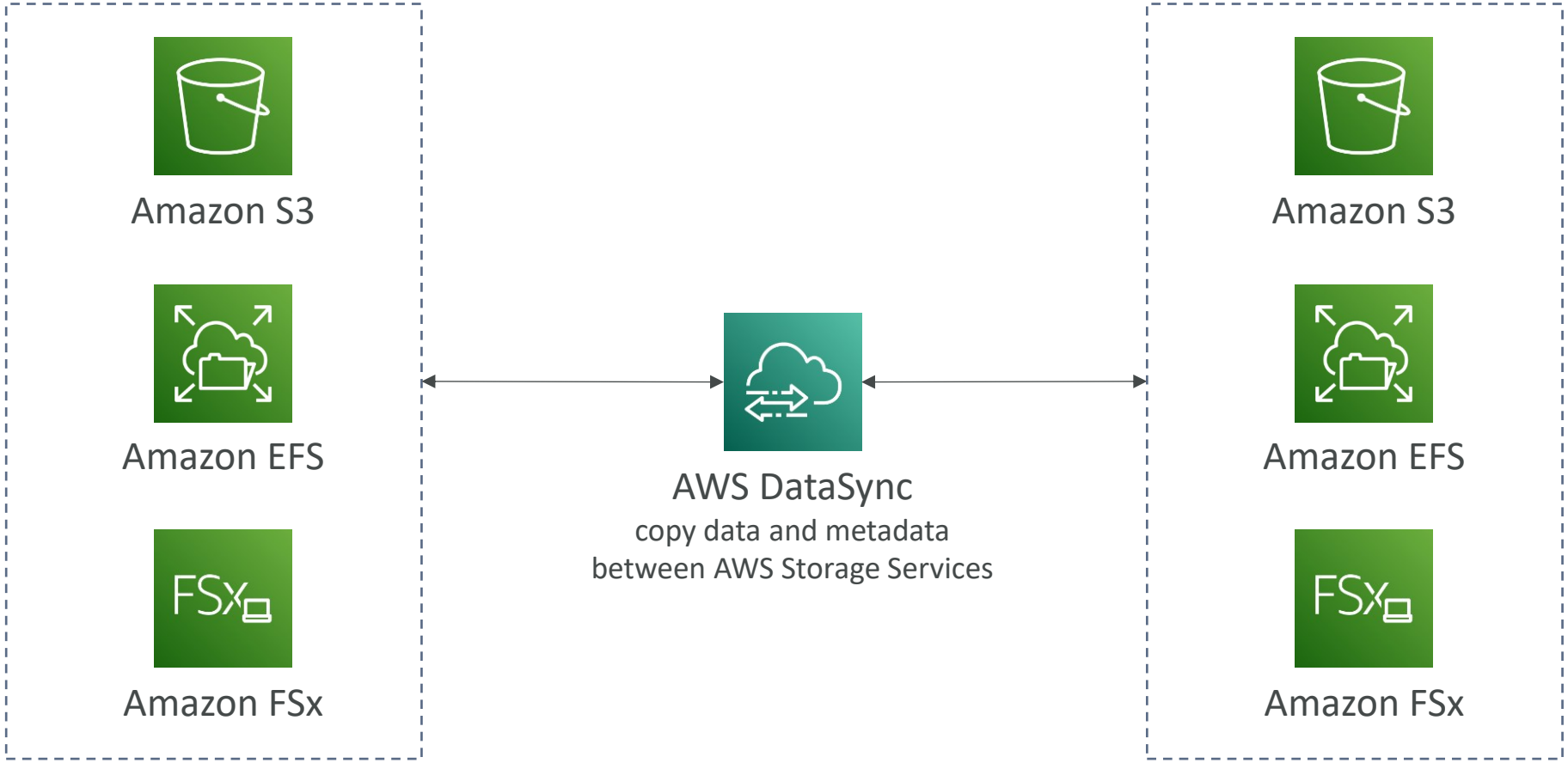
AWS DataSync
Transfer between AWS storage services



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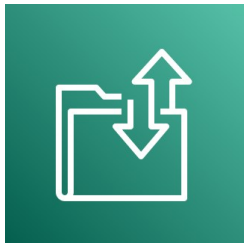
AWS Transfer Family



- A fully-managed service for file transfers into and out of Amazon S3 or Amazon EFS using the FTP protocol
- Supported Protocols
  - **AWS Transfer for FTP** (File Transfer Protocol (FTP))
  - **AWS Transfer for FTPS** (File Transfer Protocol over SSL (FTPS))
  - **AWS Transfer for SFTP** (Secure File Transfer Protocol (SFTP))
- Managed infrastructure, Scalable, Reliable, Highly Available (multi-AZ)
- Pay per provisioned endpoint per hour + data transfers in GB
- Store and manage users' credentials within the service
- Integrate with existing authentication systems (Microsoft Active Directory, LDAP, Okta, Amazon Cognito, custom)
- Usage: sharing files, public datasets, CRM, ERP, …

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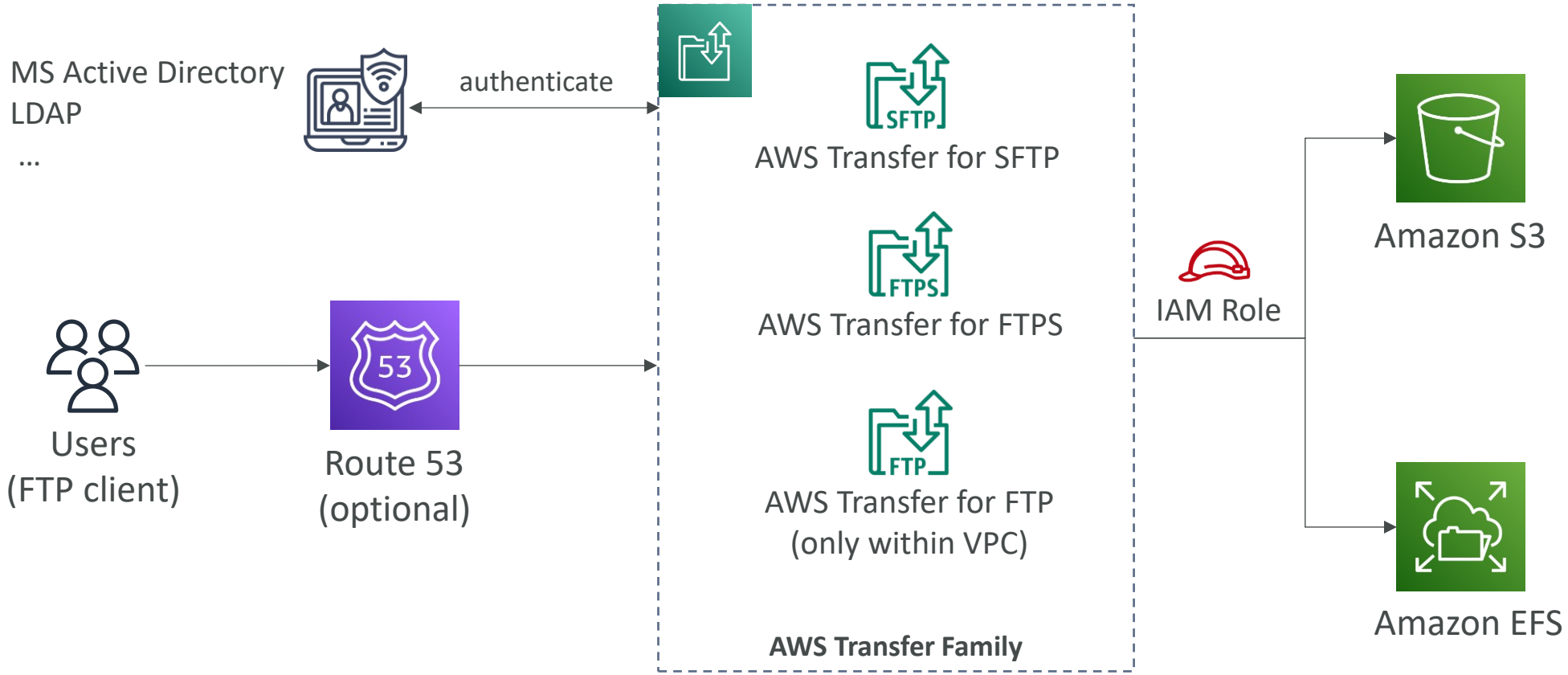
AWS Transfer Family



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Other Services You Should Know: Networking and Content Delivery

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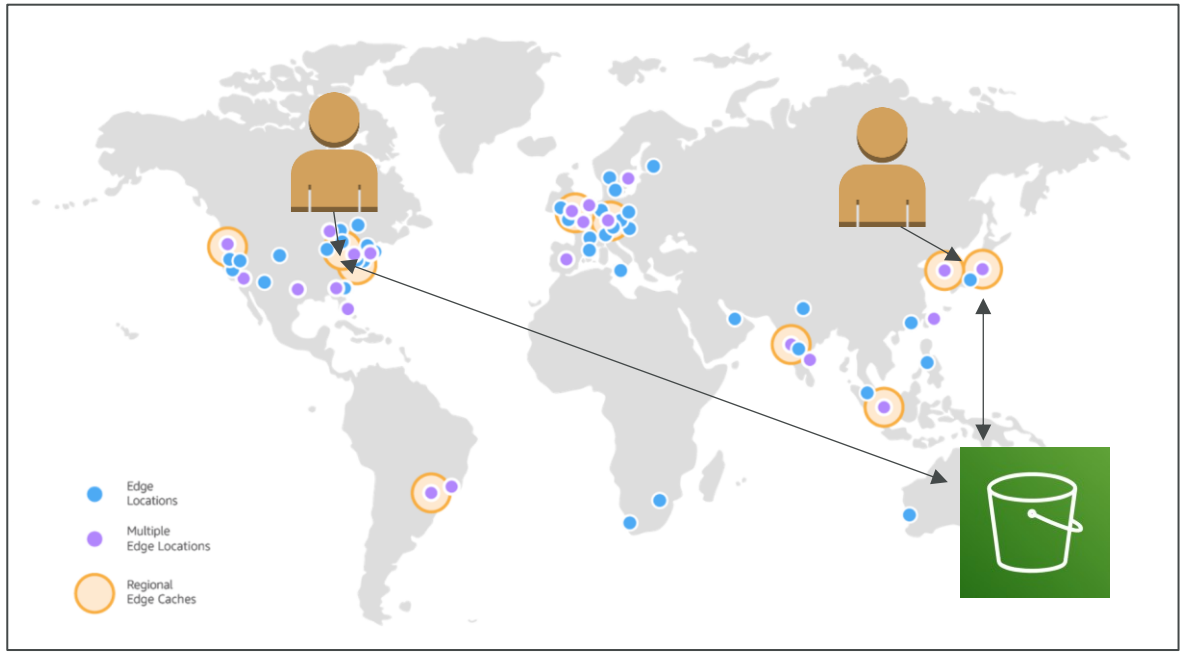
Amazon CloudFront

- Content Delivery Network (CDN)
- **Improves read performance, content is cached at the edge**
- Improves users experience
- Hundreds of Points of Presence globally (edge locations, caches)
- **DDoS protection (because worldwide), integration with Shield, AWS Web Application Firewall**



Source: https://aws.amazon.com/cloudfront/features/?nc=sn&loc=2

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CloudFront – Origins

- **S3 bucket**
  - For distributing files and caching them at the edge
  - For uploading files to S3 through CloudFront
  - Secured using Origin Access Control (OAC)

- **VPC Origin**
  - For applications hosted in VPC private subnets
  - Application Load Balancer / Network Load Balancer / EC2 Instances

- **Custom Origin (HTTP)**
  - S3 website (must first enable the bucket as a static S3 website)
  - Any public HTTP backend you want

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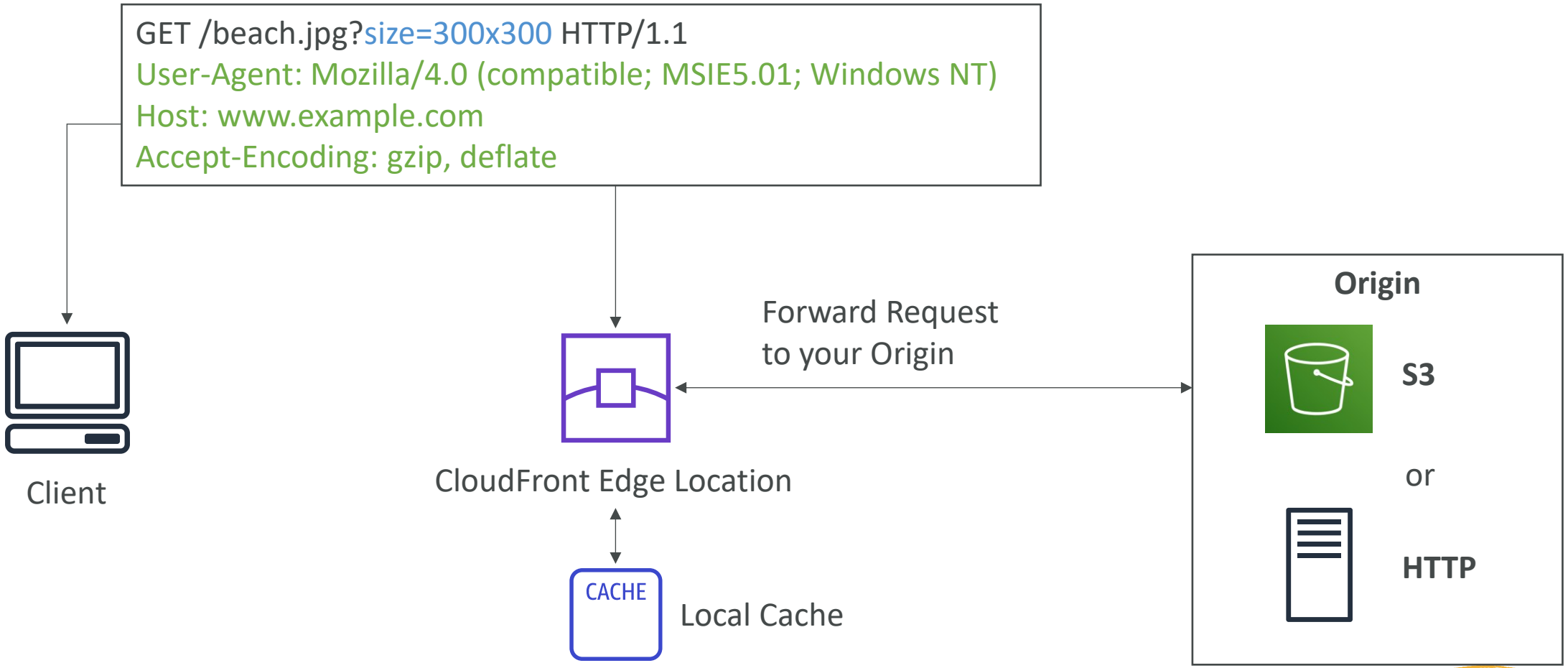
CloudFront at a high level



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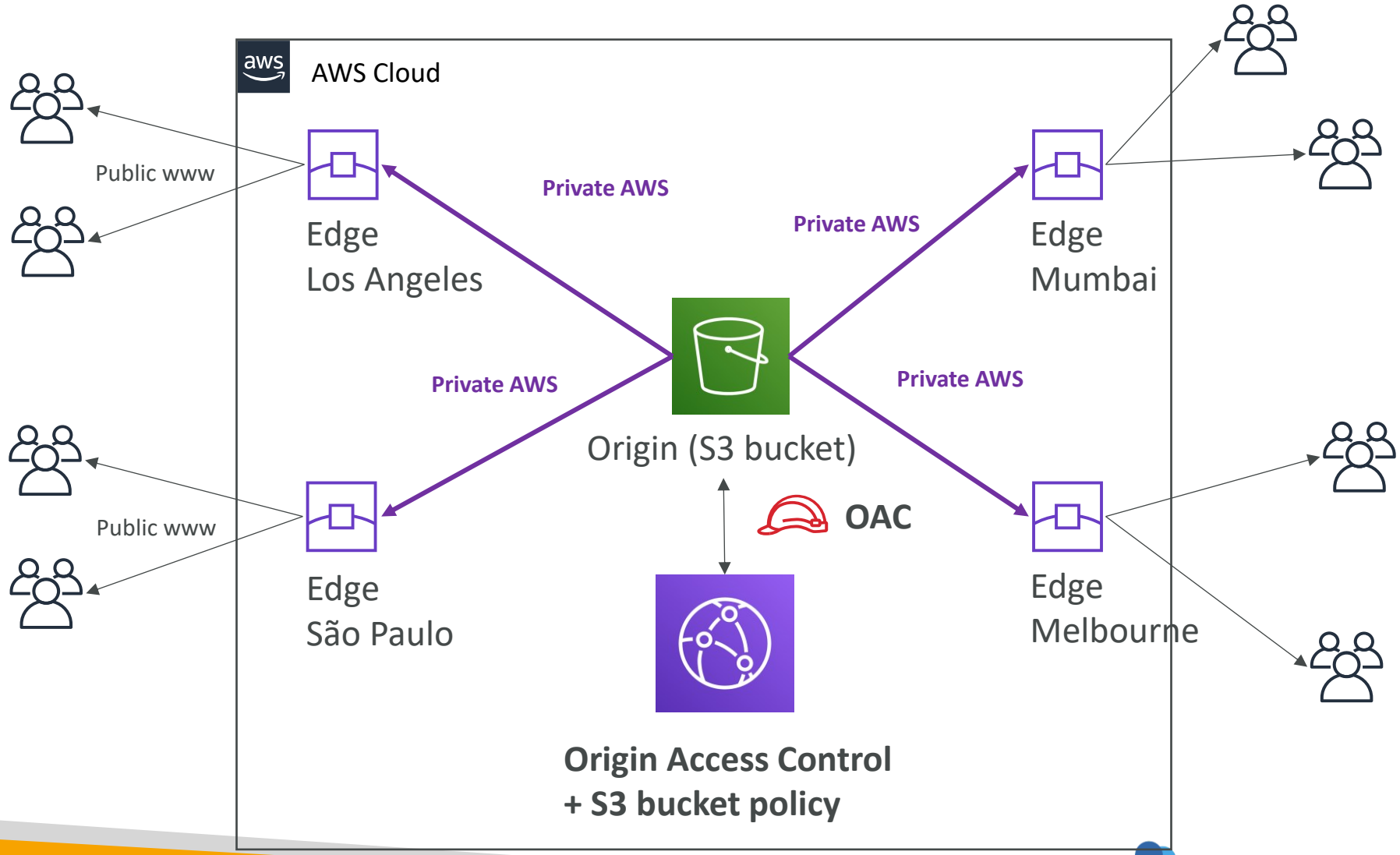
CloudFront – S3 as an Origin



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CloudFront vs S3 Cross Region Replication

- CloudFront:
  - Global Edge network
  - Files are cached for a TTL (maybe a day)
  - **Great for static content that must be available everywhere**

- S3 Cross Region Replication:
  - Must be setup for each region you want replication to happen
  - Files are updated in near real-time
  - Read only
  - **Great for dynamic content that needs to be available at low-latency in few regions**

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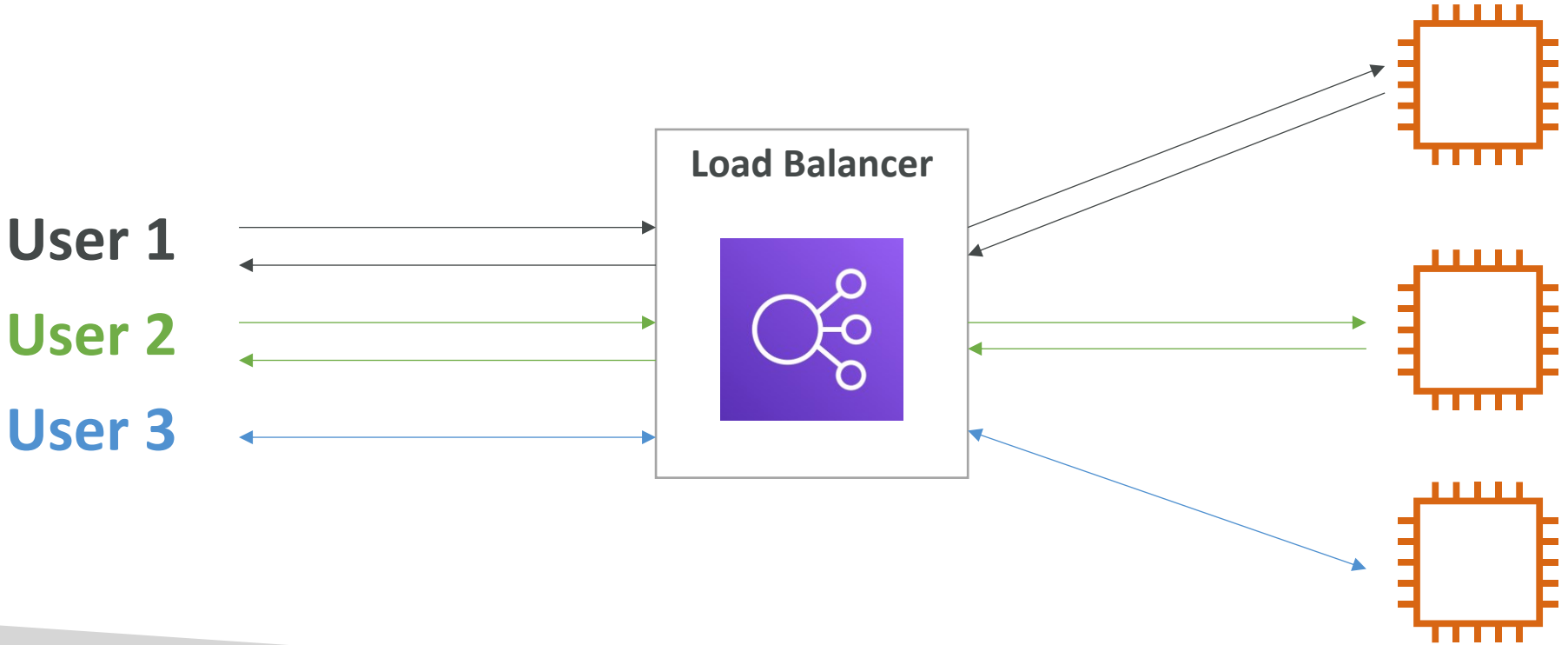
What is load balancing?

- Load balancers are servers that forward internet traffic to multiple servers (EC2 Instances) downstream.



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Why use a load balancer?

- Spread load across multiple downstream instances
- Expose a single point of access (DNS) to your application
- Seamlessly handle failures of downstream instances
- Do regular health checks to your instances
- Provide SSL termination (HTTPS) for your websites
- High availability across zones

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Why use an Elastic Load Balancer?

- An ELB (Elastic Load Balancer) is a **managed load balancer**
  - AWS guarantees that it will be working
  - AWS takes care of upgrades, maintenance, high availability
  - AWS provides only a few configuration knobs
- It costs less to setup your own load balancer but it will be a lot more effort on your end (maintenance, integrations)
- 4 kinds of load balancers offered by AWS:
  - Application Load Balancer (HTTP / HTTPS only) – Layer 7
  - Network Load Balancer (ultra-high performance, allows for TCP) – Layer 4
  - Gateway Load Balancer – Layer 3
  - Classic Load Balancer (retired in 2023) – Layer 4 & 7

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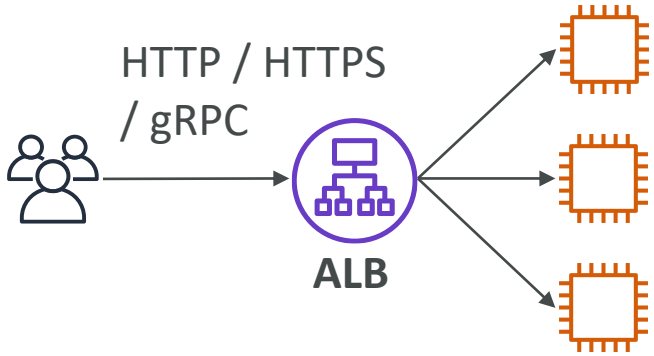
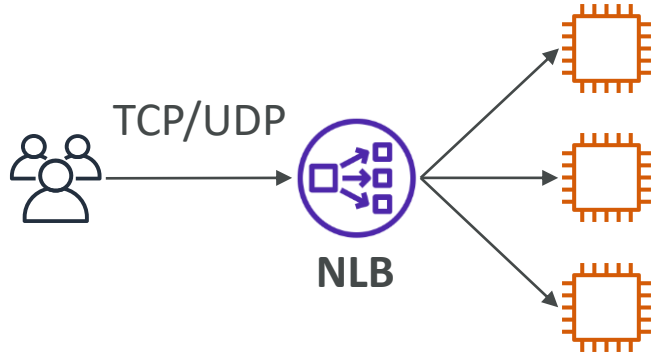
Application Load Balancer

- **HTTP / HTTPS / gRPC protocols (Layer 7)**
- **HTTP Routing** features
- **Static DNS (URL)**



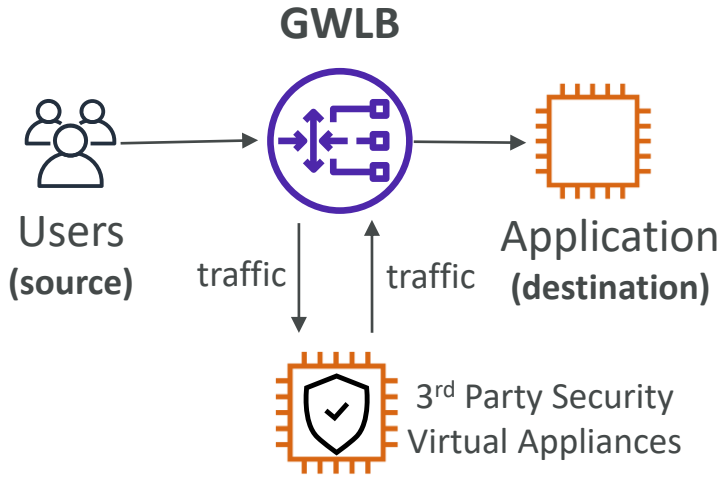
Network Load Balancer

- **TCP / UDP protocols (Layer 4)**
- **High Performance:** millions of request per seconds
- **Static IP** through Elastic IP



Gateway Load Balancer

- **GENEVE Protocol on IP Packets (Layer 3)**
- **Route Traffic to Firewalls** that you manage on EC2 Instances
- **Intrusion detection**



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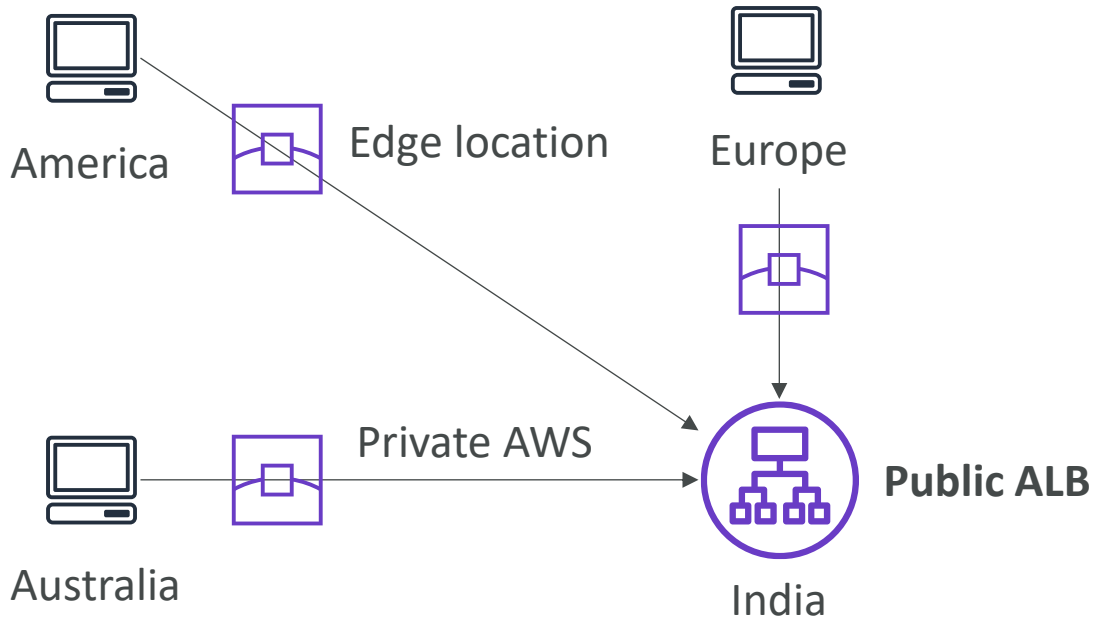
AWS Global Accelerator

- Leverage the AWS internal network to route to your application
- **2 Anycast IP** are created for your application
- The Anycast IP send traffic directly to Edge Locations
- The Edge locations send the traffic to your application



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AWS Global Accelerator

- Works with **Elastic IP, EC2 instances, ALB, NLB, public or private**
- Consistent Performance
  - Intelligent routing to lowest latency and fast regional failover
  - No issue with client cache (because the IP doesn't change)
  - Internal AWS network
- Health Checks
  - Global Accelerator performs a health check of your applications
  - Helps make your application global (failover less than 1 minute for unhealthy)
  - Great for disaster recovery (thanks to the health checks)
- Security
  - only 2 external IP need to be whitelisted
  - DDoS protection thanks to AWS Shield

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AWS Global Accelerator vs CloudFront

- They both use the AWS global network and its edge locations around the world
- Both services integrate with AWS Shield for DDoS protection.

- **CloudFront**
  - Improves performance for both cacheable content (such as images and videos)
  - Dynamic content (such as API acceleration and dynamic site delivery)
  - Content is served at the edge

- **Global Accelerator**
  - Improves performance for a wide range of applications over TCP or UDP
  - Proxying packets at the edge to applications running in one or more AWS Regions.
  - Good fit for non-HTTP use cases, such as gaming (UDP), IoT (MQTT), or Voice over IP
  - Good for HTTP use cases that require static IP addresses
  - Good for HTTP use cases that required deterministic, fast regional failover

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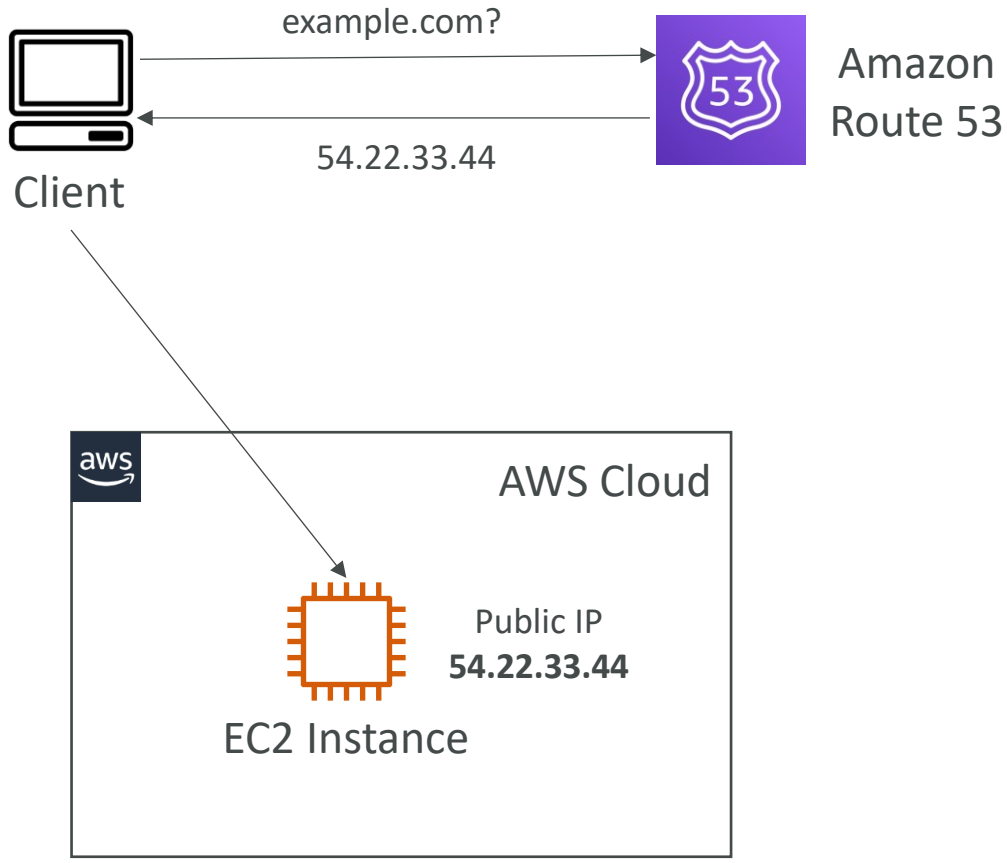
Amazon Route 53

- A highly available, scalable, fully managed and *Authoritative* DNS
  - Authoritative = the customer (you) can update the DNS records
- Route 53 is also a Domain Registrar
- Ability to check the health of your resources
- The only AWS service which provides 100% availability SLA
- Why Route 53? 53 is a reference to the traditional DNS port



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Route 53 – Records

- How you want to route traffic for a domain
- Each record contains:
  - **Domain/subdomain Name** – e.g., example.com
  - **Record Type** – e.g., A or AAAA
  - **Value** – e.g., 12.34.56.78
  - **Routing Policy** – how Route 53 responds to queries
  - **TTL** – amount of time the record cached at DNS Resolvers
- Route 53 supports the following DNS record types:
  - (must know) A / AAAA / CNAME / NS
  - (advanced) CAA / DS / MX / NAPTR / PTR / SOA / TXT / SPF / SRV

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Route 53 – Record Types

- **A** – maps a hostname to IPv4
- **AAAA** – maps a hostname to IPv6
- **CNAME** – maps a hostname to another hostname
  - The target is a domain name which must have an A or AAAA record
  - Can't create a CNAME record for the top node of a DNS namespace (Zone Apex)
  - Example: you can't create for example.com, but you can create for www.example.com
- **NS** – Name Servers for the Hosted Zone
  - Control how traffic is routed for a domain

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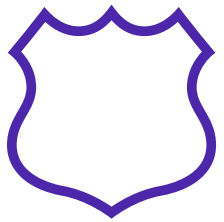
Route 53 – Hosted Zones



- A container for records that define how to route traffic to a domain and its subdomains

- **Public Hosted Zones –** contains records that specify how to route traffic on the Internet (public domain names) application1.mypublicdomain.com
- **Private Hosted Zones –** contain records that specify how you route traffic within one or more VPCs (private domain names) application1.company.internal

- You pay $0.50 per month per hosted zone

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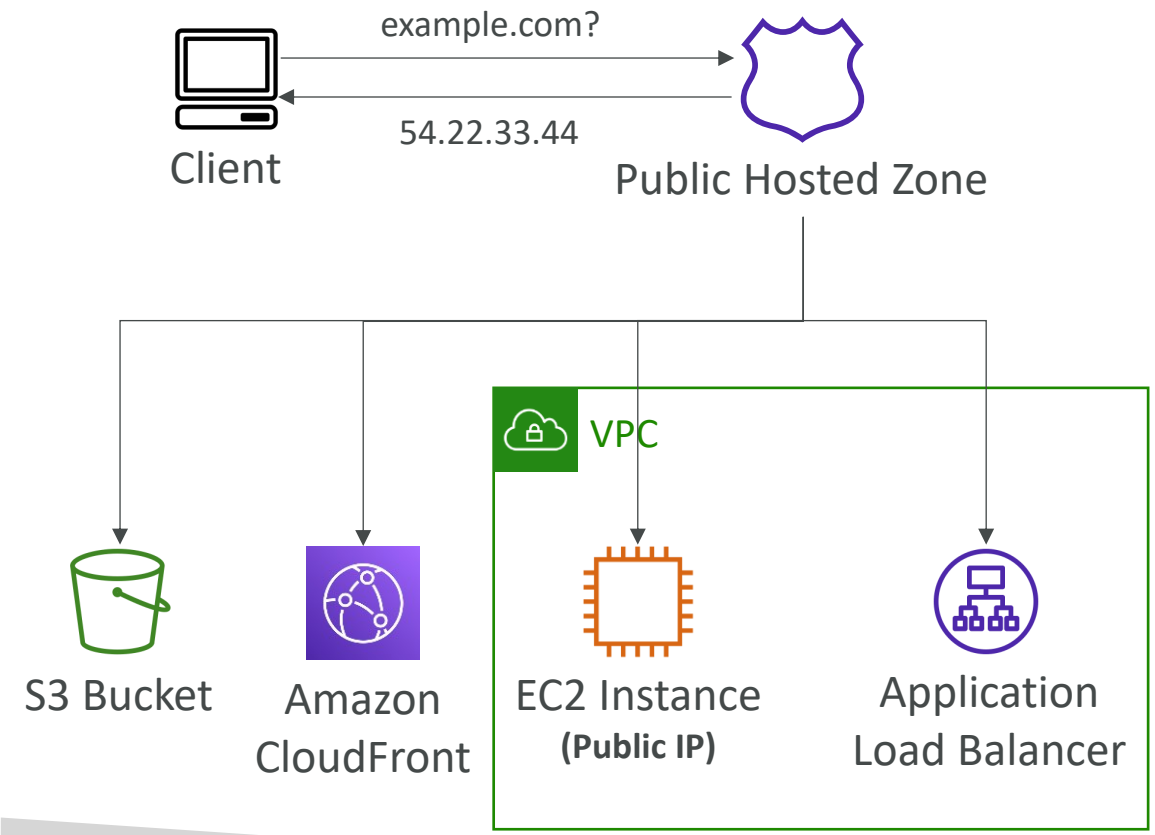
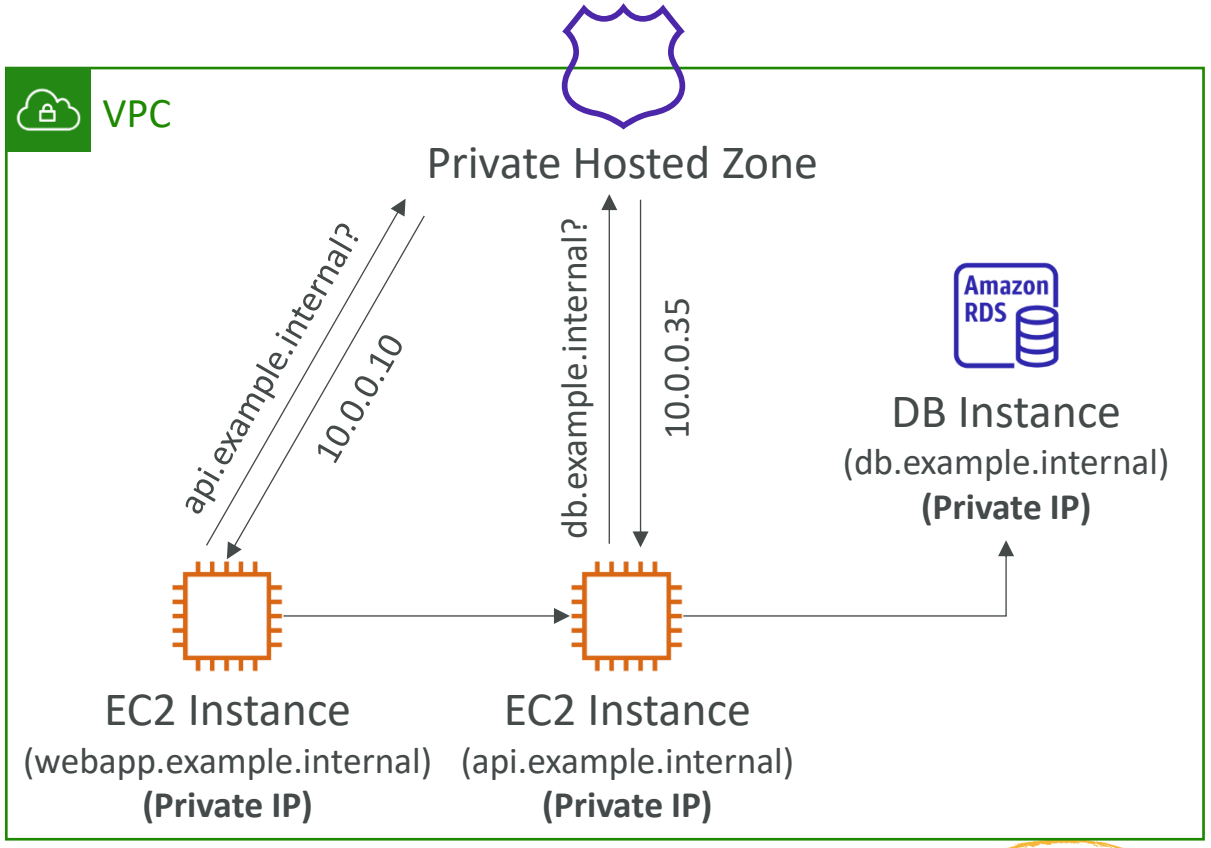
Route 53 – Public vs. Private Hosted Zones

Public Hosted Zone



Private Hosted Zone



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Other Services You Should Know: Storage

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What's an EBS Volume?

- An EBS (Elastic Block Store) Volume is a network drive you can attach to your instances while they run
- It allows your instances to persist data, even after their termination
- **They can only be mounted to one instance at a time** (at the CCP level)
- They are bound to **a specific availability zone**

- Analogy: Think of them as a "network USB stick"

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EBS Volume

- It's a network drive (i.e. not a physical drive)
  - It uses the network to communicate the instance, which means there might be a bit of latency
  - It can be detached from an EC2 instance and attached to another one quickly
- It's locked to an Availability Zone (AZ)
  - An EBS Volume in us-east-1a cannot be attached to us-east-1b
  - To move a volume across, you first need to snapshot it
- Have a provisioned capacity (size in GBs, and IOPS)
  - You get billed for all the provisioned capacity
  - You can increase the capacity of the drive over time

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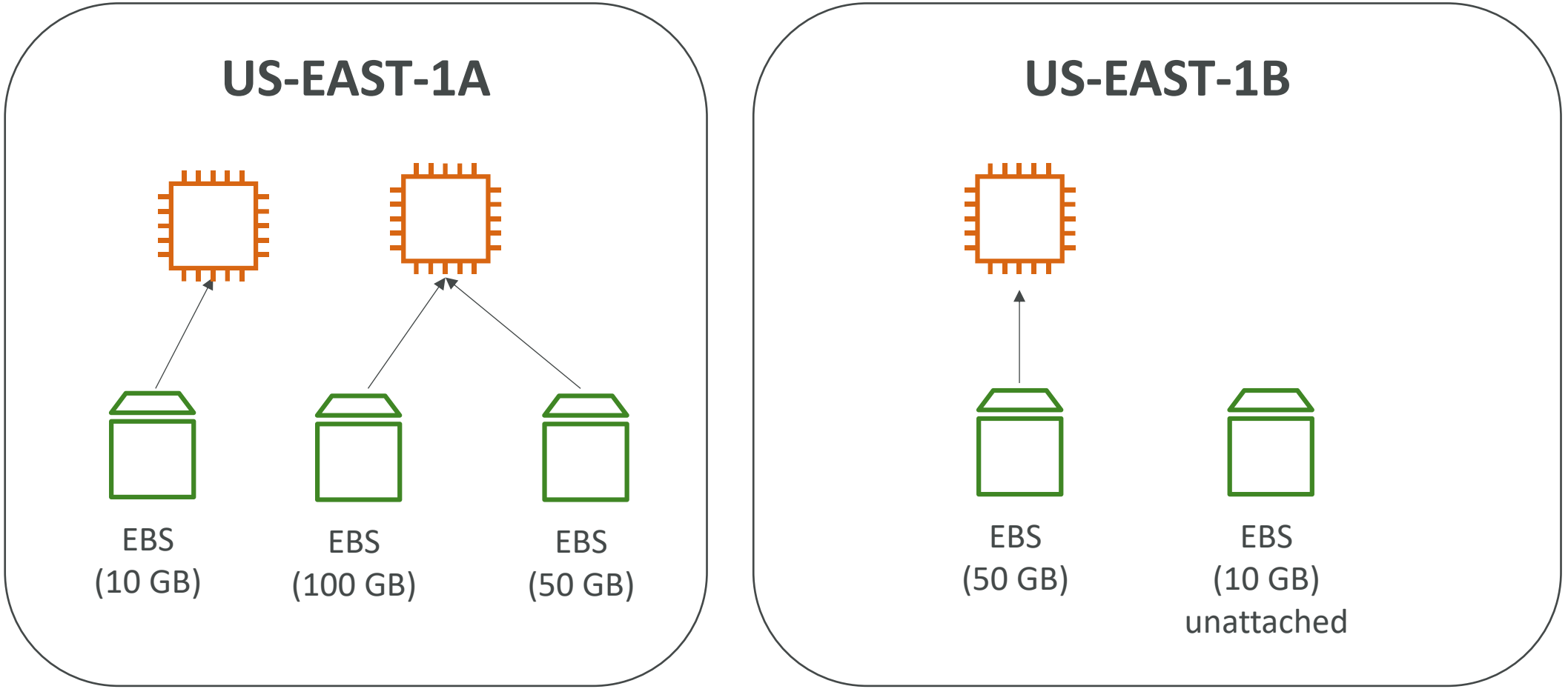
EBS Volume - Example



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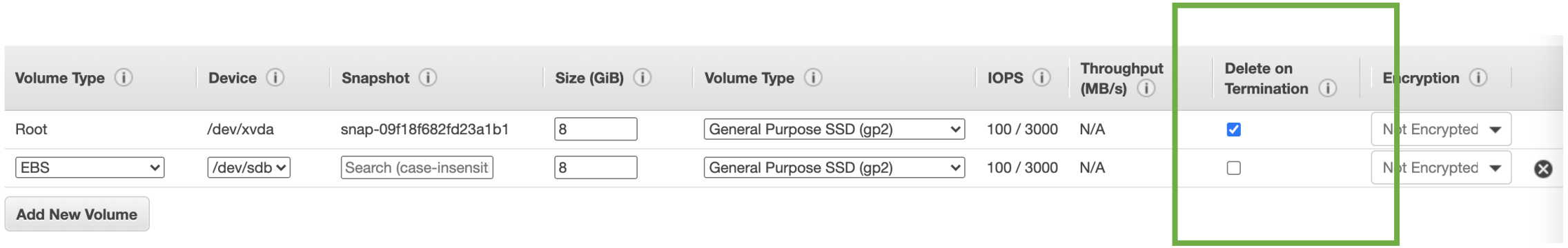
EBS – Delete on Termination attribute



- Controls the EBS behaviour when an EC2 instance terminates
  - By default, the root EBS volume is deleted (attribute enabled)
  - By default, any other attached EBS volume is not deleted (attribute disabled)
- This can be controlled by the AWS console / AWS CLI
- **Use case: preserve root volume when instance is terminated**

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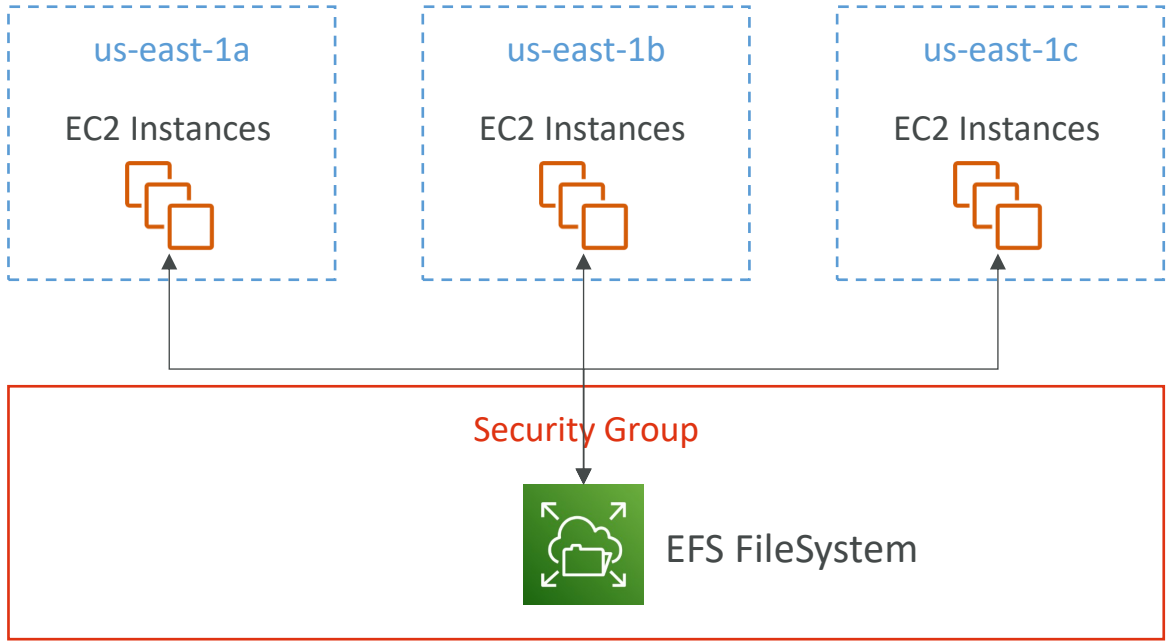
Amazon EFS – Elastic File System

- Managed NFS (network file system) that can be mounted on many EC2
- EFS works with EC2 instances in multi-AZ
- Highly available, scalable, expensive (3x gp2), pay per use



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Amazon EFS – Elastic File System

- Use cases: content management, web serving, data sharing, Wordpress
- Uses NFSv4.1 protocol
- Uses security group to control access to EFS
- **Compatible with Linux based AMI (not Windows)**
- Encryption at rest using KMS

- POSIX file system (~Linux) that has a standard file API
- File system scales automatically, pay-per-use, no capacity planning!

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EFS – Performance & Storage Classes

- **EFS Scale**
  - 1000s of concurrent NFS clients, 10 GB+ /s throughput
  - Grow to Petabyte-scale network file system, automatically
- **Performance Mode (set at EFS creation time)**
  - **General Purpose (default) –** latency-sensitive use cases (web server, CMS, etc…)
  - **Max I/O –** higher latency, throughput, highly parallel (big data, media processing)
- **Throughput Mode**
  - **Bursting –** 1 TB = 50MiB/s + burst of up to 100MiB/s
  - **Provisioned –** set your throughput regardless of storage size, ex: 1 GiB/s for 1 TB storage
  - **Elastic –** automatically scales throughput up or down based on your workloads
    - Up to 3GiB/s for reads and 1GiB/s for writes
    - Used for unpredictable workloads

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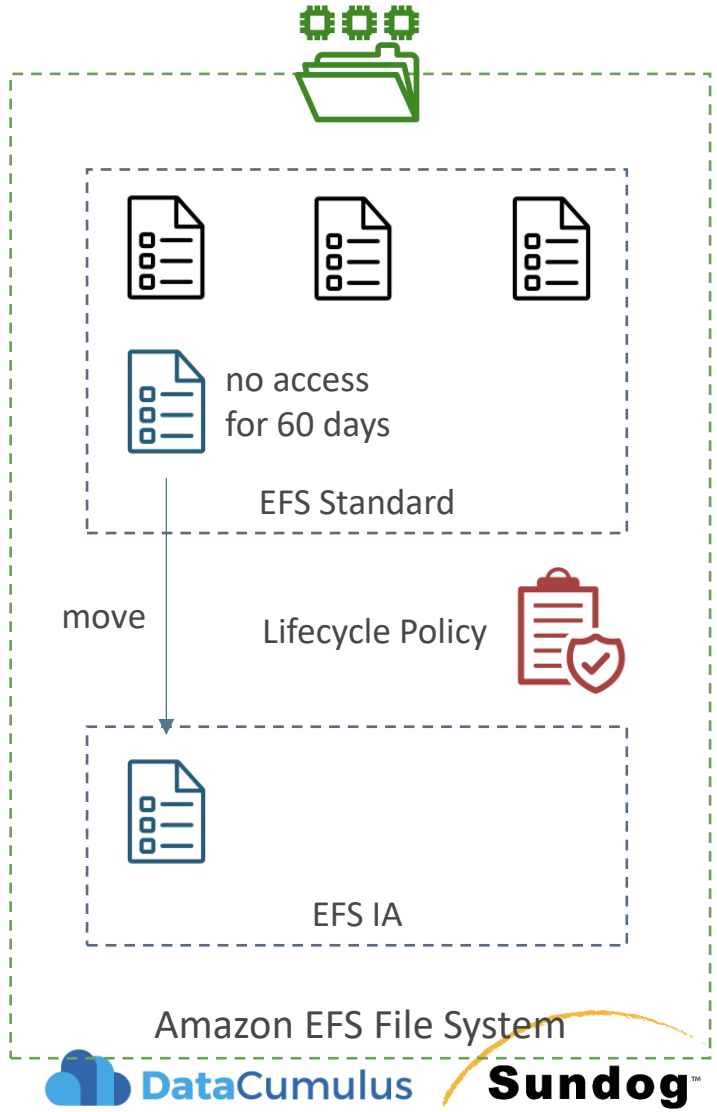
EFS – Storage Classes

- **Storage Tiers (lifecycle management feature – move file after N days)**
  - **Standard**: for frequently accessed files
  - **Infrequent access (EFS-IA):** cost to retrieve files, lower price to store.
  - **Archive:** rarely accessed data (few times each year), 50% cheaper
  - Implement **lifecycle policies** to move files between storage tiers
- **Availability and durability**
  - Standard: Multi-AZ, great for prod
  - One Zone: One AZ, great for dev, backup enabled by default, compatible with IA (EFS One Zone-IA)
- Over 90% in cost savings



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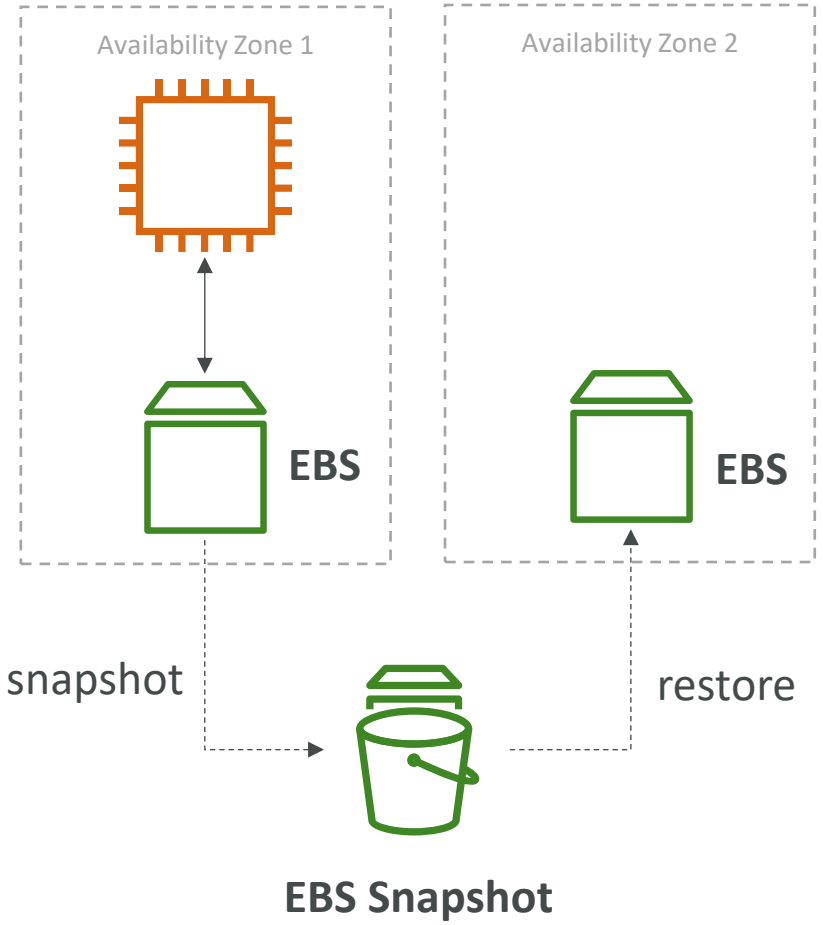
EBS vs EFS – Elastic Block Storage

- EBS volumes…
  - one instance (except multi-attach io1/io2)
  - are locked at the Availability Zone (AZ) level
  - gp2: IO increases if the disk size increases
  - gp3 & io1: can increase IO independently
- To migrate an EBS volume across AZ
  - Take a snapshot
  - Restore the snapshot to another AZ
  - EBS backups use IO and you shouldn't run them while your application is handling a lot of traffic
- Root EBS Volumes of instances get terminated by default if the EC2 instance gets terminated. (you can disable that)



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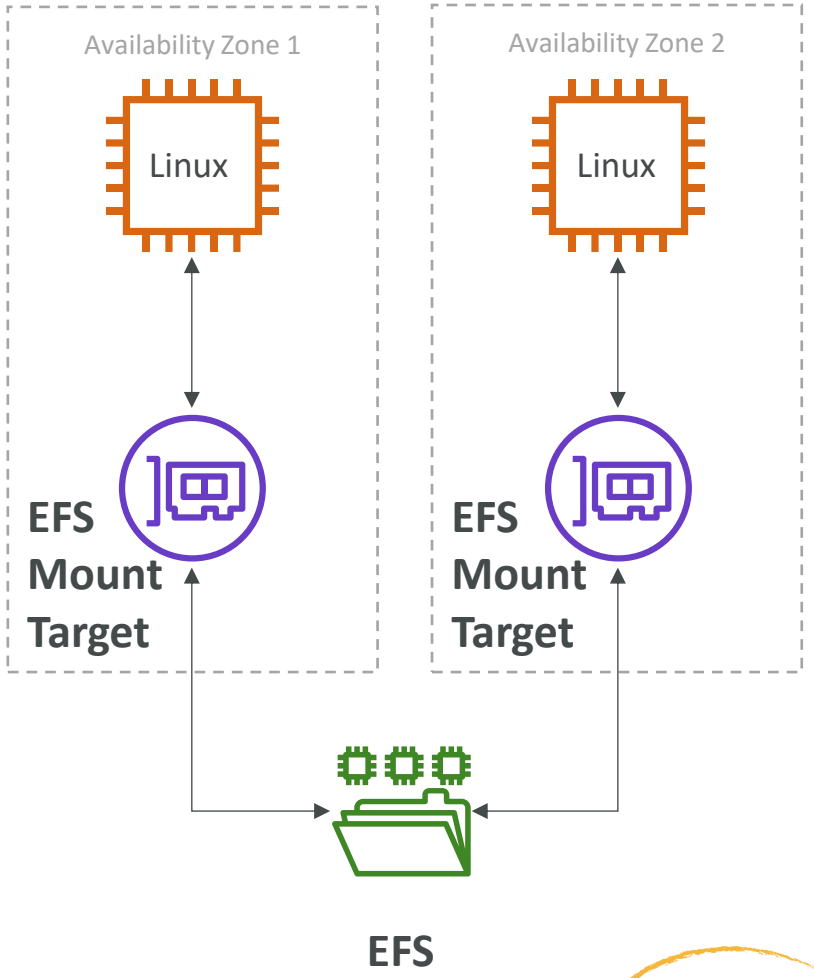
EBS vs EFS – Elastic File System

- Mounting 100s of instances across AZ
- EFS share website files (WordPress)
- Only for Linux Instances (POSIX)

- EFS has a higher price point than EBS
- Can leverage Storage Tiers for cost savings

- Remember: EFS vs EBS vs Instance Store



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Making GenAI Architectural Decisions

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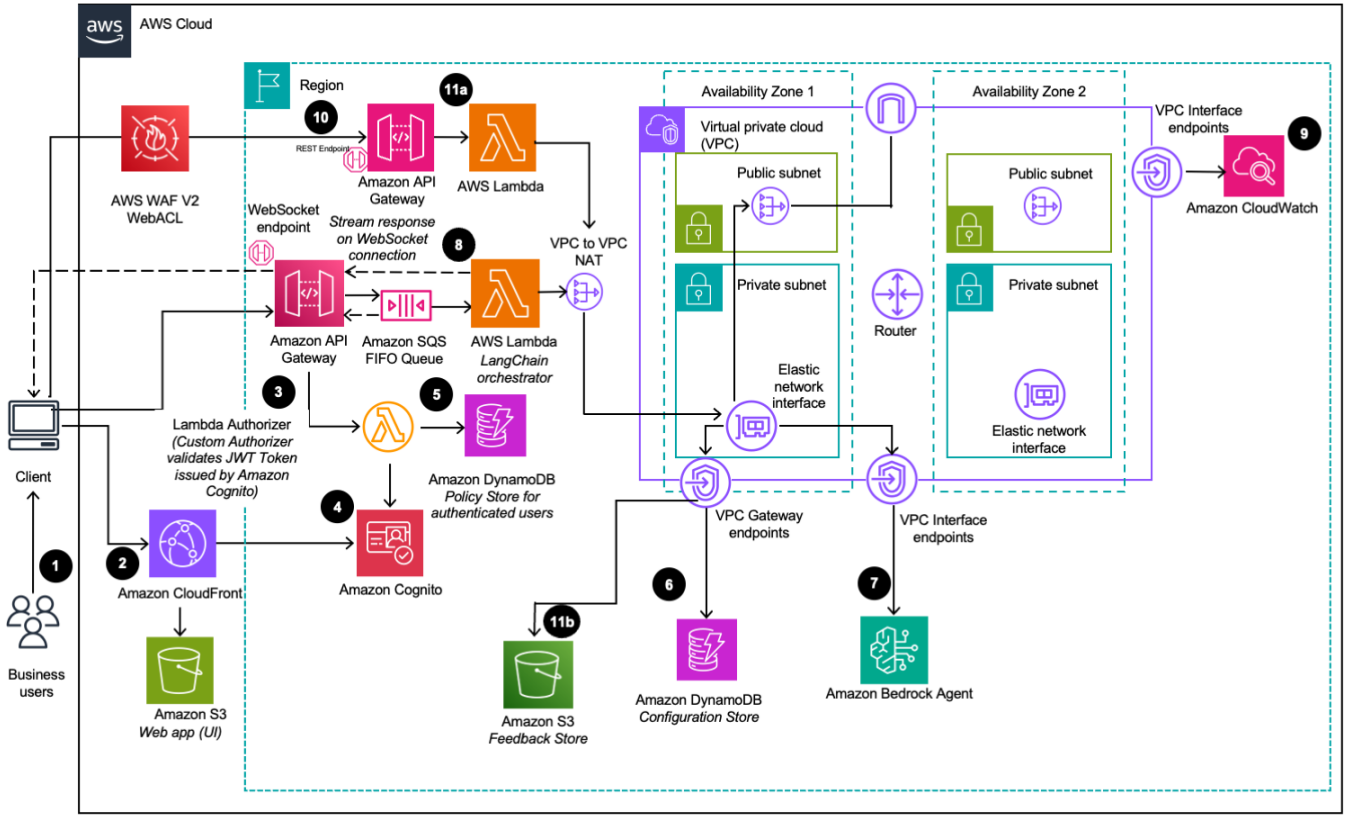
Production systems are more than just Bedrock



Source: "Generative AI Application Builder on AWS"

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Choosing a Vector Store

Option	Best fit	Why you'd choose it	Cost profile	Ops burden	Security & compliance	Key limitations / gotchas
OpenSearch Service (managed cluster)	Custom RAG at scale where you want control	Mature search features + strong control over performance tuning, index settings, networking	$$–$$$ (always-on)	Medium	VPC patterns, fine-grained access controls (varies by config)	You pay for provisioned capacity; tuning matters
OpenSearch Serverless	Spiky/variable query traffic; low-ops	Don't manage clusters; good default for "serverless RAG"	$–$$ (usage-based)	Low	IAM-centric patterns; fewer knobs than provisioned	Less tuning control; performance predictability can vary
Amazon Kendra	Enterprise document search with permissions	Built for enterprise content + **document-level ACL** enforcement (huge if SharePoint/Confluence/Box)	$$$	Very low	Strong enterprise security posture	Less "vector-db tinkering"; cost can be high for large corp deployments
Aurora PostgreSQL + pgvector	"RAG + relational data" in one system	You already have relational + metadata filters + transactions; add vector similarity in the same DB	$$ (depends on size/HA)	Medium	Strong compliance story + DB governance; works in GovCloud per AWS	Needs pgvector enabled + correct version; scaling and ANN tuning are your job
Neptune Analytics	"RAG + graph relationships" and hybrid reasoning	Best when **relationships matter** (fraud rings, supply chain, IAM graphs) + vector search inside traversals	$$–$$$	Medium	Good for regulated industries; graph governance patterns	It's an analytics engine; vector loading has constraints (e.g., not from Neptune DB snapshots)
Amazon S3 Vectors	Massive-scale, cost-optimized vector storage (especially for S3 data)	Purpose-built vector store in S3; claims big cost reduction + simple "no infra" operations, but potentially high latency	$ (very cost-driven)	Very low	Inherits S3 durability/availability model; IAM-based access patterns	Metadata / filtering limits (e.g., per-vector metadata limits when used w/ Bedrock KB) – Can be very slow

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Choosing a Vector Store

If the question mentions…

- SharePoint / Confluence / document permissions / “must respect existing ACLs” → **Kendra**
- Graph relationships (“who knows who”, fraud rings, lineage, dependency graphs) → **Neptune Analytics**
- Already on Postgres + need joins/transactions + vector search → **Aurora + pgvector**
- Huge vector corpus + cost pressure + vectors mostly tied to S3 objects → **S3 Vectors**
- Need full search platform + lots of tuning control → **OpenSearch managed**
- Unpredictable traffic + “minimize ops” → **OpenSearch Serverless**

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Choosing a Vector Store

Cost-first: S3 Vectors (massive corpora, infrequent queries)

Permissions-first: Kendra

Relationship-first: Neptune Analytics

SQL-first: Aurora + pgvector

Search-first: OpenSearch (managed/serverless)

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OpenSearch vs. OpenSearch Serverless

Dimension	OpenSearch	OpenSearch Serverless
Billing	Always-on nodes	Per-request + storage
Tuning	Full control (, shards)	Limited
Latency	Predictable	Variable
Scaling	Manual/Auto	Automatic
Exam Bias	“Need fine-grained control”	“Unpredictable traffic”

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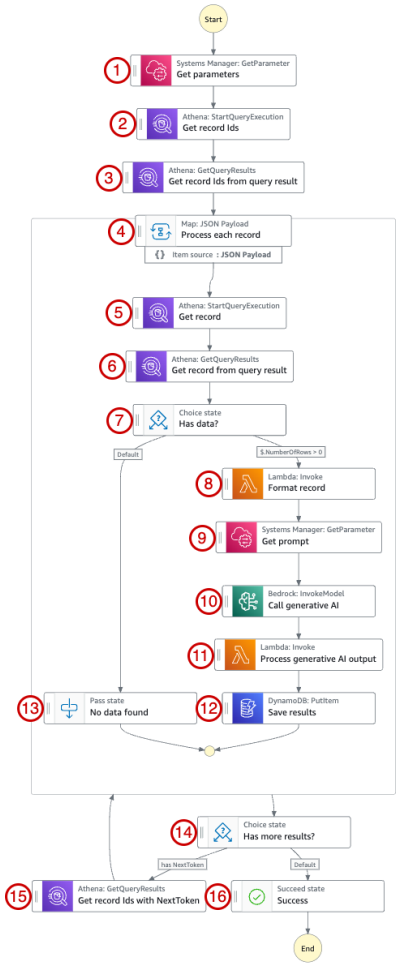
Orchestration: When do I need Step Functions?

- Auditable state transitions
- Retry & failure isolation
- Explicit approval steps
- Human approval
- Serverless is nice, but Step Functions aren't the only serverless choice.



Source: [Orchestrating Clinical Generative AI Workflows Using AWS Step Functions]

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Example: Internal Legal Q&A Chatbot

- Internal use only
  - Not worrying about CloudFront, WAF
- Kendra is appropriate here for Bedrock retrieval
  - Due to document-level access control required for Legal
  - And for Sharepoint integration
- Cognito authenticates users, user context passed through to Kendra for auth
- Bedrock used for retrieval using Kendra, generation of result

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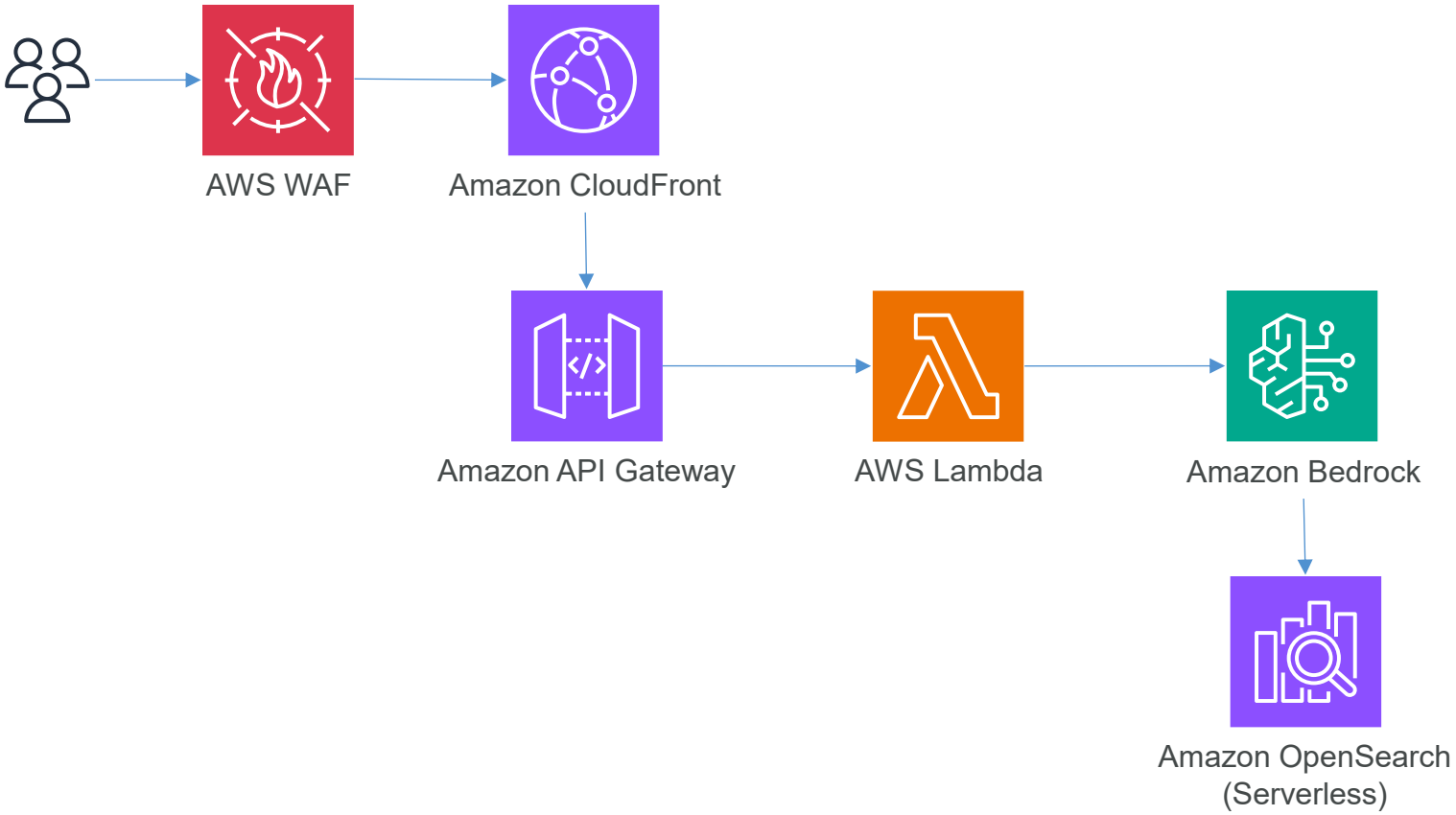
Example: Customer-facing Chatbot



- External-facing systems need WAF, CloudFront, API Gateway
- Lambda can call Bedrock, referencing data in OpenSearch
  - Maybe Lambda also does other DB queries here
- If personal / private info is involved, Cognito may also be a factor
- Requirements may also involve guardrails
- Think also about ingestion
  - S3 -> EventBridge -> Lambda -> Bedrock (for embeddings) -> OpenSearch?

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Exam Preparation

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More Prep Resources

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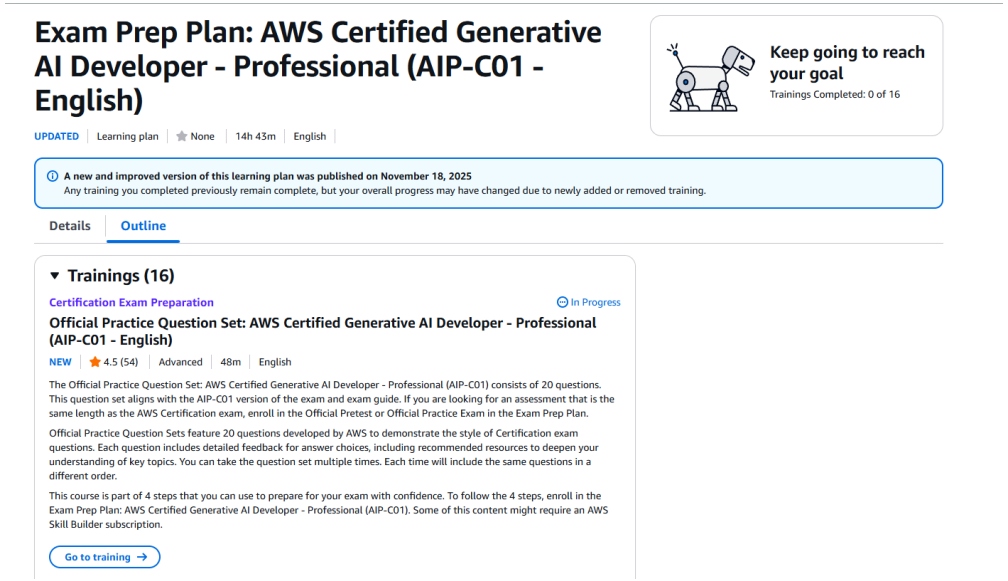
AWS Skill Builder

- The free resources are likely sufficient
  - “Standard Exam Prep Plan” includes Amazon’s own training resources
  - “Official Practice Question Set” – 20-question practice test
- https://skillbuilder.aws/learning-plan/9VXVGYT38G/



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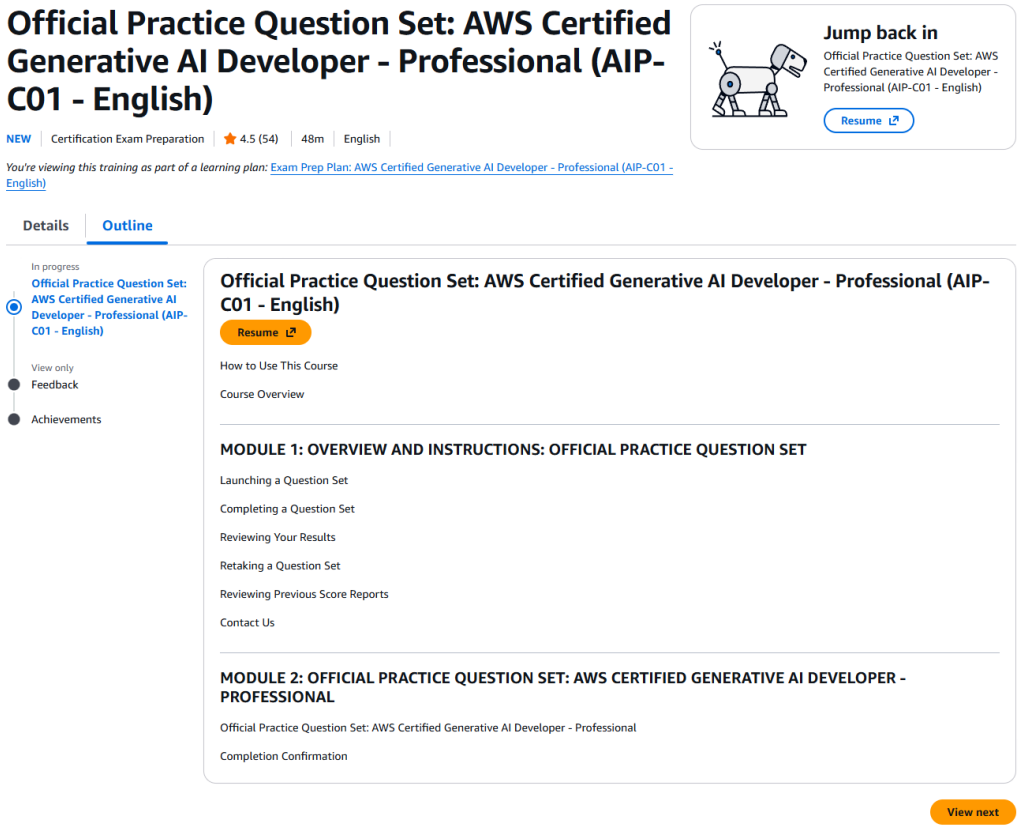
Practice Exams (including Amazon's)



There is also a full-length practice exam in the subscription-based materials.

And don't forget the one in this course!

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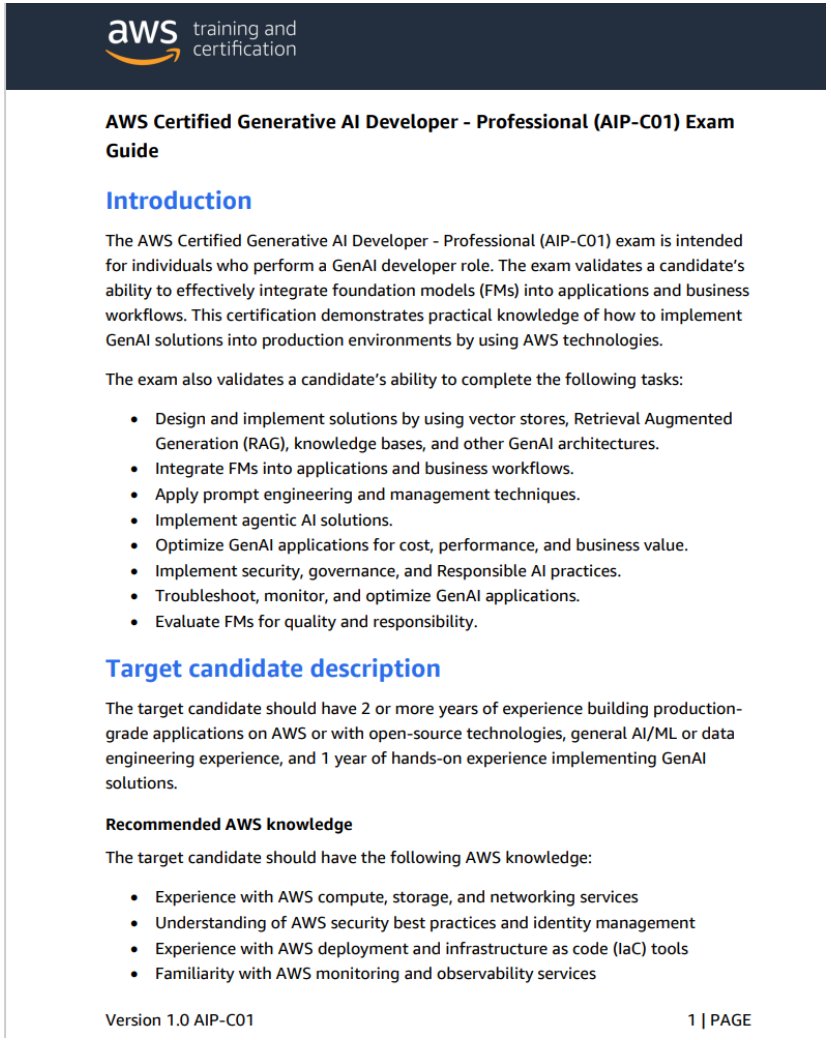
Amazon’s Exam Guide

https://d1.awsstatic.com/onedam/marketing-channels/website/aws/en_US/certification/approved/pdfs/docs-aip/AWS-Certified-Generative-AI-Developer-Pro_Exam-Guide.pdf



aws training and certification

AWS Certified Generative AI Developer - Professional (AIP-C01) Exam Guide

Introduction

The AWS Certified Generative AI Developer - Professional (AIP-C01) exam is intended for individuals who perform a GenAI developer role. The exam validates a candidate’s ability to effectively integrate foundation models (FMs) into applications and business workflows. This certification demonstrates practical knowledge of how to implement GenAI solutions into production environments by using AWS technologies.

The exam also validates a candidate’s ability to complete the following tasks:

- Design and implement solutions by using vector stores, Retrieval Augmented Generation (RAG), knowledge bases, and other GenAI architectures.
- Integrate FMs into applications and business workflows.
- Apply prompt engineering and management techniques.
- Implement agentic AI solutions.
- Optimize GenAI applications for cost, performance, and business value.
- Implement security, governance, and Responsible AI practices.
- Troubleshoot, monitor, and optimize GenAI applications.
- Evaluate FMs for quality and responsibility.

Target candidate description

The target candidate should have 2 or more years of experience building production-grade applications on AWS or with open-source technologies, general AI/ML or data engineering experience, and 1 year of hands-on experience implementing GenAI solutions.

Recommended AWS knowledge

The target candidate should have the following AWS knowledge:

- Experience with AWS compute, storage, and networking services
- Understanding of AWS security best practices and identity management
- Experience with AWS deployment and infrastructure as code (IaC) tools
- Familiarity with AWS monitoring and observability services

Version 1.0 AIP-C01 1 | PAGE

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Consider taking the Certified AI Practitioner exam first

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What to Expect

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Sitting for the exam

- 3 hours! (170 minutes technically)
  - But you'll probably only need a little over 2.
  - 75 questions (85 during beta, with 205 minutes)
  - TAKE THE TIME TO FULLY UNDERSTAND THEM
- Use the flag capability
  - You'll have time to go back and double check things.
- You can't bring anything into the room
  - You'll be given note paper or dry-erase board, earplugs, and that's it.
  - If testing at home, your room needs to be pretty much empty
- Schedule smartly
  - As soon as possible after you've mastered the practice exams
  - Choose a time of day that matches your highest energy level – you need stamina
  - Remember different testing providers may have different availability
  - Testing at a testing center can actually be better
    - Don't have to worry about having your environment validated
    - Don't have to worry about technical glitches

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Prepare

- Use the bathroom before you head in
- Eat something (but not right beforehand)
- Get a good night's sleep
- Review things you need to memorize before going in
  - Evaluation metrics
  - Services that are new to you
- Be alert
  - Whatever works for you…
- Make sure you're ready
  - You've got $300 on the line ($150 for the beta)
  - Take the practice exams! Repeatedly!

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Strategies

- Don’t rush
  - For each question, take the time to understand:
    - What you are optimizing for
    - Requirements
    - The system as a whole
- Your pace should be about 2 – 2 ½ minutes per question
  - If you’re taking longer, make your best guess, flag it and come back to it later
- Use the process of elimination
  - Even if you don’t know the correct answer, you can often eliminate several.
- Keep calm
  - You’re not going to get 100%. You don’t have to.

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Your AWS Certification journey

Foundational

Knowledge-based certification for
foundational understanding of AWS Cloud.
No prior experience needed.

Associate

Role-based certifications that showcase your knowledge
and skills on AWS and build your credibility as an AWS Cloud professional.
Prior cloud and/or strong on-premises IT experience recommended.

Professional

Role-based certifications that validate advanced skills
and knowledge required to design secure, optimized,
and modernized applications and to automate processes on AWS.
2 years of prior AWS Cloud experience recommended.

Specialty

Dive deeper and position yourself as a trusted advisor to your
stakeholders and/or customers in these strategic areas.
Refer to the exam guides on the exam pages for recommended experience.

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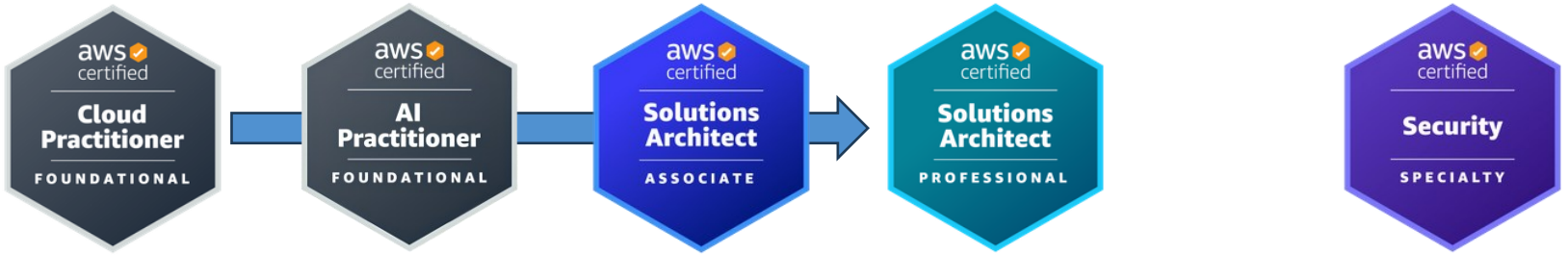
AWS Certification Paths – Architecture

Architecture

Solutions Architect

Design, develop, and manage cloud infrastructure and assets, work with DevOps to migrate applications to the cloud



optional for IT/ cloud professionals

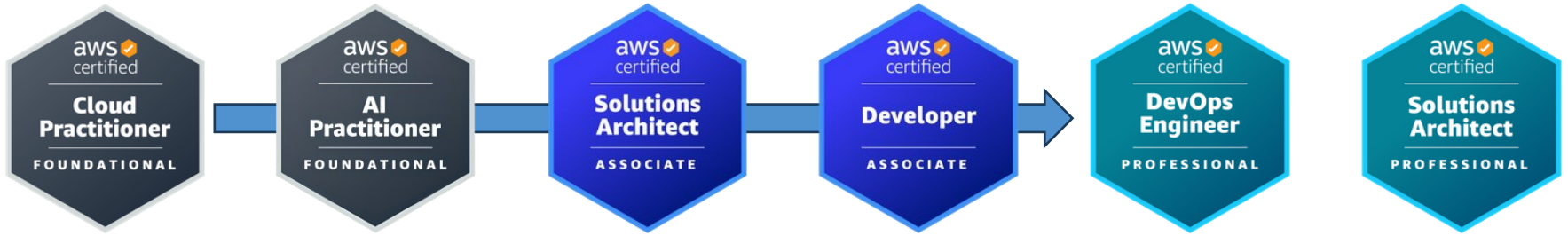
recommended for IT/cloud professionals to leverage AI

Dive Deep

Architecture

Application Architect

Design significant aspects of application architecture including user interface, middleware, and infrastructure, and ensure enterprise-wide scalable, reliable, and manageable systems



optional for IT/ cloud professionals

recommended for IT/cloud professionals to leverage AI

Dive Deep

https://d1.awsstatic.com/training-and-certification/docs/AWS_certification_paths.pdf

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AWS Certification Paths – Operations

Operations

Systems Administrator

Install, upgrade, and maintain computer components and software, and integrate automation processes

optional for IT/ cloud professionals

Dive Deep

Operations

Cloud Engineer

Implement and operate an organization's networked computing infrastructure and Implement security systems to maintain data safety

optional for IT/ cloud professionals

Dive Deep

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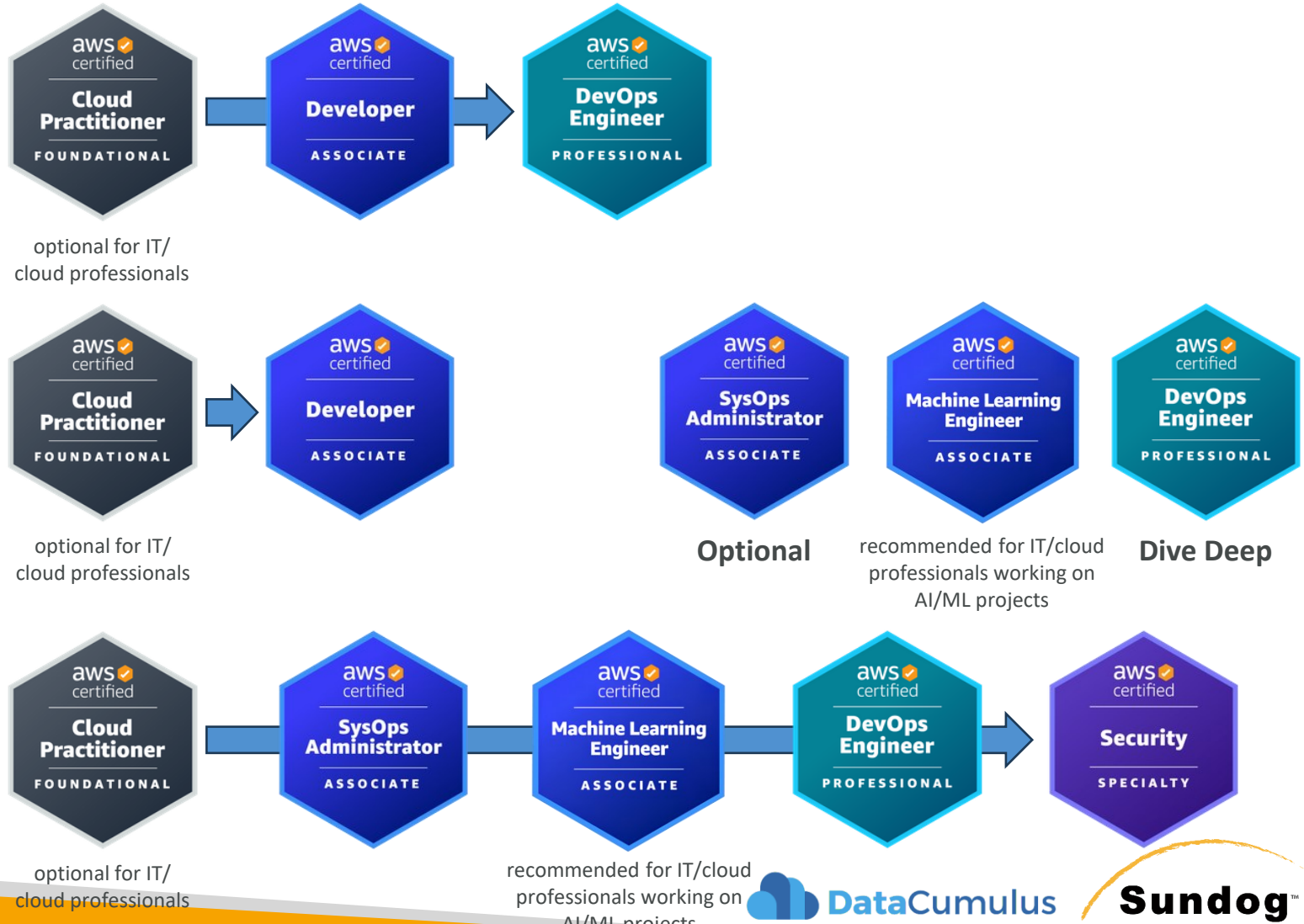
AWS Certification Paths – DevOps

DevOps
Test Engineer
Embed testing and quality best practices for software development from design to release, throughout the product life cycle

DevOps
Cloud DevOps Engineer
Design, deployment, and operations of large-scale global hybrid cloud computing environment, advocating for end-to-end automated CI/CD DevOps pipelines

DevOps
DevSecOps Engineer
Accelerate enterprise cloud adoption while enabling rapid and stable delivery of capabilities using CI/CD principles, methodologies, and technologies



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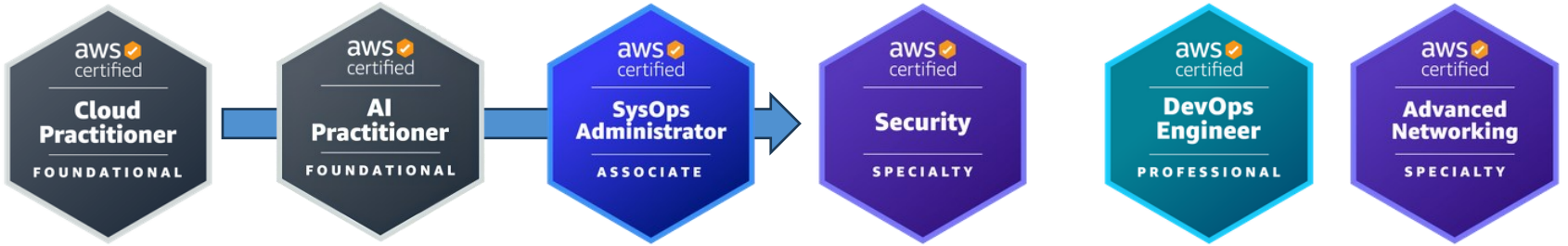
AWS Certification Paths – Security

Security

Cloud Security Engineer

Design computer security architecture and develop detailed cyber security designs. Develop, execute, and track performance of security measures to protect information



optional for IT/ cloud professionals

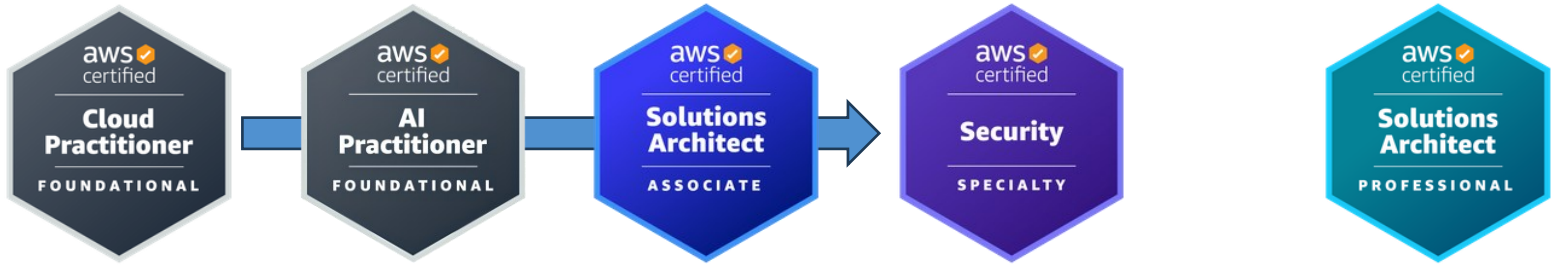
recommended for IT/cloud professionals to secure AI/ML systems

Dive Deep

Security

Cloud Security Architect

Design and implement enterprise cloud solutions applying governance to identify, communicate, and minimize business and technical risks



optional for IT/ cloud professionals

recommended for IT/cloud professionals to secure AI/ML systems

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AWS Certification Paths – Development & Networking

Development

Software Development Engineer

Develop, construct, and maintain software across platforms and devices

optional for IT/ cloud professionals

recommended for IT/cloud professionals to leverage AI

Networking

Network Engineer

Design and implement computer and information networks, such as local area networks (LAN), wide area networks (WAN), intranets, extranets, etc.

optional for IT/ cloud professionals

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AWS Certification Paths – Data Analytics & AI/ML

Data Analytics

Cloud Data Engineer

Automate collection and processing of structured/semi-structured data and monitor data pipeline performance

optional for IT/ cloud professionals

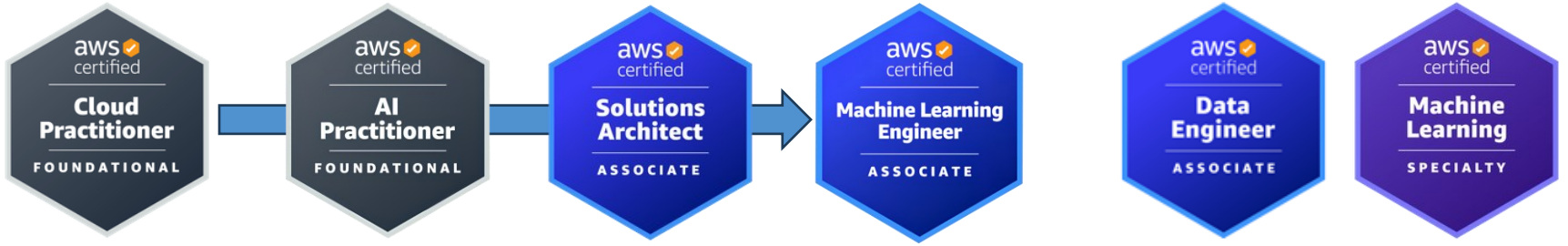
recommended for IT/cloud professionals working on AI/ML projects

Dive Deep

AI/ML

Machine Learning Engineer

Research, build, and design artificial intelligence (AI) systems to automate predictive models, and design machine learning systems, models, and schemes



optional for IT/ cloud professionals

optional for AI/ML professionals

Dive Deep

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AWS Certification Paths – AI/ML

AI/ML

Prompt Engineer

Design, test, and refine text prompts to optimize the performance of AI language models

AI/ML

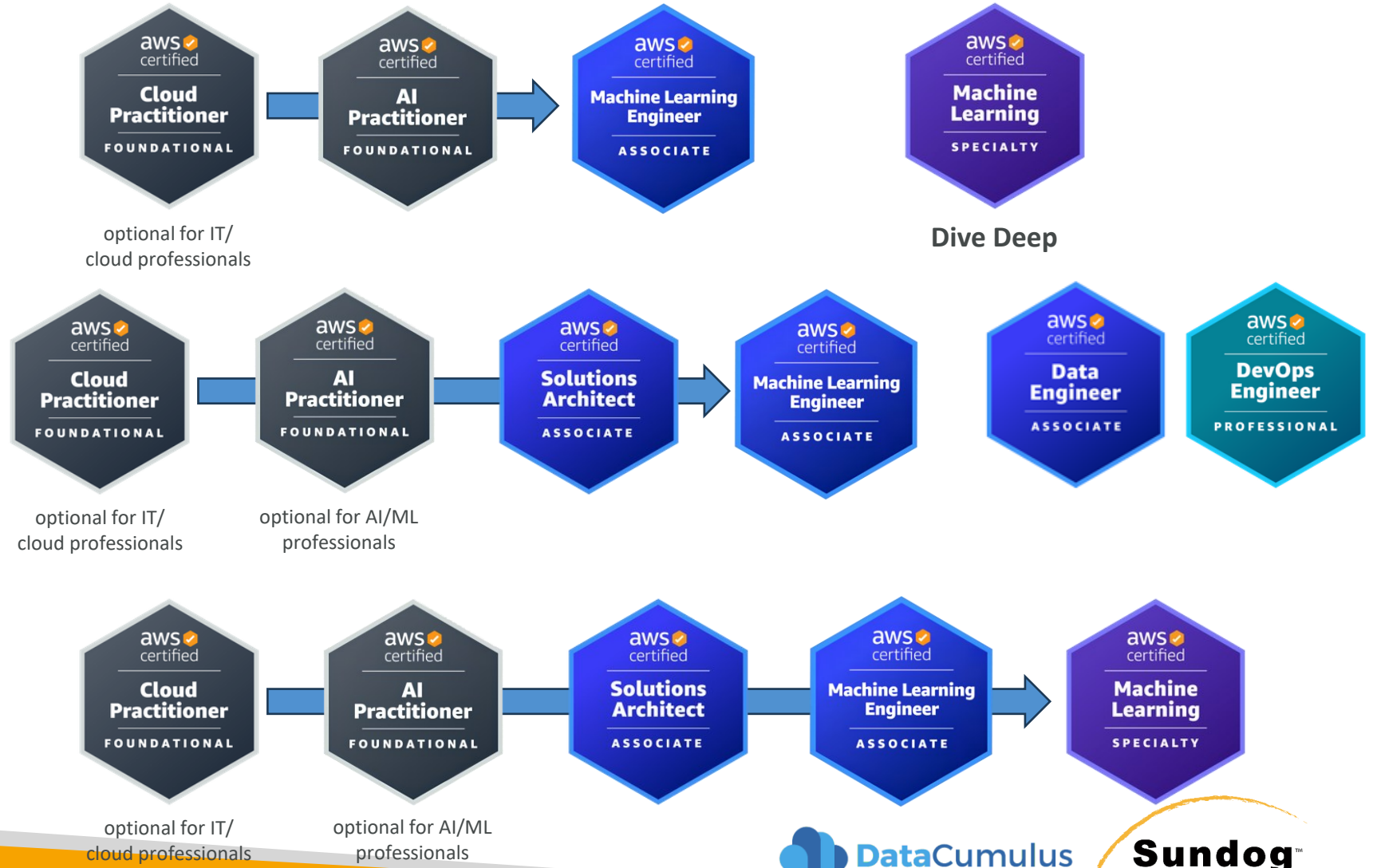
Machine Learning Ops Engineer

Build and maintain AI and ML platforms and infrastructure. Design, implement, and operationally support AI/ML model activity and deployment infrastructure

AI/ML

Data Scientist

Develop and maintain AI/ML models to solve business problems. Train and fine tune models and evaluate their performance



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Congratulations

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